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GENERAL KNOWLEDGE 2022

Manohar Pandey

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Current Affairs



NATIONAL

UNION BUDGET 2021-22

Finance Minister Nirmala Sithraman presented the first paperless Union Budget in India's history on February 1, 2021 in view of the ongoing COVID-19 pandemic. Union Budget 2021-22 lays down the vision for Atmanirbhar Bharat or Self reliant India.

The Budget proposals this year are based on six pillars namely Health and Wellbeing, Physical & Financial Capital, and Infrastructure, Inclusive Development for Aspirational India, Reinvigorating Human Capita, Innovation and R&D and Minimum Government and Maximum Governance

Focused Points

- Estimates of ₹ 34.83 lakh crore for expenditure in 2021-2022 including ₹ 5.5 lakh crore as capital expenditure, an increase of 34.5% to give required push to economy .
- The fiscal deficit in 2021-22 is estimated to be 6.8% of GDP.
- The final report covering 2021-26 of 15th finance commission was submitted to the President, retaining vertical shares of states at 41%.
- Government set up a development financial institution.
- ₹ 223846 crore outlay for **Health and Well being** in 2021-22 an increase of 137%.
- **Jal Jeevan Mission (Urban)** will be launched for universal water supply in all 4378 Urban Local Bodies with 2.86 crore household tap connections.
- ₹ 64180 crore outlay over six years for **PM Aatma Nirbhar Swasth Bharat Yojana .**
- The announcement of **National Research Foundation (NRF)** with an outlay of ₹ 5000 crore.
- No changes in IT Slabs, 80C and 80D limits.
- Senior citizens of 75 year of age and having only pension income are exempted from filing their income tax returns.
- Limit of turnover for tax audit increased to ₹ 10 crore from ₹ 5 crore for entities carrying out 95% transactions digitally
- ₹ 2.83 lakh crore announced for health and wellness sector, which includes ₹ 35000 crore for COVID-19 vaccines.
- The Foreign Direct Investment (FDI) limit in insurance has been increased from 49 % to 74 %.
- A record sum of ₹ **110055 crore for the Railways** in 2021-22. 100% electrification of Broad-Gauge routes to be completed by December, 2023.
- Government has committed ₹ 1.97 lakh crore for PLI schemes covering 13 sector.
- **7 textile parks** will be launched over three yea₹ under **Mega Investment Textiles Parks (MITRA)** scheme.
- National Rail Plan for India (2030) to create a 'future ready' Railway system by 2030
- **Ujjwala Scheme** will be extended to cover 1 crore more beneficiaries.

- **SWAMITVA Scheme** to be extended to all States/UTs
- Agricultural credit target enhanced to ₹ 16.5 lakh crore in FY22-animal husbandry, dairy, and fisheries to be the focus areas
- **100 new Sainik Schools** to be set up in partnership with NGOs/private schools/states.
- Higher Education Commission to be set up, legislation to be introduced this year for the same
- PSLV-CS51 to be launched by **New Space India Limited** (NSIL) carrying Brazil's Amazonia Satellite and some Indian satellites
- ₹ 4000 crore over five year for **Deep Ocean Mission** survey exploration and conservation of deep sea biodiverity
- ₹ 3768 crore allocated for **first digital census** in the history of India.

ECONOMIC SURVEY 2020-21

- India's Economic Survey 2020-21 was tabled in the Parliament by the Finance Minister Nirmala Sitharaman on January 29, 2021.
- The survey was prepared under **Krishnamurthy Subramanian**, the Chief Economic Advisor (CEA) to the government of India.
- A foundational theme of a survey as a result of the pandemic has been **Saving Lives and Livelihoods**.

Key Highlights

- The survey estimates nominal **GDP** growth of **15.4%** and **real GDP growth** of 11% in 2021-22.
- The survey projected a **V-shaped recovery**; GDP declined by 23.9% in the first quarter and by 7.5% in the second quarter.
- In 2020-21, the **growth rate of agriculture** is estimated to be **3.4%.** While the contribution of the sector to **Gross Value Added (GVA)** declined from 18.3% to **17.8%** between 2014-15 and 2019-20, it is estimated to increase to 19.9% in 2020-21.
- The **industrial sector** is estimated to decline by **9.6%** in 2020-21.
- Within the sector, highest decline is estimated in construction (12.6%) and mining (12.4%). The contribution of the industrial sector to GVA has declined from 32.5% in 2011-12 to 25.8% in 2020-21.
- **India** ranked **48** in **Global Innovation Index in 2020**, which makes it first among Central and South Asian countries, and third among the lower middle-income economies.
- The Survey has strongly recommended an increase in public spending on healthcare services from 1% to **2.5-3% of GDP**, as envisaged in the National Health Policy 2017.
- The Survey has constructed a **Bare Necessities Index** at rural, urban & all India level, with 26 indicators on 5 dimensions- sanitation, water, housing, micro-environment, & other facilities.
- The **monthly GST collections** have crossed the **1 lakh crore mark** consecutively for the last 3 months, reaching its' highest ever in December 2020.
- **Gross Non-Performing Assets ratio** of Scheduled Commercial Banks decreased from 8.21% at the end of March, 2020 to **7.49%** at the end of September 2020.
- India's forex reserves at an all-time high of **$ 586.1 billion** as on January 08, 2021, covering about 18 months worth of imports.

PM Modi Addressed WEF's Davos Dialogue

Prime Minister Narendra Modi addressed the World Economic Forum's Davos Dialogue through video conferencing on January 28, 2021.

The Davos Dialogues agenda marks the launch of the World Economic Forum's Great Reset Initiative in the post COVID world.

Three more Rafale Join IAF

The third batch of three Rafale fighter jets of the Indian Air Force (IAF) reached the Jamnagar airbase from France on January 27, 2021.

With the induction of these aircrafts, the number of Rafale fighter jets has increased to eleven.

72nd Republic Day

India celebrated its 72nd Republic Day with only 25000 spectators and a shorter parade on Rajpath on January 26, 2021.

Rafale fighter jets featured in the Republic Day parade and culminated the flypast by carrying out the 'Vertical Charlie' formation.

The 122-member contingent of the Bangladesh Armed Forces comprised of Bangladesh Army, sailors of the Bangladesh Navy and Air Warriors of the Bangladesh Air Force took part in India's Republic Day Parade.

Flight lieutenant Bhawana Kanth becomes first woman fighter pilot to take part in IAF's tableau.

For the first time, Cultural Tableau from Ladakh participated and the tableau of Uttar Pradesh was designed on the theme 'Ayodhya : Cultural Heritage of Uttar Pradesh' and it showcased Ayodhya and Ram mandir.

India's First Social Impact Bond Created

Pimpri Chinchwad Municipal Corporation, Pune has recently signed an MoU with UNDP India to co-create India's first Social Impact Bond (SIB).

A Social Impact Bond, also called pay-for-success bond or pay-for- success financing or simply social bond, is basically a contract with public sector authority, where it pays for better social outcomes.

Underwater Study to Ascertain Origin of Ram Setu

The government on January 25, 2021 given its nod to an underwater research project to determine the origins of the Ram Setu, a 48-km-long chain of shoals between India and Sri Lanka.

The study is to be conducted by the Council for Scientific and Industrial Research (CSIR) and National Institute of Oceanography (NIO) Goa.

Digital Voter ID Card Launched

Law Minister Ravi Shankar Prasad launched the e-EPIC programme and distributed e-EPICs and Elector Photo Identity Cards to five new voters to mark the National Voters' Day on January 25, 2021.

The e-Elector Photo Identity Card (e-EPIC) is a non-editable digital version of the elector photo identity card.

Howrah-Kalka Mail Renamed as Netaji Express

Indian Railways has renamed the Howrah-Kalka Mail as 'Netaji Express', in a tribute to freedom fighter Subhas Chandra Bose on January 23, 2021.

The Howrah-Kalka Mail runs between Howrah in West Bengal (Eastern Railway) and Kalka in Haryana(Northern Railway) *via* Delhi.

7th Edition of MASCRADE 2021

Union Minister Dr. Harsh Vardhan inaugurated the 7th edition of 'MASCRADE 2021 – Movement against Smuggled & Counterfeit Trade' on January 21, 2021.

The two-day programme was organized by FICCI CASCADE (Committee against Smuggling and Counterfeiting Activities Destroying the Economy), to combat illicit trade, especially in a post COVID era.

India Innovation Index Released

The NITI Aayog released the second edition of the India Innovation Index along with the Institute for Competitiveness on January 20, 2021.

In the 'Major States' category, Karnataka occupied the top position.

In UT and city states category, Delhi topped followed by Chandigarh.

Netaji's Birthday Declared as 'Parakram Diwas'

The government of India has decided to celebrate the birthday of Netaji Subhash Chandra Bose, on 23rd January, as 'Parakram Diwas' every year.

This day will be dedicated to honour and remember Netaji's indomitable spirit and selfless service to the nation.

Special Motor Bike Ambulance for the CRPF

The Defence Research and Development Organisation (DRDO) has launched a bike-based casualty transport emergency vehicle, 'RAKSHITA' for the Central Reserve Police Force (CRPF), on January 18, 2021.

The specially designed bike has been developed by the DRDO's Delhi-based laboratory, Institute of Nuclear Medicine and Allied Sciences (INMAS).

India Digital Summit 2021 Held

Union Minister Ravi Shankar Prasad inaugurated the 15th edition of India Digital Summit 2021through virtual platform on January 19, 2021.

The theme of the 2021 Summit was 'Aatmanirbhar Bharat – Start of New Decade'.

The summit focused to bring thought leadership on various digital initiatives.

World's Largest COVID-19 Vaccination Campaign Launched

Prime Minister Narendra Modi launched the world's largest COVID-19 vaccine drive on January 16, 2021.

The recipients include 30 million doctors, nurses and other front-line workers, to be followed by 270 million people who are either over 50 or have illnesses that make them vulnerable to COVID-19.

Pradhan Mantri Kaushal Vikas Yojana 3.0 Launched

The government of India launched the third phase of Pradhan Mantri Kaushal Vikas Yojana on January 15, 2021.

The scheme has been rolled out for the current financial year, ending March 2021, and will be implemented by the Ministry of Skill Development and Entrepreneurship.

16th Pravasi Bharatiya Divas Convention Held

PM Narendra Modi inaugurated the 16th Pravasi Bharatiya Divas Convention organized in a virtual format on January 9, 2021.

The theme of the event was 'Contributing to Aatmanirbhar Bharat', aimed at encouraging Indian diaspora to be part of socio-economic development in India.

'Digital Ocean' Application Launched

Union Minister Dr. Harsh Vardhan virtually inaugurated the web-based application 'Digital Ocean' on December 29, 2020. Digital Ocean is a first of its kind digital platform for Ocean Data Management. It can be accessed at www.do.incois.gov.in.

India's First Indigenous Pneumonia Vaccine Launched

Union Health Minister Dr. Harsh Vardhan launched **Pneumosil**, India's first Pneumococcal Conjugate Vaccine (PCV), on December 28, 2020.

The Pneumococcal Conjugate Vaccine (PCV) would be used to treat pneumonia in children and has been developed by **SII** in collaboration with other partners like the **Bill and Melinda Gates Foundation**.

India International Science Festival 2020 Held

PM Narendra Modi inaugurated the 6th India International Science Festival 2020 that took place virtually in **New Delhi** from December 22-25, 2020.

The theme of IISF-2020 was 'Science for self-reliant India and Global welfare' to support the Government's initiative of making Atmanirbhar Bharat.

ISRO to Establish RAC-S at IIT Varanasi

The ISRO and IIT-BHU signed an agreement to establish a Regional Academic Centre for Space at IIT Varanasi (BHU) to facilitate short and long term projects at the institute on December 23, 2020.

Regional Academic Centre for Space (RAC-S) of ISRO will act as a major facilitator for promoting space technology activities in Uttar Pradesh, Madhya Pradesh and Chhattisgarh.

India's First Hypersonic Wind Tunnel Test Facility Launched

Defence Minister Rajnath Singh inaugurated the advanced Hypersonic Wind Tunnel test facility of the DRDO in Hyderabad on December 19, 2020.

The state-of-the-art HWT Test facility has made India as the only third country in the world, after the US and Russia, to have such a huge facility in size and capability.

ISRO Sets up Dedicated Control Centre 'NETRA'

Indian Space Research Organization (ISRO) has set up a dedicated Space Situational Awareness (SSA) Control Centre, 'NETRA' at its ISTRAC campus at Peenya, Bangalore on December 16, 2020. The Network for space object Tracking and Analysis (NETRA) will monitor, track and protect India's space assets.

India's First Driverless Train Flagged-off

PM Narendra Modi inaugurated India's first-ever fully-automated driverless train service on the **Magenta Line** of the **Delhi Metroon** December 28, 2020.

The service will be available on Delhi Metro's Magenta Line which connects Janakpuri West in West Delhi to Botanical Garden in Noida.

Ayushman Bharat PM-JAY SEHAT Scheme Launched

PM Narendra Modi on December 26, 2020 has launched Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) SEHAT scheme via video-conferencing to extend health insurance coverage to all the residents of Jammu & Kashmir.

Cloud Storage Service 'DigiBoxx' Launched

NITI Aayog CEO Amitabh Kant has launched indigenously developed India's first Digital Asset Management and Storage platform, called 'DigiBoxx' on December 23, 2020.

It provides faculity to store and collaborate with your digital assets, like raw files, photographs, sensitive collateral, etc.

'Status of Leopards in India 2018' Report Released

Union environment minister Prakash Javadekar released a report titled the 'Status of Leopard in India 2018' on December 21, 2020.

As per the report, the population of leopard in India has increased by 60% in four years since 2014.

Nehru Zoological Park First to get ISO Certificate

The Nehru Zoological Park, Hyderabad has become the first in India to acquire ISO 9001 : 2015 Quality Management Standards Certification from **Accreditation Service for Certifying Bodies** (ASCB), United Kingdom.

The zoo was assessed on the basis of sanitisation, food processing, animal breeding, medicare, animal care, hygiene maintenance and establishment.

Lakshadweep Becomes India's First 100% Organic UT

The Ministry of Agriculture has declared Lakshadweep as India's first Union Territory to become 100% organic.

Lakshadweep is second only to **Sikkim**, which was India's first state to be declared completely organic in **2016**.

Indo-Bangladesh Summit Held

PM Narendra Modi addressed a virtual Indo-Bangladesh summit with his Bangladesh counterpart Sheikh Hasina on December 17, 2020.

During the summit, the two leaders jointly inaugurated the restored Chilahati – Haldibari rail route re-linking Bangladesh with North Bengal in India.

Communication Satellite 'CMS-01' Launched

India's communication satellite CMS-01 was successfully launched by PSLV-C50 from the Satish Dhawan Space Centre (SDSC) SHAR, Sriharikota in Andhra Pradesh on December 17, 2020.

CMS-01 is the 42nd communication satellite of India and it is envisaged for providing services in Extended-C Band of the frequency spectrum covering Indian mainland, Andaman- Nicobar & Lakshadweep Islands.

PM Modi Lights 'Swarnim Vijay Mashaal'

PM Narendra Modi lit up the 'Swarnim Vijay Mashaal' at the **National War Memorial** in Delhi to mark the 50th anniversary of India's victory in the 1971 India-Pakistan war, on December 16, 2020. Defence Minister Rajnath Singh unveils the logo for Swarnim Vijay Varsh to mark Vijay Diwas on the 50th anniversary of the 1971 India-Pakistan war.

PM Lays Foundation Stone of New Parliament Building

PM Narendra Modi laid the foundation stone for the new Parliament building at Sansad Marg in Parliament Complex on December 10, 2020. In the new building, the Lok Sabha chamber will have a seating capacity for 888 members, while the Rajya Sabha will have 384 seats for members.

Union Cabinet Approved PM-WANI Scheme

The Cabinet approved a public Wi-Fi access network called PM-WANI (Wi-Fi Access Network Interface) to enable easily accessible public Wi-Fi hotspots spread across the country on December 9, 2020.

The scheme will provide a boost to ease of doing business and boost the proliferation of broadband internet services.

First Indigenously Developed 100 Octane Premium Petrol Launched

Union Minister Dharmendra Pradhan launched India's first indigenously developed '100 Octane premium petrol' through video conferencing on December 2, 2020.

It has been developed by Indian Oil Corporation Limited (IOCL) at its Mathura refinery in Uttar Pradesh.

One Nation One Ration Card Scheme Starts in A&N Islands

The One Nation One Ration Card scheme commenced in **Andaman and Nicobar Islands** on November 30, 2020.

It is to mention that 28 states and UTs have joined on the network under One Nation One Ration Card. Rest are required to join by March, 2021.

Landline Users Must Dial Zero before a Mobile Number

The Ministry of Communications on November 25, 2020 noted that all fixed to mobile calls will be dialled with prefix '0' **from January 15, 2021**.

There will be no change in dialling plan from fixed to fixed, mobile to fixed and mobile to mobile calls.

IPPB Launched PM Jeevan Jyoti Bima Yojana

India Post Payments Bank (IPPB) has announced the launch of Pradhan Mantri Jeevan Jyoti Bima Yojana (PMJJBY) in partnership with **PNB MetLife** India Insurance Company Limited for its customers on November 24, 2020.

PM Laid the Foundation of Water Supply Projects in UP

PM Narendra Modi virtually laid the foundation stone of rural drinking water supply projects in Mirzapur and Sonbhadra districts of Vindhyachal region in Uttar Pradesh on November 22, 2020. The projects are planned to be completed in 24 months with an estimated cost of around ₹ **5555.38 crore**.

PM Modi Launched RuPay Card in Bhutan

PM Narendra Modi and Bhutan's Prime Minister, Lotay Tshering virtually launched **RuPay Card Phase-II in Bhutan** on November 20, 2020. With the launch of Phase 2 of RuPay Card, Bhutanese Card Holders will be able to easily access the RuPay network in India.

World's First Tram Library for Children Launched in Kolkata

Kolkata Young Reader's Tramcar, world's first library on tram for children was launched in Kolkata, West Bengal on November 14, 2020. The tram would run on Shyambazar- Esplanade and Esplanade-Gariahat routes, spanning across north and south Kolkata, every day from morning to evening.

Two Ayurveda Institutions Dedicated to the Nation

PM Narendra Modi dedicated two future-ready national premier Ayurveda institutions to the nation on November 13, 2020.

These are the Institute of Teaching & Research in Ayurveda (ITRA), Jamnagar as an Institution of National Importance (INI), and the National Institute of Ayurveda (NIA), Jaipur as a Deemed to be University by the University Grants Commission.

PM Modi Unveiled Statue of Swami Vivekanand

PM Narendra Modi virtually unveiled the life-size statue of Swami Vivekananda on the campus of Jawaharlal Nehru University (JNU) in Delhi on November 12, 2020.

The statue of Swami Vivekananda has been installed in the university campus with the support of JNU alumni.

Atmanirbhar Bharat Abhiyaan 3.0 Announced

Finance Minister Nirmala Sitharaman announced new stimulus measures under Atmanirbhar Bharat Abhiyaan 3.0 to boost the Indian economy amid the ongoing COVID-19 pandemic, on November 12, 2020.

Under the stimulus package, the FM made 12 big announcements and allocated funds and ensured credit guarantees to various sectors.

India's First Sandalwood Museum in Karnataka

The Mysuru Forest Division has set up India's first Sandalwood Museum at its premises at the Sandalwood Depot in Aranya Bhavan, Mysuru, Karnataka.

The museum has been conceptualised to spread awareness about the history and cultivation of sandalwood products.

Nepal Released Special Anthology on Gandhi

Nepal President Bidhya Devi Bhandari has released a special anthology on Mahatma Gandhi-My understanding about Gandhi- during a ceremony at Kathmandu on November 10, 2020.

The anthology marks the culmination of the two years long celebrations of **150 years of Mahatma Gandhi**.

Online News & OTT Platforms Under IB Ministry

The government of India has issued an order to bring video streaming **Over-The-Top (OTT) platforms** such as online films, audio-visual programmes and online news and current affairs content under the ambit of Ministry of Information and Broadcasting.

The government amended the (Allocation of Business) Rules, 1961, to make the changes, which come into effect immediately.

15th Finance Commission Submitted its Report

The Fifteenth Finance Commission, led by Chairman **NK Singh** submitted its report for the period 2021-22 to 2025-26 to the President Ram Nath Kovind on November 9, 2020.

The report titled Finance Commission in Covid Times was submitted by Singh along with members of the Commission.

India-Italy Virtual Summit 2020 Held

The Bilateral Summit between India and Italy was attended by the Indian PM Narendra Modi and **Prime Minister of Italy Prof. Giuseppe Conte** virtually on November 6, 2020.

15 MoUs/Agreements were signed during the Summit in various sectors such as energy, fisheries, ship building, design etc.

Missile Test

Missile	Fact
Akash-NG	The Defence Research and Development Organisation (DRDO) conducted successful maiden launch of Akash-NG (New Generation) Missile from Integrated Test Range off the coast of Odisha on January 25, 2021. Akash-NG is a new generation Surface-to-Air Missile for use by Indian Air Force (IAF) with an aim of intercepting high manoeuvring aerial threats.
Smart Anti-Airfield Weapon (SAAW)	The DRDO successfully conducted captive and release trial of indigenously developed Smart Anti-Airfield Weapon (SAAW) from Hawk-I aircraft of Hindustan Aeronautics Limited (HAL) off the Odisha coast on January 21, 2021.
MRSAM	India successfully test-fired a **Medium Range Surface-to-Air Missile** (MRSAM) from Integrated Test Range (ITR), off **Odisha** coast, on December 23, 2020. The MRSAM missile has been jointly developed by the DRDO in collaboration with Israel Aerospace Industries.
Prithvi II	India successfully test-fired two Prithvi-2 ballistic missiles off the eastern coast of Odisha in Balasore on December 17, 2020. The tactical nuclear capable weapons are developed by DRDO and can strike targets at a range up to 350 km.
Anti-Ship Version of BrahMos	The Indian Navy test-fired the anti-ship version of the BrahMos supersonic cruise missile from the Andaman and Nicobar Islands territory on December 1, 2020.
Land-Attack Version of BrahMos	India successfully test-fired a land-attack version of the BrahMos supersonic cruise missile from the **Andaman and Nicobar Islands** territory on November 24, 2020.

Armed Forces Exercise

Exercise	Facts
AMPHEX – 21	A large-scale tri-service joint amphibious exercise AMPHEX – 21 was held in Andaman & Nicobar group of islands from January 21-25, 2021. The exercise involved participation of troops of Indian Army, Indian Navy as well as the Indian Air Force.
Desert Knight 21	The Indo-France bilateral Air exercise 'Ex-Desert Knight 21' was held between the Indian Air Force and French Air and Space Force in Jodhpur, Rajasthan from January 20-24, 2021. From the Indian Air Force side, Rafale, Mirage 2000, Sukhoi-30 MKI among other aircrafts participated in the exercise.
Sea Vigil 21	Indian Navy coordinated the second edition of the biennial pan-India, largest coastal defence exercise 'Sea Vigil-21' along the entire 7516 Km coastline & Exclusive Economic Zone of India from January 12-13, 2021. The exercise aims at assessing India's preparedness in the domain of coastal defence and maritime security.
Kavach	Andaman and Nicobar Command (ANC) conducted a large scale Joint Military exercise 'Exercise Kavach' in the Andaman Sea and Bay of Bengal. The exercise will involve assets of Indian Army, Indian Navy, Indian Air Force and the Indian Coast Guard.
PASSEX 2020	The Indian Navy and Vietnamese Navy undertook the naval passage exercise PASSEX in the South China Sea from December 26-27, 2020. The two-day exercise was conducted s part of efforts to boost maritime cooperation between the two countries.

Exercise	Facts
SIMBEX-20	The 27th edition of India–Singapore Bilateral Maritime Exercise SIMBEX-20 has been hosted by Indian Navy in Andaman Sea from November 23-25, 2020. From Indian side, destroyer Rana with integral Chetak helicopter and indigenously built corvettes Kamorta and Karmuk, IN submarine Sindhuraj and P8I maritime reconnaissance aircraft participated in the exercise.
SITMEX-20	The 2nd edition of Singapore, India and Thailand Trilateral Maritime Exercise SITMEX-20 was hosted by Republic of Singapore Navy (RSN) in Andaman Sea from November 21-22, 2020. Indian Navy (IN) Ships including indigenously built ASW corvette Kamorta and missile corvette Karmuk participated in the exercise.
Sagar Kavach	A two-day joint coastal security exercise, 'Sagar Kavach' was held at Paradip Coast by a joint effort of the state of Odisha and West Bengal governments from November 5-6, 2020.

INTERNATIONAL

Biden Signed Executive Orders Aimed at Racial Equity

US President Joe Biden signed four new Executive Orders in an effort to foster racial equity in the country on January 26, 2021.

The four executive actions Biden signed :

1. Direct the Department of Housing and Urban Development "to take steps necessary to redress racially discriminatory federal housing policies".
2. Direct the Department of Justice to end its use of private prisons.
3. Reaffirm the federal government's 'commitment to tribal sovereignty and consultation'.
4. Combat xenophobia against Asian American and Pacific Islanders.

SpaceX Launched 143 Satellites in a Single Mission

SpaceX successfully launched its ambitious rideshare mission called Transporter-1, carrying 143 small satellites, and created a new record of launching many satellites from a single rocket on January 24, 2021 With this launch, SpaceX broke the record set by ISRO in February 2017 for deploying 104 satellites onboard PSLV in a single mission.

China Adopted Law Letting Coast Guard Fire on Foreign Vessels

China adopted a law that gives the country's coast guard a free hand to open fire on any foreign vessel deemed a threat to national sovereignty on January 22, 2021.

The bill specifies the circumstances under which different kind of weapons hand-held, ship borne or airborne can be used.

First-ever Treaty to Ban Nuclear Weapons Enters into Force

The first-ever Treaty on the Prohibition of Nuclear Weapons entered into force on January 22, 2021. The treaty states that all ratifying countries will "never under any

circumstances develop, test, produce, manufacture, otherwise acquire, possess or stockpile nuclear weapons or other nuclear explosive devices."

Dutch Government Resigned Over Child Benefits Scandal

The Netherlands Prime Minister Mark Rutte and the four-party coalition government resigned over a child benefits scandal on January 15, 2021.

The Dutch government falsely accused thousands of parents of fraudulently claiming child allowance.

China Unveils Prototype Superfast Maglev Train

People's Republic of China has unveiled the 69-foot prototype of a new domestically developed superfast maglev train from China's Chengdu city on January 13, 2021. It can reach up to a speed of 620 km per hour and has been developed with High-Temperature Superconducting (HTS) technology.

Donald Trump Becomes First US President to be Impeached Twice

Donald Trump became the first President in US history to be impeached twice as the Democratic- controlled House of Representatives formally charged him with 'incitement of insurrection" on January 13, 2021.

Ten Republicans joined Democrats to impeach Trump by 232-197.

Oman Sultan's Eldest Son to Succeed Him Under New Law

Oman's Sultan Haitham will be succeeded by his eldest son Dhi Yazan, according to a new basic law published on January 12, 2021. The law also creates a new position of crown prince and establishes succession from ruler to the eldest son.

Five New Countries Join UNSC

Five new countries namely India, Mexico, Norway, Ireland, and Kenya formally joined the United Nation Security Council (UNSC) as non-permanent Members on January 4, 2021. The new countries will hold non-permanent seats on the 15-member council for two years.

WEF Global Risks Report 2021 Released

The World Economic Forum (WEF) released the sixteenth edition of the Global Risks Report, 2021 on January 18, 2021. The report was released based on the Global Risks Perception Survey undertaken by more than 650 members of World Economic Forum. The top three risks by impact are climate action failures, infectious diseases and weapons of mass destruction.

Global Climate Risk Index 2021 Released

The Bonn-based environmental think tank 'Germanwatch' has released the 16th edition of Global Climate Risk Index 2021.

Mozambique, Zimbabwe and Bahamas were the top three most affected countries in 2019, respectively.

India ranked as the seventh worst-hit country in terms of climate change in 2019.

Henley Passport Index 2021 Released

The Henley Passport Index 2021 has been released by Henley & Partners, based on the exclusive data from the International Air Transport Association (IATA). Japan has once again topped the list of being the most powerful passport in the world, for third consecutive year.

As per the latest report, India has been ranked 85th among 110 countries with a visa-free score of 58.

Human Development Report 2020 Released

United Nations Development Programme (UNDP) has released the Human Development Index (HDI) report on December 15, 2020.

This year's report is titled as Human Development Report 2020: The Next Frontier Human Development and the Anthropocene.

According to the report, **Norway** topped the Human Development Index, followed by **Ireland**, and **Switzerland**.

US Coronavirus Relief Package Bill Signed

US President Donald Trump abruptly signed a measure providing $ 900 billion in pandemic aid on December 27, 2020.

He signed the act as part of the massive $ 2.3 trillion package for the year-end bill to provide long- delayed coronavirus relief and fund the federal government.

EU and Britain Reach over Post-BREXIT Trade

European Union and United Kingdom have struck a post-Brexit trade deal, after months of negotiations over future business rules on December 24, 2020.

The agreement will ensure Britain and the 27-nation bloc can continue to trade in goods without tariffs or quotas after the UK breaks fully free of the EU on 1st of next month.

Nepalese Parliament Dissolved

Nepal President **Bidya Devi Bhandari** dissolved Parliament on the recommendation of Prime Minister KP Sharma Oli on December 20, 2020.

The surprise move came amidst a prolonged tussle for power between the countries embattled Prime Minister Oli and former premier Pushpa Kamal Dahal 'Prachanda'.

Bill for ex-Presidents' Lifetime Immunity in Russia

Russian President Vladimir Putin signed legislation that will grant former presidents lifetime immunity once they leave office on December 22, 2020.

The bill gives former presidents and their families' immunity from prosecution for crimes committed during their lifetime.

Human Freedom Index 2020 Released

The Human Freedom Index 2020, a worldwide ranking of civil, economic and personal freedom, published by Cato Institute and Fraser Institute on December 18, 2020.

New Zealand (8.87) topped the index followed by Switzerland (8.82) and Hong Kong (8.74) at the second and third place, respectively.

Forbes 2020 List of Highest Paid Celebrities

Kylie Jenner with $590 million was crowned the highest-paid celebrity by Forbes in its Highest-Paid Celebrity list of 2020, released on December 15, 2020.

Roger Federer is the highest-paid athlete securing a third rank on the list with $ 106.3 million in earnings, while soccer superstar Cristiano Ronaldo is on fourth with $ 105 million.

Global Terrorism Index 2020 Released

Institute for Economics and Peace (IEP) has released the Global Terrorism Index 2020 to measure the impact of terrorism on the countries.

With a score of 9.592, **Afghanistan** has topped the index as the worst terror impacted nation among the 163 countries. It is followed by **Iraq** (8.682) and **Nigeria** (8.314) at second and third place respectively.

UAE becomes First Gulf Nation to Generate Electricity from Coal

The first coal-based power plant of Arab Gulf country is being developed in Saih Shuaib, Dubai, UAE.

The 2400 MW Hassyan clean coal power station will be built at an estimated cost of $ 3.4 billion.

Cambridge Dictionary Announced Word of the Year 2020

Quarantine has been named the Word of the Year 2020 by Cambridge Dictionary. As per the Cambridge Dictionary, quarantine was one of the most highly searched words on its website in 2020.

'Quarantine' was closely followed by 'lockdown'and 'pandemic' on the shortlist.

US Withdrew from Open Skies Treaty

The Trump administration has officially withdrawn from the Open Skies treaty, six months after starting the process to leave, on November 22, 2020.

The Open Skies treaty, proposed by US President **Dwight Eisenhower** in 1955, was signed in1992 and took effect in 2002.

15th G20 Summit Held

The **Kingdom of Saudi Arabia** hosted the 15th annual G20 Leaders' Summit in the capital city, Riyadh under the theme Realising the Opportunities of the 21st Century for All from November 21-22, 2020.

PM Narendra Modi participated in the Summit along with the respective Heads of State/ Government of 19 member countries, EU, other invited countries & international organizations.

UAE Cancels Lenient Penalties for 'Honour Killings'

The UAE has cancelled legal clauses that allowed judges to issue lenient sentences for 'honour killings'.

As per the Emirati government, those crimes will now be treated by courts as normal murder cases.

Nagorno- Karabakh Peace Deal Signed

Armenia, Azerbaijan and **Russia** have signed an agreement to end military conflict over the disputed enclave of Nagorno-Karabakh on November 10, 2020. Under the deal, Azerbaijan will hold on to areas of Nagorno-Karabakh that it has taken during the conflict.

Scotland to Ban Smacking Children

Scotland has become the first part of the UK to ban the smacking of children after new legislation came into effect on November 7, 2020.

The new law seeks to give children the same protection from assault as adults.

Japan to Achieve Zero Carbon Emissions by 2050

Japanese Prime Minister Yoshihide Suga on October 27, 2020 declared that the country will achieve zero carbon emissions by 2050.

It had previously made a commitment only to reduce emissions 80% by 2050 and achieve carbon neutrality in the second half of the century.

Newly Elected PM/President

Name	Post (Country)	Facts
Kaja Kallas	Prime Minister (Estonia)	Kaja Kallas of the Reform party (Renew Europe) became Estonia's first female prime minister on January 24, 2021. With this, Estonia becomes the only country in the world where both the president and the prime minister are women.
Marcelo Rebelo de Sousa	President (Portugal)	The incumbent President of Portugal, Marcelo Rebelo de Sousa has won a second five-year term after winning the 2021 Portuguese presidential election with landslide victory on January 24, 2021.
Joe Biden	President (The USA)	Joseph Robinette Biden Jr. has been sworn in as the 46th president of the United States on January 20, 2021. Apart from this, Kamala Harris took oath as 49th Vice President of the country, becoming the first women to hold the post in the history of United States.
Yoweri Museveni	President (Uganda)	Uganda's incumbent President Yoweri Museveni has been declared as the winner of the country's presidential election for 2021 on January 16, 2021. Museveni won 5.85 million votes, while main opposition candidate Wine had 3.48 million votes.
Sadyr Zhaparov	President (Kyrgyzstan)	Nationalist Sadyr Zhaparov has won the presidency of Kyrgyzstan on January 10, 2021 as the central Asian nation held its first elections since political unrest in October, 2020.
Nana Akufo-Addo	President (Ghana)	Ghana's incumbent President Nana Akufo-Addo has been declared the winner of the country's 2020 presidential election on December 10, 2020. He previously served as Attorney General from 2001 to 2003 and as Minister for Foreign Affairs from 2003 to 2007 under the Kufuor-led administration.
Maia Sandu	President (Moldova)	Maia Sandu, a former World Bank economist, has won Moldova's 2020 Presidential election with 57.72% vote to become the President-elect of the country. She is the first woman to be elected as the President of Moldova.
Roch Kabore	President (Burkina Faso)	Burkina Faso's President Roch Marc Christian Kabore has been declared the winner of the recently-held general elections after he secured over 57% of the votes on November 26, 2020.
Francisco Sagasti	President (Peru)	Peruvian lawmaker Francisco Sagasti sworn in as interim President after the departure of two presidents over the last week, on November 18, 2020. Sagasti has taken over from **Manuel Merino**, the former speaker of Congress.

SPORTS AND GAMES

TENNIS

ATP Finals 2020

The ATP Finals 2020 was a men's tennis tournament played at the O2 Arena in London, **United Kingdom**, from November 15-22, 2020. **Daniil Medvedev** (Russia) beat Dominic Thiem (Austria) to win the 2020 ATP Tour Finals tennis tournament.

Paris Masters 2020

Daniil Medvedev won the Paris Masters for the first time by beating Alexander Zverev for his eighth career title and third at a Masters event on November 8, 2020.

He became the fourth Russian to capture the Paris Masters title after Marat Safin, Nikolay Davydenko and Karen Khachanov.

TABLE TENNIS

ITTF Women's World Cup 2020

The 24th edition of the ITTF Women's World Cup table tennis title was held in **Weihai, China**, from November 8-10, 2020.

The world number one **Chen Meng** defeated Chinese compatriot Sun Yingsha to win her maiden title.

With her victory, she became the first winner of the ITTF's #RESTART series.

BADMINTON

Thailand Open 2021

Badminton World Federation (BWF) organized the Yonex Thailand Open Super 1000 badminton tournament 2021 held from January 12-17, 2021.

Viktor Axelsen (Denmark) and Carolina Marin (Spain) won men's and women's singles title respectively.

SHOOTING

International Air Rifle Championship

World number one Elavenil Valarivan of India won gold and Shahu Tushar Mane bagged a silver at the 2020 Sheikh Russel International Air Rifle Championship held virtually on October 18, 2020.

The championship was organised by Bangladesh Shooting Sport Federation (BSSF).

FORMULA ONE

Bahrain Grand Prix 2020

Lewis Hamilton (Mercedes-Great Britain) has won the 2020 Bahrain Grand Prix held at Bahrain International Circuit, Bahrain on November 29, 2020.

Red Bull Racing teammates Max Verstappen (Netherlands) and Alexander Albon (Thailand) were spotted at second and third place, respectively.

Turkish Grand Prix 2020

Lewis Hamilton (Mercedes-Great Britain) has won the Turkish Grand Prix 2020, held at Istanbul Park in Tuzla, Turkey on November 15, 2020.

Sergio Perez (Racing Point-BWT- Mexico) secured second position and Sebastian Vettel of Ferrari finished third.

CRICKET

Border- Gavaskar Trophy 2020-21

India made history by defeating Australia by three wickets in the fourth and final Test at the Gabba in Brisbane and retained the Border- Gavaskar Trophy on January 19, 2021.

India successfully chased the 328-run target to seal the four-match series 2-1.

Australia-India ODI Series 2020

The Australian cricket Team won the 3-match Dettol ODI series 2020 against India by 2-1.

Australian Batsman Steve Smith was chosen as the Player of the Series for scoring 216 runs in the series which included 2 centuries.

Indian Premier League 2020

Mumbai Indians beat Delhi Capitals in the Dream11 Indian Premier League (IPL) 2020 final by 5 wickets on November 10, 2020, held at Dubai International Cricket

Stadium to clinch their record fifth IPL title.

With this win Mumbai Indians have become the only team to win consecutive IPL trophies (2019 and 2020) since Chennai Super Kings in 2010 and 2011.

BOXING

Cologne World Cup 2020

Indian pugilists have bagged **nine medals** including three gold, two silver and four bronze at the Cologne Boxing World Cup 2020, held in Cologne, Germany from December 17-19, 2020.

Amit Panghal (52 kg), Manisha Moun (57 kg) and Simranjit Kaur (60 kg) bagged gold medals in their respective categories.

HOCKEY

National Ice Hockey Championship 2021

Ice Hockey Association of India (IHAI) organized the National Ice Hockey Championship from January 16 to 22, 2021 at Gulmarg Ice Rink, in Jammu and Kashmir (UT).

The Indo-Tibetan Border Police (ITBP) has won the 10th National Ice Hockey Championship 2021 trophy, by defeating team Ladakh with the score line (goals) 5-1.

ATHLETICS

Dhaka Marathon 2021

Jigmet Dolma from India secured fourth position in the Dhaka Marathon 2021 organised at Dhaka in Bangladesh on January 10, 2021.

In the marathon, Hicham Lakohi of Morocco came first in the elite competition of the marathon in men's category while Angela Jim Asunde from Kenya won the women's category in the same event.

The marathon was organized to commemorate the return of Bangabandhu Sheikh Mujibur Rahman from Pakistan jail to Bangladesh in 1972.

AWARDS & HONOURS

NATIONAL

Pradhan Mantri Rashtriya Bal Puraskar 2021

The Pradhan Mantri Rashtriya Bal Puraskar (PMRBP) for the year 2021 has been conferred to 32 children, hailing from from 32 districts of 21 States/UTs on January 25, 2021.

Seven awards have been given in the field of Art and Culture, nine awards for Innovation and five for Scholastic Achievements.

Subhash Bose Aapda Prabandhan Puraskar 2021

The government of India announced the Subhash Chandra Bose Aapda Prabandhan Puraskar 2021 on January 23, 2021.

The Subhash Chandra Bose Aapda Prabandhan Puraskar 2021 has been conferred upon Dr. Rajendra Kumar Bhandari in Individual category and Sustainable Environment and Ecological Development Society (SEEDS) in Institutional category.

Padma Awards 2021

The Union Home Ministry announced the recipients for Padma awards on the eve of Republic Day on January 26, 2021.

The 2021 Padma Award has been conferred upon 119 winners, which include 7 Padma Vibhushan, 10 Padma Bhushan and 102 Padma Shri Awards.

Award	Winners
Padma Vibhushan	Shinzo Abe (Public Affairs) (Japan), SP Balasubramaniam (Posthumous) (Art), Dr. Belle Monappa Hegde (Medicine), Narinder Singh Kapany (Posthumous) (Science and Engineering) (USA), Maulana Wahiduddin Khan (Others- Spiritualism), BB Lal (Others- Archaeology) and Sudarshan Sahoo (Art)
Padma Bhushan	Ms. Krishnan Nair Shantakumari Chithra (Art), Tarun Gogoi (Posthumous) (Public Affairs), Chandrashekhar Kambara (Literature and Education), Ms. Sumitra Mahajan (Public Affairs), Nripendra Misra (Civil Service), Ram Vilas Paswan (Posthumous) (Public Affairs), Keshubhai Patel (Posthumous) (Public Affairs), Kalbe Sadiq (Posthumous) (Others-Spiritualism), Rajnikant Devidas Shroff (Trade and Industry) and Tarlochan Singh(Public Affairs)
Padma Shri	Gulfam Ahmed (Art), Ms. Lakhimi Baruah (Social Work), Jagdish Chaudhary (Posthumous) (Social Work), Jai Bhagwan Goyal (Literature and Education), Maheshbhai & Shri Nareshbhai Kanodia (Duo) (Posthumous) (Art), P. Subramanian (Posthumous) (Trade and Industry) and 96 others

Digital India Awards 2020

President Ram Nath Kovind will virtually confer the Digital India Awards 2020 to honour exemplary initiatives and practices in digital governance, on December 30, 2020 via video conferencing.

Category	Winners			
	Platinum Icon	Gold Icon	Silver Icon	
Innovation in Pandemic	eSanjeevani - National Telemedicine Service	Covid-19 Sample Collection Management System	Aapda Sampoorti Portal	Pravasi Shramik and Rojgar Setu Portal
Excellence in Digital Governance Ministry/Department (Central)	eCommittee Supreme Court of India, Department of Justice	Department of Posts	Department of Fertilizers	Department of Land Resources
Excellence in Digital Governance State/UT	Haryana	Tamil Nadu	Uttar Pradesh	West Bengal
Excellence in Digital Governance District	Khargone, Madhya Pradesh	Changlang, Arunachal Pradesh	Kamareddy, Telangana	
Open Data Champion	Department of Health and Family Welfare	Research Data Management in ICAR	Food Corporation of India	Ministry of Micro, Small and Medium Enterprises
Exemplary Product	Port Community System PCS1x - National Maritime Single Window	ServicePlus	Integrated Temple Management System (ITMS)	

51st IFFI Awards

The 51st edition of International Film Festival of India (IFFI) was held at Shyamaprasad Stadium near Panaji in Goa from January 16-24, 2021.

Awards Winners

Award	Winners
Golden Peacock Award for Best Film	Into the Darkness
Silver Peacock Award for Best Actor (Male)	Tzu-Chuan Liu (The Silent Forest)
Silver Peacock Award for Best Actor (Female)	Zofia Stafiej (I Never Cry)
Special Mention Award	Kripal Kalita (Bridge)
Indian Personality of the Year Award	Biswajit Chatterjee
Lifetime Achievement Award	Vittorio Storaro (Italy)

Gallantry Awards & Defence Decorations

President Ramnath Kovind has approved awards of 455 Gallantry and other Defence decorations to Armed Forces personnel and others on the eve of 72nd Republic Day celebrations.

These include one Mahavir Chakra, 5 Kirti Chakras, 5 Vir Chakras, 7 Shaurya Chakras, 4 Bar to Sena Medal (Gallantry), 130 Sena Medal (Gallantry), and others.

Colonel Bikumalla Santosh Babu decorated with Mahavir Chakra posthumously.

Ramanujan Prize for Young Mathematicians 2020

Carolina Araujo from **Brazil** became the **first non-Indian** to win Ramanujan Prize for Young Mathematicians 2020 for her outstanding work in algebraic geometry in a virtual ceremony on December 9, 2020.

INTERNATIONAL

19th Dhaka International Film Festival Organised

The 19th Dhaka International Film Festival (DIFF) was shown 226 films from 73 countries from January 16-24, 2021.

Kyrgyz film 'The Road to Eden' jointly directed by Bakyt Mukul and Dastan Zhapar won the best film award at the festival.

Masud Hasan Ujjal won the Best Film Award in the Bangladesh Panorama section for his film 'Unponchash Batash'.

French Order of Merit

Indian physicist and Padma Shri awardee Rohini Godbole has been conferred with the Ordre National du Merite, on January 14, 2021.

She has been recognised for contributions to collaborations b/w France & India and commitment to promoting enrolment of women in science.

ICC Awards of the Decade 2020

The winners of the prestigious International Cricket Council (ICC) Awards of the Decade were announced on December 28, 2020.

Category	Winner (Men's)
ICC Spirit of Cricket Award of the Decade	MS Dhoni
Sir Garfield Sobers Award for ICC Male Cricketer of the Decade	Virat Kohli
ICC Men's ODI Cricketer of the Decade	Virat Kohli
ICC Men's T20I Cricketer of the Decade	Rashid Khan
ICC Men's Test Cricketer of the Decade	Steve Smith
ICC Men's Associate Cricketer of the Decade	Kyle Coetzer

US Legion of Merit

US President Donald Trump has presented America's highest military decoration, 'The Legion of Merit' to PM Narendra Modi on December 21, 2020.

He has been awarded for his leadership in elevating bilateral strategic partnership and accelerating emergence of India as a global power.

ATP Awards 2020

The Association of Tennis Professionals (ATP) 2020 Awards were announced on December 21, 2020 to felicitate the players and others for their performance during 2019-20 season.

Category	Winner
Player of the Year	Novak Djokovic
Doubles Team of the Year	Mate Pavic and Bruno Soares
Coach of the Year	Fernando Vicente (coached Andrey Rublev)
Comeback Player of the Year	Vasek Pospisil (Canada)
Most Improved Player	Andrey Rublev (Russia)
Newcomer of the Year	Carlos Alcaraz (Spain)

UNEP Champions of the Earth 2020

The United Nations Environment Programme (UNEP) named six young engineers, scientists, activists and entrepreneurs from all over the world as its Young Champions of the Earth 2020 awardees on December 15, 2020.

The winners are :

1. Frank Bainimarama
2. Yacouba Sawadogo
3. Nemonte Nenquimo
4. Fabian Leendertz
5. Mindy Lubber
6. Robert D. Bullard

Goldman Environmental Prize 2020

Paul Sein Twa of **Myanmar** wins 2020 Goldman Environmental Prize for Asia region in recognition of his efforts in promoting the self-determination of the Karen people in managing their natural resources.

The Goldman Environmental Prize is one of the most prestigious environmental prizes awarded annually to grassroots environmental activists.

Time Person of the Year 2020

US President-elect **Joe Biden** and Vice President-elect **Kamala Harris** have been jointly named as Time magazine's 2020 'Person of the Year'.

The TIME magazine releases the annual list since 1927, to recognise a person or organization who impacted the country and world the most, for better or worse, during the calendar year.

Rabindranath Tagore Literary Prize 2020

Indian journalist-author **Raj Kamal Jha** has been honoured with the Rabindranath Tagore Literary Prize 2020 for his novel, **The City and The Sea** on December 9, 2020.

Jha's book 'The City and The Sea' is based on the December 2012 Nirbhaya rape and murder case.

Global Teacher Prize 2020

Ranjitsinh Disale, a government teacher from Zilla Parishad Primary School, in Paritewadi village, Solapur district of Maharashtra, has won the 2020 Global Teacher Prize on December 3, 2020. He is the **first Indian to win this award**, which carries prize money of $ 1 million (₹ 7.4 crore).

Nobel Prize 2020

Category	Winners	Achievement
Economic Sciences	Paul R. Milgrom (USA) and Robert B. Wilson (USA)	For improvements to auction theory and inventions of new auction formats
Peace	World Food Programme (WFP)	For its efforts to combat hunger, for its contribution to bettering conditions for peace in conflict-affected areas and for acting as a driving force in efforts to prevent the use of hunger as a weapon of war and conflict
Literature	Louise Gluck (USA)	For her unmistakable poetic voice that with austere beauty makes individual existence universal
Chemistry	Emmanuelle Charpentier (France) and Jennifer A. Doudna (USA)	For the development of a method for genome editing
Physics	Roger Penrose (UK)	For the discovery that black hole formation is a robust prediction of the general theory of relativity.
	Reinhard Genzel (USA) and Andrea Ghez (USA)	For the discovery of a supermassive object at the centre of our galaxy
Physiology (Medicine)	Hervey J. Alter (USA), Michael Houghton (UK) and Charles M. Rice (USA)	For the discovery of Hepatitis-C virus

PERSONS IN NEWS

NATIONAL

Shrishti Goswami

19-year-old Shrishti Goswami was designated as one-day Chief Minister of Uttarakhand on the occasion of National Girl Child day on January 24, 2021.

She administers from Gairsain and also review various schemes run by the state government that include Atal Ayushman Scheme, Smart City project and other development projects.

Amanda Gorman

22-year-old Amanda Gorman made history as she became the youngest poet to read a poem at a US presidential inauguration on January 20, 2021.

She read her own poem, titled 'The Hill We Climb,' at the inauguration of President Joe Biden and Vice President Kamala Harris.

Narendra Chanchal

The iconic Indian Bhajan singer Narendra Chanchal, who specialized in religious songs and hymns, has passed away at the age of 80 on January 22, 2021.

Some of his ever popular bhajans include Chalo Bulawa Aaya Hai, Tune Mujhe Bulaya Sherawaliye, among others.

V. Shanta

Renowned oncologist Dr V Shanta has passed away, following chest pain at the age of 93 on January 19, 2021.

She was the chairperson of Adyar Cancer Institute in Chennai, which she joined in 1954. The institute is known for providing state-of-the-art healthcare to all patients.

Mata Prasad

Former Governor of Arunachal Pradesh and senior Congress leader Mata Prasad has passed away at the age of 95 on January 20, 2021.

He served as minister in the Congress government in Uttar Pradesh in 1988-89 and appointed as Governor of Arunachal Pradesh in 1993.

Madhavsinh Solanki

Former Chief Minister of Gujarat, Madhavsinh Solanki has passed away at the age of 93 on January 9, 2021.

He had served as External Affairs Minister of India from June 1991 to March 1992. He had been the Chief Minister of Gujarat for three times between 1976 to 1990.

Soma Mondal

Soma Mondal has taken over as the new Chairman of Steel Authority of India Limited (SAIL) with effect from January 1, 2021. She is the first-ever women-head of the body.She succeeds Anil Kumar Chaudhary, who superannuated on December 31, 2020.

Siddhartha Mohanty

The Appointments Committee of the Cabinet has approved the appointment of Siddhartha Mohanty as the new Managing Director (MD) of Life Insurance Corporation (LIC) with effect from February 1, 2021.

He replaced TC Suseel Kumar, who retired on January 31, 2021.

Buta Singh

Former Union Minister and senior Rajasthan Congress leader Buta Singh has passed away at the age of 86 on January 2, 2020.

He served as Home Minister of India in the Rajiv Gandhi government from 1986 to 1989 and also served as the Minister of Agriculture and Rural Development.

Suneet Sharma

Suneet Sharma has been appointed as the new Chairman & Chief Executive Officer (CEO) of Railway Board, Ministry of Railways and ex-officio Principal Secretary to Government of India on December 31, 2020.

Uday Kotak

The RBI has approved the re-appointment of Uday Kotak as Managing Director of Kotak Mahindra Bank for a further period of three years, with effect from January 1, 2021.

Kotak is founder managing director and promoter of the bank.

Motilal Vora

Veteran Congress leader Motilal Vora has passed away following post-COVID-19 complications at the age of 93 on December 21, 2020.

He served as the chief minister of Madhya Pradesh twice, once from 1985 to 1988 and then again in 1989 for 11 months.

Narinder Singh Kapany

Indian-American physicist Narinder Singh Kapany, who is considered as the **'Father of Fibre Optics'** has passed away at the age of 94 on Dec. 4, 2020. He is credited with coining of the term fibre optics.

Mahashay Dharampal Gulati

Mahashay Dharampal Gulati, the owner of MDH has passed away at the age of 97 on December 3, 2020.

He was known as the 'The king of spices', and fondly called 'dalaji' and 'Mahashayji' by his near and dear ones.

Shashi Shekhar Vempati

The CEO of Prasar Bharati, Shashi Shekhar Vempati has been elected as the Vice President of Asia Pacific Broadcasting Union (ABU) for a period of three years on December 16, 2020.

Uday Shankar

Uday Shankar has been named as the President of the Federation of Indian Chambers of Commerce & Industry (FICCI) for the year 2020-21.

Varsha Joshi

The government of India has appointed Varsha Joshi as the interim Chairperson of National Dairy Development Board (NDDB), with effect from December 1, 2020. She replaced incumbent **Dilip Rath**.

Rajeev Chaudhary

Lt. Gen Rajeev Chaudhary has been appointed as the 27th Director General of BRO with effect from December 1, 2020. He succeeded **Lt Gen Harpal Singh.**

Harpal Singh

Lt. Gen Harpal Singh has been appointed as the new Engineer-in- Chief of the Indian Army on November 25, 2020. He took over his new appointment on December 1, 2020.

Yashvardhan Sinha

President Ram Nath Kovind has administered the oath of office to Mr. Sinha as the **Chief Information Commissioner** in the Central Information Commission at a ceremony held at Rashtrapati Bhavan.

K. Sivan

Karnataka Governor Vajubhai Vala presented Doctor of Science honorary doctorate upon ISRO Chairman K. Sivan on November 21, 2020.

The Visvesvaraya Technological University has conferred the doctorate to him.

Sonu Sood

Actor Sonu Sood has been appointed as the **Punjab state icon** by the Election Commission of India (ECI) on November 16, 2020.

Tanay Manjrekar

Pune-born engineer Tanay Manjrekar became the **first Indian to ride in 'hyperloop'** at a test conducted in Nevada, Las Vegas on November 9, 2020.

The Virgin employees journeyed 500 metres in 15 seconds in a two-person pod called 'Pegasus' or XP-2, which travelled at 160 km/hour.

Ahmed Patel

Senior Congress leader Ahmed Patel, who was the political secretary to Congress President Sonia Gandhi, has passed away at the age of 71 on November 25, 2020.

He has represented Gujarat for eight terms in the Parliament of India, from 1977 to 2020.

INTERNATIONAL

Giuseppe Conte

Italian Prime Minister Giuseppe Conte tendered his resignation following widespread criticism of his handling of the coronavirus pandemic in the country on January 26, 2021.

Khurelsukh Ukhnaa

The Prime Minister of Mongolia, Khurelsukh Ukhnaa, has resigned along with his entire government, following protests and public outrage over the government's handling of the COVID-19 pandemic.

Janet Yellen

US Senate confirmed Janet Yellen as the first woman to head the US Treasury on January 25, 2021.

The former chair of the Federal Reserve and noted economist was approved by the Senate on an 84-15 vote.

Lloyd Austin

US Senate confirmed the appointment of Retired Army General Lloyd Austin as the United States (US) Secretary of Defence on January 22, 2021.

With this He is be the first African American to serve as the Secretary of Defence.

Raj Iyer

Indian-American Dr Raj Iyer has been appointed as the first Chief Information Officer (CIO) of the US Army, since the creation of the position by the Pentagon in July 2020.

He will direct the execution of policies and programmes to modernise the US Army.

Avril Haines

The US Senate has approved Joe Biden's nomination of Avril Haines, making her the first woman to be Director of National Intelligence on January 21, 2021. The Senate voted 84-10 to confirm Haines to the post.

Alan Burgess

Alan Burgess, the world's oldest living first-class cricketer, from New Zealand has passed away at the age of 100 years and 250 day on January 5, 2021.

He was a right-hand batsman and left-arm spinner. He was a tank driver in World War II.

Ludovic Orban

The Prime Minister of Romania Ludovic Orban has **stepped down** from his post after his ruling **National Liberal Party (NLP)** lost in 2020 parliamentary election on December 7, 2020.

Gitanjali Rao

The Indian-American young scientist and inventor, Gitanjali Rao, has been named as the first-ever 'Kid of the Year' by the iconic TIME magazine on December 4, 2020.

She has been named for her 'astonishing work' using technology to tackle issues ranging from contaminated drinking water to opioid addiction and cyberbullying.

BOOKS & AUTHOR

India 2030 : Rise of a Rajasic Nation Book Released

'India 2030 : The Rise of a Rajasic Nation' is a collection of essays by the likes of Bibek Debroy, Vikram Sood, Raghunath Anant Mashelkar, Ram Madhav and David Frawley, among others.

The Commonwealth of Cricket (Ramachandra Guha)

Indian writer and historian Ramachandra Guha has come out with a book on cricket titled 'The Commonwealth of Cricket : A Lifelong Love Affair with the Most Subtle and Sophisticated Game Known to Humankind'.

Abdul Kalam- Ninaivugalukku Maranamillai Book Released

Vice President M Venkaiah Naidu released 'Abdul Kalam- Ninaivugalukku Maranamillai', the biography (in Tamil) on former President APJ Abdul Kalam at Raj Bhavan in Chennai on January 17, 2021.

Manohar Parrikar – Off The Record (Waman Subha Prabhu)

Goa's Chief Minister Pramod Sawant released a book titled 'Manohar Parrikar – Off the Record', written by senior journalist Shri Waman Subha Prabhu.

Through this book, the author narrates the multi-faceted personality of late Parrikar, based on the collection of memories held by Waman Prabhu who happened to be with late Parrikar during the journey of his life.

Modi India Calling-2021 Book Released

A coffee table book titled "Modi India Calling – 2021" was released on the eve of the 16th Pravasi Bharatiya Divas (PBD) on January 9, 2021.

The 450-page book is filled with over hundreds of photographs of Prime Minister Narendra Modi from his "107 overseas and bilateral visits".

India's 71-Year Test : The Journey to Triumph in Australia (R. Kaushik)

Team India head coach Ravi Shastri launched a book titled 'India's 71-Year Test : The Journey to Triumph in Australia', at the Sydney Cricket Ground on January 6, 2021.

The Population Myth : Islam, Family Planning and Politics in India (SY Quraishi)

Former Chief Election Commissioner (CEC) SY Quraishi has come out with his book titled "The Population Myth: Islam, Family Planning and Politics in India".

The book evaluates India's demographics from a religious perspective and seeks to demolish two basic myths that 'Islam is against family planning' and 'Muslim rate of growth' is linked to capture political power.

Justice Rajindar Sachar Autobiography 'In Pursuit of Justice' Released

The autobiography of late Justice Rajindar Sachar, titled, "In Pursuit of Justice : An Autobiography", was launched posthumously by Sachar's family on the occasion of his birth anniversary on December 29, 2020.

The book is a story of a great jurist who was an even a greater human being, as he delivered a lot to the society and did not allowed politics to cloud his judgments.

Sutranivednachi Sutra-Ekanbav (Dr. Roopa Chari)

Union Minister ShripadNaik released a Konkani book titled 'Sutranivednachi sutra-ekanbav' on December 27, 2020.

The book has been written by Dr. Roopa Chari, a well-known personality in the field of Compering in Goa and is published by Sanjana Publications.

The Presidential Years

Rupa Books announced that the memoir, titled 'The Presidential Years', will be globally released in Jan. 2021.

'The Presidential Years' is Pranab Mukherjee's fourth volume of memoirs recollecting the challenges he had to face during his stint as the President.

Oh Mizoram (Sreedharan Pillai)

Vice-President M. Venkaiah Naidu released a book titled 'Oh Mizoram' through video conferencing during an event organised at New Delhi on December 19, 2020.

The book 'Oh Mizoram' has been written by **the Governor of Mizoram, PS Sreedharan Pillai**.

WHO'S WHO

President Ram Nath Kovind
Vice-President M. Venkaiah Naidu
Prime Minister Narendra Modi

Cabinet Ministers

Minister	Portfolio
Rajnath Singh	Defence
Amit Shah	Home Affairs
Nitin Jairam Gadkari	Road Transport and Highways; Micro, Small and Medium Enterprises
DV Sadananda Gowda	Chemicals and Fertilizers
Nirmala Sitharaman	Finance; Corporate Affairs
Narendra Singh Tomar	Agriculture and Farmers Welfare; Rural Development; Panchayati Raj; Food Processing Industries
Ravi Shankar Prasad	Law and Justice; Communications; Electronics and Information Technology
Thaawar Chand Gehlot	Social Justice and Empowerment
Dr. S Jaishankar	External Affairs
Ramesh Pokhriyal 'Nishank'	Education
Arjun Munda	Tribal Affairs
Smriti Zubin Irani	Women and Child Development; Textiles
Dr. Harsh Vardhan	Health and Family Welfare; Science and Technology; Earth Sciences
Piyush Goyal	Railways; Commerce and Industry; Consumer Affairs; Food and Public Distribution
Dharmendra Pradhan	Petroleum and Natural Gas; Steel
Mukhtar Abbas Naqvi	Minority Affairs
Prakash Javadekar	Environment, Forest and Climate Change; Information & Broadcasting; Heavy Industries and Public Enterprise
Prahlad Joshi	Parliamentary Affairs; Coal; Mines
Dr. Mahendra Nath Pandey	Skill Development and Entrepreneurship
Giriraj Singh	Animal Husbandry, Dairying and Fisheries
Gajendra Singh Shekhawat	Jal Shakti

Ministers of State (Independent Charge)

Minister	Portfolio
Santosh K. Gangwar	Labour and Employment (Independent Charge)
Rao Inderjit Singh	Statistics and Programme Implementation (Independent Charge); and Planning (Independent Charge)
Jitendra Singh	Development of North Eastern Region (Independent Charge); Prime Minister's Office; Personnel, Public Grievances and Pensions; Department of Atomic Energy; Department of Space
Kiren Rijiju	Youth Affairs and Sports (Independent Charge); Minority Affairs; AYUSH (Independent Charge)
Raj Kumar Singh	Power (Independent Charge); New and Renewable Energy (Independent Charge); Skill Development and Entrepreneurship
Prahlad Singh Patel	Culture (Independent Charge); Tourism (Independent Charge)
Hardeep Singh Puri	Housing and Urban Affairs (Independent Charge); Civil Aviation (Independent Charge); Commerce and Industry
Mansukh Mandaviya	Shipping (Independent Charge); Chemicals and Fertilizers

Ministers of State

Minister	Portfolio
Faggansingh Kulaste	Steel
Ashwini Kumar Choubey	Health and Family Welfare
General (Retd) VK Singh	Road Transport and Highways
Arjun Ram Meghwal	Parliamentary Affairs; Heavy Industries and Public Enterpris
Krishan Pal Gurjar	Social Justice and Empowerment
Danve Raosaheb Dadarao	Consumer Affairs, Food and Public Distribution
G. Kishan Reddy	Home Affairs
Parshottam Rupala	Agriculture and Farmers Welfare
Ramdas Athawale	Social Justice and Empowerment
Sadhvi Niranjan Jyoti	Rural Development
Babul Supriyo	Environment, Forest and Climate Change
Sanjeev Kumar Balyan	Animal Husbandry, Dairying and Fisheries
Anurag Singh Thakur	Finance; Corporate Affairs
Angadi Suresh Channabasappa	Railways
Nityanand Rai	Home Affairs
Dhotre Sanjay Shamrao	Education; Communication; Electronics and Information Technology
Rattan Lal Kataria	Jal Shakti ; Social Justice & Empowerment
V. Muraleedharan	External Affairs; Parliamentary Affairs
Renuka Singh Saruta	Tribal Affairs
Som Parkash	Commerce and Industry
Rameswar Teli	Food Processing Industries
Pratap Chandra Sarangi	Micro, Small and Medium Enterprises; Animal Husbandry, Dairying and Fisheries
Kailash Choudhary	Agriculture and Farmers Welfare
Debasree Chaudhuri	Women and Child Development

Governors and Chief Ministers

State	Governor	Chief Minister
Andhra Pradesh	*Biswabhusan Harichandan*	Jaganmohan Reddy
Arunachal Pradesh	*BD Mishra*	Pema Khandu
Assam	*Jagdish Mukhi*	Sarbananda Sonowal
Bihar	*Phagu Chauhan*	Nitish Kumar
Chhattisgarh	*Anusuiya Uikey*	Bhupesh Baghel
Goa	*Bhagat Singh Koshyari*	Pramod Sawant
Gujarat	*Acharya Dev Vrat*	Vijay Rupani
Haryana	*Satyadev Narayan Arya*	Manohar Lal Khattar
Himachal Pradesh	*Bandaru Dattatraya*	Jai Ram Thakur
Jharkhand	*Droupadi Murmu*	Hemant Soren
Karnataka	*Vajubhai R. Vala*	BS Yediyurappa
Kerala	*Arif Mohammed Khan*	P. Vijayan
Madhya Pradesh	*Anandiben Patel (Add. Charge)*	Shivraj Singh Chauhan
Maharashtra	*Bhagat Singh Koshyari*	Uddhav Thackerey
Manipur	*Najma Heptullah*	N. Biren Singh
Meghalaya	*Satya Pal Malik*	Conrad Sangma
Mizoram	*PS Shreedharan Pillai*	Zoramthanga
Nagaland	*RN Ravi*	Nephiu Rio
Odisha	*Ganeshi Lal*	Naveen Patnaik
Punjab	*VP Singh Badnore*	Amarinder Singh
Rajasthan	*Kalaraj Mishra*	Ashok Gehlot
Sikkim	*Ganga Prasad*	Prem Singh Tamang
Tamil Nadu	*Banwarilal Purohit*	EK Palanisami
Tripura	*Ramesh Bais*	Biplab Deb
Telangana	*Tamilisai Soundararajan*	K. Chandra-shekhar Rao
Uttar Pradesh	*Anandiben Patel*	Yogi Adityanath
Uttarakhand	*Baby Rani Maurya*	Trivendra Singh Rawat
West Bengal	*Jagdeep Dhankhar*	Mamata Banerjee

Note In this book, we were unable to update some new facts. Given below is the page numberwise updation of such facts.

Page 122

- According to India State of Forest Report, 2019, the total forest and tree covers 24.56% of the total land area of the country.

Page 133

- The NH 548 which is of 5 km in length, is Shortest National Highway in India.
- Amongst the states, Maharashtra has the largest length of National Highways in the country followed by Uttar Pradesh.
- In October, 2020 world's longest highway tunnel (Atal Tunnel) was inaugurated. Its length is 9.02 km. It connect Manali with Lahaul-Spiti.

Page 134

- The Government of India on 5th February, 2020 gave approval for construction of greenfield major port at Vadhavan in Maharashtra.

Page 143

Wildlife Conservation in India

Project Dolphin	2020
Project Asiatic Lion	2020

Page 179

- The 104th Amendment 2020 extended the reservation of seats for SCs and STs in the Lok Sabha and State Legislative Assemblies by 10 years. However, it discontinued the reservation of seats to Anglo-Indian community in Lok Sabha and State Legislatives.

Page 190

- Public Wi-Fi Access Network Interface (PM-WANI) was launched on 9th December 2020.
- Kisan Suryodaya Yojana was launched on 24th October 2020.
- Ghar Tak Fibre Scheme was launched on 21st September 2020.
- Pradhan Mantri Matsya Sampada Yojana (PMMSY) was launched on 10th September 2020.
- PM SVANidhi Scheme was launched on 1st June 2020.
- Pradhan Mantri Garib Kalyan Rojgar Abhiyaan was launched on 20th June, 2020.
- One Nation, One Ration Card scheme is scheduled to implemented by 1st June, 2020.

Page 197

- At present, there are 14 Navratnas.

Page 198

- At present, there are 62 Miniratna-I and 12 Miniratna-II.

Classification of MSMES

Manufacturing and Services
(1st July 2020)

	Micro	Small	Medium
Investment	< ₹ 1 cr	< ₹ 10 cr	< ₹ 50 cr
Turnover	< ₹ 5 cr	< ₹ 50 cr	< ₹ 250 cr

Page 204

- Vijaya Bank and Dena Bank merged with Bank of Baroda (BoB) on 1st April, 2019.
- In April, 2020, government has merged Indian Bank with Allahabad Bank, Oriental Bank of Commerce (OBC) and United Bank of India was merged with Punjab National Bank.

Page 207 & 208

- At present, both IMF and World Bank have 190 members each.
- The 12th WTO Conference is scheduled to be held in Nur-Sultan (Kazakhstan) in 2021.

Page 262

- **COVID-19** It is caused by Novel Corona Virus. It affected the respiratory system. Its symptoms are fever, dry cough, tiredness, aches, pains, nasal congestion, headache, conjunctivitis, sore throat, diarrhoea, loss of taste or smell or a rash on skin or discoloration of fingers or toes.

Page 270
- Param Siddhi super computer was developed in India in 2020 by CDAC.

Page 275
- Toman is the currency of Iran.

Page 276
- Sugar Bowl of the World in located in Cuba.

Page 293
- At present, there are 3 functional reactors in Kakrapar Nuclear Power Station (Gujarat), 5 functional reactors in Rawatbhata Nuclear Power Station (Rajasthan) and 2 functional reactors in Kudankulan Nuclear Power Station (Tamil Nadu).
- There are now 22 operating nuclear power reactors (two boiling water reactors and twenty PHWRs (Pressurised Heavy Water Reactors) with a total installed capacity of 6780 MW.

Page 295
- **EOS-01/RISAT-2BR2** It was launched on 7th November, 2020 through PSLV-DL C49 from Sriharikota. It is an Earth observational satellite.
- **CMS-01/GSAT-12R** It was launched on 17th December, 2020 through PSLV-XL C50 from Sriharikota. It is a communication satellite.

Page 297
- Dar Yasin, Mukhtar Khan and Channi Anand (Photography) won Pulitzer Prize in 2020.

Page 298
- In 2020, Man Booker International Prize is given to Marieke Lucas Rijneveld (Dutch) for the book Discomfort of Evening.
- The 2020 Miss India event was postponed due to COVID-19 pandemic.

Page 301
- The prize money in Rajiv Gandhi Khel Ratna Award has been increased from ₹ 7.5 lakh to ₹ 25 lakh.
- In 2020, Rohit Sharma (Cricket), Vinesh Phogat (Wrestling), Rani Rampal (Hockey), Monika Batra (Table Tennis) and Mariyappan Thangavelu (Para athlete) were awarded Rajiv Gandhi Khel Ratna Award.
- Arjun Award carries a cash prize of ₹ 15 lakh.
- Dronacharya Award carries cash prize of ₹ 10 lakh.
- In 2020, Dharmendra Tiwari (Archery), Purushottam Rai (Athletics), Shiv Singh (Boxing), Romesh Pathania (Hockey), Krishan Kumar Hooda (Kabaddi), Vijay Bhalchandra Munishwar (Power lifting), Naresh Kumar (Tennis) and Prakash Dahiya (Wrestling) were awarded.
- Dhyanchand Award carries a cash prize of ₹ 10 lakh.
- Kuldip Singh Bhullar (Athletics), Jincy Philips (Athletics), Pradeep Shrikrishna Gandhe (Badminton), Trupti Murgunde (Badminton), N Usha (Boxing), Lakha Singh (Boxing), Sukhvinder Singh Sandhu (Football), Ajit Pal Singh (Hockey), Manpreet Singh (Kabaddi), J Ranjith Kumar (Para Athletics), Satyaprakash Tiwari (Para Badminton), Manjeet Singh (Rowing), Sachin Nag (Swimming), Nandan P Bal (Tennis) and Netarpal Hooda (Wrestling) were awarded Dhyanchand Award for 2020.

Page 314
- The Commonwealth (London) Organisation has 54 members.

Page 316
- The 2022 winter olympic will be held in Beijing, China.
- Tokyo, will host 2020 summer olympic (postponed) to 2021 due to COVID-19 pandemic.

Page 318
- ICC Twenty-20 Cricket World Cup 2021 is scheduled to be held in India.

Page 319
- The 15th Men's Hockey World Cup 2023, will be held at the Kalinga Stadium in Bhubaneswar and in Rourkela, Odisha.



INDIAN HISTORY

ANCIENT INDIA

PRE-HISTORIC PERIOD

- The artefacts from **Bori** in Maharashtra suggest the appearance of early human beings in India around 1.4 million years ago.
- Early humans used tools of stone. This period is therefore, known as the **Stone Age**, *which has been divided into*
  The Palaeolithic or Old Stone Age
  The Mesolithic or Middle Stone Age
  The Neolithic or New Stone Age
  The Chalcolithic Age or Stone-copper Age

The Palaeolithic Age (5,00,000-9000 BC)

- Palaeolithic men were hunters and food gatherers.
- They had no knowledge of agriculture, or pottery and lived in cave and rock shelters.
- They used tools of unpolished, rough stones. Therefore they are also called **Quartzite men.**
- **Homo Sapiens** first appeared in the last phase of this period.
- This age is further divided into three phases : Lower, Middle and Upper Palaeolithic. According to the nature of the stone tools used by the people and the changes in the climate and environment.

The Mesolithic Age (9000-6000 BC)

- It was a transitional phase between the **Palaeolithic Age** and the **Neolithic Age**.
- In this age, the climate became warm and dry which brought about changes in fauna and flora and made it possible for human beings to move to new areas.
- The Mesolithic people lived on hunting, fishing and food-gathering. At a later stage, they also domesticated animals.
- The people of the Palaeolithic and Mesolithic ages practised painting.
- **Bhimbetka** in Madhya Pradesh, is a striking site of pre-historic painting of Mesolithic age.

The Neolithic Age (6000-1800 BC)

- The people of this age used tools and implements of **polished stone**. They particularly used stone axes.
- It is interesting that in Burzahom (kashmir), domestic dogs were buried with their masters in their graves.
- First use of hand made pottery and potter wheel appears during the Neolithic age. The men lived in caves and decorated their walls with hunting and dancing scenes.
- Neolithic settlers were earliest farming communities. They produced barley, gram and ragi. They domesticated cattle, sheeps and goats. They wove cotton and wool to make clothes.
- Main Neolithic sites in India include Burzahom, Machragarh, Daojali Hading, Chirand, Maski, Brahmagiri, Hallur, Piklihal, Utnoor, Nagarjuna konda and paiyampalli.

The Chalcolithic Age (1800BC-1000 BC)

- Chalcolithic culture was based on use of tools made of stone and copper. Copper was the first metal to be used by man.
- Important Chalcolithic sites in India are Ahar, Kayatha, Malwa, Eran, Rangpur, Navdatoli, Nevasa, Daimabad and Inamgaon.

INDUS VALLEY CIVILISATION

- Indus Valley Civilisation is one of the four earliest civilisations of the world alongwith the civilisations of Mesopotamia (Tigris and Euphrates), Egypt (Nile) and China (Hwang Ho).
- The Civilisation forms a part of the proto-history of India and belongs to the **Bronze age**.
- In its mature form, the civilisation lasted between 2600-1750 BC.
- **Dayaram Sahni** first discovered Harappa in 1921. RD Bannerjee discovered Mohenjodaro or 'Mound of the Dead' in 1922.

It can be divided into the following sub-parts

- Early Phase – 3500-2600 BC
- Middle (mature) Phase – 2600-1900 BC
- Later Phase – 1900-1400 BC

NOMENCLATURE

- **Indus Valley Civilisation**, coined by **John Marshall**, as it flourished along the Indus river.
- **Harappan Civilisation named** after the first discovered site, Harappa.
- **Saraswati-Sindhu Civilisation**, as most of the sites have been found along the Indus-Saraswati river.

TOWN-PLANNING

- A common feature was the **grid system** i.e. streets cutting across one another at right angles, dividing the town into large rectangular blocks.
- The towns were divided into two parts: the Upper part or Citadel and the Lower part.
- **Underground drainage system** connected all houses to the street drains, made up of mortar, lime and gypsum. They were covered with either brick or stone slabs and equipped with manholes. This shows a developed sense of health and sanitation.
- **The Great Bath** (Mohenjodaro) It was used for religious bathing. Steps at either end lead to the surface. There were changing rooms alongside.
- **The Granaries** (Harappa) 6 granaries in a row were found in the Citadel at Harappa.
- Houses were made up of burnt bricks.
- Lamp-posts were erected at regular intervals. It indicates the existence of street lighting.

Indus Cities At a Glance

City	*Province*	*River Bank*	*Year of Discovery*	*Archaeologist*
Harappa	Pakistani Punjab	Ravi	1921	DR Sahni
Mohenjodaro	Sind	Indus	1922	RD Bannerjee
Chanhudaro	Sind	Indus	1931	MG Majumdar
Sutkagendor	Baluchistan	Dasht	1931	Aurel Stein
Rangpur	Gujarat	Meedar	1931	MS Vats
Ropar	Indian Punjab	Sutlej	1953	YD Sharma
Lothal	Gujarat	Bhogava	1953	SR Rao
Kalibangan	Rajasthan	Ghaggar	1951	A Ghosh
Dholavira	Kachchh (Gujarat)	Luni	1967-68	JP Joshi
Banawali	Haryana	Ghaggar	1973	RS Bisht
Alamgirpur	Uttar Pradesh	Hindon	1974	YD Sharma
Rakhigarhi	Haryana	Ghaggar	1997	Amrendra Nath

AGRICULTURE

- Agriculture was the backbone of the civilisation. The soil was fertile due to inundation in the river Indus.
- They used wooden ploughshare (ploughed field from Kalibangan) and stone sickles for harvesting.
- **Crops produced** were wheat, barley, dates, peas, sesamum, mustard, millet, ragi, bajra and jowar. At Lothal and Rangpur, rice husks were found. Sugarcane was not known to the Indus people.
- They were **first to produce cotton** in the world, which Greeks called as Sindon. A fragment of woven cotton cloth was found at Mohenjodaro.

DOMESTICATION OF ANIMALS

- They domesticated buffaloes, oxens, sheep, asses, goats, pigs, elephants, dogs, cats etc.
- Camel bones are reported at Kalibangan and remains of horse are discovered from Surkotada. Remains of Rhinoceros are found from Amri.

ART AND CRAFTS

- Harappans used tools made of stone and bronze. However iron was not known to them. Bronze was made by mixing copper (from Khetri) with tin.
- Boat making, jewellery of gold, silver precious stone and bead making was practised. Cotton fabrics were used in summer and woollens in winter.
- **Pottery** Both plain and painted (red and black) pottery was made. Pots were decorated with human figures, plants, animals and geometrical patterns.
- **Metal Images** Bronze image of a dancing girl (identified as *devdasi*) and stone steatite image of a bearded man were both obtained from Mohenjodaro.
- **Terracotta Figurines** Terracotta means the fire baked clay. It was used to make toys, objects of worship, animals, toy-carts, etc.

TRADE

- Agriculture, industry and forest provided the basis for internal and external trade.
- Trade was based on barter system. Coins are not evident. Bullock carts and boats were used for transportation.

Towns and Trade

Daimabad	Bronze industry.
Lothal	Factory for stone tools and metallic finished goods.
Balakot	Pearl finished goods, bangle and shell industry.
Chanhudaro	Beads and Bangles factory. It was the only city without a citadel.

- Weights and measures were made of limestone, steatite etc. Generally in weighing mostly 16 or its multiples were used.
- Foreign trade flourished with Mesopotamia or Sumeria (Iraq), Central Asia, Persia, Afghanistan and Bahrain.

Major Imports

Imports	*From*
Gold	Kolar (*Karnataka*), Afghanistan, Persia (*Iran*)
Silver	Afghanistan, Persia (*Iran*), South India
Copper	Khetri (*Rajasthan*) Baluchistan, Arabia
Tin	Afghanistan, Central Asia
Lapis Lazuli and Sapphire	Badak-Shan (*Afghanistan*)
Jade	Central Asia
Steatite	Shahr-i-Sokhta, Kirthar Hills
Amethyst	Maharashtra

RELIGIOUS PRACTICES

- **Chief Female Diety** A terracotta figurine, where a plant is shown growing out of the embryo of a woman and represents the **mother Goddess** (Goddess of Earth).
- **Chief Male Diety** *Pashupati Mahadeva* (Proto-Siva), represented in seals, sitting in a yogic posture on a throne and having three faces and two horns. He is surrounded by an elephant, a tiger, a rhino and a buffalo, and two deers appear at his feet.
- Indus people believed in ghosts and evil forces and used amulets for protection against them. Fire altars are found at Lothal and Kalibangan.
- No temples have been found at any Harappan sites.
- Swastika symbol was originated here.

SCRIPT

- It was pictographic in nature. Fish symbol is the most represented. It was written from right to left in the first line. The style is called *Boustrophedon*.

DECLINE OF THE CIVILISATION

The Harappan culture flourished upto 1800 BC, then it began to decline. There is no unanimity among historians regarding the reason for decline of this urban civilisation. There are many different theories by the thinkers, that shows the decline of the Indus culture.

Decline of Indus Civilisation Different Views

Views	*Thinkers*
External Aggression	Wheeler, Piggot and Gordon-Childe
Inundation	MR Sahani
Epidemic	KVR Kennedy
Tectonic Disturbances (e.g. Dholavira)	Marshall and Raikes
Sudden Decline	Wheeler
Climatic Change	Aurel Stein and AN Ghosh
Deforestation, Scarcity of Resources, Ecological Imbalances	Fairservis
Flood (e.g. Mohenjodaro)	Marshall, SR Rao, Maickey
The Destruction due to Change in course of River Ghaggar	GF Holes

Important Harappan Sites

Site	*Archaeological Finds*
Harappa (*Gateway city*)	Two rows of six granaries with brick platform, work men's quarter, stone symbol of lingam and yoni, virgin-Goddess (seal), clay figures of mother Goddess, wheat and barley in wooden mortar, copper scale and mirror, vanity box, dice. Sculpture of dog chasing a deer (bronze) nude male and nude dancing female (stone), sand stone male torso.
Mohenjodaro (*Mound of the Dead*)	The great bath, the great granary (largest building), multipillared assembly hall, college, proto-Shiva seal, clay figures of mother Goddess, dice. **Sculpture Bronze** dancing girl, steatite image of bearded man.
Kalibangan (*Black Bangle*)	Decorated bricks, bangle factory, wheels of a toy cart, wells from every ploughed field house. Bones of camel, 7-fire altars.
Chanhudaro (*Lancashire of India*)	Inkpot, lipstick, imprint of dog's paw on a brick, only city without citadel, bronze toy cart, Terracotta bullock cart.
Daimabad	Bronze images of Charioteer with chariot ox; elephants and rhinoceros.
Amri	Actual remains of rhinoceros.
Alamgirpur	Impression of cloth on a trough.

Site	*Archaeological Finds*
Lothal (*Manchester of Indus Valley Civilisation*)	Rice husk, fire altars, dockyard, grinding machine, tusks of elephant, terracotta figure of horse, dying vat, painted jar (bird and fox), teracotta ship, impressions of cloth on some seals, modern day chess, instrument for measuring angles.
Ropar	Buildings made of stone and soil. Dog buried with human, oval pit burials.
Banawali	Oval shaped settlement, only city with radial streets, lack of systematic drainage pattern. Toy plough, largest number of barley grains.
Surkotada	Both Citadel and Lower Town fortified with stone wall, First actual remains of horse bones, pot burials.
Dholavira	Only site to be divided into 3 parts. Giant water reservoir; unique water harnessing system; dams and embankments; a stadium; rock-cut architecture.
Rakhigarhi	largest Indus valley site.

THE ARYANS AND THE VEDIC AGE

- The word **Aryan** literally means *of high birth*. The location of the original homeland of the Aryans is still controversial, but the most accepted theory is that they migrated from Central Asia and settled in the place which is known as the **land of seven rivers.**
- This theory was given by Max Muller.

RIGVEDIC PERIOD (1500-1000 BC)

Early Vedic people lived in the region of Sapta Sindhu. They had knowledge of about 40 rivers including Yamuna, Saraswati and Ganga. Ocean was mentioned as **Samudra**. Himalaya as Himvant, Hindukush as Munjavant and desert as Dhawa.

Economy

- Aryans possessed better knowledge in agriculture. Ploughshare is mentioned in the Rigveda. It was made up of wood and was drawn by the ox. They were acquainted with sowing, harvesting, threshing and knew about different seasons.
- The term 'ayas' used for copper or bronze shows that metal working was known
- Cow was the standard unit of exchange. Gold coins like Nishka, Krishna and Satmana were in use. **Godhuli** was used as a measure of time and **Gavyuti** as measure of distance.

Rivers mentioned in the Rigveda

Rigvedic Name	*Modern Name*
Sindhu	Indus
Vitasta	Jhelum
Asikni	Chenab
Parushni	Ravi
Vipas	Beas
Sutudri	Sutlej
Gomati	Gomal
Krumu	Kurram
Drishadvati	Ghaggar
Suvastu	Swat
Kubha	Kabul
Nandi tara	Saraswati
Sadanira	Gandak

Rigvedic Gods

- Rigvedic people believed in nature worship, and not in erecting temples or idol worship. They performed Yajnas in open areas.

- **Soma** was considered to be the God of plants and an intoxicating drink was named after him. The 9th mandala of Rigveda, i.e. *Soma Mandala* is attributed to Soma.
- Some female divinities were **Aditi** and **Usha** (Goddess of Dawn).

Gods and their Concerned Fields

God	*Concern Field(s)*
Agni	Fire God
Varuna	Water God and upholder of natural order
Surya	God with 7 horses-driven chariot
Savitri	God of light to whom Gayatri Mantra is addressed
Mitra	Solar God
Pushan	God of marriage, also guarded roads and cattle
Vishnu	One who covered Earth in three steps-Upakrama
Rudra	God of animals
Dyaus	Oldest God and Father of the World
Ashwin/ Nastya	God of health, youth and immortality
Sindhu	River Goddess
Yama	God of death
Maruts	Personified storm
Usha	Goddess of dawn
Aditi	Goddess of eternity

Polity

- The King or Rajan was the head of the tribal polity. Although his post was hereditary, we have also some traces of election by the tribal assembly called **Samiti**.
- The Samiti was the National Assembly of the people; while the **Sabha** was the Council of Elders.
- King was assisted by many functionaries. Most important is the **Purohita,** the religious advisor of the king, followed by the **Senani**, the head of the army.

Society

- Kinship was the basis of society's structure. People gave their primary loyalty to the tribe, which was called **Jana** or Vis. *Vis* was divided into **grama**. When grama clashed with one another, it caused the **Sangrama** meaning war.
- Society was mostly Patriarchal. Monogamy was prevalent. No evidence of child marriage. Levirate and widow remarriage was practised.

The Dasarajna War (Battle of ten Kings)

- This battle was fought on the banks of Parushni River (Ravi). **Sudas,** the son of Divodas and the Bharata king of Tritsu family won over an alliance of ten tribes (5 Aryans and 5 non-Aryans).
- The battle broke out due to a dispute between **Vashistha** (priest of Bharatas) and Visvamitra (priest of Alliance).

LATER VEDIC PERIOD (1000-600 BC)

- Aryans expanded from Punjab to Western UP, covered by Ganga-Yamuna Doab (Aryavrata).
- Later Vedic texts refer to rivers Narmada, Sadanira Chambal and Vindhya mountains. They gave 3 fold territorial division of India into Aryavarta (Northern India), Madhyadesa (Central India) and Dakshinapatha (Southern India).
- The expansion towards the East is indicated in the legend of Satapatha Brahmana i.e., how Videha Madhava migrated from the Saraswati region, crossed Sadanira and came to the land of Videha (modern Tirhut).

Political Organisation

- Formation of large kingdoms. Tribal assemblies lost their importance and royal power increased at their cost. Vidhata totally disappeared. Women were no longer permitted to attend assemblies.
- The term **Rashtra** indicating territory, first appeared in this period.
- **Taittariya Brahmana** refers to the theory of divine origin of kingship.
- Satapatha Brahmana refers to 12 Ratninas or civil functionaries of the time.
- There was a development of the judiciary. Kings administered the criminal court.

12 Ratninas (Satapatha Brahmin)

Purohita	The Priest
Mahishi	The Queen
Yuvraj	Crown Prince
Suta/Sarathi	The Royal Herald/The Charioteer
Bhagadugha	Collector of Taxes
Akshavapa	Courier
Palagala	Friend of King
Govikarta	Head of Forest Department
Senani	The General
Gramani	Head of the Village
Kshata	Gateman/Chamberlain
Sangrahitri	Treasurer

Social Organisation

- The four-fold division of society become more clear, initially based on occupation and later became hereditary.
- **Brahmin** The growing cult of sacrifice enormously added to the power of Brahmins.
- **Kshatriyas** They constituted the warrior class.
- **Vaishyas** They were the agriculturists, cattle rearers, traders, artisans and metal workers.
- **Shudras** Lowest in the social hierarchy and born to serve the upper three varnas.
- The Ashrama system developed in this period and the position of women declined as compared to what was in the Rigvedic age.

Religion

- Rituals became important in the cult of sacrifice.
- **Prajapati** became the supreme God. Vishnu was conceived as the preserver and protector of the people.
- **Pushan**, responsible for well being of cattle, became the God of Shudras.
- Towards the end of the Vedic age, a section of society began to resent priestly domination.

Economy

- **Agriculture** became the chief economic activity. Main crops were wheat, rice, barley and beans.
- New occupational groups emerged such as fishermen, washermen, dyers, door-keepers and footmen.
- Tin, silver and iron was now known to the people.
- Merchants were organised into Guilds, as indicated by the terms Ganas (corporations) and **Sresthins** (eldermen).
- Characteristic pottery : Painted Grey Ware.

THE VEDIC LITERATURE

- The word Veda comes from the word Vid meaning to know or knowledge.
- Vedic texts are divided between **sruti** (based on hearing) and **smriti** (based on memory).
- Vedas are divided into **Samhitas**.

Rigveda

Oldest religious text in the world. Collection of hymns, composed around 1700 BC contains 1028 hymns and 10580 verses and is divided into 10 mandalas. The last mandala contains the Purushasukta hymn which mention four Varnas. Saraswati is the deity river in the Rig Veda.

Samaveda

Samaveda derives its root from *saman*, which means **melodies**. It is a collection of melodies. The hymns of the Samaveda were recited by Udgatri. It contains Dhrupad Raga.

Yajurveda

Deals with the procedure for the performance of sacrifice. The beliefs and rituals of non-Aryan are written in it.

Atharvaveda

It is a book of magical formula. It contains charms and spells toward off evils and diseases.

Brahmanas

- These are the prose commentaries on various Vedic hymns. They explain Vedas in an orthodox way. They explain the hidden meaning behind the hymns. They are ritualistic by nature.

The most important is the **Satapatha Brahmana** attached to Yajurveda.

Rigveda : Aitareya and Kaushitiki
Yajurveda : Shatapatha and Taittariya
Samveda : Jaiminaya and chhandogya
Atharva Veda : Gopatha

The Aranyakas

The sages dwelling is the forest explained the vedic scriptures to their pupils in the form of Aranyakas. This has magical power and they form the concluding part of the Brahmanas.

The Upanishada

- Also called Vedanta, because they denote the last phase of the Vedic period (800-500 BC).
- They are spiritual and philosophical in nature.
- There are 108 Upanishadas. Vrihadaranyaka is the oldest Upanishad.

Vedangas

They are the limbs of the Vedas. *These are treatises on Science and Arts.*

1. Shiksha (deals with pronounciation)
2. Kalpa (deals with rituals)
3. Vyakarana (grammar)
4. Nirukta (etymology)
5. Chhanda (metrics)
6. Jyotisha (astronomy)

- **Panini** wrote **Ashtadhyayi** (4th century BC) on Vyakarana (grammar).

Upavedas

There are four Upavedas

Dhanurveda (Upaveda of Yajurveda) deals with the art of warfare.

Gandharvaveda (Upaveda of Samaveda) deals with art and music.

Shilpaveda (Upaveda of Atharvaveda) deals with architecture.

Ayurveda (Upaveda of Rigveda) deals with medicine.

Shad-Darshana

Six system of philosophies, given by six philosophers of ancient India.

- Nyaya (analysis) Gautama
- Vaisesika (logic) Kannada
- Sankhya (enumeration) Kapila
- Yoga (application) Patanjali
- Purva Mimansa (enquiry) Jaimini
- Uttara Mimansa (vedanta) Badarayana

Sutras

Sutra literature is divided into three classes–**Srauta Sutra** (dealing with large public sacrifice), **Griha Sutra** (dealing with rituals connected with birth, naming, marriage) and **Dharma Sutra** (explain social and local customs).

Puranas

- Puranas means 'the old'. There are 18 famous puranas. Matsya purana is the oldest. Puranas deals with 5 subjects : Sarga (creation); Pratisarga (recreation); Wansa (Genealogies of Gods); Manvantara (rule of Manus); and Wamshanucharita (Genealogies of kings).

Epic

- The **Mahabharata**, written by **Ved Vyas,** is older than the **Ramayana**. Originally, Mahabharata consisted of 880 verses, then it was raised to 24000 verses.
- Ramayana written by **Valmiki**, originally consisted 6000 verses, which was raised to 12000 and finally 24000 verses.

Smritis

- Dharma Shastra is the other name for **smritis**, which are the law books written in **sloka** form. The important smritis are Manav Dharma Shastra, Vishnu Dharma Shastra, Yajnavalkya Smriti and Narada Smriti.
- Manav Dharma Shastra or Manusmriti is the oldest and most famous. Manu is supposed to be the first King and law maker.

PRE-MAURYAN AGE

- The material advantages brought about by the use of the iron implements in Eastern UP and Bihar in 6th century BC, helped in the formation of large territorial states.
- Use of iron tools in agriculture produced surplus.
- Thus, many **Janapadas** sprung up in 6th century BC, the larger of which were called **Mahajanapadas**.

THE MAHAJANAPADAS

- The **Anguttara Nikaya** of **Suttapitaka** and **Mahavastu** (Buddhist literature) and **Bhagavati Sutta** (Jain literature) mentions the list of the 16 Mahajanapadas.
- People now owned stronger allegiance to the janapada or territory, than the Jana or tribe they belonged to.
- *Mahajanapadas were of two types*

  Non-monarchial Republican States Kamboj, Kuru, Koliyas (Ramgrama), Malla, Moriya (Pipplivana), Shakya (Kapilvastu), Vajji, Panchal, Lichchhavis (Vaishali), Bhaggas (Sumsumasa), Kalamas (Kesaputta), Videhas (Mithila), Jnatrikas (Kundalgrama).

  Monarchial States Anga, Avanti, Chedi, Kashi, Kosala, Gandhara, Magadh, Matsya, Surasenas, Vatsa.

- Asmaka was the Southernmost Mahajanapada.
- Vatsa was earlier a Kuru clan.
- Vajji was a confederacy of eight republican clans.
- **Lichchhavis** are said to be the oldest republic in the World.

Mahajanapadas and their Capitals

Mahajanapadas (Location)	*Capital(s)*
Gandhara (Between Kabul and Rawalpindi)	Taxila
Anga (Bhagalpur and Mungher district of Bihar)	Champa
Magadha (Patna and Gaya district of Bihar)	Girivrajs, Rajagriha (Bimbisara); Pataliputra (Udayin); Vaishali (Shishunaga); Pataliputra (Ashok)
Kashi (Varanasi district, UP)	Varanasi
Vajji (Vaishali district, Bihar)	Vaishali
Malla (South of Vaishali district, UP)	Kushinagara and Pava
Chedi (River Ken) (Bundelkhand area)	Suktimati or Sotthivati
Vatsa (River Yamuna) (Allahabad and Mirzapur district in UP)	Kaushambi
Kosala (Eastern UP)	Sravasti and Ayodhaya (Saket)
Kuru (Ganga Yamuna doab) (Delhi-Meerut region)	Hastinapur and Indraprastha
Panchala (Ganga-Yamuna doab) (Rohilkhand)	Ahichhatra, Kampilya
Matsya (Jaipur-Bharatpur-Alwar district)	Viratnagar/Bairath
Surasenas (Mathura region)	Mathura
Assaka/Asmaka (River Godavari) (Near Paithan in Maharashtra)	Potali/Poddana/Potana
Avanti (Malwa)	Ujjain (Northern capital), Mahishmati (Southern capital)
Kamboj (Hazara district of Pakistan)	Rajapur or Hataka

MAGADHA EMPIRE

- The period from the 6th century BC to 4th century BC saw the struggle for supremacy among the four Mahajanapadas-Magadha, Kosala, Vatsa and Avanti.
- Ultimately, Magadha emerged as the most powerful and prosperous kingdom in North India.
- The founders of Magadha were **Jarasandha** and **Brihadratha**. But the actual growth started under the Haryankas and its expansion took place under the **Sisunagas** and **Nandas** and reached its zenith under the **Mauryas**.

HARYANKA DYNASTY

Bimbisara (544-492 BC)

- He built the capital city Rajgir (Girivraja). He was contemporary to Gautam Buddha and the first king to have a standing army, for which he is known as **Seniya**.
- His three wives belonged to the royal family of Koshala (Mahakosaladevi), Lichchhavi (Chellana) and Madra clan of Punjab (Khema).
- He sent his personal physician, Jivaka to his rival Avanti king **Chandapradyota Mahasena** of Ujjain to cure him of jaundice.

Ajatashatru (492-460 BC)

- He was the son of Chellana and Bimbisara. He occupied the throne by killing his father.
- He defeated his maternal uncle Prasenjit, king of Kosala. He destroyed Vaishali after a protracted war of 16 years.
- He patronised the **First Buddhist Council** and Buddha died during his reign.

Udayin (460-444 BC)

He built the new capital city Pataliputra. Udayin was succeeded by weak rulers Anuruddha, Munda and Naga-Dasak.

SHISHUNAGA DYNASTY (412-344 BC)

Shishunaga

He was the minister of Nag-Dasak and was elected by the people. He temporarily shifted the capital to Vaishali.

The greatest achievement of the Shishunagas was the destruction of the power of Avanti with its capital at Ujjain.

Kalasoka

He transferred the capital from Vaishali to Pataliputra and convened the **second Buddhist Council** in Vaishali (383 BC).

Nanda Dynasty (344-323 BC)

It is considered to be the first non-Kshatriya dynasty.

Mahapadamananda

- The great conqueror and founder of the dynasty, also known as Ekarat, Eka-chhatra (sovereign ruler). He is also described as ''the first empire builder of Indian History.''
- He conquered Koshala and Kalinga (from here, he brought an image of the Jina as victory trophy).

Dhanananda
(The last Nanda ruler)

- Alexander invaded North-Western India during his reign (326 BC), but the huge army of Dhanananda deterred Alexander from advancing towards the Gangetic Valley.
- He is probably referred as Agrammes or Xandrames in the Greek texts.
- Chandragupta Maurya assisted by Kautilya overthrew DhanaNanda to establish the Mauryan dynasty.

Pre-Mauryan Society

Division of society into four classes necessitated the formation of **Dharmasutras**. Shudras were illtreated, restrictions were imposed on women and **Dharmasutras** condemned the Vaishyas for lending money.

Pre-Mauryan Economy

Agricultural production increased. Varihi and Sali were new varieties of rice. **Karisa, Nivartan** and **Kulyavapa** were units of land measurement.

FOREIGN INVASIONS

Iranian Invasion

- **Cyrus of Persia** was the first foreign conqueror, who penetrated well into India. He destroyed the city of **Capisa**. He enrolled Indian soldiers in the Persian army.
- His successors, Darius I and Xerxes invaded North-West India in 516 BC.

Alexander's Invasion

- Alexander, the king of Macedonia conquered Kabul in 328 BC. He moved to India through the unguarded Khyber pass.
- **Ambhi**, the ruler of Taxila, readily submitted to Alexander. He got the first and strongest resistance from Porus. Alexander defeated Porus in the **Battle of Hydaspes** on the bank of River Vitasta (Jhelum), but was impressed by his bravery, so he restored to Porus his kingdom and made himself his ally.
- After the **Battle of Sakala**, Alexander proceeded upto Beas with a view to conquer the East, but his fatigued army refused to cross the river.
- So, he was forced to retreat. He placed the North-Western India under the Greek Governor Selucus Nikator.
- He remained in India for 19 months (326-325 BC) and died in Babylon (323 BC), at the age of 33 years.

Impacts of Invasion

- Iranian contact gave an impetus to Indo-Iranian trade.
- There was a cultural exchange in the form of Kharoshthi script, from Iran to India and also in the use of bell shaped capitals in sculpture.
- Opening of new trade routes between North-West India and Western Asia.
- Indians learnt from the Greeks in the field of coinage, astronomy, architecture and sculpture (Gandhara School).
- Establishment of many Greek settlements in Kabul and Sind.

RELIGIOUS ENVIRONMENT

- The 6th century BC was the period of great religious upheaval or intellectual revolution of the various sects that emerged in this period. Jainism and Buddhism were the most prominent. This marked the beginning of Second Urbanisation in India, also known as the age of the Buddha.

Immediate Causes for the Rise of Hetrodox Sects

- Division of the society into 4 varnas.
- Reaction of the Kshatriyas to the Brahmins supremacy.
- The desire of the Vaishyas to improve their social position, with increase in their economic position.
- To preserve cattle wealth.
- Desire to go back to simple life. The use of Sanskrit in Vedic texts was not understandable to the masses.

JAINISM

- It was founded by **Rishabhnath**, (emblem -bull) the first Tirthankara (guru).
- The 23rd Tirthankara was **Parsavanath** (symbol-serpent). He was the Prince of Banaras, who had abandoned the throne, and led the life of a hermit. He died at Sammet Sikhar/Parasanath Hill in Giridih district of Jharkhand.
- The 24th Tirthankara was **Vardhaman Mahavira** (emblem-lion).

Life of Mahavira

- **Birth** Born to **Siddhartha** (Jnatrika clan) and **Trishla** (Lichchhavi Princess) in 540 BC at Kundagram near Vaishali in Bihar.
- Wife — Yashoda
- Daughter — Priyadarsana
- Son-in-law — Jamali
- Renunciation — At the age of 30
- Kaivalya — Perfect knowledge at the age of 42, under a sal tree at Jimbhikgrama.
- First Sermon — At Pava, to eleven disciples known as Ganddharas.

- **Death**, at the age of 72 in 468 BC, at Pavapuri near Rajagriha.
- **Titles Jina** or **Jitendriya** (one who conquered his senses); **Kevalin** (perfect learned), **Arihant** (blessed one) and **Mahavira** (the brave).

Teachings of Mahavira

- Rejected the authority of the Vedas and did not believe in the existence of God.
- Strict non-violence.
- Attainment of salvation by believing in penance and dying of starvation.
- Universal brotherhood (equality) and non-belief in Caste System. He believed in **karma** and transmigration of soul.

Jaina Philosophy

Syadavada It is the theory of *'may be'* and seven modes of prediction are possible.

Anekantavada It is the theory of plurality and states that reality has multiple aspects.

Three Ratnas (Way to Nirvana)

- Right faith — Samyak Vishwas
- Right knowledge — Samyak Gyan
- Right conduct — Samyak Acharan

Panch Mahavratas
(Five Cardinal Principles for monks)

- Ahimsa (Non-injury)
- Satya (Non-lying)
- Asteya (Non-stealing)
- Aparigraha (Non-possession)
- Brahmacharya (Observing continence)

Anuvratas For Householders.
(Minor Vows)
(*The first four principles Mahavratas were given by Parsvanath, while the fifth was added by Lord Mahavira*).

Five Instruments of Knowledge

- **Mati Jnana** Perception through activity of sense organs, including the brain.
- **Avadhi Jnana** Clairvoyant perception.
- **Shruta Jnana** Knowledge revealed by scriptures.
- **Manparyay Jnana** Telepathic knowledge.
- **Keval Jnana** Omniscience or Temporal knowledge.

Sects of Jainism

- After the death of Mahavira, during the reign of the king Chandragupta Maurya, a severe famine led to a great exodus of Jaina monks, from Ganga valley to the deccan.
- Bhadrabahu, who led the emigrants, insisted on the retention of the rule of nudity, which led to the establishment of the **Digambaras**.
- Sthulabhadra, the leader of the monk, who remained in the North allowed his followers to wear white garments —**Svetambaras**.

Jaina's Literature

- Sacred texts of Jainism were compiled in 6th century A D at Vallabhi. Mostly written in Prakrit language.
- 6 classes of Jain literature :
  (i) 12 Angas
  (ii) 12 Upangas
  (iii) 10 Parikarnas
  (iv) 6 Chhedasutras
  (v) 4 Mulasutras
  (vi) 2 Sutragranthas
- Other Important Jain texts are : Kalpasutra (in Sanskrit by Bhadrabahu); Parishishtaparvan (by Hemchandra) and Bhadrabahu charita.

Jain Councils

Council	*Year*	*Venue*	*Chairman*	*Royal Patron*	*Developments*
1st Jaina Council	300 BC	Pataliputra	Sthulabhadra	Chandragupta Maurya	Compilation of 12 Angas to replace 14 Purvas
2nd Jaina Council	AD 512	Vallabhi	Devridhigani Kshmasramana	—	Final compilation of 12 Angas and 12 Upangas

BUDDHISM

Founded by Gautama Buddha (Sakyamuni or Tathagat), originally known as **Siddhartha**.

Life of Buddha

- He was born in 563 BC at **Lumbini** in Sakya Kshatriya clan of Kapilvastu.
- Father : Suddhodhana; Mother : Mahamaya; wife : Yashodhara; Son : Rahul; Teachers : Alara Kalama and Udraka.

Teachings of the Buddha

(a) The fundamental principles of Buddha's techings are represented by the Four Noble Truths (arya-satyas)
   (i) Dukkha (that the world is full of sorrow).
   (ii) Dukkha Samuddaya (That there are causes of sorrow)
   (iii) Dukkha Nirodha (This sorrow can be stopped)
   (iv) Dukkha Nirodha gamini pratipada (path leading to cessation of sorrow)

   According to Buddha, roots of all human misery was 'desire' and its annihilation was the surest way of ending unhappiness. One could get out of this chain of suffering and achieve the final **salvation** (Nirvana) by *following eight fold path* (Ashtangika Marga).

(b) **Eight-Fold Path** (Ashtangika Marga)
- Right understanding
- Right thought
- Right speech
- Right action
- Right livelihood
- Right effort
- Right mindfulness
- Right concentration

(c) **Three Jewels** (Triratnas)
- Buddha (the enlightened)
- Dhamma (doctrine)
- Sangha (order)

(d) **Belief in Nirvana** Also known as **moksha** or salvation. It refers to belief in the concept of ultimate bliss, whereby the person gets freedom from the cycles of birth and death.

(e) **Belief in Ahimsa** Law of Karma and Madhya Marga/Madhyama Pratipada (The **Middle Path**).

Sect of Buddhism

- **Vajrayana** Its followers believed that salvation could be attained by magical power which they called **vajra**.
- The chief divinity of the sect is *Taras*. Bengal and Bihar (Eastern India) were the main areas of its concentration.
- **Hinayana** (the lesser vehicle).
- **Mahayana** (the greater vehicle).

Buddhist Scriptures

Hinayana Literature (in Pali)

- Tripitakas
  - **Vinaya Pitaka** Rules of monastic discipline for monks.
  - **Sutta Pitaka** Collection of Buddha's sermons.
  - **Abhidhamma Pitaka** Philosophies of Buddha's teachings.
- Some famous Buddhist Scholars– Ashvaghosha, Nagarjuna, Asanga, Vasubandhu, Buddhaghosha, Dinnaga and Dharmakirti.

Bodhisattvas

- Vajrapani (holds thunderbolt)
- Avlokitesvara/Padmapani (lotus bearer)
- Manjushri (holds books describing 10 paramitas)
- Kshitigarbha (guardian of purgatories)
- Maitreya (the future Buddha)
- Amitabha/Amitayusha (the Buddha of heaven)

Buddhist Architecture

- Stone-pillars depicting the life of Buddha at Gaya, Sanchi and Bharhut.
- Gandhara art and the beautiful images of the Buddha.
- Cave architecture in the Barabar hills at Gaya and in Western India around Nasik.
- Art pieces of Amaravati and Nagarjunakonda.
- **Stupa** relics of Buddha or some prominent monks.
- **Chaitya** Prayer hall.
- **Vihara** Residence.

Buddhist Universities

Buddhist Universities	*Location*	*Founder*
Nalanda	Bodagaon, Bihar	Kumaragupta I
Vikramshila	Bhagalpur, Bihar	Dharmapala (Pala ruler)
Somapuri	North Bengal	Dharmapala (Pala ruler)
Jagadal	Bengal	Ramapala (Pala ruler)
Odantpuri	Bihar sharif, Bihar	Gopala (Pala ruler)
Vallabhi	Gujarat	Bhattarka (Maitrak rule)

Buddhist Councils

Buddhist Council	*Year*	*Venue*	*Chairman*	*Royal Patron*	*Development(s)*
1st	483 BC	Saptaparni Cave, Rajgriha	Mahakassaapa	Ajatshatru (Haryanka Dynasty)	Compilation of Sutta Pitaka and Vinaya Pitaka by Ananda and Upali respectively
2nd	383 BC	Vaishali	Sabakami	Kalasoka (Shishunaga Dynasty)	Monks were split into Sthavirvadins and Maha sanghikas
3rd	250 BC	Pataliputra	Moggaliputta Tissa	Ashoka (Maurya Dynasty)	Compilation of Abhidhamma Pitaka
4th	AD 72	Kundalvan, Kashmir	Vasumitra (Chairman) and Ashvaghosha (Vice-Chairman)	Kanishka (Kushan Dynasty)	Division of Buddhists into Hinayana and Mahayana

Great Events of Buddha's Life, Symbols and Related Incident

Event	*Symbols*	*Related Incident*
Janma (*Birth*)	Lotus and Bull	In 563 BC at **Lumbini** (Sakya tribe of Kapilvastu) on Vaishakh Purnima day in Kshatriya clan.
Mahabhinishkramana (*Renunciation*)	Horse	At the age of 29 years, after witnessing **four scenes** in a sequence (old man, sick man, dead body and an ascetic).
Nirvana/Sambodhi (*Enlightenment*)	Bodhi tree	At 35 years of age at Uruvella (Bodh Gaya) under a pipal tree on the banks of Niranjan (Phalgu) river, on the 49th day of meditation.
Dharmachakra pravartana (*First Sermon*)	Wheel	At Sarnath, where his five disciples had settled.
Mahaparinirvana (*Death*)	Stupa	At the age of 80 years in 483 BC at Kusinagara in UP in the Malla republic.

Hetrodox Sects

Sect	*Founder*	*Theory*
Ajivikas	Gosala Maskariputra	Believed in Fate called 'Niyati'
Amoralism	Purana Kassapa	Samkaha Philosophy
Charvaka School	Ajita Keshakambalin	Uchchedavada (annihilationism)
Hindu Vaisesika School	Pakudha Katyayana	Sorrow, happiness and life are indestructible like Earth, water etc

THE MAURYAN EMPIRE

ORIGIN

- **Mudrarakshasa** Vrishal/Kulhina (of low clan).
- **Buddhist tradition** as Kshatriya.
- **Puranas** Moriya clan (low caste).
- **Junagarh rock inscription of Rudradaman** (AD 150) Vaishya origin.

SOURCES

- **Arthashashtra of Kautilya** (Chanakya or Vishnugupta) Written in Sanskrit by the Prime Minister of Chandragupta Maurya.
- It is a treatise on State Craft and Public Administration under Mauryas. The book is in 15 parts.

Buddhist Literature

- Ashokavadana and Divyavadana.
- Dipavamsa and Mahavamsa (Sri Lankan chronicles).
- Jatakas Sthaviravali Charitra or Parisisthaparvan of Hemachandra
  (A biography of Chanakya) Chandragupta's conversion to Jainism.
- Mudrarakshasa of Vishakhadatta.

Indica of Megasthenese wrote that Indians are free from slavery; and of the existence of 7-castes in India.

Puranas Chronology and lists of Mauryan kings.

Others Account of Diodorous, Pliny, Plutarch, Arrian and Justin.

Quick Digest

- Inscriptions of Skandagupta and Rudradaman are also found at Girnar.
- The Pillar Edict VII is the last edict to be issued by Ashoka.
- **Mahasthan** and **Sohgura Copper Plate** inscriptions – issued by Chandragupta Maurya, deals with relief measures adopted during famine.
- Latest discovery– 3 Ashokan minor Rock Edicts from Sannati village (Karnataka).

Ashoka's Edicts and Inscriptions

Ashoka's edicts were first deciphered by **James Princep** in 1837. It was mainly written in Prakrit language and 3 scripts *viz* Kharoshthi in North-West, Greek and Aramaic in West and Brahmi in Eastern India.

CHANDRAGUPTA MAURYA (321-298 BC)

- Also called as **Sandrocottus/Androcottus** by Greek Scholars.
- He entered into alliance with Parvartaka and with the help of Chanakya, dethroned the last Nanda ruler Dhanananda and founded the Mauryan Dynasty, with capital at Pataliputra.
- Chandragupta defeated **Selucus Nikator**, the general of Alexander in North-West India in 305 BC. There was a matrimonial alliance between them.
- Selucus also sent a Greek Ambassador, **Megasthenese**, to the court of Chandragupta Maurya. Chandragupta embraced Jainism and went to Chandragiri Hills, at Sravanbelagola with Bhadrabahu, where he died of slow starvation (Salekhna).

BINDUSARA (298-273 BC)

- Greeks called him **'Amitro Chates'** meaning, slayer of foes. He extended the kingdom, further to the Peninsular region of India, as far South as Mysore.
- **Antiochus** I, the Selucid king of Syria, sent his ambassador, Deimachus to his court.
- **Pliny** mentions that Ptolemy Philladelphus of Egypt sent Dionysius as his ambassador to the court of Bindusara.
- Taranath, the Buddhist monk, credits him for conquering the land between two seas. He patronised the Ajivika sect.

ASHOKA (273-232 BC)

- He was the greatest Mauryan ruler; and also the Governor of Taxila and Ujjain previously. His rule extended to the whole of sub-continent except to the extreme South. It also included Afghanistan, Baluchistan, Kashmir and valleys of Nepal.
- A Buddhist text **Dipavamsa** states that he usurped the throne after killing his 99 brothers, except the youngest one Tishya in the war of succession that lasted for 4 years.
- He fought **the Kalinga War** (261 BC) in the 9th year of his rule. The miseries of war caused deep remorse to Ashoka and therefore he abandoned the policy of physical conquest **(Bherighosa)** in favour of cultural conquest **(Dhamma ghosha)**.
- He embraced Buddhism under Upagupta.
- He sent his son Mahendra and daughter Sanghamitra to Ceylon as Buddhist missionaries, with a sapling of the original pipal tree.
- He inaugurated *Dhammayatras* from the 11th year of his reign by visiting BodhGaya; also appointed **Dhamma Mahamatyas** (officer of righthouse to spread the message of Dhamma.

Ashoka's Dhamma

- It was a code of conduct and a set of principles to be adopted and practised by the people to build up an attitude of social responsibility and to preserve the social order. It ordained to pay respect to elders, mercy to slaves and emphasised truth, non-violence and tolerance.

LATER MAURYAS

- Following the death of Ashoka, the Mauryan dynasty lasted for 137 years, the empire was divided into Western and Eastern parts.
- The Western part include the North-Western provinces, Gandhara and Kashmir was governed by **Kunala** (one of the sons of Ashoka) and then for a while by **Samprati**. The Eastern part of Maurya, Empire, with its capital at Pataliputra, came to be ruled by Dasharatha. After Dasharatha and Samprati came Salisuka. The successors of Salisuka, according to the puranas, were Devavarman, Shatadhanvan and finally Brihadratha.

MAURYAN ADMINISTRATION

- The Mauryan state was a welfare state with a highly centralised government. The King was the nucleus, assisted by Mantri Parishad, which included Yuvraj-the crown prince.
- Gopa was the purohit chief priest.
- Senapati, the commander-in-chief of army and other ministers.

Eighteen Tirthas

Mahamantri Purohita	Chief Minister and Chief Priest
Senapati	Commander-in-Chief
Yuvraja	Crown Prince
Dauvarika	Chamberlain
Prasastri	Inspector General of Prison
Sanidhatta	Incharge of Treasury
Nayaka	City Constable
Vyavaharika	Chief Judge
Mantri	Secretary incharge of the Office of Ministers
Parishadadhyasha	Ministers
Duvara Pala	Chief of the Home Defence i.e. warden of the palace to control entrance and exit
Antarvesika	Chief of the Harem
Samaharta	Tax Collector General
Pradeshtri	Divisional Commissioner
Paura	Governor of the Capital
Karmantika	Chief of the Industries
Dandapal	Police Chief
Antapal	Chief of the Frontier Defence

Provincial Administration

Province	*Capital*
Uttarapatha (North)	Taxila
Avantipatha (West)	Ujjain
Prachyapatha (East)	Kalinga
Dakshinpatha (South)	Suvarnagiri
Central Province	Pataliputra

Administrative Units & Their Heads

Unit	*Head*
Chakra (province)	Rashtrapala/Kumara
Ahar (district)	Pradeshika (administrative) and Rajuka (revenue)
Sangrahana (Group of 10 Villages)	Gopa
Gram (Village)	Gramika

Different Types of Taxes

Bhaga	Land revenue $\left(\frac{1}{6}\text{th}\right)$
Bali	Additional tax
Chorarajju	Tax collected for the search of thief
Pranaya	Emergency tax
Pindakara	Collected annually from the entire village
Praveshya	Import duty
Senabhaktam	Tax for the army, from the region through which it passed
Sulka	Custom duty
Vishti	Forced labour
Hiranya	A tax paid in gold
Udayabhagokal	Irrigation tax
Nishkramya	Export duty

Some Important Rock Edicts

Major Rock Edicts	*Content*
MRE I	Prohibition of animal sacrifice
MRE II	Refers to Cholas, Pandyas, Satya putras and Kerala putra (kingdom of South)
MRE III	Liberality to Brahmins
MRE IV	Non-violence; courtesy to relations
MRE V	Appointment of Dhamma Mahamatyas
MRE VII	Tolerance among all sects
MRE VIII	*Dhammayatras*
MRE IX	Charity, Kinship, Dhamma
MRE XII	Religious tolerance
MRE XIII	Kalinga war; Bheri Ghosa to Dhamma Ghosa

Some of the Important Adhyakshas

Adhyakshas	*Rank(s)*
Akshapataladhyaksha	Accountant General
Sitadhayaksha	Incharge of Crown Lands
Akaradhyaksha	Superintendent of Mines
Lavanadhyaksha	Salt Superintendent
Navadhyaksha	Incharge of State Boats
Panyadhyaksha	Controller of State trading
Sulkadhyaksha	Controller of Custom or Tolls
Suradhyaksha	Superintendent of Excise
Pautavadhyaksha	Superintendent of Weight and Measure
Bandhanagaradhyaksha	Superintendent of Jails
Ayudhagaradh-yaksha	Ordinance Superintendent
Kosthagaradhyaks-ha	Incharge of Warehouse
Pattanadhyakshya	Superintendent of Ports
Devatadhyaksha	Incharge of Religious Institutions
Samsthadhyaksha	Superintendent of Market
Suvarnadhyakshya	Superintendent of Gold
Hastyadhyaksha	Incharge of Elephant Force
Kosadhyaksha	Incharge of Treasury
Kupyadhyaksha	Incharge of Forest Produce
Madradhyaksha	Passport Incharge
Ganikadhyaksha	Incharge of Courtesan
Lakshanaadhyaksha	Superintendent of Mint
Dhyutadhyaksha	Controller of Gambling
Go-adhyaksha	Superintendent of Crown Herds
Aspalpha Lohadhyaksha	Superintendent of Metal
Khanyadhyaksha	Superintendent of Mines
Pattyadhyaksha	Commander of Infantry
Sunadhyaksha	Protector of Animals
Sutradhyaksha	Textile Commissioner
Vivitadhyaksha	Controller of Pasture Land

Major Edicts and their Location

Edicts	*Content*	*Location(s)*
Rock Edicts		
14 Major Rock Edicts	Ashoka's principle of government and policy of Dhamma	Kalsi (Dehradun, Girnar (Gujarat), Yerragudi (Andhra Pradesh), Mansehra (Pakistan), Sopara (Bombay), Dhauli and Jaugada (Odisha), Shahbazgarhi (Pakistan)
2 Separate Kalinga Edicts	Kalinga War and a new system of administration after war (All men are my children - Dhauli)	Dhauli or Tosali and Jaugada (Odisha)
Minor Rock Edicts	Personal history of Ashoka and summary of his Dhamma	South and Central parts of the empire
Pillar Edicts		
7 Pillar Edicts	Appendix to the Rock Edicts	Delhi Topra, Delhi-Meerut, Rampurva, Lauriya-Araraj, Lauriya-Nandangarh and Allahabad-Kosam
Other Edicts		
Queen's Edict	Refers to Karuvaki mother of Tivala/Tivara, the second Queen	On Allahabad Pillar
Bhabru Edict	Ashoka's faith in Buddhism	Bairat (Rajasthan)
Nigalisagar Pillar Edict	Stupa of Buddha at Kanakamuni was enlarged	Nigalisagar (Nepal)
Rummindei Pillar	Ashoka visited Lumbini and reduced land tribute	Rummindei/Lumbini (Nepal)
3 Barbara Cave Edicts	Donation to Ajivikas	Barabar Hills (Gaya, Bihar)

Art and Architecture

- Sculpture of Yaksha and Yakshini e.g., Yaksha of Parkham (Mathura); Yakshini from Vidisha (MP) and Didarganj (Patna).
- Royal Palace of Chandragupta Maurya at Kumharar, Patna. Single lion capital Rampurva and Lauriya Nandangarh.
- Four lion capital at Sarnath and Sanchi.
- Carved elephant at Dhauli and Engraved elephant at Kalsi.
- Stupas were built throughout the empire to enshrine the relics of Buddha, e.g. at Sanchi and Bharhut.

POST-MAURYAN PERIOD

Mauryans were succeeded by many ruling dynasties from Central Asia in North-West India and by native rulers (Shungas, Kanvas, Satavahanas) in the Eastern, Central and Deccan regions of India.

FOREIGN STATES

The Indo-Greeks

- A series of invasion began in about 200 BC. The weak Mauryan king failed to restrict them. The first to invade India were the Indo–Greeks, who ruled Bactria lying South of the Oxus river in the area covered by North Afghanistan.
- They occupied a large portion of North-Western India and moved upto Ayodhya and Pataliputra.
- The most famous Indo-Greek ruler was **Menander** (165-145 BC) or **Milinda**. He had his capital at Sakala (Modern Sialkot in Punjab). He was converted to Buddhism by Nagasena. Menander and Nagasena's conversations were recorded in the book *Milindapanho* or The Questions of Milinda. Indo-Greeks were the first ruler in India to issue gold coins.

The Shakas or Scythians (AD 1st to 4th Century)

- The Greeks were followed by the **Shaka**. There were five branches of Shakas ruling from Afghanisthan, (Punjab); Mathura; Ujjain; Western India and Deccan.
- A king of Ujjain, who called himself **Vikramaditya**, defeated the Shakas. An era called the *Vikram Samvat*, is recorded from the event of his victory over the Shaka i.e. 57 BC.
- The most famous Shaka ruler in India was **Rudradaman** (AD 130-150).
- Other important Shaka rulers in India were Nahapana, Ushavadeva, Chastana, Ghamatika etc.

The Parthians (AD 1st to 3rd Century)

- Originally, the Parthians lived in Iran, from where they moved to India and replaced the Shakas.
- They occupied only a small portion of North-Western India, compared to the Greeks and Shakas.
- The most famous Parthian king was **Gondophernes**, in whose reign **St Thomas** came to India for the propagation of Christianity.

Sudarshan Lake

- It was constructed by **Pushyagupta**, the Governor of Saurashtra under Chandragupta Maurya.
- **Tushapa** constructed a dam on the lake, during the reign of **Ashoka Maurya**. First reconstruction by Governor Survishakh under Saka Satrap Rudradaman and second by **Chakrapalit** under **Skandagupta.**

The Kushanas (AD 1st to 3rd Century)

- Kushanas (Yuechis or Tochanians) replaced the Greeks and Parthians. They were nomadic people from steppes of North-Central Asia.
- The first Kushanas dynasty was founded by **Kujala Kadphises**. Wima Kadphises issued gold coins in India. Kanishka founded the second Kushana dynasty.
- **Kanishka,** also known as Second Ashoka, was the most famous Kushana ruler. He had two capitals, Purushpur and Mathura.
- Kanishka started an era in **AD 78**, which is now known as **Saka era** and used by the Government of India.
- He held the Fourth Buddhist Council in Kashmir, where the doctrine of **Mahayana** form of the Buddhism was finalised.

NATIVE STATES

Shunga Dynasty (185-73 BC)

- Pushyamitra Shunga killed the last Mauryan king Brihadratha and founded the Shunga dynasty. Pushyamitra Shunga ruled from Vidisha (MP).
- He defeated the Bactrian king, Dematrius and conducted two Asvamedha Yagyas (chief priest-Patanjali).
- He built the Buddhist Stupa at Bharhut.
- The Greek Ambassador **Heliodorus** visited the court of fifth Shunga king Bhagabhadra, and set-up a pillar in honour of Lord Vasudeva, near Vidisha (Madhya Pradesh).
- Shunga king **Agnimitra** was the hero of Kalidasa's Malvikagnimitram.
- The great Sanskrit Grammarian, **Patanjali** was patronised by them.
- The famous book on Hindu Law **Manusmriti** was compiled during this period.
- Later Kings–Vasumitra, Vajramitra, Bhagabhadra, Devabhuti.

Kanva Dynasty (73-28 BC)

- Kanva was a minor dynasty founded by **Vasudeva** who killed the last Shunga King **Devabhuti** at his capital Pataliputra.
- Bhumimitra and Narayana succeeded Vasudeva.
- The last ruler, Susharman, was killed by the Andhra king Simuka.

Chedi Dynasty

- The **Hathigumpha inscription** of **Kharavela**, the third ruler of the dynasty, gives information about the **Chedis**.
- Kharavela pushed his kingdom upto Godavari in the South and recovered the Jains image from Magadha.
- He was a follower of Jainism. He constructed residential caves for Jain monks on the Udaygiri Hill near Bhubaneshwar, Orissa.

Satavahana Dynasty (60 BC-225 AD)

- The Satavahanas are considered to be identical with the Andhras, who are mentioned in the Puranas.
- The founder of this empire was **Simuka,** after the assassination of the last Kanva King **Susarman**.
- **Gautamiputra Satakarni** (AD 106-130) was the greatest ruler of this dynasty.
- He built the capital city Paithan/Pratisthan.
- **Vasishthiputra Sri Satkarni**, the 24th ruler, married the daughter of Saka Satrap Rudradaman, but was twice defeated by him.
  They issued mostly lead coins (apart from Copper and Bronze).
- Satavahanas were the first rulers to make land grants to Brahmins.
- There is an instance of grants being made to Buddhist monk, which shows that they also promoted Buddhism. Stupas at Nagarjuna konda and Amaravati in Andhra Pradesh became important seats of Buddhist culture under the Satavahanas.

SANGAM AGE

It was divided into three kingdoms : Chola, Chera and Pandya.

Sangam Literature

- The word Sangam is associated with a college or an assembly of Tamil scholars and poets, who flourished under the royal patronage of the Pandyan kings.
- The whole Sangam age is called **Golden** or the **Augustan Age**. According to Tamil sources, the father of Tamil literature is *Agastya*.

Important Sangam Works

- *Tolkappiyam* by Tolkappiyyar (Tamil Grammar).
- *Tirukural* or *Kural* by Tiruvalluvar is sometimes called the **Fifth Veda** or *Bible of the Tamil land*.
- Aggatiyam comprises grammar of letters and life, in three parts, written by Saint Aggatiyar.

Revenue Terminologies

Karai	Land tax
Irai	*Tribute paid by feudatories and booty collected in war*
Iravu	Extra demand or forced light gift
Ulgu	Custom duties
Variyar	Tax collector
Variyan	A well known unit of territory yielding tax

THE AGE OF THE GUPTAS

- The fall of Kushana empire towards the middle of AD 3rd century led to the establishment of the Empire of Guptas.
- Their period is generally regarded as the **Golden Age of Hinduism.** Guptas belonged to the Vaishya caste.
- **Sri Gupta** was the founder of Gupta dynasty. Sri Gupta was followed by his son Ghatotkacha, who was followed by his son Chandragupta. Both used the simple title of *Maharaja*.

CHANDRAGUPTA I (AD 319-335)

- He greatly raised the power and prestige of his dynasty by his matrimonial alliances and conquests.
- He married the Lichchhavi princess **Kumaradevi** and issued Chandragupta I Kumara devi type gold coins (Dinaras).
- Chandragupta I is also said to have started a new era-**Gupta Era** which starts from 26th February AD 320, the coronation date of Chandragupta I.

SAMUDRAGUPTA (AD 335-380)

- Son and successor of Chandragupta I.
- The long inscription in the pillar of Allahabad (Prayag Prasasti) enumerated by his court poet Harisena informs about the people and the countries that were conquered by Samudragupta.
- **Virasen** was his Commander-in-Chief during Southern campaign. Vasubandhu, a famous Buddhist scholar, was his minister.
- **Allahabad Pillar Inscription** mention the title Dharma Prachar Bandhu, meaning, the upholder of Brahmanical religion. His authority over Java, Sumatra and Malaya islands proves that he maintained a strong Navy.
- According to Chinese sources, **Meghavarman**, the ruler of Sri Lanka, sent a missionary to Samudragupta for permission to built a Buddhist temple at Gaya.

CHANDRAGUPTA II (AD 380-415)

- He killed the Saka ruler and his coward brother Ramgupta and married his wife, Dhruva Devi.
- He issued **silver coins** (first Gupta ruler to issue silver coins) and adopted the title Vikramaditya and Shakari in the memory of his victory.
- The court of Chandragupta II at Ujjain was adorned by 9 scholars known as Navaratna, including Kalidasa and Amarsingh. Harisena was the court poet and ministers. **Fa-hien**, the Chinese pilgrim, visited India at his time. Ujjain was made the second capital by Chandragupta II.

KUMARAGUPTA (AD 415-455)

- He founded the **Nalanda University.**
- He was the worshipper of Lord Kartikeya. In the last year of his reign, the empire was invaded by the Turko-Mongol tribe of the Hunas. During the war with the Hunas, Kumaragupta died.

SKANDAGUPTA (AD 455-467)

- He repulsed the ferocious Hunas' attack twice. The heroic feat entitled him the title **Vikramaditya** (Bhitari Stone Inscription).
- The continuous attack of the Hunas weakened the Gupta economy and the decline of empire began soon after the death of Skandagupta.

Later Guptas (AD 467-540)

Purugupta, Kumaragupta II, Buddhagupta, Narasimhagupta and Kumaragupta III.

ADMINISTRATION

Gupta administration was highly decentralised and quasi-feudal in character.

Administrative Unit	*Headed by*
Bhukti (province)	Uparika
Vishayas (district)	Vishyapati
Nagar/Peth (sub-district)	Purapala/Nagar Pati
Gram (village)	Gramika

Important Officials

Official	*Responsibility*
Maha Pratihari	Chief usher of Royal Palace
Dandapashika	Chief officer of the Police Department
Mahaprajapati	Chief officer of Elephant Corps
Vinayasthitisthapak	Chief officer of Religious Affairs
Mahashvapati	Chief of Cavalry
Mahadandanayaka	Minister of Justice

Taxes

- **Bhaga** King's share in the produce, to be paid by cultivators.
- **Bali** (Earlier a voluntary offering) An additional and oppressive tax during the Gupta period.
- **Bhoga** Periodic supplies of fruits, firewoods etc, which the villagers had to furnish to the king.
- **Uparika** An extra tax levied on all subjects.

Culture

- Over 2 m high bronze image of the Buddha of Gupta period has been recovered from **Bhagalpur**.
- For the first time, we get in the Gupta period, images of **Vishnu, Shiva** and some other Hindu Gods.
- **Paintings** Ajanta paintings and paintings at Bagh (Madhya Pradesh) are of this time. They belong to the Buddhist Art.
- Stupas Mirpur Khas (Sind), Ratnagiri (Odisha) and Dhammekh (Sarnath).
- Gupta Age marks the beginning of main styles of temple architecture in India, namely the **Nagara** and **Dravida** style (shikhar style), with the Garbhagriha (shrine room in which the image of God is kept).

Literature

- Aryabhatta, the mathematician wrote **Aryabhatiyam** and **Shunya Siddhanta**.
- Varahamihira wrote **Panchsiddhantika** and **Vrihatsamhita**.
- Brahmagupta hinted the law of gravitation in **Brahma Siddhanta**. Vagabhatta was a distinguished physician. **Dhanvantri** was famous for his knowledge of **Ayurveda**.
- Bhaskara wrote **Mahabhaskarya** and **Laghu Bhaskarya**.

Literary Works

Author	*Books*
Sudraka	Mrichchakatikam
Bharvi	Kiratarjuniya
Dandin	Dash Kumarcharitam and Kavya Darshan
Bhasa	Svapnavasavadattam, Charudatta
Vishakhadatta	Mudura Rakshasam Devichandraguptam
Vishnu Sharma	Panchtantra and Hitopodesha
Amarsingh	Amarkosh
Iswara Krishna	Sankhya Kanika
Vatsyayana	Kamasutra
Vatsabhahi	Ravan Vadha
Varahamihira	Panchasiddhantika, Vrihat Samhita

PUSHYABHUTI/ VARDHANA DYNASTY

- Harsha belonged to the Pushyabhuti dynasty, which ruled from **Thaneswar**. Pushyabhutis were the feudatories of Guptas, but had assumed independence after Hunas invasion.
- **Prabhakar Vardhana** (AD 580-605) was its first important ruler, succeeded by **Rajyavardhan** (AD 605-606).

Harshavardhana (AC 606-647)

- Harsha, also known as **Siladitya,** ascended the throne in AD 606 and from this year, started the Harsha Era.
- Harsha made **Kannauj,** his capital.
- Harsha wrote three dramas: **Priyadarshika**, **Ratnavali** and **Nagananda**.
- Banabhatta wrote **Harshacharita**, Kadambari and Parvatiparinay. The Chinese pilgrim **Hiuen Tsang** visited India during the Harsha period. He informs us that the revenue of Harsha was divided into four parts–one for the king, second for the scholar, third for the officer and fourth for religious purpose.
- Harsha used to celebrate a festival at Prayag, after every 5 years.
- In Eastern India, he faced opposition from the Shaivite king-**Shashanka of Gauda**, who cut off the Bodhi tree of Bodhgaya.
- Harsha defeated **Dhruvasena II**, the Maitraka ruler of Vallabhi.
- Harsha was defeated in **Deccan** by Pulakesin II, the Chalukyan king of Vatapi.
- The Kannauj Assembly (AD 643) was held in the honour of Hiuen Tsang, and to popularise the Mahayana sect of Buddhism.

Tripartite Struggle

The struggle for supremacy between the Palas, the Gurjara - Pratiharas and the Rashtrakutas for the possession of Kannauj (Farrukhabad District, UP) at the end of AD 8th century is known as the Tripartite Struggle in the history.

Other Dynasties and Rulers (AD 7th to 12th century)

Dynasty	*Capitals*	*Founder*	*Famous Rulers*	*Other Features*
Palas (*Eastern India*)	Pataliputra, Gaur	Gopala (AD 750)	▪ **Dharma Pala** Revived Nalanda University and founded Vikramshila University, defeated Bhoja (Pratihara) Amog varsha (Rashtrakuta) and won Kannauj ▪ **Devapala** won Orissa and Assam. ▪ **Mahikala** defeated by Rajendra Chola.	They traded with South-East Asia, and were replaced by Senas in Bengal.
Gurjara Pratiharas (*SW Rajasthan*) (AD 733-1019)	Jodhpur, Malwa	Harichandra	**Mihir Bhoja** He worshipped Lord Vishnu and adopted the title *Adi varaha.*	They originated in Geyanta region of Rajasthan.
Vakatakas (*Deccan and Central India*)	Vatsagumla Paunar	Vindhyashakti	**Pravarsena I** performed four Asvamedha Yagyas.	Chandragupta II married his daughter Prabhavati to the Vakataka king Rudrasena.
Eastern Gangas (Orissa)	Kalinganagar, Cuttack	Anantavar man Chodagong Deva	**Narshima Deva I** built the Sun temple at Konark.	Anantvarman built the Jagannath temple at Puri.

Dynasty	*Capitals*	*Founder*	*Famous Rulers*	*Other Features*
Western Gangas (AD 350-999)	Kolar, Talakal	Konganivarman Madhava	**Dunvirta**	Constructed Jaina monuments at Sravanabelagola.
Senas (Bengal)	Vikrampura Vijaypura	Vijaysena	**Ballasena** **Lakshmansena**	They were over thrown by Deva Dynasty.
Hoyasalas	DWara samudra	Vishnu Vardhan	**Vira Ballal** defeated the Chalukyan ruler Somesvara IV.	Hoysala art and architecture was of a high standard. The minute carving of Hoysala temple is their most attractive feature.
Rashtra kutas (AD 750-1142)	Manyaket or Malkhed	Dantidurga (earlier served the Chalukyas of Badami)	▪ **Amogvarsha** He is compared to Vikramaditya in giving patronage to men of letters. ▪ He wrote the 1st Kannad Poetry, Kavi Rajamarg, and also wrote Prashnottar Mallika.	▪ Rashtrakutas are credited with the building of Cave shrine of Elephanta. ▪ It was dedicated to Mahesh (as Trimurti) and counts among the most magnificent art creations of India.
			▪ **Krishna II** constructed Kailash temple at ellora in Dravidian style. ▪ **Krishna III** set-up Pillar of victory and a temple at Rameshwaram.	
Pallavas (AD 560-903)	Kanchi	Simhavishnu	▪ **Narashimhavarman I** (AD 630-668) occupied Chalukyan capital at Vatapi and assumed the title Vatapikonda.	▪ They were orthodox Brahmanical Hindus. ▪ Both Chalukya and Pallava tried to establish their supremacy over the land between Krishna and Tungabhadra.
Chalukyas (*Badami*)	Vatapi (Badami)	Pulakesin I	▪ **Pulakesin II** He was contemporary of Harsha and was able to check Harsha from conquering Deccan. ▪ The Chinese pilgrim **Huen Tsang** visited his kingdom.	Much of the paintings and sculptures of the Ajanta and Ellora caves were completed during the time of the Chalukyas.
Chalukyas (*Kalyani*)	Kalyani	Tailap II (defeated the Parmar king Munj)	▪ **Somevara I** (AD 1043-1068) He shifted the capital from Manyakhet to Kalyani. ▪ **Vikramaditya IV** (1070-1126) He started the Chalukya-Vikram era.	Bilhana, the writer of Vikramadeva Charita and Mitakshara on Yagyavalkya Smriti, adorned the court of Vikramaditya IV.
Yadavas	Devagiri	Bhillana	**Ramchandra**	Ramchandra was defeated by Malik Kafur.

MEDIEVAL INDIA

THE CHOLAS

- The capital of Cholas was **Tanjore**.
- **Vijayalaya** revived the Chola empire in AD 9th century. He took the title of *Narkesari*.
- Aditya Chola defeated the Pallava king **Aparajit**, captured Tondamandalam and took the title of **Maduraikonda**. He built a Shiva Temple at Tanjore.
- **Rajaraja** I (AD 985-1014) attacked Sri Lanka. He permitted the Shailendra king to build the *Churamani Buddhist Vihara* at Nagapattanam.
- He himself built the **Rajarajeshwara temple** (Shiva temple) at Tanjore. He is known as Rajaraja the Great in history.
- He defeated the Pala king Mahipala and took the title of Gangaikonda cholam.
- He built the Cholamandalam lake and also the city of Gangaikonda cholapuram.
- **Rajendra III** was the last king of the dynasty.
- The most important feat of Chola administration was local-self government. Each village was divided into 30 wards.

ARAB CONQUEST OF SIND

Muhammad Bin Qasim (AD 712) Muhammad Qasim of Iraq was the first Muslim to invade India. He defeated the ruler of Sind, Dahir and the province was given to Omayyad Khilafat.

Chola Temples (Dravid Style)

Temple	*Location*	*Builder*
Kailashnath Temple	Kanchipuram	—
Vrihadeshwara Temple	Tanjore	Rajaraja I
Koranganatha Temple	Srini Wasanllur	Parantaka I
Airateshwar Temple	Darasuram	Rajaraja II
Kampahreshwar Temple	Tribhuvan	Kullotung III
Gangaikonda Cholapuram	Gangaikonda Cholapuram	Rajendra I

THE RAJPUTS

They emerged as a powerful force in Northern India in the AD 9th and 10th centuries.

Rajput Kingdoms	*Capital(s)*	*Founder(s)*
Chauhan/Chahaman of Delhi-Ajmer	Delhi	Vasudeva
Pawar of Malwa	Ujjain, Dhar	Sri Harsha
Pratihara of Kannauj	Avanti, Kannauj	Nagabhata I
Rashtrakuta of Malkhand	Manyakheta	Dantidurga (Dantivarman II)
Chalukya/Solanki of Kathiawar	Aniha/Vada	Mularaja I
Kalchuri/Haihaya of Chedi	Tripuri	Kokkala I
Chandela of Jejakabhukti	Khajuraho, Mahoba, Kalinjar	Nannuk Chandela
Gadhwal/Rathor of Kannauj	Kannauj	Chandradeva
Tomars of Delhi and Haryana	Dhillika	—
Guhilot/Sisodiya of Mewar	Chittor	Bappa Rawal, Hammir I

TURKISH INVASIONS

Mahmud Ghaznavi

- Mahmud Ghaznavi, the ruler of Ghazni is said to have made **17 raids** into India. The initial raids were directed against the Hindustani rulers. In AD 1001, the Hindustani ruler Jaipal was again defeated by Mahmud. His son Anandpala was defeated in the **Battle of Waihind** (AD 1008-09).
- He plundered Somnath temple in AD 1025 (his sixteenth raid).
- In AD 1026, Mahmud defeated the Jats. He died in AD 1030.
- The objective of Mahmud's expeditions was to plunder and loot. He was not interested in expanding his empire to India.
- He patronised 3 persons, contemporary to him: **Firdausi** (court poet), **Alberuni** (scholar) and **Utbi** (court historian).
- Alberuni wrote 'Kitab-ul-Hind'.

Muhammad Ghori

- In 1173, Muizzudin Muhammad (Muhammad Ghori) ascended the throne at Ghazni. He moved towards Punjab and Ganga valley, but was defeated by Prithviraj, the ruler of Delhi in the First Battle of **Tarain** (1191).
- In the **Second Battle of Tarain** (AD 1192)**,** Prithviraj was defeated by Muhammad Ghori. Prithviraj was allowed to rule over Ajmer for sometime, but was executed on a charge of conspiracy after some time.
- **Prithviraj Raso**, written by the Chand Bardai, the court poet of Prithviraj. It depicts the life story of Prithviraj and his love for Sanyogita.
- The defeat laid the foundation of Muslim rule in India. Later on in 1194, Jaichand of Kannauj was also defeated at the **Battle of Chandawar**.
- Muhammad Ghori died in AD 1206, leaving the charge with Qutub-ud-din Aibak.

Bakhtiyar Khilji

Bakhtiyar Khilji, one of Ghori's commanders, annexed Bihar and Bengal and destroyed Nalanda and Vikramshila Universities.

THE DELHI SULTANATE

The Sultanate of Delhi had 5 ruling dynasties.

SLAVE DYNASTY (1206-1290)

Qutub-ud-din Aibak (1206-1210)

- This dynasty also knows as Mumluks or Ilbari Turk Dynasty.
- Originally a slave, Qutub-ud-din, because of his merit and loyalty was raised to the post of Viceroy by Muhammad Ghori.
- Lahore and later Delhi, was his capital.
- For his generosity, he was given the title of **Lakh Baksh**.
- He constructed two mosques. Quwwat-ul-Islam at Delhi and Adhai-din-ka Jhopra at Ajmer. He also began the construction of Qutub Minar in the honour of famous Sufi Saint Khwaja Qutub-ud-din Bakhtiyar Kaki. Aibak patronised writers like **Hasan-ul-Nizami**, author of Taj-ul-Masir and Fakhr-ud-din, author of Tarikh-i-Mubarak-Shahi.
- He fell from the horse, while playing Chaugan (Polo) and died. Aibak was succeeded by Aram Shah (AD 1210), but the throne was usurped by Iltutmish.

Iltutmish (1211-1236)

- He is considered to be the greatest slave king and the real consolidator of the Turkish conquest in India. He made Delhi, the capital in place of Lahore.
- He completed the construction of **Qutub Minar**. He constituted a group of 40 loyal slaves, known as Turkan-i-Chahalgani or **Chalisa**.

- He introduced the Silver **Tanka** and Copper Jital.
- He patronised Minhaj-us-Siraj, author of Tabaqat-i-Nasiri.
- He is called the Father of Tomb Building (built Sultan Garhi in Delhi).
- He saved Delhi Sultanate from the wrath of Chengiz Khan, the Mangol ruler, by refusing shelter to Khwarizm Shah, whom Chengiz was chasing.
- He was succeeded by Rukn-ud-din Firoz (AD 1236). Finally, the throne was given to Raziya.
- He started the Iqta system in Delhi Sultanate. This is an assignment of land in lieu of salary, which he distributed to his officers.

Raziya Sultan (1236-1240)

- She was the first and last Muslim woman ruler of Medieval India.
- She sent an expedition against Rajput state of **Ranthambhor**.
- Her affection towards Yakut was not liked by the Nobles. Yakut was killed, and she married the Governor of Bhatinda, Altunia. She was assassinated near Kaithal (Haryana).
- She was succeeded by Muiz-ud-din Bahram Shah, Ala-ud-din Masud, Shah Nassir-ud-din Mahmud and Ghiyas-ud-din Balban.

Ghiyas-ud-din Balban (1265-1286)

- He created a strong centralised army and established the military department **Diwan-i-Arz**. He ordered the separation of military affairs from finance department (diwan-i-wazarat). He appointed spies.
- He took up the title of **Zil-i-Ilahi** (shadow of God) and impressed upon the people that king was the deputy of God (Niyamat-i-khudai).
- Balban adopted a policy of **Blood and Iron** to save Delhi from Foreign aggressions.
- He was a patron of Persian literature and showed special favour to **Amir Khusro**. Kaiqubad, the last Ilbari king was killed by Jalal-ud-din Khalji in AD 1290.
- He insisted on the Iranian ceremonies of **sijda** and **paibos** (prostration and kissing the monarch's feet).
- Balban started the festival of **Nauroz.**

THE KHILJI DYNASTY (1290-1320)

- The most important aspect of Jalal-ud-din Khalji reign was **invasion of Devagiri** in 1294, by his nephew and son-in-law Ala-ud-din Khilji. He married his daughter to Ulugh Khan, a descendant of Chengiz Khan to win his goodwill.

Ala-ud-din Khilji (1296-1316)

- He came to the throne by murdering his uncle and father-in-law, Jalal-ud-din Khalji.
- He massacred the rebellious nobles, relatives, family members and the Mongols, who had settled in Delhi. Ala-ud-din gave harsh punishment even to the wives and children of all nobles.
- He first conquered Gujarat and married Raja's wife **Kamla Devi**. He acquired **Malik Kafur** from there, who later on carried Ala-ud-din's Southern expedition and built a mosque at Rameshwaram.
- Then, he captured Ranthambhor, Chittor and Malwa. Ala-ud-din strengthened the North-West frontier, under his trusted Commander **Ghazi Mallik.** He built Alai Fort, Alai Darwaja (at Qutub Minar), Hazar Situm, Hauz Khas and capital city Siri.

Queen Padmini

The Queen of Chittor (wife of Ratan Singh), Padmini committed jauhar, when his husband was defeated by Ala-ud-din Khilji.

Administrative Reforms of Ala-ud-din Khilji

- First sultan to have a permanent army, he paid soldiers in cash.
- He introduced the system of 'Dagh' or branding the horses and 'Chehra', detailed description of each soldier.

- Measured the cultivable land and fixed the land revenue. 'Bishwa' was declared to be the standard of measurement. The state demanded half of the produce.
- The post of special official called **'Mustakharaj'** was created for the purpose of collection of revenue.
  - Ala-ud-din controlled the market by some regulations.
    - – Fix the cost of all commodities.
    - – He set-up three markets at Delhi.

THE TUGHLAQ DYNASTY (AD 1320-1414)

Ghiyas-ud-din Tughlaq (1320-1325)

- He was the founder of Tughlaq dynasty and the first sultan of Delhi, who took up the title of *Ghazi* or slayer of the infidels.
- He liberalised Ala-ud-din's administrative policies and took keen interest in **construction of canals** and formulated a **famine policy**.
- He built the city of Tughlaqabad near Delhi and made it his capital.
- Sufi saint, Shaikh Nizam-ud-din Aulia, said about him "Delhi is far away."
- He died in AD 1325, after falling from a high raised pavilion. Ibn Battuta, the Moroccan traveller, who was in Delhi at that time, opined that his death was due to sabotage arranged by his son, Jauna Khan (Muhammad-bin-Tughlaq).

Muhammad-bin-Tughlaq (1325-1351)

- **Jauna Khan** ascended the throne under the name Muhammad-bin-Tughlaq. He was one of the most extraordinary kings ever sat on the throne of Delhi. He was an expert in Arabic, Persian Astronomy, Philosophy, Mathematics and Medicine.
- He applied his rational and innovative mind in every aspect of administration, but achieved little success as he was very hasty in nature.

His five ambitious projects were

1. Increased taxation in Doab (1326).
2. Transfer of capital from Delhi to Deogiri (Daulatabad—1327).
3. Introduction of token currency (1329).
4. Khurasan expedition in North-West (1329).
5. Qarachil expedition in Himalayas (1330).

- He died in **Thatta** (Sind), while campaigning against a Turkish slave Taghi.

Diwan-i-Kohi

A new department of agriculture 'Diwan-i-kohi' was set-up by Muhnammad-bin-Tuqhlag. He built fortress of Adilabad and city of **Jahanpanah**.

Ibn-Battuta

The famous Moroccan traveller Ibn-Battuta came to Delhi in 1334 and acted as the Quazi of the capital for 8 years. He recorded the contemporary Indian scene in his travel account **Rihla**.

Firoz Shah Tughlaq (1351-1388)

- He made the **Iqta system** hereditary.
- In order to encourage agriculture, the sultan paid a lot of attention to irrigation. He repaired a number of canals. First canal was from Sutlej to Ghaggar. The second canal carried water from Yamuna to Hisar. He imposed **Haque-i-Sharb** or **Hasil-i- Sharb** i.e., water tax.
- He brought two pillars of Ashoka from Topara and Meerut to Delhi and repaired Qutub Minar, when it was struck by lightening.
- A new department of **Diwan-i-Khairat** was set-up to make provision for marriage of poor girls.
- **Diwan-i-Bandagon** was created as the department for slaves.
- Introduced two new coins–**Adha** (50% Jital) and **Bikh** (25% Jital).
- He was succeeded by Ghiyas-ud-din Tughlaq Shah I, Abu Bakr Shah, Nassir-ud-din Muhammad and Ala-ud-din Sikandar Shah.

- Built new towns of Fatehabad, Hisar, Jaunpur (in memory of Muhammad Tughlaq), **Firozabad**.
- Barani, the historian, was in his court. He wrote Tarikh-i-Ferozshahi and **Fatwa-i-jahandari**.

System of Taxation

- A new system of taxation introduced according to Quran.
- **Kharaj** A land tax of 1/10th of the procedure of land.
- **Zakat** 2.5% tax on property (by Muslim only).
- **Jaziya** A tax by non-Muslims (even by Brahmins).
- **Khams** 1/6th of the loot captured during war.

THE SAYYID DYNASTY (1414-1451)

Khizr Khan (1414-1421)

- He founded the Sayyid dynasty and claimed to have descended from the Prophet of Islam.
- Khizr Khan took the title of **Rayat-i-Ala** and not of a king. The coins were struck and Khutba was read in the name of **Taimur** and his successor Shahrukh. Khizr Khan's three successors were incapable rulers.

Mubarak Shah (1421-1434)

- He led successful expeditions against Mewatis, Katehars and the Gangeti Doab area. He was killed and deposed by the nobles.
- **Muhammad Shah** (1434-1443) He ruled on a very small area, rest being governed by nobles. **Alam Shah** (1443-1451) was the last Sayyid king, who retired as a coward, descending in favour of Bahlol Lodhi. Thus, Sayyid dynasty was replaced by the Lodhi dynasty.
- Yahya-bin-Ahmed-bin-Abdullah Sirhindi wrote **Tarikh-i-Mubarak Shahi** (history of Muhammad Shah of Sayyid dynasty).

THE LODHI DYNASTY (1451-1526)

Considered as the first Afghan dynasty of India.

Bahlol Lodhi (1451-1489)

- Founder of Lodhi dynasty in India.
- He annexed entire Sharqi kingdom and issued Bahlol coins.
- His achievement was the conquest and annexation of the kingdom of Jaunpur.
- He never sat on the throne and used to sit on carpets, along with the Amins.

Sikandar Lodhi (1489-1517)

- Noblest of the three Lodhi rulers his real name was Nizam Khan.
- Conquered Bihar and Bengal in AD 1504, he built a new city Agra and made it his capital.
- He was a fanatical Muslim and broke the sacred images of Jwalamukhi temple at Nagarkot and ordered the temple of Mathura to be destroyed.
- He encouraged agriculture. For measurement of land, he introduced the **Gaz-i-Sikandari** of 32 digits for measuring cultivated fields.
- He was a poet and wrote verses in Persian under the pen-name of Gulrukhi.

Ibrahim Lodhi (AD 1517-1526)

- Ibrahim Lodi arrested the absolute power of the Sultan. As a result, some of the nobles turned against him. At last Daulat Khan Lodhi, the Governor of Punjab invited Babur to overthrow Ibrahim.
- He captured Gwalior from Tomar dynasty but was defeated by Rana Sanga of Mewar in Battle of Khatoli (1518).
- He was defeated and died at the hands of Babur in the First Battle of Panipat in AD 1526. This marked the end of Delhi Sultanate.

Administration

Designation	*Department*
Wazir	Revenue
Ariz-i-Mamalik	Military
Chief Sadr	Diwan-i-Risalat (*religious matter*)
Dahir-i-Mumalik	Diwan-i-Insha (*royal correspondence*)

Literature

Book	*Author*	*Book*	*Author*
Khazyan-ul-Futuh	Amir Khusro	Mitakshara	Vigyanesvara
Tughlaq Namah	Amir Khusro	Dayabhaga	Jimuta Vahana
Tarik-i-Alai	Amir Khusro	Ashiqa	Amir Khusro
Tabaqat-i-Nasiri	Minhaj-us-Siraj	Amuktamalyada	Krishnadeva Raya
Tarikh-i-Firozshahi	Zia-ud-din Barni	Futuhat-i-Firozshahi	Firoz Shah
Gita Govinda	Jayadeva	Parasana Kaghava	Jayadeva

Provincial Kingdoms

Kingdom	*Founder*	*Prominent Rulers*
Gujarat (1397)	Sultan Muzaffar Shah (Zaffar Khan)	Ahmad Shah I (*built city of Ahmedabad*) Mahmud Beghra
Malwa (1435)	Dihawan Khan Ghuri	Husan Shah (*built Jama Masjid Hindol Mahal and Jahaz Mahal at Mandu*) Mahmud Khilji (*defeated by Rana Kumbha*)
Kashmir (1339)	Shamsuddin Shah	Zainulabdin (*known as Akbar of Kashmir, he built an artificial island in the Wular lake*)
Mewar (1326)	Rana Hamir	Rana Kumbha, Rana Sangram Singh
Bengal (1342)	Shamsuddin Iliyas Shah	Ghiyas-ud-din Azam, Sher Shah Suri

VIJAYANAGARA KINGDOM (1336-1580)

THE SANGAMAS (1336-1485)

- The kingdom was founded by **Harihara I and Bukka** in 1336, in response to the Tughlaq authority in South India.
- Harihara and Bukka were originally feudatories of Kakatiya. They were brought to the centre by Muhammad-bin-Tughlaq, converted to Islam and were sent to South to control rebellion, but motivated by a Bhakti Saint Vidyaranya reconverted to Hinduism. Italian traveller Nicolo Conti visited Vijayanagara during the reign of Devraya I.

THE SALUVAS (1486-1505)

- **Saluva Narsimha** (1486-1491) Founder of the dynasty.
- **Vasco da Gama** came to India (Calicut) during the reign of Immadi in AD 1498.

THE TULUVAS (1505-1570)

- **Vira Narsimha** (1505-09) established Tuluvas dynasty.
- **Krishnadevaraya** (1509-1529) was the greatest ruler of the dynasty. Portuguese traveller, Domingo Paes, wrote high about him. Berbosa also came as a traveller. Also Friar Luis, the ambassador of Portuguese Governor, Albuquerque, resided in his court.
- He defeated Gajapati ruler of Orissa and took the title **Yavanaraja Sthapnachaya** (restorer of the Yadava of Bidar).
- He was contemporary to Babur.
- **Battle of Talikota** (1565) Between the alliance of Ahmadnagar, Bijapur, Golkonda and Bidar on one side and Sadasiva on the other side. Sadasiva was defeated.
- **Caesar Frederick**, the Portuguese traveller, visited his court (1567-68).

THE ARAVIDUS (1570-1650)

The dynasty was founded by Thirumala, brother of Rama Raya. He shifted the capital to Penugonda. Only empire in Medieval India, which employed women in the services. Women even went to battles.

Vijayanagara Architecture

- Important temples are **Vithalswami** and **Hazara** temples at Hampi, Varadraja and Ekambarnatha temples at Kanchipuram and Parvati temples at Chidambaram.
- The stories of *Ramayana* and *Mahabharata* were inscribed on the walls of the temples; e.g. Vithalswami and Hazara Ram temple.
- They issued gold coins called **Varahas** or **Pagodas** with the incarnation of Vishnu.

Famous Travellers Who Came to Vijayanagara Kingdom

Abu Abdullah/Ibn Battuta	▪ A Moroccan traveller left account of Harihara I's reign in his book **Rehla**, also called Tuhfat-un-Nuzzar fi Gharaib-il-Amsar Wa Ajaib-il-Asfar.
Nicolo de Conti	▪ An Italian traveller, who visited during the time of Devaraya I. He left an account in **Travels of Nicolo Conti**.
Abdur Razzaq	▪ Ambassador of Shahrukh of Samarqand at the Court of the Zamorin of Calicut. He gives an account of the reign of Devaraya II in his Matla-us-Sadain Wa Majma-ul-Bahrain.
Athanasius Nikitin	▪ A Russian merchant, who described the conditions of the Bahmani kingdom under Mohammad III in his Voyage to India.
Ludvico de Vorthema	▪ An Italian merchant, who visited India in 1502-1508 and left his memoirs in travels in Egypt, India, Syria etc.
Dominigo Paes	▪ Portuguese, who spent a number of years at Krishnadeva court and has given a glowing account of his personality.
Marco Polo	▪ The only traveller, who could rival Ibn Battuta's tag of the greatest medieval traveller arrived at India's opposite extremity, a Tamil port, enroute from China in AD 1290. Marco Polo to his surprise found that in Peninsular India there were tailors or seamstresses. He failed to get coat made for himself.

BAHMANI KINGDOM

- The Bahmani kingdom of Deccan was founded by **Hasan Gangu**, whose original name was Ismail Mukh. The capital was Gulbarga. Hasan Gangu took the title of Ala-ud-din Hasan Bahaman Shah (1347-1358), and became the first king of Bahmani in AD 1347. He built an observatory at Daulatabad. He founded city of Firozabad on the banks of river Bhima.
- **Humayun** (1458-61) was so cruel that he got the title of *Zalim*. **Muhammad Shah-III** Mahmud Gawan was the Prime Minister of Muhammad.
- **Nikitin** a Russian merchant, visited Bidar during his reign. After the break-up of Bahmani kingdom, five Muslim separate states came up.

1. Nizam Shahis of Ahmadnagar founded by **Ahmad Nizam Shah**, later annexed by Shah Jahan.
2. Adil Shahi of Bijapur (1490-1686) founded by **Yusuf Adil Shah**.
3. Imad Shahis of Berar (1490-1574) founded by **Fateullah Khan Imad-ul-Mulk**.
4. Qutub Shahis of Golconda (1518-1687) founded by **Quli Qutub Shah**. He built the famous **Charminar**.
5. Barid Shahis of Bidar (1528-1619) founded by **Ali Barid**.

MAHMUD GAWAN

- He was the Prime Minister or the Peshwa of Mohammad Shah III between 1463-81. The Bahmani kingdom saw a resurgence under his guidance. His military conquests included Konkan, Goa and the Krishana-Godavari delta. During the reign of Mohammad Shah III, Mahmud Gawan expanded the Bahmani Kingdom as never before.
- However, he was a Persian by birth, hence, an *Afaqi* or *Gharib*. *Afaqis* or *Gharibs* were the nobles, who were of foreign origins and had migrated from Arab or elsewhere. The rise of *Afaqi* noble was resented by the *Deccani* nobles, i.e., those who were of indigenous origin, hence the Deccani conspired against him and got him executed in 1481.

RELIGIOUS MOVEMENTS

THE SUFI MOVEMENT

- Mystics, who are called Sufis, had risen in Islam at a very early stage.
- One and the earliest Sufis was **Rabia**, a woman. Sufis organised 12 orders or **silsilas**. A silsila was generally led by a prominent mystic, who lived in a **Khanqah** or hospice, alongwith his disciple.
- The link between the teacher or pir and his disciple or murids, was a vital part of Sufism. Every pir nominated a successor or **wali** to carry out his work. Sufi orders are broadly divided into **Ba-shara** that is those, which followed the Islamic law (sharia) and Be-shara, that is those, which were not bound by it.

Sufi Terminology

Sufi Words	*Meanings*
Shaikh/Mureed/Pir	Spiritual teacher
Tasawwuf	Sufism
Murid	Disciple
Khanqah	The hospice
Sama	Musical recital
Raksa	Dance
Fana	Self annihilation
Khalifah	Successor

Famous Sufi Orders and Features

Sufi Orders	*Founders*	*Popular Saints*	*Other Features*
Chisti (Delhi and Doab region)	Khwaja Abdul Chisti	Khwaja Muin-ud-din Chisti (India) Bakhtiyar Kaki, Nizam-ud-din Auliya, Nasir-ud-din, Chirag-i-Delhi, Nizam-ud-din Aullia also known as Tahbul-i-Illahi (beloved of God)	They adopted musical recitation called **Sama** to create mood of nearness to God.
Suhrawardi (Punjab and Sind)	Shaikh Shihab-ud-din Suharwardi	Shaikh Bah-ud-din Zakariya	Saints had big jagirs and a close contact with the state
Firdausi	Shaikh Badr-ud-din of Samark	Shaikh Shams-ud-din Yahiya (Bihar)	Yahiya believed in pantheistic monoism and was a disciple of khwaja Nizam-ud-din Firdausi.
Qadiri	Shaikh Abdul Qadiri	Shah Niamatullah Syed Muhammad Jilani	Dara Shikoh, the eldest son of Shan Jahan was a follower of this order.
Naqshbandi	Khwaja Pir Muhammad	Khwaja Baqi Billah Shaikh Ahmad Sirhindi	

BHAKTI MOVEMENT

The real development of Bhakti took place in South India, between the 7th and the 12th century. The Bhakti saints came usually from the lower caste. They disregarded caste, encouraged women to join and taught in the local vernacular language. They considered that God has either a form **(Saguna)** or is formless **(Nirguna)**.

- **Ramanuja** (1017-1137) In 11th century Ramanuja tried to assimilate Bhakti to the tradition of Vedas. According to him, Moksha (salvation) lies through Karma, Gyan and Bhakti. He gave the concept of Vishishtadvaita.
- **Namdeva** (1270-1350) He was a Nirguna Upasaka. Some of his abhangas are included in the Guru Granth Sahib.
- **Ekanath** (1533-1599) Born in Patan in Aurangabad; published the first receivable edition of **Janesvari** (**Marathi Gita**). He condemned caste system and accepted disciples from lower caste.
- **Tukaram** (1598-1650) Contemporary of Shivaji. The greatest Marathi Bhakti poet, his views are similar to that of Kabir. Great devotee of Vithal.
- **Ramadas** (1608-1681) He was the spiritual guru of Shivaji. Established ashrams all over India. He wrote **Dasabodha**, which gave advice on all aspects of life.
- **Guru Nanak** (1469-1538) Founder of Sikh faith in India. He has born in Talwandi, now Nankana Sahib to a Khatri family. He composed hymns and sang them with the help of a **rabab**.
  In course of time, his teaching gave rise to Sikhism. His teachings are compiled in **Adi Granth**.
- **Vallabhacharya** (1479-1531) He emphasised worship of Krishna as an incarnation of the Almighty God. Lived in the court of Krishnadeva of Vijayanagara. He taught that there was no difference between Atma and Paramatma. By means of Bhakti, one can get salvation and merge in him.
- **Ramananda** (15th Century) The founder of Bhakti movement in North India. He discarded caste rules and included among his disciples men of all castes.
- **Kabir** (1440-1510) He was a weaver. Represented Nirguna Bhakti tradition. His followers organised themselves as Kabir Panthis.
- **Chaitanya** (1486-1533) Born in Nadia district of West Bengal. Regarded as the founder of modern Vaishnava Sect of Bengal. Regarded as the founder of modern Vaishnava Sect of Bengal.
- **Surdas** (1483-1563) Disciple of Vallabhacharya and devotee of Lord Krishna and Radha. He wrote Sur Suravali, Sahitya Ratna and Sursagar. (belonged to Saguna school).
- **Nimbarkacharya** Worshipper of Krishna and Radha and contemporary to Ramanuja.
- **Madhavacharya** (1238-1317)
  According to him, the release from transmigration can be secured only by means of knowledge and devotion. Jayatirtha was his successor.
- **Mirabai** (1498-1546) Married to Bhojraj, she was the Rathor Princess of Mewat and daughter-in-law of Rana Sanga of Mewar. She belonged to the Krishna cult of Vaishnavism.
- **Tulsidas** (1532-1623) Born in Brahmin family in Banda district of Uttar Pradesh and belonged to Ram Bhakti cult of Vaishnavism. He wrote *Ramcharitmanas, Gitawali, Kavitawali, Vinay Patrika* etc.
- **Vidyapati** Maithali saint poet, wrote Padavali i.e. thousands of love ballads on Radha and Krishna.
- **Narsingh Mehta** Saint from Gujarat, who wrote songs in Gujarati, depicting the love of Radha and Krishna.
- **Purandar Das** (1480-1564) **Vaishnava** saint, composer of Karnataka, who laid the foundation of Carnatic music.
- **Dadu Dayal** (1544-1603) Nirguna Bhakti saint, born in Gujarat, who preached in Rajasthan.
- **Thyagaraja** (1767-1847) Telugu saint and the greatest composer of Carnatic music.

THE MUGHAL EMPIRE

BABUR (1526-1530)

- He was the descendant of Timur on his father's side and Chengiz Khan on his mother's side. Originally ruled over **Farghana** (Afghanistan).
- He was invited to attack India by Daulatkhan Lodhi, Subedar of Punjab, Alam Khan Lodhi uncle of Ibrahim Lodhi and Rana Sanga.
- He was successful in his fifth expedition. In the **First Battle of Panipat** in 1526, he finally defeated Ibrahim Lodhi.
- Defeated Rana Sanga of Mewar in Battle of Khanwa in 1527. Babur took the title of 'Ghazi' after this.
- In 1529, he defeated Muhammad Lodhi (uncle of Ibrahim Lodhi) in the Battle of Ghaghra.
- Defeated another Rajput ruler Medini Rai in Battle of Chanderi in 1528. Babur was the first ruler to entitle himself 'Badshah'.

▪ He wrote **Tuzuk-i-Babari** or **Babarnama** in Turkish. It was translated into Persian by Abdul Rahim Khan-i-khanan and in English by Madam Bebridge.

HUMAYUN (1530-40 And 1555-56)

- He was the son of Babur and Maham Begum. His first campaign was against Kalinjhar.
- In 1533, the first siege of Chunar and the March of Gaur was stopped by Jalal Khan. Sher Khan (Sher Shah) offered nominal submission by sending his son **Qutub Khan** to Humayun's court.
- **Battle of Chausa** (1539) fought between Sher Shah and Humayun's army. Humayun was badly defeated and escaped. He was saved by **Nizam**.
- **His sister, Gulbadan Begum**, Wrote Humayun Namah.
- **Battle of Kannauj** (Bilgrama) (1540) Humayun was again defeated by Sher Shah, and had to flee.
- **Bairam Khan** his most faithful officer helped him. After the Battle of Machhiwara against the Afghans and Battle of Sirhind against Sikandar Shah, Humayun's second coronation was organised.
- In 1556, he fell from the stairs of library (Sher Mandal, Delhi) and died. **Dinpanah** was his second capital.

AKBAR (1556-1605)

- Akbar was born to Hamida Banu Begum at Amarkot in Rana Veersal's palace in 1542.
- Akbar was 14 years old, when he was crowned at Kalanaur in 1556.
- Bairam Khan represented him in the **Second Battle of Panipat** in 1556 against Hemu. Akbar defeated Hemu and re-occupied Delhi and Agra.
- He also ended the interference from Petticoat Government (1560-1562) represented by Maham Anaga and Adham Khan Junta.
- Akbar followed the policy of reconcilation with the Rajputs. In 1562, he married the eldest daughter of **Raja Bharmal** of Jaipur, Harakha Bai.
- Prince Salim was married to the daughter of Raja Bhagwan Das. He won Gujarat in 1572. In order to commemorate his victory of Gujarat, Akbar build Buland Darwaza at Fatehpur Sikri.
- At the time of Akbar's death in 1605, his empire included Kashmir, Sindh, Kandhar and extended as far as the Godavari in the Deccan.
- He was buried at Sikandara near Agra.

The Conquests of Akbar

Date	*Year*	*Remarks*
1561	Malwa	The ruler of Malwa was Baz Bahadur.
1564	Garhkatanga (*a kingdom in Gondwana*)	Rani Durgawati and her minor son. Bir Narayana fighting Mughals. The Mughal Army was led by Asaf Khan.
1567	Chittor	The storming of the fortrests of Chittor was one of the most famous military feats of Rana Udai Singh was its ruler.
1572-73	Gujarat	Akbar built the famous Buland Darwaza at Fatehpur Sikri, as a commemoration of his victory over Gujarat.
1574-76	Bihar and Bengal	Akbar personally marched against Bihar and drove out Daud from Patna and Hajipur.
1576	Battle of Haldighati	Rana Pratap, the son of Udai Singh of Mewar, was severely defeated by the Mughal Army under Man Singh and Asaf Khan.
1581	Kabul	Mirza Hakim was defeated.
1586	Kashmir and Baluchistan	
1591	Sindh	
1592	Orissa	The Mughal Army was led by Man Singh.
1595	Kandhar	The Mughal Army in this battle was commanded by Abdur Rahim Khan-i-Khanan.
1601	Asirgarh	The capture of Asirgarh marked the climax of Akbar's career of conquest.

Nine Jewels of Akbar's Court

Abul Fazl	He was the wazir of Akbar. He wrote **Akbarnama** and also led the Mughal imperial army, in its war in Deccan.
Faizi	He translated the famous book *'Lilavati'* by Bhaskaracharya into Persian
Tansen	▪ He served as the court musician to **King Ramchandra** of Mewar and was sent in Akbar's court. ▪ He accepted Islam at the hands of great Sufi Saint **Shaikh Muhammad Ghaus** of Gwalior. ▪ He used to sing **Megh Malhar** and **Deepak Ragas**.
Birbal (Mahesh Das)	He was conferred the title of **Raja** by Akbar. He frequently had witty and humorous exchanges with Akbar.
Raja Todarmal	He was Akbar's finance minister. He introduced standard weights of measurement of revenue in the districts. His revenue collection arrangement is called **Todarmal's Bandobast**.
Raja Man Singh	▪ He was the Raja of Amber. ▪ He was grand son of Akbar's father-in-law Bharmal and the adopted son of Raja Bhagwan Das. He assisted Akbar in many battles, including the well known Battle of Haldighati.
Abdul Rahim Khan-e-Khana	He was a poet and the son of Bairam Khan.
Faqir Azio Din	He was the cheif advisor of Akbar.
Mullah Do Piaza	He was among the Mughal Emperor's chief advisor.

IMPORTANT ASPECTS OF AKBAR'S RULE

- He abolished **Jaziya**, pilgrimage tax and forcible conversion of prisoners of war. The use of beef was also forbidden.
- He built an **Ibadat Khana** at Fatehpur Sikri to discuss religious matters. He invited many distinguished persons, such as Purshottam Das (Hindu), Maharaji Rana (Parsi), Harvijaya Suri (Jain), Monserate and Aquaviva (Christian).
- His liberation is reflected again in the pronouncement of Tauhid-i-Ilahi or Din-i-Ilahi, which propounded Sufi divine monotheism. Birbal, Abul Fazl and Faizi joined the order. Akbar established the painting karkhana headed by Abdus Samad.
- Ralph Fitch (1585) was the first Englishman to visit Akbar's court.
- Sufi Saint Shaikh Salim Chisti bless Akbar with a son, who was named Salim (Jahangir). Akbar shifted his court to Fatehpur Sikri from Agra in honour of the saint. Mansabdari System was another feature of administration during Akbar's reign, to organise nobility as well as the army. He was the first Mughal ruler to separate religion from politics.

JAHANGIR (1605-1627)

- Akbar's eldest son prince Salim assumed the title of Nur-ud-din Muhammad Jahangir and ascended the throne. He was born at Fatehpur Sikri near Agra in 1569. He was given proper education by his tutor Rahim Khankhana. In 1585, he married **Manbai**, daughter of his maternal uncle Raja Bhagwan Das.
- In 1587, he married **Jodhabai** or **Jagat** Gosain, daughter of Udai Singh, who gave birth to prince Khusro (Shah Jahan). He mostly lived in Lahore, which he adorned with gardens and buildings.
- His greatest failure was loss of Kandahar to Persia in 1622. Jahangir's wife **Nur Jahan** (daughter of Itamad-daulah) exercised tremendous influence over the state affairs. She was made the official **Badshah Begum**.
- Captain **William Hawkins**, a representative of East India Company visited Jahangir's Court in 1608. He was given the mansab of 400. In 1615, Sir Thomas Roe, an ambassador of king James I of England also came to his court. Though initially Jahangir resisted, later on he granted permission to the English to establish a trading port at Surat.
- **Pietxa Valle**, a famous traveller, came during his reign. Production of Tobacco (brought by the Portuguese) started in his reign. He was buried at Lahore.
- He wrote his autobiography, **Tuzuk-i-Jahangiri**, in Persian.

Nur Jahan

Nur Jahan's actual name was **Mehrunnisa**. She was the widow of Sher Afghani. Jahangir married her and conferred the title Nur Jahan to her.

Shah Jahan (1628-1658)

- Born to Jodhabai or Jagat Gosain in Lahore in 1592. His real name was **Khurram**. He was youngest prince to be appointed as Governor of Deccan at the age of 15. In 1612, he got married to Arzumand Bano Begum (known as Mumtaz Mahal) daughter of Asaf Khan. He was imprisoned by his son Aurangzeb in Agra Fort and died in captivity in 1658. He was buried at Taj Mahal (Agra) besides his loving wife.
- Two French travellers Bernier and Tavernier and the Italian traveller Nicolo Manucci visited during his reign. Peter Mundi described the famine that occurred during Shah Jahan's reign. His reign is considered as the **Golden Age of the Mughal Empire**.

War of Succession

- Battle of Bahadurgarh, February, 1658 Fought between Shuja and Dara, in which Shuja was defeated. Battle of Dharmat, was fought on April, 1658 wherein Combined forces of Aurangzeb and Murad, defeated Dara.
- Battle of Samugarh, May, 1658 Aurangzeb defeated Dara.
- **Battle of Khanjawa, December, 1658** Shuja was defeated by Aurangzeb and fled to Arakan.

AURANGZEB (1658-1707)

- Third son of Shah Jahan, born in Ujjain. He took the title of 'Alamgir' upon his coronation in 1659. He was called a **Zinda pir** or the living saint.
- Under him, the Mughal Empire reached its greatest extent and the largest single state ever known in India.
- **Muhtasib** (regulation of moral conduct) were appointed. He ended use of almanacs and weighing of the emperor. Aurangzeb compiled **Fatwa-i-Alamgiri**.
- He prohibited intoxicating drugs and destroyed the Vishwanath temple of Varanasi.
- Aurangzeb died in 1707 and was buried at **Khuldabad** (Daulatabad).

He forbade inscription of Kalma on the coins, also forbade sati and **Jharokha-darshan**. He ended the celebration of Nauraj (singing in the court) and in 1679, reimposed Jizya.

Revolts during Aurangzeb's Reign

- Sikhs revolted under Guru Teg Bahadur.
- Marathas under Shivaji and Sambhaji (were executed).
- Jats under Gokla, Rajaram and Churaman.
- Afghan rebellion by Ajman Khan.
- Ahoms under Mir Jumla; Bijapur.
- Golconda and Marwar.

Religious Policy of Aurangzeb

- Aurangzeb was a Sunni orthodox Muslim, who wanted to convert India from **Dar-ul-Harb** to **Dar-ul-Islam**.

LATER MUGHALS

Bahadur Shah-I (1707-1712)

- Real name is Muazzam. He also assumed the title Shah Alam-I.
- Pursued pacifist policy and was therefore know as Shah-e-Bekhabar.
- He granted Sardeshmukhi to Maratha and released Shahu.

Jahandar Shah (1712-1713)

- Jahandar Shah won the War of Succession, due to the support of Zulfiqar Khan.
- He was the first puppet Mughal Emperor. He introduced the evil practice of revenue farming or Ijara.
- Jai Singh of Amber was given the title of **Mirza Raja Sawai** and Ajit Singh was awarded the title of **Maharaja**. He abolished **Jaziya**.
- He was defeated by his nephew **Farrukh Siyar** in Agra.

Farrukh Siyar (1713-1719)

- Ascended the throne with the help of Sayyid brothers (Abdullah Khan and Hussain Khan), who were the Wazir and Mir Bakshi respectively.
- Surman Commission visited his court.

Muhammad Shah (1719-1748)

- Ascended the throne with the help of Sayyid brothers (king-makers). Sayyid brothers were killed under a conspiracy hatched by the nobles in 1720.
- During his reign, Bengal acquired virtual independence during the governorship of **Murshid Quli Khan**.
- **Saadat Khan** (Burhan-ul-Mulk), who was appointed as Governor of Awadh, laid down the foundation of the autonomous state.
- **Nadir Shah** invaded India in AD 1739 and defeated Muhammad in the Battle of Karnal (1739), and he took away Takht-i-Taus (Peacock throne) and Kohinoor diamond.
- Ahmad Shah Abdali raided the kingdom for the first time during his reign.
- He was a pleasure-loving king and was nick named Rangeela.

Ahmad Shah (1748-1754)

Ahmad Shah Abdali invaded Delhi many times and Punjab and Multan were ceded to him.

From 1748 to 1761, Ahmad Shah Abdali attacked many times and looted Mathura and Delhi.

Alamgir II (1754-1759)

- Ahmad Shah Abdali occupied Delhi during his reign. He defeated the Marathas in the **Third Battle of Panipat** in 1761.
- In this battle, Marathas were led by Sadashiv Rao Bhau, while the Peshwa, at that time was Balaji Bajirao.

Shah Alam II (1759-1806)

- He crowned himself under **Sujauddaula's** protection in Bihar and remained in exile for 12 years.
- He fought **the Battle of Buxar** in AD 1764, and was defeated by the British. He lived for several years at Allahabad as a pensioner of the East India Company.

Akbar II (1806-1837)

The king gave Raja Rammohan Roy, the title of **Raja**. Lord Hastings ceased to accept the sovereignty of Mughals and claimed the status of Pensioner of the East India Company.

Bahadur Shah II (1837-1857)

He was the **last Mughal Emperor**. He was confined by the British in the Red Fort. During the 1857 Sepoy Revolt, he was proclaimed the **emperor of India** by the rebels. He was deported to Rangoon and died there. He used to write *Shairis* in the pen-name of **Zafar**.

MUGHAL ADMINISTRATION

Officials

Official	*Department(s)*
Diwan	Executive, Revenue and Finance Minister
Mir Bakshi	Defence Minister
Mir Saman	Incharge of Royal Household
Sadr-us-sadar	Incharge of Religious Matter
Chief Qazi	Head of Judiciary
Barids	Intelligence Officers
Mustauf	Auditor General

Mansabdari System

Mansabdari system had the following features

- *Mansabdari* system was introduced in 1595-96, showing a noble's civil and military capacity. Twin ranks **Zat** and **Sawar** were allotted. The former indicated a noble's personal status, while the latter, the number of horsemen he had to maintain. *It had a three scale gradation*
  1. Mansabdar (500 Zat and below)
  2. Amir (between 500- 2500 Zat)
  3. Amir-i-Umda (2500 Zat and above)
- The salary of the *Mansabdar* was fixed on a month scale system. During Jahangir's reign, a du-aspa, sih-aspa system was introduced through which, a noble's Sawar rank could be increased without affecting his Zat.
- Mansab was not a hereditary system. *Mansabdars* were paid through revenue assignments (jagirs).

Jagirdari System

In this system, every Jagirdar was assigned land in proportion to his salary.

Methods of Revenue Collection

Rai Yield per unit area.

Kankut Based on estimates.

Zabti Based on the yields of crops.

Mughal Paintings

The Mughals introduced new themes, depicting the court, battle scenes and added new colours. (peacock blue and Indian red).

Mughal Paintings

Ruler	*Famous Painters*	*Other Features*
Humayun	Mir Sayyid Ali Abdus Samad	
Akbar	Abdus Samad Farrukh Beg Khusro Kuli Jamshed Basawan Daswan	Introduction of Persian style Daswan illustrated 'Razamnamah' (Persian Mahabharat) and Akbarnamah
Jahangir	Bishan Das Abdul Hassan Ustad Mansur (animal paintings)	Painting reached at its zenith, use of halo (divine light) started.

Mughal Architecture

Ruler	*Architecture*
Babur	Mosques at Kabuligarh (Panipat) and at Sambhal (Rohilkhand)
Humayun	City of Dinpannah, Jamali mosque and mosque of Isa Khan at Delhi
Haji Begum (wife of Humayun)	Humayun Tomb
Akbar	Agra fort, Jahangiri Mahal in Agra fort based on design of Man Mandir, Lahore Palace, Allahabad fort, temple of Govind-deva at Vrindavan and several buildings at Fatehpur Sikri that included Panch Mahal (planned on Buddhist Vihara), Diwan-i-khas, Jodhabais Palace, Diwan-i-Aam, Buland Darwaza (Iranian style). He began to build his own tomb at Sikandara, which was completed by Jahangir.
Jahangir (Indo-Persian style)	Moti Masjid at Lahore, own Mausoleum at Lahore.
Nur Jahan	Itamad-daulah's marble tomb at Agra in **Pietra dura** technique.
Shah Jahan	**At Agra** : Taj Mahal, Moti Masjid, Khanas Mahal, Sheesh Mahal, Musamman Burz (Jasmine Palace, where he spent his last years). **At Delhi** : Jama Masjid, Red Fort (Diwan-i-Khas and Rang Mahal) **Others** : Shalimar Bagh (Lahore), City of Shahjahanbad (Red Fort and Takht-i-Taus i.e. Peace throne), Nahor-i-Faiz
Aurangzeb	Moti Masjid at Delhi, Bibi-ka-Makbara (tomb of his wife Rabbia-ud-douna) at Aurangabad, Badshai mosque at Lahore.

SHER SHAH SURI AND AFGHAN EMPIRE (1540-1555)

- His real name was **Farid**.
- He was born to **Hasan**, a Jagirdar of Sasaram and Hajipur. Ibrahim Lodhi transferred his jagir to him.
- He joined the Babar Khan Lohanis service and then was appointed as Deputy Governor of Bihar.
- He usurped throne as **Hazarat-i-Ala**.
- He gained Chunar by marrying Lad Malika, the widow of Governor of Chunar.
- **Battle of Chausa** In 1539, he captured Chausa from Humayun. He assumed the title of Sher Shah as emperor. He also issued coins and Khutba was read in his name. The whole area from Bengal to Banaras was under his empire.
- **Battle of Kannauj** He also annexed Kannauj after defeating Humayun.

- **Battle of Samel** (1544) Defeated Rajput forces of Marwar. The campaign of Bundelkhand was the last campaign of his life. While besieging its fort at Kalinjar, Sher Shah got burned due to fire in the bundle of rockets in 1545.
- The peasants were given a Patta and a Qabuliyat, which fixed their rights and taxes. Sher Shah improved communication by building roads.
- Sarais were built on the road. He restored the old imperial road Grand Trunk from Sonargaon in Bengal to Peshawar. Built **Purana Qila**, alongwith Grand-Trunk.
- Built his tomb at Sasaram in Bihar.
- Malik Mohammad Jaisi wrote Padmavat (Hindi) during his reign.
- **Tarikh-i-Shershahi** was written by Abbas Khan Sarwani, his court historian.

Administrative Unit	*Head*
Iqta (Province)	Amin and Haqim
Sarkar (district)	Shiqdar-i-Shiqdaran and Munsif-i-Munsitan
Pargana (Taluka)	Shiqdar and Munsif
Gram (Village)	Amil and Muqaddam

THE MARATHA AGE

SHIVAJI (1674-80)

- Shivaji was born in the year 1627 in the hill fortress of Shivner. His father was **Shahji Bhonsle**, and Mother was **Jijabai**, Religious Teacher – **Samarth Ramdas**.
- Shivaji belonged to the Bhosle Clan of Marathas. Shivaji's father was a military Commander under Nizam Shahi ruler of Ahmadnagar.
- Shivaji's early career and life was influenced by Jijabai, Dadaji Kondev, the manager of his father's Jagir and his spiritual teacher.
- Dadaji Kondev gave him training in civil and military administration.
- In 1646, he conquered the fort of Torna and built forts at Raigarh and Pratapgarh. In 1647, he assumed full charge of his Jagir.
- In 1659, Shivaji killed Afzal Khan, ambassador of Ali Adil Shah (Sultan of Bijapur), in a meeting with his tiger paws. Later **Shaista Khan**, a Governor of Deccan, was deputed by Aurangzeb to put down the rising power of Shivaji in 1660. Shivaji lost Poona and suffered several defeats till he made a bold attack on Shaista's military camp at night and plundered Surat and later Ahmadnagar.
- **Raja Jai Singh** of Amber was then appointed by Aurangzeb to put down Shivaji (1665) and Jai Singh succeeded in besieging Shivaji in the fort of Purandhar. Consequently, the **Treaty of Purandhar** (1665) was signed according to which Shivaji ceded some fort to the Mughal and payed a visit to the Mughal court at Agra. Shivaji also agreed to help the Mughals in their attack on Bijapur.
- Shivaji visited Agra with his son Shambhaji in 1666. He was put on house arrest, but escaped from there.
- He very soon, conquered the forts, which he had surrendered to Mughals. He defeated Mughal forces in the **Battle of Salher** in 1672.
- He was crowned in 1674 at Raigarh, and assumed the title of **Haindava Dharmoddaraka** (protector of Hinduism) and **Chhatrapati**. He became the sovereign ruler of Maharashtra. He died in 1680.

Shivaji's Administration

- Shivaji divided his territory under his rule (Swaraj) into three provinces, each under a Viceroy. Provinces were divided into **prants**, which were sub-divided into **parganas** or tarafs. The lowest unit was village headed by Headman or **Patel**.
- Shivaji was helped by **Ashtapradhan** (eight ministers), which was unlike a Council of Ministers, for there was no collective responsibility. Each minister was directly responsible to Shivaji.
- His administrative reform were guided by Malik Ambar of Ahmadnagar.

Shivaji's Ashtapradhan

Ashtapradhan	*Department(s)*
Peshwa (*Mukhya Pradhan*)	Prime Minister, Finance
Sar-i-Naubat	Military Commander
Majumdar or Amatya	Accountant General (*revenue and finance minister during Peshwas*)
Waqenavis (Mantri)	Intelligence, posts and household affairs
Surunavis or Sachiv	Minister for correspondence
Dabir or Sumanta	Foreign Minister and master of royal ceremonies
Nyayadhish	Administration of justice
Pandit Rao (Sardar)	Charity and religious affairs

MARATHAS AFTER SHIVAJI

- There was a dispute of succession between Sambhaji and Rajaram.
- Son of Shivaji, **Sambhaji** (1680-89) succeeded the throne. He was succeeded by Rajaram in 1689.
- **Rajaram** (1689-1700) was killed in 1700 by Aurangzeb. His widow **Tarabai** put her infant son, Shivaji II on the throne.
- Rajaram created a new post of **Pratinidhi**, thus, taking the total ministers to nine.
- Except Pandit Rao and **Nyayadhish**, all ministers commanded armies and led expeditions. After Aurangzeb's death in 1707, Shahu, the grandson of Shivaji and son of Sambhaji was released by Bahadur Shah. He claimed the throne and this led to a civil war between Shahu and Tarabai.
- Shahu emerged victorious in the **Battle of Khed** with the help of Balaji Vishwanath. After this, Shahu ruled from Satara and Tarabai from Kolhapur.
- From now onwards, the rule of Peshwa started. They became virtual rulers of the state.

THE PESHWAS (1713-1818)

Balaji Vishwanath (1713-1720)

- Shahu honoured him with title of **Sena Karte** in 1708 and made him his Peshwa in 1713.
- The Peshwa concentrated all power in his office. He became the functional head of the Maratha Empire.

Baji Rao I (1720-1740)

Under his leadership, the Marathas compelled the Mughals first to give them the right to collect **chauth** of the vast areas and then to cede those areas to Maratha kingdom.

Balaji Baji Rao (Nana Sahib) (1740-1761)

In an agreement with the Mughal Emperor (Ahmad Shah), the Peshwa agreed to protect the Mughal Empire from internal and external enemies in return for the **chauth**. In the Third Battle of Panipat in 1761, between Maratha and Ahmad Shah Abdali, Viswas Rao, the son of Nana Saheb died. After his death, Peshwaship had lost its all power.

Later Peshwas

- Narayan Rao (1772-1773), Sawai Madhav Rao (1773-1795), Baji Rao II (1795-1818).

Maratha Confederacy

Under him, several Maratha families became prominent and got themselves entrenched in different parts of India.

- The Gaekwads at Baroda
- The Bhonsles at Nagpur
- The Holkars at Indore
- The Peshwas at Poona
- The Scindhias at Gwalior

MODERN INDIA

ADVENT OF THE EUROPEANS

Portuguese

- The **Cape Route** was discovered from Europe to India by Vasco-da-Gama. He reached port of Calicut on 17th May, 1498 and was received by the Hindu ruler of Calicut, Zamorin. This led to the establishment of trading station at Calicut (1st Portuguese factory in India), Cochin and Cannanore.
- Cochin (1502) was the early capital of the Portuguese in India. Nino da-Cunha transferred Capital to Goa in 1530.
- The first Governor of Portuguese in India was Francisco Almeida (1505-1509). He introduced the Policy of Blue Water.
- He was succeeded by Alfonso d' Albuquerque, who conquered Goa in 1510.
- Portuguese acquired Daman in 1559. They lost Hughly in 1631, during the reign of Shah Jahan.
- In 1661, the Portuguese king gave Bombay to Charles II of England as dowry, for marrying his sister.

The famous Jesuit Saint Francisco Xavier arrived in India with **Martin Alfonso De Souza** (1542-1545).

Dutch

- The Dutch East India Company established factories in India at **Masulipatnam** in 1605, Pulicat (1610), Surat (1616), Bimlipatam (1641), Karikal (1645), Chinsura, Kasimbazar, Patna, Balasore, Nagapatam and Cochin.
- They replaced the Portuguese as the most dominant power in European trade with the East. The Dutch conceded to British, after their defeat in the **Battle of Bedera** in 1759.

English

- Before the establishment of the East India Company, **John Mildenhall**, an English merchant came to India over land route to trade with Indian merchants in 1599.
- The English East India Company was formed by a group of merchants in 1599 known as Merchant Adventures.
- Jahangir issued a farman to Captain Hawkins (1613) permitting him to establish a factory at Surat.
- **Sir Thomas Roe** visited Jahangir's court (1615) as an ambassador of king James I, to seek permission to trade in India.
- First factory was built at **Surat** (1608). Surat was replaced by Bombay which acquired from Charles-II on lease as the headquarter on the West coast in 1687.
- In 1639, he obtained Madras from Raja of Chandragiri with the permission to build a fortified factory, which was later named **Fort St George**.
- In 1717, John Surman obtained royal farman from Mughal emperor Farrukhsiyar. This farman is also called the **Magna Carta** of the British rule in India, as it gave large concessions to the Company.

Danes

- The Danes arrived in India in 1616. They established settlement at **Tranqueber** (Tamil Nadu) in 1620, and Serampore (Bengal) in 1676. Serampore was their headquarter.
- They were forced to sell their settlements to the British in 1854.

French

- The French East India Company was formed in 1664 by Colbert under state patronage during the reign of Louis XIV.
- The first French factory was established at Surat by Francois Caron in 1668 and the second at Masulipatnam in 1669. They occupied Mahe, Yanam and Karaikal.
- The foundation of Pondicherry was laid in 1673, which afterwards, became its capital.

- They also developed a factory at Chandernagar.
- The Governors, Lenoir and Dumas revived the French power in India between 1720-1742. The Anglo-French conflict started with the arrival of Governor **Dupleix** in 1742.

INDEPENDENT STATES

Hyderabad

- Founded by Nizam-ul-Mulk Asaf Jah in AD 1724. His original name was **Chin Qilich Khan**, but Emperor Farrukhsiyar conferred on him the titles of 'Khan-i-Duran' and later 'Nizam-ul-Mulk'.
- **Puran Chand** was his Dewan.
- Carnatic was one of the Subahs of Deccan, hence it was under Nizam of Hyderabad.
- Nizam of Hyderabad became independent of Delhi and Deputy Governor of Carnatic made himself independent of Hyderabad, and hence assumed the title **Nawab of Carnatic**.
- Saadatullah Khan of Carnatic made his nephew Dost Ali as his successor at Carnatic without the approval of Nizam. This caused a rivalry between Carnatic and Hyderabad.
- Hyderabad remained independent until it became a part of independent India. The Nizam provided assistance to the British during the 1857 Revolt.

Awadh

- Founded by **Saadat Khan Burhan-ul-Mulk.**
- Emperor Muhammad Shah appointed him as Governor of Awadh in AD 1722.
- Committed suicide in 1739 and was succeeded by his nephew 'Safdar Jung'.
- The highest post of his Government was held by Maharaja Nawab Rai. This shows his religious tolerance towards the Hindu.

Mysore

Haider Ali (1761-1782)

- Haider Ali started his career as a soldier in Mysore State, was promoted to Commander-in-Chief and later to Faujdar at Dindigul.
- In 1769, he repeatedly defeated the British in First Anglo-Mysore War and reached the walls of Madras. He died in 1782 during Second Anglo-Mysore War.

Tipu Sultan (1782-1799)

- Succeeded Haider Ali. He planted the **Tree of Liberty** at Seringapatnam and became a member of the 'Jacobian Club'.
- He died in 1799, fighting at the gates of Seringapatnam during **Fourth Anglo-Mysore War.**
- He gave money for the construction of image of Goddess Sarda in the Shringeri temple. He assumed the title of ***Padshah*** in 1797.

Bengal

- **Murshid Quli Khan** (1717-1727) He was made Governor of **Bengal** (1717) and **Orissa** (1719). He transferred capital of Bengal from Dacca to Murshidabad. He died in 1727.
- **Battle of Plassey** On 23rd June, 1757, English won the battle against Siraj-ud-Daula and compelled the Nawab to concede all the demands.
- **Battle of Buxar** He formed an alliance with Nawab of Awadh, Suja-ud-Daula and Mughal Emperor Shah Alam II and fought with the British army at Buxar on 22nd October, 1764.

Punjab (The Sikhs)

- **Guru Nanak Dev Ji** Born in Talwandi, he was the first Sikh guru and established the **Nanak Panth**.
- **Guru Angad Dev Ji** Invented Gurumukhi Script for Punjabi language.
- **Guru Amardas Ji** Divided his spiritual empire into 22 parts called **Manjis**, which was put under the charge of a Sikh. Mughal Emperor Akbar visited him.
- **Guru Ramdas Ji** Founded the city of Amritsar. He dug a tank (sarovar) and constructed **Harmandir Sahib** in the midst of the tank.

- **Guru Arjun Dev Ji** He compiled the Adi Granth. Completed the construction of Amritsar and founded the city of Taran and Kartarpur. He was executed by Jahangir.
- **Guru Hargobind Ji** Transformed Sikhs into warrior class and defeated Mughal army at Sangrama. Fortified Amritsar and built Akal Takht at Golden Temple. Took the title of Padshah and founded the city of Kiratpur in Kashmir.
- **Guru Har Rai** He met Darashikoh, son of Aurangzeb.
- **Guru Har Kishan Ji** Ramraya established a separate seat of Guru at Dehradun.
- **Guru Teg Bahadur Ji** Executed by Aurangzeb at Delhi. **Sisganj Gurudwara** marks the site of his martyrdom.
- **Guru Gobind Singh Ji** (born in Patna). He was Tenth and the last Sikh Guru. He organised a community of warriors called **Khalsa** (Baisakhi Day, 1699), summoned the assembly of Sikhs at Anandpur and 5 persons were selected (Panj Piaras), who took the water of immortality.
- The Sikhs were required to keep 5 k's *viz*. Kesh, Kripan, Kachera, Kangha and Kara. He complied **Dasween Padshah ka Granth**. He was stabbed to death by a pathan in 1708.
- **Maharaja Ranjit Singh** Born in 1780 at Gujranwala, he founded the Sikh rule in Punjab.
- He occupied Lahore in 1799 and made it his capital. He annexed Amritsar (1802), Ludhiana, Kangra, Multan and attacked Kashmir and Peshawar. He died in 1839.
- **Successors of Ranjit Singh** Kharak Singh (1839-1840), Naunihal Singh (1840), Sher Singh (1841-1843) and Dalip Singh (1843-1849).

ASCENDANCY OF THE BRITISH

Carnatic wars

Fought between the British and the French over supremacy over India.

First Carnatic War (1746-48)

- Nawab of Carnatic's army was defeated by French Under Duplex in the **Battle of St Thome**.
- War ended with the Treaty of Aix-la-Chapelle.

Second Carnatic War (1749-54)

- War began when British and French aligned with opposite sides in the war of succession of Hyderabad.
- Robert Clive ultimately emerged victorious with the Siege of Arcot (1751). War ended with **Treaty of Pondicherry**.

Third Carnatic War (1758-63)

- French Governor Count De Lally captured Fort St David.
- French were defeated by British in **Battle of Wandiwash** (1760).
- War ended with the **Treaty of Paris**, establishing British supremacy in subcontinent over French.

Anglo-Sikh Wars

Began after the death of Ranjit Singh in 1839.

First war (1845-1846)

Sikhs were defeated. **Treaty of Lahore** ended the war.

Second War (1948-1949)

Dalhousie annexed Punjab. Sir John Lawrence became the first Commissioner of the Punjab.

Anglo-Maratha Wars

First War (AD 1775-1782)

English favoured Raghunath Rao to become the Peshwa, but were defeated and signed the Convention of Wadgaon. British later signed the **Treaty of Salbai**, renouncing the cause of Raghunath Rao.

Second War (AD 1803-1806)

The Peshwas signed the **Treaty of Bassein** (1802), which was a treaty for Subsidary Alliance.

Third War (AD 1817-1819)

Lord Hastings moved against the Marathas and Marathas were decisively defeated.

Economic and Commercial Policy

The gradual development of India has been traced through three stages of British colonialism by **RP Dutta**.

PHASES OF ECONOMIC POLICY

Early Phase (1600-1757)

East India Company was a purely trading company, dealing with import of goods and precious metals into India and export of spices and textiles.

Mercantile Phase (1757-1813)

They imposed their own prices and had no relations with the cost of production. The Company used its political power and monopolised trade and dictated terms to the weavers of Bengal. The Company used revenue of Bengal to finance exports of Indian goods.

Industrial Phase (1813-1858)

The British mercantile industrial capitalist class exploited India. Industrial Revolution in Britain completely transformed Britain's economy. The **Charter Act of 1813**, allowed one way free trade for British citizens, resulting in Indian markets flooded with cheap and machine made imported goods from Britain. Indians not only lost their foreign markets, but their Indian markets too.

Finance Imperialism (1858 onwards)

This phase saw export of capital from India, and also chains of British controlled banks, export import firms and managing agency houses. **Exploitation through railways** is the best example of Finance Imperialism.

> **Drain of wealth** It refers to a portion of National Product of India, which was not available for consumption of its own people.

Drain of Wealth Theory

- **Dadabhai Naoroji** first cited the Drain of Wealth Theory in his book titled **Poverty and Un-British Rule in India**.
- RC Dutt blamed the British policy for Indian economic ills in his book **Economic History of India**.
- Drain of Wealth began in 1757, after Battle of Plassey. In 1765, the Company acquired the Diwani of Bengal and began the purchase of Indian goods out of the revenue of Bengal, and exported them. These purchases were known as Company's investment.

LAND REVENUE SYSTEM

Permanent Settlement

- Introduced in Bengal, Bihar and Orissa, districts of Banaras and Northern districts of Madras by Lord Cornwallis in 1793.
- **John Shore** planned this settlement. It declared zamindars as the owners of the land. Hence, they could keep 1/11th of the revenue collected to themselves, while the British got a fixed share of 10/11th of the revenue collected. The zamindars were free to fix the rate.

Ryotwari Settlement

- Introduced in Bombay, Madras and Assam. Munro and Charles Reed recommended it. In this, a direct settlement was made between the Government and the Ryots (cultivators).

Mahalwari Settlement

- Introduced in the area of Ganga valley North-West frontier provinces, parts of Central India and Punjab. Revenue settlement was to be made by villages or estates with the landlords.
- A settlement was made with the village, which maintained a form of common ownership, known as **Bhaichara** or the **Mahals**.

Revolt of 1857

Causes of the Revolt

- **Political** Nana Sahib was refused his pension, as he was the adopted son of Peshwa Baji Rao II.
- **Awadh** (Lucknow) was annexed in 1856, on charge of maladministration, and **Jhansi** was annexed owing to the **Doctrine of Lapse**.
- **Military Discrimination** Indian soldiers were paid low salaries, they could not rise above the rank of Subedar and were racially insulted.
- **Religious Discrimination** The social reforms by the British were against the people's will (widow remarriage, abolition of sati, school for girls, Christian missionaries etc).
- Soldiers were asked to use the enfield rifles with greased (by pork or beef) cartridges.

Economic Grievances Heavy taxations, discriminatory tariff policy; destruction of traditional handicrafts that hit peasants, artisans and small zamindars.

Outbreak of the Revolt

- **Bengal resentment** in 19th native infantry of Behrampur, which refused to use the newly introduced enfield rifle.
- **Mangal Pandey** 34th native infantry fired at the sergeant major of his regiment. Mangal Pandey was hanged.
- On 10th May, 1857, the sepoys at Meerut refused to use enfield rifles and revolted. The mutiny spread throughout Uttar Pradesh and sepoys moved to Delhi crying **March to Delhi**.
- At Delhi, Bahadur Shah II was declared **'Shahenshah-i-Hindustan.'**

Centres of the Revolt and their Leaders

- Delhi–*Bahadur Shah II, General Bakht Khan*
- Kanpur–*Nana Saheb/Dhondhu Pant (adopted son of Baji Rao II) Tatya Tope, Azimullah Khan*
- Jhansi–*Rani Laxmi Bai*
- Lucknow–*Begum Hazrat Mahal, her son Birjis Qadir*
- Faizabad–*Maulavi Ahamdullah*
- Bareilly–*Khan Bahadur Khan*
- Bihar (Arrah)–*Kunwar Singh, Zamindar of Jagdishpur*

Suppression of the Revolt

- **John Lawrence** remarked, ''had a single leader of ability arisen among them we must have been lost beyond redemption.''
- Delhi was captured on 20th September, 1857 by **John Nicholson,** and Bahadur Shah II was deported to Rangoon, where he died in 1862. His sons were shot dead in Delhi.
- Jhansi was captured by **Hugh Rose** on 17th June, 1858. Rani of Jhansi died in the battlefield.
- Kanpur was recaptured on 6th December, 1857 by **Colin Campbell**.
- Nana Sahib and Hazrat Mahal, both escaped to Nepal.
- Lucknow was recaptured on 21st March, 1858 by **Colin Campbell**, **Havelock** and **Outram**.
- William Taylor and Edgre suppressed the **Revolt at Arrah**.
- Tatya Tope was betrayed by a friend. He was captured and executed on 15th April, 1859.

Rani Lakshmi Bai

Rani Lakshmi Bai, nicknamed Manu, was married to Raja Gangadhar Rao in 1842. The couple adopted a child in 1853, but Lord Dalhousie wished to annex Jhansi, under the Doctrine of Lapse. Rani did not surrender and died fighting at Kalpi near Jhansi, during the Revolt of 1857.

Governor-Generals of Bengal

Warren Hastings (1772-1785)

- Brought the dual government to an end by the **Regulating Act**, 1773.
- The **Act of 1781**, was the clear demarcation between the jurisdiction of the Governor-General in Council and that of the Supreme Court at Calcutta.
- **Pitt's India Act** (1784), Rohilla War (1774), First Maratha War (1775-1782) and Treaty of Salbai with Marathas (1782); Second Mysore War (1780- 84).
- Foundation of Asiatic Society of Bengal (1784) in Calcutta by Sir William Jones.
- The translator of **Abhigyan-shakuntlam** in English in 1789, who also wrote introduction to the English translation of the **Bhagwad Gita** by Charles Wilkins.

Lord Cornwallis (1786-1793)

- Third Anglo-Mysore War (1790-1792) and **Treaty of Seringapatnam** (1792).
- Introduced Permanent Settlement in Bengal and Bihar (1793).
- Also called the **Father of Civil Services** in India, introduced judicial reforms by separating revenue administration from judicial administration, and established a system of circles (thanas) headed by a Daroga (an Indian).

Sir John Shore (1793-1798)

- Played an important role in the introduction of Permanent Settlement and **Battle of Kharda** between the Nizams and the Marathas (1795).

Lord Wellesley (1798-1805)

- Introduction of the **Subsidiary Alliance** (1798), first alliance with Nizam of Hyderabad, followed by Mysore, Tanjore, Awadh, the Peshwa, the Bhonsle and the Scindia.
- Formation of Madras Presidency in 1801.
- He described himself as **Bengal Tiger**.
- **Treaty of Bassein** (1802) and the Second Maratha War.

Lord Minto I (1807-1813)

- Concluded the **Treaty of Amritsar** with Maharaja Ranjit Singh (1809).
- Charter Act of 1813 was passed.

Lord Hasting (1813-1823)

- Anglo Nepal War (1814-1816) and **Treaty of Sagauli** (1816).
- **Third Maratha War** (1817-1818), Dissolution of Maratha Confederacy and creation of Bombay Presidency.
- Pindari War and establishment of **Ryotwari System**, by Thomas Munro (1820).

Lord Amherst (1823-1828)

- First Burmese War (1824-1826), Treaty of Yandaboo (1826) and the capture of Bharatpur (1826).

Governor-Generals of India

Lord William Bentinck (1828-1835)

- Charter Act of 1833 was passed, and he was made the **first Governor General of India**. Before him, the designation was Governor-General of Bengal.
- Carried out the social reforms like **Prohibition of sati** (1829) and **Elimination of thugs** (1830).
- On Macaulay's recommendations, English was made the medium of higher education. Suppressed Female Infanticide and Child Sacrifice.

Lord Metcalfe (1835-1836)

- Known as the **liberator of press** in India.

Lord Auckland (1836-1842)

- **First Afghan War** (1838-1842)

Lord Ellenborough (1842-1844)

- Abolished Slavery.
- War with Gwalior (1843), **Annexation of Sind** by Charles Napier (1843).

Lord Hardinge (1844-1848)

- First Anglo-Sikh War (1845-46) and Treaty of Lahore (1846).
- Gave preference to English educated persons in employment.

Lord Dalhousie (1848-1856)

- Introduction of **Doctrine of Lapse** and annexation of Satara (1848), Jaipur and Sambhalpur (1849), Udaipur (1852), Jhansi (1853), Nagpur (1854) and Awadh (annexed in 1856, on account of maladministration).
- Laying down of **first railway line** between Bombay and Thane (1853), **Telegraph line** between Calcutta and Agra.
- **Postal reforms** (first issue of Indian stamp in Karachi in 1854) with **Post Office Act**.
- **Widow Remarriage Act, 1856** (the main force being Ishwar Chand Vidyasagar).
- Started **Public Works Department**, Grand-Trunk Road work and harbour of Karachi, Bombay and Calcutta developed.
- **Charter Act, 1853** Selection to Civil Service through Competitive Examinations.
- Started Engineering College at Roorkee.
- Made Shimla the summer capital.

Viceroys of India

Lord Canning (1856-1862)

- The **last Governor General** and the **first Viceroy.**
- Withdrew Doctrine of Lapse.
- Revolt of 1857, Mutiny took place.
- Passed the Government of India Act, 1858, which ended the rule of East India Company.
- The Universities of Calcutta, Bombay and Madras, were established in 1857.
- **Indian Penal Code 1859**, was passed.
- Indian Councils Act was passed in 1861.

Lord Elgin I (1862-1863)

Wahabi Movement was started during his period.

Lord John Lawrence (1864-1869)

- Established the **High Courts** at Calcutta, Bombay and Madras in 1865. Telegraphic communication was opened with Europe. Created the **Indian Forest Department**.

Lord Mayo (1869-1872)

- Organised the Statistical Survey of India and for the **first time** in Indian history, a **Census** was held in 1871.
- Started the process of financial Decentralisation in India. Established the Department of Agriculture and Commerce.
- Established the **Rajkot College** at Kathiawar and **Mayo College** at Ajmer for the Indian princes. He was the only Viceroy to be murdered in office by a Pathan convict, in the Andamans in 1872.

Lord Northbrooke (1872-1876)

Kuka Movement of Punjab took a rebellious turn during his period .

Lord Lytton (1876-1880)

- Known as the Viceroy of Reverse Character.
- **Royal Titles Act of 1876** and the assumption of the title of 'Empress of India' by Queen Victoria, the Delhi Durbar in January, 1877.
- **Vernacular Press Act** and the **Arms** Act (made it mandatory for Indians to acquires license in arms) of 1878.

Lord Rippon (1880-1884)

- **First Factory** Act of 1881 (prohibited labour) and to improve child labour conditions.
- **Local Self-Government was** introduced in 1882. Repealed the Vernacular Press Act in 1882.
- Finances of the Centre were divided.
- An **Education Commission** was appointed under Sir William Hunter in 1882, to improve Primary and Secondary Education.
- The **Ilbert Bill Controversy** (1883). It enabled Indian district magistrates to try European criminals.

Lord Dufferin (1884-1888)

- Third Burmese War (AD 1885-86).
- Establishment of the **Indian National Congress** in 1885.

Lord Lansdowne (1888-94)

- Factory Act of 1891, granted weekly holidays and stipulated working hours for women and children.
- Civil services were divided into Imperial, Provincial and Subordinate Services.
- Indian Councils Act of 1892.
- The **Durand Commission** defined the Durand Line between British India and Afghanistan (now between Pakistan and Afghanistan) in 1893.

Lord Elgin II (1894-1899)

- Southern uprisings of 1899.
- Great famine of 1896-1897, and **Lyall Commission** on famine was established.

Lord Curzon (1899-1905)

- A Commission was appointed under Sir Thomas Raleigh in 1902, to suggest reforms regarding Universities. The **Indian Universities Act** of **1904**, was passed on the basis of its recommendations.
- Ancient Monuments Preservation Act of 1904. Thus, **Archaeological Survey of India** was established.
- **Agricultural Research Institute** was established at Pusa in Delhi.
- **Partition of Bengal** in 1905.
- Colonel Young Husband's Expedition to Tibet in 1904.

Lord Minto II (1905-1910)

- Swadeshi Movement (1905-1908); foundation of Muslim League (1906); Surat Session and split in the Congress (1907).
- Morley-Minto Reforms (1909).

Lord Hardinge (1910-1916)

- Capital shifted from Calcutta to Delhi (1911); Delhi Durbar; Partition of Bengal was cancelled.
- The Hindu Mahasabha was founded in 1915, by Pandit Madan Mohan Malaviya.

Lord Chelmsford (1916-1921)

- Gandhi returned to India (1915) and founded the **Sabarmati Ashram** (1916), Champaran Satyagraha, Satyagraha at Ahmedabad (1918), Kheda Satyagraha (1918).
- August Declaration (1917) by Montague, the then Secretary of State, and Montford reforms or the Government of India Act, 1919.
- **Rowlatt Act** (March, 1919) and the **Jallianwala Bagh Massacre** (13th April, 1919). **Khilafat Committee** was formed and the Khilafat Movement started, (1919-1920). **Non-Cooperation Movement** started (1920-1922). Women's University was founded at Poona (1916).

Lord Reading (1921-1926)

- Repeal of the Rowlatt Act.
- Chaura-Chauri incident.
- RSS founded in 1925.
- Suppressed the Non-Cooperation Movement.
- Formation of Swaraj Party.
- Moplah Rebellion (1921) took place in Kerala. **Kakori Train Robbery** on 1st August, 1925.
- Communal Riots of 1923-1925 in Multan, Amritsar, Delhi etc.

Lord Irwin (1926-31)

- **Simon Commission** visited India in 1927.
- Congress passed the **Indian Resolution** in 1929.
- Lahore Session of Congress and **Poorna Swaraj** Declaration (1929).
- Dandi March (12th March, 1930).
- Civil Disobedience Movement (1930).
- First Round Table Conference was held in England in 1930. **Gandhi-Irwin Pact**.

Lord Willingdon (1931-1936)

- **Second Round Table Conference** in London in 1931 and third in 1932.
- **Government of India Act**, (1935) was passed. Communal Awards (16th August, 1932) assigned seats to different religious communities. Gandhiji went on an **epic fast** to protest against this division.

Lord Linlithgow (1936-1943)

- Congress ministries resignation celebrated as **Deliverance Day** by the Muslim League (1939), the Lahore Resolution (23rd March, 1940) of the Muslim League demanding separate state for the Muslims. (It was at this session, that Jinnah propounded his Two-Nation Theory.
- Outbreak of the World War II in 1939.
- Cripps Mission in 1942.
- Quit India Movement, 8th August, 1942.

Lord Wavell (1943-1947)

- Cabinet Mission Plan (16th May, 1946).
- Arranged the **Shimla Conference** on 25th June, 1945, with the Indian National Congress and Muslim League failed.
- Elections to the Constituent Assembly were held, and an Interim Government was appointed under Nehru.

Lord Mountbatten (March to August, 1947)

- Last Viceroy of British India and the **First Governor-General** of Free India. Partition of India decided by the 3rd June Plan, or Mountbatten Plan.
- Retired in June, 1948 and was succeeded by **C Rajagopalachari**, the first and the last Indian Governor-General of free India.
- **Indian Independence** Act was passed by the British Parliament on 4th July, 1947, by which India became independent on 15th August, 1947.

Social and Cultural Movements

BRAHMO SAMAJ

- Founded by **Raja Rammohan Roy** (1771-1833). He is regarded as the first great leader and reformer of modern India. He was one of the earliest propagators of modern education.
- He started the **Atmiya Sabha** in 1814, **Brahmo Sabha** in 1828 and the **Brahmo Samaj** in 1830. He laid emphasis on human dignity, and criticised social evils.
- He launched a movement for the abolition of sati through his journals **Sambad Kaumudi** (1819), and the **Precepts of Jesus** in 1820.
- He gave enthusiastic assistance to David Hare who founded the famous **Hindu College** in Calcutta.
- Established the **Vedanta College** (1825), in which courses, both Indian and Western and social and physical sciences were offered.
- He wrote **A Gift to Monotheists** or **Tuhafat-ul-Muwaihidin** in Persian in 1809.

ARYA SAMAJ

- The Arya Samaj was founded by **Swami Dayanand Saraswati** in 1875 in Bombay. Swami Dayanand Saraswati was born in 1824 in Gujarat. His original name was **Mula Shankar**.
- He was known as the earliest Neo-nationalist. He looked on the Vedas as 'India's Rock and Ages'. His motto was **Go Back to the Vedas** and **India for the Indians**.

- Arya Samaj stood for the four-fold Varna System determined by merit and not by birth; for equal rights for men and women. Opposed untouchability, caste discrimination, child marriage and supported widow remarriage and intercaste marriages.
- He wrote three books—Satyartha Prakash, Veda-Bhashya Bhumika and Veda Bhashya.
- In 1886, Lala Hansraj instituted the DAV School in Lahore. In 1902, Gurukul Pathshala was established at Haridwar.

RAMAKRISHNA MISSION

- It was established by **Swami Vivekananda**, to carry on humanitarian relief and social work after death of his Guru **Ram Krishna Paramhansa**, in 1897.
- His original name was **Narendranath Dutta**. He was born in Calcutta in 1863.
- He stressed on social action and proclaimed the essential oneness of all religions, and condemned any narrowness in religious matters.

Chicago Parliament

He attended the Parliament of religions held at Chicago in 1893, and published two papers **Prabhudha Bharata** in English, and **Udbodhana** in Bengali.

YOUNG BENGAL MOVEMENT

- During the late 1820s and the early 1830s, there emerged a radical intellectual trend among the youth in Bengal, which came to be known as the **Young Bengal Movement.**
- It was founded by **Henry Louis Vivian Derozio**. He was a teacher in **Hindu College** in Calcutta.
- They believed in truth, freedom and reason. Supported Women's education.
- Derozio edited the paper–**Calcutta Gazette** and **India Gazette**.

Other Movements

Dharma Sabha

The orthodox Hindus organised the **Dharma Sabha** under the leadership of **Raja Radhakant Dev** in 1830 to counter the Brahmo Samaj.

Paramhansa Mandali

- Founded by **Dadoba Pandurang** and **Bal Shastri Jambhekar** in 1849.
- Members took food cooked by low caste people. Believed in permitting widow remarriage and in education of women.

Veda Samaj

- Called the **Brahmo Samaj of the South**. Started by **Sridharalu Naidu**.
- He translated books of Brahmo Dharma into Telugu and Tamil.

The Prarthana Sabha

- Founded in 1867 by **MG Ranade**.
- Prominent leaders were Dr Atmaram Pandurang, and RG Bhandarkar and NG Chandavarkar.

Swami Narayan Sampradaya

Founded by **Swami Sahajaananda** in Gujarat, to protest against luxurious practices of Vaishnavism.

Namdhari/Kuka Movement

Founded by Bhai Balak Singh and Baba Ram Singh, in North-West frontier province, Ludhiana, in 1841.

Indian Reform Association

- Founded by **Keshab Chandra Sen** in Calcutta in 1870.
- Objective was to create public opinion against child marriages, and for legalising the Brahma form of marriage. Promote intellectual and social status of women.

Theosophical Society

- Founded by **Madam HP Blavatsky** and **Col HS Olcott** in **New York** in 1875.
- In 1882, the headquarters were shifted to **Adyar** (Tamil Nadu). The **Theosophical Society** of India was founded by **Annie Beasant**. She founded the Central Hindu College in 1898, which later became the **Banaras Hindu University** in 1916.

Deccan Education Society

Founded by **MG Ranade**, **VG Chibdonkar** and **GG Agarkar** in Pune, in 1884. Objective was to contribute to the cause of education and culture in Western India. The society founded the **Fergusson College**.

Seva Sadan

- Founded by **Behramji M Malabari** in Bombay, in 1885.
- Campaign against child marriage, enforced widowhood, and care for socially exploited women.

Indian National Social Conference

- Founded by MG Ranade and Raghunath Rao in Bombay, in 1887.
- Focus was to abolish polygamy and Kulinism, and promote intercaste marriages. The conference is also referred to as the social reform cell of the Indian National Congress.

Muslim Socio-Religious Movements

Movement	*Location*	*Leaders*	*Causes*
Fairazi Movement (1804)	Faridpur, Bengal	Haji Shariatullah, Dudhi Miyan	Emphasis on strict monotheism, and to rid the Muslim society of non-Islamic social customs.
Deoband Movement (1867)	Deoband	Muhammad Qasim Nanutavi, Rashid Ahmed Ganghoi	Against Western education and promoted classical studies in Islam. Supported the Indian National Congress and opposed Aligarh Movement.
Aligarh Movement (1875)	South	Sir Syed Ahmed Khan	Liberalisation of Indian Islam and modernisation of Indian Muslim through religious reinterpretation and modern education Urdu Journal-Tahzib-al-aklaq Founded Aligarh school in 1875. That grew into Aligarh Muslim University
Ahmediya Movement (1889-1890)	Faridkot	Mirza Ghulam Ahmed of Qadiyan	Believed in universal religion for all humanity, opposed to islamic orthodoxy and spread of Eastern liberal education amongst youth.
Ahrar Movement	—	Riza Khan and Ali Brothers	Against Aligarh Movement

Tribal Movements

Revolt	*Years*	*Area*
Kol Uprising	1824-1828, 1839, 1899	Gujarat
Bhils Uprising	1818-1831	Western Ghat
Khasi Rising	1846-48, 1855, 1914	Orissa
Kuki Rising under Rani Gaidinliu	1917-1819	Manipur
Singpo Rising	1830-1839	Assam
Kol Rising under Buddha Bhagat	1831-1832	Chhotanagpur
Khond Rising under Chakrabisai	1846-48, 1855, 1941	Khandmal area in Orissa
Tharo Kuti Rising under Jadonand and Rani Gaidinliu	1917-1919	Manipur
Ramsoi Revolt under Vasudeo Balwant Fadke (Robin-hood of Maharashtra)	1822-1829	Western Ghats
Munda Revolt under Birsa Munda	1899-1900	Chhotanagpur area

Peasant Movements

Movement	*Location*	*Leaders*	*Causes*
Pagal Panthi Movement (1825-1835)	Bengal	Karam Shah, Tipu Shah (Hajong and Garo tribes)	Against hike in rents in Bengal
Moplah Rebellion (1921)	Malabar region, Kerala	Sayyd Ali, Sayyid Fazl	Against the oppression and exploitation of Muslim Moplah peasants by Hindu Zamindars and British Government
Indigo Revolt (1860)	Nadia, district of Bengal	Digambar Biswas, Bishnu Bishwas, Harish Chandra Mukherjee (editor of newspaper Hindu Patriot)	Peasants were forced to grow Indigo in their fields by the European factory owners Dinbandhu Mitra had written about this revolt in his play **Neel Darpan** (translated into English by Madhusudan Datta)
Poona Sarvajanik Sabha (1870)	Pune	By MG Ranade	To popularise the peasants' legal rights
Eka Movement (1921)	Awadh	Madari Pasi	Higher extraction of rent
Champaran Satyagrah (1917)	Bihar	Gandhiji Dr Rajendra Prasad	Against the Tin Kathia System imposed by the European Indigo planters
Kheda Satyagrah (1918)	Uttar Pradesh	Gandhiji	Against ignored appeal for remission of band revenue in case of crop failure
UP Kisan Sabha (1918)	Uttar Pradesh	Indira Narain Dwivedi, Madan Mohan Malviya	—
Awadh Kisan Sabha (1920)	Oudh	JL Nehru, Baba Rama Chandra	To organise peasants
Andhra Ryots Association (1928)	Andhra Pradesh	NG Ranga	Abolition of zamindari
All India Kisan Sabha (1936)	Lucknow	Swami Sahajananda	Protection of peasants from economic exploitation
Bijolia Movement (1905, 1913, 1916, 1927)	Rajasthan	Sitaram Das, Vijay Pathhik Singh	The movement arose due to imposition of 86 different types of cesses on peasants
Tebhaga Movement (1946)	Bengal	Communists	Aganist zamindars and money-lenders
Telangana Movement (1945-1951)	Hyderabad	—	Against money-lenders and officials of Nizam of Hyderabad

GROWTH OF MODERN EDUCATION IN INDIA

FIRST PHASE (1758-1812)

- Initially, the East India Company was not interested in the development of education. Some minor exceptions were efforts by some individuals.
- The **Calcutta Madarsa** was established by **Warren Hastings** in 1781, for the study of Muslim law.
- The **Sanskrit College** was established by **Jonathan Duncan** at Banaras in 1791 for the study of Hindu law and philosophy.
- **Fort William College** was established by **Wellesley** in AD 1800 for training of civil servants of the Company in Indian languages and customs (closed in AD 1802).

SECOND PHASE (1813-1853)

- For the first time, the British Parliament included in the 1813 Charter a clause, under which the Governor-General in Council was bound to keep a sum not less than ₹ 1 lakh for education. However, the Company used this fund for promoting Indian languages and literature.
- The Charter allowed the Christian missionaries to spread their religious ideas in India.
- The greatest importance of the 1813 Act was that the Company, for the first time, acknowledged state responsibility for promotion of education in India.
- Establishment of **Calcutta College** in 1817, with the efforts of Raja Rammohan Roy for imparting Western education. Three Sanskrit colleges were set-up at Calcutta.
- In 1823, a General Committee of public instruction was appointed to look after the development of education in India. These, but, arose a controversy between the orientalists and the anglicists, over the nature of education.
- The controversy was settled by **Macaulay's Education Policy**, 1835, which was approved of Lord William Bentinck. The policy favoured English education to the traditional one.
- In 1844, **Lord Hardinge** decided to give government employment to Indians educated in English schools. This further boosted the Western education in India.

THIRD PHASE (1854-1900)

- In 1854, Charles Wood prepared a dispatch on educational system in India, which came to be called the **Magna Carta of Education** in the country. According to Wood's scheme,
  - The government needed to spread Western education through English medium for higher education. But, vernacular primary schools should be set-up in rural areas.
  - A grants-in-aid system to encourage private enterprise's involvement in education.
  - A Department of Public Instruction in each of the five provinces.
  - Universities in Calcutta (1857), Bombay (1857) and Madras (1857) were established.
  - Teacher's training institutions.
  - Promotion of education for Women.
- In 1882, Lord Ripon appointed the Hunter Commission under Sir WW Hunter. The Commission's views were restricted to primary and secondary education. It emphasised the state's role in extending education, female education and to involve private enterprise in education.
- As a result, Punjab (1882) and Allahabad (1887), Universities were established.

FOURTH PHASE (1901-1920)

- Lord Curzon appointed the **University Commission** under Sir Thomas Raleigh based on his report. **Indian Universities Act** was passed in 1904.

Sergeant Plan, 1944

The Sergeant Plan, worked out by the Central Advisory Board of Education, in 1944, called for elementary and higher secondary schools, universal, free and compulsory education for children in the 6-11 age group and a 6-years school course for the 11-17 age group. Though the plan aimed to reconstruct education in 40 years, it was later restricted to 16 years. **Sir John Sergeant** was the Educational Advisor to the Government of India.

Development After Independence

- In 1948, **Radha Krishnan Commission** was set-up to review University Education. Its main focus was over technical education.
- **Kothari Commission on Education** was appointed in 1964, which recommended for 6% of the national income to be reserved for Education, till 14 years of age.

Books, Newspapers and Journals

Author/Writer /Editor	*Work*
Aurobindo Ghosh	Karmayogi
	New Lamp for Old
	Bhawani Mandir
BC Chatterjee	Anand Nath
	Durgesh Nandini
BR Ambedkar	Mook Nayak
	Bahishkrit Bharat
Dadabhai Naoroji	Rast Goftar
	Voice of India
	Poverty and Un-British Rule in India
Dayanand Saraswati	Veda Bhasya Bhumika
	Satyartha Prakash
Gopal Krishna Gokhale	Nation
	Sudharak
JL Nehru	Discovery of India
	National Herald
	Wither India
	Soviet Asia
MK Gandhi	Navjeevan
	Young India and Harijan
	Indian Opinion
Madan Mohan Malviya	Hindustan
	Leader
RN Tagore	Letters from Russia, Gora
Raja Rammohan Roy	Sambad Kaumudi
	Mirat-ul-Akhbar
	Barga-Dutta
Vivekananda	Prabhudha Bharat Udbodhana
	Prachya aur Pashchaya
Annie Beasant	New India, Commonwealth
BG Tilak	Kesari and Maratha
BK Nanda	Biography of Gokhale
B Upadhyay	Sandhya
Bal Shastri Jambhekar	Darpan
Barindra Kumar Ghosh and Bhupendra Dutta	Yugantar
Bhartendu Harish Chandra	Kavivachan Sudha
Bipin Chandrapal	Paridarshak
Curzon	Philosophy of the East
Dayal Singh Majithia	Tribune
Deenbandhu Mitra	Neel Darpan
Derozio	East Indian
Devendranath Tagore	Indian Mirror
EV Ramaswamy Naiker	Kudi Arasu
GS Aiyar	Swadesh Mitram
GS Aiyar, Viraraghavachari and Subha Rao Pandit	The Hindu
Ghulam Hussain	Inquilab

Author/Writer /Editor	*Work*
Harish Chandra Mukherjee	Hindu Patriot
Henry Vivian Derozio	India Gazette
Hunters	Indian Musalmans
Ishwar Chandra Vidyasagar	Som Prakash
James Augustus Hicky (1780)	Bengal Gazette
Jyotiba Rao Phule	Ghulam Giri
KK Mitra	Sanjivani
KM Panikkar	The Hindustan Times
MA Azad	India Wins Freedom
MG Ranade	Essays in India Economics
MN Roy	India in Transition
Madam Bikaji Cama	Bande Mataram
Maulana Abdul Kalam Azad	Al-Hilal
Maulana Muhammad Ali	Comrade
Mukundrao Patil	Din Mitra
Muzaffer Ahmed	Navyug
PC Rai	History of Hindu Chemistry
Pherozeshah Mehta	Bombay Chronicle
RC Dutt	Economic History of British India
Robert Knight	Indian Statesman
Robert Knight and Thomas Bennett	Bombay Times
SA Dange	The Socialist
SC Bose	Indian Struggle
SN Bannerjee	Bengalee
Sachindranath Sanyal	Bandi Jivan
Shyamji Krishna Verma (London)	Indian Sociologist
Sir Syed Ahmed Khan	Tahzib-ul-Akhlaq
Sisir Kumar Ghosh and Motilal Ghosh	Amrit Bazar Patrika
Taraknath Das	Free Hindustan

THE INDIAN NATIONAL MOVEMENT

- The **National Conference** in 1883, decided to invite prominent public men and associations to discuss questions of general concern.
- **Indian National Union**–1884, by **AO Hume**. The National Conference and the Indian National Union merged to form the **Indian National Congress** (INC) in 1885.
- The first meeting of INC was organised by AO Hume at Gokuldas Tejpal Sanskrit College on 28th December, 1885 in Bombay. AO Hume was a retired British Civil Servant.
- It was presided over by Womesh Chandra Bannerjee and attended by 72 delegates. It was the first organised expression of the Indian National Movement on an all India scale. In 1886, the delegates to Congress become **436**.
- The venue of first meeting was Pune but it was changed to Bombay due to an outbreak of Cholera in Pune.
- **Kadambini Ganguly** was the first woman graduate of Calcutta University to address the Congress Session in 1889.

Important Facts

- First President of INC — WC Bannerjee
- First woman President — Annie Beasant
- First Muslim President — Badruddin Tayabji
- First English President — George Yule
- First Indian Women President — Sarojini Naidu
- Gandhi became President — 1924, Belgaum
- Jawaharlal Nehru became President — 1929, Lahore
- Subhash Bose became President — 1938, Haripura

- **Safety Valve Theory** British historians argue that Hume's main purpose was to provide a *safety valve* to the growing discontent among the educated Indians.
- **Opposition to Congress** By Syed Ahmed Khan, Raja Shiv Prasad of Banaras and Lord Dufferin (the then Viceroy).

Moderate Phase (1885-1905)

Objectives and Methods

- Development and consolidation of feeling of national unity irrespective of race, caste, religion or province.
- Peaceful constitutional agitations, prayer and petitions were the instruments of work.

Extremist Phase (1905-1917)

Causes for the Rise of Extremists

- Dissatisfaction with the methods and achievements of the moderates.
- Growing consciousness about the exploitative character of the British rule.
- Reactionary policies of the Curzon University Act (1904), Indian Official Secrets Act (1904), and the partition of Bengal.

Methods of the Extremists

- Promotion of **Swadeshi** and Boycott of foreign goods. Non-cooperation with Britishers. (Passive resistance).
- **Extremist Leaders** Lala Lajpat Rai, Bal Gangadhar Tilak**,** Bipin Chandra Pal, Sir Aurobindo Ghosh, Chakravarti Bose, T Prakasham, Chidambaram Pillai.

Partition of Bengal

- Through a royal proclamation, **Lord Curzon** ordered **Partition of Bengal** creating East Bengal and Assam out of the rest of Bengal, on 16th October, 1905.
- The objective was to set-up a communal gulf between the Hindus and Muslims.
- **Rabindranath Tagore** composed the National Song **Amar Sonar Bangla** for the occasion.

The Swadeshi and Boycott (1905-1908)

It had its origin in the Anti-Partition Movement of Bengal. Mass meetings were held all over Bengal, where **Swadeshi** or the use of **Indian goods** and the **boycott** of **British goods** were proclaimed and pledged. Public burning of foreign cloth were organised and shops selling foreign clothes were picketed.

Formation of Muslim League

- Set-up in 1906, under the leadership of Agha Khan, Nawab Salimullah of Dhaka and Nawab Mohsin-ul-Mulk.
- It supported Partition of Bengal, opposed the Swadeshi Movement, demanded special safeguards for its community and separate electorates for Muslims. **Calcutta Session of INC (1906)** Dadabhai Naoroji, the President of the Session, declared that self-government or **swaraj** like that of United Kingdom, was the goal of the Indian people.

Surat Split (1907)

The INC split into two groups during the session at Surat in 1907.

Home Rule Movement (1916)

- Hone Rule Movement was led by BG Tilak and Annie Besant.
- Tilak's movement was started in April, 1916 at Poona. Tilak's league was to work in Maharashtra, Karnataka, Central Province and Berar.
- Tilak's **Swaraj meant** formation of linguistic states and education in vernacular language.
- He gave the slogan Swaraj is my birth right and I shall have it.
- Tilak's newspapers Maratha and Kesari were the organs for home rule.
- Annie Besant's league worked in rest of India. Annie Besant's newspapers New India, Commonweal and became important for this movement. She coined the term Commonwealth.
- George Arundale was her organising secretary.

Lucknow Session of the Congress (1916)

- Presided by Ambika Charan Majumdar. The two wings of Congress: Extremists and Moderates, were once again united, at this session.

Rowlatt Act (1919)

- The 1919, **Sedition Committee** headed by **Justice Rowlatt** led to the Rowlatt Act.
- This act authorised the government to imprison any person without trial and conviction of the Court of Law for 2 years, thus, suspending the right of Habeas corpus.

Gandhi's Arrival (1915)

- Champaran Satyagraha (1917)–1st Civil Disobedience
- Ahmedabad Mill Strike (1918)–1st Hunger strike
- Kheda Satyagraha (1918)–1st Non-Cooperation

Jallianwala Bagh Massacre (13th April, 1919)

- The dissatisfaction against the Rowlatt Act led to mass agitations.
- A large, but unarmed crowd had gathered on 13th April, 1919 at **Amritsar** (Punjab) in the **Jallianwala Bagh**, to protest against the arrest of their popular leaders Dr Saif-ud-din Kichlu and Dr Satyapal.
- **General O' Dyer**, the military commander of Amritsar, with his troops. He surrounded the bagh, closed the exit and then ordered his men to shoot into the crowd. Thousands were killed and wounded.
- Rabindranath Tagore renounced his knighthood in protest.

Khilafat Movement (1920-1922)

- During the First World War, **Turkey** was allied with Germany and Austria against the British.
- The Indian Muslims regarded the Sultan of Turkey as their spiritual leader, **Khalifa**.
- After the war, the British removed the Khalifa from power and fragmented Turkey. Hence, the Muslims started the **Khilafat Movement** in India for the restoration of Khalifa's position.

Non-Cooperation Movement (1920-1922)

- It was the first mass-based political movement under Gandhi.
- The decision to not cooperate in the most peaceful manner with the government, and its laws were endorsed at the annual session of the Congress held at **Nagpur,** 1920. The Nagpur Session also made changes in the Constitution of the Congress.
- Gandhiji withdrew the movement due to Chauri-Chaura incident (1921, Gorakhpur)
- In December, 1922, **CR Das** and **Motilal Nehru** formed the Congress– Khilafat Swarajya Party with **CR Das** as President and **Motilal Nehru** as Secretary. It worked within the Congress.
- Advocated that Nationalists should end the boycott of Legislative Council, enter them, obstruct their working according to official plans, expose their weaknesses, transform them into arenas of political struggle and thus use them to arouse public enthusiasm. They were *pro-changers*.
- Sardar Vallabhbhai Patel, Dr Ansari, Babu Rajendra Prasad and others opposed Council Entry.

Simon Commission (1927)

- In 1927, the British Government appointed the **Indian Statutory Commission,** known popularly by the name of its Chairman Simon to go into the question of further Constitutional Reforms.
- Indians opposed it because all the members of the committee were white.
- On its arrival in India on 3rd February, 1928, the Commission was greeted with hartals and the black flag demonstrations, under the slogan **Simon Go Back**.

Nehru Report (1928)

- All important Indian leaders and parties tried to meet the challenge of the Simon Commission by getting together, and trying to evolve an alternative scheme of constitutional reforms. *Nehru report* was tabled in 1928 (by **Motilal Nehru**).

Lahore Session (1929)

- This session was presided over by Jawaharlal Nehru.
- Gandhi came back to active politics by that time.
- This session passed a resolution of **Poorna Swaraj** (Complete Independence) as its ultimate goal.
- On 31st December, 1929 the newly adopted tricolour, was hoisted and 26th January, 1930 was fixed as the **First Independence Day**.

Civil Disobedience Movement (1930)

- The **Civil Disobedience Movement** was started by Gandhi on 12th March, 1930, with his famous **Dandi March**.
- Together with 78 chosen followers, Gandhi walked nearly 375 kms from **Sabarmati Ashram** to Dandi, a village on the Gujarat sea-coast.

First Round Table Conference (1930)

- Congress boycotted the Conference.
- Muslim League was represented by Muhammad Ali, Agha Khan, Fazlul Haq, MA Jinnah and Hindu Mahasabha, by Moonje and Jayakar.
- Tejbahadur Sapru, Chintamani and Srinivas Shastri (Liberals) appeared.

Gandhi-Irwin Pact

- Under this pact, the Government agreed to release all those political prisoners, who had remained non-violent.
- Right to make salt for consumption.
- The Congress to suspend Civil Disobedience Movement and take part in Second Round Table Conference.

Karachi Session (1931)

It endorsed the **Gandhi–Irwin Pact**. This session is also memorable for its resolution on Fundamental Right and National Economic Programme, with the efforts of Jawaharlal Nehru and Subhash Chandra Bose.

Second Round Table Conference (1931)

Gandhiji went to England in September 1931, to attend the Second Round Table Conference. But, the British Government refused to concede the basic nationalist demand for freedom on the basis of the immediate grant of Dominion Status with complete control over defence, external affairs and finance.

Poona Pact (1932)

- McDonald announced the proposal on minority representation, known as the **Communal Award** in **1932**. Under this, the Depressed Classes (Muslims, Sikhs, Indian Christians, Anglo-Indians, Women and Backward Classes) were to be considered as a minority and the award would make them entitled to the right of separate electorate.
- Gandhi reacted strongly to the proposal. He considered the depressed class as an integral part of Hindu society.

Impact of Civil Disobedience Movement

- The Congress swept polls in most provinces in 1937.
- The left parties emerged as an alternative in politics.
- Some Congress activists formed **the Socialist group**. Nehru and Subhash Bose emerged as the leaders.

Third Round Table Conference

- Held in London in 1932.
- The Congress did not participate.
- The discussion led to the **Government of India Act, 1935.**

The Second World War and the National Movement

- Lord Linlithgow declared India to be at war without the prior assent of the **Central Legislature**.
- Congress agreed to support Britain only in return of independence being granted. The Viceroy could promise this only after the war. In October-November 1939, the Congress Ministries resigned in protest. The Muslim League observed this as the **Deliverence Day** (22nd December, 1939). The Satyagraha was kept limited, so as not to embarass Britain's war effort by a mass upheaval in India.

August Offer (1940)

The **Viceroy Linlithgow** *put forward a proposal that included*

- Dominion status in the unspecified future.
- A post-war body to enact Constitution.
- Expansion of Governor-General's Council with representation of the minorities.
- Congress rejected the proposal but the League accepted it.

Individual Satyagraha

- launched by Gandhiji in September 1940.
- Vinobha Bhave was 1st Individual Satyagrahi. JL Nehru was second.

Cripps Mission (1942)

- A mission under Stafford cripps clame to India in march 1942. He proposed a dominion status for India.
- The British Government's refusal of accepting immediately the Congress demand of forming a National Government, was the cause of failure of the mission.

Quit India Movement (1942)

- The All India Congress Committee met at **Bombay** on 8th August, 1942. It passed the famous **Quit India Resolution**, and proposed to start off a non-violent mass struggle under Gandhi's leadership.
- The Movement is also called the **Wardha Proposal** and a leaderless revolt.
- Gandhi told the British to quit and leave India in God's hand. His message was **Do or Die**. The merchant community and capitalists did not participate. The Muslim League kept aloof, and the Hindu Mahasabha condemned the movement. The Communist Party did not support the movement. C Rajagopalachari also did not participate in this movement.

Demand for Pakistan

- In 1930, Muhammad Iqbal, for the first time, suggested that the Frontier Province, Sind, Baluchistan and Kashmir be made the Muslim state within the Federation.
- The fear of Muslims to be subjugated by Hindus in free India was realised by Jinnah, and he demanded for the creation of Pakistan.
- In December 1943, the Karachi Session of the Muslim League adopted the slogan **Divide and Quit**.

Shimla Conference or the Wavell Plan (1945)

- Proposed by **Lord Wavell,** suggested to set-up a new Executive Council with only Indian members. The Viceroy and Commander-in-Chief would be the only non-Indian members of the Council.
- It would work under the existing Constitution. But, the door was open for discussion of new Constitution.
- Both Hindu and Muslims would have an equal representation.
- The Congress, headed by **Maulana Azad**, resented being characterised as a caste Hindu organisation.
- Jinnah demanded the Muslim League to have an absolute choice in choosing Muslim members, so rejected the plan.

The Cabinet Mission (1946)

- The **Attlee Government** announced in February 1946, the decision to send a high powered mission of three British Cabinet members (**Pathick Lawrence**, Secretary of State for India, **Stafford Cripps,** President of the Board of Trade and **AV Alexander**, first lord of Admirality).

- The Congress demanded that power should be transferred to one Centre, and that minorities demands should be worked out only after India is declared independent.

Failure of Cabinet Mission

The plan failed on the issue of the nature of grouping. The Congress wanted the grouping to be optional, till the formation of Constituent Assembly, but Jinnah was in the favour of compulsory grouping.

Interim Government (1946)

- Interim Government headed by **JL Nehru** was sworn on in 2nd September, 1946. The Muslim League refused to join initially. Wavell persuaded the League to join in October, 1946. **Liaquat Ali Khan** of the Muslim League was made the Finance Minister.
- The **Constituent Assembly** began its session in 9th December, 1946 and **Dr Rajendra Prasad** was elected its President, but the League did not attend.

Mountbatten Plan (1947)

- As Viceroy, Mountbatten proved more decisive and quick in taking decisions than his predecessors.
- His task was to explore the option of unity or division till October, 1947 and then advise the British Government on the form of transfer of power.
- **3rd June Plan** In case of Partition, two dominions and two Constituent Assemblies would be created.
- The plan declared that power would be handed over by 15th August, 1947.

Indian Independence Act (1947)

- On 18th July, 1947, British Parliament ratified the Mountbatten plan as the **Independence of India Act, 1947**.
- The act provided for the creation of two independent dominions of India and Pakistan.
- Each dominion was to have a Governor–General, to be responsible for effective operation of the act.
- Sovereignty of British power was to be abolished.

Indian Independence

- On 15th August, 1947, India got independence. India requested Mountbatten to continue as the Governor-General of India. Jinnah became the first Governor-General of Pakistan.

Indian National Congress Annual Sessions

Year/Place	*President(s)*	*Detail(s)*
1885 (December 28) Bombay	WC Bonnerjee	72 delegates attended
1886 (December 28) Calcutta	Dadabhai Naoroji	436 delegates
1887 (December 27-28) Madras	Badr-ud-din Tyabji	607 delegates, appeal to Muslim to join, 1st Muslim President
1888 (December 28-29) Allahabad	George Yule	1st English President
1889 (December 27-28) Bombay	William Wedderburn	
1890 (December 28-29) Calcutta	Pherozeshah Mehta	
1891 (December 26-27) Nagpur	P Ananda Charlu	
1892 (December 28-29) Allahabad	WC Bannerjee	
1893 (December 28-29) Lahore	Dadabhai Naoroji	
1894 (December 27-28) Madras	Alfred Webb	
1895 (December 28-29) Poona	Surendranath Banerjee	
1896 (December 27-28) Calcutta	Rahimatullah M Sayani	
1897 (December 22-29) Amravati	C Sankaran Nair	
1898 (December 27-28) Madras	AM Bose	

Year/Place	*President(s)*	*Detail(s)*
1899 (December 27-28) Lucknow	Ramesh Chandra Dutt	Demand for Permanent fixation of land revenue
1900 (December 27-29) Lahore	NG Chandravarkar	
1901 (December 27-28) Calcutta	DE Wacha	
1902 (December 23-26) Ahmedabad	Surendranath Banerjee	
1903 (December 28-30) Madras	Lal Mohan Ghose	
1904 (December 26-28) Bombay	Henry Cotton	
1905 (December 27-30) Banaras	GK Gokhale	Resentment against Partition of Bengal
1906 (December 26-29) Calcutta	Dadabhai Naoroji	Word 'Swaraj' was used for the first time by the President.
1907 (December 26-27) Surat	Rash Behari Ghosh	Split in Congress into moderates and extremists
1908 (December 29-30) Surat	Rash Behari Ghosh	Constitution of Congress drawn
1909 (December 27-30) Lahore	Madan Mohan Malaviya	Disapproval of Separate Electorate
1910 (December 28-29) Allahabad	W Wedderburn	
1911 (December 26-28) Calcutta	BN Dar	
1912 (December 27-28) Bankipur	RN Mudholkar	
1913 (December 26-29) Karachi	Syed Mohammed	
1914 (December 28-30) Madras	Bhupendra Nath Basu	
1915 (December 27-30) Bombay	SP Sinha	
1916 (December 26-30) Lucknow	AC Mazumdar	Reunion of Congress and Lucknow Pact with Muslim League
1917 (December 28-29) Calcutta	Annie Besant	1st Women President
1918 (December 26-31) Delhi	Madan Mohan Malaviya	
1919 (December 27-28) Amritsar	Motilal Nehru	Condemned the Jallianwala Bagh Massacre boosted Khilafat Movement
1920 (December 26-31) Nagpur	C Vijayaraghavachariar	New Constitution of Congress formed
1921 (December 27-31) Ahmedabad	CR Das (imprison) Hakim Ajmal Khan (Acting President)	
1922 (December 26-31) Gaya	CR Das	Swaraj Party formed
1923 (December 28-31) Delhi	Maulana Abul Kalam Azad	Special Session
1924 (December 26-27) Belgaum	MK Gandhi	
1925 (December 26-28) Kanpur	Sarojini Naidu	First Indian Women President.
1926 (December 26-28) Guwahati	SS Iyengar	
1927 (December 26-27) Madras	MA Ansari	Independence Resolution adopted and to boycott the Simon Commission
1928 (December 28-31) Calcutta	Motilal Nehru	First All India Youth Congress formed
1929 (December 29-31) Lahore	Jawaharlal Nehru	Passed Poorna Swaraj Resolution and launch of Civil Disobedience Movement accepted

Year/Place	*President(s)*	*Detail(s)*
1931 (March 29) Karachi	Vallabhbhai Patel	Endorsement of Gandhi Irwin Pact and Resolution of Fundamental Right and National Economic Policy.
1932 (April 24) Delhi	AR Setri	
1933 (April 7) Calcutta	N Sengupta	
1934 (October 26-28) Bombay	Rajendra Prasad	
1936 (April 12) Lucknow	JL Nehru	Congress to adopt Socialism as its goal
1937 (December 27-28) Faizpur	JL Nehru	Session held in village for the first time
1938 (February 19-21) Haripura	SC Bose	National Planning Commission set-up
1939 (March 10) Tripuri	SC Bose	Rajendra Prasad became the President after resignation of SC Bose
1940 (March) Ramgarh	Maulana Azad	
1946 (November 22) Meerut	JB Kripalani	
1948 (December 18-19) Jaipur	P Sitaramayya	
No Session	1930, 1935, 1941-1945	

Important Years of India's Freedom Struggle

Year	*Event(s)*
1905	Partition of Bengal announced to come in force from 16th October, 1906
1906	Muslim League founded at Dacca
1911	Delhi becomes the new capital of India
1912	Bomb thrown on Lord Hardinge on his state entry into Delhi
1914	Ghadar Party formed at San Francisco
1914	Outbreak of the First World War
1915	Gandhiji arrives in India
1917	Mahatma Gandhi launches the Champaran campaign in Bihar to focus attention on the grievances of indigo planters.
1918	Beginning of Trade Union Movement in India
1918	Rowlatt (Sedition) Committee submits its report. **Rowlatt Bill** introduced on 16th February, 1919.
1919	Jallianwala Bagh tragedy
1920	First meeting of the All India Trade Union Congress (under Narain Malhar Joshi)

Year	*Event(s)*
1920	The Indian National Congress (INC) adopts the Non-Cooperation Resolution
1920-1922	Non-Cooperation Movement, suspended on 12th February, 1922, after the violent incidents at Chauri Chaura on 5th February, 1922
1923	Swaraj Party formed by Motilal Nehru and others
1924	The Communist Party of India (CPI) starts its activities at Kanpur
1925	Kakori Train Conspiracy case
1928	The Nehru Report recommends principles for the New Constitution of India. All Parties Conference considers the Nehru Report, 28th-31th August, 1928.
1929	**Sharda Act** passed prohibiting marriage of girls below 14 and boys below 18 years of age, with effect from 1930
1929	All Parties Muslim Conference formulates the '**Fourteen Points**' under the leadership of Jinnah
1929	Bhagat Singh and Batukeshwar Dutt throw bomb in the Central Legislative Assembly

Year	*Event(s)*
1929	Lord Irwin's announcement that the goal of British policy in India was the grant of the Dominion Status
1929	The Lahore Session of the INC adopts the goal of Complete Independence–Poorna Swaraj for India; Jawaharlal Nehru hoists the tricolour of Indian Independence on the banks of the Ravi at Lahore
1930	First Independence Day observed
1930	Mahatma Gandhi launches the Civil Disobedience Movement with his epic Dandi March (12th March to 6th April). First phase of the Civil Disobedience Movement 12th March, 1930 to 5th March, 1931
1931	Bhagat Singh, Sukh Dev and Rajguru executed
1932	Gandhiji arrested and imprisoned without trial
1932	British Prime Minister Ramsay McDonald announces the infamous '***Communal Award***'.
1934	Gandhiji withdraws from Active Politics and devotes himself to 'constructive programmes' (1934-1939)
1935	The Government of India Act (1935) passed
1937	Elections held in India under the Act of 1935 (February, 1937). The INC contests election and forms ministries in several provinces (July, 1937).
1938	Haripura Session of INC. Subhash Chandra Bose elected Congress President
1939	Second World War (1st September). Great Britain declares war on Germany; the Viceroy declares that India too is at war
1939	The Congress ministries in the provinces resign in protest against the war policy of the British Government
1939	The Muslim League observes the resignation of the Congress ministries as 'Deliverance Day'
1940	Congress launches Individual Satyagraha Movement
1941	Subhash Chandra Bose escapes from India; arrives in Berlin (28th March)
1942	Churchill announces the Cripps Mission
1942	The INC meets in Bombay; adopts the 'Quit India' Resolution.
1942	Quit India Movement begins; the Great August Uprising
1942	Subhash Chandra Bose establishes the Indian National Army ('Azad Hind Fauj')
1943	Karachi session of the Muslim League adopts the slogan **Divide and Quit**
1946	Jawaharlal Nehru takes over as the Congress President
1946	Wavell invites Nehru to form an Interim Government; Interim Government takes office (2nd September)
1946	First session of the Constituent Assembly of India starts
1947	British Prime Minister Attlee declares that the British Government would leave India not later than June 1948
1947	Lord Mountbatten, the last British Viceroy and Governor-General of India, sworns in (24th March, 1947 to 21st June, 1948)
1947	Mountbatten plan for the partition of India and the announcement (4th June) that transfer to power would take place on 15th August.
1947	India wins freedom

ART AND CULTURE

RELIGION

Religion is a collection of belief systems, cultural systems and world views that relate humanity to spirituality and sometimes to moral values. Religion is different from private belief in its social character.

Hinduism

- It consists of a collection of intellectual and philosophical points of view rather than a rigid system of beliefs.
- There is no specific founder of the religion.
- Its roots can be traced to the historical Vedic religion of the Iron age India.
- Hinduism is the third largest religion in the world after Christianity and Islam.
- Prominent texts are Vedas, Ramayana and Mahabharata.

Islam

- Prophet Mohammed is the founder of Islam.
- Quran is the holy book of Islam.
- Islam is the second largest religion in the world.
- Most Muslims belong to two denominations–Shia and Sunni.
- Islam arrived in India in the AD 8th century.

Christianity

- Jesus is the founder of Christianity.
- Bible is the holy book of Christianity.
- Christianity is mainly divided in Catholic and Protestant.
- Christianity is the world's largest religion.
- Holy texts of the Christianity are the Old Testament and the New Testament.

Sikhism

- Guru Nanak is the founder of Sikhism.
- **Guru Granth Sahib** is the holy book of Sikhism.
- Sikhism is the third largest religion of India.
- Sikhism developed in India during 16th and 17th centuries.

Buddhism

- Mahatma Buddha is the founder of Buddhism.
- Buddhism is the indigenous religion of India.
- Holy book of Buddhism is **Tripitak**.
- The three Jewels of Buddhism are Buddha, Dhamma and Sangha.
- Buddhists generally classify themselves as either Theravada Mahayana.

Jainism

- **Mahavira Swami** is the founder of Jainism.
- Jainism is a religion indigenous to the Indian subcontinent.
- Holy book of Jainism is Kalp Sutras.
- The main doctrines of Jainism are Anekantavada, Syadvada and Nayavada.
- The two sects of Jainism are Svetambara and Digambara.
- Jain holy texts consist of Purvas, Agamas, Angas and Upangas.

LANGUAGES

Sanskrit

- Sanskrit is the mother of many Indian languages.
- The Vedas, Upanishads, Puranas and Dharmasutras are all written in Sanskrit.
- It is one of the twenty-two languages listed in the Indian Constitution.
- The Dharmasutras, the Manusmriti, Arthashastra and Gita Govinda are the famous books in Sanskrit.
- Panini, Kautilya, Kalhana and Jayadeva are the famous writers of Sanskrit.

Pali

Pali is an Indo-Aryan language, which was used for the earliest Buddhist scriptures. Pali literature is mainly concerned with Theravada Buddhism.

Telugu

- The Vijyanagara period was the golden age of Telugu literature.
- Eight Telugu literary luminaries popularly known as Ashtadiggajas.
- Ramakrishna was the author of Panduranga Mahatmayam, which was considered one of the greatest poetical works of Telugu literature.

Kannada

- Kannada language developed fully after the AD 10th century. The earliest known literary work in Kannada is Kavirajamarg written by the Rashtrakuta King Nripatunga Amoghavargha I.
- Pampa, known as the father of Kannada, wrote his great poetic works Adi Purana and Vikramarjuna Vijaya in the AD 10th century.

Malayalam

- The language of Malayalam emerged around the AD 11th century. By 15th century, Malayalam was recognised as an independent language.
- Bhasa Kautilya, a commentary on Arthashastra and Kokashndisam are two great works.
- Rama Panikkar and Ramanujan Ezhuthachan are well-known authors of Malayalam literature.

Tamil

- Tamil is the mother language of the Dravidian language family.
- The Sangam literature is a collection of long and short poems composed by various poets in praise of numerous heroes and heroines.
- There are about 30000 lines of poetry, which are arranged in eight anthologies called Ettuttokai.

Persian and Urdu

- Urdu emerged as an independent language towards the end of the AD 4th century.
- Urdu as a language was born out of the interaction between Hindi and Persian.
- Urdu became more popular in the early 18th century.
- The earliest Urdu poet is supposed to be Khusrau. Urdu has given us a new form of poem that is called a nazm.

Hindi

- Hindi is a direct descendant of the Sanskrit language through Prakrit and Apabhramsha.
- It is spoken largely in North India.
- Evolution of Hindi literature can be better understand through four stages of Adi Kal, Bhakti Kal, Riti-kavya Kal and Adhunik Kal.

LITERATURE

Bengali Literature

- Raja Ram Mohan Roy wrote in Bengali besides English, which gave an impetus to Bengali literature.
- Ishwar Chandra Vidhyasagar (1820-91) and Akshay Kumar Dutta (1820-86) were two other writers of this early period.
- Sharat Chandra Chatterji (1876-1938) and RC Dutta, a noted historian and a prose writer, all contributed to the making of Bengali literature.
- But the most important name that influenced the whole of India was that of Rabindra Nath Tagore (1861-1941). Novels, dramas, short stories, criticism, music and essays, all flowed from his pen. He won the Nobel Prize for literature in 1913 for his Geetanjali.

Assamese Literature

- Like Bengali, Assamese also developed in response to the Bhakti Movement.
- Shankardeva, who introduced Vaishnavism in Assam helped in the growth of Assamese poetry. Even the Puranas were translated in Assamese.
- The earliest Assamese literature consisted of buranjis (court chronicles). Shankardev has left several devotional poems, which people sang with rapturous pleasure, but it was only after 1827 that more interest was shown in producing Assamese literature.

Oriya Literature

- Oriya language shows the maximum influence of Sanskrit. It originated in the AD 9th century and its literary development took place in the 13th century.
- Worth mentioning in Oriya literature are Fakirmohan Senapati and Radha Nath Ray, whose writings deserve considerable attention in the history of Oriya literature.
- The works of Upendra Bhanja (1670-1720) were important as they ushered a new period of Oriya literature. In Odisha, the works of Saraladasa are regarded as the first works of Oriya literature.

Punjabi Literature

- Punjabi is a language with several shades. It is being written in two scripts, Gurumukhi and Persian.
- Guru Nanak was the first poet in Punjabi language. Some other contemporary poets, mostly Sufi saints, used to write in this language.
- Guru Gobind Singh, the 10th guru, was educated in Patna (Bihar), where he learnt Persian and Sanskrit.
- He has composed two savaiyyas in Punjabi, but these are not a part of the Adi Granth. Similar is the popularity of Bulley Shah, who was a Sufi saint. He has composed a large number of songs.

Indian Literature in English Language

- English came into India with the British and soon became a language of formal education. Some of the early Indian works in English were written by Raja Rammohan Roy, Henry Vivian Derozio and Madhusudan Dutt. Aurobindo Ghosh wrote his epic poem 'Savitri: A Legend and a Symbol' in English.
- India's only Nobel laureate in literature Rabindranath Tagore wrote some of his work originally in English and did some of his own English translation from Bengali. Sarojini Naidu and Jawaharlal Nehru also wrote in English. Nehru's 'The Discovery of India' and 'Glimpses of World History' are quite popular.
- Some notable Indian poets, who write in English are Nissim Ezekiel, Dom Moraes, Arun Kolakar and Dilip Chitre. Other notable Indian writers are Khushwant Singh, Salman Rushdie, Vikram Seth, Arundhati Roy, Kamala Markandaya, Kiran Desai, Jhumpa Lahiri etc.

MUSIC

The music of India is said to be one of the oldest unbroken musical traditions in the world.

It is said that the origin of this system go back to the Vedas (ancient scripts of the Hindus). Many different legends have grown up concerning the origin and development of Indian classical music.

Hindustani Music

Classical Music

- Hindustani classical music originated in North India around 13th and 14th centuries. In contrast to Carnatic music, the Hindustani classical music was not only influenced by ancient Hindu musical traditions and Vedic philosophy, but also by the Persian elements.
- Hindustani music is based on the Raga system. The Raga is a melodic scale, comprising of notes from the basic seven – *Sa, Re, Ga, Ma, Pa, Dha* and *Ni*.
- On the basis of notes included in it, each raga attains a different character. The form of the raga is also determined by the particular pattern of ascent and descent of the notes, which may not be strictly linear.
- Hindustani classical music is primarily vocal-centric. The major vocal forms associated with Hindustani classical music are the khayal, ghazal, dhrupad, Tappa, tarana and thumri.

Thumri

It is an informal vocal form of Hindustani classical music and is said to have begun with the court of Nawab Wajid Ali Shah, the Nawab of Oudh.

Khayal

It is a form of vocal music adopted from Medieval Persian music. It is based on imagination and improvisations of the performer.

Dhrupad

It is the oldest form of Hindustani music, traditionally performed by male singers. It is mostly a poetic form. Tansen Sang in Dhrupad style.

Tappa

It is developed in 18th century from the folk songs of camel riders of Punjab. They are essentially folklore of love and passion and are written in Punjabi. Developed as a form of classical music by Mian Ghulam Nabi Shori.

Tarana

It is a medium to fast paced song usually performed towards the end of the concert. It consists of a few lines of poetry with rhythmic syllabies.

Ghazal

It is a poetic form consisting of rhyming couplets on love and devotion.

It is an ancient form originating in 6th century Arabic poetry. It spread into South Asia in the 12th century, due to the influence of Sufi mystics.

Musical Instruments and Instrumentalists

Instrument	*Instrumentalists*
Stringed Instruments	
Rudra Veena	Asad Ali Khan, Zia Moin-ud-din Dagar
Santoor	Shiv Kumar Sharma
Sarod	Buddhadev Dasgupta, Ali Akbar Khan, Amjad Ali Khan, Bahadur Khan, Sharan Rani, Zarin S Sharma
Sarangi	Ustad Binda Khan
Sitar	Ravi Shankar, Hara Shankar Bhattacharya, Nikhil Banerjee, Vilayat Khan, Mustaq Ali Khan
Surb Ahar	Sajjad Hussain, Annapurna
Veena	Doraiswamy Iyengar, Chittibabu, Emani Sankara Shastri, Dhanammal, S Bala Chandran, KR Kumaraswamy
Violin	Gajanan Rao Joshi, MS Gopal Krishnan, TN Krishnan, Baluswamy Dikshitar, Dwaran Venkataswamy Naidu Lalyuli G Jayaraman, Mysore T Chowdiah, VG Jog
Wind Instruments	
Flute	TR Mahalingam, N Ramani, Hari Prasad Chaurasia, Pannalal Ghosh
Nadaswaran	Sheikh Chinna Moula, Neeruswamy Pillai, Rajaratanam Pillai
Shehnai	Bismillah Khan
Percussion (Strikting Thumping) **Instruments**	
Mridangam	Palghat Mani Iyer, Karaikudi R Mani, Palghat Raghu
Pakhawag	Pt Ayodhya Prasad, Gopal Das, Babu Ram Shankar Pagaldas
Tabla	Zakir Hussain, Nikhil Ghosh, Kishan Maharaj, Alla Rakha Khan, Pandit Samta Prasad, Kumar Bose, Latif Khan
Kanjira	Pudukkotai Dakshinamurthi Pillai

Instruments Vocalists

Carnatic	MS Subbulakshmi, Balamuralikrishna, Bombay Jaishri Ramnath, HK Raghavendra, Aryakudi Ramanujan Iyengar Venkataram, Sitarajam, Mani Krishnaswamy, Akhil Krishnan, ML Vasanthakumari, MD Ramanathan, GN Balasubramaniam
Dhrupad	Ustad Rahim Fahim-ud-din Dagar, Zahir-ud-din Dagar, Wasif-ud-din Dagar, Bundecha Bandhu, Pt Abhay Narayan Mallick, Pt Ritwik Sanyal, Uday Bhawalkar
Hindustani	Shubha Mudgal, Madhup Mudgal, Mukul Shivputra, Pandit Jasraj, Parveen Sultana, Naina Devi, Girija Devi, Ustad Ghulam Mustafa Khan, Gangubai Hangal, Krishna Hangal, V Rajput, Kumar Gandharva, Faiyyaz Khan, Mallikariun Mansur, Bheemsen Joshi.
Thumri	Ustad Bade Ghulam Ali Khan, Ustad Mazhar Ali Khan, Ustad Zawad Ali Khan, Poornima Chaudhary, Shanti Heerananda, Naina Devi, Rita Ganguly
Qawwali	Ghulam Hasan Niyazi, Sultan Niyazi, Ghulam Farid Nizami, Iqbal Hussain Khan Bandanawaji, Aslam Sabaari, Chand Nizami

DANCE

Indian Classical Dances

India has thousands of year old tradition of fine arts and classical and folk music and dances. Some of the world famous dance forms that originated and evolved in India are Bharatnatyam, Kathak, Kathakali, Kuchipudi, Manipuri, Mohiniattam, Odissi and Sattriya.

Bharatnatyam

- Bharatnatyam is a classical Indian dances of Tamil Nadu.
- The music accompanying this dance is carnatic music. It evolved out of the Devadasi system of South Indian temples. Two famous styles are Pandanallur and Tanjore.
- Famous dancers associated with Bharatnatyam are E Krishna Iyer, Rukmini Devi Arundale and Anna Pavlova (Russian).

Kathak

- Kathak is a classical dances of North India. Kathak is said to be derived from the word katha, meaning the art of storytelling.
- Famous centres are Lucknow and Jaipur. Lucknow school depicts Mughal Court etiquette, while the Jaipur school depicts stories of Rajput kings and Gods.
- Famous exponents are Sitara Devi, Sambhu Maharaj, Uma Sharma Shovana Narayan etc.

Kathakali

- Kathakali is the classical dance form of Kerala. The word Kathakali literally means **story play.**
- Kathakali is considered as one of the most magnificent theatres of imagination and creativity.
- Famous exponents of Kathakali are Vallathol Narayan Menon, Kunju Kurup, Guru Gopinath etc.

Kuchipudi

- Kuchipudi is one of the classical dance forms of the South India and had its origin in Andhra Pradesh.
- Kuchipudi exhibits scenes from the Hindu epics, legends and mythological tales.
- Famous exponents of Kuchipudi are Lakshmi Narayan Shastri, Raja and Radha Reddy, Swapana Sundari and Yamini Krishnamurti.

Manipuri

- Manipuri is one of the major classical dances of India.
- The most striking part of Manipuri dance is its colourful decoration, lightness of dancing foot, delicacy of abhinaya (drama), lilting music and poetic charm.
- Popular exponents are Javeri sisters, Rita Devi, Nirmala Mehta, Guru Bipin Singh etc.

Mohiniattam

- Mohiniattam from Kerala, is a solo female dance and is known for its rhythmic and unbroken flow of the body movements.
- Mohiniattam has the grace and elegance of Bharatanatyam and vigour of Kathakali.
- Famous exponents of this dance form are Kalyani Amma, Vaijayanthimala, Bharati Shivaji and Hema Malini.

Odissi

- Odissi is one of the famous classical Indian dances from Odisha state.
- It is a graceful and sensous dance style and involves the tribhanga (three bends) posture.
- The (three bends) symbolise the means to escape the limitations of the body.
- Famous **dancers** of Odissi are Indrani Rehman, Sonal Mansingh, Kiran Sengal, Rani Karna, Sharon Lowen and Myrta Barvie.

Sattriya

- Sattriya is the classical dance form from Assam. It was created by Vaishnav Saint Sremanta Sankaradeva in 15th century.
- It is traditionally performed by bhokos (male Monks) in monasteries, but now by female also. The dance is based on mythological themes.
- It is performed on Assamese music called Borgeet and instruments used are khol (Drum, Talas Cymbals) and Flute.

Folk Dances and Tribal Dances in India

State	*Dances*
Maharashtra	Kathakeertan, Lezin, Dandaniya, Tamasha, Gafa, Dahikala, Lavani, Mauni, Dasavtar
Karnataka	Huttari, Suggi Kunitha, Yakashagana
Kerala	Kaikottikali, Kaliyattam, Tappatikkali
Tamil Nadu	Kolattam, Pinnal Kolattam, Kummi, Kavadi, Karagam
Andhra Pradesh	Ghanta Mardala, Veedhi Natakam, Burrakatha
Odisha	Ghumara Sanchar, Chadya Dandanata, Chhau
West Bengal	Kathi, Chhau, Baul, Kirtan, Jatra, Lama
Assam	Bihu, Khel, Gopal, Raslila, Tabal Chongli, Canoe
Punjab	Giddha (women), Bhangra (men)
Jammu and Kashmir	Rauf, Hikat
Himachal Pradesh	Jhora, Jhali, Dangli, Mahasu, Jadda, Jhainta, Chharhi
Haryana	Jhumar, Raslila, Phag dance, Daph, Dhamal, Loor, Gugga, Khoria, Gagor
Gujarat	Garba, Dandiya Rass, Tippani, Gomph
Rajasthan	Ginad, Chakri, Gangore, Teratali, Khayal, Jhulan Leela, Jhuma, Suisini
Bihar	Jata Jatin, Jadur, Chhau, Kathaputli, Bakho, Jhijhiya, Samochakwa, Karma, Jatra, Natna
Uttar Pradesh	Nautanki, Thora, Chappeli, Raslila, Kajri
Madhya Pradesh	Karma
Meghalaya	Laho
Goa	Mando
Mizoram	Khantumm
Uttarakhand	Chdiya, Pandav Nritya, Jagars, Jhora

Classical Dancers of India

Dance	*Dancers*
Bharatnatyam	Bala Saraswati, CV Chandrasekhar, Leela Samson, Mrinalini Sarabhai, Padma Subramanyam, Rukmini Devi, Sanyukta Panigrahi, Sonal Mansingh, Yamini Krishnamurti
Kathak	Bharti Gupta, Birju Maharaj, Damayanti Joshi, Durga Das, Gopi Krishna, Kumudini Lakhia, Sambhu Maharaj, Sitara Devi
Kuchipudi	Josyula Seetharamaiah, Vempathi Chinna Sathyam
Manipuri	Guru Bipin Sinha, Jhaveri Sisters, Nayana Jhaveri, Nirmala Mehta, Savita Mehta
Odissi	Debaprasad Das, Dhirendra Nath Patnaik, Indrani Rahman, Kelucharan Mahapatra, Priyambada Mohanty
Kathakali	Mrinalini Sarabhai, Guru Shankaran, Namboodripad, Thottam Shankaran, Kutti Nayyar, Shankar Kurup, KC Pannikar, TT Ram Kulti Nayyar, etc.
Mohiniattam	Protima Devi, Sanyukta Panigrahi, Sonal Mansingh, Pankaj Charan Das, Kelucharan Mahapatra, Madhvi Mudgal, etc.
Sattriya	Jatin Goswami, Sharodi Saikia, Indira Bora, Maniram Datta, Anita Sharma, Pradip Chaliha etc.

Architecture

Indian architecture, which has evolved through centuries, is the result of socio-economic and geographical conditions of the region. Indian architecture evolved in various ages in different parts and regions of the country. Apart from the natural evolution Indian architecture was generally affected by many great and important historic developments.

Rajput Architecture

- The Rajputs were great patrons of art and architecture, the finest examples being their forts and palaces. The Rajput palaces are complex compositions built as inner citadels surrounded by the city and enclosed by a fortified wall as at Chittorgarh and Jaisalmer. Some forts, such as those at Bharatpur and Deeg, were protected by wide moats.
- The palaces of Jaisalmer, Bikaner, Jodhpur, Udaipur and Kota represent the maturity of the Rajput style.

Deccan Style

- The architecture of the Deccan is marked by its distinct originality and independence of style, unlike the architectural styles of the other provinces, which combined both the temple architecture and the Islamic building ideals.
- It derived its elements from the architectural styles of the Sultan of Delhi and that of the distant Persia.
- Some key features of this style are–presence of bulbous dome with lotus neck, military style of architecture, presence of thick walls and combination of gateway and mosque like Charminar.

Mughal Style

- The medieval period saw great developments in the field of architecture. With the coming of Muslims to India, many new features came to be introduced in buildings.
- The development of Muslim style of architecture of this period can be called the Indo-Islamic architecture or the Indian architecture influenced by Islamic Art.
- The Indo-Islamic style was neither strictly Islamic nor strictly Hindu.
- The earliest building of this period is Quwwat-ul-Islam Mosque and the Qutub Minar at Delhi.

PUPPETRY

- The early puppet shows in India dealt mostly with histories of great kings.
- Princes, heroes and also political satire in rural areas.
- There are 4 types of puppetry in India String puppetry, Shadow puppetry, Rod puppetry and Glove puppetry.

Painting

The history of Indian paintings is just about as old as the history of the people of India. The most primitive instances of paintings in India can be traced back to cave paintings.

Mughal School

- This school has a specific style of South Asian painting. Usually, it was confined to miniatures either as book depictions or as individual works to be kept in albums.
- This practice materialised from Persian miniature painting, with Indian influences of Hindu, Buddhist and Jain.
- It wonderfully blossomed during the Mughal Empire. Later, this school of painting reached other Indian courts of Muslims and Hindus and afterwards Sikhs.
- Akbar and Jahangir were exceptionally great patrons of this painting.
- Mughal School of painting revolved around court scenes, portraits, hunting scenes, lovers, battle fronts etc.
- The Mughals introduced two new colours peacock blue and Indian red.

Rajput School

- This school progressed and thrived during the 18th century in the majestic Rajputana courts.
- This school of painting flooded from the approach of Mughal painting.
- A typical style of painting with particular common characteristics came up in every Rajput realm.
- This school illustrated an assortment of themes like landscapes, events from Mahabharata, Ramayana, Krishna's life and human beings.

Bengal School

- This school was a fashion of art, which progressed during the British Raj in India of early 20th century.
- Indian Nationalism was greatly linked to this painting. British arts administrators also supported and promoted it.
- Raja Ravi Verma and Abanindranath Tagore were amongst the pioneer artists of this school. Today's renowned artists, who belong to this School include Nilima Dutta, Sudip Roy, Paresh Maiti and Bikash Bhattacharjee.

Cultural Heritage

Site	*State*
Mahabodhi Temple	Bihar
Red Fort	Delhi
Qutub Minar	Delhi
Humayun's Tomb	Delhi
Champaner Achaelogical Park	Gujarat
Churches of Old Goa	Goa
Jantar Mantar	Jaipur
Hampi	Karnataka
Pattadakal	Karnataka
Chhatrapati Shivaji Terminus	Mumbai, Maharashtra
Ajanta Caves	Maharashtra
Ellora Caves	Maharashtra
Sanchi Stupa	Madhya Pradesh
Khajuraho	Madhya Pradesh
Rock Shelters of Bhimbetka	Madhya Pradesh
Elephanta Caves	Maharashtra
Konark Sun Temple	Odisha
Mahabalipuram	Tamil Nadu
Great Living Chola Temples	Tamil Nadu
Agra Fort	Uttar Pradesh
Taj Mahal	Uttar Pradesh
Fatehpur Sikri	Uttar Pradesh
Mountain Railway of India	West Bengal
Nalanda Mahavira	Bihar
Hill forts of Rajasthan	Rajasthan
Ahmedabad City	Gujarat
Jaipur City	Rajasthan
Rani ki Vau	Gujarat
Capital complex	Chandigarh
Victorial and Art Deco ensemble of Mumbai	Maharashtra

WORLD HISTORY

MESOPOTAMIAN CIVILISATION

- It is the oldest civilisations of the World. Mesopotamia is the land between the rivers Tigris and Euphrates.
- Mesopotamians were the first to use **potter's wheel**, to make glass ware, to evolve a proper system of writing called **Cuneiform**. Cuneiform script was deciphered by Henry Rawlinson.
- Mesopotamians discovered **Sexagesimal system** of counting (based on sixties), Pythagoras theorem, lunar Calender and calculated the length of day and night.

Chinese Civilisation

- The earliest civilisation was by the Shang (Chou) Dynasty, followed by the Chin and Han Dynasty.
- In 3rd century BC, the ruler of China dynasty built the **Great Wall**.
- Chinese script was pictographic, and their calendar was a combination of Solar-Lunar calendar. **Silk** became the chief item of export during the Hangs.
- The two major religions were **Taoism** and **Confucianism**. They invented water clock, abacus, umbrella, Paper and Seismograph.

Iranian Civilisation

This civilisation developed in around 6th century BC by the Achaemenid Empire under its first ruler, Cyrus. His capital was at Pasaragadae. His successors were Darius I and Darius III. The Achaemenids introduced the use of gold and silver coins. Their main religion was **Zoroastrianism** founded by **Zarathustra** or **Zoroaster**. Their official language was Aramaic.

Greek Civilisation

- The civilisation developed around 800 BC, when the small villages clustered to form city-states.
- They worshipped **Zeus** (Sky God), **Poseidon** (Sea God), **Apollo** (Sun God), and **Athena** (Goddess of Victory) etc.
- In the **Battle of Marathon** (490 BC), Greeks defeated King Darius I. **Alexander** was the greatest Greek ruler.
- The Olympic Games originated in Greece. *Iliad* and *Odyssey* are among the best epics of the world written by Homer.

Roman Civilisation

- Italy was the centre of the civilisation. The city of Rome was founded by Romulus in 1000 BC on the bank of river Tiber.
- The war between Carthage and Rome is known as **Punic War** (264 BC to 146 BC).
- **Julius Caesar**, one of the generals murdered, Pompey, another general, and occupied the throne. He was attached to the Egyptian queen **Cleopatra**. Caesar was succeeded by Octavian and Diocletion.
- Romans worshipped the planets. They developed the Latin language. **Lucretius**, **Seneca**, **Cicero** and **Marus** were the famous Roman philosophers and **Horace** and **Virgil** were the poets. **Tacitus** and **Pliny** were the historians.
- They invented the Concrete, useful for constructing buildings.

Renaissance

- The Renaissance or ''Revival of learning'' started in the AD 14th century in Italy.
- The fall of Constantinople by the Turks in 1453 led to the dispersal of scholars from throughout Europe to Italy. Renaissance led to the revival of classical learning, art and architecture and propelled humanism.
- Famous Renaissance writers include Dante, Petrorch, Boccacciro and Machiavelli.
- Renaissance painters—Leonardo da Vinci (The last Supper and Monalisa), Michelangelo (The last Judgement and the fall of man), Raphael (Madonna).
- The movement also helped in the development of printing press.

Reformation (16th Century)

- This movement was started in Germany by Martin Luther by publicly protesting against the sale of **Letters of Indulgence**.
- It was a revolt against **Roman Catholic Church**. As a result, Western Europe was split between Roman Catholic and Protestant countries.

Glorious Revolution of England (1688)

- This revolution started against the policies of **King James II**. He tried to secure freedom of worship for Catholics. This united the Whigs and Tories of Anglican Church against him and they invited William of Orange to occupy English throne.
- James II threw the great seal into the river Thames and fled to France. The event is known as **Glorious or Bloodless Revolution**.
- It ended the despotic rule of the **Stuarts** and established the supremacy of Parliament. The **Bill of Rights** (1689) was passed that settled down the problem of succession.

Industrial Revolution

- It began in Britain in AD 1750 with the invention of **Spinning Jenny** by Hargreaves, **Water frame** by Richard Arkwright (1769), **Mule** by Samuel Crompton (1779), **Power loom** by Emmund Cartwright (1785) and **Steam engine by** James Watt (1769).
- This fastened the production of cloth and better quality yarn was produced. The economic progress also affected the cultural and social life of the people.

American Revolution

- **George Washington**, the first President of America, was the pioneer of the revolution.
- A fundamental difference of opinion had developed between British authorities and the Americans on the related issues of taxing.
- If taxes were necessary, then Americans wanted their own assemblies to impose them, **No taxation without representation** (1763).
- **Boston Tea Party** (1773) A group of citizens of Boston dumped the crates of tea, loaded on a ship of East India Company into the sea. This was because of the problem of tea tax.
- On 4th July, 1776, the Declaration of Independence was issued, authored by Thomas Jefferson. The war ended with the **Treaty of Paris** in AD 1783.
- Benjamin Franklin established the **American Philosophical Society**. Americans were the first to have a Written Constitution.

French Revolution

- The Revolution initiated on 5th May, 1989 during the kingship of Louis XVI.
- The immediate cause of the revolution was the extravagant expenditure and inefficiency by Louis XV and Louis XVI. Other causes were social inequality, heavy taxation etc.
- **Montesquieu, Voltaire and Rousseau** were the French writers and thinkers of the period.
- **Liberty, Equality and Fraternity** were the watch of the revolution.

Unification of Germany

- This was the result of **the Blood and Iron Policy** of Bismarck, the Prime Minister of King William I (Prussia).
- After the Napoleonic war (1803-15), the 38 independent states were unified under the king of Prussia. From 1815 to 1850, Austria ruled over the German confederation.
- Bismarck defeated Austria and dissolved the German confederation.
- He founded a new confederation of 22 states in 1866. The unification was completed with the **Prussia-France War** (1870), in which the French Emperor **Louis Bonaparte** was defeated.
- William I, the king of Prussia was declared as the Emperor of Germany at Versailles in France.

IMPORTANT WARS OF THE 20th CENTURY

Russo-Japanese War (1904-05)

The conflict arose from the rivalry of Russia and Japan for control of Manchuria and Korea. Russia was forced to surrender Korea, the Liaotung Peninsula and Sakhalin to Japan.

First World War (1914-18)

International conflict began between Austria and Serbia. The chief contestants were the Central Powers (Germany and Austria) and the Triple Entente (Britain, France and Russia). Many other countries joined as the war began.

The naval blockade of Germany caused severe food shortages and helped to end the war. An Armistice was agreed in November 1918 and peace treaties were signed at Versailles (1919).

Second World War (1939-45)

International conflict arising from disputes provoked by the expansionist policies of Germany in Europe and Japan in the far East. The axis powers– Germany, Italy and (after September 1940) Japan–controlled most of Europe, much of Northern Africa, China and Asia.

The United States stayed out of the war until 7th December, 1941, when Japan attacked **Pearl Harbour**, and the United States joined the Allies in fighting not only Japan but the other Axis powers as well. The war ended when the US dropped atomic bombs on **Hiroshima** and **Nagasaki** in Japan. The Allies (led by USA, Britain and USSR) emerged Victorious.

The Korean War (1950-1953)

Conflict between North Korea, supported by China and South Korea supported by UN forces dominated by the USA. Negotiations continued for two years, before a truce was agreed in July 1953.

The Vietnam War (1954-1975)

Conflict between US backed South Vietnam and the Viet Cong, who had the support of communist North Vietnam. It followed to the partition of Vietnam. In 1975, South Vietnam was overrun by North Vietnamese forces, and the country was united under Communist rule.

The Iran-Iraq War (1980-1990)

War began shortly after the Iranian Revolution of 1979. Iraq wanted control over oil-rich Iranian border territory. Iraq expressed readiness to negotiate peace, after it lost ground. In 1988, Iran agreed to a ceasefire. Iraq invaded Kuwait in 1990.

The Gulf War (1991)

Military action by a US led coalition to expel Iraqi forces from Kuwait. Kuwait was liberated (26th February, 1991) and a ceasefire was declared 28th on February.

The Bosnian War (1992-98)

Ethnically rooted war in Bosnia and Herzegovina, a republic of Yugoslavia with a multiethnic population–Muslims, Serbs and Croats. The Dayton Peace Agreement was signed in 1995. **Bosnia Herzegovina** became a single state.

The US-Afghan War (2001)

Military action by US against Afghanistan in protest against the 11th September, attack on World Trade Centre Towers. US claimed that Afghanistan was the breeding ground of terrorists, and had given refuge to Osama bin Laden.

The Second Gulf War (2003)

Military action by a US led coalition to oust Saddam Hussain from power in Iraq. It was conducted on the pretext of Iraq possessing Weapons of Mass Destruction (WMD).



GEOGRAPHY

WORLD GEOGRAPHY

UNIVERSE

- The universe is commonly defined as the totality of everything that exists including all physical matters and energy, the planets, stars, galaxies and the contents of intergalactic space.
- The study of universe is known as Cosmology.

  Cosmology = Cosmos (Universe) + Logos (Study)
- The universe comprises of billions of galaxies. The galaxies are made up of millions of stars held together by the force of gravity and these stars account for most of the masses of the galaxy.
- Our own galaxy is called the **Milky Way** (or the Akash Ganga) and it contains about 300 billion stars and one of these is the Sun. **Andromeda** is our nearest galaxy.
- Latest known galaxy is the **Dwarf Galaxy**.
- Evolution of universe is explained by the Big Bang Theory. Big Bang was an explosion that occurred 13.8 billion years ago, leading to the formation of galaxies of stars and other heavenly bodies.
- Planets and other objects go round the Sun and make up the solar system with the Sun at the centre.

STARS

- Stars are heavenly bodies made up of hot burning gases. They produce their own light.
- A neutron star is about 20 km in diameter and has a mass of about 1.4 times that of the Sun. Neutron stars are formed by supernova explosions. A pulsar is a highly magnetised and rotating neutron star.
- A star's colour indicates the temperature of its surface. Blue colour denotes maximum temperature, white, yellow, orange, red stars have sequentially declining range of temperatures.

Sirius

Brightest star outside solar system is Sirius, also called as **Dog Star**.

Proxima Centauri

Closest star to our solar system is Proxima Centauri (4.2 light years away) followed by Alpha Centauri (4.3 light years away) and Barnard's Star (5.9 light years away).

Concept of Black Hole and Chandrasekhar Limit

- A renowned Indian Physicist Chandrasekhar had predicted an upper limit to the mass of stars, which is called as Chandrasekhar limit. It is 1.44 times the mass of sun.
- Above the upper limit two conditions are possible
  1. Explosion of the star to form neutron stars.
  2. Collapse and compaction of the stars to form Black Holes.
- The black holes are formed due to collapse and compaction under gravity, at the end of the life cycle of a star.

Units of Distance

Unit	*Description*
Light year	It is the distance that light can travel in one year. It is equal to almost 9.46×10^{12} km.
Astronomical Unit	It is the average distance between the Sun and the Earth. 1AU = 150 million km.
Parsec	It is the distance from the Earth to a star that has parallax of 1 arc second. The actual length is about 3.262 light years.

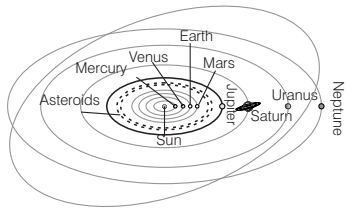
Solar System

- The solar system comprises of the Sun eight planets, their satellites and other non-stellar objects which are believed to have been developed from the condensation of gases and other lesser bodies.
- The Sun is at the centre of the solar system and all the planets revolve around it in an elliptical orbit. It is the nearest star to the Earth.
- The end of solar system is about 122AU (A stronomical units) away from the sun.

Components of Solar System

Components of solar system are

- The Sun, eight planets (excluding Pluto) and their respective satellites.
- Five Dwarf planets (Ceres, Pluto, Eris, Makemake and Haumea).
- Interstellar debris such as asteroids,meteoroids, comets.
- The electrically charged gases, called Plasma.
- Interplanetary dust particles called cosmic dust which pervade between planets in the Solar System.
- The components of solar system other than planets, dwarf planets and satellites are called as Small Solar System Bodies.



Position of Revolution of Sun and Planets

Origin of Earth

Theories/Hypothesis	*Given/Proposed by*
Nebular	Laplace
Gaseous Mass	E Kant
Binary Star	HN Russell
Inter stellar Dust	Otto Schmidt
Big Bang	Hubble

THE SUN

- The Sun accounts for more than 99% of the mass of the solar system and due to this, the Sun exerts immense gravitational pull on the planets to keep them rotating around it in definite elliptical orbit.
- The Sun is the major source of energy of the solar system. The energy is provided by the nuclear fusion reaction, that converts hydrogen into helium in the core of the Sun.
- The glowing surface of the Sun that we see is called **Photosphere**. Above the Photosphere is the red coloured **Chromosphere** and beyond it is the magnificent **corona**, which is most easily visible during eclipses.
- Superimposed on Sun's white light are hundred of dark lines called **Fraunhofer lines**. Each line indicates some elements existing as gases in the solar system.

Facts About Sun

- The Sun is at the centre of solar system.
- It has a surface temperature of about 6000°C.
- It is one of the stars in the Milky Way Galaxy.
- It takes 224 million years to complete one circle (revolution) around the centre of milky way. This is called Cosmic Year.
- The Sun is 1300000 times bigger than the Earth in terms of volume.
- The time taken by sunlight to reach the Earth is 8 min and 16.6 second.
- The average distance of Sun from the Earth is 149600000 km.

Concepts Associated with Sun

- **Solar Winds** and **Solar Flares :** The Sun is continuously emitting streams of photons in all directions either as spiral streams called Solar Winds or bouts of incandescent material called Solar Flares.
- **Aurora** The constituent particles of the solar wind are trapped by the Earth's magnetic field and enter the Earth's upper atmosphere as Aurora. It is described as **Aurora Borealis** in the Northern hemisphere and **Aurora Australis** in Southern hemisphere.
- **Plages and Sunspots** The chromosphere of the Sun is continuously changing. Bright spots on the chromosphere are called Plages and dark spots are called Sunspots. The sunspots are cold and dark regions on the Sun's surface with a periodicity of 11 years.

PLANETS

- Planets are celestial bodies, which continuously revolve around and are lighted by a star. There are **eight planets** in the solar system that revolve around sun in elliptical orbit.
- The sequence of planets according to their size (in descending order) is Jupiter, Saturn, Uranus, Neptune, Earth, Venus, Mars and Mercury.
- The sequence of planets according to their distance from the Sun is Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

Classification of Planets

1. **Inner Planets** Mercury, Venus, Earth and Mars.
2. **Outer Planets** Jupiter, Saturn, Uranus and Neptune.

Inner Planet	*Outer Planet*
They are called as Terrestrial or Rock planets.	They are called as Jovian or Gaseous planets.
They are nearer to the Sun.	They are far away from the Sun.
They have very few natural satellites (or Moons) or no satellites.	They have a large number of natural satellites (or Moons).
They have a core of molten rocks.	They are formed mainly of gases with a small rocky core.
They move faster and have a shorter period of revolution.	They move rather slowly and have a longer period of revolution.

Planets Factsheet

Biggest Planet	Jupiter
Biggest Satellite	Gannymede (Jupiter)
Blue Planet	Earth
Green Planet	Uranus
Brightest Planet	Venus
Brightest Star outside solar system	Sirius (Dog Star)
Closest Star of solar system	Proxima Centauri
Coldest Planet	Neptune
Evening Star	Venus
Farthest Planet from Sun	Neptune
Planet with maximum number of satellites	Saturn
Fastest revolution in solar system	Mercury
Hottest Planet	Venus
Densest Planet	Earth
Fastest rotation in solar system	Jupiter
Morning Star	Venus
Nearest Planet to Earth	Venus
Nearest Planet to Sun	Mercury
Red Planet	Mars
Slowest revolution in solar system	Neptune
Slowest rotation in solar system	Venus
Smallest Planet	Mercury
Smallest Natural Satellite	Deimos
Earth's Twin	Venus
Only Satellite with an atmosphere like Earth	Titan

A Comparative Study of the Planets of the Solar System

Planet	*Special Characteristics*	*Rotation and Revolution Time*	*Important Physical Properties*	*Satellite System*
Mercury	Smallest and the innermost planet. It has no atmosphere. It has a cratered surface, much like the moon.	Rotation : 58.65 days Revolution: 88 days (Fastest revolution in the solar system)	It has the maximum diurnal range of temperature.	No satellite
Venus	Also called as the veiled planet as it is surrounded by thick clouds. It is known as **Evening and Morning star** as it is seen in the East in morning and in West in the evening. It is the brightest planet in solar system because of **almost 70% albedo**. It contains 90 to 95% CO_2. The night and day temperature are almost the same.	It has the slowest rotational speed. It has almost equal rotation and revolution. **Rotation** : (Clockwise) 243 days and **Revolution** : 224.7 days.	Rotates from East to West unlike the other planets. It is the hottest planet.	No satellite
Earth	The Earth is neither too hot nor too cold. It is called as the **Blue Planet**, as 71% of surface is made up of water.	Rotation : 24 hours; Revolution : 365 days and six hour.	It is the densest of all and is unique for the presence of higher forms of life.	Moon is the only natural satellite
Mars	Known as **Red Planet**. It has a thin atmosphere comprising of nitrogen and argon.	Rotation 24.6 hour (almost equal to Earth) Revolution : 687 days	It is marked by dormant volcanoes. Olympus Mons is the highest mountain, which is three times higher than the Mount Everest.	Two satellites : Phobos and Deimos
Jupiter	▪ It is the **largest planet** in the solar system with a mass 2.5 times greater than the combined mass of all the remaining planets, satellites and asteroids put together. ▪ It contains hydrogen, helium, methane and ammonia. A great red spot is detected on it.	Fastest rotational velocity (9.8 hrs). Revolution 11.8 years.	It is too massive to solidify as a planet, but not massive enough to develop nuclear fusion and become a star. It gives off more energy than it receives from the Sun, because of the heat inside.	It has 67 satellites. Some of the prominent satellites are: Europa, Callisto and Gannymede .
Saturn	It is the second largest planet and is surrounded by a set of eight rings, which are made up of primordial dust and ice particles.	Rotation in 10.3 hours Revolution in 29.5 years.	It has the least density of all the planets. 30 times less dense than Earth.	It has 82 satellites, the largest being Titan.

Planet	*Special Characteristics*	*Rotation and Revolution Time*	*Important Physical Properties*	*Satellite System*
Uranus	It is unique as its axis of rotation is inclined at only 98° to its orbital plane.	Unlike the others, which spin on their axis, Uranus actually rolls, apparently from North to South. Rotation : 17.2 hours, Revolution : 84 years.	Surrounded by a system of 9 faints rings.	It has 27 satellites. Th e prominent are Miranda, Ariel etc.
Neptune	It is farthest planet from sun. It has a dynamic atmosphere, which contains an Earth sized spot called the Great Dark Spot that is similar to the Jupiter's Great Red Spot .	Rotation 16 hours and Revolution 165 years.	It has five faint rings. It appears vivid blue in colour	It has 14 satellites. The prominent are Triton and Nereid.

Pluto is not a Planet now

- Pluto was discovered by Clyde Tombaugh in 1930.
- The redefinition of planet by the International Astronomical Union (IAU) on 24th August, 2006, states that in the solar system, a planet is a celestial body that
  1. It is in orbit around the Sun.
  2. It has sufficient mass, so that it assumes a hydrostatic equilibrium (nearly round) shape.
  3. It has cleared the neighbourhood around its orbit.
- A non-satellite body fulfilling the first two rule is classified as a Dwarf planet. So, Pluto was reclassified as a Dwarf Planet.
- Pluto has been given the number 134340.

MOON

- The study of Moon is called selenology.
- Moon is also known as the **fossil planet**.
- The Moon is the only natural satellite of the Earth.
- The Moon has no atmosphere and no twilight.
- The size of the Moon is one-fourth the size of the Earth.
- Gravitational pull of Moon is one-sixth that of the Earth.
- Mainly silicon, iron, magnesium etc, elements are found on the Moon's surface.

Moon at a Glance

Distance from Earth	384400 km
Diametre	3475 km
Mass (with respect to Earth)	1 : 8.1
Ratio of gravitational pull of Moon and Earth	1 : 6
Highest Mountain	18046 ft (Hugyens Mountain)
Time taken by moonlight to reach Earth	1.3 s
Rotation speed	3680 kmph
Speed of revolution around Earth	3680 kmph
Revolution period around Earth	27 days, 7 h, 43 min and 11.47s
Rotation period	27 day, 7 h, 43 min and 11.47s
Atmosphere	Absent
Part of Moon not visible from Earth	41%
Maximum distance from Earth (Apogee)	406000 km
Minimum distance from Earth (Perigee)	356400 km
Circumference	11000 km

Interstellar Debris : Asteroids, Meteoroids, Comets

Various Aspects	*Asteroids or Planetoids*	*Comets*	*Meteoroids/Meteors*
Constituents and Genesis	Composed of rocks, dust and metal. They cannot retain their atmosphere due to small size.	Comets may originate in a huge cloud called the Oort cloud that is thought to surround the solar system. It is composed of frozen gases and dust.	Meteoroids are small fragments of rocks and metal. Under the Earth's gravitational field, they burn and become white hot through friction as they fall through the atmosphere and are seen as the meteors or Shooting Stars.
Shape and Size	No definite shape, rather same as of Planetoids or small planet.	Comets have a head and tail, where the tail always points away from the Sun, because of the solar wind and the radiation pressure.	No definite shape.
Orbit	They orbit the Sun in the asteroid belt, which lies between the orbits of Mars and Jupiter.	They have an extremely eccentric orbit, but with definite periodicity.	Meteoroids travel through space. Meteors are scattered in the interplanetary space of the solar system.

- **Hailey's Comets** reappears after 76 years, last seen in 1986.
- **Comet Shoemaker Levy-9** collided with Jupiter between 16th and 21st July, 1994.
- **Comet Smith-Tuttle** could collide with Earth in 2126. It will create an impact of about 1.6 million times that of Hiroshima bomb.
- **Asteroid Apophis** NASA has authenticated the impact of asteroid called *Apophis*, which is having a probability of impact on Earth is dated on April, 2036.

THE EARTH

- The Earth is a Unique and densest Planet. Earth is the third planet from the Sun and the fifth-largest of the eight planets in the solar system. It is also the largest of the solar system's four terrestrial planets. Earth is also called as **Blue Planet**.

Earth at a Glancet

Angle of Inclination	23° 30′
Atmosphere	Nitrogen – 78%, Oxygen – 21%, CO_2 – 0.03%, Argon – 0.93%
Mean temperature	22°C
Rotation	From West to East in 23 : 56 : 04 hrs.
Revolution	365. 26 days
Satellite	1 – (Moon)
Special Feature	Only known planet to support life

ROTATION OF EARTH

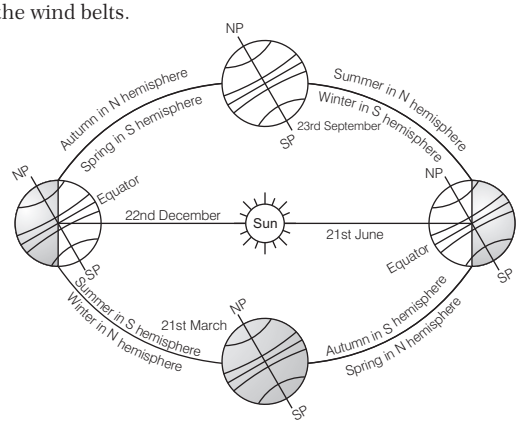
- The Earth spins on its imaginary axis from West to East in 23 hr, 56 min and 4.09 second.
- The rotational speed at equator is maximum (1667 km/hr) and then decreases towards the poles, where it is zero. The days and the nights are equal at the equator.

- The rotation of the Earth has the *following implications such as*
  1. Causation of day and night.
  2. Change in the direction of winds and ocean currents.
  3. Rise and fall of tides everyday.
  4. A difference of one hour between the two meridians, which are 15° apart.

REVOLUTION OF EARTH

It is the Earth's motion in elliptical orbit around the Sun. It takes 365 days, 5 hours, 48 minutes and 45.51 seconds. It leads to one extra day in every fourth year. *The revolution of Earth results in*

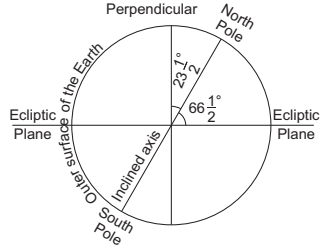
1. changes of season.
2. variation of the length of the days and nights at different times of the year.
3. shifting of the wind belts.



Revolution of Earth

Tilt of the Earth's Axis

- The axis of the Earth is inclined to the plane of ecliptic (the plane, in which the Earth orbits round the Sun) at an angle of 66½° giving rise to different seasons and varying lengths of day and night.
- If the axis were perpendicular to this plane, all parts of the globe would have equal days and nights at all times of the year.
- The Earth is tilted about 23.5° from a line perpendicular to ecliptic plane.



Equinox

- These are the days, when days and nights are equal. Under this situation, the Sun is vertically overhead at the equator. It happens on two days of the year i.e., **21st March** (Vernal Equinox) and **23rd September** (Autumnal Equinox).

Summer Solstice

- After the March equinox, the Sun appears to move Northward and is vertically overhead at the Tropic of Cancer on 21st June. This is known as Summer Solstice. On **21st June** the Northern hemisphere will have its longest day and shortest night. The Southern hemisphere will have shortest day and longest night.

Winter Solstice

- On 22nd December, the Sun is overhead at the Tropic of Capricorn. This is the Winter Solstice, when the Southern hemisphere will have its longest day and shortest night.

Summary of Daylight Hours in Northern and Southern Hemisphere

Length of Day	*Day*
Longest day in the Northern hemisphere	21st June
Shortest day in the Northern hemisphere	22nd December
Longest day in the Southern hemisphere	22nd December
Shortest day in the Southern hemisphere	21st June
Equal day/night in both hemisphere	21st March- 23rd September

Seasons

- They are the periods into which the year can be divided as a result of the climatic **conditions**, mainly due to the changes in duration and intensity of solar radiation. *There are four seasons*

Spring	*Summer*	*Autumn*	*Winter*
When the Sun is directly overhead the equator. (21st March)	When the Sun is directly overhead the Tropic of Cancer -the North temperate zone experiences summer. (21st June)	When the Sun returns to the equator and the North temperate zone experiences the season of autumn. (23rd September)	The Sun is at the Tropic of Capricorn and the North temperate zone experiences winter. (22nd December)

Difference Between Latitude and Longitude

Latitude	*Longitude*
Latitude is the angular distance of a point on Earth's surface from the centre of Earth, measured in degree.	Longitude is the angular distance of a point on the Earth's surface along the Equator, East or West from the prime meridian.
Important lines of latitudes are Equator, the Tropic of Cancer, ($23\frac{1}{2}^\circ$ N) and the Tropic of Capricorn ($23\frac{1}{2}^\circ$ S) and the Arctic and Antarctic circle ($66\frac{1}{2}^\circ$ N and S) respectively.	Important lines of longitudes are prime meridian (0°) and International Date Line (180°).

Universal Time (Standard Time) and Time Zones

- To avoid confusion about having many local times within one country, a particular meridian is chosen for the whole country, whose time is known as 'standard time'.
- The Indian Government has accepted the meridian of 82.5° degree East for standard time, which is 5 hrs 30 mins ahead of the Greenwich Mean Time.
- The Earth is divided in 24 longitudinal zones, each being 15° or 1 hour apart in time (360° = 24 hours, 360/24 =15° in 1 hour) or 1° in 4 minutes are called Standard Time Zones.
- Russia, which extends from 19° E to 169^0 W latitude has adopted 11 time zones.

International Date Line

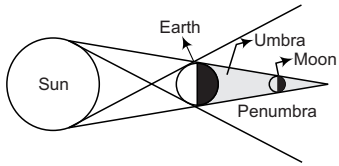
- It is the 180° meridian running over the Pacific Ocean, deviating at Aleutian Island, Fiji, Samoa and Gilbert Island.
- At the International Date Line, the date changes by exactly one day when it is crossed.
- Samoa and Tokelau shifted its position to the West of the Date line on 30th December, 2011.

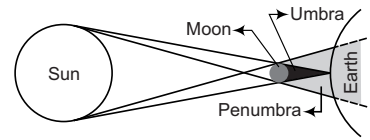
Eclipses

- An eclipse occurs, when the Sun, Moon and Earth are in a straight line.
- There are two types of eclipses-Solar and Lunar eclipse. Generally, a total of seven eclipse including solar and lunar eclipses, take place in a year.

Difference Between Lunar and Solar Eclipses

Lunar Eclipse	*Solar Eclipse*
It is the situation, when the Earth comes between Sun and Moon.	It is the situation, when the moon comes between Sun and Earth.
It occurs only on a full Moon day. But it does not occur on every full Moon day, because the Moon is so small and the plane of its orbit is tilted about 5 degree with respect to the Earth's orbital plan.	It occurs only on a new Moon day. But it does not occur on every new Moon day, because of the inclination of the Moon's orbital plan.





Structure of Earth's Interior

The interior of Earth is divided into three parts

1. Crust 2. Mantle 3. Core

The Outer Layer : The Crust

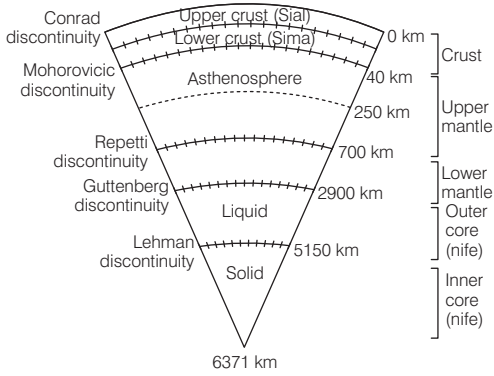
- It is the outermost and the thinnest layer of the Earth. This layer has the least density and its thickness varies about 8 to 40 km. Thickness of the crust is more in the continents and lesser in the oceans, while density of continental crust is lower than oceanic crust.
- The upper layer of crust of the Earth is rich in lighter minerals like silica and aluminium. Hence, this layer is also called as **Sial** (silica and aluminium). The average density of this layer is 2.7 gm/cm^3.
- Due to presence of minerals like silica and magnesium in the lower part of Earth's crust it is known as **Sima** (silica and magnesium).

The Middle Layer : The Mantle

- It is the intermediate layer of the Earth in terms of both its location and density. It is about 2900 km in thickness, composed of minerals in a semi solid state.
- It is divided further in two layer, upper mantle and lower mantle. The upper part of the mantle is called the Asthenosphere, which is about 250 km thick.
- The average density of this layer is about 5.68 gm/cm^3.

The Inner Layer : The Core

- The core is the innermost layer of the Earth and occupies its centre. It is about 3500 km in radius.
- The core is further divided in two layers : outer core and inner core.
- The outer part of the core is believed to have the properties of a liquid and the innermost part of the core (about 1255 km in radius) may be called solid or crystalline.
- This layer is also known **as Nife** (nickel and iron), because this layer contain large concentration of iron and nickel.
- Temperature of the core is between 2200°C and 2750° C.
- Density of this part of the Earth is 13.6 gm/cm^3 and is many times greater than the average density of the Earth 5.53 gm/cm^3.



Internal Structure of Earth

Earth's Magnetic Field

Magnetic fields are produced by the motion of electrical charges. The origin of the Earth's magnetic field is thought to be associated with electrical currents produced by the coupling of convective effects and rotation in the spinning liquid metallic outer core of iron and nickel. This mechanism is termed as the Dynamo effect.

Composition of Earth's Crust

Element	%	*Element*	%
Oxygen	46	Calcium	3.6
Silicon	28	Sodium	2.8
Aluminium	8	Potassium	2.6
Iron	6	Magnesium	1.5

Continental Drift Theory

- This theory is given by **Alfred Wagener**, in 1915, to explain the origin and evolution of the continents and the oceans.
- According to this theory, about 250 million years ago, there was only one huge continental landmass named Pangaea means *All Earth* and it was surrounded by one huge mass of water body, named **Panthalassa**.
- The present shape of the continents and oceans is due to the break up of Pangaea.
- The Northern drift cuts Pangaea from East to West creating **Laurasia** in the North and **Gondwana land** in South. A shallow sea called **Tethys** was situated between the Laurasia and the Gondwana land.

Plate Tectonic Theory

- Plate tectonic theory is a scientific theory that describe the large-scale motions of Earth's lithosphere.
- The theory of plate tectonics states that the lithosphere is divided into several rigid segments, which include both oceanic and continental crusts. These segments are called plates and they are moving on the asthenosphere, which is not a liquid, but a semisolid, which flows under stress.
- There are 7 major plates and more than 15 small plates consists whole Earth's lithosphere.

The world's major plates are

North American Plate
South American Plate
African Plate
Eurasian Plate
Pacific Plate
Antarctic Plate
Indo-Australian Plate

- Depending upon the type of movement, *plate margins are of three types*
  1. **Divergent Plates** (Constructive margins).
  2. **Convergent Plates** (Destructive margins).
  3. **Parallel Plates** (Conservative margin or Transform Boundary Plate).

Forces and Processes Affecting the Earth's Crust

- Appearance of the surface of the Earth keeps changing.
- These changes are produced under the influence of *two type of forces*

Exogenetic or External Forces

The forces occurring on the surface of the Earth are called the external or exogenetic forces. **Weathering** and **Erosion** are the examples of external forces.

Weathering

It is the process of disintegration or decomposition of rocks in-situ by natural agents. *There are three types of weathering*

- **Physical Weathering** (or mechanical weathering) It involves rock disintegration without any change in the chemical constituents of the rocks.
- **Chemical Weathering** It involves the decomposition due to chemical changes. There are various chemical processes, which cause chemical weathering such as solution, oxidation, carbonation, hydration, hydrolysis and chelation.
- **Biological Weathering** It is mainly controlled by plants and animals and human beings.

It is divided into three types

1. Faunal weathering
2. Floral weathering
3. Anthropogenic weathering

Erosion

It involves removal of rock material and their transportation. Erosion is performed by mobile agents such as rivers, streams, winds, waves and underground water.

Endogenetic or Internal Forces

The forces originating in the interior of the Earth are called the internal or the endogenetic forces. These forces are of two types

(a) Catastrophic Forces

- Catastrophic or sudden endogenetic forces are result of long periods of preparation deep within the Earth. However, their cumulative effects on Earth's surface are quick and sudden.
- Volcanoes and Earthquakes are sudden endogenetic forces.

(b) Diastrophic Forces

- Diastrophic forces include both vertical and horizontal movements, which are caused due to forces deep within the Earth.

These are of two types

- **Epeirogenetic Force** These are vertical movements. It causes upliftment and subsidence of continental masses through upward movements.
- **Orogenetic Forces** Orogenetic movement is caused due to endogenetic forces working in horizontal movement. When it operates in opposite direction, it is called tensional force. Tensional force create **faulting**, **cracking** and **fracture**. Tensional forces are also called as divergent forces. The force, when operates face to face, is called compressional force or convergent force. Compressional force create **folding** and **warping**.

EARTHQUAKES

- It refers to the vibration of the Earth's surface caused by the catastrophic endogenetic forces. **Seismology** is the special branch of geology that deals with the study of earthquake.
- The magnitude of an earthquake; usually expressed by the Richter scale, is a measure of the amplitude of seismic waves. The magnitude of an earthquake is a measure of the amount of energy released, an amount that can be estimated from seismograph reading.

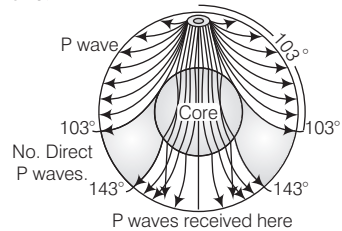
- place of origin of earthquake is called **focus**. The place on the ground surface, which is perpendicular to the focus or hypocentre is called **epicentre**.
- The waves generated by earthquake are called seismic waves and *they are classified into 3 types such as*
  1. **Primary Wave** (P Waves) These are the waves of short wave length and high frequency. They are longitudinal waves and can travel through solid, liquid and gases.
  2. **Secondary Wave** (S Waves) These are the waves of short wave length and high frequency. They are transverse waves, which travel through only solid.
  3. **Surface Waves or Long Waves** (L Waves) They are the waves of long wavelength, and they travel through the Earth's crust. It causes most of the structural damage on surface of Earth during earthquake.

The traditional zones of earthquakes are

(a) Circum Pacific belt
(b) Mid Continental belt
(c) Mid Atlantic belt

Shadow Zone

- 103° to 143° of area is called shadow zone.



VOLCANISM

- Volcanism includes all phenomena connected with the movement of heated material from the interior towards the surface of the Earth.
- A volcano is a vent or opening, through which heated molten materials consisting of gases, water, liquid lava, fragments of rocks are ejected from the highly heated and melted interior to the surface of the Earth.

Classification of Volcanoes

Active Volcanoes	*Dormant Volcanoes*	*Extinct Volcanoes*
Those volcanoes that frequently erupt or they have erupted in recent times. e.g. Mount Etna in Italy.	Those volcanoes that have been known to erupt and show signs of possible eruptions in future. e.g. Mt. Kilimanjaro in Africa.	Volcanoes that have not erupted at all in historic times but they retain the features of a volcano. e.g. Mt. Chimborazo in Ecuador.

Important Volcanic Mountains

Name	*Country*	*Name*	*Country*
Visuwius	Italy	Cotopaxi	Ecuador
St. Helens	USA	Fujiyama	Japan
Krakatoa	Indonesia	Popo Ceteptl	Mexico
Mauna Loa	USA (Hawaii)	Ojas del Salado	Chile-Argentina

ROCKS

- A rock can be defined as an aggregate of one or more minerals found on surface of earth. Rocks may be hard, soft and varied in colours.
- *Rocks are classified in three main types depending on the process of their formation*
  1. Igneous 2. Sedimentary
  3. Metamorphic

Igneous Rocks

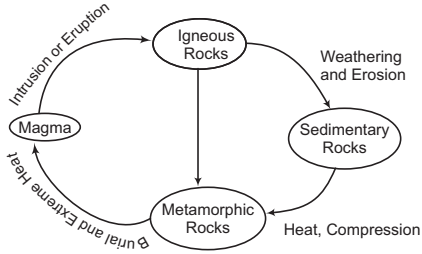
- Formed due to the cooling, solidification and crystallisation of hot and molten magma.
- They are known as the **primary rocks** as all the other rocks are formed directly or indirectly from the igneous rocks.
- They are hard, granular and crystalline rocks, less affected by chemical weathering.

- Moreover, it does not have any fossil or does not form any strata or layers of lava. E.g., of Igneous rocks—Granite, Basalt, Gabbro, pegmatite and peridotite.

Sedimentary Rocks

It is formed due to the aggregation and compaction of sediments derived from the older rocks or dead remains of plants and animals and contains fossils of plants.

- The sedimentary rocks can be classified on the basis of the nature of sediments : Mechanically, chemically and organically formed rocks.



Rock's Cycle

Metamorphic Rocks

- These are the changed form of igneous and sedimentary rocks.
- These are the rocks, which change either in form or composition without disintegration.
- Already formed igneous, sedimentary or even metamorphic rocks are changed to other forms and this process is called **metamorphosis**.
- The agents of metamorphism are: heat, compression and solution.

Original Rocks	*Metamorphic Rocks*
Sandstone	Quartzite
Limestone	Marbal
Shale and mudstone	Slate
Granite	Gneiss
Coal	Diamond
Clay	Slate

LANDFORMS

- There are four major landforms formed on earth's surface. These are mountains, hills, plateau and plains.

Landform	*Total Global Surface Area in%*
Plain	41
Plateau	33
Hill	14
Mountain	12

Mountains

Based on their mode of formation four main types of mountain can be distinguished

Fold Mountains

- It is formed due to face to face orogenetic movement generated by endogenetic forces. Example of fold moutains : Himalayas, Alps, Andes, Rockies, Atlas etc.

  On the basis of age, fold mountains are grouped into

  (a) **Young/New Fold Mountains** It came into existence after the continental drift and plate movements. e.g., Himalayas, Andes, Rockies, Alps. Himalayas are regarded as the youngest mountains in the world.

  (b) **Old Mountains** They were once young fold mountains. But now they have denuded due to erosional processes. e.g., Pennines (Europe), Appalachians (US), Aravalis (India) etc.

Block Mountains

- It is formed when great block of Earth's crust may be raised or lowered due to tectonic activities. The land between the two parallel faults either rises forming block mountains or horsts or subsides into a depression termed as Rift Valley or Graben.
- Examples-Vindhya and Satpura in Madhya Pradesh, the Vosges in France and Black forest in Germany (through which Rhine river flows).

Volcanic Mountains

- They are formed due to the accumulation of volcanic material. They are also called mountains of accumulation.
- Examples-mount Fuji (Japan), Cotopaxi in Andes, Vesuvius and Etna in Italy, Mount Mayon (Philippines) etc.

Residual or Dissected Mountains

- They are formed as a result of erosion of plateaus and high planes by various agents of erosion.
- Examples Catskill mountains of New York, Sierras of Central Spain, Highlands of Scotland and Parasnath, Girnar, Rajmahal of India.

Major Mountain Ranges

Range	*Location*	*Length (km)*
Andes	South America	7200
Himalayas, Karakoram and Hindu Kush	South Central Asia	5000
Rockies	North America	4800
Great Dividing Range	East Australia	3600
Atlas	North-West Africa	1930
Caucasus	Europe	1200
Alps	Europe	1200
Alaska	USA	650

Plains

- A relatively low-lying and flat land surface with the least difference between its highest and lowest points is called a Plain.

Classification of Plains

Structural Plain	*Erosional Plain*	*Depositional Plain*
Formed due to the uplift of a part of the sea floor.	Formed, when the elevated tract of land is worn down to a plain by the process of erosion.	Formed by filling up of sediments into depressions along the foothills, lakes and seas.

Plateaus

- Tabular upland having relief of more than 500 feet may be defined as plateau.
- Tibetan plateau (5000 m) is the highest plateau in the world.

Important Plateaus of the World

Plateau of Iran	Iran
Plateau of Tibet	China-Tibet
Castilian Plateau	Spain
Anatolian Plateau	Turkey
Air Plateau	Africa
Katanga Plateau	Democratic Republic of Congo.
Laurentian Plateau	Canada (Eastern side)
Colombia Plateau	USA (Western side)
Edwards Plateau	USA (Mexico)
Borborema Plateau	Brazil (Eastern)
Deccan Plateau	South India
Mato Grosso Plateau	Brazil
Kimberley Plateau	Australia

ATMOSPHERE

- Atmosphere is a thick gaseous envelope surrounding the Earth from all sides and attached to the Earth through the force of gravity.

Significance of Atmosphere

- Acts as a filter because it absorbs the various harmful ultraviolet radiation.
- Source of various life supporting gases such as oxygen, carbon dioxide, etc.
- Supports life forms in biosphere.

Extent of Atmosphere

- Though the exact altitude of atmosphere is not known, but it is estimated that it extends till 10,000 km above the Earth's surface.
- The vertical distribution of atmospheric layers is not uniform.
- Almost 98% of atmospheric mass is limited to an altitude of 30 km from the surface of Earth.

Composition of Atmosphere

- The atmosphere is composed of gases, water vapour and particulates. Other gases such as Helium, Ozone and Hydrogen etc, are present in traces. Ozone gas absorbs the ultraviolet radiations and protects the biosphere from its adverse impact.
- The percentage composition of various gases in atmosphere upto 50 km is given below

Gases	*Percentage Composition*	*Significance*
Nitrogen	78.08 %	Acts as dilutent and is generally chemically inactive.

Gases	*Percentage Composition*	*Significance*
Oxygen	21%	Inhaled by biotic components for survival.
Argon	0.93%	Inert gas.
Carbon dioxide	0.03%	Being a greenhouse gas, it maintains the temperature of the lower atmosphere.
Neon	0.0018%	Inert gas.
Helium	0.005%	Inert gas.

STRUCTURE OF ATMOSPHERE

Troposphere

- It extends up to an average altitude of 18 km from the Earth's surface. Thickness varies from 8 km at the poles to 18 km at the equator.
- **Tropopause** separates troposphere from stratosphere.
- This layer accounts for practically the entire water vapour, all dust particles and most of the carbon dioxide contained in the atmosphere. Due to this all weather phenomena such as condensation, precipitation and storms etc. occur in the troposphere only.

Stratosphere

- The stratosphere extends up to about 50 km, where **stratopause** separates it from the mesosphere.
- In this layer, the temperature increases with increase in height. This phenomenon is known as **temperature inversion**.
- The temperature rises in this layer from about–60°C at the tropopause to 0°C at stratopause.
- The part of the stratosphere, in which there is a concentration of ozone is often called **ozonosphere**. It absorbs ultraviolet radiation, which is harmful for lifeforms in the biosphere.
- Stratosphere is free from dust particles and also from atmospheric turbulences. Hence, this layer is considered ideal for flying of jet aircrafts.

Mesosphere

- Mesosphere extends above the strato- pause up to a height of about 80 km.
- In this layer, the temperature decreases with height like in the troposphere and it falls from about 0° C at its base to about – 100° C at height of 80 km.
- It is considered the coldest layer of the atmosphere.
- The upper limit of the mesosphere is marked by the **mesopause**, a transitional layer separating it from the ionosphere.
- Meteoroids burn in this layer which prevents them from colliding with Earth's surface.

Ionosphere

- Ionosphere is located above the mesosphere and extends up to about 600 km. This layer is called ionosphere because it contains electrically charged ions that reflect the radiowaves back to the Earth, which makes radio communication possible.
- Absorption of solar radiation by ionised particles causes an increase in temperature with increasing height in the ionosphere.

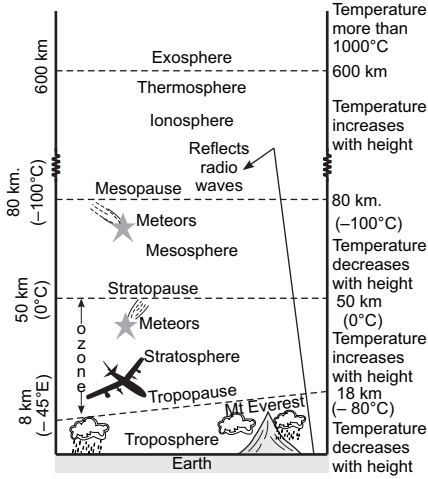
Thermosphere

- The zone between the 85 and 600 km above the surface is often called thermosphere. In this layer, the temperature increases with increasing altitude. The upper limit of the thermosphere, the **thermopause** is generally taken at an altitude of about 600 km.
- The day temperature at 600 km altitude exceed 1400° C, while night temperature remain about 225° C .
- The upper part of the thermosphere contains only the lighter gases like helium and hydrogen.

Exosphere and Magnetosphere

- The outermost part of the atmosphere of the Earth is called **exosphere**.
- This zone of the atmosphere extends beyond 640 km.

- The upper limit of the exosphere is uncertain as this layer acts as a transitional layer between the Earth's atmosphere and the outer space. The outer part of the exosphere is called **magnetosphere**.



Structure of the Atmosphere

Greenhouse Effect and Global Warming

- The greenhouse gas (GHG) is a gas in the atmosphere that absorbs solar radiations raising the temperature of the planet. The process is the fundamental cause of the greenhouse effect.
- The primary greenhouse gases in the Earth's atmosphere are water vapour, carbon dioxide, methane, nitrous oxide and ozone.
- **Global Warming** It is the increase of Earth's average surface temperature, due to effect of greenhouse gases, such as carbon dioxide emissions from burning fossil fuels or from deforestation. This is a type of greenhouse effect.
- **Kyoto Protocol** The Kyoto Protocol is a protocol to the United Nations Framework Convention on Climate Change (UNFCCC), aimed at fighting global warming, adopted in 1997.

Ozone Layer : The Blanket of Globe

- The ozone layer is located within the Stratosphere, about 24 km above the Earth's surface. The layer consist of ozone gas molecules that are formed as the sunlight reacts with oxygen.
- The ozone layer is very important as it protects life on Earth by filtering the Sun's dangerous ultraviolet radiation.
- Due to increased pollution on Earth, chemicals such as chlorofuorocarbons (CFCs) are destroying this protective ozone layer, which could lead to increased health risks and damage agricultural and acquatic ecosystem.

Montreal Protocol

Montreal Protocol on substances that deplete the ozone layer is an international treaty designed to protect the ozone layer from Chlorofluorocarbons (CFCs), effective from 1989.

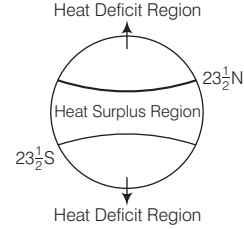
Insolations

- Solar radiation that is intercepted by the Earth is known as Insolation.
- Insolation is measured with the help of **Pyranometers**.

The amount of insolation depends on following factors

(a) The area and nature of the surface.
(b) The inclination of the rays of the Sun.
(c) Distance between the Earth and the Sun.

- On an average, insolation is highest near the equator, marginally lower at the tropics and lowest at the poles.



- The Earth's surface does not absorb all the energy that it receives. The proportion of the solar radiation reflected back from the surface is called **Albedo**.

Heat Budget of the Earth

- The Earth receives energy continuously from the Sun, but its temperature is still almost constant.
- This is because the atmosphere loses an amount of heat equal to the amount of heat gain through insolation. This mechanism of maintaining the balance between incoming and outgoing heat in the atmosphere is called the **Heat Budget** or **Heat Balance**.

Atmospheric Pressure

- Air is a mixture of several gases having their own weight. The pressure exerted by air due to its weight is called atmospheric pressure on the Earth's surface.
- Atmospheric pressure is neither the same for all the regions nor the same for one region all the time.
- Atmospheric pressure is affected by various factors such as altitude, temperature and Earth's rotation.

Influence on the Atmospheric Pressure

Factor	*Influence on the Atmospheric Pressure*
Altitude	Air pressure increases, when air descends. This is due to the decrease in volume of the air. When air rises its volume increases and the pressure of its molecules is spread over a larger area. So, its pressure decreases.
Temperature	The pressure of air rises, when its temperature falls. Low temperature at the poles cause the air to contract–high pressure develops; whereas the high temperature along the equator cause the air to expand–low pressure develops.

The Global Pressure Belts

Thermal Effect : Equatorial Low Pressure Belt

- It is located on either side of the geographical equator in a zone extending between 5°N and 5°S.
- Its location is not stationary and there is a seasonal drift of this belt with the Northward (summer solstice) and Southward (winter solstice) migration of the Sun.
- It is thermally induced because of the intense heating of the ground surface by the almost vertical Sun rays.
- It represents the zone of convergence of North-East and South-East trade winds. This convergence zone is characterised by light and feeble winds. And because of the frequent calm conditions, this belt is called a **belt of calm** or **doldrums**.

Dynamic Effect : Sub-Tropical High Pressure Belt

- It extends between 30° to 35° latitudes in both the hemispheres.
- It is not thermally induced, but dynamically induced as it is formed by the rotation of the Earth and sinking and settling down of winds.
- Here the convergence of winds at higher altitude above this zone results in the subsidence of air. Subsistence results in concentration of their volume and ultimately causes high pressure in this zone.
- This zone of high pressure is also called **Horse Latitude**.

Dynamic Effect : Sub-Polar Low Pressure Belt

- It extends between 60° to 65° latitude in both the hemisphere.
- The low pressure belt does not appear to be thermally induced, because there is low temperature throughout the year.
- Naturally, there should have been high pressure belt instead of low pressure belt. But as it is dynamically induced low pressure is formed.

- It is more developed and regular in Southern hemisphere, while it is broken in Northern hemisphere because of the over dominance of water (ocean) in the Southern hemisphere.

Thermal Effect : Polar High Pressure Belt

- High pressure persists at the poles throughout the year, because of the prevalence of very low temperature all the year round.

Wind System

- The pressure difference is the major cause of the genesis of the wind system. The air moves from high pressure to low pressure.
- The slope of pressure from high to low is called as **pressure gradient**, which is also called as **barometric slope**.
- The imaginary line joining the points having same pressure is called **isobars**.
- Ideally, the direction of air movement should be perpendicular to the isobars, but the winds are deflected due to **Coriolis force** caused by the rotation of the Earth.

Factors Controlling Wind Motion

- Pressure gradient
- Rotation of Earth and Coriolis force
- Frictional force
- Centrifugal action of wind

Geostrophic Wind

The winds blowing parallel to the isobars generally at the height of 600 m is called geostrophic wind.

CLASSIFICATION OF WINDS

Primary Wind Movement

Permanent Winds

- These winds include trade wind, westerlies and polar winds.

Trade Winds

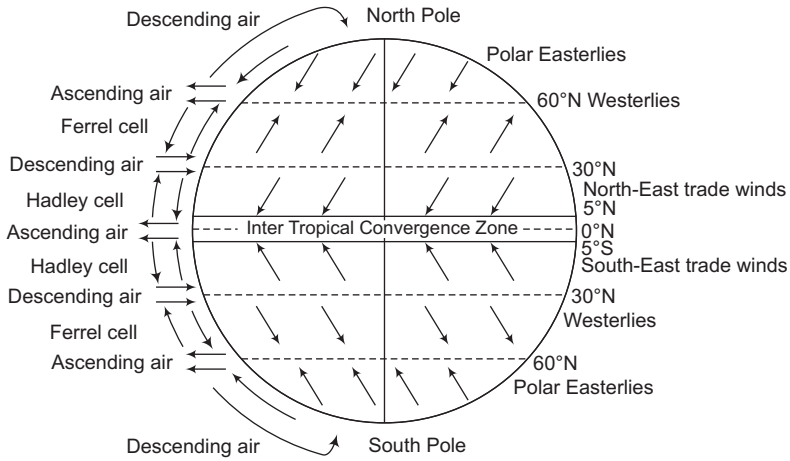
- These are steady currents of air blowing from the sub-tropical high pressure belt towards the equatorial low pressure belt.
- Under the influence of the Coriolis forces, they flow from the North-East in the Northern hemisphere and from South-East in the Southern hemisphere.

Westerlies

- The permanent winds blowing from the sub-tropical high pressure belt to the sub-polar low pressure belt in both the hemisphere is called Westerlies.
- The general direction of the Westerlies is South-West to North-East in the Northern hemisphere and North-West to South-East in the Southern hemisphere.
- Because of the dominance of the land masses in the Northern hemisphere, the Westerlies become more complex and complicated. They become less effective during summer seasons and more vigorous during winter seasons.
- The westerlies become more rigorous in Southern hemisphere because of lack of land and dominance of water surface. Their velocity increases Southward and they become stormy that they are called **roaring forties** between 40°-50° of Southern hemisphere, **furious fifties** at 50° South latitude and **shrieking sixties** at 60° South latitudes.

Polar Winds

- It blows from polar high pressure belt to sub-polar low pressure belt. They are North-Easterly in Northern hemisphere and South-Easterly in the Southern hemisphere.



Prevailing Wind and the Impact of Pressure Belt

Secondary Wind Movement

Secondary wind movements, wind patterns that are not permanent like the primary winds. Secondary wind movement includes seasonal or periodic winds.

Seasonal Winds The winds, which reverse their direction completely every 6 months with change in season. The best example is monsoon winds.

Tertiary Movement

Local Winds

Wind	*Nature*	*Region*
Land Breeze	Warm	Blows from land to sea
Chinook	Warm	Rockies (USA and Canada)
Fohn	Warm	Alps/Europe
Khamsin	Hot	Egypt
Harmattan	Hot	Sahara to Guinea Coast (Ghana, Nigeria etc)
Brick Fielder	Hot	Australia
Norwester	Hot	New Zealand
Black Roller	Hot	North America
Kalbaisakhi	Hot	North India
Sirocco	Hot	North Africa
Mistral	Cold	France to Mediterranean (Rhine valley)
Levanter	Cold	Spain
Blizzard	Cold	Siberia, Canada, USA
Bora	Cold	Yugoslavia
Southerly Buster	Cold	Australia
Sea Breeze	Cold	Blows from sea to land

Cyclones

- Cyclones are the centres of low pressure surrounded by closed isobars having increasing pressure outward.
  - Air blows inward in anticlockwise direction in the Northern hemisphere.
  - Air blows inward in clockwise direction in the Southern hemisphere.
- Cyclones are mainly of two types

(i) Tropical cyclone

(ii) Temperate cyclone

Differences Between Tropical and Temperate Cyclones

Tropical Cyclone	*Temperate Cyclone*
They are found in the trade wind belt, between 8-20 degree North and South. But not found between 0 to 8 degree as there is no coriolis force there.	Normally found between 30 to 65 degree North and South in the sub-polar frontal zone, where cold polar air mass meets the warm tropical air mass.
It travels from East to West in the Easterly wind belt.	It moves from West to East embedded in the Westerly wind belt.
Tropical cyclones are much smaller with a diameter of about 200 to 500 km.	They form over a much large area with a diameter 300 to 1500 km.
Tropical cyclones are non frontal in nature.	Temperate cyclones are frontal in nature.
It is formed only over the oceans.	It is formed either over continents or over the oceans.
Tropical cyclone can form only in the summer.	They can form both in summer as well as in winter.
It does not have definite lifecycle and they dissipate as they pass over the land surface	It takes more time to dissipate as it has a more definite lifecycle.

Naming of Cyclones in Different Regions

Name of Cyclone	*Region*
Typhoons	South China Sea
Tropical Cyclones	Indian Ocean
Hurricanes	Caribbean Sea
Tornadoes	USA
Willy Willies	Northern Australia

Anticyclones

- They are the wind systems, which have the highest air pressure at the centre and lowest at the outer margins.
- The wind blows from centre to outward in clockwise direction in Northern hemisphere and from centre to outward in anti- clockwise direction in Southern hemisphere.

JET STREAM

- The strong and rapidly moving circumpolar Westerly air circulation in a narrow belt of a few hundred kilometres width in the upper limit of troposphere is called Jet Stream.

Humidity

- Humidity of air refers to the content of the water vapour present in the air at a particular time and place.
- Humidity is measured by an instrument called **hygrometer**.

Humidity Related Concepts

Measurement of Humidity	*Definition*
Humidity Capacity	The capacity of air of certain volume at certain temperature to retain maximum amount of moisture content.
Absolute Humidity	The total weight of moisture content per volume of air at definite temperature is called absolute humidity.
Specific Humidity	The mass of the water vapour in grams contained in a kilogram of air. It represents the actual quantity of moisture present in a definite amount of air.
Relative Humidity	It is the ratio of the amount of water vapour actually present in the air having definite volume and temperature (i.e. absolute humidity) to the maximum amount the air can hold (i.e. humidity capacity). $\text{Relative humidity} = \frac{\text{Absolute humidity}}{\text{Humidity capacity}} \times 100$

CLOUDS

- Clouds are the masses of small water droplets or tiny ice crystals.

Classification of Clouds

Clouds are classified according to their appearance, form and height. There are four groups

High Clouds (6-20 km)	*Middle Clouds (2.5-6 km)*	*Low Clouds (below 2.5 km)*	*Clouds with Great Vertical Extent (upto 9000 m)*
Cirrus Composed of small ice crystal, white, transparent and fibrous in appearance.	**Alto-cumulus** Composed of water droplets in layers and patches.	**Strato-cumulus** Large globular masses, bumpy looking, soft and grey in appearance forming a pronounced regular and sometimes wavy pattern.	**Cumulus** Round topped and flat based forming a whitish grey globular mass, consists of individual cloud units.
Cirro-cumulus Composed of ice crystals, but globular or rippled in appearance.	**Alto-stratus** Composed of water droplets, forming sheets of grey or blue colour clouds.	**Nimbo-stratus** Dark grey and rainy looking, dense and shapeless, often gives continuous rains.	**Cumulonimbus** They have a great vertical extent, white or black globular masses, whose rounded tops often spread out in the form of anvil. It is characterised by convectional rain, lightning and thunder.
Cirro-stratus Looks like a thin white, almost transparent sheet, which causes the Sun and Moon to have halos.		**Stratus** These are low, grey and layered, almost fog like in appearance, bringing dull weather and often accompanied by drizzle.	

Rainfall

On the basis of its origin, rainfall may be *classified into following types*

- **Convectional Rainfall** It occurs due to thermal convection currents caused due to insolational heating of ground surface.
- **Frontal or Cyclonic Rainfall** It occurs due to upward movement of air caused by convergence of cold air masses against warm air masses.
- **Orographic Rainfall** It occurs due to ascent of air forced by a mountain barrier.

WEATHER AND CLIMATE

- Weather refers to the sum total of the atmospheric conditions in terms of temperature, pressure, wind, moisture, cloudiness, precipitation and visibility of a particular place at any given time.
- The average weather conditions over a large area and for a long time are called the **climate of a place**.

Climatic Division of the World

Climatic Zone	*Climatic Types*	*Rainfall*	*Natural Vegetation*
Equatorial Zone (10°N-10°S)	▪ Hot, wet equatorial	Rainfall all the year (200 cm)	Equatorial rain forests
Tropical Zone (10°N-30°N) (10°S-30°S)	▪ (a) Tropical Monsoon	Heavy summer rain (150 cm)	Monsoon forest
	▪ Sudan type	Rain mainly in summer (180 cm)	Savanna (*Tropical grassland*)
	▪ Desert (a) Saharan type (b) Mid-latitude type	Little rain (10 cm)	Desert vegetation and scrub
Warm Temperate Zone	▪ Western Margin (*Mediterranean type*)	Winter rain (90 cm)	Mediterranean forests
(30°N-45°N) (30°S-45°S)	▪ Central Continental type (*Steppe type*)	Light summer rain (50 cm)	Steppe, temperate grassland
	▪ Eastern Margin (a) China type or Gulf type	Heavier summer rain (120 cm)	Warm, wet forests and bamboo
Cool Temperate Zone	▪ Western Margin (*British type*)	Rain in autumn and winter (75 cm)	Deciduous forests
(45°N-65°N) (45°S-65°S)	▪ Central Continental (*Siberian type*)	Light summer rain (60 cm)	Coniferous forests
	▪ Eastern Margin (*Lauritian type*)	Moderate summer rain (100 cm)	Mixed forests
Frigid Zone (65°N-90°N)	▪ Arctic or Polar	Very light summer rain (25 cm)	Tundra, mosses, lichens
(65°S-90°S)	▪ Mountain Climate	Heavy rainfall (variable)	Alpine, fern, conifers pastures, snow

Great Deserts

Desert Name	*Country/ Region*	*Desert Name*	*Country/ Region*
Sahara, Libyan, Nubian	North Africa	Dasht-e-Lut (Barren Desert)	Iran
Gibson, Simpson, Great Victoria, Great Sandy	Australia	Desert Sechura	Peru
Rub-al-Khali, An-Nafud	Arabia	Atacama	North Chile
		Patagonia	Argentina
Dasht-e-Kavir (Salt-e-Kavir)	Iran	Kalahari	Botswana
		Namib	Namibia

Famous Grasslands

Grassland	*Country*	*Grassland*	*Country*
Steppe	Eurasia	Veld	South Africa
Pustaz	Hungary	Downs	Australia
Prairie	USA	Canterbury	New Zealand
Pampas	Argentina		

Some Important Isopleths

Isopleth	*Represents*	*Isopleth*	*Represents*
Isobath	Equal depth in sea	Isohyets	Rainfall
Isobars	Equal pressure	Isohaline	Salinity
Isobronts	Thunder storm at same time	Isonif	Snow
Isocline	Slope	Isotherms	Temperature
Isodapane	Equal transportation cost	Isohypse (or contour lines)	Elevation above sea-level
Isohels	Sunshine		

OCEANS AND RELATED ASPECTS

- The water component of the Earth is called hydrosphere. It includes the oceans, seas, lakes, ponds, rivers and streams. The hydrosphere covers about 70% of the surface of Earth.

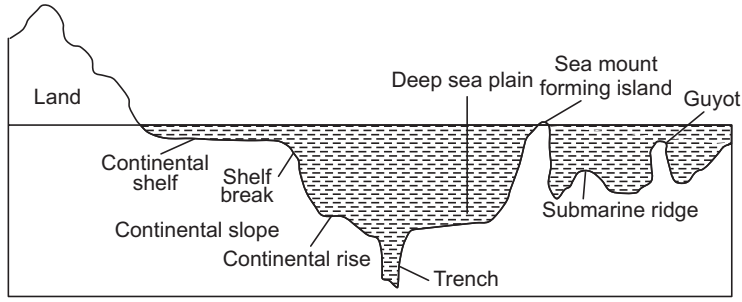
Oceans

- There are four oceans. In the order of their size they are: Pacific ocean, Atlantic ocean, Indian ocean and Arctic ocean. The average depth of oceans on earth is about 4 km.

Salient Features of Major Oceans

Ocean	*Salient Features*
Pacific	It is the largest and deepest ocean. Most of the islands of the ocean are of volcanic or coral origin. Marina Trench, the world's deepest trench with depth 11033 km is in Pacific Ocean.
Atlantic	It has the longest coast line. It is the busiest ocean for trade and commerce, since its sea routes connect the two most industrialised regions, namely Western Europe and North-Eastern United States of America. Atlantic ocean is still widening.
Indian	Only ocean named after a country. It is deeper than the Atlantic ocean. Here the number of continental islands is more than that of volcanic islands.
Arctic	It is the smallest ocean and lies within the Arctic circle. The North pole lies in the middle of the Arctic ocean. Most of the part of the ocean is frozen and hence it is least saline and the shallowest among all the oceans.

Relief of the Ocean Basin



Relief of the Ocean Basin

Continental Shelf

- The shallow sub merged extension of the continent is called the **continental shelf**.
- Extends to a depth of 100 fathoms (1 fathoms = 1.8 m).
- Average width 70 km; average slope of about 1°.
- Continental shelf covers 8.6% of total area of the oceans. It extends over 13.3% of the Atlantic ocean, 5.7% of Pacific ocean and 4.2 % of Indian ocean.
- In this regions, the mountain extend along the coast, thus the shelf is narrower.
- About 20% of oil and gas are found here. They also provide the richest fishing grounds in the world.

Continental Slope

- Extends seawards from the continental shelf.
- Depth-200 to 2000 metres.
- Average slope 2 to 5 degrees.
- The boundary between shelf and slope is known as shelf break which is a seaward edge with an abrupt drop.
- They cover about 8.5% of the total ocean area.

Continental Rise

- It is an area at the foot of the slope, slightly rising due to the accumulation of debris transported over the slope.
- Average slope-0.5° to 1°.
- Oil deposits occur here.

Abyssal or the Deep Sea Plains

- It is an underwater vast and flat plain on the oceanic floor.
- Average depth 3000 to 6000 m.
- It covers about 40% of total oceanic area.
- Parts of the Abyssal plains are occupied by raised ridges or submarine mountains and by very deep trenches or canyons.

Deeps/Trenches

- Trenches are narrow and steep side depressions.
- Trenches are formed, when one plate of Earth's crust is subducted below the other.
- Marina trench (challenger deep) which is the deepest point in the world situated in the North-West Pacific ocean, near Philippines, is more than 11 km deep.

Oceanic Ridges

- Oceanic ridges are formed by the volcanic activity along the divergent boundary of plates.
- It is a thousand kilometer long and hundred kilometer wide mountain range on the oceanic floor.
- Their summits may rise above the sea level in the form of islands i.e., Azores Island of Iceland.

Sea Mounts and Guyots

- A ridge rising more than 1000 m above the ocean floor is called sea mount. Flat topped sea mounts are called guyots.
- Both of them are formed by the volcanic activity.
- Largest number of sea mounts and guyots are found in the Pacific ocean.

Submarine Canyons

- Submarine canyons are the deep gorges on the ocean floor and are restricted to the continental shelves, slopes and rises.
- Many submarine canyons are found along the mouths of major rivers e.g., Hudson canyon.

Coral Reefs

- Coral reefs are formed due to accumulation and the compaction of skeletons of lime secreting organisms known as Coral Polyps.
- Corals are found mainly in the tropical oceans and seas, because they require high mean annual temperature ranging between 20°C to 25°C.
- Corals do not live in deeper waters due to lack of sufficient sunlight and oxygen.
- Coral reefs are formed mainly along Western coast.

Coral Bleaching

When corals are stressed by changes in conditions such as temperature, light or nutrients, they expel the symbiotic algae living in their tissues, causing them to turn completely white, this phenomenon is known as **coral bleaching**.

SALINITY

- Salinity is defined as the total amount of solid material in grams contained in 1 kg of sea water and is expressed as Parts Per Thousand (PPT). The oceanic salinity not only affects the marine organism and plant community, but also affects the physical properties of the ocean such as temperature, pressure, density, waves and currents.
- Average salinity of ocean water is 35 Parts per thousand.
- The factors controlling salinity are Temperature, Evaporation, Rainfall and Influx of river.
- Average salinity in Southern Hemisphere is more than in Northern Hemisphere.
- Iso-halines represent the salinity distribution on the surface of the sea. These are the lines joining places having an equal degree of salinity.
- The main source of salinity is dissolution of the rocks of oceanic crust, which contain various salts.
- Salinity on an average decreases from equator to poles. The highest salinity is recorded near the tropics rather than the equator because of the heavy precipitation in the equatorial region.

Most Saline Water Bodies

Water Body	*Percentage Composition*
Lake Assal (Djibouti)	34.8%
Dead Sea (West Asia)	33.7%
Great Salt Lake (USA)	32.0%
Lake Van (Turkey)	23%

Movements of Oceanic Water

Waves

- Waves are the oscillatory movements in water mainly produced by winds, manifested by an alternate rise and fall of the sea surface.
- The waves are the important agents of erosion in the coastal regions, where they carve out various landforms like caves, bays, gulfs, capes and cliffs.

- Seismic waves or Tsunamis are the waves caused by earthquakes or volcanic eruptions at the sea bottom.
- The tsunamis is also called as harbour wave and can be quite dangerous.
- The tsunamis which hit the coast in South-East Asia on 26th December, 2004, caused very heavy damage in that region.
- Tsunamis in the Indonesian Islands of Sulawesi, Java and Sumatra in 2018 killed many and injured a large number of people.

Ocean Currents

- An ocean current is a continuous, directed movement of ocean water generated by the forces acting upon it, such as breaking waves, wind, coriolis effect, temperature and salinity differences and tides caused by the gravitation pull of the Moon and the Sun.
- Ocean currents circulate in clockwise direction in Northern hemisphere and in anti-clockwise direction in Southern hemisphere.

Ocean currents are of two types

1. **Warm Currents** The ocean currents flowing from lower latitude to higher latitude are called warm currents.
2. **Cold Currents** The ocean currents flowing from higher latitude to lower latitude are called cold currents.

Currents of Pacific and Atlantic Oceans

Current	*Nature*
North Pacific Ocean	
North Equatorial Current	Warm
Kuroshio Current	Warm
Oyashio Current	Cold
Okhotsk Current	Cold
Californian Current	Cold
South Pacific Ocean	
East Australian or Great Barrier Current	Warm
South Equatorial Current	Warm
Counter Equatorial Current	Warm
Peruvian Current/Humboldt Current	Cold
West Wind Drift Current	Cold
North Atlantic Ocean	
North Equatorial Current	Warm
Florida Current	Warm
Antilles Current	Warm
Cayenne Current	Warm
Gulf stream	Warm
North Atlantic Drift	Warm
Labrador Current	Cold
Irminger or Greenland Current	Cold
Canaries Current	Cold
South Atlantic Ocean	
South Equatorial Current	Warm
Brazilian Current	Warm
Falkland Current	Cold
Benguela Current	Cold
West Wind Drift	Cold
Guinea Current	Cold

Currents of the Indian Ocean

- The Asiatic monsoon influences the currents of the North Indian Ocean, while the currents of South Indian Ocean are influenced by the atmosphere's anticyclonic circulation.
- **Equatorial Current** This current flows from East to West and upon reaching the East coast of Africa, a good portion turns Southward crosses the equator and becomes the Mozambique current.
- **Mozambique Current** It flows South along the East coast of Africa near the equator to about 35°S, where it becomes Agulhas stream.
- **Agulhas Stream** It flows Westward along the South-West of Madagascar and joins the Mozambique current along the East African coast.
- **North-East Monsoon Drift** It flows along the coast of India during winter.
- **South-West Monsoon Drift** It flows along the coast of India during summer.
- **Somalia Current** It is an ocean boundary current that runs along the coast of Somalia and Oman in the Western Indian ocean. It is similar to the Gulf Stream in the Atlantic ocean.

- **West Australian Current** It flows along west coast of Australia from poles of equator.
- There is no counter-equatorial current in the Indian Ocean rather only monsoonal currents which are changing its direction with respect to seasons.

TIDES

- The periodic rise and fall of the sea level as a result of the gravitational forces between the Earth, the Moon and the Sun is called a tide.
- The time interval between two tides is 12 hours and 26 minutes.

Spring Tides

- When the Earth, the Moon and the Sun are in a straight line (SYZYGY), the Sun assists the gravitational pull of the Moon, creating a condition of higher high tides and lower low tides known as spring tides.

Neap Tides

- When the Sun and Moon are at right angles to the Earth (Quadrature), the Sun partially reduces the gravitational pull of the Moon, producing lower high tides typical of neap tides.

CONTINENTS

- The word continent was derived from the term continent land,meaning continuous or connected land. A continent is one of Earth's seven main divisons of land. There are seven continents in the world.

Area of the Continents

Continent	*% of Earth Area*
Asia	29.5
Africa	20.4
North America	16.3
South America	11.8
Antarctica	9.6
Europe	6.7
Australia	5.3

ASIA

Physiography

- It has 49 countries.
- **Latitude** 10° S and 80° N.
- **Longitude** 25° E and 170° W.
- **Oceans and Seas** Arctic ocean, Pacific ocean, Indian ocean, Red sea, Gulf of Aden, Persian Gulf, Gulf of Oman, Arabian sea, Bay of Bengal, China sea, Yellow sea, Okhotsk sea and Bering sea.
- **Straits** Strait of Malacca, Bering strait Hormuz strait, Dardenelly strait.



- **Islands** Kurile, Sakhalin, Honshu, Shikoku, Hokkaido, Taiwan, Borneo, Sumatra, Java, Celebes, New Guinea, Philippines, Sri Lanka, Bahrain, Cyprus.

Finer Points

- Caspian sea is the world's largest lake and five times larger than the Lake Superior.
- **Hwang Ho** is known as China's sorrow. It flows through loess land, hence it is also called Yellow sea.
- **Amur river** forms the boundary between Russia and China.
- **Yangtze Kiang** is the longest river of Asia.
- **Laos** is the only landlocked country in South-East Asian peninsula.
- Irawaddy river is known as the lifeline of Myanmar. In Gulf of Martaban where pearls are found.
- Fujiyama, a volcanic mountain is the highest peak of Japan.

- Japan is the most industrialised nation of Asia. Japan is called Land of Rising Sun.
- Myanmar is called land of mountains and rivers.
- Pakistan is called country of canals.
- Dead sea, one of the saltiest water body in the world, is a landlocked sea. It lies between Israel and Jordan.
- Osaka is called the Manchester of Japan.
- Largest island of Asia is Borneo.
- Longest river of Asia is Yangtze of China.
- Largest desert of Asia is Gobi.
- Most densely populated country in Asia is Singapore.
- Biggest producer and exporter of rubber in the world is Thailand.
- Largest gold mine in the world is Grasberg.

Structural Aspects of Asia

Mountains	Himalayas, Karakoram, Kailash, Kunlun shan, Tien shan Altai, Sayan, Yablonovy, Stanvoy, Kolyama, Verkhyoyansk, Pegu Yoma, Arakan Yoma, Hindukush, Elburz, Sulaiman, Kirthar, Makran, Zagros, Pontic, Taurus
Mountains from Pamir Knot	South-West: Sulaiman South-East: Himalaya, Karakoram and Kunlun North-East: Tien shan North-West:Hindukush
Mountains from Armenian Knot	South-West : Taurus South-East: Zagros ; North-East: Elburz ; North-West: Pontic
Rivers	Ob, Yenisey, Amur, Yalu, Hwang Ho, Si kiang, Mekong, Tigris, Eupharates, Amu Darya and Syr Darya
Lakes	Baikal, Balkash, Van Golu, Asad, Dead Sea, Tonle Sap Toba, Lop Nor, Caspian sea and Sea of Gallilee
Plains	Manchurian, Great Plain of China, West Siberian Plain, Mesopotamian
Deserts	Rub-al-Khali, Al Nafud, Dasht-I-Kavir, Dasht -I-Lut, Gobi, Thar, Takla Makan, Syrian Desert
Plateaus	Ladakh, Tibet, Yunan, Pamir, Armenian, Iranian, Mongolia, Indo-China, Shan, Deccan, Baluchistan, Arabian, Anatolia, Loess

Socio-Cultural Aspects

- Asia is the birth place of great religions like Judaism, Christianity and Islam from the Mediterranean lands; Confucianism from China; Hinduism, Buddhism and Jainism from India.
- The ancient river valley civilisations of Asia are the Indus valley, the Mesopotamian and the Chinese.
- There are also contrasts in the development of languages. eg., Tamil, Japanese, Chinese and Arabic are rich in literature while some have only dialects with no scripts.
- *The three most prominent racial groups in Asia are as follow*

1. **The Arabian Group** Arabs, Iranians, Jews, Turks.
2. **The Indian Group** India, Pakistan and Bangladesh.
3. **The Mongoloid** Chinese, Japanese, Koreans and most of the people of South-East Asia.

Main Tribes

- Abhors, Afridis, Khirgiz, Kurds, Nicobaris and Mundas, etc.

Economic Aspects

- Asia's wealth is in its agricultural resources which include fertile soil and water for irrigation.
- China, India, Thailand are the leading producers of rice, wheat and other coarse cereals.
- Gulf countries are known for its rich oil and gas resources.

AFRICA

Physiography

- The planet's second largest continent includes 54 individual countries.
- **Latitude** 35° S and 37° N.
- **Longitude** 50° E and 17° W.



- **Oceans and Seas** Indian ocean, Red sea, Atlantic ocean, Madagascar sea, Gulf of Guinea, Mediterranean sea.
- **Straits** Strait of Bab-el-Mandeb, Strait of Gibraltar, Mozambique channel.
- **Islands** Madagascar, Cape Verde Islands, the Comoros, Zanzibar, Mauritius, Seychelles.
- **Plateaus** The entire continent is a plateau.

Structural Aspects of Africa

Mountains and Plateau	Atlas, Ethiopian Highlands, Mt Kenya, Mt Elgon, Mt Kilimanjaro, Drakensberg, Mt Cameroon, Mt Rouwenzori, Katanga Plateau, Jos Plateau
Rivers	Nile, Zaire or Congo, Niger, Orange, Limpopo, Zambezi
Lakes	Kariba, Nayasa, Mweru, Tanganyika, Edward, Tana, Nasser, Chad, Volta, Assal, Victoria
Deserts	Sahara, Libyan, Arabian, Nubian, Namib, Kalahari

Finer Points

- Lake Victoria is the largest lake of Africa, which is located between Uganda, Kenya and Tanzania. It is source of White Nile river. The equator passes through it.
- Nasser lake is a man-made lake, which lies on Nile river and located between Egypt and Sudan.
- Nile river is the longest river of the world and life line of Egypt.
- The White Nile and the Blue Nile meet at Khartoum (Sudan) to form river Nile.
- South Sudan is 54th and new born country of Africa and its capital is 'Zuba'.
- **Congo river** cuts equator twice.
- The Zambezi river includes the Victoria fall, one of the largest falls in the world. It makes the natural political boundary between Zambia and Zimbabwe.
- The Orange river forms the natural boundary between South Africa and Namibia.
- The Limpopo river cuts the Tropic of Capricorn and it separates South Africa from Botswana and Zimbabwe.
- The highest peak of Africa is Mt Kilimanjaro and is located in Tanzania.
- The three points where Africa almost touches Eurasia are Gibraltar, Suez and Bab-el-Mandeb.
- The highest temperature in the world has been recorded at **Al-Aziziyah** (Libya) at 58°C making it the hottest place in the world.

Socio-Cultural Aspects

- The blacks, living in the equatorial and tropical region form about 70% of the population.
- Swahili is a famous language of Africa apart from French, English and other regional/tribal languages.
- The pygmies of Congo basin are notable for their short stature.

Main Tribes

- Bushmen, Masai, Berbers, Zulu, Nubia, Dogon, Bemba, Afar, Amhara, etc.

Economic Aspects

- Africa is rich in mineral deposits and these minerals are mostly found on the plateau region, South of the equator. The minerals found in Africa are diamond, gold, copper, bauxite, platinum, iron ore, petroleum, manganese, etc.
- Cotton is the chief cash crop of Africa. Egypt and Sudan grow the best quality long staple cotton in the world.
- Ethiopia and Kenya are the leading producers of coffee.

- Oil palm is a part of the natural vegetation and large quantities of palm oil is produced in the West African countries.
- Zanzibar leads in the export of cloves.

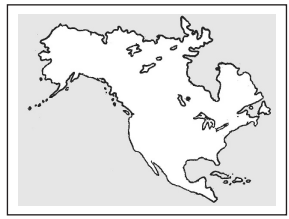
Leading Producers in Africa

Agricultural Produce	*Leading Producer*
Tea	Kenya
Wheat	Ethiopia
Rubber	Ghana
Mineral	*Leading Producer*
Gold	Ghana
Copper	Zaire
Phosphates	Morocco

NORTH AMERICA

Physiography

- North America, the planet's third largest continent includes 23 countries and more than 12 small islands.
- **Latitude** 7° N and 84° N.
- **Longitude** 70° W and 180° W.



- **Major Deserts** Chihuahuan, Colorado, Mojave, Sonoran.
- **Oceans and Seas** Atlantic ocean, Caribbean sea, Gulf of California, Gulf of Alaska, Bering sea, Hudson Bay.
- **Islands** Greenlands, Baffin, Victoria, New Foundland, Cuba, Jamaica, Haiti, Bermuda, Hawaii.

Structural Aspects of North America

Mountains	Brook's range, Alaska and Aleutian range, Cascade range Rocky mountain, Coast range, Sierra Nevada, etc
Rivers	Mississippi-Missouri, St Lawrence, Colorado, Columbia, Sacramento, Rio Grande, Yukon, Mackenzie, Nelson, Saskatchewan, Peace, etc
Lakes	Great Bear, Great Slave, Athabasca, Winnipeg, Superior, Michigan, Huron, Erie, Ontario, Great Salt, Mead
Plateaus	Columbia-Snake, Colorado, Mexican
Peninsula	Ungava, Yucatan, Kenai

Finer Points

- 49° latitude parallel forms the boundary between Canada and USA (longest boundary in the world) and 100° W longitude divides the North America into more or less two equal parts.
- Canada has the longest coastline in the world and it is the second largest country (in area) in the world.
- USA is the fourth largest country in area and third largest country (in population) in the world.
- Erie canal, Great lakes waterways, St Lawrence seaway are some of the most important canals of North America.
- North America is rich in mineral resources such as iron ore, petroleum and natural gas, copper, gold, silver, nickel, sulphur, coal, etc.
- The prairie region of North America is ideally suited for the cultivation of wheat.
- The coniferous forests of the North America provides large quantities of soft wood. The prominent trees are fir, pine, larch and spruce, etc.
- Black hills, Blue hills and Green hills are located in USA.
- Longest river of North America is Missouri.
- **Lake Superior** is the largest fresh water lake in the world.
- Canada is the largest producer of newsprint in the world.
- A worker in the logging industry in North America is called as lumberjack.
- The Panama canal connects Caribbean sea and Pacific ocean. By using Panama canal, the distance from New York to San Francisco can be shortened by a nearly 23,200 km.

Socio-Cultural Aspects

- Eskimos who live in the frozen North belong to the Mongoloid race.
- The Spanish, Portuguese, French, Dutch and British set-up colonies along the coast of North America.
- English, Spanish and French are popularly spoken languages in this continent.
- Most of the people belong to Christianity.

Main Tribes

Abenaki, Crow, Innu, Shawnee, Yurok, Cree, Omaha and Seneca.

Economic Aspects

- Anglo-America is rich in iron ore. It is mined in the USA as well as in Canada.
- Grand Bank of Newfoundland coast is famous for fishing activites.
- North America is a leading producer of wheat, maize and meat.
- The Canadian Shield is a storehouse of ores of nickel, lead, zinc, gold, platinum and copper.
- Hollywood is the centre of the film industry, in Los Angeles, California.
- The Silicon valley is the centre of the electronic and computer industry.

SOUTH AMERICA

Physiography

- South America, the fourth largest continent has 13 countries.
- **Latitude** 12° N and 55° S.
- **Longitude** 35° W and 81° W.
- **Straits** Strait of Magellan, Drake Passage.
- **Deserts** Atacama, Patagonia.



Structural Aspects of South America

Mountains	Western Mountains, Andes, Brazilian highlands, Guiana highlands
Rivers	Amazon, Magdalena, Orinoco, Parana, La Plata
Lakes	Maracaibo, Titicaca, Popo
Plateaus	Bolivian, Patagonia, Altiplano

Finer Points

- South America has been divided into three physical regions: mountains and highlands, river basins and coastal plains.
- **Cotopaxi** in Ecuador is one of the highest active volcanoes in the world.
- Paraguay and Bolivia are the only landlocked countries.
- **Pampas** is the most fertile region of South America and alfa-alfa grasses are found here.
- South America as well as Mexico, Central America and West Indies are collectively known as **Latin America**.
- It has the world's highest waterfall i.e., **Angel fall** in Venezuela on Orinoco river.
- It has the world's second longest river after Nile and the largest river by volume i.e., Amazon river.
- The longest mountain range of the world i.e., the Andes lies in South America.
- Moreover, the driest place on Earth i.e., **Atacama desert**, the largest rain forest i.e., the Amazon rain forest, the highest capital city i.e., Lapaz (Bolivia), the highest commercially navigable lake i.e., Lake Titicaca are situated in South America.
- **Llanos** and **Campos** are the Savanna grassland in South America. Selvas are the equatorial rainforests of Amazon basin.
- Brazil is the only country, through which both equator and one of the tropics (Tropic of Capricorn) pass.

Socio-Cultural Aspects

- Different tribes of American Indians (Amerinds) were the original inhabitants of South America.

- **Machu Picchu** is an Incancitadel set high in the Andes Mountain in Peru.

Indigenous People

- Aymara, Awa, Cocama, Juris, Shipibo, Yagua, Shuar, Wai-Wai.

Economic Aspects

- Some of the countries of South America are specialised in coffee plantation (Brazil, Ecuador, Columbia and Venezuela, etc), maize cultivation and wheat cultivation.
- South America is rich in minerals such as petroleum, aluminium, bauxite, copper, gold, lead, nickel, nitrates, diamond, coal, etc.
- Brazil has the world's largest reserves of iron is Serra Dos Carajas mountains.
- Brazil is also known as the 'coffee bowl of the world' because it is the largest producer of coffee.
- Peru is the leading producer of lead and Colombia is the leading producer of coal.

EUROPE

Physiography

- Europe, the planet's sixth largest continent includes 50 countries and is part of the Peninsula of Eurasia.
- **Latitude** 35° N and 73° N.
- **Longitude** 25° W and 65° E.



- **Ocean and Seas** Atlantic ocean, Arctic ocean, Mediterranean sea, Caspian sea, Black sea, White sea, North sea, Norwegian sea, Baltic sea, Gulf of Bothnia, Gulf of Finland, Bay of Biscay, Aegean sea, Adriatic sea.
- **Straits** Dardenelles Strait, English Channel Straits of Gibraltar.
- **Lakes** Lake Ladoga, Onega, Peipus, Vanern, Vattern.
- **Islands** British Isles, Iceland, Sardinia, Sicicly, Crete.

Important Gulf and Bays

Gulf/Bay	*Location*
Gulf of Bothnia	Sweden and Finland
English Channel	Britain and France
Gulf of Lions	South of France
Bay of Biscay	France and Spain

Structural Aspects of Europe

Mountains	Vesuvius, Appennine, Vosges, Alps, Mt Etna, Dinaric Alps, Carpathian, Pindus, Ural, Caucasus, Black Forest, Mt Stromboli, Kjolen, Pennines, Cantabrian, Pyrenees, Sierra Morena, Sierra Nevada
Rivers	Po, Tiber, Rhone, Ebro, Garonne, Loire, Seine, Weser, Elbe, Wista, Onega, Dvina, Mezen, Volga, Don, Dniepe, Dnister, Danube, Rhine
Peninsula	Iberian, Balkan, Kola

- Europe is situated between Ural mountains in the East and Atlantic Ocean in the West of Asia and North of Africa. To the North of Europe lies Arctic ocean, to the South lies the Mediterranean sea, the Black sea and the Caucasus mountain and to the East Ural mountains and the Caspian sea.
- **Greenland** is the world's largest island and it belongs to Denmark.
- **Balkan States** It is a group of 9 countries namely Serbia, Montenegro, Slovenia, Croatia, Bosnia- Herzegovina, North Macedonia, Bulgaria, Greece, Romania, Albania and Kosovo. However, Montenegro became an independent state in 2006 and Kosovo in 2008.
- **Baltic States** It is a group of three countries namely Estonia, Lithuania and Latvia.
- Norway, Sweden and Denmark are collectively called as **Scandinavian** countries.
- Finland, Iceland, Norway, Sweden and Denmark are collectively called as Nordic countries.
- United Kingdom is the name given to the combination of Great Britain and Northern Ireland. Great Britain consists of England, Scotland and Wales.

Socio-Cultural Aspects

- Europeans belong to the white race known as the Caucasian race.
- There are three distinctive people in Europe : the Nordic people, the Mediterranean and the Mongoloid race.
- English, French, Spanish, Russian, German and Italian are most infeuential languages. Maximum number of people belong to Christianity.

Economic Aspects

- Both intensive and extensive cultivation are done in Europe with a great variety of crops due to the occurrence of different kinds of soil and climate. On the steppes, wheat, sugar beet and sunflowers are the major crops, whereas in the Mediterranean region, fruit cultivation is prominent. Moreover, cattle, sheep and pigs are reared for meat, milk and other products.
- The continental shelf areas around Europe including Dogger Bank are major fishing zone.
- Russia is the largest country of the world and the highest populated country of Europe.
- Vatican city is surrounded by Rome, Italy. It is the smallest country of the world both in terms of area and population.
- UK is one of the most industrialised nations of the world, though it lacks the raw material. It is self-sufficient in oil production.

AUSTRALIA

Physiography

- It is an island continent.
- **Latitude** 10°41'S and 39°S.
- **Longitude** 114°E and 154°E.
- **Islands** Tasmania.



Structural Aspects of Australia

Mountains	Great Dividing range, Darling range, Musgrave, Flinders, Australian Alps
Deserts	Gibson, Great Sandy, Great Victoria, Simpson, Little Sandy, Western desert, Rangipo, Tanami, etc
Rivers	Murray, Darling, Swan

Finer Points

- Australia is known as forgotten land. It was discovered by Captain James Cook in 1770.
- It is the smallest continent of the world. It lies entirely in the Southern hemisphere. The Tropic of Capricorn runs almost through the middle of the continent and divides the continent in two equal parts.
- It is the only continent that is also a country and it lies between Pacific and Indian ocean.
- To the South-East of the mainland lies the mountainous island of Tasmania.
- Largest city of Australia is Sydney .
- The highest peak of Australia is **Mt Koscuisko** (2228 m).
- **Great Barrier Reef** is the world's longest coral reef.
- The highest peak of New Zealand is Mt Cook (3724 m).

Socio-Cultural Aspects

- Aboriginal Australians were the original inhabitants of the continent. The gold rush brought new settlers from the European countries.

Economic Aspects

- Agriculture is carried out in the Murray Darling basin, around the river Swan in Perth and along the rainy East coast.
- Coal, iron ore, bauxite, uranium, gold, petroleum are the major mineral resources of Australia.
- Tropical forests are found in North coastal region, which gets monsoon rain.
- The Australian trade is based on agriculture, animal rearing and minerals.

ANTARCTICA

Physiography

- It is the fifth largest and least populous of the continents.
- It was discovered in 1820. Roald Amundsen was the first man to reach geographical South pole in Antarctica.
- It is known as the continent for Science.



- Mt Erebus is the only active volcano on Antarctica.
- Mt Vinson (4892 m) is the highest peak of Antarctica.
- It is the only continent, which is completely frozen. It is, therefore, known as White Continent.
- Scientific studies have shown Antarctica to be rich in gold, platinum, coal, nickel, hydrocarbon and iron ore.
- Some animals do manage to survive in the cold and dry climate of antarctica. These animals include penguins, seals and the snow petrel (a bird).
- About 98% of Antarctica is covered by ice that averages 1.9 km in thickness.

Highest/Lowest Points of Various Continents

Continent	*Highest Point (m)*	*Lowest Point (m)*
Asia	Mt Everest (8848)	Dead Sea (–427)
Africa	Mt Kilimanjaro (5895)	Lake Assal (–155)
North America	Mt Mckinley (6198)	Death Valley (–86)
South America	Mt Aconcagua (6960)	Laguna Del Carbon (–105)
Antarctica	Vinson Massif (4892)	Deep Lake, Vestfold Hills (–50)
Europe	Mt Elbrus (5642)	Caspian Sea (–28)
Australia	Mt Koscuisko (2228)	Lake Eyre (–16)

Busiest Straits

Strait	*Water Bodies Joined*
Bab-al-Mandeb	Red Sea and Arabian Sea
Bering	Arctic Ocean and Bering Sea
Bosporus	Black Sea and Sea of Marmara
Dover	North Sea and English Channel
Florida	Gulf of Mexico and Atlantic Ocean
Gibralter	Mediterranean Sea and Atlantic Ocean
Malacca	Andaman Sea and South China Sea
Palk	Bay of Bengal and Palk Bay
Magellan	South Pacific and South Atlantic Ocean
Sunda	Java sea and Indian Ocean

Major Rivers of the World

River	*Origin*
Nile	Victoria Lake
Amazon	Andes (Peru)
Yangtze	Tibetan Plateau, (Tanggula Mountain)
Mississippi Missouri	Itaska lake (USA)
Yenisei	Mungaragiyn Gol Mountain, Mongolia
Huang Ho	Kunlun Mountains
Ob	Altai Mountains, Russia
Congo	Lualaba and Luapula rivers
Amur	Onon River, Shilka River
Lena	Baikal Mountains
Mekong	Tibetan Highlands
Niger	Guinea Highlands

Riverside Cities

Town	*River*
Yangon (Myanmar)	Irrawaddy
Baghdad (Iraq)	Tigris
Basra (Iraq)	Tigris and Euphrates
Belgrade (Serbia)	Confluence of Sawa and Danube
Berlin (Germany)	Spree

Town	*River*
Bristol (UK)	Avon
Budapest (Hungary)	Danube
Cairo (Egypt)	Nile
Alexandria (Egypt)	Nile
Glasgow (Scotland)	Clyde
Kabul (Afghanistan)	Kabul
Karachi (Pakistan)	Indus
Khartoum (Sudan)	Nile
Lahore (Pakistan)	Ravi
Lisbon (Portugal)	Tagus
London (UK)	Thames
Lucknow (India)	Gomti
Montreal (Canada)	Ottawa
Nanking (China)	Yangtze
New Castle (UK)	Tyne
New Orleans (USA)	Mississippi
New York (USA)	Hudson
Paris (France)	Seine
Philadelphia (USA)	Confluence of Schuylkill and Delaware
Rome (Italy)	Tiber
Shanghai (China)	Yangtze
Srinagar (India)	Jhelum

Important Canals of the World

Canal	*Connects*
Panama	Pacific Ocean with Atlantic Ocean
Suez	Mediterranean Sea to Red Sea
Erie	Atlantic Ocean to Great Lakes
Kiel	North Sea to Baltic Sea

Major Lakes of the World

Lake	*Location*
Caspian	Europe and Asia
Superior	North America
Victoria	Africa
Huron	North America
Tanganyika	Africa
Baikal	Asia (Russia)
Great Bear	North America
Aral	Asia
Great Slave	North America

Major Waterfalls

Waterfall	*Location*
Angel Falls	Venezuela
Tugela Falls	South Africa
Olo'upena Falls	United States
Catarata Yumbilla	Peru

Major Islands

Island	*Area* $(km)^2$	*Country*
Greenland	2130800	Denmark
New Guinea	785753	Indonesia and Papua New Guinea
Borneo	743330	Indonesia, Malaysia and Brunei
Madagascar	587041	Madagascar
Baffin Island	507451	Canada
Sumatra	473481	Indonesia

Leading Mineral Producers

Mineral	*Leading Producer*
Gold	China
Bauxite	Australia
Copper	Chile
Platinum	South Africa
Chromium	South Africa
Antimony	China
Tungsten	China
Phosphate	Morocco
Manganese	South Africa
Diamond	Russia
Iron ore	China
Petroleum	USA

Industrial Cities

City	*Industry*
Anshan (China)	Iron and Steel
Baku (Azerbaijan)	Petroleum
Belfast (Northern Ireland)	Ship Building
Birmingham (UK)	Iron and Steel
Chicago (USA)	Meat Packing
Detroit (USA)	Automobile
Havana (Cuba)	Cigars
Hollywood (USA)	Films
Johannesburg (South Africa)	Gold Mining
Kansas City (USA)	Meat Packing
Kawasaki (Japan)	Manufacturing
Kimberley (South Africa)	Diamond Mining
Krivoi Rog (Ukraine)	Iron and Steel
Leeds (UK)	Woollen Textiles

City	*Industry*
Saint Petersburg (Russia)	Ship Building
Los Angeles (USA)	Petroleum
Lyon (France)	Silk Textiles
Magnitogorsk (Russia)	Iron and Steel
Manchester (UK)	Cotton Textile
Milan (Italy)	Automotive
Multan (Pakistan)	Pottery
Munich (Germany)	Lenses
Nagoya (Japan)	Automobiles
Philadelphia (USA)	Locomotives
Rourkela (India)	Iron and Steel
Sheffield (UK)	Cutlery
Vladivostok (Russia)	Ship Building
Wellington (New Zealand)	Dairying

Major Mines

Mine	*Mineral*
Kimberley (South Africa)	Diamond
Lubumbashi (Democratic Republic of Congo)	Copper
Hwange (Zimbabwe)	Coal
Witwatersrand (South Africa)	Gold
Katanga (Democratic Republic of Congo)	Copper and Cobalt
Kasai Province (Democratic Regpublic of Congo)	Diamond

Important International Boundary Lines

Name of Boundary Line	*In Between Countries*
Radcliffe Line	India and Pakistan
Mc Mohan	India and China
Durand Line	Pakistan and Afghanistan
Hindenburg Line	Germany and Poland
Maginot Line	France and Germany
Oder Neisse Line	Germany and Poland
Siegfried Line	Fortification between Germany and France
38th Parallel Line	North and South Korea
49th Parallel Line	USA and Canada
24th Parallel Line (Sir Creek)	Pakistan claims that it is the boundary between India and Pakistan in Rann of Kachchh

Changed Names of Countries

Old Name	*New Name*
Abyssinia	Ethiopia
Angora	Ankara
Batavia	Jakarta
British Guiana	Guyana
Burma	Myanmar
Ceylon	Sri Lanka
Constantinople	Istanbul
Dutch East Indies	Indonesia
Gold Coast	Ghana
Holland	The Netherlands
Malaya	Malaysia
Nippon	Japan
Northern Rhodesia	Zambia
Persia	Iran
Rhodesia	Zimbabwe
Siam	Thailand
South-West Africa	Namibia

Distinctive Tribes

Tribe	*Residing Area*
Abhors	Mongoloids living in North-Eastern Part of India
Afridis	Tribes residing in the North-West Frontier (Pakistan)
Bantus	Negroes living in the Central and South Africa
Boers	The Dutch settlers of South Africa
Flemings	A term used for the people native to Belgium
Magyars	Inhabitants of Hungary
Maoris	Inhabitants of New Zealand
Negroes	Dark skinned group of people originally native to Africa South of Sahara
Pygmies	Short stature people found in Congo basin in Africa
Red Indians	Original inhabitants of North America
Semites	Caucasian people of ancient times

INDIAN GEOGRAPHY

INDIA AT A GLANCE

- India is the **seventh largest country** in the world with an area of 3287263 sq km, which is 2.42% of world's area.
- India is the **second most populous country** in the world with a population of 1.21 billion (2011), which is 17.44% of the world's population.
- Indian **sub-continent** is located in the Northern and Eastern hemisphere.
- India shares **longest boundary** with Bangladesh (4096 km), followed by China (3488 km), Pakistan (3323 km), Nepal (1751 km), Myanmar (1643 km), Bhutan (699 km) and Afghanistan (106 km).
- In India, the **Tropic of Cancer** (23.5° N latitude) passes through 8 states (Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram).
- **Islands** Andaman and Nicobar island group in the Bay of Bengal; Lakshadweep island group in the Arabian sea.
- **Indian Standard Time** (IST) The 82°30' E longitude is taken as the Standard Meridian of India, as it passes through middle of India (from Naini, near Allahabad).
- The 82° 30′ E Meridian also decides the time in Sri Lanka and Nepal.
- On the South-Eastern side, the Gulf of Mannar and the Palk Strait separates India from Sri Lanka.

Indian States Situated on the Border

Neighbouring Country	*Indian States Sharing their Borders*
Pakistan (5)	Gujarat, Rajasthan, Punjab, Jammu and Kashmir and Ladakh
Afghanistan (1)	Ladakh
China (5)	Ladakh, Uttarakhand, Himachal Pradesh, Sikkim, Arunachal Pradesh
Nepal (5)	Uttar Pradesh, Uttarakhand, Bihar, West Bengal, Sikkim
Bhutan (4)	Sikkim, West Bengal, Assam, Arunachal Pradesh
Bangladesh (5)	West Bengal, Assam, Meghalaya, Tripura, Mizoram

India Facts

Latitudinal extent	8°4' North to 37° 6' North
Longitudinal extent	68°7' East to 97° 25' East
North-South extent	3214 km
East-West extent	2933 km
Land frontiers	15200 km
Total coastline	7516.6 km
Number of states	28
Number of Union Territories	8
Land neighbours (7)	Pakistan, Afghanistan, China, Nepal, Bhutan, Bangladesh and Myanmar
State with longest coastline	Gujarat
Active volcano	Barren Island situated in Andaman and Nicobar islands
Southernmost point	Indira point or Pygmalion point in Great Nicobar
Southernmost tip of mainland	Kanyakumari
Northernmost point	Indira Col (Border between China and the Siachen Glacier)
Westernmost point	West of Guhar Mota in Gujarat
Easternmost point	Kibithu in Arunachal Pradesh

Physical Features

Physiographic divisions of India are

(a) The Himalayan Range of Mountains
(b) The Peninsular Plateau
(c) The Great Plains of India
(d) The Coastal Plains
(d) The Islands of India

The Himalayas

- Himalayas means 'Abode of snow'. These are young fold mountains of tertiary period, which were folded over Tethys sea due to intercontinental collision.
- They are one of the youngest fold mountain ranges in the world and comprise mainly sedimentary rocks.
- They stretch from the Indus river in the West to the Brahmaputra river in the East.
- The Pamir, popularly known as the **Roof of the World** is the connecting link between the Himalayas and the high ranges of Central Asia.
- *They consists of three parallel ranges such as*
  (a) Himadri (Greater Himalayas)
  (b) Himachal (Lesser Himalayas)
  (c) Shiwalik (Outer Himalayas)

Greater Himalaya (Himadri)

- Northernmost part of the Himalayan range, it is one of the world highest region with an average altitude of 6100 m above sea level.
- It includes world's highest peak, Mt Everest (8848 m) located on border of Nepal and China. It is known as Sagarmatha in Nepal and Chomolungma in China.
- Zanskar range is situated on the Western part of Greater Himalayas. It includes Nanga Parbat (8126 m) in Gilgit Baltistan and Dhaulagiri (8172 m) in Nepal.
- **Important Peaks** Kanchenjunga (8598m, Sikkim), Makalu (8481 m, Nepal) Manaslu (8156 m, Nepal), Nanga Parbat (8126 m, Gilgit Baltistan), Kamet (7756 m, Uttarakhand) and Nanda Devi (7816 m, Uttarakhand).

Important Passes in Greater Himalayas

Pass	*Location*	*Connectivity*
Karakoram	Ladakh	India to China
Burzil	PoK	Kashmir valley to Gilgit
Zoji La	Ladakh	Srinagar to Leh
Shipki La	Himachal Pradesh	Shimla to Gartok (Tibet)
Jelep La	Sikkim	Sikkim to Lhasa (Tibet)
Yang Yap	Arunachal Pradesh	Entry of Brahmaputra river

Middle Himalayas or Lesser Himalayas (Himachal)

- *From West to East, middle Himalaya is divided into following ranges*
  (a) Pirpanjal Range (Jammu and Kashmir). It is longest range of the middle Himalaya
  (b) Dhauladhar Range (Himachal Pradesh)
  (c) Mussoorie Range (Uttarakhand)
  (d) Nag tibba Range (Uttarakhand)
  (e) Mahabharat Range (Nepal)

Important Passes in Middle Himalayas

Pass	*Location*	*Connectivity*
Pirpanjal	Jammu and Kashmir	Jammu-Srinagar road passes from this pass
Banihal	Jammu and Kashmir	Jammu-Srinagar NH-44 passes from this pass. Jawahar tunnel (India's longest road tunnel)
Rohtang	Himachal Pradesh	Kullu valley with Lahaul and Spiti valley in Himachal Pradesh

- Average altitude of the mountain is 3700-4500 km.
- Important hill resorts are Shimla, Ranikhet, Almora, Nainital, Darjeeling.

Outer Himalayas (Shiwalik)

- These are the Southern most Himalayan Mountain Range.
- Their average altitude varies from 600 to 1500 metres.
- They are known as Jammu Hills in Jammu and Dafla, Miri, Abor, Mishmi Hills in Arunachal Pradesh.
- They are the youngest part of Himalayas.
- They form the foothills of Himalayas.

Trans Himalayan Zones

- This zone lies to the North of the Greater Himalayas.
- Some important ranges of this zone are Karakoram and Ladakh etc. The highest peak in this region is K_2 or Godwin Austin or Qagir (8611m, in Pak occupied Kashmir). K_2 is also the second highest peak of the world and the highest peak of India, located in Karakoram range.
- **Mt Rakaposhi** is the highest peak in Ladakh range and the steepest peak in the world.
- **Siachin Glacier** is 2nd longest glacier of the world outside the polar region (more than 75 km) and is located in Nubra valley (Ladakh). Baltoro, Biafo, Batura, Hispar are the other important glaciers in this region.
- The Northernmost point of LOC (NJ 9842) is located in the Karakoram ranges.

Himalayan Mountain Peaks

Peak	*Height (m)*	*Country*
Mt Everest	8848	Nepal-China
K_2	8611	PoK
Kanchenjunga	8586	Nepal-India
Lhotse	8516	Nepal-China
Makalu	8481	Nepal-China
Cho oyu	8201	Nepal-China
Dhaulagiri	8172	Nepal
Manaslu	8163	Nepal
Nanga Parbat	8125	Pakistan
Annapurna	8091	Nepal
Gasherbrum I	8080	Pakistan-China
Broad peak	8051	Pakistan-China

Mountain Peaks in India

Peak	*Height (m)*	*State*
K_2	8611	PoK
Kanchenjunga	8586	Sikkim
Nanda Devi	7817	Uttarakhand
Saltoro	7742	Jammu and Kashmir
Kangto	7090	Arunachal Pradesh
Reo Purgil	6816	Himachal Pradesh
Saramati	3841	Nagaland
Sandakphu	3636	West Bengal
Khayang	3114	Manipur
Anaimudi	2695	Kerala
Dodda Betta	2636	Tamil Nadu

Highest Peaks in India

Hill/Range	*Peak*
Himalaya	K_2 (8611 m, PoK)
Aravali	Gurushikhar (Mount Abu) (1722 m)
Western Ghat	Anaimudi (2695 m)
Eastern Ghat	Mahendragiri (1501 m)
Nilgiri	Doda Beta (2637 m)
Andaman and Nicobar	Saddle peak (731 m)
Naga hills	Saramati (3826 m)
Mizo hills	Blue mountain (2157 m)

The Great Plains

- To the South of the Himalayas and to the North of the peninsula lies the great plains of North India.
- It extends from West to East for 2400 km having an average width in between 150-300 km.
- The plains are formed by depositional works of three major river systems i.e., Indus, Ganga and Brahmaputra.
- The vast plains of North India are alluvial in nature and the Westernmost portion is occupied by the desert.
- It is composed of Bhangar (old alluvium), Khadar (new alluvium) in the river Bed, Bhabhar (plains containing porous gravel and pebbles where the streams disappear) and Terai (damp, thick forest area, where Bhabhar stream reappears).

Difference between Bhangar and Khadar

Bhangar	*Khadar*
These are low plains formed of older alluvium	The deposit of fresh alluvium every year brought by the Himalayan rivers makes this belt of world's most fertile plains
It contains calcareous deposits known as Kankar	It does not contain calcareous deposits of calcium

Difference between Terai and Bhabar

Terai	*Bhabar*
Terai is a broad long zone South of Bhabar plain	Bhabar is a long narrow plain along the foothills of Himalayas
It is a marshy damp area covered with dense forest	It is a pebble studded zone of porous rocks
It is 20-30 km wide	It is 9-16 km wide
It is most suitable for agriculture	It is unsuitable for agriculture

Doab

The Punjab - Haryana plain is drained by five rivers and the intervening area between the river is known as **doab**.

From South to North doabs are as follow

Doab	*Region*
▪ Bist	▪ Between Beas and Sutlej
▪ Bari	▪ Between Beas and Ravi
▪ Rachna	▪ Between Ravi and Chenab
▪ Chaj	▪ Between Chenab and Jhelum
▪ Sind	▪ Between Jhelum, Chenab and Indus

The Peninsular Plateau (Lava Plateau)

- Rising from the height of 150 m above the river plains upto the average elevation of 600-1000 m is the irregular triangle known as the peninsular plateau.
- It is composed of the old cystalline, igneous and metamorphic rocks.
- Narmada, which flows through a rift valley divides the region into two parts: the Central Highlands in the North and the Deccan plateau in the South. The Deccan plateau is the largest plateau in India.

Some Plateaus oft Peninsular India

The Central Highland

- The Central Highlands of the plateau lie to the North of the Naramada river covering a major area of the Malwa plateau.

The Deccan Plateau

- This plateau is a triangular land lying to the South of the river Narmada. It is made up of lava flows in the cretaceous era through the fissure eruptions.

Meghalaya Plateau

- This plateau is separated from main block of the peninsular plateau by a gap called **Garo-Raj-Mahal gap**. From East to West, the plateau compresses Garo, Khasi, Jaintia and Mikir hills.

The Bundelkhand Upland

- It is located to the South of Yamuna river between Central India Plateau and the Vindhyan range. It is composed of granites and gneiss.

Telangana Plateau

- It is located in Western Andhra Pradesh, comprising the North- Eastern part of the Deccan plateau and has an area of about 148,000 sq km. The plateau is chiefly made of ancient pre-Cambrian gneiss. The plateau is drained by Godavari and Krishna rivers.

Karnataka Plateau(Mysore Plateau)

- It is located in Karnataka. The plateau has an area of 73000 sq miles. It is composed of ancient volcanic rocks, crystalline schists and granites. The plateau is drained by Godavari, Krishna, Kaveri, Tungabhadra, Sharavati and Bhima.

Chota Nagpur Plateau

- It covers mostly Jharkhand, Chhattisgarh and Purulia region of West Bengal.

Hill Ranges of the Peninsula India

Aravali Range

- Aravalis are the world's oldest fold mountains running in North-East to South-East direction. It is an example of **relic mountain**.

Satpura Range

- It is a series of seven mountains running in East-West direction, South of Vindhya and in between the Narmada and Tapi. It comprises of Rajpipla hills, Mahadeo hills and Maikal range.

Vindhyan Range

- This range acts as a water divide between Ganga river system with the river system of South India. The Maikal range forms a connecting link between Vindhya and Satpura.

Eastern Ghat

- It comprises the discontinuous and low hills that are highly eroded by the rivers such as the Mahanadi, the Godavari, the Krishna, the Cauveri etc.
- Some of the important ranges include the Javadi hills, the Velikonda range, the Nallamala hills, the Mahendragiri hills etc.

Western Ghat

- They are locally known by different names such as Sahyadri in Maharashtra, Nilgiri hills in Karnataka and Tamil Nadu and Anaimalai hills and Cardamom hills in Kerala.
- It runs from the South of the valley of river Tapi to Kanyakumari.
- The Eastern and the Western Ghats meet each other at the Nilgiri hills, which is also called blue mountain.

Passes in Sahyadris

There are three important passes in the Sahyadris

- Thalghat (between Mumbai and Pune)
- Palghat (between Palakkad and Coimbatore)
- Bhorghat (between Mumbai and Nasik)

The Coastal Plains

- *On the basis of location and active geomorphological processes, it can be broadly divided into*
  (a) Eastern coastal plains
  (b) Western coastal plains

Eastern Coastal Plains

- **Length** 1100 km
- **Width** 120 km
- Utkal Plain—Ganga plain to Mahanadi.
- Andhra Coastal Plain—Utkal plain to Pulicat lake (Andhra Pradesh).
- Tamil Nadu Plain—North of Chennai to Kanyakumari. Coromondal Coast is part of this plain.

Western Coastal Plains

- **Length** 1500 km
- *It is divided into*
  - Kathiawar Plains—Rann of Kachchh to Daman.
  - Malabar/Kerala Plains—Mangalore to Kanyakumari.
  - Konkan Plains—Daman to Goa.
  - Karnataka Plains—Goa to Mangalore.
  - Gujarat Plains lies to the East of Kachchh and Kathiawar

Difference between Eastern and Western Coast

Eastern Coast	*Western Coast*
Smooth outline	Dissected outline
Occurrence of deltas	Occurrence of estuaries
Less rainfall	More rainfall
Broader	Narrower
Long rivers	Short rivers

Islands

- *India has large number of islands, most of which are located in two groups*
  1. Andaman and Nicobar group
  2. Lakshadweep group
- Group of island is called archipelago.

Andaman and Nicobar Group

- It is located in the Bay of Bengal.
- There are nearly 325 islands in Andaman group, whereas the Nicobar group of islands consist of 247 islands.
- **Ten degree channel** separates Andaman group from Nicobar group.
- **Duncan passage** lies between South Andaman and little Andaman group.

Lakshadweep Group

- It is located in Arabian sea.
- **Minicoy** is the second largest and Southern most island and Androtт island is largest of this group.

- **Minicoy** is separated from rest of the Lakshadweep by **Nine degree** channel.
- **Eight degree** channel separates Lakshadweep group from Maldives.

Drainage in India

- Water drains in two directions of the main water divide line of India. 90% of water drains into Bay of Bengal and the rest drains into Arabian sea.
- Those Himalayan rivers, which originated before the formation of Himalaya are known as **antecedent rivers** such as Indus, Brahmaputra and Sutlej.
- *In India, the rivers can be divided into two main groups*

1. Himalayan rivers
2. Peninsular rivers

The Himalayan Rivers System

- *Himalayan river system is divided into three major river system, which are discussed below*

The Indus System

- The Indus also known as Sindhu, is the Westernmost of Himalayan rivers in India.
- It originates from a glacier near Bokhar Chu in the Tibetan region near Mansarovar lake.
- In Tibet, it is known as '**Singi Khamban**' or **Lion's mouth**.
- In Jammu and Kashmir, its Himalayan tributaries are Zanskar, Dras, Gortang, Shyok, Shigar, Nubra, Gilgit etc.
- Its most important tributaries which join Indus at various places are Jhelum, Chenab, Ravi, Beas and Sutlej.

The Ganga System

- The Ganga system is the second major drainage system of India.
- It rises in the Gangotri glacier near Gaumukh (3900 m) in the Uttarakashi district of Uttarakhand. Here, it is known as the Bhagirathi. At **Dev Prayag**, the Bhagirathi meets the **Alaknanda**, hereafter, it is known as the Ganga. The Alaknanda has its source in the Satopanth glacier above Badrinath.
- The Alaknanda consists of the Dhauli and the Vishnu Ganga, which meet at Joshimath or **Vishnu Prayag**.
- The other tributaries of Alaknanda such as the Pindar joins it at **Karna Prayag**, while Mandakini or Kali Ganga meets it at **Rudra Prayag**. At **Nand Prayag**, Alaknanda meets Nandakini. Dev Prayag, Karn Prayag, Nand Prayag, Rudra Prayag and Vishnu Prayag are collectively known as **Panch Prayag**.
- The left bank tributaries of Ganga are Ramganga, Gomti, Kali or Sarda, Gandak, Kosi, the Ghaghra.
- The right bank tributaries of Ganga are Yamuna and Son. Yamuna joins the Ganga at Allahabad.
- Kosi is called as *Sorrow of Bihar*, while Damodar is called as *Sorrow of Bengal* as these cause floods in these regions. Hooghly is a distributary of Ganga flowing through Kolkata.

The Brahmaputra System

- It is one of the largest rivers of the world.
- It is known as Tsangpo in Tibet, Dihang or Siang in Arunachal Pradesh, Brahmaputra in Assam and Jamuna in Bangladesh.
- Brahmaputra forms large number of riverine islands. **Majuli** is the largest riverine island in the world.
- The combined stream of Ganga and Brahmaputra forms the biggest delta in the world, the Sunderbans, covering an area of 58752 sq km. Its major part is in Bangladesh.
- Brahmaputra is volume wise largest river of India, whereas lengthwise Ganga is the longest in India.

The Peninsular River System

- *Peninsular river system can be divided in two groups*

East Flowing Rivers (or Delta Forming Rivers)

- East flowing rivers do not form estuaries except some small rivers.
- East flowing rivers drain into Bay of Bengal.
- East flowing rivers form Delta.

East Flowing Rivers

River	*Source*	*Length*	*Tributaries*
Mahanadi	North foothills of Dandakarnaya	857 km	Seonath, Hasdeo, Ib, Mand, Tel, Ong and Jonk
Godavari	Trimbak plateau of North Sahyadri near Nasik	1465 (longest river of peninsular India)	Penganga, Wardha, Wainganga Indravati, Sabari, Manjira
Krishna	North of Mahabaleshwar in the Western Ghats	1400 km	Bhima, Tungabhadra, Ghat Prabha, Malaprabha, Musi and Koyna
Cauveri	Rise in Brahmgiri range in Western Ghats	800 km	Herangi, Hemavati, Shimsa, Arkavati Kabani, Bhavani and Amravati etc

West Flowing Rivers (or Estuaries Forming Rivers)

- West flowing rivers form estuaries and do not form delta. West flowing rivers drain into Arabian sea.

West Flowing Rivers

River	*Source*	*Length*	*Description*
Sabarmati	Mewar in Aravali range	320 km	It falls into Gulf of Khambat and its tributaries are Hathmati, Sedhi, Wakul.
Mahi	Rises from Vindhyan range	533 km	Flows in Madhya Pradesh, Rajasthan and Gujarat.
Narmada	Amarkantak plateau	1312 km	It flowes through Madhya Pradesh, Maharashtra and Gujarat. Its tributaries are Hiran, Tawa, Banjar, Shar, Shakkar and Burhner. It flows into Gulf of Khambat.
Tapi	Rises from Multai on the Satpura range in Betul district	730 km	Also known as Twin or handmaid of Narmada. Its tributaries are Purna, Betul, Arunavati, Ganjal, etc.
Luni	Rises from Aravalis	482 km	Also called salt river. It is finally lost in the marshy grounds at the head of the Rann of Kachchh.

Comparison of Delta and Estuary

Delta	*Estuary*
It is the triangular deposition of alluvium at the mouth of river at it falls into the sea.	It is the funnel shaped mouth of river.
Excess deposition of slit and soil forms a delta.	Tidal waves remove the silt out and an estuary is formed.
It is more fertile land.	They provide critical habitats for many fish species.

Highest Waterfalls of India

Waterfall	*Height (m)*	*State*
Barehipani	400 m	Odisha
Nohkalikai	340 m	Meghalaya
Dudhsagar	310 m	Karnataka and Goa
Jog or Gersoppa	253 m	Karnataka
Duduma	157 m	Odisha and Andhra Pradesh
Bishop	135 m	Meghalaya
Hundru	100 m	Jharkhand
Shivasamundram	100 m	Karnataka
Langshiang	85 m	Meghalaya

Important Indian Towns on Rivers

Town	*River*	*Town*	*River*
Jamshedpur	Confluence of Kharkai and Subarnarekha	Ayodhya	Saryu
Delhi	Yamuna	Ahmedabad	Sabarmati
Kanpur	Ganga	Patna	Ganga
Surat	Tapti	Kota	Chambal
Firozpur	Sutlej	Jabalpur	Narmada
Allahabad	At the confluence of the Ganga and Yamuna	Panji	Mandavi
		Ujjain	Kshipra
Varanasi	Ganga	Guwahati	Brahmaputra
Haridwar	Ganga	Kolkata	Hooghly
Badrinath	Alaknanda	Cuttack	Mahanadi
Ludhiana	Sutlej	Hyderabad	Musi
Srinagar	Jhelum	Nashik	Godavari
Lucknow	Gomti	Coimbatore	Noyyal

River Projects and their Beneficiary States

Project	*River*	*Purpose*	*Beneficiary States*
Bhakra Nangal Project	Sutlej	Power and irrigation	Punjab, Himachal Pradesh, Haryana and Rajasthan
Damodar Valley	Damodar	Power, irrigation and flood control	Jharkhand and West Bengal
Hirakud	Mahanadi	Power and irrigation	Odisha
Tungabhadra Project	Tungabhadra	Power and irrigation	Andhra Pradesh and Karnataka
Nagarjunasagar Project	Krishna	Power and irrigation	Andhra Pradesh and Telangana
Gandak River Project	Gandak	Power and irrigation	Bihar, Uttar Pradesh, Nepal (joint venture of India and Nepal)
Kosi Project	Kosi	Flood control, power and irrigation	Bihar
Farakka Project	Ganga, Bhagirathi	Power, irrigation, removing accumulation of silt from Kolkata port	West Bengal
Beas Project	Beas	Irrigation and power	Rajasthan, Haryana, Punjab and Himachal Pradesh
Indira Gandhi Canal Project	Sutlej, Beas and Ravi	Irrigation	Rajasthan, Punjab and Haryana
Chambal Project	Chambal	Power and irrigation	Madhya Pradesh and Rajasthan
Kakrapara Project	Tapi	Irrigation	Gujarat
Ukai Project	Tapi	Power and irrigation	Gujarat
Tawa Project	Tawa (Narmada)	Irrigation	Madhya Pradesh
Poochampad Project	Godavari	Irrigation	Telangana

Project	*River*	*Purpose*	*Beneficiary States*
Malaprabha Project	Malaprabha	Irrigation	Karnataka
Durgapur Barrage	Damodar	Irrigation and navigation	Jharkhand and West Bengal
Iddukki Project	Periyar	Hydro electricity	Kerala
Koyna Dam	Koyna	Hydro electricity	Maharashtra
Sardar Sarovar Project	Narmada	Power and irrigation	Gujarat, Madhya Pradesh, Maharashtra, Rajasthan
Tipaimukh Dam	Barak	Flood control and Hydroelectric power generation	Manipur
Kishanganga Project	Kishanganga	Power	Jammu and Kashmir
Ramganga Multipurpose Project	Chui sot stream near Kalagarh	Power and irrigation	Uttarakhand
Matatilla Project	Betwa	Power and irrigation	Uttar Pradesh and Madhya Pradesh
Tehri Dam Project	Bhagirathi	Hydro electricity, Irrigation	Uttarakhand
Rihand Dam	Rihand	Hydro electricity	Uttar Pradesh
Mayurkashi Project	Mayurkashi	Irrigation	West Bengal
Nizamsagar Project	Manjira	Irrigation	Telangana
Shivasamudram Project	Cauveri	Irrigation	Karnataka

THE CLIMATE OF INDIA

- India has tropical monsoon type of climate. It is greatly influenced by the presence of Himalayas in the North as they block the cold air masses from Central Asia. It is because of Himalayas that the monsoon winds cause rainfall in India.
- The Tropic of Cancer (23.5°N) divides India into two almost equal climatic zones, namely, the Northern zone (subtropical) and the Southern zone (tropical).
- The Southern zone has the mid-day Sun almost vertically overhead at least twice every year and the Northern zone does not have the mid-day Sun vertically overhead during any part of the year.
- The sub-tropical climate of the Northern zone is responsible for cold winter season and hot summer season.
- The Southern tropical climate zone is warmer than the North and does not have a clear cut winter season.

Western Disturbances

These are the depressions generated over the Mediterranean sea and enter India after crossing over Iraq, Iran, Afghanistan and Pakistan, under the influence of Westerly jet stream. After reaching India, they move Eastwards causing light rain in the Indo- Gangetic plains and snowfall in Himalayan belt.

The factors influencing the climate of India are

(a) Physiography

Physiography affects temperature, atmospheric pressure, direction of wind and amount of rainfall.

(b) Monsoon Wind

It is the most dominating factor of Indian climate. The South-West summer monsoon from the Arabian sea and the Bay of Bengal bring rainfall to the entire country.

Besides the North-East winter monsoon travelling from land to sea causes rainfall along the Coromandel coast after acquiring moisture from the Bay of Bengal.

(c) El Nino

It is a narrow warm current, which occasionally appears off the coast of Peru in December by temporarily replacing the cold Peru current.

(d) La Nina

It is the reverse of El Nino. It is a harbinger of heavy monsoon showers in India.

Seasons in India

Season	*Temporal Aspect*	*Precipitation*	*Other Climatic Features*
Winter	November to March	Rainfall in sub- Himalayan belt by Western disturbances	Clear sky, low temperature and high humidity
Summer (Pre-monsoon)	March to June	Only 1% rainfall of total Indian rainfall mostly by storms by convective currents	Dry season, high temperature and low humidity
Rainy	June to September	Rainfall through South-West monsoon	High heat, high humidity, extensive cloud and several spells of moderate to heavy rainfall
Cool	Mid- September to November	Retreating monsoon causes rainfall in Tamil Nadu and adjoining areas of Andhra Pradesh	No clouds, severe and devastating tropical cyclones, clear sky

Local Winds and their Significance

Local Wind	*Significance*
Norwester	It is a spring storm shower and causes rainfall in Assam and West Bengal
Mango Shower	Thunderstorm causing rainfall in Southern plateau, helps in mango ripening
Cherry Blossoms	Thunderstorm causing rainfall in Karnataka, helps in flowering of coffee
Kalbaisakhis	Storms in West Bengal during summer due to the strong convective movement

Precipitation Pattern in India

- *India can be divided into the four regions depending upon the annual average rainfall received by these regions*

Areas of Very High Rainfall (above 200 cm)

- It includes almost whole of Assam, Nagaland, Meghalaya, Mizoram, Arunachal Pradesh, Sikkim, parts of Manipur, West Bengal and the Western coast from Mumbai to Thiruvananthapuram.

Areas of High Rainfall (100-200 cm)

- It includes Eastern slopes of Western ghats, major parts of Northern plain, Odisha, Madhya Pradesh, Andhra Pradesh and Tamil Nadu.

Areas of Low Rainfall (50-100 cm)

- It includes large parts of Gujarat, Maharashtra, Western Madhya Pradesh, Andhra Pradesh, Eastern Rajasthan etc.

Areas of Very Low Rainfall (below 50 cm)

It includes arid and semi-arid area and includes large areas of Western Rajasthan, Kachchh and most of the region of Ladakh (Jammu and Kashmir).

Climatic Regions of India

Type	*Areas*	*Characteristics*
Tropical Rain Forests Climate	Western ghats, West coastal plains, Parts of Assam and Tripura	High temperature throughout the year, heavy seasonal rainfall, average annual rainfall 200 cm (*May to November*)
Tropical Savana Climate	Most of peninsular region (*except leeward side of Western ghats*)	Dry winters, annual rainfall varies from 76 cm to 100 cm
Tropical Semi-Arid Steppe Climate	Rain shadow belt running Southward from Central Maharashtra to Tamil Nadu	Low rainfall, varies from 38 cm to 80 cm and temperature from 20°C to 30°C.
Tropical and Sub-tropical Steppes Climate	Punjab, Haryana and Kachchh region	Temperature varies from 12°C to 35°C. Annual rainfall declines to 50 cm
Tropical Desert Climate	Western parts of Barmer, Jaisalmer and Bikaner districts of Rajasthan and parts of Kachchh	Scanty rainfall (*mostly in form of cloud burst*), high temperature
Humid Sub-tropical Climate with dry winters	South of Himalayas, in the northern plains	Mid winters and extremely hot summers
Mountain Climate	Mountainous region (*above 6000 m)*	Rainfall varies from 60 cm to 250 cm (*mostly during South-West monsoon*)

FORESTS AND NATURAL VEGETATION IN INDIA

Forests in India

- According to state records, the total forest and tree covers **24.56%** of the total land area of the country.
- Madhya Pradesh has the largest areas under forests followed by Arunachal Pradesh. Arunachal Pradesh has the highest per capita forest area.
- The lowest forest cover in terms of percentage of geographical area is in Haryana and Punjab, because of the extensive agriculture.

Mangroves of India

Region	*State/UT*
Sunderbans	West Bengal
Godavari and Krishna	Andhra Pradesh
Ratnagiri	Maharashtra
Gulf of Kachchh	Gujarat
Udupi	Karnataka
Cauveri delta	Tamil Nadu
Kendrapara	Odisha
North and South Andaman and Nicobar	Andaman and Nicobar

CLASSIFICATION OF FORESTS

Administrative Classification

Reserved Forests

- These forests are under the direct supervision of the government where no public entry is allowed for collecting timber and grazing of cattle. About 53% of forest area falls under this category.

Protected Forests

- These forests are looked after by the governments, but local public are allowed to collect timber/firewood and graze their cattle without causing serious damage to the forests. These forests occupy about 29% of the total forest cover.

Unclassified Forests

- These are unclassified forests where there is no restriction on cutting trees or grazing animals. About 18% of total forest area is under this category.

Constitutional Classification

State Forests

- These are under the full control of the government (State/Central) and include almost all the important forest areas of the country. About 94% of total forest area is under this category.

Commercial Forests

- These forests are owned and administered by the local bodies (Municipal corporation, Municipal board, Town area, District board, Village Panchayat etc) and occupy about 5% of the country's forest area.

Private Forests

- These are under the private ownership and cover only 1.3% of the forest area. Now these forests are rapidly converted to waste lands. Odisha has highest private forests in the country.

Forest Cover in India

- Forest cover is defined as "All lands more than one hectare in area, with a tree canopy density of more than 10% irrespective of ownership and legal status. Such lands may not necessarily be a recorded forest area. It also includes orchards, bamboo and palm."

Very Dense Forest

- All lands with forest cover having canopy density of 70% and above. It spreads over 3.02% of the geographical area in India.

Moderately Dense Forest

- All lands with forest cover having canopy density between 40% and 70%. These area spreads over 9.38% of the total geographical area.

Open Forest

- All lands with forest cover having canopy density between 10% and 40%. It covers 9.26% of total geographical area.

Scrub

- Degraded forest lands with canopy density less than 10%. In India, it is upto 1.41% of total area.

Non-Forest

- It is any area not included in the above classes. It is 76.92% of total geographical area in the country.

AGRICULTURE IN INDIA

- India is essentially an agricultural land. Two-thirds of its population still lives on agriculture. Agriculture is primary activity, which includes farming, animal rearing and fishing. There are three crop seasons in India : Kharif, Rabi and Zaid.

Green Revolution

- It is a phrase generally used to describe the spectacular increase in the production and productivity of foodgrains that took place during 1970s.
- The *components of Green Revolution are*
  (a) High yielding variety seeds
  (b) Irrigation
  (c) Use of fertilizers
  (d) Use of insecticides and pesticides
  (e) Command area development programme
  (f) Consolidation of holdings
  (g) Land reforms
  (h) Supply of agricultural credit
  (i) Rural electrification
  (j) Rural roads and Marketing
  (k) Farm mechanisation
  (l) Agricultural Universities

Forests of India

Forest Type	*Distribution*	*Climatic Conditions*	*Characteristics*	*Species*
Tropical Evergreen Forests	▪ Rainy slopes of Western ghats. ▪ North-East India except Arunachal Pradesh. ▪ Eastern part of West Bengal and Odisha. ▪ Andaman and Nicobar Islands.	▪ Rainfall > 200 cm ▪ Relative humidity > 70% ▪ Average temperature is about 24° C. ▪ Hot and humid climate.	▪ Height of trees is 40 to 60 m. ▪ Leaves are dark green and broad.	Mahogany, Mahua, Bamboo, Canes, Ironwood, Kadam, Irul, Jamun, Hopea, Rubber tree, Toon, Telsur etc.
Tropical Moist Deciduous Forests	▪ Eastern parts of Sahyadris (Western ghats). ▪ North Eastern part of Peninsula. ▪ Middle and lower Ganga valley. ▪ Foothills of Himalayas in Bhabar and Tarai region.	▪ 100 to 200 cm rainfall per annum. ▪ Moderate temperature.	▪ 30 to 40 m high trees. ▪ Due to deficiency of water, they shed their leaves in spring (onset of summer). ▪ These cover about 20% India's forest area.	Sal, Teak, Arjun, Mulberry, Kusum, Sandalwood, Siris, Haldi, Khair, Mango, Banyan tree etc.
Tropical Dry Deciduous Forests	▪ Large parts of Maharashtra, Telangana and Andhra Pradesh. ▪ Parts of Punjab, Haryana and Eastern parts of Rajasthan. ▪ Northern and Western parts of Madhya Pradesh. ▪ Tamil Nadu. ▪ Southern parts of Uttar Pradesh.	▪ 50 to 100 cm rainfall. ▪ Moderate humidity.	▪ 6 to 15 m high trees. ▪ Roots are thick and long.	Teak, Sal, Bamboo, Mango, Acacia, Neem, Shisham etc.
Dry Forests or Arid Forests	▪ Rajasthan and adjoining areas of Haryana, Gujarat and Punjab. ▪ Rainshadow areas of Peninsular India.	▪ Low rainfall (less than 50 cm per annum). ▪ Relative humidity is less.	▪ Thorny vegetation. ▪ Roots are very long. ▪ Leaves are small.	Cactus, Thorny bushes, Kikar, Babool, Date palm, Acacia, Khair, Euphorbias etc.
Mountainous Forests or Himalayan Forest	▪ In Himalayan regions	▪ Due to increase of altitude the temperature decreases, hence Himalayan forests contain all the varieties of world except equatorial forest.	▪ Each vegetation belt occurs at a height of 300 or more in Eastern Himalayas.	Sal, Teak, Chir, Deodar, Oak, Olive, Chestnut, Conifers, Spruce and Larch etc.

Soils in India

Types of Soils	*States where found/occurrence*	*Composition*	*Crops grown*
Alluvial	Punjab, Haryana, Uttar Pradesh, Bihar and Jharkhand	Rich in phosphorus, potash and lime but deficient in nitrogen	Large variety of Rabi and Kharif crops such as wheat, rice, sugarcane, cotton and jute
Black (or Regur soil)	Deccan plateau, valleys of Krishna and Godavari, Andhra Pradesh, Madhya Pradesh and Tamil Nadu	Rich in iron, lime, aluminium, magnesium, calcium but lacks in nitrogen, phosphorus and humus	Cotton, sugarcane, jowar, tobacco, wheat and rice
Red	Eastern parts of Deccan Plateau, Tamil Nadu, Goa, Odisha and Meghalaya	Rich in iron and potash but deficient in lime, nitrogen, phosphorus and humus	Wheat, rice, cotton, sugarcane and pulses
Laterite	Summits of Eastern and Western ghats, Assam hills, Andhra Pradesh, Karnataka, West Bengal and Odisha	Rich in iron but poor in silica, lime, phosphorus, potash and humus	Tea, coffee, rubber, cashew and millets
Desert	West and North-West India, Rajasthan, North Gujarat and Southern Punjab	Rich in soluble salts, but deficient in organic matter	Generally unsuitable for cultivation, but with irrigation useful for cultivation of drought-resistant lime, millets, barley, cotton, maize and pulses
Mountain	Hills of Jammu and Kashmir, Uttarakhand and Assam hills	Rich in iron and humus, but deficient in lime	With the use of fertilizers, tea, fruits and medicinal plants can be grown
Saline and Alkaline	Drier parts of Bihar, Jharkhand, Uttar Pradesh, Haryana, Punjab, Rajasthan and Maharashtra	Many salts such as sodium, magnesium and calcium	Unfit for agriculture
Peaty and Marshy	Kerala, coastal regions of Odisha, Tamil Nadu and Suderbans of West Bengal	Contain large amount of soluable salts and organic matter, but lacks in potash and phosphates	Useful for rice and jute cultivation

Major Crops and Producing States

Crop Type	*Crop Name*	*Major Producers*
Cereals	*Wheat*	Uttar Pradesh, Punjab, Haryana and Madhya Pradesh
	Rice	West Bengal, Uttar Pradesh, Punjab and Andhra Pradesh
	Barley	Maharashtra, Uttar Pradesh and Rajasthan
	Bajra	Madhya Pradesh, Gujarat and Rajasthan
Cash Crops	*Sugarcane*	Uttar Pradesh and Maharashtra
	Tobacco	Gujarat and Andhra Pradesh
Oil Seeds	*Coconut*	Kerala and Tamil Nadu
	Linseed	Maharashtra, Rajasthan, Madhya Pradesh and Haryana
	Groundnut	Gujarat, Andhra Pradesh and Tamil Nadu
	Rape Seed and Mustard	Rajasthan, Madhya Pradesh and Haryana
	Sunflower	Karnataka, Andhra Pradesh and Maharashtra
Fibre Crops	*Cotton*	Punjab, Haryana, Maharashtra, Gujarat and Tamil Nadu
	Jute	West Bengal, Bihar and Assam
	Silk	Karnataka and Kerala
Plantations	*Coffee*	Karnataka, Kerala and Tamil Nadu
	Rubber	Kerala and Karnataka
	Tea	Assam and West Bengal
Spices	*Pepper*	Kerala, Karnataka and Tamil Nadu
	Ginger	Meghalaya, Andhra Pradesh, Kerala
	Turmeric	Andhra Pradesh, Karnataka and Tamil Nadu

Growing Conditions of Some Crops

Crop	*Temperature*	*Rainfall*	*Soil*
Rice	Not above 35°C	150-300 cm	Clayey or loamy
Wheat	10°–15°C (winter), 21°–26°C (summer)	75-100 cm	Well-drained loams and clay loam
Cotton	21°–30°C, but not below 21°C, 200 frost free days	50-75 cm	Black soil (regur), alluvial or laterite soils
Sugarcane	20°–26°C	75-150 cm	Any type of soil that can retain moisture
Jowar	Not below 16°C (Rabi) 26°C-33°C (Kharif)	<100 cm	Variety of soils including clayey and sandy
Jute	24°–35°C	Rainfall of 150 cm, with 90% humidity	Light sandy or clayey loams
Tea	24°–30°C	150-300 cm, grown on hill slopes	Well-drained, deep friable loams or forest soils, rich in organic water
Coffee	15°–28°C	150-200 cm	Rich, well-drained friable loams containing good amount of humus and minerals like iron and calcium
Bajra	25°–30°C	40-50 cm	Sandy loams, red and black soils
Ragi	20°–30°C	50-100 cm	Red, light black and sandy loams
Pulses	20°–25°C	50-75 cm	Loamy soil
Rubber	25°–35°C	150-200 cm	Rich, well-drained loamy soils
Tobacco	16°–40°C	50-100 cm	Well-drained loamy soil, alluvial or black soils

MINERAL RESOURCES OF INDIA

Three types of minerals found in India are

1. **Metallic** Iron-ore, copper, aluminium, tin, lead, gold and silver.
2. **Non-metallic** Coal, mica, manganese, petroleum and sulphur.
3. **Radioactive** Uranium and thorium.

Metallic Mineral Mines

Metallic Mineral	*Mines*
Iron	Kemmangundi, Sandur and Hospet(Karnataka)
	Gorumahisani, Badampahar (Odisha), Bailadila and Dalli-Rajhara (Chhattisgarh), North Goa
Manganese	Found in Karnataka, Odisha, Madhya Pradesh, Maharashtra
Chromite	Found in Odisha, Bihar, Karnataka, Maharashtra and Andhra Pradesh
Copper	Malanjkhand Belt (Balaghat ,Madhya Pradesh), Khetri-Singhana Belt (Jhunjhun), Singhbhum (Jharkhand)
Bauxite	Found in Odisha, Gujarat , Jharkhand, Maharashtra, Chhattisgarh
Gold	Kolar and Hutti (Karnataka), Ramgiri in Anantapur (Andhra Pradesh)

Non-Metallic Mineral Mines

Non-Metallic Mineral	*Mines*
Limestone	Found in Karnataka, Andhra Pradesh, Rajasthan, Madhya Pradesh, Gujarat, Chhattisgarh, Meghalaya
Dolomite	About 90% of the dolomite is found in Madhya Pradesh, Andhra Pradesh, Chhattisgarh, Odisha, Gujarat, Rajasthan and Maharashtra
Asbestos	Rajasthan , Andhra Pradesh and Karnataka
Gypsum	Found in Rajasthan and Jammu and Kashmir
Graphite	Found in Kalahandi, Bolangir (Odisha) and Bhagalpur (Bihar)

Atomic Mineral Mines

Atomic Mineral	*Mines*
Uranium	Hazaribagh (Jharkhand) , Gaya (Bihar), Saharanpur (Uttar Pradesh), monazite sands of Kerala coast. Found as by-product in copper mines of Udaipur. Tummalapalle (Andhra Pradesh) has highest reserve of Uranium in Inda
Thorium	Derived from monazite. Found in Kerala, Jharkhand , Bihar, Tamil Nadu and Rajasthan
Lithium	Found in Lepidolite and Spodumene. Lepidolite is found in Jharkhand, Madhya Pradesh, Rajasthan and Bastar region of Chhattisgarh
Zirconium	Found along Kerala coast and alluvial rocks of Ranchi and Hazaribagh district of Jharkhand
Beryllium	Occurs in Nellore district, Andhra Pradesh, Sikkim, Jammu and Kashmir
Antimony	Occurs is Lahaul and Kangra districts of Himachal Pradesh and some district of Madhya Pradesh

Nuclear Power Plants and Reactors Under Construction

Power Station	*Location*
Kalpakkam	Tamil Nadu
Kakrapar Unit 4	Gujarat
Gorakhpur	Haryana
Rawatbhata Unit 7 and 8	Rajasthan
Kudankulam Unit 3 and 4	Tamil Nadu

Refineries in India

Refinery / Company	*State*
Digboi, IOC	Assam
Noonmati, IOC	Assam
Barauni, IOC	Bihar
Koyali, IOC	Gujarat
Chennai, CPCL	Tamil Nadu
Haldia, IOC	West Bengal
Mathura, IOC	Uttar Pradesh
Panipat, IOC	Haryana
Nagapattinam, CPCL	Tamil Nadu
Mumbai, HPCL	Maharashtra

Refinery / Company	*State*
Vishakhapatnam, HPCL	Andhra Pradesh
Trombay, BPCL	Maharashtra
Kochi, BPCL	Kerala
Bongaigaon, IOCL	Assam
Numaligarh, NRL	Assam
Jamnagar, RPL	Gujarat
Mangaluru, MRPL, ONGC	Karnataka
Tatipaka, ONGC	Andhra Pradesh
Jamnagar, Essar	Gujarat

Nuclear Power Stations (Operational) in India

Location	*State*
Tarapur	Maharashtra
Kaiga	Karnataka
Kalpakkam	Tamil Nadu
Kakrapar	Gujarat
Rawatbhata	Rajasthan
Narora	Uttar Pradesh
Kudankulan	Tamil Nadu

Industry	*Details*
Cotton Textile Industry	▪ The first successful modern cotton textile mill was established in Mumbai in 1854 by local Parsi entrepreneur CN Dewar with the name of Bombay Spinning and Weaving Company. ▪ Mumbai is called cottonopolis of India. ▪ Ahmedabad is called Manchester of India. ▪ Coimbatore is called Manchester of South India. ▪ Kanpur is called Manchester of Uttar Pradesh. ▪ **Distribution** Maharashtra (Mumbai, Sholapur, Pune, Kolhapur, Satara, Wardha, Aurangabad and Amravati), Gujarat (Ahmedabad, Vadodra, Rajkot, Surat, Bhavnagar, Porbandar, Maurvi and Viramgam), Tamil Nadu (Chennai, Tirunelvelli, Madurai, Tuticorin, Salem, Tirucchchirappali and Perambur.
Woollen Textile Industry	▪ The first woollen textiles mill was set-up in 1876 at Kanpur. ▪ Jammu and Kashmir is an important producer of handloom and powerloom woollen goods. ▪ **Distribution** Punjab (Dhariwal, Amritsar, Ludhiana, Kharar), Maharashtra (Mumbai), Karnataka (Bengaluru, Belgaum, Mangaluru, Chitradurga, Gulbaraga and Mysore), Uttar Pradesh (Kanpur, Mirzapur, Tanakpur, Modinagar, Bareilly, Agra, Meerut and Varanasi), Madhya Pradesh (Indore, Gwalior, Ujjain, Bhopal), Rajasthan (Alwar, Jaipur, Nagaur, Bhilwara and Pushkar).
Jute Textile Industry	▪ First modern jute mill was set-up in 1855 at Rishra near Kolkata. ▪ India is the second largest producer of raw jute and jute goods, whereas it is second largest exporter of jute goods after Bangaldesh. ▪ **Distribution** West Bengal, Bihar, Uttar Pradesh, Assam, Odisha, Chhattisgarh and Tripura.
Silk Textile Industry	▪ India is the second largest producer of natural silk, after China and is the only country producing all four varieties or natural silk *viz*. Mulberry, Tasar, Eri and Muga, of which Golden yellow Muga silk is unique in India. ▪ **Distribution** Karnataka is the leading producer followed by West Bengal and Jammu & Kashmir.
Rubber Industry	▪ The first factory of synthetic rubber was set-up at Bareilly. ▪ **Distribution** Bareilly (UP), Baroda (Gujarat) Synthetic Rubber Units, Mumbai, Ahmedabad, Amritsar-Reclaimed Rubber Units.
Tea Industry	▪ Tea cultivation in India was first started in the mid-19th century in Darjeeling, Assam and Nilgiris. ▪ Nearly, 98% of the tea production comes from Assam, West Bengal, Tamil Nadu and Kerala, while the rest of it comes from Karanataka, Terai region of Uttar Pradesh, Himachal Pradesh, Arunachal Pradesh, Manipur and Tripura.

Industry	*Details*
Sugar Industry	▪ Uttar Pradesh and Maharashtra are leading producer of sugar **Distribution** Uttar Pradesh (Gorakhpur, Deoria, Basti, Gonda, Meerut, Saharanpur, Muzaffarnagar, Bijnor and Moradabad), Bihar (Darbhanga, Saran, Champaran and Muzaffarpur), Punjab (Phagwara and Dhuri), Haryana (Ambala, Rohtak and Panipat), Maharashtra (Nasik, Pune, Satara, Sangli, Kolhapur and Solapur) and Karnataka (Munirabad, Shivamogga and Mandya).
Paper Industry	▪ One of the first paper mill in the country was set-up at Serampore (Bengal) in 1832, which failed. In 1870, a first successful venture was started at Ballygunj near Calcutta. ▪ **Raw Material** Bamboo (70%), Salai wood (12%), Sabai (9%), Bagasses (4%) and Waste paper and Rags (5%). ▪ **Distribution** Madhya Pradesh (Nepanagar), Hindustan Paper Corp, Vellore, Mysore Paper Mill, Bhadravati, Maharashtra, (Mumbai, Pune, Ballarpur and Kamptee produce paper and Vikhroli), Andhra Pradesh (Rajahmundry and Sirpur), Madhya Pradesh (Indore, Bhopal and Shahdol), Karnataka.
Iron and Steel Industry	▪ **Distribution** Bhadrawati (Karnataka), Jamshedpur (Jharkhand), Durgapur, Burnpur (West Bengal), Bokaro (Jharkhand), Rourkela (Odisha), Bhilai (Chhattisgarh), Salem (Tamil Nadu) and Vishakhapatnam (Andhra Pradesh).
Ship Building Industry	▪ **Distribution** Cochin Shipyard, Mumbai (Mazagon Dock), Hindustan Shipyard at Vishakhapatnam and Hooghly Dock and Port Engineers Limited Kolkata. Mazagon Dock Ltd. at Mumbai builts vessels for Indian Navy.
Aircraft Industry	▪ **Distribution** Hindustan Aeronautics India Limited was formed by merging two aircraft factories at Bengaluru and Kanpur. Four other factories are at Nasik, Lucknow, Koraput (Odisha) and Hyderabad
Fertilizer Industry	▪ The Fertilizer Corporation of India (FCI) was set-up in 1961. ▪ National Fertilizer Limited (NFL) was set-up in 1974. ▪ **Distribution** Sindri (Bihar), Nangal (Punjab), Trombay (Maharashtra), Gorakhpur (Uttar Pradesh), Durgapur (West Bengal), Namrup (Assam), Cochin (Kerala), Rourkela (Chattisgarh), Neyveli (Tamil Nadu), Varanasi (Uttar Pradesh), Vadodra (Gujarat), Kanpur (Uttar Pradesh), Vishakhapatnam (Andhra) and Kota (Rajasthan).
Machine Tool Industry	▪ It forms the basis for the manufacturing of industrial, defence equipments, automobiles, railway engines and electrical machinery. ▪ **Distribution** Hyderabad, Bengaluru, Pinjore (Haryana), Kalamassery (Kerala), Ajmer and Srinagar.
Heavy Electrical Equipments Industry	▪ **Distribution** Bengaluru, Bhopal, Jammu, Tiruchirapalli, Ramchandrapuram (Hyderabad) and Haridwar.
Photo Films Industry	▪ The Hindustan Photo Films Manufacturing Company at Udagamandalam (Tamil Nadu) is the only factory in the public sector, producing photo paper and films.
Glass Industry	▪ **Distribution** Uttar Pradesh (Firozabad, Hathras, Naini), Maharashtra (Mumbai, Telogaon, (Pune) Kolhapur, Tamil Nadu (Salem, Chennai, Coimbatore) and Karnataka (Bengaluru).

TRANSPORT

Railways

- India has the second largest railway network in Asia and the fourth largest in the world after the USA, Russia and China.
- It is the largest public sector undertaking of the country and it is the world's second largest railway network under single management.
- Indian railways have 18 railway zones.
- The first Indian railway line in India was operated for public traffic in 1853 between Mumbai and Thane over a distance of 34 km.
- On 3rd February, 1925, the first electric passenger train in India ran between victoria Terminus and Kerala.
- The first Metro rail was introduced in Kolkata on 24th October, 1984. As of **December, 2020**, thirteen cities have operational metro in India.
- The Delhi Metro is the busiest as well as the largest metro system while Ahmedabad metro system is Smallest (6 km) among all.
- The Government is constructing two Dedicated Freight Corridors (DFC). The Eastern dedicated freight Corridor (1760 km) stretchs from Ludhiana (Punjab) to Dankuni (West Bengal) while the Western dedicated freight corridor (1468 km) operates from Dadri in Uttar Pradesh to Jawaharlal Nehru Port in Mumbai.

Important Facts of Indian Railways

- Vivek Express which runs from Dibrugarh in Assam to Kanyakumari in Tamil Nadu (4286 km) is the longest train of India.
- India's is first indigenously built engine-less train-Vande Bharat is the fastest train in India. It is operational on two routes–New Delhi to Varanasi and Katra.
- The Lucknow-Delhi, Tejas Express is the India's first 'private' train run by its subsidiary IRCTC.
- On 7th August, 2020 the First Kisan Rail of the country was flagged off from Devlali in Maharashtra to Danapur in Bihar.
- Government with the objective of upgrading the railway system and boosting investment has allowed private players to operate trains in India. Under this, through, open and transparent bidding process, private players will be allowed to operate 151 trains on long busy routes for 35 years.

Konkan Railways

- It runs from Mangalore to Roha (40 km South of Mumbai). The network involves 3 states such as Maharashtra, Goa and Karnataka.
- It has a total length of 756 km. Almost 10% of the line passes through tunnels.

Railway Zones

Indian Railways is geographically organised into zones. At present there are 18 Zonal Railways (including Metro Railway, Kolkata)

Railway Zones

Zones	*Headquarters*
Central (CR)	Mumbai
East Central (ECR)	Hajipur
East Coast (ECoR)	Bhubaneswar
Eastern (ER)	Kolkata
North Central (NCR)	Allahabad
North Eastern (NER)	Gorakhpur
North Western (NWR)	Jaipur
North-East Frontier (NFR)	Guwahati
Northern (NR)	Delhi
South Central (SCR)	Secunderabad
South East Central (SECR)	Bilaspur
South Eastern (SER)	Kolkata
South Western (SWR)	Hubbali
Southern (SR)	Chennai
West Central (WCR)	Jabalpur
Western (WR)	Mumbai
Metro Railway	Kolkata
South Coast	Visakhapatnam

Indian Railways Recognised by UNESCO

Railway	*Speciality d*
Darjeeling Himalayan Railways (1999)	Narrow gauge railway from Siliguri to Darjeeling in the state of West Bengal
Nilgiri Mountain Railways (2005)	It connects the town of Mettupalayam with the hill station of Udagamandalam in the Nilgiri hills
Chhatrapati Shivaji Terminus (2004)	It was completed in 1887 marking 50 years of Queen Victoria's rule
Kalka-Shimla Railways (2008)	Narrow gauge railway in North-West India travelling along a most mountainous route from Kalka to Shimla

** The year in bracket, represents the year, in which UNESCO has added the railway line to the World Heritage Site list.*

Airways

- JRD Tata was the first person to take a solo flight from Mumbai to Karachi in 1931. In 1953, all the private airline companies were nationalised and Indian Airlines and Air India came into existence.
- On 1st March, 2007, the Union Government merged both Indian Airlines and Air India. The brand name of the new airlines is Air India (or Indian) and its logo is *Maharaja*.

International Airports in India

International Airport	*City*
Rajiv Gandhi International Airport	Hyderabad (Telangana)
Calicut International Airport	Kozhikode (Kerala)
Chhatrapati Shivaji International Airport	Mumbai (Maharashtra)
Kempegowda International Airport	Bengaluru (Karnataka)
Goa (Dabolin) Airport in Vasco de Gama City	Goa
Netaji Subhash Chandra Bose International Airport	Kolkata (West Bengal)
Thiruvananthapuram International Airport	Thiruvananthapuram (Kerala)
Lokpriya Gopinath Bordoloi International Airport	Guwahati (Assam)
Sardar Vallabhbhai Patel International Airport	Ahmedabad (Gujarat)
Indira Gandhi International Airport	Delhi
Chennai International Airport	Chennai (Tamil Nadu)
Shri Guru Ram Das Jee International Airport	Amritsar (Punjab)
Cochin International Airport	Cochin (Kerala)
Coimbatore International Airport	Coimbatore (Tamil Nadu)
Lal Bahadur Shastri Airport	Varanasi (Uttar Pradesh)
Chaudhary Charan Singh Airport	Lucknow (Uttar Pradesh)
Ambedkar Airport	Nagpur (Maharashtra)
Jaipur International Airport	Jaipur (Rajasthan)
Srinagar International Airport	Srinagar (Jammu and Kashmir)
Tiruchirapalli International Airport	Tiruchirapalli (Tamil Nadu)
Veer Savarkar International Airport	Port Blair (Andaman Nicobar Island)

Roadways

India has road network of 5.8 million km, which is second largest in the world. The total length of the National Highways in the country is 142, 126 km as of April, 2019. This comprises only 2.7% of the total road network but carries over 40% of the total traffic.

Indian Road Network

Types of Roads	*Length (in km)*
Expressways	1583
National Highways	142,126
State Highways	176,166
Major District Roads	561,940
Rural and Other Roads	2650000
Total Length	56 lakhs kms (Approx.)

National Highways

Some of the important information regarding the National Highways

- NH 44 is the longest National Highway in India.
- The NH 548 which is of 5 km in length, is Shortest National Highway in India.
- NH 4 is located in Andaman and Nicobar Islands. It is known as Andaman Trunk Road.
- NH 27 is an East-West National Highway starting at Porbandar and terminating at Silchar.
- Amongst the states, Maharashtra has the largest length of National Highways in the country followed by Uttar Pradesh.
- In October, 2020 world' longest highway tunnel (Atal Tunnel) was inaugurated. Its length is 9.02 km. It connect Manali with Lahaul-Spiti.

National Highways

NH No.	*States through which it passes*
NH 1	Jammu and Kashmir
NH 2	Assam, Nagaland, Manipur, Mizoram
NH 3	Punjab, Himachal Pradesh, Jammu and Kashmir
NH 4	Andaman & Nicobar
NH 5	Punjab, Chandigarh, Haryana, Himachal Pradesh
NH 6	Meghalaya, Assam, Mizoram
NH 7	Punjab, Chandigarh, Haryana, Himachal Pradesh, Uttarakhand
NH 8	Assam
NH 9	Punjab, Haryana, Delhi, Uttar Pradesh
NH 10	Sikkim, West Bengal
NH 11	Rajasthan
NH 12	West Bengal
NH 19	Delhi, Haryana, Uttar Pradesh, Bihar, Jharkhand, West Bengal
NH 20	Bihar, Jharkhand
NH 21	Rajasthan, Uttar Pradesh
NH 24	Gujarat, Rajasthan, Madhya Pradesh, Uttar Pradesh, Bihar, West Bengal, Assam
NH 44 (Longest NH in India)	Jammu & Kashmir, Himachal Pradesh, Punjab, Haryana, Delhi, Uttar Pradesh, Madhya Pradesh, Maharashtra, Telangana, Andhra Pradesh, Karnataka, Tamil Nadu
NH 48	Delhi, Haryana, Rajasthan, Gujarat, Maharashtra, Karnataka, Tamil Nadu
NH 53	Gujarat, Maharashtra, Chhattisgarh, Odisha

Water Transport

- Approximately 95% of the country's trade by volume and 70% by value is moved through maritime transport. India is among the 20 leading merchant fleets all over the world. The country has 12 major ports and approximately 200 minor ports.
- Indian port sector has seen certain major milestones in the recent past, including the commissioning of the first phase of operations at International Container Transhipment Terminal, Vallarpadam, Solid Cargo Port Terminal, Dahej, Coal terminal, Mundra, Bulk terminal, Hazira and a Greenfield port, Dhamra.

Sethusamudram Project

- This Project envisages dredging of a ship channel across the Palk straits between India and Sri-Lanka.
- This will allow ships sailing between the East and West coasts of India to have a straight passage through India's territorial waters, instead of having circumvent Sri Lanka. This will lead to a saving of up to 424 nautical miles (780 km) and up to 30 hours in sailing time.
- Two channels will be created, one across North of Adam's bridge (the chain of Islets and shallows linking India with Sri Lanka), South-East of Pamban Island and another through the shallows of Palk bay, deepening the Palk straits. The total length of these two channels would be 89 km.

Inland Waterways of India

Waterways	*Stretch of the Waterways*	*Specification*
NW 1	Allahabad-Haldia (1620 km)	Along Ganga river
NW 2	Sadiya- Dhubri (891 km)	Along Brahmaputra river
NW 3	Kottapuram-Kollam (205 km)	Along Champakara and Udyogmandal canal
NW 4	Bhadrachalam to Rajahmundri and Wazirabad to Vijayawada (1095 km)	Along Godavari and Krishna river
NW 5	Mangalgarhi to Paradeep and Talcher to Dhamara (623 km)	Along Mahanadi and Brahmini river system
NW 6	Between Lakhipur and Bhanga (121 km)	Barak river

- The Inland Waterways Authority of India, headquartered at Noida was constituted on 27th October, 1986 as a statutory body. It is the incharge of National Waterways in India.
- Under National Waterway Act, 2016, Government of India has designated new 106 Inland Waterways. Thus, now, there are 111 designated National Waterways in India.

Sagarmala Project

The Sagarmala project was launched by the Central Government in 2015 to enhance the performance of India's logistics sector through port led development. It is a flagship program of the Ministry of Shipping.

Under the Sagarmala, many projects have been envisaged. These include port modernisation, new port development, port connectivity enhancement, port linked coastal economic cone industrialisation and coastal community development.

Ports in India

- The Inland waterways authority in India, divides Indian ports into three categories: major, minor and intermediate ports.
- Largest container port of India is Jawaharlal Nehru (Nhava Sheva) port in Mumbai.
- The largest natural port is in Mumbai.
- Kandla in Gujarat is a **tidal port**. It has been converted into a free trade zone.
- New Mangaluru port is also called the '**Gateway of Karnataka**'.
- Mumbai port is the busiest port of India.
- Kandla port is called the '**offspring of partition**' as it was developed after the partition as a substitute of Karachi port. It is a tidal port and a **free trade zone** located in Rann of Kachchh.
- The Government of India on 5th February, 2020 gave approval for construction of greenfield major port at Vadhavan in Maharashtra.

Ports of Western Coasts

Port	*Important Fact*
Mumbai	It handles maximum cargo. It is a natural harbour, it handles mostly petroleum and dry cargo.
Kandla (Renamed as Deendayal Port)	Tidal port and important traffic handled are crude oil, petroleum, edible oil, food grains.
Marmagao	It handles iron ore. It has a naval berth.
New Mangaluru	It is an all weather port, and handles iron-ore export from Kudremukh.
Cochin	Major natural port in Willingdon island in Kerala.
Jawaharlal Nehru	It is called as Nhava Sheva Port. It is most mechanised port of India.

Ports of Eastern Coasts

Port	*Important Fact*
Kolkata (Renamed as Shyama Prasad Mukherjee)	Oldest port, India's riverine port having two dock system.
Paradeep	It handles iron ore and some amounts of coal and dry cargo.
Chennai	All weather port having deep drafted berth, oil jetties, iron ore terminals etc.
Vishakhapatnam	It is in Andhra Pradesh and is well known for its outstanding performance. It serves the Bhilai and Rourkela Steel plant.
Tuticorin	Artificial deep sea harbour, all weather port offer direct weekly container service to USA. Also known as VO Chidambaranar Port.
Ennore	First corporatised major port in India.

Major Tribes of India

Tribal Group	*Found in*	*Tribal Group*	*Found in*
Abhors	Areas surrounding Assam Valley		
Abhuj Maria	Chhattisgarh(*Bastar district*)	Kol	Madhya Pradesh and Uttar Pradesh
Angami	Nagaland	Kolam	Andhra Pradesh
Apatanis	Arunachal Pradesh	Kotas	Telangana, Chhattisgarh, Madhya Pradesh
Bakarwals	Tamil Nadu	Kuki	Manipur
Baigas	Madhya Pradesh	Lahaulas	Himachal Pradesh
Bakarwals	Jammu and Kashmir	Lepchas	Sikkim
Bhils	Madhya Pradesh and Rajasthan	Lushai	Mizoram
Bhotias	Uttarakhand	Muria	Chhattisgarh
Bharia	Madhya Pradesh	Meenas	Rajasthan
Birhors	Madhya Pradesh and Jharkhand	Moplahs	Kerala
Chang	Nagaland	Mundas	Jharkhand
Chenchus	Andhra Pradesh, Telangana and Odisha		
Chutia	Assam	Mishmi	North-East
Gaddis	Himachal Pradesh	Nagas	Nagaland
Galaog	North-East	Oarons	Jharkhand and Odisha
Garos	Assam and Meghalaya	Onges	Andaman and Nicobar
Gonds	Madhya Pradesh and Maharashtra	Singpho	North-East
Gujjars	Jammu and Kashmir and Himachal Pradesh	Santhals	West Bengal, Odisha, Jharkhand and Bihar
Irula	Tamil Nadu	Sangtam	Nagaland
Jaintias	Meghalaya	Sema	Nagaland
Jarawa	Little Andaman	Sentinelese	Andaman and Nicobar
Kanikaran	Tamil Nadu	Shompens	Andaman and Nicobar
Katkari	Maharashtra and Madhya Pradesh	Todas	Tamil Nadu
Kharia	Jharkhand, Madhya Pradesh and Odisha	Uralis	Kerala
Khond	Andhra Pradesh, Madhya Pradesh, Bihar, Chhattisgarh and Andhra Pradesh, West Bengal	Wancho	North-East
Tharu	Uttar Pradesh	Warlis	Maharashtra
Khasi	Assam and Meghalaya		

Indian Towns Associated with Industries

Town	*State*	*Industry*
Ahmedabad	Gujarat	Cotton textiles
Agra	Uttar Pradesh	Leather, Marble, Carpets, Dairy
Aligarh	Uttar Pradesh	Locks, Cutlery
Ankleshwar	Gujarat	Oil Refinery
Amritsar	Punjab	Woollen clothes
Anand	Gujarat	Milk and its products
Bokaro	Jharkhand	Steel plant
Bengaluru	Karnataka	Telephones, Aircrafts, Machine Tools, Cotton textiles, IT
Batanagar	West Bengal	Shoes
Bhilai	Chhattisgarh	Steel plant
Barauni	Bihar	Chemical fertilizer, Oil refinery
Bhagalpur	Bihar	Silk Industries
Bongaigaon	Assam	Petroleum
Cochin	Kerala	Ship building, Coconut oil, Rubber
Calicut	Kerala	Coffee, Coconut
Coimbatore	Tamil Nadu	Cotton industries
Durgapur	West Bengal	Steel plant, Oil Refinery
Digboi	Assam	Oil Refinery
Delhi	Delhi	Textiles, Electronic, DDT
Darjeeling	West Bengal	Tea
Firozabad	Uttar Pradesh	Bangle works, Glass Industry
Gwalior	Madhya Pradesh	Pottery, Tobacco
Gomia	Jharkhand	Steel Plant
Haldia	West Bengal	Chemical fertilizer, Jute
Jamshedpur	Jharkhand	Iron and Steel, Locomotives, Railway coaches
Jalandhar	Punjab	Surgical goods and Sports articles
Jaipur	Rajasthan	Cloth, Printing, Brass ware
Jharia	Jharkhand	Coal mines
Jabalpur	Madhya Pradesh	Tobacco, IT, Garment Manufacturing
Kanpur	Uttar Pradesh	Cotton and Woollen mills, Leather, Sugar
Korba	Chhattisgarh	Aluminium factory, Thermal plant
Kolar	Karnataka	Gold Mining center
Kolkata	West Bengal	Jute, Leather, Electric goods
Kota	Rajasthan	Stone Polishing, Power Plants
Kanchipuram	Tamil Nadu	Silk clothes
Kandla	Gujarat	Chemical fertilizers, Famous Port
Ludhiana	Punjab	Hosiery
Lucknow	Uttar Pradesh	Embroidery work, Chikan work
Chennai	Tamil Nadu	Leather, Cigarette, Integral Coach Factory
Madurai	Tamil Nadu	Automobile Rubber, Chemical and Granite Manufacturing
Mathura	Uttar Pradesh	Oil refinery
Mysore	Karnataka	Sandalwood oil, Silk goods
Meerut	Uttar Pradesh	Publication work, Sports goods, Scissors making
Mumbai	Maharashtra	Cinema industries, Cotton textiles
Neyveli	Tamil Nadu	Lignite industries

STATES/UNION TERRITORIES

Indian States /Union Territories At a Glance

S. No.	*Area-Wise Ranking States*	*Capital*	*Date of Formation*	*State Animal*	*State Bird*	*State Tree*	*State Flower*
1.	Rajasthan	Jaipur	30 March, 1949	Chinkara	Godawan	Khejari	Rohida
2.	Madhya Pradesh	Bhopal	26 January, 1950	Barasingha	Paradise Fly Catcher	Banyan	White Lily
3.	Maharashtra	Mumbai, Nagpur (Winter)	1st May, 1960	Giant Squirrel	Green Imperial Pigeon	Mango	Jarul
4.	Andhra Pradesh	Hyderabad (de juse) Amarawati (de facto)	1 November 1956	Black Buck	Rose Ringed Parakeet	Neem	Jasmine
5.	Uttar Pradesh	Lucknow	24th January, 1950	Barasingha	Sarus Crane	Ashok	Palash
6.	Gujarat	Gandhinagar	1st May, 1960	Asiatic Lion	Great Flamingo	Banyan	Marigold
7.	Karnataka	Bengaluru	1st November, 1956	Elephant	Indian Roller	Sandal wood	Lotus
8.	Odisha	Bhubaneswar	1st April, 1936	Elephant	Peacock	Banyan	Lotus
9.	Chhattisgarh	Raipur	1st November, 2000	Wild Buffalo	Hill Myna	Sal	—
10.	Tamil Nadu	Chennai	1st November, 1956	Nilgiri Tahr	Emerald Dove	Palm	Shen Kandhal
11.	Bihar	Patna	26th January, 1950	Gaur	Indian Roller	Peepal	Kachnar
12.	West Bengal	Kolkata	26th November, 1950	Fishing Cat	White throated King Fisher	Chatim	Night Flowering Jasmine
13.	Arunachal Pradesh	Itanagar	20th February, 1987	Mithun	Great Hornbill	Hollong	Lady Slipper orchid
14.	Jharkhand	Ranchi	15th November, 2000	Elephant	Koel	Sal	Palash
15.	Assam	Dispur	26th January, 1950	Rhinoceros unicornis	White wing wood duck	Hollong	Foxtail orchid
16.	Himachal Pradesh	Shimla (Summer) Dharmshala (Winter)	25th January, 1971	Musk Deer	Monal	Deodar	Rhododendron
17.	Uttarakhand	Dehradun	9th November, 2000	Musk Deer	Himalayan Monal	Burans	Brahma Kamal

Indian States /Union Territories At a Glance

S. No.	*Area-Wise Ranking States*	*Capital*	*Date of Formation*	*State Animal*	*State Bird*	*State Tree*	*State Flower*
18.	Punjab	Chandigarh	1st November, 1966	Black Buck	Baaz	Sheesham	Lilium
19.	Haryana	Chandigarh	1st November, 1966	Black Buck	Black Francolin	Peepal	Lotus
20.	Kerala	Thiruvananthapuram	1st November, 1956	Elephant	Great Hornbill	Coconut	Kanikonna
21.	Meghalaya	Shillong	21st January, 1972	Clouded Leopard	Hill Myna	Gamhar	Lady Slipper Orchid
22.	Manipur	Imphal	21st January, 1972	Sangai	Nongin	Uningthou	Shirui Lilly
23.	Mizoram	Aizawl	20th February, 1987	Hillock Gibbon	Mrs Hume's Pheasant	Iron Wood	Red Vanada
24.	Nagaland	Kohima	1st December, 1963	Mithun	Blyth's Tragopan	Alder	Rhodo Dendron
25.	Tripura	Agartala	21st January, 1972	Phayre's Leaf Monkey	Green Imperial Pigeon	Agar Wood	Indian Rose Chestnut
26.	Sikkim	Gangtok	16th May, 1975	Red Panda	Blood Pheasant	Rhododen Dron	Noble Dendrobium
27.	Goa	Panaji	30th May, 1987	Gaur	Black Crested Bulbul	Matti	—
28.	Telangana	Hyderabad	2nd June, 2014	Spotted Deer	Indian roller	Jammi Chettu	Tangedu Puvvu
	Union Territories						
29.	Andaman and Nicobar Islands	Port Blair	1st November, 1956	Dugong	Andaman Wood Pigeon	Andaman Padauk	—
30.	Delhi	New Delhi	1st February, 1992	Nilgai	House sparrow	—	—
31.	Dadra and Nagar Haveli and Daman and Diu	Daman	Proposed from 26th January, 2020	—	—	—	—
32.	Puducherry	Puducherry	1st November 1954	Squirrel	Asian Koel	Bael Fruit Tree	Cannon Ball Tree flower
33.	Chandigarh	Chandigarh	1st November, 1966	Indian Grey Mongoose	Indian Grey Horn bill	Mango	Dhak
34.	Lakshadweep	Kavaratti	1st November, 1956	Butterfly Fish	Noddy Tern	Bread Fruit	—
35.	Jammu and Kashmir	Srinagar (Summer) Jammu (Winter)	31st October, 2019	—	—	—	—
36.	Ladakh	Leh	31st October, 2019	—	—	—	—

RANK OF INDIAN STATES AND UTs (CATEGORY-WISE) Based on the Census 2011

Literacy Rate

Rank	*Overall Literacy*		*Male Literacy*		*Female Literacy*	
	State/UT	*%*	*State/UT*	*%*	*State/UT*	*%*
1.	Kerala	93.91	Lakshadweep	96.11	Kerala	91.98
2.	Lakshadweep	92.28	Kerala	96.02	Mizoram	89.40
3.	Mizoram	91.58	Mizoram	93.72	Lakshadweep	88.25
4.	Tripura	87.75	Goa	92.81	Tripura	83.15
5.	Goa	87.40	Tripura	92.18	Goa	81.84
6.	Daman and Diu	87.07	Puducherry	92.12	Andaman and Nicobar Islands	81.84
7.	Puducherry	86.55	Daman and Diu	91.48	Chandigarh	81.38
8.	Chandigarh	86.43	NCT of Delhi	91.03	Puducherry	81.22
9.	Delhi	86.34	Himachal Pradesh	90.83	NCT of Delhi	80.93
10.	Andaman and Nicobar Islands	86.27	Chandigarh	90.54	Daman and Diu	79.59
11.	Himachal Pradesh	83.78	Andaman and Nicobar Islands	90.11	Nagaland	76.69
12.	Maharashtra	82.91	Maharashtra	89.82	Himachal Pradesh	76.60
13.	Sikkim	82.20	Uttarakhand	88.33	Sikkim	76.43
14.	Tamil Nadu	80.33	Sikkim	87.29	Maharashtra	75.48
15.	Nagaland	80.11	Gujarat	87.23	Tamil Nadu	73.86
16.	Manipur	79.85	Tamil Nadu	86.81	Meghalaya	73.78
17.	Uttarakhand	79.63	Manipur	86.49	Manipur	73.17
18.	Gujarat	79.31	Dadra and Nagar Haveli	86.46	Punjab	71.34
19.	Dadra and Nagar Haveli	77.65	Haryana	85.38	West Bengal	71.34
20.	West Bengal	77.08	Nagaland	83.29	Gujarat	70.73
21.	Punjab	76.68	Karnataka	82.85	Uttarakhand	70.70
22.	Haryana	76.64	West Bengal	82.67	Karnataka	68.13
23.	Karnataka	75.60	Odisha	82.40	Assam	67.27
24.	Meghalaya	75.48	Punjab	81.48	Haryana	66.77
25.	Odisha	73.45	Chhattisgarh	81.45	Dadra and Nagar Haveli	65.93
26.	Assam	73.18	Madhya Pradesh	80.53	Odisha	64.36
27.	Chhattisgarh	71.04	Rajasthan	80.51	Chhattisgarh	60.59
28.	Madhya Pradesh	70.63	Uttar Pradesh	79.24	Madhya Pradesh	60.02
29.	Uttar Pradesh	69.72	Assam	78.81	Andhra Pradesh	59.74
30.	Jammu & Kashmir	68.74	Jharkhand	78.45	Arunachal Pradesh	59.57
31.	Andhra Pradesh	67.66	Jammu & Kashmir	78.26	Uttar Pradesh	59.26
32.	Jharkhand	67.63	Meghalaya	77.17	Jammu & Kashmir	58.01
33.	Rajasthan	67.06	Andhra Pradesh	75.56	Jharkhand	56.21
34.	Arunachal Pradesh	66.95	Arunachal Pradesh	78.69	Bihar	53.33
35.	Bihar	63.82	Bihar	73.39	Rajasthan	52.66
	National Average	**74.04%**		**82.14%**		**65.46%**

Sex Ratio

Rank	*Adult Sex Ratio*		*Child Sex Ratio (0-6)*	
	State/UT	*Ratio*	*State/UT*	*Ratio*
1.	Kerala	1084	Mizoram	970
2.	Puducherry	1037	Meghalaya	970
3.	Tamil Nadu	996	Andaman and Nicobar	968
4.	Andhra Pradesh	993	Puducherry	967
5.	Chhattisgarh	991	Chhattisgarh	969
6.	Manipur	992	Arunachal Pradesh	972
7.	Meghalaya	989	Kerala	964
8.	Odisha	979	Assam	962
9.	Mizoram	976	Tripura	957
10.	Himachal Pradesh	972	West Bengal	956
11.	Goa	973	Tamil Nadu	943
12.	Karnataka	973	Nagaland	943
13.	Uttarakhand	963	Jharkhand	948
14.	Tripura	960	Sikkim	957
15.	Assam	958	Andhra Pradesh	939
16.	West Bengal	950	Karnataka	948
17.	Jharkhand	949	Odisha	941
18.	Maharashtra	929	Manipur	936
19.	Lakshadweep	947	Bihar	935
20.	Nagaland	931	Dadra and Nagar Haveli	926
21.	Madhya Pradesh	931	Goa	942
22.	Rajasthan	928	Madhya Pradesh	918
23.	Arunachal Pradesh	938	Daman and Diu	904
24.	Gujarat	919	Lakshadweep	911
25.	Bihar	918	Himachal Pradesh	909
26.	Uttar Pradesh	912	Uttar Pradesh	902
27.	Punjab	895	Uttarakhand	890
28.	Sikkim	890	Gujarat	890
29.	Jammu and Kashmir	889	Rajasthan	888
30.	Andaman and Nicobar Islands	876	Maharashtra	894
31.	Haryana	879	Chandigarh	880
32.	Delhi	868	Delhi	871
33.	Chandigarh	818	Jammu and Kashmir	862
34.	Dadra and Nagar Haveli	774	Punjab	846
35.	Daman and Diu	618	Haryana	834
	National Average	**940**		**919**

ENVIRONMENT AND ECOLOGY

Man in earlier times, believed in 'nature-nurture'concept. However, since the last 50 years there has been drastic change in perspective about man-environment relation.

Environment

- Environment is defined as the social, cultural and physical condition that surround, affect and influence the survival, growth and development of people, animals or plants.
- Environment includes everything around us. It encompasses both the living (biotic) and non-living (abiotic) components of Earth.
- It consists of four segments: atmosphere, hydrosphere, lithosphere and biosphere.

Ecosystem

The term "ecosystem" was first coined by British ecologist Tansley. ''An ecosystem is defined as a natural unit that consists of living and non-living components which interact to form a stable system''.

Functions of Ecosystem

(a) It regulates flow rates of biological energy.
(b) It regulates the rates of nutrients.
(c) It helps in biological regulation including photoperiodism and nitrogen fixing organisms.

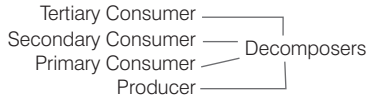
Components of Ecosystem

The components of an ecosystem are grouped as abiotic (non-living) and biotic (living).

(a) The abiotic factors are inert factors of the ecosystem and includes air, water, soil, temperature, chemical products etc.
(b) Biotic factors includes plants, animals and micro-organisms. This living component can be further divided into:
  (i) Producers make their own food. Plant, algae and green bacteria are all producers.
  (ii) **Consumers** depend on producers for their food. Consumers are futher classified into primary, secondary and tertiary consumers.
  (iii) **Decomposers** include bacteria and fungi, that feed on decaying matter.

Food Chain and Food Web

- Food chain is a feeding hierarchy in which organism in an ecosystem are grouped into nutritional (tropic) levels and are shown in a succession to represent the flow of food energy and the feeding relationship between them.
- Complex network of inter connected food chain called a **Food Web**



Grazing Food Chain

(a) Grass—Grasshopper—Frog—Snake —Hawk
(b) Green Plant—Goat—Wolf—Lion

Detritus Food Chain

(a) Dead Plant—Soil mites—Insects —Lizards
(b) Dead Organic Matter—Bacteria—Protozoa—Rotifers.

Ecology

The term 'Ecology' first coined by German biologist Ernst Haeckel.

It was derived from Greek word *Okologie*, means 'home or surrounding'.

It was difined as "the study of the interaction between life and its physical environment; the relationship between animals and plants and how one species affect another."

Environmental Pollution

- Pollution may be defined as addition of undesirable material into the environment as a result of human activities.
- The agents which cause environmental pollution are called **Pollutants**. Primary pollutants are emitted directly from identifiable sources. Secondary pollutants are produced due to chemical reaction among primary pollutants.

Types of Environmental Pollution

- Pollution may be of the following types Air pollution, Noise pollution, Water pollution, Soil pollution and Thermal pollution.

Air Pollution

It may be defined as the presence of any solid, liquid or gaseous substance including noise and radioactive radiation in the atmosphere in such concentration that may be directly and indirectly injurious to humans or other living organisms, plants, property or interferes with the normal environmental processes. Air pollutants are of two types (1) Suspended particulate matter (Dust, smoke, soot, pollen and soil particles) and (2) Gaseous pollutants like Carbon Dioxide (CO^2), NOx etc.

Noise Pollution

Noise by definition is 'sound without value' or any noise that is unwanted by the recipient.

Noise level is measured in terms of decibels (dB). WHO (World Health Organisation) has prescribed optimum noise level as 45 dB by day and 35 dB by night. Anythings above 80 dB is hazardous.

Common Sounds Level

Noise Source	*Decibels*
Whispering	34
Conversation	60
Vacuum cleaner	69
Heavy traffic	90
Jet aircraft	115
Human pain threshold	120

Water Pollution

Water Pollution is caused by a variety of human activities such as industrial, agricultural and domestic. Agricultural run off laden with excess fertilizers and pesticides, industrial effluents with toxic substances and sewage water with human and animal wastes lead to water pollution.

Soil Pollution

Addition of substances which adversely affect the quality of soil or its fertility is known as soil pollution. Soil can be polluted by pathogenic organims, organic and inorganic chemicals and toxic metals.

Thermal Pollution

The waste heat from the boilers and heating processes increases the temperature of water. Discharge of hot water may increase the temperature of the receiving water by 10 to 15°C above the ambient water temperature.

OZONE Layer Depletion

- Ozone is a highly reactive molecule containing three oxygen atoms. The Earth's atmosphere between 20 to 3km above the surface contains a thin layer of ozone. This ozone layer serves as a natural filter for blocking incoming UV radiation from the sun.
- Ozone depletion means thinning of the ozone layer. It has been noted that there is a depletion of ozone ozyer in recent times. The significant cause of depletion of ozone layer is presence of chlorofluoro carbons (CFCs). CFCs are used in refrigeration, air conditioning, fire extinguishers etc.
- Montreal protocol on substances that deplete ozone layer (1987) is a landmark international agreement designed to protect stratospheric ozone layer. Parties to the protocol were required to phase down the use of CFCs, halons and other man-made ODCs.

Climate Change And Global Warming

- Climate change is a long term shift in weather condition identified by changes in temperature, precipitation, wind and other indicators.
- Atmospheric gases like carbon dioxide, methane, nitrous oxide and water vapour are capable of trapping outgoing infrared radiation from Earth that causes global warming. Greenhouse effect is naturally occuring phenomenon that is responsible for heating of Earth's surface and atmosphere.

- United Nations Framework Convention on Climate Change (UNFCCC) is an international agreement (1992) that aims at stabilizing greenhouse gas concentration in the atmosphere to manageable level. Kyoto protocol was negotioted under the UNFCCC. It entered into force in 2005.

BIODIVERSITY

- Biodiversity is the variety of species and ecosystems on Earth and the ecological processes of which they are a part, including ecosystem, species and genetic diversity components.

Threats to Biodiversity

- The unsustainable harvesting of natural resources, including plants, animals and marine species and their degradation or fragmentation of ecosystems through land conversion for agriculture, forest clearing etc.
- Invasive non-native or alien species being introduced to ecosystems to which they are not adapted.
- A changing global climate threatens species and ecosystems as the distribution of species is largely determined by climate, as is the distribution of ecosystems and plant vegetation zones.

Hotspots of Biodiversity

(i) The areas on Earth which exhibit high species richness as well as species endemism are termed as hotspot of biodiversity.

(ii) To qualify as a hot spot.
  (a) It must support 0.5% of the global plant species.
  (b) It must have lost more than 70% of its original habitat.

(iii) There are 34 hot spot of biodiversity on a global level, out of which following are present in India :

- The Western Ghat
- The Eastern Himalayas
- Indo-Burma region
- Sundalands

Threatened Species of India

Birds	Great Indian Bustard, Forest Owlet, Vulture, Bengal Florican, Himalayan Quail, Siberian Crane
Mammals	Flying Squirrel, Red Panda, Pygmy Hog, Kondana Rat, Snow Leopard, Asiatic Lion, One-Horned Rhinoceros
Reptiles	Gharial, Hawksbill Turtle, River Terrapin
Amphibians	Tiger Toad

Wildlife Conservation in India

Project	*Year*
Project Hangul	1970
Project Gir Lion	1972
Project Tiger	1973
Project Turtles	1975
Crocodile Breeding Scheme	1975
Project Manipur Thamin	1977
Project Rhino	1987
Project Elephant	1992
Project Red Panda	1996
Project Sea Turtle	1999
Project Vulture	2006
Project Snow Leopard	2009
Project Dolphin	2020
Project Asiatic Lion	2020

Tiger Conservation

"The Project Tiger was launched in India in 1972 as conservation programme for saving the Indian Tiger. Project Tiger Scheme has been under implementation since 1973 as a centrally sponsored scheme of Government of India.

Tiger Census 2018

- According to the census, the total count of tigers has risen to 2,967 from 2,226 (in 2014 census). India has also achieved the target for doubling the tiger count four years ahead of the deadline of 2022. Highest increase in tiger count was accounted in Madhya Pradesh followed by Maharashtra and Karnataka.

Name of Tiger Reserve	State
Kaziranga, Orang, Manas, Nameri	Assam
Kamlang, Pakke, Namdapha	Arunachal Pradesh
Nagarjunsagar	Andhra Pradesh
Valmiki	Bihar
Indravati	Chhattisgarh
Bandipur	Karnataka
Palamau	Jharkhand
Nagarahole	Karnataka
Corbett, Rajaji	Uttarakhand
Kanha	Madhya Pradesh
Bor	Maharashtra
Periyar	Kerala

Name of Tiger Reserve	State
Dudhwa	Uttar Pradesh
Similipal	Odisha
Ranthambore	Rajasthan
Melghat, Tadoba-Andheri	Maharashtra
Sariska	Rajasthan
Pench	Madhya Pradesh
Satpura	Madhya Pradesh
Sathyamangalam	Tamil Nadu
Kawal	Telangana
Mukandra Hills	Rajasthan
Pilibhit	Uttar Pradesh

Important Sanctuaries and National Parks Hills

Name	Location
Achanakmar Sanctuary	Chhattisgarh
Bandhavgarh National Park	Madhya Pradesh
Bandipur Sanctuary	Karnataka
Banerghatta National Park	Karnataka
Bhadra Sanctuary	Karnataka
Chandraprabha Sanctuary	Uttar Pradesh
Corbett National Park	Uttarakhand
Dachigam Sanctuary	Jammu & Kashmir
Dandeli Sanctuary	Karnataka
Dudhwa National Park	Uttar Pradesh
Gandhi Sagar Sanctuary	Madhya Pradesh
Ghana Bird Sanctuary	Rajasthan
Gir National Park	Gujarat
Gautam Buddha Sanctuary	Bihar and Jharkhand
Jaldapara Sanctuary	West Bengal
Kaziranga National Park	Assam

Name	Location
Khangchendzonga National Park	Sikkim
Nagarhole National Park	Karnataka
Namdapha Sanctuary	Arunachal Pradesh
Pachmarhi Sanctuary	Madhya Pradesh
Ranganathittu Bird Sanctuary	Karnataka
Simlipal National Park	Odisha
Sunderban National Park	West Bengal
Sonai Rupai Sanctuary	Assam
Tungabhadra Sanctuary	Karnataka
Velovadar Blackbuck National Park	Gujarat
Vedanthangal Bird Sanctuary	Tamil Nadu
Indian Wild Ass Sanctuary	Gujarat

Biosphere Reserves of India

Name	State	Type
Great Rann of Kachchh	Gujarat	Desert
Gulf of Mannar (UNESCO)	Tamil Nadu	Marine
Sunderbans (UNESCO)	West Bengal	Gangetic Delta
Cold Desert	Himachal Pradesh	Western Himalayas
Nanda Devi (UNESCO)	Uttarakhand	West Himalayas
Nilgiri (UNESCO)	Tamil Nadu, Kerala and Karnataka	Western Ghats
Dehang-Dibang	Arunachal Pradesh	East Himalayas
Pachmarhi (UNESCO)	Madhya Pradesh	Semi-Arid
Seshachalam Hills	Andhra Pradesh	Eastern Ghats
Simlipal (UNESCO)	Odisha	Deccan Peninsula
Achanakmar-Amarkantak (UNESCO)	Madhya Pradesh, Chhattisgarh	Maikala Range
Manas	Assam	East Himalayas
Khangchendzonga (UNESCO)	Sikkim	East Himalayas
Agasthyamalai (UNESCO)	Kerala, Tamil Nadu	Western Ghats

Name	*State*	*Type*
Great Nicobar (UNESCO)	Andaman and Nicobar Islands	Islands
Nokrek (UNESCO)	Meghalaya	East Himalayas
Dibru-Saikhowa	Assam	East Himalayas
Panna (UNESCO)	Madhya Pradesh	Catchment area of Ken River

Environment Related Important International Agreements/Conference

UN Conference on the Human Environment	Stockholm (1972)
Convention on Migratory Species	Bonn (1979)
Convention for the Protection of the Ozone Layer	Vienna (1985)
Protocol on Substances that Deplete the Ozone Layer	Montreal (1987)
Convention on the Transboundary Movement of Hazardous Wastes	Basel (1989)
Earth Summit (UN Conference on Environment and Development)	Rio-de-Janiro (1992)
Convention on Prior Informed Consent	Rotterdam (1998)
UN Conference on Sustainable Development	Rio-de-Janiro(2012)
Nagoya Protocol on Genetic Resources	Nagoya (2010)
Convention on Biological Diversity (CBD-CoP-11)	Hyderabad (2012)
Lima Climate Change (CoP-20) Conference	Lima (2014)
Paris Climate Change Conference (CoP-21)	Paris (2015)
Marrakech Climate Change Conference (CoP-22)	Marrakech
Bonn Climate Change Conference (CoP-23)	Bonn (2017)
Katowice Climate Conference (CoP-24)	Poland (2018)
UN Climate Change Conference (CoP-25)	Madrid (2019)
UN Climate Change Conference (Cop-26)	Glasgow (2020)

Terminology

- **Biome** Terrestrial regions characterised by certain types of vegetation and other forms of life. Examples include various types of deserts, grasslands and forests.
- **Wetland** Land that is always covered with saltwater or freshwater, excluding streams, lakes and the open ocean.
- **Biosphere** Zone of the Earth where life is found. It consists of parts of the atmosphere (the troposphere), hydrosphere (mostly surface water and groundwater) and lithosphere (mostly soil and surface rocks and sediments on the bottoms of oceans and other bodies of water) where life is found.
- **Wildlife** All free, undomesticated species. Sometimes the term is used to describe animals only.
- **Threatened Species** Wild species that is still abundant in its natural habitat but is likely to become endangered because of decline in its numbers.
- **Smog** Originally, a combination of smoke and fog but now used to describe other mixtures of pollutants in the atmosphere.
- **Acid Rain** It refers to the precipitation with elevated levels of hydrogen icons or a low pH. When fossil fuel is burnt, oxides are formed in the atmosphere. The oxides formed are carbon oxides, sulphur oxides and nitrogen oxides. All of these oxides form the acidic solution when they dissolve in water.
- **Ecomarks** The Ministry of Environment and Forests, Government of India instituted a scheme, that is operating on a national basis and provides accreditation and labeling for household and other consumer products which meet certain environmental criteria.
- **Coral Bleaching** Coral bleaching occurs when the relations between the coral host and zooxanthellae, which give coral much of their colour, breaks down. Without the zooxanthellae, the tissue of the coral animal appears transparent and the coral's bright white skeleton is revealed.
- **Sustainability** Ability of Earth's various systems, including human cultural systems and economies, to survive and adapt to changing environmental conditions indefinitely.



INDIAN POLITY

CONSTITUTION OF INDIA

- The Constitution is a set of Fundamental Principles, according to which state or other organisation is governed.
- The idea to have a Constituent Assembly for India, for the 1st time given by **MN Roy.**
- The Constitution declares India to be a sovereign, socialist, secular, democratic republic and assures its citizens of justice, equality and liberty and endeavours to promote fraternity among them.

Making of the Constitution

- The Constituent Assembly was formed in 1946, under the scheme formulated by **Cabinet Mission Plan**.
- The total strength of the Assembly was 389, out of these 296 were elected to represent the British India and 93 seats were alloted to the princely states.
- Out of 296 members, 292 members were to be elected by the provincial legislatures while 4 members were to represent the Chief Commissioner's provinces of Delhi, Ajmer-Merwara; Coorg and British Baluchistan.
- Seats were allotted to each province and each Princely state proportional to their respective population roughly in the ratio of one to a million.
- The members from provinces were to be elected by the representatives of three principal communities–Muslims, Sikh and Gereral, by the method of proportional representation. The members from the princely states were nominated by the respective rulers.
- The Constituent Assembly, held its first meeting on **9th, December, 1946,** and reassembled on 14th August, 1947, as the Sovereign Constituent Assembly for the dominion of India.
- It took **2 years, 11 months** and **18 days** to finalise the Constitution.
- **Objective Resolution** was moved in the first session of the Constituent Assembly (on 13th December, 1946), by **Pandit Jawaharlal Nehru**. It was adopted after considerable deliberation and debate in the Assembly on 22nd January, 1947.
- **Dr Sachchidanand Sinha** became the provisional President of the Constituent Assembly, when it met on 9th December, 1946, while later **Dr Rajendra Prasad** and HC Mukherjee were elected the permanent President and Vice-President of the Assembly respectively.
- **BN Rau** was appointed as the Constitutional Advisor of the Assembly.
- The Constituent Assembly appointed a Drafting Committee on 29th August, 1947. **Dr BR Ambedkar** who was the Chairman of the Drafting Committee, submitted a Draft Constitution of India to the President of the Assembly on 21st February, 1948.
- The other members of the drafting committee were N Gopal Swamy Ayyangar, Alladi Krishna Swami Ayyar, KM Munshi, Mohammad Saad-ul-lah, N Madhava Rau and TT Krishnamachari. Mahatma Gandhi was not a member of the Constituent Assembly.

Enactment of the Constitution

- On **26th November, 1949**, the Constitution was declared as passed. The provisions related to citizenship, elections and provisional Parliament etc were implemented with immediate effect, that is from the 26th November, 1949. The rest of the provisions came into force on 26th January, 1950.
- Our Constitution had a Preamble, 395 Articles, 18 Parts and 8 Schedules at the time of enactment. Since then, the Constitution has undergone 103 Amendments and in its current form, consists of a Preamble, 25 Parts, 448 Articles and 12 Schedules.

Committees of the Constituent Assembly

Constituent Assembly appointed 8 major committees and some minor committees to deal with different tasks of Constitution making. *Major committees are*

Name	*Headed by*
Union Powers Committee	Jawaharlal Nehru
Union Constitution Committee	Jawaharlal Nehru
Provincial Constitution Committee	Sardar Patel
Drafting Committee	Dr BR Ambedkar
Advisory Committee on Fundamental Rights, Minorities and Tribal and Excluded Areas	Sardar Patel
Rules of Procedure Committee	Dr Rajendra Prasad
State Committees (committee for negotiating with states)	Jawaharlal Nehru
Steering Committee	Dr Rajendra Prasad

THE PREAMBLE

- The *Preamble* means Introduction or Preface of the Constitution or essence of the Constitution. NA Palkhivala, an eminent jurist and Constitutional expert, called the Preamble as the **Identity card of the Constitution.**
- The Preamble to the Indian Constitution is based on the **Objectives Resolution** drafted and moved by Pandit Nehru and adopted by the Constituent Assembly.

Preamble as in Constitution

We, the People of India, having solemnly resolved to Constitute India into a **Sovereign Socialist, Secular, Democratic, Republic** and secure to all its citizens :

Justice, social, economic and political.

Liberty of thought, expression, belief, faith and worship.

Equality of status and of opportunity and to promote among them all.

Fraternity assuring the dignity of the individual and the **unity and integrity** of the nation. In our Constituent Assembly this twenty-sixth day of November, 1949, do hereby adopt, enact and give to ourselves this Constitution.

Sources of Indian Constitution

The Government of India Act, 1935, formed the basis or 'blue print' of the Constitution of India. Federal scheme, Office of Governor, Judiciary, PSCs are borroweded from it. The Constitutions of following countries are the sources of Indian Constitution :

Britain First past the post system, Parliamentary Government, rule of law, law making procedure, office of the CAG, single citizenship, Bicameralism, Cabinet system.

United States Fundamental Rights, Judicial Review, Independence of Judiciary, written Constitution, Preamble, Post of Vice-President, Impeachment of President.

Ireland Directive Principles of State Policy (Ireland borrowed it from Spain), methods of election of the President, nomination of members in the Rajya Sabha.

Canada Federation with a strong centre, Residuary Powers with centre, appointment of State Governors by the centre, advisory jurisdiction of the Supreme Court.

USSR Fundamental Duties, Idea of justice in Preamble.

Australia Concurrent list, provision regarding trade, commerce and intercourse, Joint sitting of the Parliament.

Germany Suspension of Fundamental Rights during the emergency.

South Africa Procedure of Constitutional Amendment. Election to the Rajya Sabha Members.

France Republic and the ideals of liberty, equality and fraternity in the Preamble.

Part-I (Articles 1-4)	Deals with territory of India, formation of new states, alterations of boundary, names of existing states.
Part-II (Articles 5-11)	Deals with various provisions of citizenship.
Part-III (Articles 12-35)	Deals with Fundamental Rights of Indian citizens. (Article 31- dealing with the right to property was deleted by 44th Amendment).
Part-IV (Articles 36-51)	Deals with Directive Principles of State Policy.
Part-IV-A (Article 51A)	Added by 42nd Amendment in 1976. Contains the Fundamental Duties of the citizens.
Part-V (Articles 52-151)	Deals with Government at the Union Level (Duties and Functions of Prime Minister, Minister, President, Attorney General, Parliament- Lok Sabha and Rajya Sabha, Comptroller and Auditor General).
Part-VI (Articles 152-237)	Deals with Government at State Level. (Duties and Functions of Chief Minister and his Ministers, Governor, State Legislature, High Court, Advocate General of the State).
Part-VII (Article 238)	Deals with states, was deleted in 1956, by the 7th Amendment.
Part-VIII (Articles 239-241)	Deals with Union Territories.
Part-IX (Articles 243-243-O)	Deals with Panchayats. Added by 73rd Amendment Act in 1992.
Part IX-A (Articles 243 P-243ZG)	Deals with the Municipalities. Added by 74th Amendment in 1992.
Part IX-B	The Co-operative Societies (243-ZH to 243-ZT). (added by 97th Amendment, 2011)
Part-X (Articles 244- 244A)	Deals with Scheduled and Tribal Areas.
Part-XI (Articles 245-263)	Deals with relation between Union and States.
Part-XII (Articles 264-300A)	Deals with distribution of revenue between Union and States, Appointment of Finance Commission (Article 280), contracts liabilities etc.
Part-XIII (Articles 301-307)	Relates to Trade, Commerce and Intercourse within the Territory of India.
Part-XIV (Articles 308-323)	Deals with UPSC and state Public Service Commission.
Part-XIV A (Articles 323A-323B)	Deals with tribunals. (Added by 42nd Amendment)
Part-XV (Articles 324-329A)	Deals with Elections (Also Election Commission).
Part-XVI (Articles 330-342)	Deals with special provisions for Scheduled Castes and Scheduled Tribes and Anglo-Indian Representation.
Part-XVII (Articles 343-351)	Relates to Official Language.
Part-XVIII (Articles 352-360)	Deals with Emergency Provisions.
Part-XIX (Articles 361-367)	Exemption of Criminal Proceeding for their Official Acts of President and Governors and Miscellaneous.
Part-XX (Article 368)	Deals with Amendment of Constitution.
Part-XXI (Articles 369-392)	Temporary, Transitional and Special Provision (Article-369 gives temporary powers to the Parliament to make laws for state list).
Part-XXII (Articles 393-395)	Concerns the short title, commencement, authorative text in Hindi and repeals of Constitution.

Amendability of the Preamble

- The Preamble can be amended under **Article 368** or not, this question arose for the first time in **Kesavananda Bharati Case** (1973). In this case Supreme Court held that Preamble is the part of the Constitution and can be amended, subject to the condition that no amendment is done to the basic features of the Constitution.
- The Preamble has been amended only once so far, in 1976, by 42nd Constitutional Amendment Act, which added three new words **Socialist, Secular** and **Integrity**. This amendment was held to be valid.

Welfare State

A Welfare State is one that provides social justice to all its citizens. It is based on the principles of equality of opportunity, equitable distribution of wealth and public responsibility for those unable to avail themselves of the minimal provisions for a good life. Indian Constitution has many features that aim at providing a Welfare State. The Preamble to the Constitution was amended in 1976 by the 42nd Amendment Act to insert the goal of socialism. Directive Principles of State Policy also aim at establishing a Welfare State.

UNION AND ITS TERRITORY

- **Articles 1-4** under Part-I of the Constitution deals with the Union and its Territories.
- **Article 1,** describes India, that is Bharat, as a Union of States.
- Territory of India can be *classified into three categories*
  1. Territories of the States
  2. Union Territories
  3. Territories that may be acquired by the Government of India at any time
- At present, there are **28 States** and **9 Union Territories,** after unification of Daman and Diu and Dadra and Nagar Haveli.

The 'Territory of India' is a wider expression than the 'Union of India' because the latter includes only states while the former includes not only states, but also UTs and territories that may be acquired by the Government of India at any future time.

- **Article 2** empowers the Parliament to admit into the Union of India or establish, new states on such terms and conditions as it thinks fit.
- **Article 3** authorises the Parliament to
  (a) form a new state by separation from any state or by uniting two or more states or parts of states or by uniting any territory to a part of any state;
  (b) increase the area of any state;
  (c) diminish the area of any state;
  (d) alter the boundaries of any state;
  (e) alter the name of any state.

Schedule	*Subject*
Ist Schedule	Names of States and UTs and their Territorial Extent.
2nd Schedule	Salaries and Emoluments of President, Governor etc.
3rd Schedule	Forms of Oath and Affirmations of Members of Legislatures, Ministers and Judges
4th Schedule	Allocation of Seats in the Rajya Sabha
5th Schedule	Administration and control of Scheduled Areas and Scheduled Tribes
6th Schedule	Administration of Tribal Areas in the state of Assam, Meghalaya, Tripura and Mizoram
7th Schedule	Distribution of Power between the Union and the State Government
8th Schedule	Languages (22 scheduled languages)
9th Schedule	Validation of certain Acts and Regulations (added by 1st Amendment, 1951)
10th Schedule	Anti-Defection Law (added 52nd Amendment, 1985)
11th Schedule	Powers, authority and responsibilities of Panchayats. It has 29 matters. (added by 73rd Amendment, 1992)
12th Schedule	Powers, authority and responsibilities of Municipalities. It has 18 matters (added by 74th Amendment, 1992)

- A Bill seeking to create a new state or alter boundaries of existing states can be introduced in either House of the Parliament, only on the recommendation of the President.
- President has to refer the Bill to the State Legislature concerned for expressing its opinion, within a specified time.
- Parliament is not bound to accept or act upon the views of the State Legislature on a State Reorganisation Bill.
- **Article 4** Says that laws made under Articles 2 and 3 are not to be considered as amendments of this Constitution under Article 368. Thus, such laws can be passed by a simple majority in both Houses of Parliament.

EVOLUTION OF STATES AND UNION TERRITORIES

Dhar Commission

- The Constituent Assembly appointed the SK Dhar Commission in June 1948, to study the feasibility of reorganising the states on linguistic basis.
- The Dhar Commission categorically rejected the basis of linguistic formation of states.

JVP Committee

- The Congress in 1948, appointed a three member committee to consider the recommendation of the Dhar Commission, members were Jawaharlal Nehru, Vallabhbhai Patel and Pattabhi Sitarammaiah.
- The committee rejected language as the basis for the reorganisation.

Fazl Ali Commission

- In 1953, Government appointed a three-member State Reorganisation Commission under the Chairmanship of Fazl Ali. KM Panikkar and HN Kunzru were other two members.
- It broadly accepted language as the basis of reorganisation of states.
- But it rejected the theory of one language one state.
- Fazl Ali Commission's recommendations were broadly accepted and it was followed by the **States Reorganisation Act** (1956) and the 7th Constitutional Amendment Act (1956).
- After reorganisation, 14 States and 6 Union Territories were created on 1st November, 1956.

Evolution of States and Union Territories

In **1956**, there were fourteen States and six Union Territories.

In **1953,** Andhra Pradesh was created as the 1st linguistic state.

In **1960,** Bombay was bifurcated into Gujarat and Maharashtra.

In **1963,** Nagaland was created as separate state.

In **1966,** Haryana was created out of Punjab and Chandigarh became a Union Territory.

In **1970,** the Union Territory of Himachal Pradesh was elevated to the status of a state.

In **1971,** Manipur, Tripura and Meghalaya were granted statehood.

In **1974,** Sikkim became an associate state of the Indian Union. By the 36th Constitutional Amendment Act (1975), Sikkim became a full fledged state of the Indian Union.

In **1987,** Mizoram, Arunachal Pradesh & Goa became states.

In **2000,** three more new states; Chhattisgarh, Uttarakhand and Jharkhand were created.

On **2nd June, 2014** 'Telangana' state came into existence after reorganisation of Andhra Pradesh.

On **31st October, 2019** the state of Jammu & Kashmir was reorganised into two Union Territories of J & K and Ladakh.

On **26th January, 2020** two UTs Daman and Diu and Dadra and Nagar Haveli got merged into single UT.

Union Territories

- Jammu & Kashmir, Ladakh, National Capital Territory of Delhi, Puducherry, Adaman & Nicobar are headed by the Lieutenant Governor. Daman and Diu, Dadra and Nagar Haveli have a common administrator.
- From 26th of January both the UTs have been merged into single UT named 'Dadra and Nagar Haveli' and 'Daman and Diu'.
- Lakshadweep, Chandigarh is also governed by an administrator.
- Jammu & Kashmir Delhi and Puducherry have Legislative Assemblies.
- There are total 8 Union Territories-Delhi, Puducherry, Daman and Diu & Dadra and Nagar Haveli, Chandigarh, Lakshadweep, Andaman and Nicobar Islands, Jammu and Kashmir and Ladakh
- By the **69th Constitutional Amendment Act 1991**, Delhi was given the status of National Capital Territory of India.

CITIZENSHIP

- Indian Constitution deals with the citizenship from Articles 5-11 under Part II. Articles 5 to 8 deal with the citizenship of (a) persons domiciled in India; (b) persons migrated from Pakistan, (c) persons migrated to Pakistan but later returned and (d) persons of Indian origin residing outside India.

Acquisition of Citizenship

- A citizen is a person who enjoys full membership of the country in which he lives.
- Indian Constitution provides a single and uniform citizenship for the entire country.
- The Citizenship Act of 1955, prescribes five ways of **acquiring citizenship** *viz* birth, descent, registration, naturalisation and incorporation of territory.

By Birth

- Every person born in India on or after 26th January, 1950 but before 1st July 1987 shall be a citizen of India by birth, irrespective of the nationality of his parents.
- The children of foreign diplomats posted in India and enemy aliens cannot acquire Indian citizenship by birth.

By Descent

- Persons born outside India on or after 26th January, 1950, but before 10th December, 1992, are citizens of India by descent if their father was a citizen of India at the time of their birth.

By Registration

- The Central Government may, on an application, register as a citizen of India any person if he *belongs to any of the following categories*
  - (a) A person of Indian origin, residing in India for 7 years.
  - (b) A person of Indian origin, who is ordinarily resident in any country or place outside undivided India.
  - (c) Persons, who are married to citizens of India and resident of India for 7 years.
  - (d) Minor children of persons, who are citizen of India.

By Naturalisation

- It can be acquired by foreigner, who has resided in India for 12 years.

By Incorporation of Territory (Foreign Territory)

- If any new territory become a part of India, the Government of India specifies the person of that territory to be citizens of India. For example, Pondicherry in 1962.

Loss of Citizenship

- The Citizenship Act, 1955 prescribes three ways of **losing citizenship**.

By Renunciation

- If a person gives up his Indian citizenship.

By Termination

- When an Indian citizen voluntarily acquires the citizenship of another country.

By Deprivation

- Deprivation of citizenship by the Government of India on the basis of acquisition of citizenship by fraud, helping an enemy during a war or being disloyal to the Constitution.

Overseas Citizens of India

- Citizenship Act has been amended in 2003, by which people of Indian origin of 16 specified countries, except Pakistan and Bangladesh, becomes eligible to be registered as the **Overseas Citizens of India** (OCI).
- OCIs are entitled to some benefits like multiple entry, multipurpose life long visas, they can live and work in India or their country of naturalisation.
- They are not entitled to hold Constitutional posts and employment with the government and they can't vote.
- All Persons of Indian Origin (PIO) card-holders are deemed to Overseas Citizens of India (OCI) card-holders with effect from 9th January, 2015.

The Citizenship (Amendment) ***Act, 2019***

- It amends the Citizenship Act, 1955.
- The amended Act provides that the Hindus, Sikhs, Buddhists, Jains, Parsis and Christians from Afghanistan, Bangladesh and Pakistan who entered India on or before 31st December, 2014 will not be treated as illegal migrant and will be eligible for grant of citizenship.
- It also reduces the period of naturalisation for such group of persons from 11 year to 5 years.

FUNDAMENTAL RIGHTS

Right to Equality (Articles 14-18)

- Equality before law and equal protection of laws. (Article 14)
- Prohibition of discrimination on grounds of religion, race, caste, sex or place of birth. (Article 15)
- Equality of opportunity in matters of public employment. (Article 16)
- Abolition of untouchability. (Article 17)
- Abolition of titles. (Article 18)

Right to Freedom (Articles 19-22)

- Article 19 of the Constitution guarantees six Fundamental Rights/ freedoms to the citizens of India.
- Freedom of speech and expression, to assemble peacefully and without arms, to form associations or union, to move freely throughout the territory of India, to practice any profession or to carry on any occupation, trade or business.
- The state can impose reasonable restrictions on the exercise of these rights on the ground of security of the state, friendly relations with foreign states, public order, decency, morality, contempt of court, defamation etc.
- Protection in respect of conviction for offences. (Article 20)
- Protection of life and personal liberty (Article 21).
- **Article 21A** states that the state shall provide free and compulsory education to all children of the age of 6-14 years. Protection against arrest and detention in certain cases **(Article 22).**

Article19 automatically gets suspended during National Emergency. Article 21 & 22 can't be suspended even during National Emergency (Article 352).

Right to Property It was repealed, from Article 31, by the 44th Amendment Act, 1978, and was made a legal right under Article 300A.

Right against Exploitation (Articles 23-24)

- Prohibition of trafficking in human beings and forced labour (Article 23).
- Prohibition of employment of children in factories etc (Article 24).

Right to Freedom of Religion (Articles 25-28)

- Freedom of conscience and free profession, practice and propagation of religion (Article 25).
- Freedom to manage religious affairs (Article 26).
- Freedom as to payment of taxes for promotion of any particular religion (Article 27).
- Freedom from attendance at religious instructions or religious worship in certain educational institutions (Article 28).

Cultural and Educational Rights (Articles 29-30)

- Protection of language, script, and culture of minorities (Article 29).
- Right of minorities to establish and administer educational institutions (Article 30).

Right to Constitutional Remedies (Article 32)

- Right to move to the Supreme Court (Article 32) and the High Courts (Article 226) in case of violation of the Fundamental Rights. Termed by **BR Ambedkar** as the **Heart and Soul of the Constitution**.
- The Supreme Court and High Courts have the power to issue writs under Article 32, if Fundamental Rights are violated.
- Supreme Court has ruled that Article 32 is the basic feature hence cannot be taken away through amendment

Writs	*Meaning*	*Intended Purpose*
Habeas Corpus	You may have the body	To release a person who has been detained unlawfully whether in prison or in private custody.
Mandamus	We Command	To secure the performance of public duties by lower court, tribunal or public authority.
Certiorari	To be certified	To quash the order already passed by an inferior court, tribunal or quasi judicial authority.
Prohibition	The act of stopping something	To prohibit an inferior court from continuing the proceedings in a particular case where it has no jurisdiction to try.
Quo Warranto	What is your authority	To restrain a person from holding a public office to which he is not entitled.

Directive Principles of State Policy

- The Directive Principles of State Policy are enumerated in Part IV of the Constitution from **Articles 36 to 51**.
- Dr BR Ambedkar, described these principles as 'novel features' of the Constitution.
- The Directive Principles of State Policy along with Fundamental Rights contain the **philosophy** and the **soul** of the Constitution.

Features

- These are **constitutional instructions** to the state in legislative, executive and administrative matters.
- They resembles the 'Instrument of Instructions' enumerated in the Government of India Act of 1935.
- They promotes social and economic democracy. They embody the concept of 'a welfare state'.
- These are fundamental in the governance of the country.
- They are non-justiciable.

Some of the Important DPSP's are

- State to secure a social order for the promotion of welfare of the people, (Article 38), Organisation of Village Panchayats. (Article 40)
- To secure the right to work, education and public assistance in cases of unemployment, old age etc. (Article 41); Uniform Civil Code for the citizens. (Article 44)
- Promotion of educational and economic interests of Scheduled Castes, Scheduled Tribes and other weaker sections, (Article 46)
- Separation of Judiciacy from Executive. (Article 50)
- Promotion of international peace and security. (Article 51)

Fundamental Rights	*Directive Principles*
These are negative as they prohibit the state from doing certain things.	These are positive as they require the state to do certain things.
These are **justiciable,** i.e., they are legally enforceable by the courts in case of their violation.	These are **non-justiciable**, *i.e.*, they are not legally enforceable by the courts for their violation.
They aim at establishing political democracy in the country.	They aim at establishing social and economic democracy in the country.
These have legal sanctions.	These have moral and political sanctions.
They promote the **welfare of the individual**. Hence, they are personal and individualistic.	They promote the **welfare of the community.** Hence, they are societarian and socialistic.
The courts are bound to declare a law **violative** of any of the Fundamental Rights as unconstitutional and invalid.	The courts cannot declare a law **violative** of any of the Directive Principles as unconstitutional and invalid. However, they can uphold the validity of a law on the ground that it was enacted to give effect to a directive.

FUNDAMENTAL DUTIES

- The Fundamental Duties are the obligations of all citizens to help and promote a spirit of patriotism and to uphold the unity of India.
- Originally, Constitution did not contain Fundamental Duties. In 1976, the FDs of citizens were added by 42nd **Constitutional Amendment Act** on the recommendations of **Swarn Singh Committee**.
- Originally, ten FDs were added. In 2002, one more duty was added through 86th Amendment.

List of Fundamental Duties

According to **Article 51 A**, *it shall be the duty of every citizen of India*

(a) to abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem;

(b) to cherish and follow the noble ideals that inspired the national struggle for freedom;

(c) to uphold and protect the sovereignty, unity and integrity of India;

(d) to promote harmony and the spirit of common brotherhood amongst all the people of India transcending religious, linguistic and regional or sectional diversities and to renounce practices derogatory to the dignity of women;

(e) to defend the country and render national service when called upon to do so;

(f) to value and preserve the rich heritage of the country's composite culture;

(g) to protect and improve the natural environment including forests, lakes, rivers and wildlife and to have compassion for living creatures;

(h) to develop scientific temper, humanism and the spirit of inquiry and reform;

(i) to safeguard public property and to abjure violence;

(j) to strive towards excellence in all spheres of individual and collective activity so that the nation constantly rises to higher levels of endeavour and achievement; and

(k) to provide opportunities for education to his child or ward between the age of 6 and 14 years. This duty was added by the 86th Constitutional Amendment Act, 2002.

UNION GOVERNMENT

PRESIDENT

Article 52 provides for the office of the President of India. The President is the head of the Indian State. He is the first citizen of India and acts as the symbol of unity, integrity and solidarity of the nation.

Qualification

Under **Article 58**, a person to be eligible for election as President *should fulfil the following qualifications*

- He should be a citizen of India.
- He should have completed 35 years of age.
- He should be qualified for election as a member of the Lok Sabha.
- He should **not hold any office of profit** under the Union Government or any State Government or any local authority or any other public authority.

Election

The President is elected **not directly** by the people but by *members of electoral college consisting of*

- the elected members of both the Houses of Parliament;
- the elected members of the Legislative Assemblies of the states and
- the elected members of the Legislative Assemblies of the Union Territories of Delhi, Puducherry and Jammu & Kashmir.
- Nominated members don't participate in the election of the President.
- When Assembly is dissolved, the members cease to be qualified to vote in the Presidential election.
- **Value of the Vote of an MLA**

$$= \frac{\text{Total Population of State}}{\text{Total Number of Elected MLAs}} \times \frac{1}{100}$$

- **Value of the Vote of an MP**

$$= \frac{\text{Total Value of Votes of All MLAs of All States}}{\text{Total Number of Elected MPs}}$$

- The President's election is held in accordance with the system of proportional representation by means of single transferable vote and the voting is through secret ballot.
- All disputes regarding election of the President is **adjudicated** by the Supreme Court.
- Nomination for election of President must be proposed by at least 50 electors as proposers and 50 electors as seconders.
- Security deposit for the nomination as President is ₹ 15000 in RBI.

Conditions of President's Office

Under **Article 59**, the Constitution lays down the *following condition of the President's office*

- He should not be a member of either House of Parliament or a House of the State Legislature.
- He should not hold any other Office of profit.
- He is entitled to such emoluments, allowances and privileges as may be determined by Parliament.
- His emoluments and allowances cannot be diminished during his term of office.
- The **Oath** of the President is administered by the Chief Justice of India and in his absence, the senior most Judge of the Supreme Court. (Article 60)
- Any person acting as the President also undertake similar oath.

Term of the President

Under **Article 56**, the President shall hold office for a **term of 5 years** from the date on which he enters upon his office.

- He can resign from his office by writing letter to the Vice-President.

Name	*Tenure*		*Important Facts*
Dr Rajendra Prasad	26.01.1950	13.05.1962	▪ First President and also had the longest tenure (12 years)
Dr S Radhakrishnan	13.05.1962	13.05.1967	▪ Was also first Vice-President of India
Dr Zakir Hussain	13.05.1967	03.05.1969	▪ Shortest tenure; First Muslim President; First President to die in harness
V V Giri (Vice-President)	03.05.1969	20.07.1969	▪ First acting President of India
Justice M Hidayat-ul-lah	20.07.1969	24.08.1969	▪ Was also the Chief Justice of India and Second Acting President
V V Giri	24.08.1969	24.08.1974	—
F Ali Ahmed	24.08.1974	11.02.1977	▪ Died in office
BD Jatti	11.02.1977	25.07.1977	▪ Acting President
N Sanjeeva Reddy	25.07.1977	25.07.1982	▪ Youngest President (64 years)
Giani Zail Singh	25.07.1982	25.07.1987	▪ First Sikh President
R Venkataraman	25.07.1987	25.07.1992	▪ Oldest President (76 years)
Dr SD Sharma	25.07.1992	25.07.1997	
KR Narayanan	25.07.1997	25.07.2002	▪ First Dalit President
Dr APJ Abdul Kalam	25.07.2002	25.07.2007	▪ First scientist to become President
Pratibha Patil	25.07.2007	24.07.2012	▪ First woman President
Pranab Mukherjee	25.07.2012	25.07.2017	—
Ram Nath Kovind	25.07.2017	Till date	—

**As on 31st December, 2020*

Impeachment

Under **Article 61**, President can be impeached from office for 'violation of the Constitution'.

- The impeachment can be initiated by either House of the Parliament.
- These charges should be signed by one-fourth members of the House (that framed the charges) and a **14 days' notice** should be given to the President. After the impeachment bill is passed by a majority of two-thirds of the total membership of that House, it is sent to the other House, which should investigate the charges.
- The President has the right to appear and to be represented at such investigation. If the other House also sustains the charges and passes the impeachment bill by a majority of two-thirds of the total membership, the President stands removed from his office from the date on which the bill is so passed.
- No President has so far been impeached.

Vacancy of the President's Office

Under **Article 62** a vacancy in the President's office *can occur in any of the following ways*

- On the expiry of his tenure of 5 years.
- By his resignation or his removal by the process of impeachment By his death.
- When he becomes disqualified to hold office or when his election is declared void.
- An election to fill the vacancy (due to expiration of term) must be held before the expiry of the term.
- If the office fall vacant by resignation, removal, death or otherwise the Vice-President acts as President. The election to fill such vacancy should be held within six month from the date of occurrence of vacancy.
- The newly-elected President remains in office for a full term of five years from the date he assumes charge of his office.

Quick Facts

V V Giri In the History of Presidential Election, is the only person, who won the election of the President as an independent candidate in 1969.

NS Reddy In July 1977, Neelam Sanjeeva Reddy was elected unopposed as no one else filed nomination for the post of the President.

Justice Hidayat-ul-lah Only once in the History of India, Justice Mohd Hidayat-ul-lah, Chief Justice of Supreme Court, discharged the duties of the President in 1969.

Justice Mohd Hidayat-ul-lah is the only person to perform the function of the President two times in two different capacities, the first time in 1969, being the Chief Justice of the Supreme Court and the second time being the Vice-President of India in 1982.

Powers and Functions of the President

Executive Powers

Under **Article 77**, the Executive *powers of the President are*

- All Executive actions of the Government of India are **formally** taken in his name.
- He appoints Prime Minister, other Ministers, Chief Justice and Judges of Supreme Court, High Courts, the Attorney General of India, the Comptroller and Auditor General, Chairman and Members of UPSC, Chief Election Commissioner and other Members of Election Commission, Governors, Members of Finance Commission etc.

Legislative Powers

- He can summon or prorogue the Parliament and can dissolve the Lok Sabha.
- He can summon a joint sitting of both the Houses of Parliament, which is presided by Speaker of the Lok Sabha.
- Address the Parliament at the commencement of the first session after each general election and the first session of each year.
- He can appoint any Member of the Lok Sabha to preside over its proceedings when the offices of both the Speaker and the Deputy Speaker fall vacant. Similarly, He can also appoint any member of the Rajya Sabha to preside over its proceedings when the offices of both the Chairman and the Deputy Chairman fall vacant.
- **Nominate 12 members** of Rajya Sabha (from amongst persons who have special knowledge in respect of Literature, Science, Art and Social service) and **2 members** of Anglo-Indian Community in the Lok Sabha.
- He decides on questions as to disqualifications of members of the Parliament, in consultation with the Election Commission.
- His **prior recommendation** or permission is needed to introduce certain types of bills in the Parliament, e.g., Money Bill, Bill for creation/recreation of new States.
- When a bill is sent to the President after it has been passed by the Parliament, *he can*
  - (a) give his assent to the bill or
  - (b) withhold his assent to the bill or
  - (c) Return the bill (if it is not a Money Bill) for reconsideration of the Parliament.
- However, if the bill is passed again by the Parliament, with or without amendments, the President has to give his assent to the bill.
- He can **promulgate ordinances** (Article 123) when the Parliament is not in session. The ordinances must be approved by the Parliament within **six weeks** from its reassembly. He can also withdraw an ordinance at any time.

Financial Powers

- Money Bills can be introduced in the Parliament only with his prior recommendation.
- No demand for a grant can be made except on his recommendation. He causes to be the laid the annual financial statement before the Parliament. (i.e., the Union Budget).
- He can make advance out of the Contingency Fund of India to meet any unforeseen expenditure.
- He constitutes a Finance Commission after every five years to recommend the distribution of revenues between the centre and the states.

Judicial Powers

- He appoints the Chief Justice and the Judges of Supreme Court and High Courts.
- He can seek advice from the Supreme Court on any question of law or fact. However, the advice tendered by the Supreme Court is not binding on the President. (Article 143)
- The President also has various powers of Pardon, Reprieve, Remission, Respite and Commutation. (Article 72)
- The president enjoys legal immunity and is not accountable to any court of law for anything done in the exercise of his official duties.

Diplomatic Powers

- The international treaties and agreements are negotiated and concluded on behalf of the President.
- He sends and receives diplomats like ambassadors, high commissioners.

Military Powers

- Supreme Commander of the defence forces of India appoints the chiefs of the Army, the Navy and the Air Force.
- Declares war or concludes peace, subject to the approval of the Parliament.

Emergency Powers

- The President of India can proclaim emergency in three conditions after getting the written recommendation of the Cabinet.
- **National Emergency** (Article 352) arising out of war, external aggression or armed rebellion within the country.
- **Constitutional Emergency** (Article 356) arising out of the failure of the Constitutional machinery in the states. It is also known as **President's Rule** or State Emergency.
- **Financial Emergency** (Article 360) arising out of a threat to financial stability or credit of India.

Declaration of National Emergency

- First emergency was declared in 1962, due to Chinese aggression.
- Second emergency was declared in 1971, due to Indo-Pakistan War.
- Third emergency was declared in 1975, on grounds of internal disturbance.

Position of President

- The Constitution of India has provided for a Parliamentary form of Government and the President has been made only a nominal executive, the real executive being the Council of Ministers headed by the Prime Minister.

VICE-PRESIDENT

- Article 63 There shall be a Vice- President of India. He occupies the second highest office in the country.

Election

- The Vice-President is indirectly elected by an electoral college consisting of the members (both elected and nominated) of both Houses of Parliament.
- The election is held in accordance with the system of proportional representation by means of single transferable vote.

Qualification

- Should be a citizen of India.
- Should have **completed 35** years of age.
- Should be qualified for election as a member of Rajya Sabha.
- Should not hold any office of profit.

Oath

- Under **Article 69**, the oath of office of the Vice-President is administered by the President or some person appointed in that behalf by him.

Condition of Office

- Should not be a member of either House of the Parliament or State Legislature.
- Should not hold any office of profit.

Term of Office

- Can resign from his office at any time by addressing the resignation letter to the President.
- Holds office for a term of 5 years from the date on which he enters upon his office.

Name	*Tenure*	*Notes*
Dr S Radhakrishnan	1952 to 1962	1st Vice-President; had the longest tenure (10 years, elected twice)
Dr Zakir Hussain	1962 to 1967	
V V Giri	1967 to 1969	Shortest tenure so far (2 years)
GS Pathak	1969 to 1974	
B D Jatti	1974 to 1979	
Mohd Hidayat-ul-lah	1979 to 1984	
R Venkataraman	1984 to 1987	
Dr SD Sharma	1987 to 1992	
KR Narayanan	1992 to 1997	
Krishan Kant	1997 to 2002	Died in office
Bhairon Singh Shekhawat	2002 to 2007	
Mohammad Hamid Ansari	2007 to 2017	
Venkaiah Naidu	2017 to Till date	

**As on 31st December, 2020*

- Can be removed by a resolution of the Rajya Sabha passed by an absolute majority and agreed by the Lok Sabha. Can be elected for any number of terms.

Powers and Functions

- He acts as the *ex-officio* Chairman of Rajya Sabha. In this capacity, his powers and functions are similar to those of the speaker of Lok Sabha.
- He acts as President when a vacancy occurs in the office of the President due to his resignation, removal, death or otherwise. He can act as President only for a maximum period of six months.
- While acting as President or discharging the functions of President, the Vice-President does not perform the duties of the office of Chairman of Rajya Sabha, those duties are performed by the Deputy Chairman of Rajya Sabha.

Acting President

When two Presidents, Dr Zakir Hussain and Fakr-ud-din Ali Ahmed, died in office, then respective Vice-Presidents, V V Giri and BD Jatti acted as President.

Discharging The Functions of President

- The Vice-President Dr S Radhakrishnan discharged the functions of the President in June 1960, when the President Dr Rajendra Prasad was on a 15 days tour to the USSR and again in July 1961, when Dr Rajendra Prasad was very ill.

PRIME MINISTER

- In the scheme of Parliamentary system of government, the President is the nominal executive authority (dejure executive) and **Prime Minister is the real executive** authority (de facto executive).
- Prime Minister is the head of the government while President is the head of the state. Article 75, says that the Prime Minister shall be appointed by the President.

Oath, Term and Salary

- President administers to him the oaths of office and secrecy.
- The **term of the Prime Minister is not fixed** and he holds office during the pleasure of the President. However, this does not mean that the President can dismiss the Prime Minister at any time.

- So long as the Prime Minister enjoys the majority support in the Lok Sabha, He cannot be dismissed by the President. However, if he loses the confidence of the Lok Sabha, He must resign or the President can dismiss him.
- The **salary** and allowances of the Prime Minister are determined by the Parliament from time to time.

Powers

- He recommends persons who can be appointed as Ministers by the President.
- The Prime Minister can recommend the dissolution of the Lok Sabha before its expiry.
- He allocates portfolios among Ministers. He can ask a Minister to resign or get him dismissed by the President. The Prime Minister advices the President on appointments of important officials.
- The Prime Minister along with his cabinet advises the President on declaration of emergency on various grounds.

Union Council of Ministers

- Article 74 says that there shall be a Council of Ministers to aid and advise the President.
- The total number of Ministers shall not exceed 15% of total strength of Lok Sabha. The Council of Ministers is collectively responsible to Lok Sabha (Article 75).
- The Council of Ministers (other than Prime Minister) are appointed by President on the advice of the Prime Minister. They hold office during the pleasure of the President

Name	*Tenure*		*Notes*
Jawaharlal Nehru	15.08.1947	27.05.1964	First Prime Minister of India, died in office; also had the longest tenure (17 years)
Gulzari Lal Nanda	27.05.1964	09.06.1964	First acting Prime Minister
Lal Bahadur Shastri	09.06.1964	11.01.1966	Only Prime Minister to die abroad during an official tour
Gulzari Lal Nanda	11.01.1966	24.01.1966	First to become acting Prime Minister twice
Indira Gandhi	24.01.1966	24.03.1977	First woman Prime Minister of India; First Prime Minister to lose an election, First Member of Rajya Sabha become PM
Morarji Desai	24.03.1977	28.07.1979	Oldest Prime Minister (81 years) and the first to resign from office, First CM to become PM
Charan Singh	28.07.1979	14.01.1980	Only Prime Minister who did not face the Parliament
Indira Gandhi	14.01.1980	31.10.1984	First Prime Minister to be assassinated
Rajiv Gandhi	31.10.1984	01.12.1989	Youngest Prime Minister (40 years)
VP Singh	21.12.1989	10.11.1990	First Prime Minister to step down after vote of no-confidence
Chandra Shekhar	10.01.1990	21.06.1991	
PV Narasimah Rao	21.06.1991	16.05.1996	First Prime Minister from southern India
Atal Bihari Vajpayee	16.05.1996	01.06.1996	Shortest tenure of a Prime Minister
HD Deve Gowada	01.06.1996	21.04.1997	
IK Gujral	21.04.1997	19.03.1998	
Atal Bihari Vajpayee	19.03.1998	13.10.1999	
Atal Bihari Vajpayee	13.10.1999	22.05.2004	
Dr Manmohan Singh	22.05.2004	25.05.2014	First Sikh Prime Minister, Longest tenure after Jawaharlal Nehru
Narendra Modi	26.5.2014	Till date	First Prime Minister born after independence

**As on 31st December, 2020*

- **No Legal Responsibility** There is no provisions in the Constitution for a system of legal responsibility of a Minister in India. While in Britain, there is legal responsibility of a minister.

Deputy PMs	*Tenure*
Sardar Vallabhbhai Patel	1947-50
Morarji Desai	1967-69
Charan Singh & Jagjiwan Ram (Jointly)	1977-79
YB Chavan	1979-80
Devi Lal	1989-91
LK Advani	2002-04

Types of Ministers

- **Cabinet Ministers** They are the real policymakers. The Cabinet's consent is necessary for all important matters.
- **Ministers of State** can hold either independent charge or attached to a Cabinet Minister.
- **Deputy Ministers** do not hold separate charge.

PARLIAMENT

- The Constitution of India provides a Parliamentary form of government, both at the centre and in the states.
- The Parliament of India consists of the President, the Lok Sabha and the Rajya Sabha (Article 79).
- Although President is not a member of either of the House, He is an integral part of it.

Rajya Sabha

- Rajya Sabha is a permanent body and **not subject to dissolution**. Its maximum strength is 250 out of which 238 are representatives of States & UTs and 12 are nominated by the President.
- The total membership of the Rajya Sabha, at present is 245. Of these, 225 represent the States & 8 represent the UTs.
- Out of 8 UTs, only three (J & K, Delhi and Puducherry) have representation in the Rajya Sabha. The population of other five are too small to have any representative in the Rajya Sabha.
- There are no seats reserved for SCs and STs in Rajya Sabha.
- The members of Rajya Sabha are indirectly elected by the elected members of State Legislative Assemblies.
- One-third members retire every second year and their seats are filled by fresh elections at the beginning of every third year.
- Representation of People Act, (1951) provided the term of office of a member of the Rajya Sabha shall be six years.

Lok Sabha

- Lok Sabha is the Lower House of Parliament. Its maximum strength is 552. At present, the strength of Lok Sabha is 545
- The members are directly elected by the people from territorial constituencies in States & UTs.
- The Constitution provides for reservation of seats for SCs and STs in Lok Sabha on the basis of their population.
- Its **normal term is five years** after which it automatically get dissolves.

Note *The Constitution provided for nomination of two Anglo-Indians to the Lok Sabha. However, this provision has not been extended after 2020.*

States/UTs	*In Rajya Sabha*	*In Lok Sabha*
States		
Andhra Pradesh	11	25
Arunachal Pradesh	1	2
Assam	7	14
Bihar	16	40
Chhattisgarh	5	11
Goa	1	2
Gujarat	11	26
Haryana	5	10
Himachal Pradesh	3	4
Jharkhand	6	14
Karnataka	12	28
Kerala	9	20
Madhya Pradesh	11	29
Maharashtra	19	48
Manipur	1	2

Meghalaya	1	2
Mizoram	1	1
Nagaland	1	1
Odisha	10	21
Punjab	7	13
Rajasthan	10	25
Sikkim	1	1
Tamil Nadu	18	39
Telangana	7	17
Tripura	1	2
Uttarakhand	3	5
Uttar Pradesh	31	80
West Bengal	16	42
Union Territories		
Andaman and Nicobar Island	—	1
Chandigarh	—	1
Dadra and Nagar Haveli	—	1
Daman and Diu	—	1
Delhi (The National Capital Territory of Delhi)	3	7
Jammu & Kashmir	4	5
Ladakh	—	1
Lakshadweep	—	1
Puducherry	1	1
Nominated Members	12	2
Total	**245**	**545**

- The President is authorised to dissolve Lok Sabha (on the advice of Prime Minister) at any time even before the completion of five years and this cannot be challenged in the Court of Law.
- Lok Sabha can be **extended** during the National Emergency by a law of Parliament for one year at a time for any length of time. But this extension cannot go beyond a period of six months after the emergency has ceased to operate.
- When th Lok Sabha is dissolved, the bills pending in it lapses. Also the bills passed by the Lok Sabha but pending in the Rajya Sabha, lapses.

Speaker of Lok Sabha

- The Speaker of the Lok Sabha conducts the business in the House and maintains decorum and order in the House. He/She decides whether a bill is a Money Bill.
- He/She permits the moving of various kinds of motions and resolutions like the motion of no confidence, motion of adjournment, motion of censure and calling attention notice as per the rules.
- He can be removed by a resolution passed by majority of all the members of Lok Sabha.

Name	*Tenure*	
GV Mavalankar	15.05.1952	27.02.1956
MA Ayyangar	08.03.1956	10.05.1957
MA Ayyangar	11.05.1957	16.04.1962
Hukam Singh	17.04.1962	16.03.1967
N Sanjeeva Reddy	17.03.1967	19.07.1969
Dr GS Dhillon	08.08.1969	19.03.1971
Dr GS Dhillon	22.03.1971	01.12.1975
Bali Ram Bhagat	05.01.1976	25.03.1977
N Sanjeeva Reddy	26.03.1977	13.07.1977
K S Hegde	21.07.1977	21.01.1980
Dr Balram Jakhar	22.01.1980	15.01.1985
Dr Balram Jakhar	16.01.1985	18.12.1989
Rabi Ray	19.12.1989	09.07.1991
Shiv Raj Patil	10.07.1991	22.05.1996
PA Sangma	23.05.1996	23.03.1998
GMC Balyogi	24.03.1998	19.10.1999
GMC Balyogi	22.10.1999	03.03.2002
Manohar Joshi	10.05.2002	20.06.2004
Somnath Chatterjee	04.06.2004	31.05.2009
Meira Kumar	04.06.2009	04.06.2014
Sumitra Mahajan	06.06.2014	16.06.2019
Om Birla	19.06.2019	Till date

**As on 31st December, 2020*

Speaker	*Lok Sabha*
Shri GV Mavalankar	First
Shri B Das	First
Shri Hukam Singh	First
Seth Govind Das	Second, Third, Fourth & Fifth
Shri DN Tiwari	Sixth
Shri Jagjivan Ram	Seventh & Eighth
Professor NG Ranga	Ninth
Shri Inderjit Gupta	Tenth, Eleventh, Twelfth & Thirteenth
Shri Somnath Chatterjee	Fourteenth
Shri Balasaheb Vikhe Patil	Fourteenth
Shri Manikrao Hodlya Gavit	Fifteenth
Shri Kamal Nath	Sixteen
Virendra Kumar	Seventeenth

Members of Parliament

Qualification

The Constitution lays down the *following qualifications for person to be chosen a Member of the Parliament*

- Must be a citizen of India.
- Must not be less than **30 years** of age in the case of **Rajya Sabha** and not less than **25 years of age** in the case of the **Lok Sabha**.
- Must possess other qualification as prescribed by Parliament.

Disqualification

Under the Constitution, a person shall be disqualified for being elected as a Member of Parliament

- If he holds any office of profit under the Union or State Government (except that of a minister or any other office exempted by the Parliament).
- If he is of unsound mind and declared so by a court.
- It he is an undischarged insolvent.
- It he is not a citizen of India.

Oath and Salaries

- Oath is administered by the President or some other person appointed by him for this purpose. Salaries and allowances are determined by Parliament.

Leader of Opposition

- The leader of the largest opposition party having at least one-tenth seats of total strength of the house is recognised as leader of opposition in that House.
- He was accorded Statutory recognition in 1977. He is given the status equivalent to a Cabinet Minister.

Joint Session

- Joint sessions take place on the order of the President if
  - (a) a bill passed by one House and rejected by another.
  - (b) the amendments made by the other House are not acceptable to the House, where the bill originated.
  - (c) other house do not take any action for six months on bill.
- Joint session (Article 108) is presided over by the Speaker of the Lok Sabha.
- The deadlock over a bill is resolved by a majority of the total numbers of the members of both the Houses present and voting.

Difference of Powers Between two Houses

- A Money Bill can be introduced only in the Lok Sabha on the recommendation of the President and not in the Rajya Sabha.
- The final power to decide whether a particular bill is a Money Bill or not, is vested to the Speaker of the Lok Sabha. The Speaker of Lok Sabha presides over the joint sitting of both the Houses.
- A resolution for the discontinuance of the National Emergency can be passed only by the Lok Sabha and not by the Rajya Sabha.
- The Rajya Sabha cannot remove the Council of Ministers by passing a no-confidence motion.

Sessions of Parliament

- The Parliament generally meets in three sessions in a year i.e., **Budget session** (February-May) (longest session), **Monsoon session** (July-September), **Winter session** (November-December) shortest session.
- There should not be a gap of more than six months between two sessions of Parliament.

Term	*Meaning*
Quorum	A Quorum is the minimum number of members of a deliberative assembly necessary to conduct the business of that group. Quorum for either House (Article 100 (c)) is 1/10 of the total number of members.
Penalty	If a person sits or votes as a member of either House of the Parliament before he has complied with the requirements of Article 99 (oath) or when he knows that he is not qualified or that he is disqualified for membership thereof, he shall be liable in respect of each day on which he so sits or votes to a penalty of ` 500 to be recovered as a debt to the union.
Question Hour	The first hour of every sitting in Lok Sabha (11 am to 12 pm). In the Rajya Sabha timing of question hour is 12-1 pm. In this, questions are asked by members and answered by ministers. Question hour is an important mechanism through which executive's accountability is brought about. *There are three types of questions* ▪ **Starred Questions** require oral answers. ▪ **Unstarred Questions** ten days notice is required and written answer is needed. Supplementary question cannot follow. ▪ **Short Notice Questions** are the ones, which relates to matters of urgent public importance and can be asked by members with notice shorter than ten days prescribed for an ordinary question.
Zero Hour	The time gap between question hour and fixed agenda of the day is known as **zero hour.** This time is allotted everyday for miscellaneous business call-attention notices, questions on official statements and adjournment motions.
Lame Duck Session	This refers to the last session of the old Parliament, which is held after a new Lok Sabha has been elected after the general election. The lame duck under it are those members who were part of the old House, but could not get re-elected for the new House.
Snap Vote	Refers to a vote taken unexpectedly without voters being briefed or informed about it in advance.
Speaker Pro Tem	As soon as the new Lok Sabha is constituted, the President appoints a Speaker Pro Tem, who is usally the senior-most member (seniority in terms of number of year he/she served as a member) of the House. In case the two members are equally qualified, the weightage is given to member's age.
Whip	A directive issued by any political party to ensure the support of its members voting in favour or against a particular issue on the floor of the house. A person may lose the membership of the party and the legislature if he votes against the whip or abstains from voting.
Guillotine	When due to lack of time, demand for grants are put to vote whether they are discussed or not in the House on the last day of the alloted time, it is called Guillotine and it concludes the discussion on demands for grants.
Public Bill/ Private Bill	A bill introduced in the Parliament by a Minister is a Public Bill whereas the bill introduced by one member other than a Minister is called Private Bill.

Consolidated Fund of India

Under **Article 266**, it is a fund to which all receipts are credited and all payments are debited. *In other words,*

(a) all revenues received by the Government of India.
(b) all loans raised by the government by the issue of treasury bills, loans or ways and means of advances.

(c) all money received by the government in repayment of loans from the Consolidated Fund of India.

- All the legally authorised payments on behalf of the government are made out of this fund. No money out of this fund can be appropriated (issued or drawn) except in accordance with a Parliamentary Law.

Public Accounts of India (Article 266)

- It is a fund to which all public money (except related to consolidated fund of India) received by Government of India is credited.
- This includes provident fund deposits, judicial deposits, remittance, savings bank deposits etc. It is operated by executive action.

Contingency Fund of India (Article 267)

- The Constitution authorised the Parliament to establish a 'Contingency Fund of India', into which amounts determined by law are paid from time to time. Accordingly, the Parliament enacted the Contingency Fund of India Act in 1950.
- This fund is placed at the disposal of the President and he can make advances out of it to meet unforeseen expenditure pending its authorisation by the Parliament.

Committee System

- Committees have been created so that Members of Parliament can discuss and debate on the working of a certain department of the government.
- Most of the committee's functions under the direction of the **Speaker** and are essentially committees of the Lok Sabha. **Committee's are classified under two heads** Standing Committee and Adhoc Committees. **Adhoc Committees** are created for a temporary period.
- The **Standing Committees** are broadly classified into **six categories** Committees to Enquires, Committees to Scrutinise, Financial Committees, Committees related to day-to-day business. Housekeeping or Service Committees and Departmental Standing Committees.
- The **Financial Committees** of **Parliament are** Estimates Committee, Public Accounts Committee, Committee on Public Undertaking.
- The **Public Accounts Committee** was set-up first in **1921**, under the provisions of the Government of India Act of 1919. At present, it consists of 22 members (15 from the Lok Sabha and 7 from the Rajya Sabha). Since, 1967, a convention has developed whereby the Chairman of the Committee is selected invariably from the opposition.
- The first Estimates Committee was set-up in 1950. On the recommendation of John Mathai. It has **thirty members,** all from the Lok Sabha only. It is the largest Parliamentary Committee.
- The **Committees on Public Undertakings** was created in 1964 on the recommendations of the Krishna Menon Committee. It has 22 members (15 from the Lok Sabha and 7 from the Rajya Sabha).
- In 1993, **17 Departmental Related Standing Committees** were set-up. In 2004, 7 more committees were set-up. Thus, total 24 committees were set-up.
- Members of the Rajya Sabha are associated with all the committees except the Estimates Committee.
- The Chairman of all the Committees (except the Joint Committee on salaries and allowances of MPs) are appointed by the Speaker amongst the Member of the Committee.
- The Speaker is the ex-officio Chairman of Business Advisory Committee, Rules Committee and General Purpose Committee.

SUPREME COURT

- Supreme Court stands at the apex of the 'Judicial System of India'. It is the ultimate interpreter of the Constitution and the laws of the land.
- **Article 124** states the establishment and Constitution of Supreme Court.
- Supreme Court was inaugurated on 28th January, 1950.

Name	Tenure	
	From	To
Hiralal J Kania	26.01.1950	06.11.1951
M Patanjali Shastri	07.11.1951	03.01.1954
Mehar Chand Mahajan	04.01.1954	22.12.1954
BK Mukherjee	23.12.1954	31.01.1956
SR Das	01.02.1956	30.09.1959
BP Sinha	01.10.1959	31.01.1964
PB Gajendragadkar	01.02.1964	15.03.1966
AK Sarkar	16.03.1966	29.06.1966
K Subba Rao	30.06.1966	11.04.1967
KN Wanchoo	12.04.1967	24.02.1968
M Hidayatullah	25.02.1968	16.12.1970
JC Shah	17.12.1970	21.01.1971
SM Sikri	22.01.1971	25.01.1973
AN Ray	26.04.1973	28.01.1977
MH Baig	29.01.1977	21.02.1978
YV Chandrachud	22.02.1978	11.07.1985
PN Bhagwati	12.07.1985	20.12.1986
RS Pathak	21.12.1986	18.06.1989
ES Venkataramth	19.06.1989	17.12.1989
S Mukharjee	18.12.1989	25.09.1990
Ranganath Mishra	25.09.1990	24.11.1991
KN Singh	25.11.1991	12.12.1991
MH Kania	13.12.1991	17.11.1992
LM Sharma	18.11.1992	11.02.1993
MN Venkatachaliah	12.02.1993	24.10.1994
AM Ahmadi	25.10.1994	24.03.1997
JS Verma	25.03.1997	17.01.1998
MM Punchhi	18.01.1998	09.10.1998
AS Anand	10.10.1998	31.10.2001
SP Bharucha	01.11.2001	05.05.2002
BN Kirpal	06.05.2002	07.11.2002
GB Pattanaik	08.11.2002	18.12.2002
VN Khare	19.12.2002	01.05.2004
S Rajendra Babu	02.05.2004	31.05.2004
RC Lahoti	01.06.2004	31.10.2005
YK Sabharwal	01.11.2005	13.01.2007
KG Balakrishnan	14.01.2007	12.05.2010
SH Kapadia	12.05.2010	28.09.2012
Altamas Kabir	29.09.2012	18.07.2013
P Sathasivam	19.07.2013	26.04.2014
Rajendra Mal Lodha	27.04.2014	27.09.2014
H.L. Dattu	28.09.2014	02.12.2015
T.S. Thakur	03.12.2015	31.01.2017
J.S. Khehar	4.01.2017	27.08.2017
Dipak Misra	28.08.2017	02.10.2018
Ranjan Gogoi	03.10.2018	17.11.2019
SA Bobde	18.11.2019	Till date

**As on 31st December, 2020*

- At present, the Supreme Court consists of 34 judges (1 CJI and 33 judges).

Appointment

- Only senior most Judge of the Supreme Court is appointed by the President as CJI. Other Judges of the Supreme Court are appointed by the President in consultation with the Chief Justice. He may also consult other Judges of the Supreme Court and High Court, while appointing a Judge of the Supreme Court.

Woman Judge of Supreme Court

The first woman Judge of the Supreme Court was Justice Fatima Beebi in 1987 and the second woman Justice was Sujata V Manohar in 1994. Indu Malhotra is the first women judge of Supreme Court to be directly appointed to the Supreme Court from the bar.

Qualification

Under **Article 124** (3), a person to be appointed as a Judge of the Supreme Court *should have the following qualifications*

- He should be a citizen of India.
- He should have been a Judge of a High Court (or High Courts in succession) for five years or He should have been an advocate of a High Court (or High Courts in succession) for ten years or He should be a distinguished jurist in the opinion of the President.
- The Constitution does not prescribed a minimum age for appointment as a Judge of the Supreme Court.

Oath or Affirmation

- Administered by the President or some person appointed by him for this purpose.

Tenure of Judges

- Holds office until he attains the **age of 65 years.**
- Can resign his office by writing to the President.
- Can be removed from his office by the President on the recommendation of the Parliament.

Removal of Judges

- Under **Article 124** A Judge of the Supreme Court shall not be removed from his office except by an order of the President passed after an address by each House of the Parliament by special majority.
- On the ground of proved misbehaviour or the incapacity, the judges can be removed from their office.
- Under **Article 124** (5) Parliament may regulate by law the procedure relating to the removal of a Judge of the Supreme Court. Parliament provides the procedure for removal by an act, known as The Judges Enquiry Act (1968).
- No judge of the Supreme Court has been impeached so far. The first such case of impeachment is that of justice V Ramaswami of the Supreme Court in Lok Sabha (1991-93). But, finally did not get impeached.
- The second case of impeachment is that of **Justice Saumitra Sen** of the Calcutta High Court (became the first judge in Indian History against whom an impeachment motion was passed in the Rajya Sabha).

Seat of Supreme Court

Under **Article 130,** the Constitution declares **Delhi as the seat of the Supreme Court**. But, it authorises the CJI to appoint any other place or places as seat of the Supreme Court. He can take decision only with the approval of the President.

Salaries and Allowances

- Under **Article 125**, The salaries, allowances, privileges, leave and pension of the Judges of the Supreme Court are determined from time to time by the Parliament by law.
- Salary of Chief Justice of India is ₹ 2.8 lakh per month.
- Salary of other judges of Supreme Court is ₹ 2.5 lakh per month.
- The retired Chief Justice and judges are entitled to 50% of their last drawn salary as monthly pension.

Independence of Supreme Court

- Judges are appointed by the President in consultation of the Judiciary itself.
- **Security of tenure** removed only by the President on the grounds mentioned in the Constitution.
- Fixed service conditions and salaries etc (charged on Consolidated Fund of India).
- Conduct of judges cannot be discussed in Parliament or State Legislature.
- Ban on practice after retirement

Jurisdictions and Powers

Original Jurisdiction

The Supreme Court *decides the dispute between*

- The centre and one or more states;
- The centre and any state or states on one side and one or more states on the other or between two or more states.
- In the above disputes the Supreme Court has exclusive original jurisdiction.

Writ Jurisdiction

- Every individual has the right to move the Supreme Court directly for the enforcement of his Fundamental Rights.

Appellate Jurisdiction

- The Supreme Court is primarily a court of appeal and hears appeals against the judgements of the lower courts.

Advisory Jurisdiction

- The Constitution (Article 143) authorises the President to seek the opinion of the Supreme Court. Its opinion is not binding on President.

Power of Judicial Review

- Judicial review is the power of the Supreme Court to examine the **constitutionality** of **legislative enactments** and executive order of both Central and State Government.
- On examination, if they found to be violative of the Constitution, they can be declared as illegal, unconstitutional and invalid.
- Under **Aticle 137,** the Supreme Court can review its own judgement to remove any error that might have been given in its order.

  The Supreme Court used the power of judical review in following cases.

  (a) Golaknath Case (1967)
  (b) Bank Nationalisation Case (1970)
  (c) Privy Purse Abolition Case (1971)
- Kesavananda Bharti Case (1973)
- Minerva Mills Case (1980)
- **Public Interest Litigation** (PIL) Any person can now initiate a proceeding on behalf of the aggrieved person (if the aggrieved persons cannot do so on their own) in either the High Court or the Supreme Court for the enforcement of Fundamental Rights.
- It was during **Chief Justice PN Bhagwati's tenure**, that PIL was started in India.

The National Judicial Appointments Commission Act, 2014

- The Bill has been introduced in conjunction with the Constitutional (121st Amendment) Bill, 2014. It amends provisions related to appointment and transfer of judges to the higher judiciary. It establishes a National Judicial Appointments Commission (NJAC) to make recommendations to the President on appointment and transfer of judges to the higher judiciary.
- On October 16, 2015 the Constitution Bench of Supreme Court by 4 : 1. Majority upheld the collegium system and struck down the NJAC as unconstitutional.

STATE GOVERNMENT

GOVERNOR

- The Governor is the Constitutional Head of the state and the same Governor can act as Governor of more than one state (Articles 153).
- The Governor is appointed by the President (Articles 155) Article 156 states that the Governor holds office during the pleasure of the President.

Qualification

Under **Article 157**, the Constitution lays down the *following qualifications for the Governor's office*

- Must be citizen of India.
- Completed 35 years of age.

Conditions of Office (Article 158)

- Shouldn't be a member of either House of Parliament or State Legislature.
- Shall not hold any other office of profit.
- Use his official residence (Raj Bhawan) without payment. When same person is appointed as the Governor of two or more states, the emoluments and allowances payable to him are shared by states as determined by the President.

Oath (Article 159)

- His oath is administered by the Chief Justice of the concerned State High Court and in his absence, the seniors most judge of that court.

Tenure (Article 156)

(a) hold office for a period of 5 years.
(b) the Governor shall hold office during the pleasure of the President;
(c) may resign by writing a letter addressed to the President;

Powers and Functions of Governor

Executive Powers

- The executive power of the state shall be vested with the Governor and shall be exercised by him either directly or through officers subordinate to him in accordance with the Constitution.
- All executive actions of the Government of a State are formally taken in his name. He can make rules for more convenient transaction of the business of a State Government.
- Appoints the **Advocate General** of a State, State Election Commissioners, Chairman and Members of the State Public Service Commission, VCs of Universities.
- He can seek any information relating to the administration of the affairs of the state and proposals for legislation from the Chief Minister.

Legislative Powers

- Governor is an integral part of the State Legislature. He has the right of addressing and sending messages and of summoning, proroguing and dissolving the State Assembly. He has the power to nominate one member of Anglo-Indian Community to the Legislative Assembly of the state. He appoints 1/6 members of Legislative Council. His other powers are mostly similar to the President.

Financial Powers

- He enables the State budget to be laid before the State Legislature, he constitutes a Finance Commission after every 5 years to review the financial position of the Panchayats and the municipalities.

Judicial Powers

- Under Article 161, he can grant pardons, reprives, respites and remissions of punishment or suspend, remit and commute the sentence of any person convicted of any offence against any law relating to a matter to which the executive power of the state extends.
- He cannot pardon a death sentence.

Emergency Powers

- The Governor has no emergency power to counter external aggression or armed rebellion. He reports to the President if the State Government is not running constitutionally and can recommend the imposition of President's rule (Article 356).
- When President's rule comes into force in any state, the Governor runs the state with the help of advisers on behalf of the President.

CHIEF MINISTER

- Chief Minister is the real Executive, head of the State Government. His position at the state level is analogous to the position of the Prime Minister at the Centre.

Appointment

- **Article 164**, says that Chief Minister shall be appointed by the Governor.
- The Governor has to appoint the leader of majority party or coalition in the State Legislative Assembly as Chief Minister.

Oath, Term and Salary

- Oath of the office of Chief Minister is administered by the Governor or person appointed by him for this purpose.
- A person, who is not a member of State Legislature can be appointed but he has to get himself elected within 6 months otherwise he is removed.
- The **term of the CM is not fixed** and he holds office during the pleasure of the Governor. However, this does not mean that the Governor can dismiss him at any time. He cannot be dismissed by the Governor as long as he enjoys the majority support in the Legislative Assembly.
- But, if he loses the confidence of the assembly, he must resign or the Governor can dismiss him.
- The **salary and allowances** of the Chief Minister are determined by the State Legislature.

STATE LEGISLATURE

- There is no uniformity in the creation of State Legislatures. Most of the states have unicameral system (single house), only 6 states are having Bicameral (Two Houses).

Legislative Assembly

- The **Legislative Assembly** (Vidhan Sabha) consists of not more than 500 members and not less than 60 members. However, the Legislative Assembly of **Sikkim, Goa, Mizoram**, and **Puducherry** have less than 60 members.

Legislative Council

- As per **Article 169**, if the Legislative Assembly passes a resolution for abolishing or creating of the **Legislative Council** by a majority of the total membership of the assembly and by a majority of not less than two-third of the members present and voting, the Parliament may approve the resolution by a simple majority.

Strength of Legislative Council

- The total strength cannot exceed one-third of the strength of the Legislative Assembly, subject to minimum of 40 members.

States having Legislative Council

- Uttar Pradesh
- Maharashtra
- Bihar
- Andhra Pradesh
- Karnataka
- Telangana

Chairman of Legislative Council

- The Chairman is elected by the council itself from amongst its members.

Qualification of Member

Under **Article 173**, *he must fulfil the following conditions*

(a) He must be a citizen of India.

(b) He must make and subscribe before the person authorised by the Election Commission an oath or affirmation according to the form prescribed in the Third Schedule.

(c) He must be **not less than 30 years** of age in the case of the Legislative Council and **not less than 25 years** of age in the case of the Legislative Assembly.

(d) He must possess other qualifications prescribed by Parliament , under Representation of People Act (1951).

Duration of the Two Houses

Legislative Assembly- Same as Lok Sabha
Leglislative Council- Same as Rajya Sabha

S.N.	*State*	*Number of Seats*
1.	Andhra Pradesh	175
2.	Arunachal Pradesh	60
3.	Assam	126
4.	Bihar	243
5.	Chhattisgarh	90
6.	Goa	40
7.	Gujarat	182
8.	Haryana	90
9.	Himachal Pradesh	68
10.	Jharkhand	81
11.	Karnataka	224
12.	Kerala	140
13.	Madhya Pradesh	230
14.	Maharashtra	288
15.	Manipur	60
16.	Meghalaya	60
17.	Mizoram	40
18.	Nagaland	60
19.	Odisha	147
20.	West Bengal	295
21.	Punjab	117
22.	Rajasthan	200
23.	Sikkim	32
24.	Tamil Nadu	235
25.	Telangana	119
26.	Tripura	60
27.	Uttarakhand	70
28.	Uttar Pradesh	403

1.	Delhi	70
2.	Puducherry	30
3.	Jammu & Kashmir	114 (24 in Pakistan Occupied Kashmir, POK)

State	*Number of Seats*
Uttar Pradesh	100
Andhra Pradesh	58
Maharashtra	78
Karnataka	75
Bihar	75
Telangana	40

HIGH COURT

- Every High Court (whether exclusive or common) consists of a Chief Justice and such other judges as the President may from time to time deem necessary to appoint.
- Thus, the Constitution **does not specify the strength of a High Court** and leaves it to the discretion of the President. Accordingly, the President determines the strength of a High Court from time to time depending upon its workload.
- The Constitution of India provides a High Court for each state, but the 7th Amendment Act of 1956, authorised the Parliament to establish a common High Court for two or more states or and a union territory.
- The territorial jurisdiction of a High Court is co-terminus with the territory of a state. Similarly, the territorial jurisdiction of a common High Court is co-terminus with the territories of the concerned states and union territory.
- At present, there are **25 High Courts** in the country. Out of them, three are common High Court, Delhi is the only union territory that has a Court of its own (since 1966). In 2013, separate High courts were created for the three North-Eastern states of Manipur, Meghalaya and Tripura. In 2019, Telangana HC has been declared as 25th HC.

Appointment

- Under **Article 217**, the judges of the High Court are appointed by the President.
- The Chief Justice of the High Court is appointed by the President after consultation with the Chief Justice of Supreme Court and Governor of the concerned state.
- For the appointment of other judges, the Chief Justice of concerned High Court is also consulted.

Qualification

- He should be a citizen of India.
- He should have held a judicial office in the territory of India for 10 years or he should have been an advocate of a High Court (or High Courts in session) for 10 years.

Tenure

- He holds office until he attains the **age of 62 years**.

Removal

- He can resign his office by writing to the President.
- He can be remove from his office by the President on the recommendation of the Parliament (same as Judge of Supreme Court).
- He vacates his office when he is appointed as a Judge of the Supreme Court or when he is transferred to another High Court.

Salaries and Allowances

- Determined by Parliament from time to time. Present salary of the Chief Justice of High Court is ₹ 2,50,000 per month; and that of a Judge of the High Court is ₹ 2,25,000.

Independence of High Court

The provision for judicial independence for High Court is very much similar to Supreme Court.

Jurisdiction and Powers of High Court

At present, a High Court enjoys the following jurisdiction and powers

- Original jurisdiction
- Writ jurisdiction (Article 226)
- Appellate jurisdiction

- Supervisory jurisdiction
- Control over subordinate courts.
- A court of record.
- Power of judicial review.
- The Supreme Court can issue writ jurisdiction only where a Fundamental Right has been infringed. High Court can issue these writs not only in such cases, but also where an ordinary legal right has been infringed.
- High Court does not have advisory power as similar to Supreme Court.

JAMMU AND KASHMIR

- **Article 370** of the Indian Constitution accorded special status to the State of Jammu and Kashmir.
- Jammu and Kashmir had its own Constitution, which was framed by a special Constituent Assembly and came into being on the 26th January, 1957.
- However, On 5th August 2019, the special status of J & K was revoked when the President promulgated the Constitution (Application to Jammu & Kashmir) Order 2019. It repealed Article 370 {except Article 370 (1)}, and made applicable, all the provisions of the Constitution to J&K.
- On 31st October 2019, the State of Jammu & Kashmir is reorganised into two separate Union Territories of Jammu & Kashmir (with Legislature) and Ladakh (without Legislature).

Official Language

- **Part XVII** of the Constitution deals with the official language in **Articles 343** to **351.**
- Hindi written in **Devanagari Script** is to be the official language of the Union.
- The official language Act, 1963 provides that English, in addition to Hidi, to be used for offical purposes of the Union.
- **In 1955**, the President appointed an Official Language Commission, under the **Chairmanship of BG Kher.**
- Originally, there were fourteen languages in Eight Schedule, but eight were added through amendments—1. Assamese 2.Bengali 3. Gujarati 4. Hindi 5. Kannada 6. Kashmiri 7. Konkani 8. Malayalam 9. Manipuri 10. Marathi 11. Nepali 12. Oriya 13. Punjabi 14. Sanskrit 15. Sindhi 16. Tamil 17. Telugu 18. Urdu 19. Santhali 20. Bado 21. Maithili 22. Dogri
- Sindhi was added by the **21st Amendment Act, 1967**; Konkani, Manipuri and Nepali were added by the **71st Amendment Act, 1992;** and Bodo, Dogri, Maithili and Santhali were added by the **92nd Amendment Act, 2003**.

PANCHAYATI RAJ

PANCHAYATS

- Panchayats constitute functional institutions of grassroots governance in villages. The **Balwant Rai Mehta Committee** (1957) recommended a three-tier Panchayati Raj structure.
- The three tier system of Panchayati Raj was first adopted by Rajasthan (Nagaur District on 2nd October, 1959).
- In 1977, the government appointed the **Ashok Mehta Committee** to examine the measures to strengthen PRIs. The **LM Singhvi Committee** (1986) recommended Constitutional Status for local bodies.
- The Panchayati Raj system was constitutionalised through 73rd Constitutional Amendment Act of 1992. It added Part-IX (Articles 243 to 243O) and schedule 11 (29 functional items of Panchayats) to the Constitution.
- The salient features of Part IX *of the Constitution are as follows*
  - The term of the Panchayats is 5 years unless dissolved earlier.
  - Seats shall be compulsorily reserved for Scheduled Castes and Scheduled Tribes.
  - Seats to be reserved for backward classes is left at the discretion of the State Government. One-third seat are reserved for women.

- There is provision of Finance Commission to review the financial position of Panchayats and recommend grant-in-aids.
- A State Election Commission headed by the State Election Commission shall conduct elections for the Panchayats. The Constitution has provide for **three tier system** of Panchayats at the village, intermediate and district levels.

MUNICIPALITIES

- The system of Urban Local Government was constitutionalised by 74th Constitutional Amendment Act of 1992. It added Part -IX A (Article 243 P to 243 ZG) and Schedule 12 (18 functional items of Municipalities).
- The Constitution provides for three types of Municipalities, **Nagar Panchayat** (for an area in transition from rural to urban). **Municipal Council** (for smaller urban area). **Municipal Corporation** is for larger urban area. All the members of a municipality shall be elected directly by the people of Municipal area.
- Seats shall be reserved for Scheduled Castes and Scheduled Tribes. One-third of the seats shall be reserved for women.
- Municipalities have the power to impose taxes, duties, tolls and fees in accordance with law. The Constitution provides for a State Finance Commission to review the financial position of the Municipalities and recommend measures to augment their funds.
- Under **Article 243 ZD, a District Planning Committee** shall be constituted to consolidate the plans prepared by the Panchayats and Municipalities in the district.
- Under **Article 243 ZE, a Metropolitan Planning Committee** shall be constituted to prepare a draft development plan for the metropolitan area as a whole.

MISCELLANEOUS

Comptroller and Auditor General of India

- The Constitution of India (Article 148) provides for an independent office of the Comptroller and Auditor General of India (CAG).
- **Article 148 to 151** of the Constitution deals with CAG's appointment, powers and audit reports.
- The CAG of India is appointed by the President. He holds office for **six years or till sixty five years** of age, whichever is earlier.
- His removal process is similar to that of a judge of the Supreme Court.
- **Shri V Narahari Rao**, was the first Comptroller and Auditor General of India (1948-1954).

Guardian of Public Purse

CAG controls the entire financial system of the country at both the levels- the centre and the state, through audit reports. His duty is to uphold the Constitution of India and laws of Parliament in the field of financial administration.

Attorney General of India

- The Constitution (Article 76) has provided for the office of the Attorney General for India.
- He is the highest law officer in the country. The Attorney General of India must be a person, who is qualified to be appointed a judge of the Supreme Court. He is appointed by the President.

Functions

- The **term of office** of the Attorney General is **not fixed** by the Constitution. Further, the Constitution does not contain the procedure and grounds for his removal. He holds office during the pleasure of the President.
- The Attorney General's duty is to give advice to the Government of India upon such legal matters, which are referred to him by the President.
- The Attorney General of India is **not a member of the Cabinet**.
- He has the Right to Speak in both the Houses of Parliament, but he has no Right to Vote.
- Mr MC Setalvad was the first Attorney General of India.

ELECTION COMMISSION

- An independent Election Commission has been established, under the Constitution (Article 324) in order to carry out elections for Parliament, State Legislatures and the Offices of President and Vice-President. The Election Commission was established in accordance with the Constitution on 25th January, 1950.
- The Election Commission prepares, maintains and periodically updates the electoral rolls, supervises the nominations of candidates, register political parties, monitors the election campaign. It also organises the polling booths, counting of votes, and declaration of results, to ensure the orderly and fair manner of elections.

Composition of Election Commission

At present Election Commission consists of Chief Election Commissioner and two Election Commissioners.

- The **Chief Election Commissioner** and other Election Commissioners are appointed by the President.
- They hold office for a term of 6 years or till the age of 65 years, whichever is earlier (Article 324).
- The Chief Election Commissioner can be removed on ground similar to that of a judge of the Supreme Court.
- The general election is held on the basis of **Universal Adult Suffrage**.

NOTA

- The None of the Above (NOTA) is an option on the EVMs, through this voters get the right to reject all the candidates contesting the elections. A 'NOTA' does not require the involvement of the presiding officer.
- NOTA was first used in the assembly elections of November, 2013.
- In August 2018, the Supreme Court abolished the use of NOTA in the elections of Rajya Sabha and State Legislative Councils.

POLITICAL PARTIES

- Political parties are voluntary associations or organised groups of individuals, who share the same political views and who try to gain political power through constitutional means and who desire to work for promoting the national interest.

Party	*Symbol*	*Year of Formation*
Bahujan Samaj Party (BSP)	Elephant	1984
Bhartiya Janata Party (BJP)	Lotus	1980
Communist Party of India (CPI)	Ears of Corn and Sickle	1925
Communist Party of India-Marxist (CPM)	Hammer, Sickle and Star	1964
Indian National Congress (INC)	Hand	1885
Nationalist Congress Party (NCP)	Clock	1999
All India Trinamool Congress	Two Flowers	1998
National People's Party (NPP)	Book	2019

- To be **recognised as a National Party,** a party needs to secure atleast 6% of the **valid votes** polled in any four or more states in a general election to the Lok Sabha or State Assembly. In addition to it, it has to win atleast four seats in the Lok Sabha from any state or states as well.
- For getting **recognition as a State Party**, a political party has to be poll atleast 6% of the valid votes in the state concerned during a general election, either to that of the Lok Sabha or the State assembly.
- In addition to this, the party should also win minimum two seats in the Assembly or the state concerned.

Deposits and Nominations

Every candidate has to deposit ₹ 25,000 for Lok Sabha and Rajya Sabha elections. SC and ST members have to pay half amount of this.

e-GOVERNANCE

- e-governance simply means electronic governance. Governments are providing various information of their website about their working. Citizen's interaction with government became easier. With the help of information technology, citizens are able to get land certificate, property certificate act very quickly, citizens are able to pay various bills online or in common service centres established by Central/Sate Governments. This is the impact of E-governance.
- The **National e-governance Plan** was approved by the cabinet in May, 2005.
- Varous e-governance projects o Central Government are MCA21, Pensions, Income Tax, Central Excise, Passport Seva Project, Aadhar, e-courts, E-Procurement and E-ofice.
- Besides, the Central Government, various State Governments are undertaking various projects for e-governance under the National e-governance plan.
- **E-district** It is a mission Mode Project under e-governance. Its objective under National e-governance Policy is computerisation of services. Under it, different Programmes are conducted in following states
  - – Gyandoot Project – Madhya Pradesh
  - – Compact 2020 – Andhra Pradesh
  - – Land Programme – Karnataka
  - – Friends – Kerala
  - – Disha – Haryana

Union Public Service Commission

- The Union Public Service Commission is the central recruiting agency in India. It is an independent constitutional body in the sense that it has been directly created by the Constitution.
- Article 315 to 323 in part XIV of the Constitution contain elaborate provisions regarding UPSC.

The Structure of UPSC

- The Chairman and other members of the UPSC are appointed by the President and they hold office for a term of **6 years** or until they attain the age of 65 years. They are independent of the Executive and Legislature in the same manner as the judges of the Supreme Court.
- Age of retirement for a member of a State or Joint Public Service Commission is 62 years.

Removal of Members

- UPSC members can resign by addressing their letter of resignation to the President. President can remove them by issuing orders. Supreme Court makes such recommendation on the basis of an inquiry.

Functions

- To conduct exams for appointment to services under the Union, Maintains continuity of administration, Advise the President (not obligatory on him) in matters relating to appointment, promotions and transfers from one service to another of civil servants.
- All disciplinary matters affecting persons in the Service of Union.
- Matters regarding award of pension and awards in respect to injuries sustained during service under the government.

Finance Commission of India (Article 280)

- The Finance Commission consists of a Chairman and four other members.
- The Chairman is selected from among the persons, who had experience in public affairs. *While the members are selected from among persons who*
  - (a) are or have been qualified to be appointed judges of the High Court.
  - (b) have special knowledge of finance and accounts of government.
  - (c) wide experience in financial matters and in administration.
  - (d) having special knowledge on economics.

- The President constitutes a Finance Commission **every five years,** to
  (a) recommend to the President distribution of net proceeds of the taxes which are divisible between union and states.
  (b) recommend the principles, which should govern the grants of the revenues of the states out of the Consolidate Fund of India.
- The Commission's recommendations are of advisory nature and hence, not binding on the government.

Planning Commission

- It was established in March 1950 by an executive resolution of the Government.
- It was responsible for formulation of five year plans and ensuring its effective implementation.
- It was abolished in August 2014 and replaced by NITI Aayog.

NITI Aayog

- The Government has replaced Planning Commission with a new institution named NITI Aayog (National Institution for Transforming India), (est. on 1st January, 2015) The institution serve as 'Think Tank' of the Governments directional and policy dynamo.
- NITI Aayog provide Governments at the Central and State levels with relevant strategic and technical advice across the spectrum of key elements of policy, this includes matters of national and international import on the economic front, dissemination of best practices from within the country as well as from other nations, the infusion of new policy ideas and specific issue-based support.

Functions

- NITI Aayog seeks to provide a critical directional and strategic input into the development process.
- To develop mechanisms to formulate credible plans at the village level.
- To pay special attention to the sections of our society that may be at risk of not benefitting adequately from economic progress.
- To actively monitor and evaluate the implementation of programmes and initiatives.
- To pay special attention to the sections of our society that may be at risk of not benefitting adequately from economic progress.
- To provide advice and encourage partnerships between key stakeholders and national and international like-minded Think Tanks, as well as educational and policy research institutions.
- To create a knowledge, innovation and entrepreneurial support system through a collaborative community of national and international experts, practitioners and other partners.
- To offer a platform for resolution of inter-sectoral and inter-departmental issues in order to accelerate the implementation of the development agenda.
- To actively monitor and evaluate the implementation of programmes and initiatives, including the identification of the needed resources so as to strengthen the probability of success and scope of delivery.

Structure of NITI Aayog

The NITI Aayog comprises the following

- Prime Minister of India as the Chairperson.
- Governing Council comprising the Chief Ministers of all the States and Lt. Governors of Union Territories.
- The Regional Councils are convened by the Prime Minister and comprises of the Chief Ministers of States and Lt. Governors of Union Territories in the region. These are chaired by the Chairperson of the NITI Aayog or his nominee.
- Experts, specialists and practitioners with relevant domain knowledge as special invitees nominated by the Prime Minister.

National Development Council (NDC)

- It was established in August 1952 by an executive order of the government to take decisions on matters related to five year plans. Prime Minister is the ex-officio Chairman of NDC.
- After the inception of NITI Aayog, no work has been assigned to NDC. The government has proposed to abolish it soon.

Central Information Commission (CIC)

- Right to Information became an act in 2005. The aim is to make the government more transparent in its working.
- Under the act, a Central Information Commission and State Information Commission need to be constituted.
- The Chief Information Commissioner and other Information Commissioners shall be **appointed by the President** on the **recommendation of a committee** consisting of the Prime Minister, leader of oppositon in Lok Sabha and a Union Minister.
- The Central Information Commission and State Information Commission hear complaints from any person, who has been denied information by any government authority.
- In its six years of working, the RTI Act has given citizens the power to make its government accountable.

National Human Rights Commission

- It is an statutory body. It was established in **1993** under a legislation enacted by the Parliament.
- The commission is a multi-member body consisting of a chairman and five members. The chairman should be **a retired Chief Justice of India** or a **retired Judge of Supreme Court.**
- Chairman and members are appointed by the President on the recommendations of a six members committee.

Functions of NHRC

- To inquire into any violation of human rights or negligence in the prevention of such violation by a public servant, either suo motu or on a petition presented it or on an order of a court.
- To intervene in any proceeding involving alegation of violation of human rights pending before a court.
- Function of the NHRC are mainly recommendatory in nature. It has no power to punish the violators of human rights.

Lokpal and Lokayuktas

The Lokpal and Lokayuktas Act 2013 was passed by the Parliament to provide for the establishment of a body of Lokpal for the union and Lokayukta for state to inquire into allegation of corruptions against certain public functionaries and related matters.

Salient Features of Lokpal and Lokayuktas Act

- Lokpal consists of a Chairperson and a maximum of eight members of which 50% shall be Judicial member.
- 50% of Members of Lokpal shall be from SC/ST/OBCs, minorities and women.
- The selection of Chairperson and Members of Lokpal shall be through a selection committee consist of
  - (a) Prime Minister
  - (b) Speaker of Lok Sabha
  - (c) Leader of opposition in Lok Sabha
  - (d) Chief Justice of India or a sitting Judge of Supreme Court nominated by CJI.
  - (e) Eminent Jurist on the basis of recommendations of the first four members of the selection committee.
- Prime Minister has been brought under the purview of the Lokpal.
- Justice P. N. Ghose was appointed as the 1st Lokpal of India.

CONSTITUTIONAL AMENDMENTS

Under **Article 368** of the Constitution, Parliament has the power of amending the Constitution.

There are three methods

- **Method of Simple Majority** The Constitution can be amended by simple majority in matters relating to citizenship, abolishing or creating second chambers in the states etc. However these amendments doesn't come under purview of Article 368.
- **By Special Majority of Parliament** Constitutional Amendments must be passed by each House by a majority of the total membership of that House and by a majority of not less than two-thirds of the members of that House present and voting.
- **By Special Majority of Parliament and Consent of States** In this method, a special majority of Parliament and also the consent of half of the State Legislature (by a simple majority), is needed.

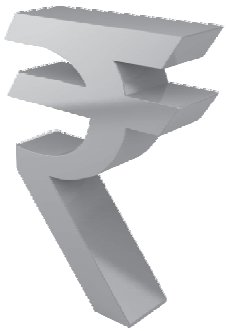
Important Amendments

- **The 1st Amendment 1951,** to overcome certain practical difficulties related to Fundamental Rights. It made provision for special treatment of educationally and socially backward classes and added Ninth Schedule to the Constitution.
- **The 7th Amendment 1956,** was necessitated on account of reorganisation of states on a linguistic basis and changed First and Fourth Schedules.
- **The 8th Amendment 1960**, extended special provision for reservation of seats for SCs, STs and Anglo-Indians in Lok Sabha and Legislative Assemblies for a period of 10 years from 1960 to 1970.
- **The 9th Amendment 1960,** transferred certain territories to Pakistan following September, 1958, Indo-Pak Agreement.
- **The 10th Amendment 1961,** incorporated the territories of Dadra and Nagar Haveli in Indian Union.
- **The 12th Amendment 1962,** incorporated the territories of Goa, Daman and Diu in Indian Union.
- **The 13th Amendment 1962,** created Nagaland as a State of the Union of India.
- **The 14th Amendment 1962,** incorporated former French Territory of Puducherry in Indian Union.
- **The 18th Amendment 1966,** was made to facilitate reorganisation of Punjab into Punjab and Haryana and also created the UT of Chandigarh.
- **The 21st Amendment 1967,** included Sindhi as the 15th regional language in the Eighth Schedule.
- **The 22nd Amendment 1969,** created a sub-state of Meghalaya within Assam.
- **The 23rd Amendment 1970,** extended the reservation of seats for SC/ST and nomination of Anglo-Indians for a further period of 10 years (upto 1980).
- **The 27th Amendment 1971,** provided for the establishment of the States of Manipur and Tripura, the formation of the Union Territories of Mizoram and Arunachal Pradesh.
- **The 31st Amendment 1973,** increased elective strength of Lok Sabha from 525 to 545. Upper limit of representatives of state became 525 from 500.
- **The 36th Amendment 1975,** made Sikkim a state of the Indian Union.
- **The 42nd Amendment 1976,** provided supremacy of Parliament and gave primacy to Directive Principles over Fundamental Rights and also added 10 Fundamental Duties. New **words-socialist, secular and integrity**, were added in **the Preamble**.
- **The 44th Amendment 1978,** restored the normal duration of Lok Sabha and Legislative Assemblies to 5 years, **Right to Property** was deleted from Part III; it limited the power of the government to proclaim internal emergency.
- **The 45th Amendment 1980,** extended reservation for SC/ST by 10 years (upto 1990).
- **The 52nd Amendment 1985,** inserted the Tenth Schedule in the Constitution regarding provisions as to disqualification on the grounds of defection.
- **The 55th Amendment 1986,** conferred statehood on Arunachal Pradesh.
- **The 56th Amendment 1987,** Hindi version of the Constitution of India, was accepted for all purposes and statehood was also conferred on the UT of Goa.

- **The 61st Amendment 1989,** reduced voting age from 21 to 18 years for Lok Sabha and assemblies.
- **The 73rd Amendment 1992,** (Panchayati Raj Bill) provided Gram Sabha in villages, direct elections to all seats in Panchayats and reservation of seats for the SC and ST and fixing of tenure of 5 years for Panchayats.
- **The 74th Amendment 1992,** (Nagar Palika Bill) provided for Constitution of three types of Municipalities, reservation of seats for SC and ST woman and the backward class.
- **The 86th Amendment 2002,** deals with the insertion of a new **Article 21A** after **Article 21**. The new **Article 21A** deals with Right to Education "the state shall provide free and compulsory education to all children from the age of 6 to 14 years, in such a manner as the state may, by law determine."
- **The 88th Amendment 2003,** provides for the insertion of a new **Article 268A**. Service tax levied by union and collected and appropriated by the union and the states. Amendment of **Article 270**, Amendment of Seventh Schedule.
- **The 89th Amendment 2003,** provides for the Amendment of **Article 338**. There shall be a National Commission for the SCs/STs.
- **The 91st Amendment 2003,** amended the anti-defection laws and provided for Amendment of **Article 75**. The total number of ministers, including the Prime Minister, in the Council of Ministers shall not exceed 15% of the total number of members of the House of the people.
- **The 92nd Amendment 2003,** provided for the Amendment of Eigth Schedule by adding four new regional languages (Bodo, Maithili, Santhali and Dogri), thus extending the list to 22 languages. **The 93rd Amendment 2005,** (came into effect on 20th January, 2006) provided for special provision by law, for the advancement of any socially and educationally backward classes of citizens or for the SCs/STs in so far as such special provisions relate to their admission to educational institutions including private educational institutions.
- **The 94th Amendment 2006,** to provide for a Minister of Tribal Welfare in newly created Jharkhand and Chhattisgarh.
- **The 95th Amendment 2010,** extended reservation for the SC/ST for further period of 10 years, that is upto 25th January, 2020.
- **The 96th Amendment 2011** substituted 'Odia' for 'Oriya'.
- **The 97th Amendment 2011,** provided for the Co-operative Societies in Part IX B of the Constitution of India. It also amended **Article 19 (1) (c)** and inserted **Article 43B**.
- **The 98th Amendment 2012,** provides for special provisions for the Hyderabad-Karnataka region of the State of Karnataka.
- **The 99th Amandment 2014** regulates the procudere to be followed by the NJAC for recommanding for appointment and transfer of CJI and Judges of Supreme Court and High Courts. But Supreme Court declared this unconstitutional and void.
- **The 100th Amendment 2015** deals with the acquiring of territories by india and transfer of certain territories to Bangladesh in pursuance of the agreements and its protocol between india and Bangladesh.
- **The 101th Amendment 2017** deals with Goods and Services Tax (GST) Act. GST is a comprehensive indirect tax levy on manufacture, sale and consumption of goods and services at national level.
- **The 102 Amendment 2018** provides the constitutional status to National Commission for Backward Classes.
- **The 103 Amendment Act 2018** provides the 10% reservation for economically weaker sections of society.
- **The 104th Amendment 2020** extended the reservation of seats for SCs and STs in the Lok Sabha and State Legislative Assemblies by 10 years. However, it discontinued the reservation of seats to Anglo-Indian community in Lok Sabha and State Legislatives.

GLOSSARY

- **Address of President** The President of India addresses to both Houses of Parliament assembled together at the commencement of the first session, after each general election to Lok Sabha and at the commencement of the first session of each year.
- **Adjournment of House** The speaker of the Lok Sabha determines, when the sitting of the House is to be adjourned sine die or to a particular day or to an hour or part of the same day. In the Rajya Sabha, Chairman decides, when Rajya Sabha needs to be adjourned.
- **Breach of Privilege** If a person disregards the privileges, rights and immunities of the Members of Parliament, then he commits breach of Privilege.
- **Closure** It is the parliamentary procedure, by which debate is closed and the measure under discussion brought up for an immediate vote.
- **Contempt of Court** It is disobedience to or disregard of the rules, orders, process or dignity of a court , which has power to punish such offence committed.
- **Caretaker Government** It is the government in the interregnum, which comes into existence as soon as the Council of Ministers resigns or loses confidence or the Prime Minister dies. It lasts till the next Council of Ministers is formed. It is a constitutional necessity under Article 74.
- **Delegated Legislation** The Parliament gives the Executive, the power to make rules and regulations covering an act of the Parliament.
- **Dissolution** Under Article 85, the President of India dissolves the House of the People as per the procedure fixed by the Constitution. The dissolution ends the very life of the existing House and fresh election is essential to form a new House.
- **Electoral Roll** It is commonly known as voter's list. It gives the names of all those people, who are eligible to vote.
- **Extradition** It is the surrender by a foreign state of a person accused of a crime to the state, where it was committed.
- **Floor Crossing** It refers to the defection of a Member of Parliament from the party, he was elected, to another political party.
- **Hung Parliament** When in a general election, no political party or coalition of the political parties is in a position to form a majority government, such a Parliament is called a Hung Parliament.
- **Locus Standi** It means on what grounds can a person file a case. Earlier a person, who did not have locus standi, could not file a case on behalf of aggrieved person on his own. Later, when the concept of Public Interest Litigation (PIL) started, locus standi was waived and any citizen could file a PIL and bring to court's notice, violation of rights of people.
- **Point of Order** It is an extra-ordinary process, which when raised, has the effect of suspending the business before the House. This is meant to assist the Presiding Officer in enforcing the Rules, Directions and Provisions of the Constitution for regulating the business of the House.
- **Rule of Law** The Rule of Law theory was given by English Jurist Dicey. It has three meanings (a) Absolute supremacy of law (b) Equality before law (c) Constitution is the result of the ordinary law of the land.
- **Subordinate Legislation** The rules and regulations made by the government within the purview of the authority delegated by the Legislature are called Subordinate Legislation.
- **Untouchability** It means social disabilities historically imposed on certain classes of people by reason of their birth in certain castes.
- **Vote on Account** It is an estimate of an advance payment to enable government departments to carry on their work from the beginning of financial year till the passing of the Appropriation Act.
- **Vote of Credit** The Lok Sabha can grant vote of credit to meet an expenditure, whose amount or details cannot be precisely stated on account of magnitude or the indefinite character of service.



INDIAN ECONOMY

ECONOMY

- Economy is the financial condition of the different sectors of the country.
- The study of economy of any country helps us in finding out the financial condition of the population as well as the different working sectors of the economy. It also helps in comparing the economic condition of two different countries.

Economic Growth

- It may be defined as a state of expansion that can move an underdeveloped country from a near subsistence mode of living to substantially higher levels over a period of time.

Economic Development

- Till the 1960s, economic development was often used as a synonym of economic growth. It is now taken as mean economic growth with change, which refers to the qualitative changes in the economy. These changes are in the form of improvement in the level of living, reduction in inequality, rise in efficiency and improvement in technique.

INDIAN ECONOMY

- Indian economy is the 5th largest in world (as per December, 2020) by nominal GDP and 3rd largest on Purchasing Power Parity (PPP) basis in the year, 2019.
- **Purchasing Power Parity** (PPP) is a theory which states that exchange rates between two currencies are in equilibrium when their purchasing power is the same in each of the two countries.
- India has a share of around 17.9% in world population, but accounts for only (2.6%) of world's GNI (Gross National Income) on exchange rate basis.
- India is one of the G-20 major economies and a member of BRICS.

Characteristics of Indian Economy

Main characteristics of Indian economy are

- **Agrarian Economy** In an agrarian or primary economy, agricultural dominance prevails in both the GNP and employment.
- **Mixed Economy** The nature of Indian economy is a mixed economy, where both public and private sector coexist. Mixed economy are often under government regulation 'JM Keynes' supported the mixed economy, which is guided mainly by the private sector but partly operated by the government.
- **Developing Economy** *Following features show that Indian economy is a developing economy*
  (a) Low per capita income
  (b) Heavy population pressure
  (c) Prevalence of chronic unemployment and underemployment
  (d) Steadily improving rate of capital formation
  (e) Low human development indicators.

Broad Sectors of Indian Economy

- **Primary Sector** Agriculture, forestry, fishing, mining and quarrying.
- **Secondary Sector** Manufacturing, electricity, gas and water supply, construction etc.
- **Tertiary Sector** Business (also called service sector) transport, telecommunication, banking, insurance, real estate, community and personnel services etc.

NATIONAL INCOME OF INDIA

- National Income is the net value of all the final goods and services produced in a country during a financial year.
- In other words, National Income is the measurement of the production power of an economic system in a given time period.
- National Income is a flow concept. In India the financial year is from 1st April to 31st March. It is calculated annually.
- National Wealth is the measurement of the monetary value of assets, possessed by the citizens of a nation at a particular point of time.

NATIONAL INCOME AGGREGATES

Gross National Product (GNP)

GNP refers to the money value of total output of production of final goods and services produced by the nationals of a country during a given period of time, generally a year. *Symbolically*,

$$GNP = C + G + I + (X - M) + (R - P)$$

Where, C = Consumption expenditure

G = Government expenditure

I = Investment expenditure

(X – M) = Net exports

(R – P) = Net Factor Income from Abroad

Gross Domestic Product (GDP)

It is the total money value of all final goods and services produced within the geographical boundaries of the country during a given period of time.

Symbolically, $GDP = GNP - (R - P)$

When, $R - P = 0$ then GDP = GNP

Gross Value Added (GVA)

- It is a measure of the value of goods and services produced in an area, industry or sector of an economy. In national accounts, GVA is the money value of output minus cost of intermediate consumption.
- RBI dumps GVA model in August, 2018 and switches back to GDP to measure economy.

Net National Product (NNP)

- NNP refers to the money value of total final goods and services produced by the nationals of a country minus depreciation (wear and tear) during a given period of time.

  NNP = GNP – Depreciation
- When NNP is calculated at Factor Cost (FC) it is called National Income. This measure is calculated by deducting Indirect Taxes and adding subsidies in NNP at Market Price (MP).

$$NNP_{FC} = NNP_{MP} - \text{Indirect Taxes} + \text{Subsidies}$$

or $NNP_{FC} = GNP_{MP}$ – Depreciation – Indirect Taxes + Subsidies

Net Domestic Product (NDP)

It is calculated by subtracting the value of depreciation from Gross Domestic Product (GDP).

NDP = GDP – Depreciation

Personal Income (PI)

Personal Income = National Income – Undistributed Profits of Corporations – Payment for Social Security Provisions – Corporate Taxes + Transfer Payments + Net Interest paid by the government.

CSO and NSSO

- In 1951, Central Statistical Office (CSO) was constituted to publish National Income data.
- NSSO (National Sample Survey Organisation) was set up in 1950, for conducting large scale sample survey on various socio-economic issues.
- In May, 2019, Ministry of Statistics and Programme Implementation has merged the both CSO and NSSO in a single body named as National Statistical Office (NSO).

Personal Disposable Income (PDI)

- When personal direct taxes are subtracted from personal income, the obtained value is called personal disposable income. *Symbolically*,

  PDI = Personal Income – Direct Taxes

  PDI = Consumption + Saving

Estimates of National Income in India

- In 1868, the first attempt was made by **Dadabhai Naoroji** in his book *Poverty and Un-British Rule in India*. He estimated the per capita annual income to be ₹ 20.
- The first scientific attempt to measure National Income in India was made by **Prof VKRV Rao** in 1931-32.
- He divided the Indian economy into 13 sectors.
- In 1949, National Income Committee under the Chairmanship of **Prof PC Mahalanobis** was constituted.

National Statistical Commission (NSC) It was set-up on 1st June, 2005 for promoting statistical network in the country. It was then headed by **Prof SD Tendulkar**.

Measurement of Growth and Development

Human Development Report

The Human Development Report (HDR), published by the UNDP since 1990, captures the essence of human development. Human development report rank countries based on their ranking on Human Development Index (HDI). HDI was devised by Pakistani Economist Mahbub ul Haq along with Indian Economist Amartya Sen.

In 2020 HDR, India has been placed at 131st position with 0.645 score in medium human development category. Norway, Ireland and Switzerland occupies first three positions.

Meaning of HDI Value

Very high Human Development	=	0.800 and above
High Human Development	=	0.700 and above
Medium Human Development	=	0.550 and above
Low Human Development	=	0.350 and above

Happy Planet Index (HPI)

HPI measures the ecological efficiency with which human well-being is delivered. It is calculated by multiplying indices of life satisfaction (estimated by compiling responses to international surveys) and life expectancy and dividing that product by ecological footprint.

Gross National Happiness (GNH)

The term Gross National Happiness was coined in 1972, by Bhutan's then King Jigme Singye Wangchuck. GNH was designed in on attempt to define an indicator that measures quality of life or social progress in more holistic and psychological terms than the economic indicator of GDP.

Various Index of Human Development Report 2020

Index	*Variables*	*Country's Rank*
Human Development Index (HDI) 2020	Life expectancy at birth index. Education index comprises mean year of schooling and expected year of schooling. GNI Per Capita (PPPUS $) Index for decent standard of living.	▪ Norway ▪ Ireland ▪ Switzerland
Inequality Adjusted HDI (IHDI)	Introduced in HDR-2010.Measures the average level of human development after adjusting for inequality.	▪ Norway ▪ Iceland ▪ Switzerland
Gender Inequity Index (GII)	Introduced in HDR-2010 Gil exposes differences in the distribution. It measures gender inequality based on three dimensions and five indicators.	▪ Switzerland ▪ Denmark ▪ Sweden
Multidimensi-onal Poverty Index (MPI)	MPI launched by Oxford Poverty and Human Development Initiative (OPHI) and UNDP in July 2010.	_

PLANNING OF INDIA

NITI Aayog

NITI Aayog National Institution for Transforming India Aayog came into existence on 1st January, 2015; policy making **think-tank** of government that replaces Planning Commission and aims to involve states in economic policy making. It will be providing strategic and technical advice to the Central and the State Government.

15 Years Vision Document by NITI Aayog

The 1st 15 year vision document will come into effect from 2017-18 after the end of the 12th five year plan. It will be formulated with centre objective of eradication of poverty. These will be framed keeping in mind the country's social goals and the sustainable development agenda. According to NITI Aayog, the issue was discussed at length and a decision was taken at the highest level 15 year Vision Documents divided into two parts.

- **7-years National Development Agenda** The first 15-year vision document will start from 2017-18, along with a 7-year National Development Agenda which will lay down the schemes, programmes and strategies to achieve the long-term vision.
- **3-years National Development Agenda** The long vision documents will comprise of three year mass economic framework. National Development Agenda will be reviewed after a gap of every three years to ensure that it was aligned with financial needs and requirements.
- For the first Development Agenda, the review would be done in 2019-20, in line with the termination year of the 14th Finance Commission.
- 2017-18 to 2032-33 Vision Document.
- 2017-18 to 2024-25 National Development Agenda.
- 2017-18 to 2019-20 Review of Development Agenda (to be repeated after every three years).

National Development Council (NDC)

- The NDC was Constituted on 6th August, 1952, with Prime Minister as the Ex-officio Chairman.
- NDC aims to make cooperative environment for economic planning between states and the Planning Commission. It gives final approval to the Five Year Plan and evaluates the management of plans from time to time.
- With establishment of NITI Aayog, the NDC has no work assigned to it and been proposed to be abolished.

Planning Commission

- National Planning Committee was set up under the Chairmanship of Jawaharlal Nehru in 1938. But Planning Commission was set-up in the year 1950.
- It is the central body for making plans in India and Prime Minister is the Ex-officio Chairman. It is a non-statutory, extra-constitutional and advisory body and finds no mention in the Constitution of India. It was replaced by NITI AAYOG on 1st January, 2015.

History of Planning

1944	Bombay Plan (A Plan of Economic Development) by eight industrialists
1944	Gandhian Plan (S N Agrawal)
1945	People's Plan (MN Roy)
1950	Sarvodaya Plan (Jai Prakash Narayan)

Five Year Plans *At a Glance*

Plan	*Objective*	*Assessment*
First Plan (1951-56)	▪ Highest priority accorded to agriculture in view of large-import of foodgrain and inflation. ▪ Increasing the rate of investment from 5% to 7%. ▪ 31% of total plan outlay on agriculture followed by transport and communication, social services, power and industry.	▪ Agricultural production increased dramatically. ▪ National Income went up by 18% and per capita income by 11%. ▪ Targeted growth rate was 2.1% and 1st Plan achieved 3.6%. ▪ Price level was stable.

Five Year Plans *At a Glance*

Plan	*Objective*	*Assessment*
Second Plan (1956-61)	▪ Rapid industrialisation with particular emphasis on the development of basic and heavy industry, also called Mahalanobis Plan. ▪ To promote a socialistic pattern of society as envisaged at Avadi Summit of Indian National Congress. ▪ To increase National Income by 25% expansion of employment and reduction of inequality. ▪ To increase the rate of investment from 7% to 11% of GDP.	▪ Moderately successful, targeted growth rate was 4.5%, but achieved 4.2%. ▪ Durgapur (UK), Bhillai (USSR) and Rourkela (W Germany) Steel plants set-up. ▪ Acute shortage of foreign exchange reserves lead to primary development targets. ▪ Inflation and low agricultural production and Suez crisis.
Third Plan (1961-66) (Gadgil Yojana)	▪ Indian economy entered **take off** stage (WW Rostow). ▪ Self-reliant and self-generating economy was the goal. ▪ Priority to agriculture and development of basic industries. ▪ To increase the National Income by 30% and per capita income by 17%. ▪ The situation created by Indo-Pakistan Conflict (1965), two successive years of severe drought evaluation of currency by **57%** general rise in prices and erosion of resources for plan delayed.	▪ A failure because of worst famine (1965-66) in 100 years. ▪ Indo-China (1962) and Indo-Pakistan (1965) conflict diverted the resources from development to defence. ▪ Postponement of Fourth Plan by 3 years. ▪ Targeted growth-**5.6** % achieved growth-2.8%.
Fourth Plan delayed because between 1966 to 1969 three Annual Plans were formulated.Relied heavily on foreign aid (International Monetary Fund).		
Three Annual Plans (1966-69)	▪ Due to miserable failure of Third Plan, the government was forced to declare 'plan holidays'. Three Annual Plans were drawn during this intervening period.	The prevailing crisis in agricultural and serious food shortage necessitated the emphasis on agriculture during the Annual Plans.
Fourth Plan (1969-74)	▪ Objective was 'growth with stability' and 'progressive achievement of self-reliance'. ▪ Provision of national minimum for weaker sections. ▪ Nationalisation of 14 major Indian banks.	▪ Adoption of import-substitution policy and export-promotion policy widened the industrial base. ▪ Targeted growth 5.7% however achieved growth 3.3%. ▪ The plan was failure on account of run way inflation (due to 1972 oil crisis or supply shock); huge influx of refugees from Bangladesh post 1972 Indo-Pak War.
Fifth Plan (1974-79)	▪ Original approach to plan prepared by C Subramanium who proposed economic growth along with direct attack on poverty. ▪ However, final draft prepared by DP Dhar with objectives of **'removal of poverty'** (*Garibi Hatao*) and attainment of self-reliance. ▪ To step up domestic rate of saving. ▪ Introduction of minimum needs programme.	▪ Targeted growth 4.4% achieved 4.8%. ▪ Fifth Plan cost calculations based on 1971-72 prices proved to be wrong. ▪ Fifth Plan terminated one year before the plan period, in March, 1978. ▪ Brought to the core problem associated with coalition government making a mockery of formulation of Five Year Plan.

Five Year Plans *At a Glance*

Plan	*Objective*	*Assessment*
▪ **Rolling Plan** (Gunar Myrdal) was brought out by Janata Party Government under Morarji Desai in 1978. The focus of the plan was enlargement of the employment potential in agriculture and allied activities to raise the income of the lowest income classes through minimum needs programme.		
Sixth Plan (1980-85)	▪ Removal of poverty through strengthing of infrastructure for both agriculture and industry. ▪ The emphasis was laid on greater management, effeciency and monitoring of various schemes. ▪ Involvement of people in formulating schemes of development at local level.	▪ Indian economy made an all round progress and most of the targets fixed by the percentage was achieved. ▪ Targeted growth 5.2% ▪ Achieved growth 5.4%
Seventh Plan (1985-90)	▪ To accelerate foodgrains production. ▪ To increase employment opportunities. ▪ To raise productivity level of industries by upgrading of technology.	▪ Foodgrain production grew by 3.23% as compared to a long-term growth rate of 2.68% between 1967-68 and 1988-89. ▪ The Indian economy finally crossed the barrier of **'the Hindu rate of growth'** (Prof Raj Krishna). ▪ Average annual growth rate was 6.0% as against the targeted 5.0% and average of 3.5% in the previous plans.
▪ **Annual Plan** The Eighth Plan could not take off due to fast changing political situations at the centre. Therefore, from 1990-1992, Annual Plans were formulated.		
Eighth Plan (1992-97)	▪ Process of fiscal reforms and economic reforms initiated by Narsimha Rao Government to prevent another major economic crisis. ▪ To increase the average industrial growth rate to 7.5%. ▪ To provide a new dynamism to the economy and improve the quality of life of the common man. ▪ Modernisation of industries.	▪ Higher economic growth rate of 6.6% achieved as against the targeted 5.6%. ▪ Improvement in trade and current account deficit. ▪ Significant reduction in fiscal deficit and agricultural growth and industrial growth increased. ▪ Unshackled private sector and foreign investment control was the prime reason for high growth. ▪ Overall socio-economic development indicators low. ▪ The growth became jobless and fruitless.
Ninth Plan (1997-2002)	"Growth with Social Justice and Equality" ▪ Emphasis on seven Basic Minimum Services (BMSs), which included safe drinking water, universalisation of primary education, streamlining PDS among others. Pursued the policy of fiscal consolidation. ▪ Decentralisation of planning with greater reliance on States and PRI (Panchayat Raj Institute). ▪ Ensuring food and nutritional security to all. Empowerment of women, SC/STs/OBCs.	▪ Global economic slowdown and other factors led to revision of targeted growth rate from 7% to 6.5% that too was not achieved. The economy grew at 5.4% only. ▪ Agriculture grew by 2.4% as against the target of 4.2% per annum. ▪ Industrial growth was 4.5% as against the target of 3% per annum.

Five Year Plans *At a Glance*

Plan	*Objective*	*Assessment*
Tenth Plan (2002-2007)	▪ The Tenth Plan aimed at achieving 8% GDP growth. Assuming that ICOR (Incremental Capital Output Ratio) will decline from 4.53% to 3.58%. ▪ It aimed at increasing domestic saving rate from 23.52% to 29.4% of GDP and gross capital formation to 32.2% from 24.4% of GDP. ▪ To improve the overall framework of governance. ▪ Reduction of poverty ratio.	▪ Increase in GDP growth to 7.5% compared to 5.4% in the Ninth Plan. The lower than targeted growth rate of 8% was due to low growth of 3% in the first year of Tenth Plan. ▪ Increase in gross domestic saving and investment. ▪ Reduction in ICOR to 4.2% though higher than targeted, but less than Ninth Plan's ICOR of 4.53%. ▪ Increase in foreign exchange reserves to US $ 287 billion. ▪ However, Tenth Plan fared worst on socio-economic indications and the agricultural growth rate was meagre 2.1%.
Eleventh Plan (2007-2012)	▪ Average GDP growth of 8.1% per year. ▪ Agricultural GDP growth of 4% per year generation of 58 million employment opportunities. ▪ Sex ratio for age group 0-6 years to be raised to 935 by 2011-12 and to 950 by 2016-17. ▪ 11th plan document was filled with motto of faster and more inclusive growth.	▪ The growth rate during the 11th plan period was about 7.9%, which is higher than the 7.5% growth rate achieved in the 10th plan. ▪ As against the target of 4% growth in the agricultural sector the plan could register a growth of only 3% during 2007-12 period. ▪ The services sector continued to register a growth rate of more than 10% However, the industrial growth rate showed at 7.9%
Twelfth Plan (2012-17)	▪ Main theme was ''Faster, More inclusive and Sustainable Growth.'' ▪ Real GDP growth rate of 8.2%. ▪ Agriculture growth rate of 4.0%. ▪ Manufacturing growth rate of 10%.	▪ The plan period was extended by 6 months. (i.e. September, 2017) ▪ First 4 years of the plan is estimated 6.8%.

INDIA'S NEW ECONOMIC POLICY (1991)

- New economic policy is related to economic reforms, which was revised and implemented, for the first time in the year 1985 during the period of Prime Minister Rajiv Gandhi.
- The second wave of new economic reforms came in the year 1991, during the period of PV Narsimha Rao Government widely known as LPG (Liberalisation, Privatisation and Globalisation) reforms.
- Macro-economic crisis of the early 1990's, necessitated economic reforms in India. *The crisis had three aspects*
  - (i) Fiscal imbalance or increasing fiscal deficit
  - (ii) Fragile balance of payment situation
  - (iii) Inflationary pressures in the economy
- The Balance of Payment (BoP) situation was such that India did not have enough foreign exchange reserves to pay even for 3 months imports.
- Two distinct strands of reform measures was prescribed by the **World Bank** and the IMF. *These were*

Macro-economic Stabilisation-Demand Management

It comprised of measures, *viz.,*

- control of inflation,
- fiscal adjustment and
- Balance of Payment adjustment.

Structural Adjustments-Supply-side Management

It comprised of measures namely,

- trade and capital flow reforms,
- industrial deregulation,
- disinvestment and public enterprises reforms and
- financial sector reforms.

The following four main steps were taken under the Fiscal Policy, 1991:

- To control public expenditure strictly
- To expand Tax Net
- To observe discipline in management of funds of Central and State Governments.
- To curtail grants (subsidy)

Measures implemented under the Industrial Reforms Policy, 1991 were:

- Number of reserved industries decreased to 8. Presently, these are only 2 that are Atomic energy, Railway operations (as per August, 2019).
- The work of rehabilitation of sick industries handed over to Board for Industrial and Financial Reconstruction (BIFR). Industries were made powerful with the help of Memorandum of Understanding (MoU).
- Voluntary Retirement Schemes started to cut down the size of work force.

MACRO-ECONOMIC PROBLEMS IN INDIA

POVERTY

- Poverty is a social phenomenon wherein section of society is unable to fulfil even its basic necessities of life.
- In India traditionally, the poverty line used to be defined on the basis of calorie intake. According to this, 2100 calories a day was fixed for urban areas and 2400 calories in rural areas.
- It stipulated a benchmark daily per capita expenditure of ₹ 27 and ₹ 33 in rural and urban areas respectively to arrive at a poverty line based on the methodology of Suresh Tendulkar.
- At present, poverty estimation in India is now carried out by NITI Aayog's task force through the calculation of poverty line based on the data captured by the National Sample Survey Office under the Ministry of Statistics and Programme Implementation (MoSPI).

UNEMPLOYMENT

- An economic condition marked by the fact that individual actively seeking jobs remain unemployed.
- Unemployment is expressed as a percentage of the total available workforce.

Types of Unemployment

Types of unemployment are as follows

Structural Unemployment

This type of unemployment is associated with economic structure of the country, i.e. productive capacity is inadequate to create a sufficient number of jobs.

Under Employment

Those labourers are underemployed who obtain work, but their efficiency and capability are not utilised at their optimum and as a result they contribute in the production up to a limited level.

Open Unemployment

When the labourers don't find any work to do, they come under the category of open unemployment. The migration from rural to urban area in search of work is very often found in India, which is an example of open unemployment.

Disguised Unemployment

If a person does not contribute anything in the production process or in other words, if he can be removed from the work without affecting the productivity adversely, he will be treated as disguised unemployed.

Frictional Unemployment

Frictional unemployment is the time period between jobs when a worker is searching for or transitioning from one job to anothor.

Seasonal Unemployment

It appears due to a change in demand based on seasonal variations. Labourers do not get work round the year.

Cyclical Unemployment

The demand for labour increases with the economy in the boom phase. Again, when the economy passes though recession, demand for labour contracts and the surplus is released as the unemployed labour force.

Measurement of Unemployment

- **Usual Status Approach** A person is unemployed if he had no gainful work for a relatively larger period (more than 183 days) during the 365 days preceding the date of survey.
- **Weekly Status Approach** A person who did not have gainful work even for a day or some days of the week preceding the date of survey would be deemed as unemployed.
- **Daily Status Approach** A person having no gainful work even for 1 hour in a day will be termed as 'unemployed' for that day.

FLAGSHIP PROGRAMMES OF GOVERNMENT OF INDIA

The flagship programmes were launched by the Government of India, to bridge education, health, employment and infrastructure divides. The ultimate objective of the flagship programme is to achieve broad-based improvement in the living standards of our people.

New Social Welfare Schemes

- **Public Wi-Fi Access Network Interface (PM-WANI)** It was launched on 9th December 2020. It aims to revolutionise the tech sphere and will substantially improve the wireless connectivity of the country. It will boost the 'Ease of Doing Business' and 'Ease of Living.'
- **Kisan Suryodaya Yojana** It was launched on 24th October 2020 to provide day-time power supply for irrigation to the farmers of Gujarat.
- **Ghar Tak Fibre Scheme** It was launched on 21st September 2020. The scheme aims at connecting all 45,945 villages of Bihar with high-speed optical fibre internet by March 31, 2021.
- **Pradhan Mantri Matsya Sampada Yojana (PMMSY)** It was launched on 10th September 2020 to focus on the sustainable development of the fisheries sector in India. It aims to enhance fish production in India by an additional 70 lakh tons and increase fisheries export earnings to ₹ 1,00,000 crore by FY 2024-25.
- **PM SVANidhi Scheme** It was launched on 1st June 2020 by the Central Government to help the street vendors resume their livelihood activities who were impacted by COVID-19. The tenure of the scheme is up to March 2022. It aims at facilitating working capital loan up to ₹ 10,000 at a subsidized rate of interest, incentivizing regular repayment of the loan and to reward digital transactions.
- **Pradhan Mantri Garib Kalyan Rojgar Abhiyan** This scheme was launched on 20th June, 2020 to boost employment and livelihood opportunities for migrant workers returning to villages, in the wake of COVID-19 outbreak.
- **One Nation, One Ration Card** This scheme is scheduled to implemented by 1st June, 2020. The scheme allows the beneficiaries can avail of the benefits across India, that is, poor migrant workers can buy subsidised rice and wheat from any ration shop in the country.
- **Atal Bhujal Yojana** Atal Bhujal Yojana (or Atal Jal) is groundwater management scheme launched by PM Modi on 25th December, 2019. It improve groundwater management scheme in seven states i.e. Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh under the Jal Jeevan Mission.
- **Laghu Vyapari Mann Dhan Scheme** This scheme was launched on 19th August, 2019. This pension scheme for small traders functions under the Ministry of Labour and Employment. Under the scheme, traders aged between 18 to 40 who have an annual turnover of less then 1.5 crore are eligible. Traders should contribute a monthly amount than they turning 60, the subscribers will get ₹ 3000 monthly pension.
- **Pradhan Mantri Kisan Maan Dhan Yojana** Launched in 2019, it is a pension scheme for small and marginal farmers. Under this, a monthly pension of ₹ 3000 will be provided to the farmers who attains the age of 60 years. The farmers will have to make a monthly contribution of ₹ 55 to ₹ 200, depending on their age of entry.
- **Pradhan Mantri Shram Yogi Maan Dhan Yojana** Similar to PM Kisan Maan Dhan Yojana, it is also a pension scheme launched in 2019 for unorganised sector workers with monthly income upto ₹ 15000 per month
- They will get assured pension of ₹ 3000 per month after attaining the age of 60 years.

- **Pradhan Mantri Krishi Sinchayee Yojana** (PMKSY) (set-up in July, 2015) The scheme is aimed to give assured irrigation to farmers.
- **Swachh Bharat Abhiyan** (SBA) (Set-up in 2nd October, 2015) Total sanitation by 2019, was the slogan of this programme. It is successfully ended in 2019. The year 2019 also marks the 150th Birth anniversary of Mahatma Gandhi.
- **Deendayal Upadhyaya Gram Jyoti Yojana** (DDUGJY) (set up in July, 2015) Its long-term aim was to provide 24×7 uninterrupted power supply to all homes. It is successfully ended in 2018 with the electrification of Leisang Village in Manipur.
- **Pradhan Mantri Jan Dhan Yojana** (PMJDY) (Set-up in August, 2014) Pradhan Mantri Jan Dhan Yojana is a scheme for comprehensive financial inclusion launched by the Prime Minister of India, Narendra Modi in August, 2014.
- Account holders will be provided zero-balance bank account with Rupay debit card, in addition to accidental insurance cover of ₹ 2 lakh.
- **Beti Bachao, Beti Padhao Yojana** It was introduced for generating awareness and improving the efficiency of delivery of welfare services meant for women with an initial corpus of ₹ 100 crore. The government would focus on campaigns to sensitise people of this country to wards the concerns of the girl child and women. The process of sensitisation must begin early and therefore the school curriculum must have a separate chapter on gender main streaming.
- **Deendayal Upadhyay Antyodaya, Yojana** This Yojana replaced the National Rural Livelihood Mission (Ajeevika) and National Urban Livelihood Mission. It is an overeaching scheme for uplift of urban and rural poor through enhancement of livelihood opportunities through skill development and other means.
- **Pradhan Mantri Kaushal Vikas Yojana** It is a demand-driven, reward-based skill training scheme. PMKVY is formed to provide skill training to class 10 and 12 dropout youths across the country.
- Under the scheme, besides assessing and certifying 10 lakh youth for the skills they already possess, around 24 lakh youth will be skilled over the next year.
- **Atal Pension Yojana** The Atal Pension Yojana (APY) will focus on all citizen in the unorganised sector who join the National Pension System (NPS) administered by the Pension Fund Regulatory and Development Authority (PERDA) and who are not members of any statutory social security scheme. It is available to people between 18 and 40 year of age with bank accounts. The subscribers are required to opt for a monthly pension from ₹ 1000 to ₹ 5000.
- **Pradhan Mantri Jeevan Jyoti Bima Yojana** The PMJJBY is available to people in the age group of 18 to 50 and having a bank account people who join the scheme before completing 50 years can, however, continue to have the risk of life over upto the age of 55 years subject to payment of premium Aadhar would be the primary Know Your Customers (KYC) for his bank account. Life insurance of ₹ 2 lakh with a premium of ₹ 330 per year.
- **Pradhan Mantri Suraksha Bima Yojana** The scheme will be a one-year cover, renewable from year to year. It is available to people between 18 and 70 year of age with bank accounts. It has an annual premium of ₹ 12 for ₹ 2 lakh accidental and ₹ 1 lakh full disability.
- **Startup Standup India** Startup India is a revolutionary scheme that has been started on August, 2015 to help the people who wish to start their own business. Standup India Scheme facilitates bank loan between ₹ 10 lakh and ₹ 1 crore to at least one Scheduled Caste (SC) or Scheduled Tribe (ST) borrower and at least one woman borrower per bank branch for setting up a greenfield enterprise.
- **Pradhan Mantri Krishi Sinchai Yojana** The primary objectives of PMKSY are to attract investments in irrigation system at field level, develop and expand cultivable land in the country. The primary objective is to enhance rain water use in order to minimise wastage of water, enhance crop per drop by implementing water saving technologies and precision irrigation.
- **Pradhan Mantri Ujjwala Yojana** Prime Minister Narendra Modi has launched

Pradhan Mantri Ujjwal Yojana on 1st May, 2016 (Labour Day) at Ballia (UP) by providing cooking gas connections to 10 women.

- The objective of the scheme is to provide cooking gas connections to 5 million beneficiaries below the poverty line in the next 3 years (till the year 2019).
- **Ujala Yojana** It was launched by Union Minister for State (IC) for Power, Coal and Renewable Energy Piyush Goyal in Bhopal, Madhya Pradesh on 30th April, 2016.

  The main motive of this policy is energy efficiency inthe country. Consumers can buy the bulbs from distributor by showing any identification card.
- **Pradhan Mantri Fasal Bima Yojana** It is the new crop damage insurance scheme started on 18 Feb, 2016. It will replace the existing two crop insurance schemes National Agricultural Insurance Scheme (NAIS) and Modified NAIS.
- **Pradhan Mantri Sahaj Bijli Har Ghar Yojana** The scheme aims for electrifying all the households in rural and urban areas which are still living without power.
- **Kisan Samman Nidhi Scheme** It was launched in 2019 to provide ₹ 6000 per year financial assistance to those farmers have cultivable area upto 2 hectare. ₹ 6000 per year will be paid in three instalments.

Employment Generation Programmes

Van Dhan Internship Programme

- Union Minister of Tribal Affairs launched the Van Dhan Internship Programme on 16th October, 2019. It was organised by the TRIFED under the Ministry of Tribal Affairs. The programme will help the tribal population to become self-reliant and entrepreneurs.

Swarna Jayanti Shahari Rozgar Yojana (SJSRY) (Set-up in 1997)

- To provide gainful employment to urban unemployed and under employed poor through self employment or wage employment. It has been revamped with effect from April, 2009.

Employment Assurance Scheme (EAS) Jawahar Gram Samridhi Yojana (JGSY) merged into it.

- To provide wage employment and food security in rural areas and also to create durable economic and social assets.

Mahatma Gandhi Rural Employment Guarantee Act (MGNREGA, launched on 2nd February, 2006) The National Rural Employment Guarantee Act (NREGA) 2005, envisages securing the livelihood of people in rural areas. *The main provisions of the act are*

- Employment to be given within 15 days of application for work.
- The scheme provides a legal guara- ntee for 100 days of employment in every financial year to adult members of any rural household willing to do public work-related unskilled manual work at the statutory minimum wage of 120 per day. If employment is not provided within 15 days, daily unemployment allowance in cash has to be paid.
- Employment within 5 km radius, else extra wages to be paid. At least one-third beneficiaries have to be women.

- **Prime Minister's Employment Generation Programme** (Set-up in 2008) To generate employment opportunities in rural as well as urban areas through setting up of new self-employment ventures/ projects/ micro enterprises.

Rural Development Programmes

- **National Broadband Mission** The government has launched the National Broadband Mission on 17th December, 2019. The mission aim to provide broadband access to all villages by 2022.
- **Indira Awas Yojana** (IAY, launched in 1999) Indira Awas Yojana (IAY) is a social welfare programme to provide housing for rural poor in India.
- This scheme, operating since 1985, provides subsidies and cash assistance to people in villages to construct their houses themselves.
- **Pradhan Mantri Gram Sadak Yojana** (PMGSY) (Set-up in 2000) To line all villages with pakka road having population of 500 and above in general areas and 250 and above in tribal and general areas.
- **Nirmal Gram Puruskar** (Set-up in October 2003) It is an incentive scheme to encourage PRIs to take up sanitation promotion.
- **Bharat Nirman Yojana** It was launched on 16th December, 2005, with the aim of developing rural infrastructure. The duration of implementing this scheme has been fixed for 4 years.

- **Rajiv Gandhi Grameen Vidyutikaran Yojana** (RGGVY launched in 2005) Aims at providing electricity in all villages and habitations and access to electricity to all rural households. Connections to BPL families are given free of cost. 90% cost of the scheme is released as grant where as 10% as loan.
- **Rajiv Awas Yojana** (RAY) (Set-up in 2010) It aims at *slum-free* India in next five years.

Urban Development Programme

- **Housing for All by 2022** Government has proposed to set up a Mission on Low Cost Affordable Housing to be anchored in the National Housing Bank with a view to increase the flow of cheaper credit for affordable housing to the urban poor/EWS/LIG segment.
- The Narendra Modi Government has renewed the 10 years old Jawaharlal Nehru National Urban Renewal Mission (JNURM) and named it after the first BJP Prime Minister.

  The renewed scheme is known as Atal Mission for Rejuvenation and Urban Transformation (AMRUT). AMRUT for 500 Tier 2 and Tier 3 cities will also be launched alongwith smart city project.
- The Ministry of housing and urban affairs has now extended teh mission by two years till March, 2022.
- **National Heritage City Development and Augmentation Yojana** (HRIDAY) (Set-up in January, 2015) The programme called Heritage City Development and Augmentation Yojana (HRIDAY) is to be launched for conserving and preserving the heritage characters of these cities.

Women Empowerment Programmes

- **Indira Gandhi Matritva Sanyog Yojana** (IGMSY) **2010** To improve the health and nutritional status of pregnant, lactating women and infants.
- **Ujjwala** (Set-up in December, 2007) A comprehensive scheme for prevention of trafficking with five specific components-prevention, rescue, rehabilitation, reintegration and repatriation of victims.
- **Dhan Laxmi** (Set-up in March, 2008) Conditional cash transfer scheme for the girl child to encourage families to educate girl children and to prevent child marriage.
- **National Mission for Empowerment of Women** (NMEW) (Set-up in 2010) To achieve empowerment of women socially, economically and educationally by securing convergence of schemes.
- **Rajiv Gandhi Scheme for Empowerment of Adolescent Girls** (RGSEAG)— 'Sabla' (Set-up in November, 2010) It aims at empowering adolescent girls of 11 to 18 years by improving their nutritional and health status, upgradation of home skills, life skills and vocational skills.
- **Sukanya Samriddhi Yojana** (Set-up in January, 2015) This scheme encompasses all the girls besides their economic strata can open Sukanya Samridhi Account in Post office and in the banks. It is launched along BBBP Yojana Campaign.
- **Pradhan Mantri Matru Vandana Yojna** It is a maternity benefit programme being implemented in all districts of the country with effect from 1st January, 2017. Cash incentive of ₹ 5000 will be provided in three installments to all pregnant women and lactating mothers.

Education Oriented Programmes

- **Atal Innovation Mission** The government has set up the Atal Innovation Mission at NITI Aayog in March, 2019. It encourages the school to provide exposure to science and technology to student. The main objective is to create scientific character and innovative thoughts among the young minds of the country (students between grade 6th to 12th).
- **Mid Day Meal** (MDM, launched in 1995) Under the scheme, hot cooked meal of a minimum 300 calories and 8-12 gms of protein is being provided to children studying in primary schools/Education Guarantee Scheme (EGS)/Alternative and Innovative Education (AIE) centres.
- **Sarva Shiksha Abhiyan** (SSA, launched in 2001) The main objective of this programme was to provide educational facility to all children of 6-14 age group in the state, to complete the primary education by 2007 and upper primary education by 2010, of all enrolled children and to ensure universal stay of all children up to the year 2010.

- **Kasturba Gandhi Balika Vidyalayas** (KGBVs) (Set-up in 2004) To set-up residential schools at upper primary level for girls belonging to SC/ST/OBC/Minority communities. The scheme is being implemented in rural areas and urban areas with female literacy below 30% and national average respectively.
- **Rashtriya Madhyamik Shiksha Abhiyan** (RMSA) **or Scheme for Universalisation of Access for Secondary Education** (SUCCESS) (Set up in March, 2009) Aims at raising the enrollment rate at secondary stage from 52.26% in 2005-06, to 75% in next 5 years by providing a secondary school within, reasonable distance of 5 km of any habitation; ensure universal access by 2017 and universal retention by 2020.

Health Oriented Programmes

- **SUMAN Scheme** The government launched the Surakshit Matritva Aashwasan (SUMAN) scheme on 10th October, 2019. Pregnant women, mother upto 6th month after delivery and all sick newborns will be able to free healthcare benefits under this scheme.
- **Bharatiya Poshan Krishi Kosh** The Union Government has launched Bharatiya Poshan Krishi Kosh with aim of reducing malnutrition in India on 18th November, 2019. It aims to reduce malnutrition among women and children across the country, through a multi-sectoral results-based framework, including agriculture.
- **National Rural Health Mission** (NRHM) (Set-up in April, 2005) To provide effective healthcare to rural population with special focus on 18 states with weak health indices/infrastructure; to raise public spending on health from 0.9% of GDP to 2.3% of GDP; reduction of IMR and MMR and universal access to healthcare with emphasis on women.
- **National Urban Health Mission** (NUHM 2013) The Union Cabinet gave its approval to launch a National Urban Health Mission (NUHM) as a new sub-mission on May, 2013 under the overarching National Health Mission (NHM).
- Centre-State funding pattern will be 75:25 except for North-Eastern states and other special States of Jammu and Kashmir, Himachal Pradesh and Uttarakhand for whom the funding pattern will be 90:10.
- **National Ayush Mission** (NAM) (Set-up in September, 2014) Ayurveda, Yoga, Unani Siddha and Homeopathy (AYUSH).
- This mission is aimed at addressing the gaps in health services particularly in vulnerable and far-off areas of India.
- **Mission Indradhanush** Launched in 2014, this mission is launched to achieve full immunisation coverage for all children by 2020.
- **Ayushman Bharat Yojana** It was launched in 2018 to provide ₹ 5 lakh health insurance to 10 crore poor families.

Food Security Programme

- **National Food Security Act (NFSA) 2013** It aims to provide food and nutritional security to the whole India. It provide access to adequate quality food at affordable prices to people and a life with dignity.

AGRICULTURE

Agriculture is the mainstay of the Indian economy and has a crucial role to play in the country's economic development. *Its importance to the Indian economy can be gauged from the following facts*

- **Contribution to GDP** Although, its share in national income is declining but still it contributed 17% (Approx) to the total GDP in 2017-18 (Aggregate Economy).
- **Contribution to Employment** Agriculture provides livelihood to more than half of the population. It contributes around 52% to the total employment in the country.
- **Contribution to Industry** Many of our small scale industries are dependent on agriculture. It has been the source of supply of raw material to industries like, handloom, cottage etc.
- **Contribution to Trade** India has emerged as significant agri exporter in a few crops *viz.* cotton, rice, pepper, sugar etc. The share of agriculture sector in the country's total export is 11.76% for the year 2018-19.

GREEN REVOLUTION

- Green Revolution was a part of New Agricultural Strategy which included, initially, the Intensive Agriculture District Programme (IADP) and later the **High Yielding Varieties Programme** (HYVP).
- It was launched in 1960's and was the brainchild of Norman Borlaug, though in India, it was made successful by **Dr MS Swaminathan**. The term 'Green Revolution' was coined by Dr William Gaud.
- The key pillars of this revolution were High Yielding Variety (HYV) seeds, chemical fertilizer and pesticides and improved irrigation facilities.
- The achievement of Green Revolution were rise in cereal production especially wheat and rice, change in cropping pattern in favour of wheat, and increase in employment opportunities.
- A National Commission on Farmers was appointed in 2004, under the Chairmanship of Dr MS Swaminathan which suggested an **Agricultural Renewal Action Plan** (ARAP).
- The ARAP comprised of soil health enhancement, irrigation water supply augmentation and demand management, credit and insurance, technological reforms and assured and remmunerative marketing.

Second Green Revolution

- The call for Second Green Revolution was given by Prime Minister Manmohan Singh at the 93rd Science Conference **in 2006**. The second green revolution seeks to build up on the achievements of Ist green revolution and bridge the regional and crop imbalance which were not addressed by first Green Revolution.
- The Second Green Revolution seeks to cover dryland farming and concenrate on the small and marginal farmers. It seeks to raise the foodgrain production to 400 million tonnes by 2020.

Operation Green

Operation Green was launched on the lines of Operation Flood for enhancing the production of tomato, onion and potatoes in budget 2018-19. A sum of ₹ 500 crore has been allocated for this new measure. It may help in doubling the income of farmers by the end of 2022.

Tricolour Revolution

The reference to a Tricolour Revolution was made by Prime Minister Narendra Modi. This phrase has three components. *These are as follows.*

- **Saffron Energy Revolution** for promotion and better utilisation of solar energy.
- **White Revolution** to ensure cattle welfare and further the goals of White Revolution.
- **Blue Revolution** for fishermen's welfare, cleansing river and sea and conserving water.

Major Agricultural Revolutions

Revolution	*Production*
Black Revolution	Petroleum Production
Blue Revolution	Fish Production
Brown Revolution	Leather/non-conventional (India)/Cocoa production
Golden Fibre Revolution	Jute Production
Golden Revolution	Overall Horticulture development/Honey Production
Green Revolution	Foodgrain (Cereals, Wheat and Leguminious plant) production
Grey Revolution	Fertiliser Revolution
Pink Revolution	Onion production/ Pharmaceutical (India)/ Prawn production
Rainbow Revolution	Holistic development of agriculture sector
Red Revolution	Meat and Tomato Production
Round Revolution	Potato Revolution
Silver Fibre Revolution	Cotton Revolution
Silver Revolution	Egg/Poultry Production
White Revolution	Milk/Dairy production (In India-Operation Flood) started by Dr. Verghese Kurien
Yellow Revolution	Oil Seeds production
Evergreen Revolution	Increase in productivity and prosperity without ecological harm

CROP PRODUCTION

- India's production of foodgrains has been increasing every year, and India is among the top producers of several crops such as wheat, rice, pulses,

sugarcane and cotton with record rice (115 million tonnes) and wheat (101 million tonnes) production.
- Indian ranks second in wheat and rice production after China.
- As per estimates, India is the fifth largest producer of natural rubber with a share of 7.9% in world production.
- India consumes coffee comprising both Arabica (32%) and Robusta (68%) coffee. Indian coffee is primarily an export-oriented commodity with about 70% of production being exported.
- India is second largest producer of **tea** in the world after China. Tea is grown in 16 states in India. Assam, West Bengal, Tamil Nadu and Kerala, account for around 95% of total tea production.

Fisheries Sector

- India is the third largest producer of fish and second largest producer of inland fish in the world.
- At present, the fisheries sector is a source of livelihood of over 11 million people engaged fully, partially or in subsidiary activities pertaining to the sector.

National Agriculture Mark, or e-NAM

- It is an online trading platform for agricultural, commodities in India. e-NAM provides inter-connectivity to e-mandis and help farmers get betters prices.

Regional Rural Banks (RRBs)

- RRBs formally launched in 2nd October, 1975, at Moradabad and Gorakhpur (Uttar Pradesh), Bhiwani (Haryana), Jaipur (Rajasthan) and Malda (West Bengal).
- The **objective** of the RRBs was to provide credit and other facilities particularly to small and marginal farmers, agricultural labourers etc so as to develop agriculture.

NABARD

- **National Bank for Agriculture and Rural Development** (NABARD) was set-up in July, 1982, as the Apex Bank with a paid-up capital of ₹ 100 crore contributed equally by RBIs and Government of India. Its headquarter is at Mumbai.
- NABARD serves as apex financing agency to the institutions providing investment and production credit for promoting various developmental activities in rural areas.
- **Rural Infrastructure Development Fund** (RIDF) was set up in 1995-96, under NABARD for holistic rural development.

Lead Bank Scheme

- The lead bank scheme, introduced towards the end of 1969, on the basis of recommendations of Prof. Dr Gadgil.
- Under the scheme, each district had been assigned to different banks (public and private) to act as consortion leades to coordinate the efforts of bank in the district.

Agriculture Insurance Company of India Limited (AIC)

- AIC was incorporated under the Companies Act, 1956 on 20th December, 2002, as a specialised insurer with the capital participation from GIC, four public sector General Insurance Companies and NABARD. AIC has an authorised share capital of INR 15 billion and paid-up capital of INR 2 billion.
- The other specialised insurer is **Export Credit Guarantee Corporation** (ECGC). It was established in 1957.

Mega Food Park Scheme

- The Tenth Plan Scheme of Food Parks was renamed as the **Mega Food Park Scheme** (MFPS) in 2008. The objective of the scheme is a link farm produced to the market by bringing together farmers, processors and retailers.

INDUSTRY

- Industry sector comprises of manufacturing, electricity and gas and water supply, construction etc.
- It contributes around 29.6 of the Indian GDP in 2018-19. The long-term average annual growth of industries during the post-reform period between 1991-92, to 2011-12, averaged 6.7%.
- The industrial growth rate in terms of index of Industrial Production (IIP) during 2018-19 stood at 3.6% mainly due to slowdwon in manufacturing industry.

Industrial Policies

- Industrial policies were launched in 1948, 1956, 1977, 1980 and 1991.
- The Industrial Policy Resolution of 1948, marked the beginning of the evolution of the Indian Industrial Policy.
- The IPR 1956, called the **Economic Constitution** of India, gave the public sector a strategic role in the economy.
- The objective of the IPR 1956, was establishment of **socialistic pattern of the society** in the country.

New Industrial Policy, 1991

Formed the basis for the economic reforms in India, which proved to be a watershed in the history of Indian economy. *The main aim of the new industrial policy 1991, was*

- To unshackle the Indian industries from the cobweb of unnecessary bureaucratic control;
- To introduce liberalisation with a view to integrate Indian economy with the world economy;
- To remove restrictions on FDI and to abolish MRTP Act, 1969 and
- To shed the load of the public enterprises.

Public Sector Enterprises

These companies are primarily owned by Union Government or states or both.

- There were only five Central Public Sector Enterprises (CPSEs) in 1951 which increased to nearly 300 in total in the year 2019.
- The second five year plan gave top priority to the industrial growth emphasising on development of heavy industries in public seftor.
- To impart greater managerial and commercial autonomy to the PSEs, the concept of **Navratnas** and **Miniratnas** was started in 1997. The category of Maharatnas were introduced in the year 2009.

Maharatna

- In 2009, the government established the **Maharatna** status, which raised the PSEs investment ceiling from ₹ 1000 crore to ₹ 5000 crore. The Maharatna firms can now decide on investments of upto 15% of their net worth.

Eligibility Criteria for Maharatna

- *The six criteria for eligibility of Maharatna are*
  1. Having Navratna status;
  2. Listed on Indian stock exchange;
  3. An average annual turnover of more than ₹ 20000 crore during the last 3 years;
  4. An average annual net worth of more than ₹ 10000 crore during the last 3 years;
  5. An average annual net profit after tax of more than ₹ 5000 crore during the last 3 years;
  6. Should have significant global presence.

List of Maharatna

There are eight Maharatnas in India

1. Oil and Natural Gas Corporation (ONGC)
2. Steel Authority of India Limited (SAIL)
3. Indian Oil Corporation Limited (IOCL)
4. Bharat Heavy Electricals Limited (BHEL)
5. Gas Authority of India Limited (GAIL)
6. National Thermal Power Corporation (NTPC)
7. Coal India Limited (CIL)
8. Bharat Petroleum Corporation Limited (BPCL)
9. Hindustan Petroleum Corporation Limited
10. Power Grid Corporation of India Limited

Navratna

To be qualified as a Navratna Company

- The company must obtain a score of 60 (of the total 100).
- The score is based on six parameters which included net profit to net worth, total manpower cost to total cost of production, Profit Before Depreciation, Interest and Taxes (PBDIT) to capital employed, PBDIT to turnover, earning per share and inter-sectoral performance.
- The company must first be a Miniratna-1 and must have four independent directors on its board.
- The Navratna status empowers a company to invest upto ₹ 1000 crore or 15% of their net worth overseas without government approval.
- At present, there are **14 Navratnas**.

List of Navratna

- Bharat Electronics Limited
- Hindustan Aeronautics Limited

- Mahanagar Telephone Nigam Limited
- National Aluminium Company Limited
- National Mineral Development Corporation
- Nevyeli Lignite Corporation Limited
- Oil India Limited
- Power Finance Corporation Limited
- Rashtriya Ispat Nigam Limited
- Rural Electrification Corporation Limited
- Shipping Corporation of India Limited
- Engineers India Limited
- National Building Corporation Limited
- Container Corporation of India Limited

Miniratna

Category-I Miniratna

- Public Sector Enterprises (PSEs) that have made profit continuously for the last three years or earned a net profit of ₹ 30 crores or more in one of 3 years.
- At present, there are 62 Miniratna-I.
- Miniratnas-I has granted certain autonomy like incurring capital expenditure without government approved upto ₹ 500 crore or equal to their networth.

Category-II Miniratna

- PSEs that have made profit for the lst three years and should have a positive net worth.
- At present, there are 12 Miniratna-II.
- Miniratnas-II can incur the capital expenditure ₹ 300 crore or 50% of their net worth whichever is lower.

SMALL-SCALE INDUSTRIES

- A new thrust in favour of small-scale industries was given in the Industrial Policy Resolution of 1977. With effect from 2nd October, 2006 government enacted the Micro, Small and Medium Enterprises Development Act.
- The **MSMED Act, 2006** clearly defines, for the first time, not only the **medium** enterprises, but also extends it to the services sector too.
- MSMED (Amendment) Bill, 2018 proposes to reclassify all MSMEs on the basis of their annual turnover.

Classification of MSMES

Manufacturing and Services (1st July 2020)

	Micro	*Small*	*Medium*
Investment	< ₹ 1 cr	< ₹ 10 cr	< ₹ 50 cr
Turnover	< ₹ 5 cr	< ₹ 50 cr	< ₹ 250 cr

- MSME is second largest employment provider after agriculture sector. It also contributes about 31% to the nation's GDP. 45% share in the overall exports and 34% share in the manufacturing output as per 2017 report.
- **Abid Hussain Committee** was set-up to look into the problems of small-scale industries.
- Ministry of Micro, Small and Medium Enterprises acts as the nodal agency for growth and development of SSIs in India.
- Under public procurement policy for MSMEs, it is mandatory for Central Government and PSUs to purchase atleast 25% of their goods and services from MSME sector.
- **Small Industry Development Organisation** (SIDO) It is the apex body for assisting the government in formulating and overseeing the implementation of its policies.
- **Small Industries Development Bank of India** (SIDBI) Formed in 1990, It acts as apex institution for financing SSIs through various credit schemes.

LARGE-SCALE INDUSTRIES

Iron and Steel Industry

- First steel industry at Kulti, West Bengal–Bengal Iron Works Company was established in 1870. First large scale steel plant-TISCO at Jamshedpur (1907), was followed by IISCO at Burnpur (1919). The first public-owned steel plant was Rourkela integrated steel plant set-up in 1954, with the help of Germany.
- India is the fourth largest producer of crude steel in the world after China, with production of 106.5 Mt in 2018. India is the largest producer of sponge iron since 2002. **Steel Authority of India Limited** (SAIL), was established in 1974, for the development of the steel industry.

Iron and Steel Plants

Location	*Assistance*
Rourkela (Odisha)	Germany
Bhilai (Chhattisgarh)	Russia
Durgapur (West Bengal)	Britain
Bokaro (Jharkhand)	Russia
Vishakhapatnam (Andhra Pradesh)	Russia

Textile Industry

- It is the largest industry in India accounting for about 7% of industrial output in terms of value, provides employment to more than 3.5% million persons and contributes around 15% to total export earnings in 2018-19.
- The first Indian modernised cotton cloth mill was established in 1818, at fort Gloster near Kolkata, but this was unsuccessful. The second mill was established in 1854, at Bombay by KGN Daber.

Jute Industry

- Jute industry was started in 1855, at Rishra and India is the largest producer and second largest exporter of jute in the world. Jute Technology Mission was launched on 2nd June, 2006.

Gems and Jewellery

- Gems and jewellery is an important emerging sector in the Indian economy. As per recent reports, india is the second largest counsumer of gold after China in 2019.
- This sector contributes around 7 per cent of country's GDP and 15 per cent to India's total merchandise exports.

Paper Industry

- Paper Industry in India is the 15th largest paper industry in the world.
- The first paper mill in India was set-up at Sreerampur, West Bengal, in the year of 1862.

Silk Industry

- India is the second largest (after China) silk manufacturer contribu- ting to 18% of the total raw silk production. The majority of silk is produced mainly in Bhoodan Pochampally (also known as silk city), Kanchipuram, Dharamvaram and Mysuru.

Sugar Industry

- India is the second largest producer of sugar in the world with a 22% share.
- It is the second largest agro-based industry in the country.
- **BB Mahajan Committee** was set-up to study the sugar industry.
- Uttar Pradesh is the leading producer state of sugarcane and sugar in the year 2019-20 with Maharashtra at 2nd position.

Cement Industry

- The foundation of stable Indian cement industry was laid in 1914 at Porbandar in Gujarat.
- India is the second largest producer of cement in the world.

Fertilizer Industry

- The first fertilizer industry was set-up in 1906, in Ranipet near Chennai.
- Urea is the only fertilizer under statutory price control.

Automotive Industry

- India is the second largest manufacturer of motorcycle and fourth largest manufacturer of commercial vehicles in the world, as per latest reports.

FOREIGN TRADE

- Trade between two or more nations is called foreign trade or international trade.
- With the liberalisation of the economy in 1991 and adoption of *export promotion* policy measures has led to substantial growth in exports and diversification of our exports.
- With a view to boost exports, the government announced several measures in **Foreign Trade Policy** (FTP) 2015-20 including Merchandise Export from India Scheme (MEIS) and Services Export from India Scheme (SEIS).

Features of FTP

- India to be made a significant participants in world trade by 2020.
- Merchandise Exports from India Scheme (MEIS) to promote specific services for specific markets FTP.
- All existing service related schemes has been replaced with Service Exports from India Scheme (SEIS).
- **Make in India** is an initiative of the government to encourage multinational as well as domestic companies to manufacture their products in India. It was launched in September 2014.

Trade Composition

Export Composition

- As per Economic Survey 2018-19, the top four items in India's export basket are petroleum products (14.5%), pearl, precious and semi-precious stones (7.9%), Drug formulations and Biologicals (4.4%) and Gold and other metal jewellery (3.9%).
- India's largest export destination country is USA with 16% of India's export followed by UAE.
- There has been a gradual shift in India's manufactures exports from labour-intensive sectors like textiles, leather manufactures, handicrafts and carpets to capital and skill intensive sectors. For the year 2018-19, it was exports of organic chemicals which grew the highest at 30.6 per cent.

Import Composition

- In the import basket of 2018-19, crude petroleum (22%) had the largest share followed by gold and other precious metal jewellery (6.4%). Import of pearls and precious stones comes at third position (5.3%) in import basket.
- China continues to be the largest source of imports for India (13.7%) followed by USA (6.9%).

Balance of Trade (BoT)

The difference between a nation's imports of goods and its exports is known as **Balance of Trade**. It is the most important element of the Balance of Payments. There are three possibilities in the Balance of Trade (BoT)

1. Balanced BoT, i.e., exports = imports
2. Adverse BoT, i.e., exports < imports
3. Favourable BoT, i.e., exports > imports

Balance of Payment (BoP)

The Balance of Payment (BoP) records the transactions in goods, services and assets between residents of a country with the rest of the world for a specified time period typically a year. There are two main accounts in the BoP; the Current Account and the Capital Account. In addition to that BoP comprises-errors and omissions and change in foreign exchange reserves.

Foreign Exchange Reserves in India

These are the main Foreign Exchange Reserves in India

- Foreign Currency Assets (FCAs);
- Gold Stock of RBI;
- Special Drawing Rights (SDRs) and
- Reserve Tranche Position (RTP) in the IMF.

- **FERA** (Foreign Exchange Regulation Act) was enacted in 1973, to consolidate and regulate dealings in foreign exchange so as to conserve the foreign exchange and utilise it to promote economic development.
- **FEMA** (Foreign Exchange Management Act) was enacted in 1999, to replace existing FERA. This act seeks to make offences related to foreign exchange, a civil offence not a criminal offence.

Special Economic Zone (SEZ)

- A Special Economic Zone (SEZ) is a geographical region that has economic and other laws that are more free-market oriented than a country's typical national laws.
- Asia's first Export Processing Zone (EPZ) was set-up in **Kandla**, in **1965**.
- The first SEZ policy was announced in April **2000**, to make SEZ an engine of growth supported by quality infrastructure backed by attractive fiscal package.
- To overcome the shortcomings experienced on account of the multiplicity of controls and clearances and an unstable fiscal regime and with a view to attract foreign investments in India, SEZ Act, 2005 was enacted with effect from 10th February, 2005.

SEZ Act, 2005

- Duty free import/ domestic procurement of goods for development, operation and maintenance of SEZ units.
- 100% Income Tax exemption on export income of SEZ units, exemption from Central Sales Tax and Service Tax. Single window clearance mechanism for establishment of units.

FOREIGN DIRECT INVESTMENT (FDI)

- FDI occurs when a company invests in a business that is located in another country and it is investing not less than 10% of shares belonging to the foreign company.
- It is a non-debt capital flow. Foreign portfolio investment occurs when foreign investment in the form of shares, equities and bonds, is made by a foreign company.
- *The different forms of foreign investments are*
  - Direct foreign investment;
  - Foreign collaboration;
  - Inter-governmental loans;
  - Loans from international institutions;
  - External commercial borrowings.

INDIAN FINANCIAL SYSTEM

- Financial system refers to the borrowing and lending of funds for the demand and supply of funds of all individuals institutions, companies and of the government.
- The Indian Financial System consists of two parts, *viz*, Indian Money Market and the Indian Capital Market.

Indian Money Market

- Money Market is a market for 'near money' or it is the market for borrowing and lending of short-term funds.

Indian Capital Market

- Capital Market is the market where medium and long-term funds are borrowed and lent. It does not deal with capital goods, but is concerned with raising of money capital for investment purposes.
- Gilt edged market refers to the market for government and semi-government securities backed by the RBI.
- Industrial securities market refers to the market for shares and debentures of old and new companies.

Share Market in India

- Stock exchange or share market deals in shares, debentures and financial securities.
- Bombay Stock Exchange (BSE) is the oldest stock exchange in Asia which was established in 1875.
- It is synonymous with **Dalal Street**. BSE was corporatised and renamed BSE Limited in 2005.
- In 1894, the Ahmedabad Stock Exchange was started to facilitate dealing in the shares of textile mills. There are around 24 stock exchanges in India.
- Among them two are national level stock exchanges namely Bombay Stock Exchange (BSE) and National Stock Exchange (NSE).
- The rest are Regional Stock Exchanges (RSE).

SEBI

- Securities and Exchange Board of India was set-up in 1988 and made a statutory body in 1992.
- Forward Market Commission (FMC) has been recently merged with SEBI.

Main Functions of SEBI

- To protect the interests of investor in securities.
- To promote the development of securities market and to regulate the securities market.
- To prohibit unfair trade practices in the market.
- On 25th October, 2007, SEBI allowed creation of a separate exchange for Small and Medium Enterprises (SMEs).

Payment Bank

- Payment Banks differ from conventional banks as they are not allowed to lend to customers or issue credit cards.
- They can accept deposits of up to ₹ 1 lakh and can offer current and saving account deposits. They can also issue debit cards and offer internet banking.

Important Share Price Indices of India

- **BSE SENSEX** This is the most sensitive share index of the Bombay Stock Exchange. This is the representative index of 30 main shares. Its base year is 1978-79. BSE is the oldest stock exchange of India, founded in 1875.
- **BSE 200** This represents 200 shares of Bombay Stock Exchanges. Its base year is 1989-90.
- **DOLLEX** Index of 200 BSE Dollar Value Index is called Dollex. Its base year is 1989-90. **NSE-50**. National Stock Exchange launched a new share Price Index, NSE-50 in place of NSE-100 in April 1996. NSE-50 includes 50 companies shares. This stock exchange was founded on Ferwani Committees's recommendation in 1994.

Index	*Country*
Hang Seng	Hongkong
JCI	Indonesia
Nikkei 225	Japan
Kospi	South Korea
Kuala Lumpur Composite	Malaysia
TSEC Weighted Index	Taiwan
SSE Composite Index	China
SET	Thailand
FTSE 100	UK
NASDAQ Composite Index	US
STOXX	Europe
Dow Jones	US (New York)

CREDIT RATING AGENCIES

- A Credit Rating Agency (CRA) is a company that assigns credit for issuance of certain types of debt obligation as well as the debt instruments themselves.
- **CRISIL,** set up in 1988, is a credit rating agency. It undertakes the rating fixed deposit programmes, convertible and non-convertible debentures and also credit assessment of companies.
- Some other Credit Rating Agencies in India are ICRA, CARE and India Rating.

BANKING IN INDIA

- The first bank of limited liability managed by an Indian was Oudh Commercial Bank in 1881. Subsequently, PNB was established in 1894. The largest bank Imperial Bank of India was nationalised in 1955 and renamed as State Banks of India followed by formation of its 7 associates in 1959.
- The step toward *Social banking* was taken with the nationalisation of 14 Commercial Banks on 19th July, 1969. Six more Commercial Banks were nationalised on 15th August, 1980.

Reserve Bank of India (RBI)

- RBI is the Central Bank of the country.
- RBI was set up on the basis of Hilton Young Commission recommendation in April 1935 with the enactment of RBI Act, 1934. Its first Governor was Sir Osborne Smith.
- RBI was nationalised in 1949, and its first Indian Governor was CD Deshmukh.
- Its headquarter is located in Mumbai.

Functions

- *The main functions of the RBI includes*
  (i) Regulation of credit (ii) Issue of notes and coins
  (iii) Regulation of foreign exchange (iv) Lender of last resort
  (v) Banker's bank (vi) Agent of the government
  (vii) Banker to the government and commercial bank

Monetary Policy of RBI

- Since 1952, Monetary Policy of the RBI emphasise on twin goals. *These are*
  (i) Economic growth (ii) Inflation control
- The instruments of Monetary Policy of RBI is of two types :–

Quantitative/General Credit Control

- Quantitative credit control are used to control the volume of credit and indirectly to control the inflationary and deflationary pressures caused by expansion and contraction of credit.

Quantitative Credit Control Instruments

- **Bank Rate** It is also called the rediscount rate. It is the rate at which the RBI gives finance to Commercial Banks.
- **Cash Reserve Ratio** (CRR) It is the amount of funds that the banks have to keep with RBI. If RBI increases CRR, the available amount with the banks comes down.
- **Statutory Liquidity Ratio** (SLR) It is the ratio of liquid asset which all commercial banks have to keep in the form of cash, gold and government approved securities with itself before providing credit to customers.
- **Repo Rate** It is the rate at which RBI lends short term money to the banks against securities. Decrease in repo rate generally injects liquidity in the market.
- **Open Market Operations** (OMOs) It is the buying and selling of government securities in the open market by the RBI to regulate the flow of money in the market.
- **Reverse Repo Rate** It is the rate at which banks park short-term excess liquidity with the RBI. Increase in reverse repo rate withdraws liquidity from the market.
- **Marginal Standing Facility** (MSF) It is a window for banks to borrow from the Reserve Bank of India in an emergency situation when inter-bank liquidity dries up completely.

Qualitative/Selective/Direct Credit Control

- Qualitative/Selective credit controls are used to control and check the rising tendency of the prices of certain individual commodities of common use.
  Generally, the RBI uses three kinds of selective credit controls
  - Minimum margins for lending against specific securities;
  - Ceiling on the amount of credit for certain purposes;
  - Discriminatory rate of interest charged on certain types of advances.

Bank Board Bureau

The Bank Board Bureau (BBB) is, an autonomous body of Union Government of India, is constituted on 28th February, 2016. The BBB was formed to improve the governance of public sector bank.

Indradhanush Scheme, 2016

Indradhanush Scheme is for the banking reforms in India. The seven key reforms of indradhanush mission include appointments, destressing, BBB, capitalisation, empowerment, frame work of accountability and governance reforms.

State Bank of India

- State Bank of India (SBI) was previously called Imperial Bank of India in 1921, which was created by amalgamation of 3 presidency banks *viz* Bank of Bengal, Bank of Bombay and Bank of Madras.
- It was nationalised in 1955. In 2017, the five associates banks were merged into SBI.

Bharatiya Mahila Bank

- India's first all women bank, Bharatiya Mahila Bank (BMB) opened in Mumbai in November, 2013.
- The main objective of the bank was to focus on the banking need of women and to promote their economic empowerment. In 2017, it was merged in SBI.

Top 5 Private Sector Banks

- ICICI Bank, HDFC Bank, Axis Bank, Kotak Mahindra Bank, Yes Bank.

Merger of Nationalised Bank

- Vijaya Bank and Dena Bank merged with Bank of Baroda (BoB) on 1st April, 2019.
- In 1st April, 2020, Government has merged of Indian Bank with Allahabad Bank; Oriental Bank of Commerce (OBC) and United Bank of India was merged Punjab National Bank; Syndicate Bank with Canara Bank and Andhra Bank along with Corporation Bank will be merged in Union Bank of India. Post merger there will be 12 Public Sector Banks in India.

INSURANCE

- Insurance industry includes two sectors-Life Insurance and General Insurance. Life insurance in India was introduced by Britishers. A British firm in 1818, established the Oriental Life Insurance Company at Calcutta now Kolkata.
- Since the opening up, the number of participants in the Insurance Industry has gone up from 7 insurers (including LIC, four public sector general insurers, one specialised insurer and the GIC as the national re-insurer) in 2000 to 53 insurers in, 2019.
- Insurance Regulatory and Development Authority of India (IRDAI) is the regulating authority of insurance sector.

Life Insurance Corporation of India (LIC)

- LIC was established on 1st September, 1956, which set the pace for nationalisation of life insurance under the guidance of CD Deshmukh. It has head office at Mumbai and eight zonal offices, the most recent being at Patna.

General Insurance Corporation of India (GIC)

GIC was established on 1st January, 1973 with its four subsidiaries, *viz* (i) National Insurance Company Limited, Kolkata; (ii) The New India Assurance Company Limited, Mumbai; (iii) The Oriental Fire and General Insurance Company Limited New Delhi; and (iv) United India General Insurance Company Limited, Chennai.

Unit Trust of India

It was set up in 1964 and was bifurcated in 2003 into specified undertaking of Unit Trust of India (SUUTI) and UTI Mutual Fund (UTIMF) .

Suresh Mathur Committee

IRDA has set up a ten member committee headed by Suresh mathur to look into having 100% FDI in insurance intermediaries and Third party Administrators (TPAs), on 12th January, 2014.

At a Glance

Financial Institutions	*Purpose*
Industrial Finance Corporation of India (IFCI) 1948, New Delhi	▪ To grant loans advances to industrial concerns and subscribe to debentures floated by them.
State Finance Corporations (SFCs) 1951	▪ To finance the needs of the small-scale and medium sized industries in respective states. ▪ 28 SFCs are in operations along with 28 State Industrial Development Corp. (SIDC).
Industrial Credit and Investment Corporation of India (ICICI) 1955, Mumbai	▪ To stimulate the promotion of new industries and assist in modernization of existing industries. ▪ In 2002, ICICI merged with ICICI Bank no more a DFI.
Industrial Development Bank of India (IDBI) 1964 Mumbai	▪ To meet the financial needs of industrialization and co-ordinate with all other agencies concerned with industrial development finance. ▪ Wholly owned subsidiary of RBI till 1976. ▪ Merged with IDBI Bank in 2004.
Industrial Investment Bank of India (IIBI)1985	▪ To assist the industrial units in Eastern region.
Small Industries Development Bank of India (SIDBI), Lucknow 1990	▪ To promote, finance and development industry in small scale sector.
Unit Trust of India (1964) (UTI)	▪ Set up as an investment institution to stimulate and pool the savings of the middle and low income groups. ▪ *UTI bifurcated into two parts* (i) UTI Mutual Fund (ii) Specified Undertaking of UTI (SUUTI)
National Housing Bank (NHB) 1988, New Delhi	▪ It is the apex institution of housing finance in India and is a wholly owned subsidiary of the RBI. ▪ Launched RESIDEX for tracking prices of residential properties in India in 2007. At present, RESIDEX covers 15 cities in India. ▪ It is a refinance institution.
EXIM Bank (1982), Mumbai	▪ To finance, facilitate and promote foreign trade in India. ▪ It is a specialized financial institution.
Tourism Finance Corporation of India Ltd (TFCI)—1989, New Delhi	▪ Set up on the recommendation of Yunhs Committee on Tourism. ▪ It is a specialised finance institution providing assistance to tourism related activities/projects. ▪ As on March 2010 four institutions namely EXIM Bank, NABARD, NHB and SIDBI regulated by the RBI as all India financial institutions.
Mutual Funds	▪ Mutual Funds are the most important among the newer capital market institutions. MFs functions is to mobilise the savings of the general public and invest them in stock market securities.
Venture Capital Funds	▪ Venture Capital Funds (VCFs) essentially give commercial support to new ideas and for the introduction and adaptation of new technologies. ▪ Government launched the National Venture Fund for Software and IT Industry (NVFSIT) to provide technology development especially for small and medium enterprises. NVFSIT is managed by SIDBI.

INDIAN CURRENCY SYSTEM

- The present monetary system of India is based on inconvertible paper currency and is managed by the RBI.
- The present currency system is based on **minimum reserve system.**
- It was adopted in 1957. Under the minimum reserve system, minimum of gold and foreign securities to the extent of ₹ 200 crore (of which gold should be of value ₹ 115 crore) and the balance in rupee securities is to be maintained.
- Coins are minted at four places, *viz* Mumbai, Kolkata, Hyderabad and Noida.

Security Press	*Station*	*Related to*
▪ India Security Press (1922)	Nasik (Maharashtra)	Postal material, Postal stamps, etc
▪ Security Printing Press (Estd 1982)	Hyderabad (AP)	Union excise duty stamps
▪ Currency Notes Press (1928)	Nasik (Maharashtra)	Bank notes from ₹ 1 to ₹100
▪ Bank Notes Press (1974)	Dewas (MP)	Bank note of ₹ 20, ₹ 50, ₹ 100, ₹ 500, ₹ 2000
▪ Modernised Currency Notes Press (1995)	Mysore (Kar); Salbani (West Bengal)	
▪ Security Paper mill (Estd 1967-68)	Hoshangabad (MP)	Banks and currency notes paper

Indian Rupee Sign

- The symbol of Indian rupee came into vague on 15th July, 2010.
- The new symbol is an amalgamation of Devanagri 'Ra' and the Roman 'R' without the stem.
- The new symbol designed by D. Udaya Kumar, a post-graduate of IIT Bombay was finally selected by the Union Cabinet on 15th July, 2010.
- The symbol '₹' will be printed or embossed on currency notes or coins, it would also be included in the 'Unicode Standard' and major scripts of the world to ensure that it is easily displaced and printed in the electronic and print media.

INFLATION

Inflation means a persistent rise in the price levels of goods and services leading to a fall in the currency's purchasing power.

Causes of Inflation

- Printing too much money.
- Increase in production cost.
- Tax rises.
- Decline in exchange rates.
- War or other events causing instability.
- Increase in money supply in the economy.

Effects of Inflation

- It is economically disastrous for lenders. Balance of trade can become unfavourable. Severely impacts the common man by reducing their real income. Persistent high level of inflation leads to economic stagnation.

Deflation

- A general decline in prices, often caused by a reduction in the supply of money or credit. Deflation can be also caused by a decrease in government, pesonal or investment spending. It also leads to the unemployment since there is a lower level of demand in the economy, which can lead to an economic depressson.

Stagflation

- When you have a slow economy with high inflation rates and unemployment, stagflation is usually resultant. When the economy does not grow and prices continue to rise have a Stagflation Cycle in the economy.

Measures to Control Inflation

- Increasing the bank interest rates.
- Regulating fixed exchange rates of the domestic currency. Controlling prices and wages. Providing cost of living allowances to citizens.
- Regulating black and speculative market.

TAX STRUCTURE

- **Tax** is a compulsory payment by the citizens to the government to meet the public expenditure.
- It is legally imposed by the government on the taxpayer and in no case taxpayer can deny to pay taxes to the government.
- **Direct Tax** A direct tax is that tax which is born by the person on whom it is levied. A direct tax cannot be shifted to other person.
- **Indirect Tax** Indirect taxes are those taxes which are levied on one entity (manufacturer of goods), but is passed on to the final consumer by the manufacturer.

Direct Tax	*Indirect Tax*
Personal Income Tax	Excise Duty
Corporation Tax	Custom Duty
Wealth Tax	Sales Tax
Gift Tax	Service Tax
Land Revenue	Value Added Tax
Profession Tax	Passenger Tax
Stamp Duty and Registration Charges	Entertainment Tax
Securities Transaction Tax	Electricity Duty
Banking Cash Transaction Tax	Motor Vehicles Tax

GST (Goods and Services Tax)

The Goods and Service Tax (GST) has been implemented from July 1, 2017. It incorporates many of the indirect taxes levied by states and the Central Government. Some of the taxes GST replaced include sales tax, Central Excise Duty, Octroi, Service tax etc.

GST has three components :

- CGST (Central Goods and Services Act)
- SGST (State Goods and Services Act)
- IGST (Integrated Goods and Service s Act)

Finance Commission

Finance Commission is constituted to define financial relations between the centre and the state. 15th Finance Commissions have been set up for financial year 2020-2025. Former Planning commission member NK Singh was appointed as the Chairman of the 15th Finance Commission.

Financial Relations between Centre and States

- India possesses a federal structure in which a clear distinction is made between the Union and the State functions and sources of revenue. Article 264 and 293 explain the financial relations between the Union and State Government.
- Although, the states have been assigned certain taxes which are levied and collected by them, they also have a share in the revenue of certain union taxes and there are certain other taxes which are levied and collected by the Central Government but whole proceedings are transferred to the states.

INTERNATIONAL ECONOMIC ORGANISATIONS

International Monetary Fund (IMF)

- IMF was conceived on 22nd July, 1944 and came into existence on 22nd December, 1945 when 29 countries signed the agreement. It originally had 45 members. India is the founding member.
- IMFs stated goal is to stabilise exchange rates and assist the reconstruction of the worlds international payment system.
- IMF at present has 190 members and head quartered at Washington DC. The capital resources of the IMF comprise Special Drawing Rights (SDRs) and currencies that member pay under quotas calculated for them when they join the IMF.

Special Drawing Rights

The SDRs were created in 1969 are supplementary foreign exchange reserves assets maintained by the IMF. SDRs are not a currency, SDR, value is determined by weighted currency basket of five major currencies : the Euro, the US dollar, the British pound, Japanese Yen and Chinese Yuan. SDRs are also called ***paper gold***.

- The quotas determine the amount of foreign exchange, a member may borrow from the IMF and its voting power on IMF policy matters. Quotas are denominated in SDRs.

World Bank Group

- World Bank is one of the five institutions created at the Bretton Woods Conference in 1944. Along the IMF, it constitutes *twin-sister* of Bretton Woods.
- World Bank has 190 members and is headquartered in Washington DC.
- The World Bank provides loans to developing countries for capital programme and its official goal is reduction of poverty.

World Trade Organisation (WTO)

- It was constituted on 1st January, 1995 under the **Marrakesh Agreement** and took the place of GATT (General Agreement on Trade and Tariffs) as an effective formal organisation. GATT was an informal organisation which regulated World trade since 1948. Its headquartered is at Geneva. At present, it has 164members.
- The WTO is currently endeavouring to persist with a trade negotiation called Doha Development Agenda (DDA) which was launched in 2001, to enhance equitable participation of poor countries which represent a majority of the world's population.

Functions of WTO

- To oversee, implementing and administering WTO agreements.
- To provide a forum for negotiations.
- To provide a dispute settlement mechanism. To provide facilities for implementation, administration and operation of multilateral and bilateral agreements of the world trade.

Conference	*Year*	*Place*
First	1996	Singapore
Second	1998	Geneva
Third	1999	Seattle (USA)
Fourth	2001	Doha (Qatar)
Fifth	2003	Cancun (Mexico)
Sixth	2005	Hong Kong
Seventh	2009	Geneva
Eighth	2011	Geneva
Ninth	2013	Bali (Indonesia)
Tenth	2015	Nairobi (Kenya)
Eleventh	2017	Buenos Aires (Argentina)
Twelfth	2021	Nur-Sultan(Kazakhstan Scheduled)

Asian Development Bank (ADB)

It was established in December, 1966 with the aim to accelerate economic and social development in Asia and Pacific region. It is headquartered at Manila, Philippines.

DEMOGRAPHY

Demographics

- First synochronised census in India took place in 1881. Since 1881, it has been taking place after every decade.
- Census 2011 is the 15th Census and 7th after Independence.
- The slogan of census 2011 is **Our Census, Our Future**.
- India was the first country to adopt family planning in world.
- According to the census 2011, there are 50 million plus cities in India as compared to 35 in census 2001.

Total Population	121,08,54,977
Male	623724248 (51.54%)
Females	586469174 (48.46%)
Density	382 per sq km
Adult Sex Ratio	943
Child Sex Ratio	914

Population Trend in India

- **1891-1921** Period of stagnant population
- **1921-1951** Period of steady growth
- **1951-1981** Period of high growth
- **1981-2011** Period of declining rate
- The year 1921 is known as the Year of **Great Divide.**

National Population Policy, 2000

This policy outlined the following objectives to be achieved

- To lower down the Total Fertility Rate (TFR) to achieve replacement level by 2010.
- Population stabilisation by 2045.
- Reduce MMR (Maternal Mortality Rate) to below 100 per 100000 births.
- Reduce IMR (Infant Mortality Rate) to below 30 per thousand live births.
- Making school education compulsory.
- Promote delayed marriage of girls.
- Promote and control communicable diseases.

(in Population)

Top States/UTs		*Bottom States/UTs*	
Uttar Pradesh	199281477	Lakshadweep	64429
Maharashtra	112372972	Daman and Diu	442911
Bihar	103804637	Dadra and Nagar Haveli	342853
West Bengal	91347736	Andaman and Nicobar Islands	379944
Andhra Pradesh	84665533	Sikkim	607688

(2001-11)

	2001 (%)	*2011 (%)*	*Difference*
Persons	64.83	73.0	9.2
Males	75.26	82.14	6.9
Females	53.67	65.46	11.8

Top 2 Districts		*Bottom 2 Districts*	
	As Per Population		
Thane (Maharashtra)	11054131	Dibang Valley (Arunachal Pradesh)	7948
North Twenty Four Paraganas (West Bangal)	10082852	Anjaw (Arunachal Pradesh)	21089
	As Per Sex Ratio		
Mahe (Puducherry)	1176	Daman (Daman and Diu)	533
Almora (Uttarakhand)	1142	Leh (Ladakh, J and K)	583

	2001		*2011*	
	Population (in mn)	*Proportion* (in %)	*Population* (in mn)	*Proportion* (in %)
▪ Males	532.2	51.74	623.7	51.51
▪ Females	496.5	48.26	586.4	48.46
Adult Sex Ratio	**933**		**943**	
▪ Males	85.0	51.89	82.9	52.24
▪ Females	78.8	48.11	75.8	47.76
Child Sex Ratio	**927**		**914**	

State/UT	*Total Population*	*Males*	*Females*	*Sex Ratio*	*Density*	*Decadal Growth Rate*
State						
Andhra Pradesh	84665533	42509881	42155652	992	308	11.10
Arunachal Pradesh	1382611	720232	662379	920	17	25.92
Assam	31169272	15954927	15214345	954	397	16.93
Bihar	103804637	54185347	49619290	916	1102	25.07
Chhattisgarh	25540196	12827915	12712281	991	189	22.59
Goa	1457723	740711	717012	968	394	8.17
Gujarat	60383628	31482282	28901346	918	308	19.17
Haryana	25353081	13505130	11847951	877	573	19.90
Himachal Pradesh	6856509	3473892	3382617	974	123	12.81
Jammu and Kashmir	12548926	6665561	5883365	883	124	23.71
Jharkhand	32966238	16931688	16034550	947	414	22.34
Karnataka	61130704	31057742	30072962	968	319	15.67
Kerala	33387677	16021290	17366387	1084	859	4.86
Madhya Pradesh	72597565	37612920	34984645	930	236	20.30
Maharashtra	112372972	58361397	54011575	925	365	15.99
Manipur	2721756	1369764	1351992	987	122	18.65
Meghalaya	2964007	1492668	1471339	986	132	27.82
Mizoram	1091014	552339	538675	975	52	22.78
Nagaland	1980602	1025707	954895	931	119	-0.47
Odisha	41947358	21201678	20745680	978	269	13.97
Punjab	27704236	14634819	13069417	893	550	13.73
Rajasthan	68621012	35620086	33000926	926	201	21.44
Sikkim	607688	321661	286027	889	86	12.36
Tamil Nadu	72138958	36158871	35980087	995	555	15.60
Tripura	3671032	1871867	1799165	961	350	14.75
Uttar Pradesh	199581477	104596415	94985062	908	828	20.09
Uttarakhand	10116752	5154178	4962574	963	189	19.17
West Bengal	91347736	46927389	44420347	947	1029	13.93
Chandigarh	1054686	580282	474404	818	9252	17.10
NCT of Delhi	16753235	8976410	7776825	866	11297	20.96
Daman and Diu	242911	150100	92811	618	2169	53.54
Dadra and Nagar Haveli	342853	193178	149675	775	698	55.50
Lakshadweep	64429	33106	31323	946	2013	6.23
Puducherry	1244464	610485	633979	1038	2598	27.72
Andaman and Nicobar Islands	379944	202330	177614	878	46	6.68
India	**121,08,54,977**	**623724248**	**586469174**	**943**	**382**	**17.7**

GLOSSARY

- **Absolute Advantage** (AA) The ability to produce more units of a good or service than some other producer, using the same quantity of resources.
- **Aggregate Demand** (AD) A schedule (or graph) that shows the value of output (real GDP) that would be demanded at different price levels.
- **Aggregate Supply** (AS) A schedule (or graph) that shows the value of output (real GDP) that would be produced at different price levels. In the long run, the schedule shows a constant level of real GDP at all price levels, determined by the economy's productive capacity at full employment. In the short run, the aggregate supply schedule may show different levels of real GDP as the price level changes.
- **Automated Teller Machine** (ATM) A machine that provides cash and performs banking services (for deposits and transfers of funds between accounts) automatically when accessed by customers using plastic cards coded with Personal Indentification Numbers (PINs).
- **Balance of Payments Deficit** An imbalance in a nation's balance of payments where more currency is flowing out of the country than is flowing in. This unequal flow of currency is considered unfavourable and can lead to a loss of foreign currency reserves.
- **Balance of Payments Surplus** An imbalance in a nation's balance of payments in which more currency is flowing into the country than is flowing out. This unequal flow of currency is considered favourable and can lead to an increase in foreign currency reserves.
- **Balance of Trade** The part of a nation's balance of payments accounts that deals only with its imports and exports of goods.
- **Balanced Budget** A financial plan in which income is equal to expenses.
- **Blue Chip Stocks** Stocks in large, nationally known companies that have been profitable for a long time and are well-known and trusted.
- **Command Economy** An economy in which most economic issues of production and distribution are resolved through central planning and controlling.
- **Consumer Price Index** (CPI) A price index that measures the cost of a fixed basket of consumer goods and services and compares the cost of this basket in one time period with its cost in some base period.
- **Cost-Push Inflation** Inflation caused by rising costs of production.
- **Crowding-Out** Increased interest rates and decreased private investment caused by government borrowing.
- **Currency Devaluation** A government adjusts the value of the nation's currency so that it buys less of foreign currencies than before.
- **Current Account** It is a part of a nation's balance of payments accounts; records exports and imports of goods and services, net investment income and transfer payments with other countries.
- **Current Account Balance** It compares a country's net trade in goods and services, plus net earnings, and net transfer payments to and from the rest of the world during a specified period.
- **Demand-Pull Inflation** Inflation caused by increasing demand for output or "too much money chasing too few goods."
- **Depreciation** A reduction in the value of capital goods over time due to their use in production.
- **Depreciation of Currency** A decline in the price of one currency relative to another.
- **Depression** A severe, prolonged economic contraction.
- **Fixed Exchange Rate** An exchange rate that is set and therefore prevented from rising or falling with changes in supply and demand for a nation's currency.
- **Flexible Exchange Rate** An exchange rate that is determined by the international demand for and supply of a nation's money; a rate free to rise or fall (to float).
- **Gilt-edged Market** The gilt-edged market refers to the market for government and semi-government securities, backed by the reserve Bank of India. Government securities are tradable debt instruments issued by the Government for meeting its financial requirements.

- **Hindu Growth Rate** The form was coined by Indian Economist, **Raj Krishna**. The term refers to low annual growth rate of socialist economy of India before 1991, which stagnated around 3.5% from 1950-1980, while per capital income growth averaged 1.3%.
- **Hyperinflation** A very rapid rise in the overall price level.
- **Imperfect Competition** Any market structure in which firms are not price takers, but instead must seek the price and output levels that maximise their profits.
- **Initial Public Offering** (IPO) A company's first sale of stock to the public. When a company 'goes public', it sells blocks of stock shares to an investment firm that specialises in initial offerings of stock and resells them to the public.
- **Market Failures** The systematic overproduction or underproduction of some goods and services that occurs when producers or consumers do not have to bear the full costs of transactions they undertake. Usually related to externalities or the need for public goods.
- **Mixed Economy** It refers to an economy, which is neither fully capitalist *i.e.,* some means of production are in government ownership and some are in private ownership. In reality, all economies are mixed economies where some important decisions are taken by the government and the economic activities are by and large conducted through the market.
- **Monopolistic Competition** A market structure in which slightly differentiated products are sold by a large number of relatively small producers and in which the barriers to new firms entering the market are low.
- **Monopoly** A market structure in which there is a single supplier of a good or service. Also, a firm that is the single supplier of a good or service for which there are no close substitutes; also known as a monopolist.
- **Monopsony** A market situation in which there is only one buyer of a resource. Also, a firm that is the only buyer of a resource; also known as a monopsonist.
- **Mutual Fund** A Mutual Fund is a company that brings together money from many people and invests it in stocks, bonds or other assets. The combined holdings of stocks, bonds or other assets the fund owns are known as its portfolio. Each investor in the fund owns shares, which represent a part of these holdings.
- **Natural Monopoly** An industry in which the advantages of large-scale production make it possible for a single firm to produce the entire output of the market at a lower average cost than a number of firms each producing a smaller quantity.
- **Non-Price Competition** Competition by firms trying to attract customers by methods other than reducing prices; examples include advertising and promotional gifts.
- **Oligopoly** A market structure in which a few, relatively large firms account for all or most of the production or sales of a good or service in a particular market, and where barriers to new firms entering the market are very high.
- **Open Market Operations** The buying and selling of government bonds by the Federal Reserves to control bank reserve and the money supply.
- **Pegged Exchange Rate** An exchange rate that is fixed within a certain range or against a major currency or basket of currencies.
- **Perfect Competition** A market structure in which a large number of relatively small firms produce and sell identical products and in which there are no significant barriers to entry into or exit from the industry.
- **Progressive Tax** A tax that take a larger percentage of income from people in higher-income groups than from people in lower-income ones.
- **Recession** A decline in the rate of national economic activity, usually measured by a decline in real GDP for atleast two consecutive quarters (i.e. six months).
- **Regressive Tax** A tax that takes a larger percentage of income from people in lower-income groups than from higher-income ones.
- **Velocity of Money** The average number of times each dollar is spent on final goods and services in a year.



GENERAL SCIENCE

PHYSICS

UNITS

- Measurement of any physical quantity involves, comparison with a certain basic, arbitrarily chosen, widely accepted reference standard called unit.

SI System of Units

- It is based on the following seven basic units and two supplementary units

Basic Units

Name of Quantity	*Unit*
Length	Metre (m)
Mass	Kilogram (kg)
Time	Second (s)
Electric current	Ampere (A)
Thermodynamic temperature	Kelvin (K)
Luminous intensity	Candela (cd)
Amount of substance	Mole (mol)

Supplementary Units

Name of Quantity	*Name of Unit*
Plane angle	Radian (rad)
Solid angle	Steradian (sr)

Derived Units

- These units are expressed in terms of two or more fundamental units.

Bigger Units

1 light year $= 9.46 \times 10^{15}$ m

1 parsec $= 3.086 \times 10^{16}$ m $= 3.26$ ly

1 AU $= 1.5 \times 10^{11}$ m

1 metric tonne $= 10^3$ kg

1 quintal $= 10^2$ kg

- **Dimensions** All physical quantities can be expressed in terms of powers of M,L,T,θ,A are called dimensions.

PHYSICAL QUANTITIES

Scalar Quantities

- Physical quantities, which have magnitude only and no direction are called scalar quantities.
  e.g. mass, speed, volume, work, time, power, energy etc.

Vector Quantities

- Physical quantities, which have magnitude and direction both and which obey triangle law of vector addition are called vector quantities. e.g. displacement, velocity, acceleration, force, momentum, torque etc.

MOTION

Distance

- The length of the actual path covered by a body is called distance. It is always positive.
  Its SI unit is metre.

Displacement

- The shortest distance between the final and the initial position of an object is called displacement. It may be positive, negative or zero.
- Its SI unit is metre.
- Displacement ≤ Distance

Speed

- Speed is the distance covered by a moving body in unit time interval.
- It is always equal to or greater than the magnitude of the velocity.
- The **average speed** of a particle for a given interval of time is defined as the ratio of total distance travelled to the time taken.

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Time taken}}$$

Velocity

- The rate of change of displacement is called velocity.
- Velocity is a vector quantity, which has both magnitude and direction. Its SI unit is m/s.
- It may be positive or negative.

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time taken}}$$

- If a body is moving on a circular path, then after completing one complete cycle, its average velocity is zero.

Relative Velocity It is the rate of change of displacement of one body relative to that of another and *vice-versa*.

Acceleration

- It is the rate of change of velocity. Its SI unit is m/s^2. It is a vector quantity.
- If velocity of a body is decreasing, then the acceleration is called retardation or deceleration.
- Acceleration of an object is zero, if it is at rest or moving with uniform velocity.

Equation of Motion For a straight line motion with initial velocity (u), final velocity (v) and constant acceleration (a), equations of motion are

(i) $v = u + at$

(ii) $s = ut + \frac{1}{2}at^2$

(iii) $v^2 = u^2 + 2as$

Projectile Motion

- When a particle is so projected that it makes certain angle with horizontal then the motion of the particle is said to be projectile motion.
- Path of projectile is a parabola.

$$\text{Range } (R) = \frac{u^2 \sin 2\theta}{g};\ H_{\max} = \frac{u^2 \sin^2 \theta}{2g};$$

$$\text{Time } (T) = \frac{2u \sin \theta}{g}$$

(θ = angle of projection.)

- To achieve maximum range, the body should be projected at an angle of 45° and it is given by $\frac{u^2}{g}$.
- To achieve maximum height, the body should be projected at angle of 90° and it is given by $\frac{u^2}{2g}$.
- The horizontal range is the same whether the body is projected at θ and $90° - \theta$.

Newton's Laws of Motion

First Law

- "Everybody retains its state of rest or state of motion, until an external force is applied on it." This law is also known as law of inertia or **law of Galileo**.
- Inertia is the virtue, by which a body tries to retain its original state. *Inertia is of two types*
  1. Inertia of rest
  2. Inertia of motion
- A person sitting in a moving car falls forward, when the car stops suddenly.
- Athlete runs some distance, before taking a long jump.

Second Law

- The force applied on a body is equal to the product of mass of the body and the acceleration produced in it, i.e. $F = ma$.
- SI unit of force is Newton (N). It is a vector quantity.

Inertial Frame

- This frame of reference is moving with uniform velocity or is at rest.

Non-Inertial Frame

- Whenever the frame of reference is accelerating or retarding or rotating is called non-inertial frame of reference.

Pseudo Force

- When we study the motion of an object in a non-inertial frame of reference we have to apply a force on the object equal to mass times, the acceleration or retardation of the frame in opposite direction of acceleration or retardation of the frame.
- Bogies of the trains are provided with buffers to avoid severe jerks during shunting of trains. Buffers increase the time-duration of jerks during shunting. This reduces the force with which bogies push or pull each other and severe jerks are avoided.

Momentum

- The product of the mass and the velocity of a body is called the momentum of the body.
  $\therefore$ Momentum = Mass $\times$ Velocity
- It is a vector quantity. Its SI unit is kg-m/s.
- A heavier body has a larger linear momentum than a lighter body moving with the same velocity.
- In the absence of external forces, the total momentum of the system is conserved. It is also known as the principle of conservation of linear momentum.

Conservation of Momentum

- When a man jump from a boat to the shore, the boat slightly moves away from the shore.
- Rocket works on the principle of conservation of momentum.
- When a bullet is fired from a gun, the gun recoils or takes a sharp jerk in backward direction.

Impulse

- If a large force acts on a body for a very short time Δt, then the product of force and time is called the impulse.
- Impulse = Change in momentum
  = Force $\times$ Time interval
- Its SI unit is N-s or kg-m/s.

Concept of Impulse

- A cricketer moves his hands backwards, while catching a ball.
- A person jumping from a height on a 'concrete' floor receives more injury than when jumping on a spongy floor.
- Vehicles like cars, buses and scooters are provided with shockers.

Third Law

- "Every action have equal and opposite reaction." Action and reaction always act on the different bodies and are in opposite directions to each other.
- In a rocket, gases are ejected with a great speed from the rocket in backward direction and rocket is pushed forward.
- While swimming, a person pushes the water backwards (action). The water pushes the swimmer forward with the same force (reaction).

FRICTION

- If we slide or try to slide a body over a surface, the motion is resisted by a bonding between the body and the surface. This resistance is called friction force.

Types of Frictional Forces

(a) **Static Frictional Force** Whenever there is no relative motion between a body and the surface takes place and the body only tries to move when force is applied, Static friction acts on the body.

(b) **Kinetic or Sliding Frictional Force** When there is relative motion between the surfaces occurs.

(c) **Rolling Friction Force** If a body rolls over another body.

Advantages and Disadvantages of Friction

- Walking is possible due to friction.
- The transfer of motion from one part of a machine to other part through belts is possible due to friction.
- Brake works on the basis of friction.
- Friction causes wear and tear of the parts of machinery in contact. Thus, their lifetime reduces.

Methods of Reducing Friction

- By polishing, by lubrication, by proper selection of material, by using ball bearing.

CIRCULAR MOTION

- When an object moves along a circular path, then its motion is called circular motion.

Centripetal Force

- The external force required to maintain the circular motion of the body is called centripetal force. Any of the forces found in nature (such as, gravitational force, magnetic force etc.) may act as a centripetal force.
- Cyclist bends his body towards the centre on a turn while turning.
- Generally, in rain the scooter slips at the turning of a road, because the friction between tyre and road is reduced. Due to this, necessary centripetal force is not provided.

Centrifugal Force

- Centrifugal force is a Pseudo force. It is equal and opposite to centripetal force.
- Cream separator, centrifugal drier work on the principle of centrifugal force.

WORK, ENERGY AND POWER

Work

- When a body is displaced by applying a force on it, then work is said to be done. If a body is displaced by a distance (s) on applying a force (F) on it, then work done $W = F \cdot s = Fs \cos \theta$
  where, θ is the angle between the direction of force and that of displacement.
- It is a scalar quantity. Its unit is joule (J).
- A body falling freely under gravitational pull is an example of positive work by gravitational force.
- If you lift an object to a certain height from the surface of the Earth, the work done by gravitational force is negative.
- If displacement or force is zero, then work done is zero. It **s** and **F** are perpendicular to each other then work done is zero.

Energy

- It is defined as capacity of doing a work. Its SI unit is joule and erg in CGS system.
- **Mechanical Energy** is in two forms; kinetic energy and potential energy.

Kinetic Energy

- It is the energy possessed by a body by virtue of its motion.
- If a body of mass (m) is moving with velocity (v), then kinetic energy

$$\text{KE} = \frac{1}{2} mv^2 = \frac{p^2}{2m} \qquad [p = mv]$$

  where, p is the momentum.
- When momentum is doubled, kinetic energy becomes four times.

- Kinetic energy of air is used to run wind mills.
- Kinetic energy of running water is used to run the water mills.
- A bullet fired from a gun can pierce a target due to its kinetic energy.

Potential Energy

- It is the energy possessed by a body by virtue of its position.
- Suppose a body is raised to a height (h) above the surface of the Earth, then change in potential energy of body $= mgh$.
- When, a body is falling downwards, then its potential energy goes on changing into kinetic energy.
- Potential energy of a spring with spring constant (k), having elongation or compression (x) is $\frac{1}{2}kx^2$.
- The potential energy of the wounded spring of a clock is used to drive the needles of the clock.
- For an isolated system in presence of conservative forces the sum of kinetic and potential energies at any point remains constant throughout the motion. This is known as the conservation of mechanical energy.

Law of Conservation of Energy

- Energy can neither be created nor be destroyed but it can only be transformed from one form to another.

Transformation of Energy

- In a heat engine, heat energy changes into mechanical energy.
- In the electric bulb, the electric energy is converted into light energy.
- In burning coal, oil etc., the chemical energy changes into heat energy.
- In solar cell, solar energy changes into electrical energy.
- In playing sitar, mechanical energy changes into sound energy.
- In microphone, sound energy changes into electrical energy.
- In loud speaker, electrical energy changes into sound energy.
- In electric motor, electrical energy changes into mechanical energy.
- In candle, chemical energy changes into light and heat energy.

Power

- Rate of doing work is called power.

  i.e. $\text{Power} = \frac{\text{Work}}{\text{Time}}$
- SI unit of power is watt (W) named after James Watt.

  $1\ \text{W} = 1\ \text{J/s}$

  $1\ \text{kW} = 10^3\ \text{W}$

  1 horse power = 746 W

  1 kilowatt hour (kWh) = 3.6×10^6 J

Simple Machines

- It is based on moment of force.

  Moment of force

  = Force × Length of arm
- Moment of inertia of a particle is defined by $I = mr^2$ where m is mass and r is the perpendicular distance of the mass from the axis of rotation.
- Kinetic energy associated with a rotating body is $\frac{1}{2} I\omega^2$.

  where, ω is angular speed.
- Lever, inclined plane, screw guage etc., are simple machine.
- Scissors, sea saw, brakes of cycle, hand pump, plass are levers of first kind.
- Nut cracker and waste carrying machine are lever of second kind.
- Tong, man's hand and tiller are lever of third kind.

GRAVITATION

In 1686, Newton stated that in the universe each particle of matter attracts every other particle. This universal attractive force is called gravitational force and the event is known as gravitation.

Gravitational Force

- It is always attractive in nature.
- It is the weakest force, but is a long range force. Mathematically, it is represented as

$$F_G = \frac{GMm}{r^2}$$

where, F_G is gravitational force, G is gravitational constant, M is the mass of first particle, m is the mass of second particle and r is the distance between them.

- This is called Newton's Universal Law of Gravitation.
- The value of G is

  $6.67 \times 10^{-11}\ \text{N-m}^2/\text{kg}^2$.

Gravity

- It is the force, by which Earth attracts a body towards its centre.
- The value of acceleration due to gravity, g at the surface of Earth is $9.8\ \text{m/s}^2$ or $32\ \text{ft/s}^2$.
- Earth is surrounded by an atmosphere of gases due to gravity.
- The value of g on the Moon is 1/6th of that on the Earth.

Variation in *g*

- Value of g decreases with height or depth from Earth's surface.
- g is maximum at poles and g is minimum at equator.
- g decreases due to rotation of Earth.

Satellite

- The heavenly body, which revolves round the planets is called satellite. Moon is a natural satellite of Earth. While INSAT-IB is an artificial satellite of Earth.
- Everybody inside the satellite is in a state of weightlessness.
- A satellite revolving very close to Earth's surface has a period of revolution about 84 min and its speed $\sqrt{gRe}$ is nearly 8 km/s.
- The satellite, whose time period is 24 h, is called **Geostationary Satellite**. It is used to telecast TV programs from one part of the world to another. This satellite revolve around the Earth at a height of 36000 km.
- Polar satellite revolves around the Earth in polar orbits at a height of approximately 800 km. The time period of this satellite is 84 min. These satellite are used for weather forcasting.

- The Earth rotates on its axis from West to East. This rotation makes the Sun and stars appear to be moving across the sky from East to West.

Escape Velocity

- The minimum velocity of the body that should be given to the body which enables it to escape away from Earth's gravitational field is called escape velocity. Its value for Earth's surface is 11.2 km/s.
- The escape velocity at the Moon surface is 2.38 km/s. So, there is no atmosphere around the Moon.
- Escape velocity is $\sqrt{2}$ times the orbital velocity, i.e. $\sqrt{2gRe}$.

Kepler's Laws

- All planets move around the Sun in elliptical orbits having the Sun at one foci of the orbit. The areal speed of a planet around the Sun is constant.
- The square of the period of revolution (T) of any planet around the Sun is directly proportional to the cube of its mean distance (a) from the Sun.

  i.e. $T^2 \propto a^3$

Weight of a Body in a Lift

- If lift is stationary or moving with uniform speed (either upward or downward), the apparent weight of a body is equal to its true weight.
- If lift is going up with an acceleration, the apparent weight of a body is more than the true weight.
- If lift is going down with an acceleration, the apparent weight of a body is less than the true weight.
- If the cord of the lift is broken, it falls freely. In this situation, the weight of a body in the lift becomes zero. This is the situation of weightlessness.
- While going down, if the acceleration of lift is more than acceleration due to gravity, a body in the lift goes in contact of the ceiling of lift.

General Properties of Matter

Elasticity

- Elasticity is the property of material of a body by virtue of which the body acquires its original shape and size after the removal of deforming force.
- Steel is more elastic than rubber.

Stress and Strain

- The internal restoring force acting per unit area of cross-section of the deformed body called stress. Its unit is N/m^2.
- The change occurred in the unit size of the body is called strain. It is unitless.
- The ratio of stress to strain is a constant for the material and is called **modulus of elasticity**.

$$E = \frac{\text{Stress}}{\text{Strain}}$$

- It is also called **Hooke's Law**, which states that within the limit of elasticity, the strain produced in a body is directly proportional to the stress applied on it.
- If strain is longitudinal, then the modulus of elasticity is called **Young's modulus**.

Bulk Modulus (K)

- The ratio of normal stress to volume strain is called Bulk modulus of the material of the body.

Rigidity Modulus (η)

The ratio of tangential stress to shearing strain is called modulus of rigidity.

Density

- The density of a substance (d) is defined as the ratio of its mass (M) to its volume (V).

  i.e. $d = \frac{M}{V}$

  Its unit is $\mathbf{kg/m^3}$.
- Density of water is maximum at 4°C.
- The relative density is defined as the ratio of the density of the substance to the density of water at 4° C.
- Relative density has no unit.
- Relative density is measured by hydrometer.
- When ice floats on water, its $\frac{1}{10}$ th part remain outside the water.
- If ice floating on water in a vessel melts, the level of water in the vessel does not change.
- The density of sea water is more than that of normal water. This explains why it is easier to swim in sea water.

Pressure

- It is defined as force acting normally on unit area of the surface.

  $$\text{Pressure} = \frac{\text{Normal force}}{\text{Area}}$$
- Its unit is N/m^2 also called pascal. It is a scalar quantity.
- Pressure in liquid is given by $p = hdg$, where h is the height, d is the density of liquid and g is the acceleration of gravity.
- In equilibrium condition, pressure is same at two points in the same horizontal level.
- Atmospheric pressure 1 atm

  $= 1.01 \times 10^5$ N/m^2

  = 760 torr or mm of Hg
- Atmospheric pressure decreases with altitude. That is why (i) it is difficult to cook on the mountain, (ii) the fountain pen of a passenger leaks in aeroplane at height.
- Atmospheric pressure is measured by barometer. The slow rise in the barometric reading is the indication of clear weather.
- Sudden fall in barometric reading is the indication of storm.
- Slow fall in barometric reading is the indication of rain.

Pascal's Law

- If an external pressure is applied to an enclosed fluid, it is transmitted undimished to every direction.
- Hydraulic lift, hydraulic press, hydraulic brake works on the Pascal's law.

BUOYANCY

- The upward force exerted by a fluid on the immersed body is called buoyant force or upthrust. The upthrust acts at the centre of gravity of the displaced fluid, which is called the **centre of buoyancy**.
- The centre of gravity of a body is a point, through which the entire weight of the body acts.

Archimedes' Principle

When a solid body is immersed wholly or partially in a liquid, then there is some apparent loss in its weight. This loss in weight is equal to the weight of the liquid displaced by the body.

Law of Floatation

- If density of material of a body is equal to density of liquid, the body floats fully submerged in liquid in neutral equilibrium. When body floats in neutral equilibrium, the weight of the body is equal to the weight of displaced liquid. The centre of gravity of the body and centre of gravity of the displaced liquid should lie on one vertical line.

Surface Tension (T)

- It is the force (F) acting normally on unit length (l) of imaginary line drawn on the surface of liquid.

  i.e. $T = \frac{F}{l}$.

- Its unit is N/m.
- The surface tension decreases with rise in temperature and becomes zero at the critical temperature.

- **Cohesive Force** It is the intermolecular force of attraction acting between the molecules of same substance.
- **Adhesive Force** It is intermolecular force of attraction acting between the molecules of different substance.

Some Phenomena based on Surface Tension

- Small drops of **mercury** are spherical, while large ones are flat.
- Formation of **lead shots.**
- Bigger bubbles can be formed from the soap solution than from water.
- Detergent helps in cleaning the clothes.
- Floatation of needle on water.

Capillarity

- **Capillary Tube** It is a long glass tube of very (fine bore) small radius (in mm order).
- The phenomenon a of rise or depression of liquids in a capillary tube is called capillarity.

Some Practical Examples of Capillarity

- The kerosene oil in a lantern and the melted wax in a candle, rise within the capillaries formed in the cotton wick.
- Writing nib is split in the middle, so that a fine capillary is formed in it. When it is dipped in ink, the ink rises in the capillary.

VISCOSITY

- It is the property of the liquid by virtue of which it opposes the relative motion between its adjacent layers.
- Viscosity of an ideal fluid is zero.
- Fluids are the materials, which can flow.
- Viscosity is the property of liquids and gases both. With rise in temperature, viscosity of liquids decreases and that for gases increases.
- Viscosity of gases is much less than that of liquids. Viscosity of liquid increases with increase in pressure.
- Viscosity of a fluid is measured by its coefficient of viscosity. Its SI unit is decapoise (kg/m-s) or pascal-second.
- Viscous force is proportional to velocity gradient in perpendicular direction between the layers.

Terminal Velocity

- When a body falls in a viscous medium its velocity first increases and finally becomes constant. This constant velocity is called terminal velocity.

HYDRODYNAMICS

Streamline Flow

- In this type of flow, velocity at every point in the fluid remains same.

Turbulent Flow

- The fluid does not maintain constancy of the velocity of its fluid particle.

Critical Velocity

- The velocity of the fluid upto which the flow remains streamline.

Equation of Continuity

- When fluid flow through a tube, the mass of liquid flowing out equals the mass flowing in, holds in all cases.
  Av = constant ; A = area of cross-section, v = velocity of fluid

Bernoulli's Theorem

- When an incompressible and non-viscous liquid (or gas) flows in streamline motion from one place to another, then at every point of its path, the total energy per unit volume (pressure energy + kinetic energy + potential energy) is constant.
- Venturimeter, Pitot tube, Bunsen's burner, atomiser, filter, pump and magnus effect are based upon the Bernoulli's theorem.

Periodic Motion

- Any motion, which repeats itself after regular interval of time is called periodic or harmonic motion.

Oscillatory Motion

- If a particle repeats its motion after a regular time interval about a fixed point, motion is said to be oscillatory or vibratory.

Simple Harmonic Motion (SHM)

- If a body moves to and fro on a straight line about a fixed position, then its motion is called simple harmonic motion.
- When a particle executing SHM passes through the mean position
  (a) no force acts on the particle.
  (b) velocity is maximum.
  (c) acceleration is zero.
  (d) kinetic energy is maximum.
  (e) potential energy is minimum.
- When a particle executing SHM is at the extreme end, then
  (a) velocity of particle is zero.
  (b) acceleration of the particle is maximum.
  (c) kinetic energy of particle is zero.
  (d) potential energy is maximum.
  (e) restoring force acting on particle is maximum.
  (f) In case of spring block system the restoring force $F = -kx$.
  (g) In case of spring block system, time period of oscillation is given by $T = 2\pi\sqrt{\frac{m}{k}}$.

Simple Pendulum

- It is a heavy point mass suspended from a rigid support by means of an elastic inextensible string.
- Time period of simple pendulum, $T = 2\pi\sqrt{\frac{l}{g}}$
  where, l is the length of string of simple pendulum.
- The maximum time period of a simple pendulum is 84.6 min.
- A simple pendulum with time period 2s is known as second pendulum.

- A pendulum clock goes slow in summer and fast in winter.
- If a simple pendulum is suspended in a lift descending down with acceleration, then time period of pendulum will be increase. If lift is ascending, then time period of pendulum will decrease.
- If a lift falling freely under gravity, then the time period of the pendulum is infinite.
- At Moon, the time period of simple pendulum increases, because acceleration due to gravity at Moon is $\frac{g}{6}$.

SOUND WAVES

Waves

- A wave is a disturbance, which propagates energy from one place to the other without the transport of matter. Waves of very short duration are known as pulses.
- *These are of two types*
  (i) Mechanical waves
  (ii) Electromagnetic waves

Mechanical Waves

- The waves, which require material medium (solid, liquid or gas) for their propagation are called mechanical waves or elastic waves.
- *These are of two types*
  (i) Longitudinal waves
  (ii) Transverse waves
- If the particles of the medium vibrate in the direction of propagation of wave, the wave is called longitudinal wave. Waves on springs or sound waves in air are examples of longitudinal waves.
- If the particles of the medium vibrates perpendicular to the direction of propagation of wave, the wave is called **transverse wave**. Waves on strings under tension, waves on the surface of water are examples of transverse waves.
- A wave can be represented by trough and crest or compression and rarefaction. The distance between consecutive trough (crest) or compression (rarefaction) is known as wavelength.
- **Amplitude** Maximum displacement from mean position.
- **Frequency** The number of oscillations in one second.
- **Time Period** Time taken for one oscillation.

Electromagnetic Waves

- The wave, which do not require medium for their propagation, i.e. which can propagate even though the vacuum are called electromagnetic wave. They propagate as transverse wave.
- The wavelength range of electromagnetic wave is 10^{-4} m to 10^{4} m.
- Cathode rays, canal rays, α-rays, β-rays are not electromagnetic waves. Light and heat radiation are examples of electromagnetic waves.

Note Velocity of Wave = Frequency × Wavelength

Sound Waves

- It is longitudinal mechanical waves.
- The longitudinal mechanical waves, which lie in the range 20 Hz to 20000 Hz are called **audible** or **sound waves**. These are sensitive to human ear.
- The longitudinal mechanical waves having frequencies less than 20 Hz are called **infrasonic**. These are produced by earthquakes.
- The longitudinal mechanical waves having frequencies greater than 20000 Hz are called **ultrasonic waves.** Human ear cannot detect these waves.
- Ultrasonic waves are used for sending signals, measuring the depth of sea, cleaning clothes and machinery parts of clocks, removing lamp shoot from chimney of factories and in ultrasonography.
- Musical instruments uses vibration in string. In open organ pipe (or string fixed at both ends) all harmonics will present and for closed organ pipe (string fixed at one end) only odd harmonics will present.

Speed of Sound

- Speed of sound is maximum in solids and minimum in gases.
- When sound enters from one medium to another medium, its speed and wavelength changes, but frequency remains unchanged.
- Speed of sound remains unchanged by the increase or decrease of pressure.
- The speed of sound increases with the increases of temperature of the medium. The speed of sound in air increases by 0.61 m/s, when the temperature is increased by 1°C.
- The speed of sound is more in humid air than in dry air, because the density of humid air is less than the density of dry air.
- The ratio of the speed of a body to the speed of sound in the surrounding medium is called mach number. It is often used with a numeral ''Mach'' to indicate the speed of Sound.

- Mach number $= \frac{\text{Velocity of source}}{\text{Velocity of sound}}$
- When a gun is fired at a visible distance, the sound is heard a little after the smoke is seen, because the velocity of light is much higher than that of sound.

Echo

- The sound waves received after being reflected from a high tower or mountain is called echo.
- To hear echo, the minimum distance between the observer and reflector should be 17 m. Persistence of ear is 1/10 s.
- At the Moon, the echo is not heared due to absence of atmosphere.

Doppler's Effect

- If there is a relative motion between source of sound and observer, the apparent frequency of sound heard by the observer is different from the actual frequency of sound emitted by the source. This phenomenon is called Doppler's effect.
- When the distance between the source and observer decreases, the apparent frequency increases and *vice-versa*.

HEAT

- It is a form of energy, which produces the sensation of warmth in us.
- Its units are calorie (cal), kilocalorie (kcal) or joule (J).

1 cal = 4.18 joule
1 kcal = 1000 cal

Temperature

- It is a form of energy, which flows from hotter to colder body by virtue of temperature difference.
- An instrument used to measure the temperature of a body is called a **thermometer**.
- The normal temperature of a human body is 37° C or 98.4°F.
  – 40° is the temperature, at which Celsius and Fahrenheit thermometers read same.
- Scales of temperature measurement

$$\frac{C}{5} = \frac{F-32}{9} = \frac{R}{4} = \frac{K}{5}$$

- Freezing point of mercury is – 39°C. Hence, to measure temperature below this temperature, alcohol thermometer is used. Freezing point of alcohol is – 115°C.
- **Pyrometer** measures the temperature of a body by measuring the radiation emitted by the body. It cannot measure temperature below 800°C, because at low temperature emission of radiation is very small and cannot be detected.
- Bolometer and thermopile are also used to measure the amount of heat radiated by a body. The temperature of Sun is measured by the Pyrometer.

Range of Different Thermometers

- **Mercury Thermometer** from – 30° C to 350°C.
- **Constant Volume Gas Thermometer** from – 200° C to 500° C (with H_2), below – 200° C upto – 268° C (with He) above 1000° C upto 1600° C (with N_2 gas and bulb of glazed porcelain).
- **Platinum Resistance Thermometer** from – 200° C to 1200° C.
- **Thermocouple Thermometer** from – 200° C to 1600° C.

Expansion of Gases

- When heat is supplied to the gases their temperature, volume and pressure changes.

  Boyle's law, pV = a constant
  (at constant temperature)

  Charle's law, $\frac{V}{T}$ = a constant
  (at constant pressure)

  combining above two laws

$$\frac{pV}{T} = 8.31J\ K^{-1}mol^{-1} = R$$

Thermal Expansion

- It is the increase in size of an object on heating.
- A solid can undergo *three types of expansions*
  1. Linear expansion (expansion in length). $\Delta l = l_0 \alpha\, \Delta\theta$
  2. Superficial expansion (expansion in area). $\Delta A = A_0 \beta \Delta\theta$
  3. Cubical expansion (expansion in volume). $\Delta V = V_0 \gamma \Delta\theta$
  4. The coefficients of the expansions are in the ratio of $\alpha : \beta : \gamma = 1 : 2 : 3$

Note *Almost every liquid expands with the increase in temperature. But when temperature of water is increased from* 0°C *to* 4°C*, its volume decreases, after this volume increases.*

Some Practical Applications

- Telephone wires are given enough gap to allow the wires for contraction in winter.
- A gap is provided between the iron rails of the railway track so that rails can easily expand during summer and do not band.

Specific Heat

- It is the amount of heat required to raise the temperature of a unit mass of the substance of 1°C. Its unit is J/kg/k.
- For a mass, m having specific heat S and temprature difference ΔT, heat required is

$$\Delta Q = mS\Delta T$$

- Specific heat of water is maximum and is 4200 J kg^{-1} K^{-1} (approx)
- Mercury has low specific heat.
- Specific heat of gold is 130 J/kg/k.

Latent Heat

- It is the amount of heat absorbed by a unit mass of substance to change its state without change in temperature. Its unit is joule per kilogram. For a mass (m) having latent heat (L), $Q = mL$.
- Latent heat of fusion of ice is 80 cal/g. Latent heat of vaporisation of water is 536 cal/g.

- Hot water burns are less severe than that steam burns, because steam has high latent heat.
- Ice at 0°C appears colder than water at 0°C, because ice take more heat.

Humidity

- The amount of water vapour in air is called as humidity.
- The ratio of amount of water vapour (m) actually present in a certain volume of air at a given temperature to the amount of water vapour (M) required to saturate it, is called **relative humidity (R_H).**

$$R_H = \frac{m}{M} \times 100\%$$

- Relative humidity is measured by **hygrometer**.
- Relative humidity increases with the increase of temperature.

Transmission of Heat

- Transfer of heat from one system to another or from one part of a system to another part.

Conduction

- In this process, heat is transferred without actual movement of the particles of medium. In this process, path of heat transfer is irregular. In solid, transmission of heat takes place by **conduction process**.

Important Points

- Ebonit, rubber, glass and air are bad conductor of heat.
- Metals are very good conductor of heat.
- Silver is the best conductor of heat.
- Cooking utensils are made up of aluminium, brass and steel, because these substances have low specific heat and high conductivity.
- If the door of refrigerator is kept open, it will not cool the room, it will increases the temperature of the room, because heat rejected by the refrigerator to the room will be more than the heat taken by the refrigerator from the room.
- In pitcher, the water remain cold due to loss of heat because of vapourisation from the pores of the pot.

Convection

- In this process, heat is transferred by the bodily movement of particles of the medium due to difference in densities of different parts of the medium.
- In liquid and gases transmission of heat takes place by convection process. This process is also slow.

Radiation

- It is the quickest way of transmission of heat, in which there is no need of medium for transfer of heat. Heat from the Sun reaches the Earth by radiation (electromagnetic waves). In this process, heat is transferred at the speed of light.

Newton's Law of Cooling

- The rate of loss of heat by a body is directly proportional to the difference in temperature between the body and the surrounding. For the same temperature difference and the fixed temperature of the surrounding, the rate of cooling is greater at high temperature of the cooling object.

Black Body

- Which absorbs completely all the radiations incident upon it.
  Emissive and absorptive power are radiation emitted and absorbed per second per unit area (of the body) respectively.

Kirchhoff's Law

- Kirchhoff's law signifies that good absorbers are good emitter.

Stefan's Law

- Total radiant energy emitted per second from unit surface area of a black body is proportional to the fourth power of its absolute temperature.

THERMODYNAMICS

- Thermodynamics deals with transformation of heat into mechanical work.
- **Zeroth Law of Thermodynamics** If two bodies A and B are separately in thermal equilibrium with third body C, then A and B will be in thermal equilibrium with each other.

- **First Law of Thermodynamics** The amount of heat given to a system is used up in two ways first to increase the internal energy and second to do external work.

LIGHT

- It is the radiation, which makes our eyes able to see the object. Its speed is 3×10^8 m/s.
- It is the form of energy. It is a transverse wave.
- It takes 8 min 19 s to reach on the Earth from the Sun.
- The light reflected from Moon takes 1.28 s to reach the Earth.

Reflection of Light

- The return of light into the same medium after striking a surface is called reflection.

There are two laws of reflection

(i) The angle of incidence is always equal to angle of reflection.

(ii) The incident ray, normal and reflected ray all lie in same plane.

Reflection from Plane Mirror

- Size of image is always equal to size of object.
- If the object is displaced by a distance (a) towards or away from the mirror, then its image will be also displaced by a distance (a) towards or away from the mirror.
- The minimum size of the mirror required to see the full image of an observer is half the height of the observer.
- Focal length of plane mirror is infinity, i.e. power is zero.
- The number of images formed by a plane mirror, $n = \frac{360^\circ}{\theta} - 1$

  when, $\left(\frac{360}{\theta}\right)$ is even number

  where, θ is the angle between two plane mirrors.

  $$n = \frac{360^\circ}{\theta}$$

  when $\left(\frac{360}{\theta}\right)$ is odd number
- If speed of object is v, then speed of image will be 2v with respect to the object for a plane mirror.

Reflection at Spherical Surface

- Spherical mirrors are the mirrors in which reflecting side is spherical.

There are two types of spherical mirrors

1. **Convex Mirror** Reflecting surface is outward
2. **Concave Mirror** Reflecting surface is inward

- Image formed by a convex mirror is always virtual erect and diminished.

Mirror formula is $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Where, u = object distance,
v = image distance,
f = focal length.

Magnification (m)

$$= \frac{\text{Height of image}}{\text{Height of object}} = \frac{-v}{u}$$

Uses of Mirrors

1. Plane mirror as looking glass.
2. Concave mirror as shaving mirrors, used by doctors, shades of table lamp, for search lights.
3. Convex mirror as back view mirrors in vehicles, in street lamps etc.

Refraction of Light

- When a ray of light passes from one medium to other, it bends from its path towards or away from the normal. This phenomenon of bending of light is called as refraction of light. When a ray of light travels from one medium to another, the wavelength and velocity of light changes, but the frequency does not change.

Laws of Refraction (Snell's Law)

- The incident ray, the refracted ray and the normal at the point of incidence, all lie in the same plane.
- The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for a given medium $\frac{\sin i}{\sin r} = {}_1\mu_2 = \frac{\mu_2}{\mu_1}$

  where, ${}_1\mu_2$ is called **refractive index** of second medium with respect to first medium.

- The refractive index of a medium is maximum for violet colour of light and minimum for red colour of light. i.e. refractive index depends slightly on the wavelength. Twinkling of stars is based upon refraction.
- Due to refraction, rivers appear shallow, coin in a beaker filled with water appears raised, pencil in the beaker appear broken.
- At sunset and sunrise, due to refraction, Sun appears above horizon, while it is actually below horizon.

Total Internal Reflection of Light

- When a light ray goes to rarer medium from denser medium, then as we increase the angle of incidence, angle of refraction also increases, so the angle of incidence (in denser medium), for which the angle of refraction becomes 90° (in rarer medium) is called **critical angle**.
- If the angle of incidence in denser medium is greater than critical angle (C), then the ray is reflected back into the same medium, this phenomenon is called **total internal reflection**.
- Sparkling of diamond, mirage and looming, shining of air bubble in water and optical fibre are examples of total internal reflection.

Optical Fibre

- It is a device based on total internal reflection used for telecommunication and various medical purposes like endosurgery.

Lens

Lens is a transparent medium bounded by two curved surfaces.

Lenses are of two types

1. Concave or divergent lens.
2. Convex or convergent lens.

Lens formula is $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Formula for magnification by lens is

$$m = \frac{\text{Length (height) of image}}{\text{Length (height) of object}} = \frac{v}{u}$$

Important Points

- If the lens is immersed in a medium having refractive index more than that of lens, then the nature of the lens changes i.e. convex behaves as concave and *vice-versa*.
- If the lens is immersed in a medium having refractive index equal to that of the lens, the lens behaves as a plane glass plate.
- An air bubble in water behaves as a concave Tens.

Power of Lens

- It is the reciprocal of focal length of lens. $P = \frac{1}{f(m)}$
- Its unit is diopter (D).
- In determination of power (for a lens) in D we assume f in metre.
- Equivalent power for combination of two lenses placed in contact with each other is $P = P_1 + P_2$.
- Equivalent power for combination of two lenses placed d apart $P = P_1 + P_2 - dP_1P_2$.

Optical Instruments

- **Camera** An optical device with which photographs of various objects are taken on the photographic plate.

  Microscope consists of two convex lens of short focal length. It is used to see for details of very small objects.

  Telescope used to see for details of the distant objects. In an astronomical telescope the objective lens is a convex lens of large focal length but eye-piece is a convex lens of short focal length.

Human Eye

- It is an optical instrument just like a photographic camera.
- It forms the real image of the object on retina of the eye.
- For the normal eye, the range of vision is from 25 cm to infinity.

Defect of Eye

Myopia In this case, the person cannot see the distant object clearly. Image is formed before the retina. Concave lens is used for correcting myopia.

It arises due to increase in size of eyeball or/and the increase in thickness of eye lens.

Hypermetropia In this case, the person cannot see near object clearly. And Image is formed behind the retina. Convex lens is used for correcting hypermetropia.

It arises due decrease in size of eyeball and/or decrease in thickness of eye lens.

Astigmatism In this case, the curvature of cornea becomes irregular and image is not clear. Cylindrical lens is used for correcting astigmatism, which are of different radius of curvature. It arises, when different parts of lens have different curvatures and hence the focal lengths along different planes. Its remedy is painless.

Presbyopia In old age the power of accommodation of the eye lens decrease, therefore, neither near nor distant objects are clearly seen.

It can be removed by using bifocal lenses.

Dispersion of Light

- When a ray of white light is passed through a prism, it gets splitted into its constituent colours. This phenomenon is called dispersion of light. The seven colours of band are violet, indigo, blue, green, yellow, orange and red. (VIBGYOR).
- Red, green and blue are called **primary colours or basic colours.**

Scattering of Light

- When light waves fall on extremely small bodies (particles) such as dust particles, water droplets, these are thrown out in all direction this phenomenon is called scattering.
  Scattering of light is maximum in case of violet colour and minimum in case of red colour of light, for which wavelength is greater. Blue colour of sky is due to scattering of light. The brilliant red colour of Sun during rising and setting is due to scattering of light.

Mixing of Colours

- Red + Green + Blue = White
- Red + Blue = Magenta
- Blue + Green = Peacock blue (or Cyan)
- Red + Green = Yellow
- If all the colours of white light are reflected back from the object, then it appears white.
- A rose appears red in a day time because it absorbs all the constituent colours of white light falling on it, except red, which it reflects to us.

Interference of Light Wave

- Interference is the phenomenon of sustained cancellation or reinforcement of two waves when they meet under certain specific conditions. The layer of kerosene oil over water surface appear coloured in the presence of sunlight due to proper interference of light.

Diffraction of Light

- Bending of light round the corner of an opaque obstacle.
- When vibration of electric field is confined in a direction perpendicular to the direction of propagation of the wave is called polarisation.

ELECTRICITY

Charge

- It is something that a body attains when it loses or gains the electrons.
- The positive and negative labels and sign for electric charge were chosen arbitrarily by **Benjamin Franklin**.
- Similar charges repel each other and opposite charges attract each other.
- **Conductors** are those substances, which allow passage of electrical charges through them and have very low electrical resistance. Metals like silver, iron, copper are conductor. Human body and the Earth acts like a conductor. Silver is the best conductor.
- **Insulators** are those substances, which do not allow passage of charge. Rubber, wood, mica, glass, ebonite etc. are insulators.

- There are some materials, which are not a conductor at the normal temperature, but when heated, it acts as a conductor is called semiconductors.

Coulomb's Law

The force of attraction or the force of repulsion acting between the two point charges is proportional to the product of the magnitudes of the two charges and inversely proportional to the square of the distance between them.

Electric Field

- Electric force per unit positive test charge is known as electric field intensity,

$$E = \frac{F}{q_o}$$

Electric Potential

- Electric potential at any point of the electric field is the amount of work done in bringing a unit positive charge from infinity to that point.

Capacitance

- When a conductor is given a charge, its potential (i.e. work done per unit positive test charge) increase proportional to the charge given.
  i.e. $Q \propto V \Rightarrow Q = CV$, the constant C is capacitance. SI unit is coulomb per volt called farad (F).

Gauss law

- Electric flux $\phi = \oint \mathbf{E}.d\mathbf{s}$ through a closed surface is equal to charge enclosed by the surface divided by permittivity of the medium (ε).

Electric Current

- It is amount of charge that flows per second through a cross-sections of a conductor.
- Current is scalar quantity. Its SI unit is ampere (A).

Ohm's Law

- If there is no change in the physical state of a conductor, then the ratio of potential difference across its ends and the current flowing through it is constant

$$\frac{V}{i} = R$$

  R is the **resistance** of the circuit.

Resistance

- The resistance of a conductor is directly proportional to its length (l) and inversely proportional to its cross section area (A) of the conductor.

  i.e. $R \propto \frac{l}{A} \Rightarrow R = \rho \frac{l}{A}$

  ρ is called resistivity
  SI unit at resistance is ohm (Ω)
  conductivity $\sigma = \frac{1}{\rho}$
  SI unit is $\Omega^{-1}m^{-1}$
- On increasing the temperature of the metal, its resistance increases.
- On increasing the temperature of semi conductor, its resistance decreases. Super conductor have its resistance zero at a temperature, which is called the critical temperature for that super-conductor.
- **Specific resistance or resistivity** depends only on the material of conductor and its temperature and not on the dimensions of the material. Resistivity increases with temperature.
- If a wire is stretched its resistance will change, but its specific resistance will remain unaffected.

Important Points

- A lightning conductor is fixed to tall building to protect them from the distractive effect of the lightning.
- The filament of an electric bulb is made of tungsten because it has a high melting point and can be heated to a high temperature to emit light.
- Electric fuse is a small conducting wire of alloy of copper, tin and lead having low melting point. It is connected in series.
- All electrical appliances like bulbs, fans etc., are connected in parallel across the live wires and the neutral wires.

- **Galvanometer** It is a device used to detect and measure electric current in a circuit. It can measure current upto 10^{-6}A. A galvanometer can be converted into a voltmeter by connecting a very high resistance in its series, while converted into an ammeter by connecting a very low resistance in parallel.
- **Ammeter** It is a device, which is used to measure electric current in a circuit. It is connected in series in the circuit. The resistance of an ideal ammeter is zero.
- **Voltmeter** It is a device used to measure the potential difference between two points in a circuit. It is connected in parallel in the circuit. The resistance of an ideal voltmeter is infinite.

Electric Power

- It is the electric work done per unit time, i.e. $P=\frac{W}{t}$. Its SI unit is watt. $1\text{kW}=10^3$ W
- 1 kilowatt hour $= 3.6\times 10^6$ joule.

Electric Cell

- Electric cell is a device which converts chemical energy into electrical energy.
  It is basically of two types
  (i) Primary cell (ii) Secondary cell

Lorentz Force

- Force on a moving charge in magnetic field (**B**) : $\mathbf{F} = q\mathbf{v}\times\mathbf{B}$

Ampere's Law

- It states that for any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the permeability times the electric current enclosed in the loop.

Magnet

- The material, which can attract the magnetic substances (such as cobalt, iron and nickel) is called a magnet and the property of attracting the magnetic substance by a magnet is called magnetism. When poles of two magnets are brought close together, they exert force on each other. This force is called interaction between the poles.
- Similar poles repel each other and dissimilar poles attract each other.
- The area surrounding the magnet in which another magnet experience a force on it is called **magnetic field**. The unit of magnetic field newton/ ampere-metre or weber/ metre^2 or tesla. The Earth is like a magnet situated at its centre.

Electromagnetic Induction

- The production of emf due to change of magnetic flux is called induction.
- The principle of electromagnetic induction is given by Faraday in 1831. i.e. the current is induced in such a way that it opposes the cause.
- The direction of current is determined by the Lenz's law (i.e. the current is induced in such a way that it opposes the cause). The value of induced emf will be given by rate of change of magnetic flux.

Important Points

- Transformer is a device, which converts low voltage AC into high voltage AC and high voltage AC into low voltage AC. It is based on electromagnetic induction. During the operation of transformer, electric power is theoretically constant.
- Permanent magnets are made of steel, cobalt steel, ticonal, alcomax and alnico. Electromagnets, cores of transformers, telephonediaphragms, armatures of dynamos and motors are made of soft iron, mu-metal and stalloy.
- AC Dynamo (or generator) is a device which is used to convert mechanical energy into electrical energy. It works on the principle of electromagnetic induction. Microphone converts sound energy into electrical energy and works on the principle of electromagnetic induction.

ATOMIC AND NUCLEAR PHYSICS

Cathode Rays

- Cathode rays are the stream of high speed negatively charged particles moving from cathode to anode in a discharge tube.
- Cathode rays are used in cathode ray oscilloscope and in the production of X-rays.

Properties of Cathode Rays

1. Cathode rays travel in straight line.
2. Cathode rays can produce X-rays.
3. Cathode rays can produce fluorescence.
4. Cathode rays are deflected in magnetic field and electric field.

Anode Rays

- Anode rays are the stream of positive charged particles. It is used to produce flashes of light on zinc sulphide screen.

Properties of Anode Rays

1. Anode rays travel in straight line and cast shadow.
2. These rays posses kinetic energy.
3. The rays produces flashes of light on zinc sulphide screen.
4. These rays are deflected by electric and magnetic field.

X-rays

- These rays were discovered by Rontgen. These rays are electromagnetic in nature.

Properties of X-rays

- X-rays travels in straight line.
- Speed of X-rays is equal to speed of light. These are not deflected by electric and magnetic fields.
- X-rays produce illumination on falling on fluorescent substances.
- X-rays shows photoelectric effect.
- X-rays are used in surgery, radio-therapy, engineering department and searching.
- The intensity and the penetrating power of X-rays can be controlled independently.

Photoelectric Cell

- It is a device based on phenomena of photoelectric effect, which converts light energy directly into electric energy.
- Photoelectric effect is based on the law of conservation of energy.
- Photoelectric cells are used in automatic door, automatic switches etc.

RADIOACTIVITY

- **Henry Becqueral**, **Madame Curie** and **Pierre Curie** discovered the phenomenon of radioactivity in 1896 and for this they jointly won Nobel Prize.The rays emitted by radioactivity were first recognised by Rutherford.
- Radioactivity is a nuclear phenomenon. It is spontaneous emission of radiation from the nucleus. Becqueral is the SI unit for radioactive disintegration, which means 1 disintegration per second. The nucleus having 83 protons or more are unstable. They emit α, β and γ-rays and become stable. The elements of such nucleus are called **radioactive elements** and the phenomenon of emission of α, β and γ-rays is called radioactivity.
- The penetrating power for α-particle is minimum and for γ-rays is maximum. The end product of all natural radioactive element after emission of radioactive rays is lead.
- **Nuclear Reactor** It is a device, wherein the nuclear energy produced is utilised for constructive purpose, e.g. for producing electricity. It is based on controlled nuclear chain reaction. A fissionable material (U^{233}, U^{235}, Pu^{239} etc.) are used as nuclear fuel for the fission process to take place. To start the nuclear reaction, the cadmium rods are slowly removed and to stop, they are inserted.

Fast Breeder Reactor

A nuclear reactor, which can produce more fissile fuel than it consumes is called a fast breeder reactor.

Semiconductor

- The substance in which electric conduction is not possible at a low temperature but on increasing the temperature electric conduction is possible.
- ***p-n* Junction** When *p*-type semiconductor brought into close contact of *n*-type semiconductor is called a *p-n* junction.
- **Transistor** It is a combination of two *p-n* junction in series.

Note *The first nuclear reactor was made by Fermi, who used* U^{235} *as fuel and Apsara was the first Indian nuclear reactor.*

Mass-Energy Relation

- Albert Einstein establised a relation between mass and energy on the basis of special theory of relativity in 1905. According to this, mass can be converted into energy and *vice-versa*.

  i.e. $E = mc^2$

  where, c is the velocity of light and E is the energy equivalent of mass m.

Fluorescence and Phosphorescence

- Fluorescence is the phenomena of emission of light of low frequency from a substance when some light from a source is incident on it. While in phosphorescence, the substance can emit light for some time even after the source is removed. The inner surface of the fluorescent take giving white light is painted with magnesium, tungsten and zinc berillium.
- Zinc sulphide exhibit the phenomena of phosphore- scence.

Important Laws/Theories and their Scientist

Laws/Theories	*Scientist*	*Laws/Theories*	*Scientist*
Gravitation Law, Law of Motion	Newton	Kinetic Theory of Temperature	Kelvin
Theory of Relativity	A Einstein	Theory of an Atom	Dalton
Discovery of X-ray	WC Rontgen	Laws of Electrolysis	M Faraday
Principle of Lever, Relative Density	Archimedes	Script of Writing and Reading for Blind Persons	Louis Braille

Units of Measurement

Quantity	*Unit* (SI)	*Quantity*	*Unit* (SI)
Area	Square metre	Viscosity	Poise
Volume	Cubic metre	Surface tension	Newton/metre
Velocity	Metre/second	Heat	Joule
Acceleration	Metre/second square	Absolute temperature	Kelvin
Density	Kilogram/metre cube	Resistance	Ohm
Momentum	Kilogram metre/second	Electromotive force	Volt
Work	Joule	Electrical conductivity	$\frac{1}{\Omega - m}$ = mho/meter
Energy	Joule	Electric energy	Kilo watt hour
Force	Newton	Electric power	Kilo watt or watt
Pressure	Pascal or Newton/sq. metre	Magnetic intensity	Ampere/meter
Frequency	Hertz	Charge	Coulomb
Power	Watt	Magnetic induction	Gauss
Weight	Newton or Kilogram	Intensity of sound	Decibel
Impulse	Newton-second	Power of lens	Dioptre
Angular velocity	Radian /second	Depth of sea	Fathom

CHEMISTRY

- Chemistry, is the study of the composition, properties and behaviour of matter. Lavoisier is known as the 'Father of modern chemistry'.

IMPORTANT CONCEPTS OF CHEMISTRY

Matter

- Matter is anything which occupies space and has mass.
- It exists in five phases, *viz*, solid, liquid, gas, plasma, Bose Einstein Condensate (BEC). The former three are commonly seen and among the last two, plasma is in the form of ionised gases containing super energetic and super excited particles, while BEC is extremely low density gas at extremely low temperatures.

Particles of Matter

Atom

- It is the smallest particle of matter that takes part in chemical reactions. (by Dalton's Atomic Theory).
- It can neither be created nor destroyed (law of conservation of mass given by Lavoisier). e.g., iron (Fe), gold (Au), silver (Ag) etc.

Molecules

- These are the smallest particle of a substance (element or compound) which has properties of that substance and can exist in the free state.
- These are formed by the joining of two or more atoms in fixed ratio (law of multiple proportions given by Dalton). e.g., water (H_2O), ammonia (NH_3), carbon dioxide (CO_2) etc.

States of Matter

- **Solids** have definite shape, size and volume. e.g. book, paper, wood, etc., while **liquids** have definite volume but no definite shape. e.g., water, milk, etc. And the **gases** have neither definite shape nor definite volume e.g., oxygen, hydrogen etc.
- Each of these states are inter-convertible.

Classification of Matter

Elements

- An element is a substance which cannot be split up into two or more simpler substances by the usual chemical methods of applying heat, light or electrical energy.
- These contain only single type of atoms. Elements known at present are 118, out of which 94 are natural.
- Elements which are liquid at room temperature are mercury (Hg) and bromine (Br_2).
- Elements have the following order of abundance in earth crust : oxygen > silicon > aluminium (metal) > iron > calcium.
- Elements have the following order of abundance in human body : oxygen > carbon > hydrogen > nitrogen.

Ununseptium

- Recently discovered, a superheavy chemical element with atomic number 117, is a member of group-17 in the periodic table below the five halogens (fluorine, chlorine, bromine, iodine and astatine). Its synthesis was claimed in Dubna, Russia by a joint Russian-American collaboration.
- In 2014, the GSI Helmholtz Centre for Heavy Ion Research in Germany also claimed to have successfully repeated original experiment. Ununseptium is a temporary systematic name that is intended to be used before a permanent one is commonly called 'element-117' instead of ununseptium.

Compounds

- These contain more than one kind of atoms. These cannot be separated into constituent atoms by simple physical methods e.g., Silica (SiO_2), water (H_2O) etc.

Mixtures

- These are obtained by mixing two or more substances in any proportion.
- Mixtures can be homogeneous, i.e., have uniform composition throughout (e.g., salt solution, sugar solution, air, etc) or heterogeneous, i.e., have non-uniform composition (e.g., mixture of salt and sugar, colloidal solutions etc).

Boiling Point

- It is a temperature at which vapour pressure of a liquid becomes equal to atmospheric pressure.
- It usually decreases at high altitudes, that's why at high altitudes, the boiling point of water is less than 100°C and more time is required to cook a food.
- Boiling point of water in pressure cooker is high due to high pressure and hence, less time is required to cook the food.
- Boiling point increases in the presence of impurity.

Separation Methods

Evaporation

- It is the process of conversion of a liquid into vapours at any temperature below its boiling point.
- It produces cooling effect. That's why we feel cool when some nail polish remover or spirit is kept on our palm.
- **Sublimation** In this process, a solid substances convert directly into its vapours on application of heat.
- **Crystallization** This method is mostly used for separation and purification of solid substances.
- **Fractional distillation** This process is similar to the distillation process except that a fractionating column is used to separate two or more volatile liquid which have different boiling points.

Physical Change

- It is the change which only affects the physical properties like colour, hardness, density, melting point etc. of matter. e.g., sublimation, boiling, vaporisation, cutting of trees, burning of candle etc.

Chemical Change

- These affect the composition as well as chemical properties of matter and result in the formation of a new substance, e.g., burning of fuel, burning of candle, electrolysis of water, burning of paper, photosynthesis, ripening of fruits etc.

SOLUTIONS

True Solutions

- These are homogeneous mixtures of two or more substances. e.g., sugar solution, tincture of iodine (i.e., solution of iodine in alcohol), aerated drinks (like lemonade, soda water), air, alloys etc.
- The substance which is dissolved in a liquid to make a solution is called solute and the liquid in which **solute** is dissolved is called **solvent**.

Colloidal Solutions

- These are heterogeneous mixtures in which the size of solute particles is between those in true solution and those in suspension. These are of following types on the basis of dispersed phase and dispersion medium.
- Colloidal solutions can scatter light because of the presence of large solute particles, i.e., these show **Tyndall effect** and **Brownian movement**. These are separated by special techniques like centrifugation.
- Colloidal solutions are coagulated by adding an electrolyte.
- Colloidal solutions are purified by dialysis, which is also used in the purification of blood with the help of artificial kidney machine.
- Alum is used to purify water as it coagulates (settle down) the impurities present in the water.
- $FeCl_3$ is used to stop bleeding as it coagulates the blood. Reverse osmosis is used for desalination of sea water.
- The concentration of a solution is defined as the mass of solute in grams present in 100 g of the solution.

Blue colour of sky is also due to scattering of light by dust particles suspended in air.

Dispersed Phase	*Dispersion Medium*	*Type of Colloid*	*Examples*
Liquid	Gas	Aerosol	Fog, clouds, mist
Solid	Gas	Aerosol (solid)	Smoke, automobile exhausts
Gas	Liquid	Foam	Shaving cream
Liquid	Liquid	Emulsion	Milk, butter, face-cream.
Solid	Liquid	Sol	Milk of magnesia, mud,
Gas	Solid	Solid foam	Foam, rubber, sponge, pumice
Solid	Liquid	Gel	Jelly, cheese, butter
Solid	Solid	Solid sol	Coloured gem stone, milky glass

Suspensions

- A suspension is a heterogeneous mixture in which the small particles of a solid are spread throughout a liquid without dissolving in it. Examples chalk-water mixture, muddy water, milk of magnesia, flour in water etc.
- Milk of magnesia is a suspension of magnesium hydroxide in water.
- Normality, molality and molarity are different ways of expressing concentration. Out of these, molality remains unaffected by change in volume.

ATOMIC STRUCTURE AND RADIOACTIVITY

Constituents of Atom

- Atoms are made up of three subatomic particles, these are electrons, protons and neutrons.
- Atom contains a central nucleus (having positively charged protons and neutral neutrons) surrounded by a cloud of negatively charged electrons.
- Besides other subatomic particles of atom also have positron, meson, neutrino, antiprotons.

- Hydrogen is the only atom in which neutrons are not present.
- Atom having same number of electrons and protons is neutral. If electrons are less than proton, the atom carries positive charge and if electrons are more than proton, the atom carries negative charge.
- Planck's gave quantum theory. He gave the following relation : $E = h\nu$
  where, E = energy
  h = Planc's constant
  ν = frequency (= 6.620×10^{-34} Js),
- Dual nature of matter was given by de-Broglie.

CHARACTERISTICS OF AN ATOM

Atomic Number (Z)

- It is equal to the number of protons.
- It is written as a subscript to the left of the symbol of the atom, e.g., ${}_6C$, here 6 is the atomic number of carbon (C).

Mass Number (A)

- It is equal to the sum of number of protons and number of neutrons.
- It is written as a superscript to the right of the symbol of the atom, e.g., C^{12} here 12 is the mass number of carbon (C).

Mass number = Number of protons + Number of neutrons = Number of electrons + number of neutrons (in case of neutral atom)

ATOMIC SPECIES

Isotopes

- These have same atomic number but different mass number. Examples of isotopes are hydrogen, i.e., ${}_1H^1$, (protium), ${}_1H^2$ or D (deuterium) and ${}_1H^3$ or T (tritium). T is radioactive.

Isobars

- These have same mass numbers but different atomic numbers, e.g., ${}_{18}Ar^{40}$ and ${}_{20}Ca^{40}$ are isobars.

RADIOACTIVITY

- It was discovered by **Henry Becquerel** but term radioactivity was given by **Madam Curie.**
- It is the process of spontaneous disintegration of nucleus and is measured by Geiger counter.

Particle	*Discovery*	*Charge* (C)	*Mass* (kg)	*Position*
Electron ($_1e^0$)	JJ Thomson	-1.6×10^{-19}	9.1×10^{-31}	Around the nucleus
Proton ($_1p^1$)	Goldstein	$+1.6\times10^{-19}$	1.65×10^{-27}	Inside nucleus
Neutron ($_0n^1$)	Chadwick	0	1.67×10^{-27}	Inside nucleus

- It remains unaffected by external factors like temperature, pressure etc. It involves emission of α, β and γ rays/particles and has units Curie, Becquerel, Rutherford.

Alpha (α) Particles

- These are positively charged helium nuclei ($_2He^4)^{2+}$.
- These have +2 unit charge and 4 unit mass.
- They have low penetrating power but very high ionising power and kinetic energy.

Beta (β) Particles

- These are negatively charged electrons ($_{-1}e^0$) and have – 1 unit charge and 0 mass. These are more dangerous than α-rays. These have high penetrating power as compared to α-rays.

Gamma (γ) particles

- These are the electromagnetic radiation and have very high penetrating power. These have low ionising power and kinetic energy.

Nuclear Fission

- It is a process in which a heavy nucleus is broken down into two or more lighter fragments.
- It is used in nuclear reactor and atom bomb.
- Nuclear reactor is a device that is used to produce electricity and permits a controlled chain nuclear fission.
- It contains fuels e.g., $_{92}U^{235}$, moderator e.g., graphite and heavy water, D_2O to slow down neutrons and control rods (made up of boron steel or cadmium) to absorb neutrons. It may also contain liquid sodium as coolant.

Nuclear Fusion

- It is a process which involves fusion of two or more lighter nuclei to give a heavier nuclei.
- It is used in hydrogen bomb. Energy of Sun is also a resultant of a series of nuclear fusion reaction.

Uses of Radioisotopes

1. **Iodine-131** is employed to study the structure and activity of thyroid gland. It is also used in internal radiation therapy for the treatment of thyroid disease.
2. **Iodine-123** is used in brain imaging.
3. **Cobalt-60** is used in external radiation therapy for the treatment of cancer.
4. **Sodium-24** is injected along with salt solution to trace the flow of blood.
5. **Phosphorus-32** is used for leukemia therapy.
6. **Carbon-14** is used to study the kinetics of photosynthesis.

- Radiocarbon dating is used in determining the age of carbon bearing materials such as wood, animal fossils etc.
- Uranium dating is used to determine the age of Earth, minerals and rocks.

CHEMICAL BONDING AND CHEMICAL REACTIONS

Electrovalent Bond or Ionic Bond

It is formed between a cation, which is usually a metal and an anion, which is usually a non-metal e.g., salt (NaCl), lime (CaO) etc. Compounds containing ionic bond can conduct electricity in molten state or in solution but not as a solid.

Covalent Bond

- It is formed by sharing of pairs of electrons between atoms.
- Covalent compounds
  (a) are bad conductors of electricity.
  (b) are soluble in organic solvents like benzene, acetone, ether etc.
  (c) have low melting and boiling points (diamond and graphite have very high melting points)

van der Waals' Interaction/Force

- These are relatively weak as compared to normal chemical bonds.
- The ability of geckos lizard which can hang on a glass surface using only one toe to climb on sheer surfaces has been attributed to the vander Waals' forces between these surfaces and spatulae or microscopic projections, which cover the hair-like setae found on their footpads.

TYPES OF REACTIONS

Exothermic Reactions

- These are the reactions in which energy is released in the form of light or heat. e.g., burning of natural gas, respiration, decomposition of vegetable matter into compost, combustion reactions etc.

Endothermic Reactions

- These are the reactions in which energy is consumed, e.g., digestion.

Redox Reactions

- These are the reactions in which oxidation and reduction occurs simultaneously.

Oxidation

- It involves addition of oxygen or any other electronegative element like fluorine (F), chlorine (Cl), nitrogen (N) etc. It involves removal of hydrogen (H) or any other electropositive element like sodium (Na). It involves loss of electrons, i.e., increase in the positive charge of ion.

Reduction

- It involves addition of hydrogen or any other electropositive element. It involves removal of oxygen or any other electronegative element. It involves gain of electron, i.e., decrease in oxidation state.

Catalysis

- It was discovered by Berzelius.
- It is a term used for the reactions/ processes which occur in the presence of certain substances that increase the rate of the reaction without being consumed. Such substances are called catalysts.

ACIDS, BASES AND SALTS

Acids

- These are the substances which have sour taste and turns blue litmus red.
- These are the substances which gives H^+ ions in their aqueous solutions (Arrhenius concept) e.g., HCl.
- In aqueous solution, these are good conductor of electricity.

Acid	*Source*
Citric acid	Lemon, orange, grapes
Maleic acid	Unripe apple
Tartaric acid	Tamarind
Acetic acid	Vinegar
Lactic acid	Milk
Hydrochloric acid	Stomach
Oxalic acid	Tomato

- Generally most of the acids contain hydrogen.
- Pickles are always kept in glass jar because acid present in them reacts with the metal of metallic pot.

S.No.	Process	Catalysts
1.	Manufacture of ghee from vegetable oils	Nickel
2.	Contact process for the manufacture of sulphuric acid	Pt powder
3.	Decon's process for the manufacture of chlorine	Cupric chloride
4.	Conversion of proteins into peptide	Pepsin enzyme
5.	Conversion of glucose into ethyl alcohol	Zymase enzyme
6.	Conversion of starch into maltose	Diastase enzyme
7.	Formation of vinegar from sugar cane	Micoderma aceti
8.	Conversion of sucrose into glucose and fructose	Invertase enzyme
9.	Conversion of milk into curd	Lactase (Lactobacilli)

Aqua-regia

- It is a mixture of concentrated hydrochloric acid (HCl) and concentrated nitric acid (HNO_3) in the ratio of 3 : 1. It is used to dissolve noble metals like gold and platinum.

BASES

- These are the substances which have bitter taste and turn red litmus blue. Bases give OH^- ions in their aqueous solutions (Arrhenius concept), e.g., NaOH, KOH, $Mg(OH)_2$ etc.
  Water soluble bases are called alkali, *e.g.*, NaOH, KOH.

pH VALUE

- It is a measure of acidity or basicity of a solution. pH scale is used to measure pH value. It consists of value from 0 to 14.
- It is 7 for neutral solution, greater than 7 for basic solution and less than 7 for acidic solution.

Substance	pH	Substance	pH
Gastric juice	1.0-3.0	Rain water	6.0
Soft drinks	2.0-4.0	Tears	7.4
Lemon	2.2-2.4	Sea water	8.5
Vinegar	2-3	Milk of magnesia	10.5
Urine (human)	4.8-8.4	Milk (cow)	6.3-6.6
Saliva (human)	6.5-7.5	Blood plasma (human)	7.30-7.42

BUFFER SOLUTIONS

- These are the solutions in which pH value is almost maintained on adding a little amount of an acid or a base to it.

SALTS

- These are the products of neutralisation reaction between an acid and a base e.g., Nacl.

Washing Soda

- It is chemically sodium carbonate decahydrate ($Na_2CO_3 \cdot 10H_2O$) and is used for removing permanent hardness of water.

- Magnesium hydroxide is used as a remedy for acidity.
- Ant or bee sting contains methanoic or formic acid due to which victim feels pain and irritation. Use of a mild base like baking soda is a remedy for it.

Baking Soda

- It is sodium bicarbonate ($NaHCO_3$).
- When mixed with a mild edible acid such as tartaric acid it is called baking powder and is used to make bread or cake soft and spongy.
- It is used as antacid and in soda-acid fire extinguishers.

Bleaching Powder

- It is calcium oxychloride $CaOCl_2$ and it is used for disinfecting drinking water. It is a mixed salt while alum is a double salt.

Plaster of Paris

- It is chemically calcium sulphate hemihydrate $\left(CaSO_4 \cdot \frac{1}{2} H_2O\right)$ and obtained by heating gypsum $(CaSO_4 \cdot 2H_2O)$. On mixing with water, it changes into gypsum.
- It is used to plaster the fractured bones, for making toys, materials for decoration and for making surfaces smooth.

Copper Sulphate

- Copper sulphate when anhydrous, is white and when associated with water of crystallisation (i.e ., $CuSO_4 \cdot 5H_2O$), is blue, so it is called blue vitriol. It is used to test the presence of water.

Potassium Nitrate

- It is used as a fertilizer, in gun powder ($C + S + KNO_3$), in matchsticks etc.

METALS, NON-METALS AND METALLURGY

Metals

- These are the elements which are hard, lustrous, ductile, malleable, sonorous and conductor of heat and electricity in their solid as well as molten state.
- Highly reactive metals displace the less reactive metals from their salts. The order of reactivity is : potassium (K) > sodium (Na) > calcium (Ca) > magnesium (Mg) > aluminium (Al) > zinc (Zn) > iron (Fe) > lead (Pb) > hydrogen (H) > copper (Cu) > mercury (Hg) > silver (Ag) > gold (Au) (Thus, gold is less reactive metal).

- When iron nails are kept in copper sulphate solution (blue), iron being more reactive displaces the copper from copper sulphate soloution and thus, the blue colour of solution disappears.
- Sodium and potassium are soft and highly reactive metals. These react with air and water. That's why these are kept in kerosene oil.
- Sodium and potassium burn in water, while calcium floats over it.
- Copper(Cu) is the first metal used by man.
- Pb (lead) is a bad conductor of electricity.
- Tin is called strategic metal.

Non-Metals

- These may be solid, liquid or gas (bromine is the only liquid non-metal).
- These form oxide with oxygen which are generally acidic. The order of hardness of some substances is diamond > corundum > topaz > quartz

Helium

- It is used for filling balloons and other lighter aircraft.
- He when mixed with O_2, is used by deep-sea divers for breathing and for respiratory patients.
- Gas ballooon contains helium (He) gas and neon sign bulb contains neon (Ne) gas.

Argon

- It is used to generate inert atmosphere for welding and to fill incandescent light bulbs.
  - A mixture of mercury vapours and argon gas is filled in tube lights.
  - Xenon is called stranger gas.

SOME IMPORTANT COMPOUNDS

Ferric oxide (Fe_2O_3) is used in jeweller rough.

Silver nitrate ($AgNO_3$) is used to prepare the voter's ink.

Silver iodide (AgI) is used for artificial rain.

Hydrogen peroxide (H_2O_2) is used as an oxidising agent, bleaching agent, as an insecticide and for washing old oil paintings.

METALLOIDS

- These have properties of metals as well as non-metals, e.g., arsenic, antimony, germanium, boron, silicon, tellurium.
- Inert gas : An inert gas can be detined as a gas which does not undergo chmical reaction under a set of given conditions.

METALLURGY

- It is the process of extraction and refining of metals from their mineral forms called ores.
- In metallurgy, firstly the impurities of ore are removed. Later on the concentration of ore is increased and it is then concentrated into its oxide form which further get reduced to free metal. This free metal is then, purified by several methods like poling, vapour-phase refining, zone-refining etc.
- In electrolytic refining, anode is made up of impure metal and cathode is made by thin strip of pure metal.

Minerals and Ores

- Minerals are the substance in the form of which metal is found in nature.
- The minerals from which metal can be obtained conveniently and beneficially, are called ores.

Note All ores are minerals but all minerals are not ores.

Alloys

- These are mixtures of two metals or a metal and a non-metal. Alloys of mercury are called amalgam.

Alloy	*Composition*	*Uses*
Brass	Copper (70%) + Zinc (30%)	In making utensils
Bronze	Copper (90%) + Tin (10%)	In making coins, bell, utensil
Gun metal	Copper (88%) + Tin (10%) +Zinc (2%)	In making gun, barrels, gears and bearings
German silver	Copper (60%) + Zinc (20%) + Nickel (20%)	In making utensils
Solder	Lead (50%) + Tin (50%)	For soldering
Bell metal	Copper (80%) + Tin (20%)	For casting bells, statues
Muntz metal	Copper (60%) + Zinc (40%)+ Trace of iron	In making coins
Magnalium	Aluminium (95%) + Magnesium (2%)+ Copper	For making aircraft and automobile parts
Duralumin	Iron+Aluminium (94%) + Copper + Magnesium and manganese	For making utensils, for frame of aeroplane
Type metal	Lead (80%) + Antimony (15%) + Tin (5%)	In printing industry
Stainless steel	Iron (75%) + Chromium (15%), Nickel (10%) + Carbon (0.5%)	For making utensils and surgical instruments
Babbit metal	Tin (89%) + Antimony (9%)+ Cu (2%) Nickel + Chromium	In making bearings, in large marine diesel engines.

ENERGY RESOURCES

Natural Resources

These are given by nature like air, water, mineral, sunlight etc. *These are of two types*

Renewable Natural Resources

- These are available in excess amount, i.e., never end. e.g., air, sunlight etc.

Non-Renewable Natural Resources

These are available in limited quantity, e.g., minerals, coal, petroleum, natural gas etc.

Coal

It is available in different varities : peat (60% C), lignite or brown coal **(70% C)**, bituminous coal **(80% C)**, anthracite coal **(90% C).** Bituminous is the most common variety of coal.

Charcoal

It can be wood charcoal, animal charcoal and activated charcoal depending upon the source from which it is obtained. Wood charcoal absorbs colouring matter and gases, thus used in decolourising sugar solutions and in gas masks.

- It is dangerous to have charcoal fire burning in a closed room because it produces carbon monoxide gas, which is suffocating.

Lamp Black or Carbon Black It is the most pure amorphous form of carbon (contain about 98-99% carbon). It is used for making printer ink, black paint, varnishes and carbon papers.

Fuels These are the substance, which produce heat and light on combustion.

Petroleum It is a dark coloured oily liquid with offensive odour. It is also called rock oil, mineral oil, crude oil or black gold.

- When subjected to fractional distillation, it gives petrol at 70°-250° and diesel at 250°-350°. TEL (tetraethyl lead) is used as antiknock compound.

- LPG (Liquified Petroleum Gas) is a mixture of butane and propane. A strong foul smelling substance called ethyl mercaptan, is added to LPG to detect the gas leakage.
- CNG mainly contains methane.

Fuel	*Composition*	*Sources*
Water gas	Carbon monoxide (CO) + Hydrogen (H_2)	By passing steam over red hot coke
Producer gas	Carbon monoxide (CO) + Nitrogen (N_2)	By passing insufficient air over red hot coke
Coal gas	Hydrogen (H_2) + Methane (CH_4) + Ethylene + Acetylene + CO	By fractional distillation of wood
Biogas or Gobar gas	Methane (CH_4) + Carbon dioxide (CO_2) + Hydrogen (H_2) + Nitrogen (N_2)	From organic wastes
Hydrogen Fuel	Dihydrogen (H_2)	In fuel cell for generating electric power

FLAME & ITS REGIONS

It is the hot part of fire and has three parts.

Innermost Region

- It is black because of the presence of unburned carbon particles. It has the lowest temperature.

Middle Region

- It is yellow luminous due to partial combustion of fuel.

Outermost Region

- It is blue (non-luminous) due to complete combustion of fuel. It is the hottest part of flame and is used by the Goldsmith to heat the gold.

Fire Extinguisher

- In case of electric fires and oil fires, water cannot be used as an extinguisher as it is a conductor of electricity and oil being lighter comes above the water. Such fires are extinguished by carbon dioxide.

SAFETY MATCHSTICK

- Safety matchstick contains a mixture of antimony trisulphide and potassium chlorate at its one end.
- The box side contains a mixture of powdered glass and red phosphorus.

ELECTROCHEMISTRY

- It is the branch of chemistry which deals with the study of interconversion of chemical energy and electrical energy.

Battery

- It is an arrangement of one or more cells connected in series.
- It is basically a galvanic cell.
- *These are of two types*
  - (i) Primary batteries (non-rechargeable) e.g., dry cell, mercury cell etc.
  - (ii) Secondary batteries (rechargeable) e.g., lead storage battery, nickel-cadmium battery.

Battery	*Anode*	*Cathode*	*Electrolyte*	*Uses*
Leclanche or dry cell (Potential 1.5 V)	Zinc	Graphite	Paste of NH_4Cl and $ZnCl_2$ (Zinc chloride)	Transistor, clocks
Mercury cell (Potential 1.35 V)	Zinc-mercury amalgam	Paste of HgO and carbon	Mixture of KOH and ZnO	Hearing aids, camera
Lead storage battery (contains 6 cells each of 2 V potential)	Lead	Lead grid packed with lead dioxide	38% sulphuric acid	Automobiles and inverters

Corrosion

- It is the process of oxidative deterioration of a metal surface by the action of environment to form unwanted corrosion products. e.g., conversion of iron into rust [$Fe_2O_3 \cdot x\ H_2O$], tarnishing of silver (due to the formation of Ag_2S), development of green coating [of $Cu(OH)_2 \cdot CuCO_3$ (basic copper carbonate)] on copper and bronze. It is basically an electrochemical process.
- **Corrosion of iron is called rusting.** It is accelerated by the presence of impurities, H^+, electrolytes such as NaCl, gases such as CO_2, SO_2, NO, NO_2 etc.
- It is prevented by electroplating or galvanisation (deposition of thin layer of zinc metal on iron objects) surface coating (i.e., coating of surface with oil, grease, paint galvanisation of iron (process of deposition of a thin layer of zinc over iron surface).

Note A sliced apple turns brown if kept open for some time due to the oxidation of iron present in the apple.

AIR

- It is a homogeneous mixture of different gases.
- It contains 78% nitrogen; 21% oxygen, 0.03-0.05% carbon dioxide (CO_2), argon etc.
- It has different densities at different heights from sea level. Thus, several layers are formed.
- These layers are troposphere, stratosphere, mesosphere, thermosphere, exosphere Ionosphere. Some of them are discussed here.

Troposphere

- It is the lowest layer of atmosphere, extend up to a height of 10 km.
- It is a turbulent and dusty zone, which contains air (N_2, O_2, CO_2) much water vapours and clouds.

Stratosphere

- It is the layer which contains ozone layer (which protects us from harmful UV rays coming from the Sun), thus called ozonosphere.

- It extends up to 10-50 km above sea level.
- Mesophere, thermosphere exosphere and ionosphere are the other layers of atmosphere.
- The International Space Station Orbit is also in thermosphere.

CONSTITUENTS OF AIR

Oxygen (O_2)

- It was discovered by **K Scheele**.
- It is non-combustible, colourless, odourless and neutral gas.
- It is used for artificial respiration and in oxy-hydrogen flame, oxygen-ethylene flame and oxygen acetylene flame (used for welding) and as a rocket fuel.
- Metal oxides are generally basic, but alumina (Al_2O_3), zinc oxide (ZnO) and tin oxide (SnO_2) are amphoteric oxides.
- Carbon dioxide (CO_2) is acidic, while carbon monoxide (CO) is neutral.

Nitrogen (N_2)

- It was discovered by Rutherford.
- It is a colourless, odourless, non-combustible and non-poisonous gas.
- It is filled in sealed packets and bulbs to create inert atmosphere.

Carbon Dioxide (CO_2)

- It is used by plants for photosynthesis.
- It turns the lime water milky, which disappears in the excess of CO_2 due to conversion of milky calcium carbonate ($CaCO_3$) into soluble calcium bicarbonate. It is used to extinguish fire and for artificial respiration, when mixed with oxygen (carbogen).

▪ ENO used as an antraid and reliever of bloatedness produces efferve scence, if dissolved in water due to evolution of CO_2 gas.

WATER

- It is a universal solvent and maintains the body temperature due to its high specific heat.
- Its density is maximum at 4°C.
- 22nd March is World Water Day.
- Density of ice is lower than that of water. That's why it floats over water.

Types of Water

Soft Water

- Soft water easily lather with soap.

Hard Water

- It does not lather with soap. *It is of two types*

1. **Temporary Hard Water** It contains bicarbonates of calcium and magnesium. It is converted into soft water by boiling or by adding calculated quantity of calcium hydroxide. (Clarke's process).
2. **Permanent Hard Water** It contains sulphates and chlorides of calcium and magnesium. It is converted into soft water by adding sodium carbonate (Na_2CO_3) or calgon or zeolite. Calgon is sodium hexametaphosphate, $Na_2[Na_4(PO_3)_6]$. Permutit or zeolite is hydrated sodium aluminium silicate, $Na_2Al_2Si_2O_8 \cdot H_2O$.

Heavy Water

It is deuterium oxide, D_2O (molecular mass 20). It is used as a moderator in nuclear reactors.

POLLUTION

It is the introduction of contaminants into the natural environment that cause adverse change.

Types of Pollution

Air Pollution

It is a mixture of solid particle and gases in the air. Car emissions, chemical, dust, pollen spores may be suspended as particles.

- These are the substances that contaminate the environment.
- The order of different pollutants to cause pollution is carbon monoxide (CO), SO_2 > Hydrocarbon > Particulates > Nitrogen oxides.

Stratospheric Pollution

- Stratospheric pollution means depletion of ozone layer (ozone hole) by certain compounds like chloro fluoro carbons (CFCs), oxides of nitrogen (which are released into upper atmosphere from engines of supersonic transport planes).

pH of normal rain water is 5.6.

Yellowing of white marble of Taj Mahal is the outcome of acid rain.

Water Pollution

- It is due to the presence of foreign substances in water.
- It can also be due to metals. e.g.,

Element	*Disease*
Chromium and arsenic	Cancer
Cadmium	Itai-itai disease
Mercury	Minamata disease

Soil Pollution

- It is alteration in soil. It is caused by pesticides like insecticides (e.g., DDT, BHC etc) herbicides (e.g., sodium chlorate, sodium arsenate benzipam) and fungicides (e.g., organomercury compounds bordeaux mixture).

Hazardous Effects of Pollution

Greenhouse Effect

- It is the heating of Earth and its objects because of the trapping of IR radiations by carbon dioxide (CO_2), methane (CH_4), NO, ozone (O_3), CFC's and water vapours.

Global Warming

- It is a result of increased concentration of greenhouse gases.
- It may result in melting of ice caps and glaciers, spreading of several infectious diseases like malaria, sleeping sickness etc.

Acid Rain

- It has pH less than 5. It is due to oxides of nitrogen and sulphur.
- It damages the buildings and other structures made up of limestone as marble, corrodes metal pipes, results in several diseases.

Green Chemistry

- It is a way of thinking and is about utilising the existing knowledges and principles of chemistry and other sciences to reduce the adverse impact on environment.

CARBON AND ITS COMPOUNDS

Carbon

- It is a member of group 14 and has following crystalline allotropes.

Graphite

- It is opaque and black.
- It is a very good conductor.
- It is also called **black lead**.

Diamond

- It is highly transparent.
- It is the hardest material known.
- It has a very low electrical conductivity.
- Under normal conditions, it has the highest thermal conductivity of all known materials.

Fullerenes

- An allotrope of carbon containing clusters of 60 carbon atoms joined together to form spherical molecules. It was discovered by Buckminster, so it is also known as buckminsterfullerene (C_{60}).

ORGANIC COMPOUNDS

- These are the compounds of mainly carbon and hydrogen or compounds of carbon and hydrogen with other elements like phosphorus, oxygen, nitrogen, sulphur, halogens etc.

USES OF SOME IMPORTANT ORGANIC COMPOUNDS

- **Ethylene** (C_2H_4) is used to prepare mustard gas (war gas) and for ripening of fruits.
- **Acetylene** (C_2H_2) is used to generate light, to weld metals as oxy-acetylene flame and to prepare synthetic rubber (neoprene).
- **Ethyl alcohol** (C_2H_5OH) is used in making liquours, medicines and as a fuel in aircrafts.
- **Diethylether** ($C_2H_5OC_2H_5$) is an anaesthetic agent better than chloroform.
- **Acetic Acid** (CH_3COOH) is used for making vinegar and rayons, photographic films (in form of cellulose acetate).
- **Oxalic Acid** ($C_2H_2O_4$) is used in removing ink spots, in photography and in the synthesis of coaltar.
- **Chloroform** ($CHCl_3$) is used as an anaesthetic and to preserve substances obtained from plants and animals. It converts into poisonous phosgene ($COCl_2$), when exposed to sunlight. So, it is kept in dark bottles.
- **Glycol** ($C_2H_6O_2$) is used as a antifreeze mixture in car radiator and to prevent the freezing of fuel in space crafts.
- Urea is the first synthesised organic compound (by Wohler).
- Acetic acid was the first organic compound synthesised in the laboratory from its elements.

MAN MADE MATERIALS

Drugs and Medicines

- A drug is any type of substance that alters body functioning upon intake, whereas a medicine is a chemical substance that is used for treatment of a disease.
- Ranitidine, histamine; cimetidine are used as antacids to cure acidity.
- Equanil, valium, cerotonin, etc, are used as tranquilizers (medicines use to treat mental illness).
- Analgesics are used to relief pain. e.g., codein, aspirin and paracetamol.

FOOD PRESERVATIVES

- Aspartame, sucralose, alitame, saccharine are well-known artificial sweeteners which are used to control calorie-intake. Sodium benzoate (C_6H_5COONa), common salt, sugar etc., are used as food preservatives.

SOAPS

- These are sodium and potassium salts of higher fatty acids. e.g., sodium palmitate, sodium stearate etc.

DETERGENTS

- These are sodium or potassium salts of long chain alkyl or aryl sulphonates or sulphates e.g., sodium alkyl sulphonate, sodium alkyl benzene sulphonate etc.
- These are also called **soapless soap**.
- The cationic detergents are used as fabric softeners and germicides, while non-ionic detergents are used as liquid dish washing detergents.
- Detergents lather with hard water.

FERTILIZERS

- These substances increase the fertility of soil. e.g., basic calcium nitrate $[CaO \cdot Ca(NO_3)_2]$, ammonium sulphate $[(NH_4)_2SO_4]$. These two increases the acidity of soil, which is removed by adding lime.
- Other examples are calcium cyanamide or nitrolim ($CaCN_2$), urea or carbamide (it does not affect the pH of soil), calcium super phosphate or super phosphate of lime $[Ca(H_2PO_4)_2 + 2CaSO_4 \cdot 2H_2O]$.

GLASS

- It is an amorphous or transparent solid, also called supercooled liquid.
- It contains mainly silica (SiO_2).

Colour	*Substance added*
Red	Copper (I) oxide (Cu_2O)
Green	Chromium oxide (Cr_2O_3)
Violet	Manganese oxide (MnO_2)
Blue	Calcium oxide (CaO)
Brown	Iron oxide (Fe_2O_3)

- *It can be of the following types*
  (a) **Soda or soft glass** is sodium calcium silicate ($Na_2O \cdot CaO \cdot 6SiO_2$). It is the ordinary glass and used for making bottles, window panes etc.
  (b) **Potash glass or hard glass** is used for making chemical apparatus like beakers, flasks, funnel etc.
  (c) **Crook's glass** contains caesium oxides. It is used for spectacles as it absorbs UV rays.
  (d) **Jena glass** contains B_2O_3 and alumina. It is stronger and more resistant to acids and alkalies, that's why used for making laboratory bottles for keeping acids and alkalies.
  (e) **Glass laminates** is made by fixing polymer sheets between layers of glass. It is used to make windows and screens of cars, trains and aircrafts. Specially manufactured glass laminates are used as bulletproof material.

CEMENT OR PORTLAND CEMENT

- It has the following composition calcium oxide (CaO) = 50 – 60%, silica (SiO_2) = 20 – 25%, alumina (Al_2O_3) = 5 –10%; magnesium oxide (MgO) = 2 – 3%.
- It is manufactured from limestone and clay.
- Gypsum decreases the rate of setting of cement.

Polymers (Poly-many ; mer-units)

- These are compounds of high molecular mass, made up of monomers.
- These are organic polymers of high molecular mass but they may contain other substances.
- PVC, polystyrene are synthetic plastics.

Polymers	*Monomers*
Polyethylene	Ethylene
Polyvinyl chloride	Vinyl chloride
Polytetrafluoro ethylene or teflon	Tetrafluoro ethene
Bakelite	Formaldehyde + phenol

Fibres

- These are the polymers having strong intermolecular forces like hydrogen bonding. *e.g.*, nylon-66, dacron, orlon etc.

Natural Rubber

- It is an elastomer. It is a polymer of *cis*-isoprene. Synthetic rubber (neoprene) is a polymer of chloroprene.
- It is heated with sulphur compounds at 373K in the presence of ZnO to improve its properties.
- This process is called **vulcanisation** of rubber.
- If vulcanised with 5% sulphur, it is used for making tyres and if with 30% sulphur, it is used in making battery cases.
- *Trans*-polyisoprene is called *gutta- percha*.

Explosives

- Some examples of explosives are trinitrotoluene (TNT), nitroglycerine or trinitroglycerine, cyclotrime- thylene trinitroamine (RDX also called cyclonite and hexogen).

Fibre	*Monomers*	*Uses*
Nylon-66,	Adipic acid +hexamethylene diamine	In making bristles for brushes, synthetic fibres, parachutes.
Terylene	Ethylene glycol and terephthalic acid	For making wash and wear fabrics, tyre cords, safety belts, tents etc.
Kevlar	Terephthalic acid + 1,4-diamino benzene	For making bulletproof vests.
Lexan or polycarbonate	Diethyl carbonate + *bis*-phenol-A	In making bulletproof windows.
Polyurethanes	Toluene diisocyanate + ethylene glycol	For making washable and long lasting mattresses, cushions.

Industrial Name	*Chemical Name and Formula*
Alum	Potassium aluminium sulphate ($K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$)
Borax	Sodium tetraborate decahydrate ($Na_2B_4O_7 \cdot 10H_2O$)
Caustic soda	Sodium hydroxide (NaOH)
Dry ice	Solid carbon dioxide (CO_2)
Epsom salt	Magnesium sulphate ($MgSO_4 \cdot 7H_2O$)
Glauber's salt	Sodium sulphate decahydrate ($Na_2SO_4 \cdot 10H_2O$)
Hypo	Sodium thiosulphate pentahydrate ($Na_2S_2O_3 \cdot 5H_2O$)
Laughing gas	Nitrous oxide (N_2O)
Marble or chalk or pearl	Calcium carbonate ($CaCO_3$)
Pearl ash	Potassium carbonate (K_2CO_3)
Spirit	Methyl alcohol (CH_3OH)
Soda ash	Sodium carbonate (Na_2CO_3)
Vinegar	Acetic acid (CH_3COOH)
Vermellium	Mercuric Sulphide (AgS)
T.N.T	Tri-nitro toluene $C_6H_2CH_3(NO_2)_3$

BIOLOGY

- Biology is a natural science which deals with the study of life and living organisms, *i.e.*, plants and animals. Biology has two branches
- Study of plants is called **Botany** and study of animals is called **Zoology**. Zoology and Botany are collectively called **Biology**. The term 'Biology' was coined by **Lamarck** and **Treviranus**.
- The scientist, who gave his thought for the first time about the life of plants and animals was **Aristotle**. That's why, he is known as the Father of Biology. He is also known as the Father of Zoology.
- **Theophrastus** is known as Father of Botany.

Characteristics of Living Beings

- Cellular organisation, metabolism, reproduction (growth) and response to surroundings (irritability) are the basic characteristics of life, while respiration, excretion, circulation etc, are other important characteristics of living beings.

THE CELL

- The **Cell** is the basic structural and functional unit of all known living organisms.
- The branch of biology, which deals with the study of cell, is called **Cytology**.
- **Robert Hooke** coined the term cell, when he saw honey comb-like structures in the section of cork. The name was given as he compared these structures with the compartments (cells) of a prison. However, he only discovered cell wall.
- A typical cell consists of living (Protoplasm) and non-living (Deutoplasm) matter. Protoplasm is called physical basis of life. Besides these two, a cell also contains many membrane bound structures called cell organelles. The most important cell organelle is nucleus which controls all the activities of a cell.
- Protoplasm within the nucleus is called **nucleoplasm** and outside the nucleus is called **cytoplasm**.

- The largest known cells are unfertilised ostrich egg cells (6 inches weight 3.3 pound).
- The smallest cell is of PPLO (*Mycoplasma gallisepticum,* size 0.1-0.3 μm. PPLO is Pleuro Pneumonia Like Organisms.
- The largest human cell is the female ovum and the smallest human cell is the red blood cell. The longest cell is neuron.

Types of Cells

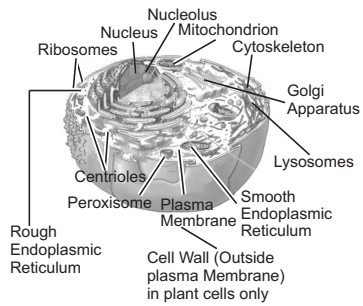
On the basis of nuclear structure cells can be of following two types

Prokaryotic Cells

- These are pro = primitive, *karyos* = nucleus primitive cell, lacking a true nucleus and most of the other cell organelles, *e.g.,* bacterial cell.

Eukaryotic Cells

- These have (*Eu* = true; *karyos* = nucleus) true nucleus and membrane bound cell organelles. These are present in various unicellular and multicellular plants and animals. *Cell organelle in a nut shell are as follows*



The Cell

All except lysosomes and ribosomes are absent in prokaryotic cells.

Functions of Cell Organelles

- Cell wall provide shape and rigidity to the cell.
- **Mitochondria** is called the **power house** of a cell, because in it, stepwise oxidation of food occurs, which results in release of energy. This energy is stored in the form of ATP (Adenosine Tri Phosphate).

- Plastids are present only in plant cells and are of three types–chloroplasts (green), leucoplasts (white) and chromoplasts (of various colours except green). Chloroplast is the site of photosynthesis as it contains chlorophyll. Chloroplast is called the **kitchen of the cell**.
- The red colour of tomatoes is due to the presence of lycopene pigment *i.e.*, chromophore in chromoplast.
- The colour of carrot is due to carotene.
- **Lysosomes** are polymorphic (*poly* = many; *morph* = forms of structure) organelles having hydrolytic enzymes. These enzymes function at acidic pH = 5. Lysosomes are sometimes called **suicidal bags of the cell** as these enzymes can dissolve the complete cell also. Lysosome may also helps in carcinogenesis, *i.e.*, conversion of a normal cell into cancerous cell.
- **Nucleus** was discovered by **Robert Brown**. It contains nucleoplasm, nucleolus and chromatin material. Nucleus is the **controlling centre of cell** as it contains all information of life in the form of nucleic acids within the chromosomes (chromatin material).

Chromosomes

- Chromosomes are thread-like structures, found in nucleus and made up of DNA along with histone proteins. Bead-like structures found on chromosome are called as genes.
  Fix number of chromosome is considered as an identification characteristic for living organisms.

Organisms	*Number of pairs of Chromosomes*
Human	23 (= 46)
Dog	39 (= 78)
Monkey	21 (= 42)
Onion	8 (= 16)

Nucleic Acids

- The chromosome or chromatin material is made up of acidic substances called nucleic acids, which contains all information about life in the form of codes (genetic codes). These are of two types

DNA and RNA Both of these are made up of long chains of units called Nucleotides, each of which contains a Nucleoside (*i.e.*, nitrogenous base + deoxyribose or ribose sugar) and a phosphate (PO_4^{--}) group. The bonds (attractive binding forces) of nucleic acids are called Phosphodiester bonds.

Deoxyribose Nucleic Acid (DNA)

- This acid is responsible for carrying our characters from one generation to the other (genetic material). It contains deoxyribose sugar and four nitrogenous bases as Adenine (A), Guanine (G) [both purines] and Cytosine (C), Thymine (T) [both Pyrimidines].
- DNA was discovered by **James D Watson** and **Francis Crick**, who got Nobel Prize for this discovery.

Ribo Nucleic Acid (RNA)

- RNA is the genetic material of some viruses only. In our body it helps in decoding the information of DNA (protein synthesis). It also contains four nitrogenous bases (Adenine (A), Guanine (G), Cytosine (C) and Uracil (U)) along with ribose sugar in its nucleotide.
- Both the nucleic acids are considered as biomolecules. Other biomolecules are proteins carbohydrates, lipids and vitamins.

Plant Cell	*Animal Cell*
It has cell wall (made from cellulose).	Cell wall is usually absent.
Plastids	Plastids are usually absent.
A big vacuole is present.	Vacuole is absent or very small in size.

Cell Division

- It is the process, by which a cell increase in number.
- It is of two types: mitosis and meiosis
  - **Mitosis** It occurs in unicellular organism during a sexual reproduction. Mitosis also occurs in

somatic cells of multicellular organism. Each mitotic cell division results in the formation of two daughter cell having number of chromosomes equal to the parent cell. *Tumour or cancer is a result of uncontrolled mitosis.*

- **Meiosis** It occurs in reproductive cells of multicellular organism and is called reduction division because of the presence of single set of chromosomes in daughter cells. It results in the formation of four daughter cells. Meiosis is a source of new genetic variation and evolution.

Nobel Prize in Medicine 2016

The Nobel Prize in Physiology or Medicine 2016 was awarded to ''Yoshinori Ohsumi'' for his discoveries of mechanisms for autophagy.

The word ''autophagy'' originates from the greek words *auto*, meaning 'self and phagein, meaning to 'eat'.

It is an evolutionarily conserved process in which the enkaryotic cell can recycle part of its own content.

BIOMOLECULES

- Biomolecules are the substances that are essential for a living being to perform its basic function.

Carbohydrates

- These are the organic compounds of carbon, hydrogen and oxygen. Chemically, these are the polyhydroxy aldehydes and ketones or their derivatives.
- These are the source of energy and about 50-75% energy of our body is obtained by the oxidation of carbohydrates.
- The main end product of carbohydrate digestion is glucose, which is called **blood sugar.**
- Main sources of carbohydrates are wheat, maize, rice, potato etc.
- The carbohydrates are divided *into three types*

Monosaccharides

- These are simple sugars, which cannot be hydrolysed further, e.g., ribose, glucose, fructose, galactose etc.
- In human beings, blood glucose level is 100-120 mg/ml.
- D-fructose is the sweetest of all sugars and found in fruit juice, honey etc.

Oligosaccharides

- They release 2-10 monosaccharides on hydrolysis. These may be disaccharides, e.g., maltose, or trisaccharides, e.g., raffinose etc.
- Sucrose is also called **invert sugar**. It gives glucose and fructose, when subjected to hydrolysis.

Polysaccharides

- They release more than ten monosaccharides on hydrolysis. These are non-sugar, i.e., do not have sweet taste, e.g., cellulose, glycogen and starch.
- Cellulose is found in plant cell wall and is digested by ruminants (like cow, goat, buffalo etc), but not by human beings. Thus, it acts as roughage in case of human beings.
- Starch is the stored food material in food grains like wheat, maize.
- Glycogen is found in animal muscles and also called animal starch.

Proteins

- These are the compounds of carbon (C), hydrogen (H), oxygen (O), nitrogen (N) and sulphur (S). These form 15% part of human body.
- Their main sources are groundnuts, soyabean, pulses, fish, meat etc.
- Proteins are called the building blocks of our body.
- Deficiency of protein causes **kwashiorkor** (a disease, in which hands and legs of children get slimmed and the stomach comes out) and **marasmus** (a disease, in which muscles of children are loosened).

Lipids (Fats)

- These are also the compounds of carbon (C), hydrogen (H) and oxygen (O). A minimum quantity of fats is necessary to maintain the normal functioning of plasma membrane of our cells.

- The main sources of fats are ghee, butter, almond, cheese, egg yolk etc.
- Fats are digested, by an enzyme called lipase, in the small intestine.
- Excess of saturated fats raises the level of blood cholesterol and may cause **atherosclerosis** (hardening of arteries). This may lead to heart attack.
- These protect different body parts from injury.
- Deficiency of fat results in dry skin and weight loss.
- The skin fat, in case of whales and seals, forms a thick layer called the **blubber**. It acts as a food reserve and also maintains the body temperature.

Vitamins

- Vitamins were first discovered by **FG Hopkins**. However, the term vitamin was coined by **C Funk**.
- *These are divided into two groups*
  (i) Fat soluble vitamin, *viz*., vitamin-A, D, E and K.
  (ii) Water soluble vitamin, *viz*., vitamin-B and C.
- Vitamin-D is synthesised in our skin by the action of ultraviolet rays of the sunlight.
- A fixed quantity of vitamins is required for the normal functioning of our body cells. Their deficiency and excess (for fat soluble vitamins only) both causes different diseases in our body. However, excessive quantity of water soluble vitamins is excreted through urine.

Water

- About 70% of the human body consists of water. Two-third of water exists inside the cells, the other one-third is outside the cells in tissue fluid and blood plasma. It is essential for digestion, transportation, excretion and to regulate body temperature.

Vitamin	*Chemical Name*	*Deficiency Disease*	*Source*
Vitamin-A	Retinol, carotenoids	Night-blindness (Nyctalopia), Hyperkeratosis, Keratomalacia	Oranges, green vegetables, carrots, pumpkin, squash and spinach
Vitamin-D	Calciferol	Rickets and osteomalacia	Fish, eggs, liver and mushrooms
Vitamin-E	Tocopherols, tocotrienols	Infertility	Many fruits and vegetables
Vitamin-K	Phylloquinone,	Non-clotting of blood	Green vegetables
Vitamin-B_1	Thiamine	Beriberi, Wernicke-Korsakoff syndrome	Oatmeal, rice, vegetables, cauliflower, potatoes, liver and eggs
Vitamin-B_2	Riboflavin	Cracking of skin, reddish eye, cracking of tongue.	Dairy products, bananas, popcorn and green beans
Vitamin-B_3	Niacin or nicotinic acid	Pellagera, glossitis	Meat, broccoli, avocados
Vitamin-B_5	Pentothenic acid	Premature graying of hair (Achromotrichia), Burning feet syndrome	Meat, fish, eggs, many vegetables, mushrooms and tree nuts
Vitamin-B_6	Pyridoxine	Abnormal protein metabolism	Meat, vegetables, tree nuts and bananas
Vitamin-B_7	Biotin	Dermatitis, enteritis and anaemia	Raw egg yolk, liver, peanuts and vegetables
Vitamin-B_9	Folic acid, folinic acid	Megaloblastic anaemia and birth defects	Leafy vegetables, pasta, bread, cereal and liver
Vitamin-B_{12}	Cyanocobalamin	Pernicious anaemia	Meat and other animal products
Vitamin-C	Ascorbic acid	Scurvy (Sailers disease)	Peppers, citrus fruits and sprouts

ZOOLOGY

The science or branch of Biology dealing with animals.

Origin of Life

- The first scientific account of the origin of life was given by Russian scientist **A Oparin** in his book 'Origin of life'.
- The primitive atmosphere contained hydrogen, methane, ammonia and water vapour. In it, oxygen and ozone were absent. Thus, it was believed that life is originated from inorganic substances by a series of complex reactions in water. This thought was supported by **Miller** and **Urey** in 1953.

Organic Evolution

- More and more creation of organism by gradual changes from lower simple organisms to higher complex organisms is called **organic evolution**. This process is considered as the main basis for the development of complex forms like human being from very simple forms like bacteria, *Amoeba* etc. There are several evidences regarding organic evolution.

Homologous Organs

- The organs, which are similar in basic structure and origin but almost similar or dissimilar in function are called **homologous organs**, e.g. wings of bat, cat's paw, front foot of horse, human's hand and wings of birds.
- These show divergent evolution.

Analogous Organs

- Organs with dissimilar origin and structure but same function e.g. wings of insects, birds and bats, eyes of octopus and mammals are **analogous organs**. These are the result of convergent evolution.

Vestigial Organs

- These are degenerate, non- functional organs, which were considered as functional earlier.
- Human body has been described to possess about 90 vestigial organs. Some of these are muscles of ear pinna, canine teeth, third molar teeth, body hairs, vermiform appendix, nictitating membrane of eye and caudal vertebrae (coccyx or tail bone) etc.

Evidence from Connecting Links

- Connecting links are those living organisms which show characteristics of more than one groups thus connecting those groups. *Some such links are*

Organism	*Name of Groups*
Virus	Living and non-living
Euglena (*Protozoa*)	Plants and animals
Balanoglossus (*Chordata*)	Non-chordata and chordata
Dipnoi (*Lung fish*)	Fishes and Land Vertebrates
Archaeopteryx (*Aves*)	Reptiles and Birds
Prototheria (*Mammalia*)	Reptiles and Mammalia

THEORIES OF EVOLUTION

Lamarckism

- Lamarck gave the idea that an organism passes characteristics that it acquires during its lifetime to its offspring (also known as inheritance of acquired characters).
- Lamarck's theory of evolution was published in *Philosophie Zoologique (1809)*

Darwinism

- Darwin's theory of evolution is about Origin of Species by means of Natural Selection and it was published in 1859.
- According to this theory, it is the nature which selects certain organisms from a common pool of variable ones and it is the continuation of forms which leads to the favoured formation of new species (group).

Mutation Theory

- **Hugo de Vries** proposed the theory of mutation. Mutations are discontinuous variation. These are due to changes in chromosomes, genes and DNA and may or may not be inherited.
- According to him, these mutations serve the basis for natural selection of Darwin.

CLASSIFICATION OF ANIMALS

Storer and **Usinger** classified animals into following phylums

Phylum–Protozoa

- These are unicellular or acellular animals, *i.e.*, their body is made up of only one cell. These can be seen from microscope only.
  Example *Amoeba, Paramecium, Euglena etc.*

Phylum–Porifera

- These have a porous body. Their skeleton is made up of minute calcareous or siliceous spicules.
- Presence of water canal system fulfills all the needs of these sedentary animals.
  Example *Sycon*, Sponge etc.

Phylum–Coelenterata

- These are first proper multicellular animals and their body plan is called blind sac body plan. They have specialised cnidoblast cell to help in catching the food.
  Example *Hydra*, Jelly fish, Sea Anemone etc.
- *Hydra* has a great tendency of regeneration of body organs. Phenomenon of polymorphism (many forms) and metagenesis (alternation of generation) are associated with coelenterates.

Phylum–Platyhelminthes

- These are also called flat worms and exist as parasitic (mostly) and free living (some) forms. They are the first animals with true muscles and incomplete alimentary canal.
- Organ systems is absent and these animals are mostly monoecious (hermaphrodite) with common reproductive system.
  Example *Planaria,* Liver fluke, Tape worm etc.

Phylum–Aschelminthes

- These are long cylindrical, unsegmented worms also called round worms.
- Most forms are parasitic but some are free living in soil and water.
- Complete digestive tract is seen first time in them along with separate male and female individuals.
- Proper sensory organs are also found first time in them.
- Their body plan is called tube within the tube plan (one tube signifies cylindrical body and the other tube signifies digestive tract).
  Example *Ascaris*, Thread worm etc.

Phylum-Annelida

- Their body is long, thin, soft and metamerically (truely) segmented.
- Nervous system is normal and blood (called haemolymph) is red (iron-rich haemoglobin). These are first to have proper organ systems.
- Their blood flows in closed vessels out of which five pairs of blood vessels are called hearts.
- Excretion by nephridia.
  Example Earthworm, *Nereis*, Leech etc.

Phylum–Arthropoda

- Jointed leg and body is their main feature.
- Circulatory system is open type. Cockroach's heart has 13 chambers.
- Arthropoda is the largest phylum (contains maximum number of animals) and its existance is recorded for maximum period over the Earth. The speciality of arthropods lies in their very hard exoskeleton made up of chitin.
  Example—Cockroach, Prawn, Crab, Bug, Fly, Mosquito, Bees etc.
- Insects generally have six feets and four wings.
- Ant is a social insect which reflects division of labour.

Phylum–Mollusca

- Their body is soft divided into head and muscular foot. Blood is colourless.
- This phylum contain largest invertebrates on Earth i.e. squids.
- Eyes of *octopus* are comparable with chordate eyes.

  Example *Pila*, *Aplysia* (Sea rabbit), *Doris* (Sea lemon), *Octopus* (Devil-fish), *Sepia* (Cuttle-fish).

Phylum–Echinodermata

- All the animals in this group are marine and have a special capacity of regeneration.
- These are the only invertebrate animals which contain proper bone like structures (ossicles).
  Example Star fish, Sea urchin, Sea cucumber etc.

Phylum-Chordata

- They have notochord, a dorsal hollow tubular nerve cord and paired pharyngeal gill slits at some stage of their life span.
- In advanced forms, notochord changes to vertebral column, nerve cord develops into brain and spinal cord and pharyngeal gill slits to structures of jaw attachment.

  5 main classes of chordates are

Pisces (Fishes)

- These are aquatic animals which cannot regulate their body temperature (cold blooded).
- Respiration takes place through gills.
  Example *Trygon, Scoliodon, Torpedo* etc.

Amphibia (First Land Vertebrates)

- These are found both on land and water. All these are cold-blooded.
- Respiration takes place through gills, skin and lungs. They require water for reproductive purposes.
  Example Frog, Toad and *Salamander.*

Reptilia (First True Land Vertebrates)

- These are crawling animals.
- These are cold-blooded and possess two pair of limbs.
- Their eggs are covered with shell made up of calcium carbonate (cledroic eggs).
  Example Lizard, Snake, Tortoise, Crocodile, Trutle, *Sphenodon* etc.
- Cobra is the only snake which makes nests.
- *Heloderma* is the only poisonous lizard.
- Sea snake is also called *Hydrophis Belcheri*. It is the world's most poisonous snake.

Aves (Aerial Vertebrates – Birds)

- The animals of this group are warm-blooded tetrapod vertebrates with flight adaptation. Their fore-feet are modified into wings to fly.
- **Example** Crow, Peacock, Parrot etc.
- Flightless birds are Kiwi and Emu.
- Largest alive bird is Ostrich.
- Smallest bird is Humming bird.

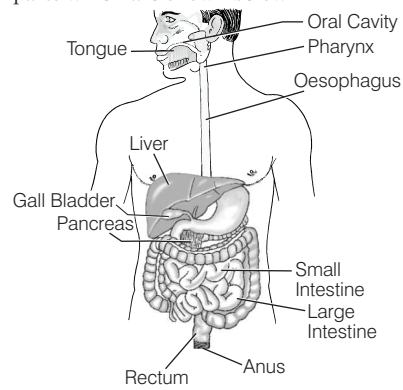
Mammalia

- These are warm-blooded animals
- Teeth come twice in the life time of these animals (diphyodont).
- There is no nucleus in red blood cells of mammals (except in camel and llama).

SYSTEMS OF HUMAN BODY

Digestive System

- The human digestive system have different parts which are shown below



Human Digestive System

Teeth

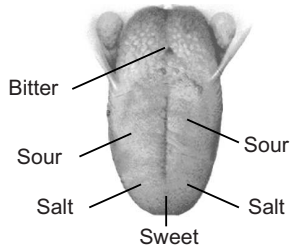
- With the help of teeths the food is chewed.
- *Teeths are of four types*
  1. Incisors (for cutting)
  2. Canines (for tearing)
  3. Premolars (for chewing)
  4. Molars (for chewing and grinding).
- The dental formula in human is

$$\frac{2I,1C\ 2Pm\ 3M}{2I,1C\ 2Pm\ 3M} \times 2 = 32$$

- Hardest part in the body is tooth enamel.
- In elephants, the tusks are the incisors of upper jaw.
- Maximum number of teeth are present in horse and pig.

Tongue

- Tongue contains taste buds due to which we sense bitter, sour, salty or sweet taste.



Taste buds on the Human Tongue

- Man and other animals have, holozoic nutrition i.e. ingestion of **solid form of food.** Parotid gland is the largest salivary gland.
- In mouth, salivary amylase (ptyalin) acts on starch and converts it into maltose.
- Pepsin of stomach digests proteins and converts them into peptones.
- Rennin of stomach converts milk to curd. Its presence is suspicious in human.
- Trypsin of intestine acts upon proteins and breaks them into polypeptides.
- Lactase of intestine converts lactose into glucose.
- Sucrase of intestine converts sucrose into glucose.
- pH of saliva, gastric juice, pancreatic juice and intestinal juice is respectively 6.8, 2.0, 7.0 and 8.5 i.e. intestinal juice is alkaline in nature.

Digestive Glands Liver

There are two main digestive glands in human body.

- It is the largest gland of the human body and secretes bile juice, which is stored in gall bladder.
- It regulates the quantity of glucose in the blood, destroys dead RBC, regulate body temperature and converts excess of amino acid into ammonia.
- Bile juice is responsible for digestion of fats. It basically emulsifies (breaks up into small droplets) fat to ease out attack of lipases (fat digesting enzyme).
- If there is found any obstruction in bile duct, by gall stones, jaundice occurs and is called obstructive jaundice. In this bile is absorbed into the blood instead of going to the duo-demum and cause yellowing of eyes and skin.

Pancreas

- It is the second-largest gland of the human body and contains islet of Langerhans. The most common of all cell, the β-cells secretes insulin. The hormone deficiency of which causes diabetes mellitus. Other cells of pancrease secrete digestive enzymes.

Mineral Nutrients

- These are metals, non-metals and their salts other than the four elements—carbon, hydrogen, nitrogen and oxygen and constitute about 4% of total body weight.
- *Minerals are of two types*
  - **Macronutrients** These are required in large amount e.g. calcium (Ca), phosphorus (P), potassium (K) etc.
  - **Micronutrients** These are required in very small amount (less than 1 g) e.g. iodine (I), iron (Fe), zinc (Zn) etc.

Roughage

- Roughage is another term for dietary fibre e.g. natural food, dalia etc.
- It does not provide energy, but only helps in retaining water in the body.

Mineral	*Major Food Sources*	*Needed for*	*Deficiency Disease*
		Macronutrient elements	
Calcium (Ca)	Milk, cheese, bread and vegetables.	Muscle contraction, nerve action, blood clotting and the formation of bone.	Tetany rickets and Osteomalacia.
Phosphorus (P)	Cheese, eggs, pea nuts and most foods.	Bone and tooth formation, energy transfer from foods, DNA, RNA and ATP formation.	Tetany and rickets.
Potassium (K)	Potatoes, meat and chocolate.	Muscle contraction, nerve action and active transport.	Nervous disorder, weak muscles leading to paralysis.
		Micronutrients Trace Elements	
Iron (Fe)	Liver, kidney, red meat, cocoa powder and water-cress	Formation of haemoglobin, myoglobin and cytochromes.	Anaemia and low immunity.
Fluorine (F)	Water supplies, tea, sea food, meat, liver and beans.	Resistance to tooth decay.	Weak teeths (if present in excess) causes mottling of teeths
Iodine (I)	Seafood, iodised salt and fish.	Thyroxine production.	Goitre.
Cobalt (Co)	Meat and yeast.	Synthesis of vitamin-B_{12}, formation of red blood cells.	Anaemia.

RESPIRATORY SYSTEM

- Respiration is a catabolic process in which the respired oxygen is used in the oxidation of food resulting in the release of energy. It is brought about by respiratory organs.

Respiratory Organ	*Animal*
Lungs	Reptiles like lizards, mammals like man, camel, cattle etc.
Skin	Frog, earthworm and leeches
Gills	Fishes, tadpoles and prawns
Air sacs/ lungs	Birds

Types of Respiration

Aerobic Respiration

- When respiration takes place in the presence of oxygen, it is known as aerobic respiration.
- The oxygen taken oxidises the respiratory substances (food) into carbon dioxide, water and energy, e.g.,

$$\text{Glucose} + O_2 \longrightarrow \text{Carbon dioxide} + \text{Water} + \text{Energy}.$$

Anaerobic Respiration

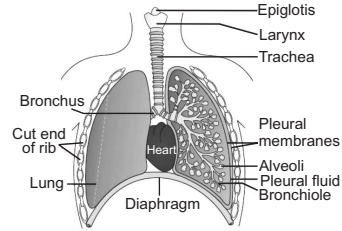
- The respiration which takes place in the absence of oxygen is known as anaerobic respiration.
- In this process, the respiratory substances are incompletely oxidised to carbon dioxide and alcohol.

$$\text{Glucose} \xrightarrow{\text{yeast}} \text{Carbon dioxide} + \text{Ethyl alcohol} + \text{Energy (in plants)}$$

$$\text{Glucose} \longrightarrow \text{Lactic acid} + \text{Energy (in animals)}$$

Human Respiratory System

The human respiratory organs are shown in the following diagram.



Human Respiratory System

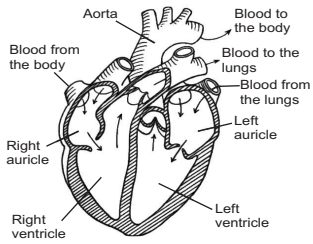
- Overall passage of air in humans is as follows
  Nostrils → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli Cells → Blood.
- Breathing rate in human = 18 - 20 per minute. Transportation of oxygen takes place by haemoglobin of blood.
- Whereas transportation of only 10-20% carbon dioxide takes place by haemoglobin of blood. Most of the CO_2 is transported through blood buffers. Larynx or voice box or Adam's apple produces sound.

CIRCULATORY SYSTEM

It is a transport system to supply the useful material and to remove the wastes from the body cells. The circulatory system mainly consists of three part. These are heart, blood vessels and blood.

Heart

- Heart is a thick, muscular, contractile, automatic pumping organ of blood vascular system.
- A fish has only two-chambered heart (single circuit).
- In amphibians and most of the reptiles (except crocodilia), heart is three-chambered (incomplete circuit).
- In crocodile**s,** birds and mammals the heart is divided into four chambers (double circuit).
- Single circuit means blood flows only once through the heart while double circuit means blood flows twice through the heart as shown below in human heart. A new born baby's heart weight about 20 g.
- The chambers which receive the blood from body tissues are called **auricles** and the chambers of heart which pump blood to body tissues are called **ventricles**.



Structure of Heart

- Human heart beat is myogenic in nature, i.e. initiated by a patch of modified heart muscles itself without requiring an external stimulation. This patch is called **SA node** (sino-auricular node) or **pacemaker**.
- When SA-node becomes defective, i.e., it does not generate cardiac impules, it can be cured by surgical grafting of an artificial pacemaker (an electric device,) in the chest of the patient. It stimulate the heart electrically at regular intervals.
- The normal rate of heart beat at rest is about 70-72 per minute. In a newly born baby, heart rate is about 140 per minute.
- During heavy exercises it may be high as 170-200 per minute. Heart is the only organ which can function for some time outside our body.

Blood Vessels

Blood vessels are of three types

Arteries

- These carry the blood away from the heart to various body parts.
- Usually these carry oxygenated (pure) blood in them, except the pulmonary artery which carries deoxygenated blood to the lungs.

Veins

- These carry the blood from various body parts to the heart.
- Usually these carry deoxygenated (impure) blood in them except the pulmonary vein which carries oxygenated blood to the heart.

Capillaries

- These are the thinnest blood vessels which connect arteries to the veins.
- These help in the exchange of materials like the nutrients, gases, waste products etc, between blood and cells.

Blood

- Blood is a fluid connective tissue and composed of blood corpuscles, plasma and plateletes. It is slightly alkaline in nature (pH 7.4).

- Its volume in an adult is 5 L.
- Blood contains RBC, WBC and Platelets.
  - Red blood cells (RBCs) contain the blood's haemoglobin and distribute oxygen.
  - WBCs are part of the body's immune system.
  - WBCs are much less in number than RBCs (1 : 600).
- People who live at high altitudes have more RBCs than those who live in low regions.
- The haemoglobin contents of adult female amount from 13.5 to 14.5% whereas in adult male its amount from 14.5 to 15.5%
- Haemoglobin count is highest in the foetus and is about 23 g per 100 mL of blood at birth.

Blood Pressure (BP)

- It is measured by **sphygmomanometer**.
- Blood pressure of a normal person = 120 / 80 mm Hg.

Bradycardia	Decreased heart rate.
Tachycardia	Increased heart rate.
Heart attack	Breathlessness, palpitations, pain in the chest, unconsciousness appear suddenly (nitroglycerine is used for the treatment).
Heart block	When heart beat is not passed to the ventricles properly. This is the defect of conducting system of the heart.
Coronary thrombosis	Formation of clot in coronary artery.

SKELETON SYSTEM

- The human skeleton consists of both fused and individual bones supported and supplemented by ligaments, tendons, muscles and cartilage. The total human skeleton contains 206 bones. *Human skeleton system is divided into two parts*

Axial Skeleton (80 Bones)

- It includes skull, vertebral column and bones of chest.
- The total number of bones of head 29.
- The total number of bones in vertebral column, initially are 33 and after development become 26.
- The total number of cranial bones are 8 and facial bones are 14.

Appendicular Skeleton (126 Bones)

- It consists of the girdles and the skeleton of the limbs.
- The girdles are pectoral girdle (4 bones) and pelvic girdle (2 bones).
- The limbs are forelimbs (60 bones) and hindlimbs (60 bones).
- Their functions are to make locomotion possible.

- Birds have spongy bones with air filled spaces, called pneumatic bones. These make the body lighter which helps them in flight.
- Fatigue is the inability of a muscle to contract due to depletion of its chemicals and lactic acid accumulation by repeated contraction. A completely fatigued muscle refuses to respond to nervous stimuli.

EXCRETORY SYSTEM

- The process of removal of nitrogenous wastes and excess water from the body is called excretion. The organs of excretion are called excretory organs. Excretion is deeply associated with water salt balance in our body.

Human Excretory System

The human urinary system includes—the kidneys (two), ureters (two), urinary bladder (one) and urethra (one). Kidneys form the urine and control osmotic pressure within the organism with respect to external environment. The filtering of nitrogenous wastes is done with the help of blood pressure in our body.

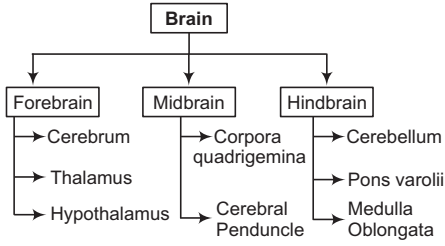
Urine

- It is pale yellow coloured fluid due to presence of urochrome pigment.
- It is acidic in nature (pH 6.0) and is slightly heavier than water. It has a faint aromatic odour due to urinoid. Daily urine output in normal adult is 1.5-1.8 L.
- Chemical composition of urine is water 95-96%, urea 2% and some other substance like uric acid, creatinine, etc., are 2-3%.

- The urine on standing gives a pungent smell. It is due to conversion of urea into ammonia by bacteria.
- Haemodialysis is the process of removal of excesss urea from the blood of patient using artificial kidney.
- Kidney stone is generally calcium oxalate crystals.
- Aquatic animals generally excrete ammonia.
- Man and shark excrete urea.
- Desert animals and birds excrete uric acid.

NERVOUS SYSTEM

- The nervous system provides the fastest means of communication within the body so that suitable response to stimuli can be made at once.
- The nervous system consists of nerve cells or neurons and sensory cells.
- Neurons are main communicating and processing cells. These form our nerves, brain and spinal cord.
- Sensory cells are situated in specialised structures, situated at the extremities of our body. These structures are called sensory organs.
- Our nervous system can be categorised into central, peripheral and autonomic nervous system. The central nervous system consists of brain and spinal cord.
- Brain is present in skull and covered with 3 membranes (meninges). These are duramater (outer), arachnoid (middle) and piamater.



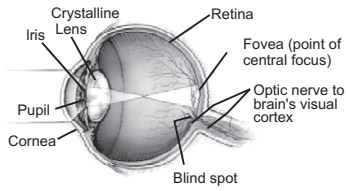
- **Cerebrum** is the major part of human brain. It controls and coordinates almost all the activities of our body.
- Broca's area of cerebrum is related with speech production while Wernicke's area of cerebrum is related with understanding speech.
- **Thalamus** deals with pain, pressure and temperature.
- **Hypothalamus** deals with water balance in the body, behavioural patterns of sex, sleep, stress emotions, thirst, hunger etc. It also regulates pituitary hormones along with metabolism of fat, carbohydrate and water.
- **Medulla** connects brain with spinal cord and have centres for the control of many autonomous function ranging from vomiting to sneezing and respiration.
- **Spinal cord** deals with impulses to and from the brain and is the centre for reflex actions like blinking of the eye when an object comes near to our eyes suddenly, rapid withdrawal of hand while being burned or pricked, knee jerk reflex etc.
- **Eye** Eyes are the sensitive detectors of light. The human eye can distinguish about 10 million colours. *It consists of three layers these are as follows*

Sclerotic Layer

- It is the outermost, bony layer, which includes — cornea and conjunctiva.

Choroid Layer

- It is the middle layer and consists of
  (a) **Pupil** It is the black hole in the centre of the iris. It changes size as the amount of light changes.
  (b) **Iris** It is the coloured part of the eye. It controls the amount of light that enters the eye by changing the size of the pupil.
  (c) **Lens** It is a biconvex transparent circular solid structure located just behind the iris. It focusses light onto the retina.



The Eye

Retina

- Light sensitive tissue that lines the back of the eye. It contains millions of photoreceptors (rods and cones) that convert light rays into electrical impulses that are relayed to the brain *via* optic nerve. The image formed on the retina is real and inverted.
- **Rods** are highly sensitive to dim light. Owl's eyes contains rods only.
- **Cones** are sensitive to bright light, and differentiate the colours as well. Hen's eyes contain cones only.

Eye Defects

Nearsightedness (Myopia)

- A condition in which nearby objects are seen more clearly than distant objects because light is focussed before retina.
- It can be corrected by using concave lens.

Farsightedness (Hypermetropia)

- A condition in which distant objects are seen more clearly than nearby objects because light is focussed behind or after the retina. It can be corrected by using convex lens.

Astigmatism

- A condition in which the lens is warped, causing images not to focus properly on the retina.
- Cylindrical glasses can correct this defect. Colour blindness, also called Daltonism is caused due to the deficiency of pigment in the cones. While night blindness is caused due to the deficiency of rhodopsin in rods.
- Eyes are most sensitive to yellow and green colour.

Ear

- Ears are meant for both balancing and hearing. Hearing is controlled by auditory area of temporal lobe of cerebral cortex.
- Mammalian ear has 3 parts i.e. external middle and internal ear. External ear comprises a pinna, external anditory meaths (canal) and tympanic membrane.
- Middle ear contains flexible chain of three small bones called ear ossicles, the malleus, the incus and the stapes.
- The auditory function is sensed through cochlea while balancing function is sensed through semi circular canals. Both of these are situated in internal ear.
- Human ear can listen sounds in the range of 60-80 decibel.
- Human ear is sensitive to sounds frequency 50-20000 cycles/sec.

REPRODUCTIVE SYSTEM

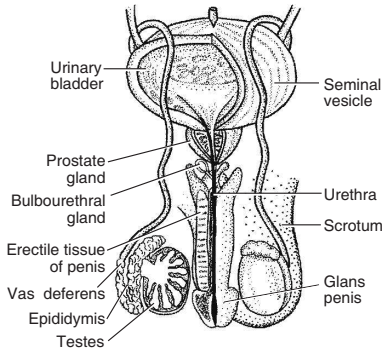
- The process by which new individuals are produced from their parents is called **reproduction** and the organs which are used for this process, collectively constitute the reproductive system. Reproduction is of two types, i.e., asexual and sexual.
- In **asexual reproduction**, only one parent is involved and sex cells are not involved. It is more common in lower animals.
- In **sexual reproduction**, two parents are involved and formation and fusion of gametes takes place.

Male Reproductive System

- Males can produce spermatozoa (sperm) throughout their life from the age of 13-14 years.
- *The various organs which constitute the male reproductive system and their functions are tabulated below.*

Reproductive Organ	*Number*	*Function*
Testes	2	Produce sperm and testosterone
Sperm duct	2	Conduct the sperm from the testes to urethra
Seminal vesicles	2	Secrete seminal plasma
Prostate gland	2	Secrete an alkaline fluid to neutralise the acidity of urethra and make the spermactive.
Penis (urethra) (have rich blood supply)	2	Pass urine and deposite sperm in female genital tract.

- The growth of hairs on body is due to masculine hormones.



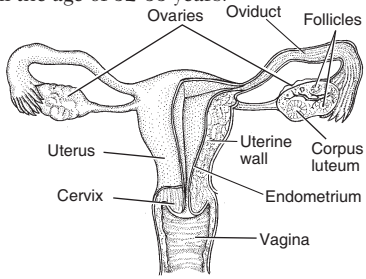
Male Reproductive System

Female Reproductive System

- *Female reproductive system includes following organs which are tabulated below.*

Reproductive Organ	*Number*	*Function*
Ovaries	2	To produce ova and hormones (estrogen and progesterone).
Oviducts	2	To move the ovum towards uterus.
Uterus	1	To provide space for developing child.
Vagina	1	To receive the sperms.

- After the age of 13-14 years, the ovary releases ovum (egg cell) after every 28 days till the age of 52-55 years.



Female Reproductive System

- This cycle of approximately 28 days is called menstruation cycle.
- Its first appearence in female at puberty is called menarch condition and its termination at about 52-55 years of age is called menopause condition.
- The fertilization of egg takes place in the fallopian tube after which the developing embryo becomes attached to the uterus (implantation).
- The connection between developing embryo and mother is called **placenta**. It supplies different materials like food, hormones, dissolved oxygen etc to embryo.
- Birth of child is technically called parturition.

> Menstruation cycle is the characteristic feature of mammals like humans only, while other mammals like cat, dog etc have seasonal oestrous cycle in them for the same purpose.

ENDOCRINE SYSTEM

It includes endocrine or ductless glands. Their secretions are known as **hormones**. *Different types of hormones are*

- **Steroids**, e.g. estradiol, testosterone, androsterone, aldosterone, cortisol and cortisone.
- **Peptides** e.g. insulin, glucagon, pituitary hormones, parathormone and relaxin.
- **Amino acid derivatives,** e.g. adrenaline, noradrenaline and thyroid hormones.
- Pituitary gland is also known as **master gland** as it secretes hormones controlling the activities of other glands of the body. Pituitary is inturn controlled by hypothalamus.

Some Other Endocrine Glands

Thyroid Gland

- It is the largest endocrine gland and situated in neck region.
- It is the 'pace setter' of our life as it controls metabolism of our cells. It secretes thyroxine which is iodine rich. Thyroid gland contains parathyroid gland over it, which is involved in the balancing of calcium and phosphate quantity within our blood.

Gonads

- Hormones of gonads are steroids.

Ovarian Hormones (Oestrogens)

- It helps in the development of primary and secondary sexual characters (oestradiol, oestriol and oestrone).

Testicular Hormones (Androgens)

- These are produced by Leydig's cells of testes.
- It stimulate growth, maturation and maintenance of male gonads and development of secondary sexual characters, e.g. testosterone, androsterone, etc.

Human Diseases

A 'disease' is a condition of the body or its part in which functions are disturbed or deranged.

- BCG (Bacillus Calmette Guerin) vaccine is given to protect TB (Tuberculosis).
- DPT (diphtheria, Pertussis and Tetanus) vaccine is given to babies within first 6 weeks of their birth.
- Milk is pasteurised (preserved) by boiling it at 62.8° for 30 min (Low Temperature Holding method LTH) or at 71.7.°C for 15 sec (high) temperature holding method. This treatment kills all bacteria.

Ebola Virus

According to World Health Organisation WHO's **19th August, 2015 Situation Report,** there were three confirmed cases of Ebola reported in the week up to 16th August all of which were reported from **Guinea**. For the first time since the beginning of the outbreak in **Sierra Leone**, a full epidemiological week has passed with no confirmed cases reported. A total of 72 cases remain under monitoring in Sierra Leone.

Thyroxine Secretions	*Growth Stimulating Hormones (of pituitary) Secretions*	*Parathormone Secretions*	*Insulin of pancrease Secretions*
Lower secretion causes cretinism (young), Hashimoto's disease and myxoedema in adult.	Lower secretion caused dwarfism in young	Lower secretion causes tetany.	Lower secretion leads to diabetes mellitus.
Higher secretion causes exophthalmic goitre (Grave's disease).	Higher secretion causes gigantism in adults.	Higher secretion causes neurofibromatosis.	**Aldosterone** hypersecretion causes Conn's syndrome.

Disease	*Pathogen* (Bacteria)	*Affected organ*	*Symptom*
Tuberculosis	*Mycobacterium tuberculosis*	Lungs	Repeated coughing, high fever.
Diphtheria	*Corynebacterium diphtheriae*	Respiratory tract	Difficulty in respiration (mainly in child of age 2-5 yrs).
Whooping cough or pertussis	*Bacillus pertussis*	Respiratory system	Continuous coughing.
Cholera	Vibrio cholerae	Intestine	Vomiting, acute diarrhoea, muscular cramps, dehydration, etc.
Tetanus (lock jaw)	*Clostridium tetani*	Central nervous system	Painful contractions of neck and jaw muscles followed by paralysis of thoracic muscles.
Pneumonia	*Diplococcus pneumoniae*	Lungs	Sudden chill, chest pain, cough, high fever and swelling in lungs.
Typhoid	*Salmonella typhi*	Intestine	High fever, diarrhoea, headache and gastro intestinal disorder
Syphilis (sexual disease)	*Treponema pallidum*	Urinary path	Wounds in urinogenital tract.

Disease	*Pathogen*	*Affected Part*	*Symptom*
Measles (Rubella disease)	Rubella virus	Whole body	Loss of appetite, reddish eruption on the body.
Chicken pox	Vericella virus	Whole body	High fever, reddish eruption on the body.
Small pox	Variola virus	Whole body	Light fever, eruption of bile on the body
Polio or poliomyelitis	Polio virus	Throat, back-bone and nerve	Fever, backbone and intestine wall cells are destroyed. It leads to paralysis.
Influenza (flu) (spread from person to person)	Influenza virus (Family Orthomyxoviridae)	Whole body	Inflammation of upper respiratory tract, nose, throat and eyes.
Rabies (hydrophobia)	RNA virus called rabies virus	Nervous system	Encephalitis, fear of water, high fever, headache, spasm of throat and chest leading to death.
Hepatitis A (epidemic Jaundice)	Hepatitis A virus	Liver	Loss of appetite, nausea, whitish stool and jaundice.
Dengue fever (transmitted by female *Aedes aegypti* mosquito*)*	RNA containing dengue virus	Whole body, particularly head, eyes and joints	High fever, backache, headache, retro-orbital pain behind the eye ball.
AIDS (Acquired Immune Deficiency Syndrome)	Human immuno deficiency virus (HIV)	White blood cells	Weak immune system.
Herpes	Herpes virus	Skin	Swelling of skin.
Ebola virus disease	Ebola virus (filovirus)	Whole body	Fatal hemorrhagic fever, liver and kidney disfunction, vomiting, headache.
Swine influenza (flu)	H_1 N_1 flu virus	Whole body (muscles)	Headache, tiredness, sore throat, vomiting, breathing problems.
COVID 19	Novel Corona Virus	Respiratory System	Fever, dry cough, tiredness, aches and pains, nasal congestion, headache, conjunctivitis, sore throat, diarrhoea loss of taste or smell or a rash on skin or discoloration of fingers or toes.

Disease	*Pathogen (Causative agent)*	*Vector*	*Parts Affected and Symptoms*
Pyorrhoea	*Entamoeba gingivalis*	None, infection by lip kissing.	Bleeding of gums.
African trypanosomiasis (sleeping sickness, gambian fever)	*Trypanosoma gambiense*	Tse-tse fly *(Glossina palpalis)*	Blood and nervous tissue. Man feels sleepy, may cause death.
Amoebic dysentery (Amoebiasis)	*Entamoeba histolytica*	None, infection by contamination.	Colon (intestine). Develop loose motion with blood, pain in abdomen
Diarrhoea	*Giardia*	None, infection by contamination	Digestive system causes loose motions, vomitting.
Kala azar or dumdum fever	*Leishmania donovani*	Send flies *(Phlebotemus)*	Spleen and liver enlarge and high fever develops.
Filariosis or elephantiasis	*Wuchereria bancrofti*	*Culex* mosquito	Swelling of legs, testes and other body parts.
Malaria	*Plasmodium* sp.	Female *Anopheles* mosquito	Periodical attacks of high fever, pain in joints, accompanied by chill, heavy perspiration and fast pulse.

- Fishes like cat fish, *Gambusia* and aquatic birds eat mosquito larvae.
- Quinone, a product of *Cinchona* tree, is administered for treatment of malaria.

BOTANY

- It is the branch of science which deals with the study of plants, which are considered as multicellular organisms. The cells of these organisms contain a cell wall made up of cellulose and other polysaccharides. Plants have the ability to synthesise their own food (autotrophic) nutrition in the presence of sunlight, via the process of photosynthesis.
- Plants have two main groups i.e. **cryptogams** (lower plants without well defined flowers and seeds) and **Phanerogams** (higher plants with well defined flowers and seeds.)
- **Cryptogams** further contains thallophytes and pteridophytes with bryophytes in between. Here, thallophytes means those plants which have thallus like body i.e. without roots, leaves and stem e.g. algae, fungi, etc.
- Bryophytes include mosses, liverworts and hornworts. Pteridophytes contain ferns and their allies.
- All the plants of cryptogams are considered as primitive as these do not have well defined features of plants like presence of phloem (a tissue for food transport) and presence of xylem (a tissue for water transport etc).
- Phanerogams are well defined advanced plants with proper roots, leaves and stem and well differentiated tissue system. These can be categorised in to **Gymnosperms** (Naked seed) and **Angiosperms** (Covered seed) plants. Out of these Angiosperms are considered as most advanced plants. These are also called vascular plants.
- *Sequoia gigantea*, the tallest plant of plant kingdom belong to gymnosperm.
- It is also known as red wood of California.

- Algae like *Nostoc, Anabaena* etc are used as manure. (biofertilizers)
- Bryophyta *Sphagnum* moss is used as fuel and antiseptic.

- Some plants catch insects to fulfill their nitrogen requirement. Such plants are called insectivorous plants e.g. bladder wort or *Utricularia*, *Drosera*, *Nepenthes* or pitcher plant *Aldrovanda, Dionea* or venus fly-trap. These plants belong to angiosperms.

PLANT MORPHOLOGY

- It is the study of both the **vegetative** (somatic) structures, as well as the **reproductive** structures of plants. Vegetative structures of vascular plants include the root stems and leaves.
- The reproductive structures on the other hand are included as flowers.

Root

- It is non-green, positively geotropic and negatively phototropic.
- Roots help in fixation, absorption of water and minerals storage of food, etc.
- *Some modifications of roots are as follows*

Modification of Root	*Example*
Conical	Carrot
Fusiform	Radish
Napiform	Turnip and beet

Stem

- This is the part of a plant which grows towards light, i.e. positively phototropic and negatively geotropic. It is one of the main structural axis of a vascular plant.
- Stems are often specialised for storage, asexual (vegetative) reproduction, protection and photosynthesis.
- *Some modification of stems are as follows*

Stem Type	*Example*
Rhizome	Banana, ginger, lotus, turmeric etc
Bulb	*Allium cepa* (onion)
Tuber	Potato

Leaf

- It is a green plant part, its main function is to make food through photosynthesis and transpiration.
- Some leaves are modified to form the flower (reproductive part) of a plant.

Flower

- This is the reproductive part of the plant.
- In the flower calyx, corolla, androecium and gynoecium are found. Out of these androecium contain (pollen grains) male sex organ and the gynoecium contains ovary with style and stigma.
- The transfer of pollen grains from anther of a flower to the stigma of same or other flower is known as **pollination**.
- Pollination is done by wind, insects, water, birds, bats and snake.

Fruit

- Fruit is a matured or ripened ovary developed after fertilization.
- In the development of some fruits, calyx, corolla and thallus also take part. Such fruits are called **false fruit**, e.g. apple, jack fruit etc.

PLANT PHYSIOLOGY

- Plants absorb capillary water (i.e. the water held in the pore spaces of soil particles) through root system and transport it upto the leaves *via* xylem. The processes for such transport includes **diffusion, osmosis** etc.
- A large amount of water is absorbed by the plants, but only a very small quantity (generally much less than 1%) is utilised by them for metabolism.
- The excess of water is lost via aerial portions and leaves of plants *via* the process called **transpiration** (thus, considered as a necessary evil for plants).

Common Name	*Botanical Name*	*Type*	*Edible Parts*
Simple Fruits			
Mango	*Mangifera indica*	Drupe	Fleshy mesocarp
Walnut	*Juglans regia*	Drupe	Cotyledons
Apple	*Pyrus malus*	Pome	Fleshy thalamus
Cashew nut	*Anacardium occidentale*	Nut	Cotyledons and fleshy thalamus
Litchi	*Litchi chinensis*	Nut	Aril
Water chestnut	*Trapa bispinosa*	Nut	Seeds
Banana	*Musa paradisiaca*	Bery	Mesocarp and endocarp
Wheat	*Triticum sativum*	Caryopsis	Starchy endosperm and embryo
Aggregate Fruits			
Lotus	*Nelumbo nucifera*	Etaerio of achenes	Fleshy thalamus and seeds
Strawberry	*Fragaria vesca*	Etaerio of achenes	Fleshy thalamus and seeds
Custard apple	*Annona reticulata*	Etaerio of berries	Inner layer of pericarp and thalamus
Multiple or Composite Fruits			
Pineapple	*Ananas comosus*	Sorosis	Fleshy axis, bracts, fused perianth pericarp
Jack fruit	*Artocarpus heterophyllus*	Sorosis	Fleshy axis, bracts, perianth and seeds
Fig	*Ficus carica*	Synconus	Fleshy receptacle or thalamus

- All those chemical elements which are necessary for plant growth, are studied under plant nutrition.
- Plants capture solar energy and convert it into chemical energy through photosynthesis or carbon assimilation. During the process CO_2 and H_2O are transformed into simple carbohydrates with the liberation of O_2 in atmosphere. Oxygen is the by product of this reaction. The process of photosynthesis is represented as

$$6CO_2 + 6H_2O \xrightarrow[\text{green plant}]{\text{Light}} 6O_2 + \underset{\text{Glucose}}{C_6H_{12}O_6}$$

- The transport of food from leaves to its consumption centre (apices, roots, fruits etc) occurs through phloem tissues and is called as translocation of food.

GENETICS

Introduction

- It is the process of transfer of heredity characters from one generation to next through genes.
- A gene is the molecular unit of heredity (genetic vehicles) and are incorporated within the nucleus (as chromosomes of DNA + Histones).
- **G J Mendel** is known as the Father of Genetics.

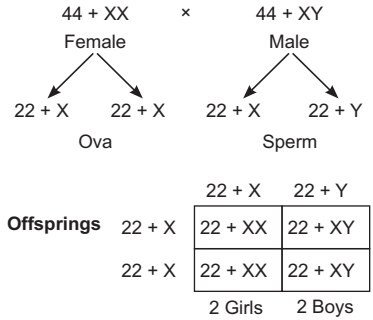
> - Term Genetics was first used by William Bateson in 1905.
> - Histones are considered as genes repressors as these mask the gene activity in nonspecific matter.

Mendel's Laws

- Mendel summarised his findings in three laws– the **Law of Dominance**, the **Law of Segregation** and the **Law of Independent Assortment**.
- **Karyotype** is the number and appearance of the chromosomes in the nucleus of a cell. The term is also used for the complete set of chromosome in a species of an individual organism. The karyotype of an individual when shown as a photograph then it is called an **idiogram**.

Sex Determination in Human

- In humans, each cell contains 46 chromosomes (44 autosome + 2 sex chromosome).
- In humans, male sex chromosome are XY, whereas female sex chromosome are XX.
- During gamete formation, in female all gametes contain only one type of chromosome, i.e. X and in male, half of the sperm contain X,–chromosome, while half contain Y- chromosome



Determination of Sex in Human Beings

- Thus, when a male gamete, i.e. sperm carrying X-chromosome fertilizes an ovum, the zygote develops into a female.
- When a sperm carrying (Y) chromosome fertilizes an ovum, the zygote developes into a male.
- Any increase or decrease in the number of sex chromosomes or autosomes causes genetic disorder.

Mutation

- It is the new sudden inheritable discontinuous variation which appear in an organism due to a permanent change in the genotype.
- *It is generally of two types*: **Germinal Mutations** Mutations appearing in reproductive cells are called germinal mutation. They are passed onto the offsprings.
- **Somatic Mutations** Mutation appearing in body cells (other than germ cells) are called somatic mutation. They are not passed onto the offsprings.

Some Hereditary Diseases

Haemophilia A

- It is the most common X-linked genetic disorder associated with serious bleeding. It is caused by a reduction in the amount or activity of coagulation factor VIII.

Colour Blindness

- It is an X-linked genetic disease.
- It is the inability to perceive differences between some of the colours under normal lighting conditions.

Sickle-Cell Anaemia

- It is a genetic blood disorder characterised by an abnormal, sickle-shape red blood cells.
- It caused by a mutation in the 11th pair of autosomal chromosome.

Turner's Syndrome (44+XO)

- It is due to the decrease in the sex-chromosome by 1 (monosomy). It happens by union of abnormal egg (egg with no sex chromosome– 22+0) and sperm having 'X' chromosome or normal egg with abnormal sperm (sperm with no sex chromosome– 22+0)
- Such persons are sterile females who have rudimentary undeveloped breast, short stature, webbed neck and abnormal intelligence.

Klinefelter's Syndrome (44+XXY)

- It is due to the increase in the sex chromosome number by 1 (trisomy). It is formed by the union of abnormal egg (XX) with normal Y sperm or normal egg with abnormal XY sperm.
- Such persons are non-functional male with undeveloped testes, mental retardation, knock kness and enlarged breast (gynaecomastia).

Down's Syndrome

- It is also called mongolism. It occurs due to increase in the chromosome number syndrome 1 (trisomy of 21st chromosome).
- It is characterised by open mouth, projected lower lip and IQ below 40.

Cri-du-chat Syndrome

- It is also called cri-du-chat syndrome because new born cries like a cat.
- It occurs due to the deletion of half part in the short arm of the chromosome number 5.

BIOTECHNOLOGY

- **Biotechnology** is a field of applied biology that involves the use of living things in engineering, technology, medicine and other useful applications.
- **Genetic Engineering** Insertion of a foreign gene fragment into another DNA molecule to produce DNA clones.
- **Gene Therapy** It is the insertion of genes into an individual's cells and tissue to treat diseases especially hereditary diseases.

Test Tube Baby (IVF)

- The IVF (in vitro fertilization) technology is a boon to childless couples.
- Test tube baby is fusion of ovum and sperm outside the body followed by implantation in uterus at 32 celled stage and further normal development to birth.
- First attempt to produce a test tube baby was made by an Italian scientist Dr. Petrucci in 1959. But this human embryo survived for only 29 days.
- The **World's first test tube baby** (a baby girl) named as Louise Joy Brown was born on 25th July, 1978 in Great Britain.
- India's first official test tube baby was born in Mumbai on 6th August, 1986. Her name is Harsha. However, first test tube baby of India and second of world was infact born on 3rd October, 1978. Her name is Durga. This baby was not accepted officially.

Cloning

- Cloning in biology is the process of producing similar populations of genetically identical individuals that occurs in nature when organisms such as bacteria, insects or plants reproduce asexually.

There are two types of cloning

1. **Reproductive Cloning** for the production of offspring.
2. **Therapeutic Cloning** It is used for development of embryonic stem cells.

- **Dolly** a sheep, the first mammal clone synthesized by **Dr. Ian Wilmut**, UK.
- **Noori** The world's first cloned Pashmina goat. Funded by World Bank, the clone project was jointly worked by SKUAST-K and Karnal based National Dairy Research Institute (NDRI).

Bt Crops

- Crop plants that contain genes for *Bt* toxins.
- *Bt* toxin gene has been cloned from the bacteria (*Bacillus thuringiensis*) and been expressed in plants to provide resistance to insects without the need of insecticides. e.g. *Bt*-cotton (first GM crop in India), *Bt*-corn, golden rice etc.

AGRICULTURE AND ANIMAL HUSBANDRY

Agriculture

- Agriculture provides food to population, the raw material to agro-based industries, and providing employment to public.
- **Green Revolution** is rapid increase in agriculture production.
- **Dr. NE Borlaug** is called *Father of Green Revolution* in world, while **Dr. MS Swaminathan** is called *Father of Green Revolution in India*.
- The periodicity of the season usually allows three harvest in a year.
    1. **Kharif Crop** Sown in June or early July and harvested by the end of September/October. e.g. rice, jowar, maize, cotton, jute, hemp, pulses etc.
    2. **Rabi Crop** Sowing is done in November and crops are harvested in March/April. e.g. wheat, gram and mustard.
    3. **Zaid Crop** These crops are grown between March and June. e.g. urad, moong, cucumber, watermelon, vegetables etc.

Animal Husbandry

- Animal husbandry is the science of rearing, caring, feeding, breeding, improvement and utilisation of domestic animals.
- The domestic animals kept for use or profit are called livestock.

Cattle (*Bos indicus*)

- Cattle include cow, bull, bullock or ox.
- They are used for agriculture, milk, transport etc.

Milk Breed	*Distribution*
Gir	Gujarat
Sahiwal	Punjab, Haryana, UP
Red Sindhi	Andhra Pradesh
Deoni	Andhra Pradesh

Buffalo (*Bos bubalus/Bubalus bubalis*)

- They are used for milk transport, meat, glue and gelatin.

Name	*Distribution*
Murrah	Punjab, Haryana and UP
Bhadawari	UP and MP
Jafarabadi	Gujarat
Surti	Rajasthan and Gujarat

Breed	*Distribution*
Gaddi and Chamba	Himachal Pradesh
Jamunapari	UP and MP
Beetal	Punjab
Marwari	Rajasthan

Breed	*Distribution*	*Uses*
Lohi	Punjab and Rajasthan	Wool and milk
Rampur-Bushair	UP and Himachal Pradesh	Brown coloured fleece
Nali	Haryana, Punjab and Rajasthan	Carpet wool
Bhakarwal	J&K	Under coat

COMPUTER

A computer is an electronic machine which stores, reads and processes data to produce meaningful information as output.

Components of Computer

The computer system comprises of the following four main components

Input Unit

- Input unit consists of those devices through which user can enter the data into a computer. Examples of input devices are Keyboard, mouse, scanner etc.

CPU (Central Processing Unit)

- CPU is considered as the 'Brain of Computer'. It is responsible for all the manipulations and processing of the data provided to the computer. It is further categorised basically into two main components
  - **Arithmetic Logical Unit** (ALU) This unit performs both arithmetical and logical operations. Arithmetic operations involves addition, subtraction, multiplication, division etc. and the logical operations involves AND, OR, NOT, NOR, NAND etc.
  - **Control Unit** (CU) It is an important part that instructs, maintains and controls the flow of information but does not stores the data. It tells the memory, ALU and I/O devices that how they have to respond to the program's instructions.

Memory Unit

- Memory unit stores the data and instructions, intermediate results or the processed data. It consists of two types
  - **Primary Memory** It is considered to be the main memory of computer that stores the data which is currently in use by the computer. Types of Primary Memory
    (a) **RAM** (Random Access Memory) It is a volatile memory, which losses the data when the power gets switched off. There are basically two types: Static RAM and Dynamic RAM.
    (b) **ROM** (Read Only Memory) It is a non-volatile memory, which retains the data even when the power gets switched off. There are basically three types: PROM, EPROM, EEPROM.
  - **Secondary Memory** The computer system uses secondary memory to store data, program instruction and information. It stores the data permanently. Types of Secondary Memory: Magnetic Tape, Magnetic Disk (Floppy Disk, Hard Disk), Optical Disc (CD, DVD, Blu-ray Disc) and Solid State drive (Pen drive, Memory cards).

Output Unit

- This unit contains those devices that provides the desired output (results) in the human acceptable format. Examples of output devices are monitor, printer, speakers etc.

Hardware

- Hardware can be defined as the physical components of a computer i.e. the parts that can be seen and touched. The four main categories of hardware are input devices e.g. keyboard, mouse etc, output devices e.g. printer, monitor etc, storage devices e.g. hard disk, floppy disk etc and Processing devices e.g. CPU etc.

Software

- The term software can be defined as the set of programs and procedures that enables a computer to perform a specific task or to process the information. Software can be classified into three types

System Software

- It is a set of one or more programs designed to control the operations of a computer system including hardware components and implementations of application software. e.g. Operating system, Device driver, Language Translator etc.

Utility Software

- It is a type of system software, which is used to support, secure and enhance the existing programs and data in the computer system. Examples of utility programs are antivirus software, backup software etc.

Application Software

- It is a set of one or more programs designed to carry out operations for a specific application. It cannot run on itself, but it is dependent on system software to get executed. e.g. word processors, spreadsheets, accounting programs etc.

Networking

Computer networking relates to the communication between a group of two or more computers linked together. Most common example of networking is Internet. According to scale or size, computer network can be categorised in three ways

- **Local Area Network** (LAN) Graphical area spread over 1km to 10km or within a same building.
- **Metropolitan Area Network** (MAN) Graphical area spread over a city or town.
- **Wide Area Network** (WAN) Graphical area spread over countries.

Security Threats

- **Worm** It is a self contained program and does not need to be a part of another program to propagate itself.
- **Spam** Spam is an unsolicited message sent over the Internet in the form of e-mails, to a large no. of users for the purpose of spreading malware, advertising publishing etc.
- **Spyware** It is a type of malicious software installed on computers and collects information about users without their knowledge and may send such information to another entity.
- **Malware** A software which is specifically designed to disrupt or damage a computer system. It is a superset of computer viruses, worms, spyware, trojan horses etc.
- **Virus** A virus is defined as a program or a piece of code that gets loaded onto the computer without users knowledge and replicates itself. e.g. Creeper, Stuxnet, Melissa, Conficker, etc.

INTERNET

It allows users to connect to server all around the world. It is a global system of interconnected computer networks, through which exchange of information such as data, news etc. Terms related to Internet

Antivirus

Antivirus is a software consisting of computer programs that attempt to identify, detect and prevent the malware from the computer.

- **HyperText Transfer Protocol** (HTTP) sends an HTTP command to the web server directing it to fetch and transmit the requested web page. It is used by the World Wide Web (WWW).
- **Uniform Resource Locator** (URL) specifies the address of a file and every file on the Internet has a unique address.
- **HyperText Markup Language** (HTML) It is a computer's language used to create hypertext documents for the WWW. Web pages are created using HTML.
- **Domain** are specific names used in URLs to identify particular Web pages.
- **IP Address** is a 32-bit numeric address separated by four number. Each number can be 0 to 255.

Wireless Communication

- It is the transfer of information over a distance without the use of electrical conductors or wires. e.g. GPS units, wireless computer mice, keyboards and headsets, satellite television etc.

Generations of Wireless Communication

- **0G** (The Zeroth Generation) 0G refers to pre-cellular mobile telephony technology in 1970s.
- **1G** (The First Generation) It is a wireless telephone technology and mobile telecommunication introduced in 1980s. 1G networks use analog signals as opposed to digital signals used by all the successive generations.
- **2G** (The Second Generation) It was commercially launched for the GSM (Global System For Mobile Communication) standard in 1991 by Radiolinja in Finland. It was allowed for enhanced data services and also introduced the Short Messaging Services (SMSs).
- **3G** (The Third Generation) It was introduced by NTT DoCoMo in Japan in 2001. Its data transfer rates are 384 K bits/sec to 2 M bits/sec. So, it allows for never before services like video calls, video conferencing, mobile, TV etc.
- **4G** (The Fourth Generation) TeliaSonera was the first operator in the world to commercially launch 4G in late 2009.

Bharti Airtel launched India's first 4G service using TDLTE (Time Division Long Term Evolution) technology in Kolkata on 10th April, 2012.

- **5G** is the fifth generation wireless technology for digital cellular networks that began wide deployment in 2019. As per the Telecom Regulatory Authority of India (TRAI), 5G is set to transform communication networks by 2020 and will bring massive growth in Indian economy by 2021. 5G will also support all spectrum types (licensed, shared, unlicensed) and bands (low, mid, high), a wide range of deployment models (from traditional macro-cells to hotspots), as well as new ways to interconnect.

Some Commonly Used Terms

- **Cache Memory** It is a temporary storage, where frequently accessed data can be stored for rapid access.
- **Registers** These are defined as the special memory units used by the CPU to speed up the rate of accessing information.
- **Operating System** It is a system software, consisting of an integrated set of programs that controls computer resources and provides common services for efficient execution of various application software.
- **Compiler** It is a computer program that transforms human readable source code into the Machine readable code at one go.
- **Interpreter** It transforms source code into the machine readable code by converting it line by line.
- **Assembler** It converts assembly language program into machine language program.
- **Modem** (Modulator-Demodulator) An electronic device used to convert computer (digital) electronic signals to communication channel (analog) electronic signals and *vice-versa*.
- **Cloud Computing** is the delivery of on-demand computing resources, everything from applications to data centres, over the Internet. e.g. Google.
- **Dual Core Processor** is the processing technology in which two processors are scheduled together and when one is busy the other takes over.
- **Biometric Device** A device used for user authentication that verifies some physical characteristics of a user such as the person's appearance, hand geometry etc.
- **Cryptography** It is a method of storing and transmitting data in a particular coded form so that only those can read and process it, for whom it is intended. It includes encoding and decoding of data.

Super Computers

A super computer can be defined as the most powerful computer in terms of performance and storage capacity. They are highly expensive and are employed for specialised applications such as for weather forecasting, several scientific researches etc.

Super Computer	*Year*	*Mft Company*
Param Siddhi	2020	CDAC Pune
Param Shivay	2019	IIT, BHU
Pratyush	2017	IITM, Pune
PARAM Kanchenjunga	2016	C-DAC and NIT Sikkim
PARAM ISHAN	2016	C-DAC and IIT Guwahati
Aaditya	2013	Indian Institute of Tropical Meteorology
PARAM YUVA II	2013	C-DAC, Pune
SAGA-220	2011	ISRO
ANUPAM-Adhya	2010-11	BARC
PARAM YUVA	2008	C-DAC, Pune
EKA	2007	Computational Research Laboratories, Pune
PARAM SARITA	2007	C-DAC, Pune

Name	*Year*	*Country*	*OS*
Frontera	2019	America	Linux
IBM Summit	2018	America	IBM
Sunway Taihulight	2016	China	Linux
Tianhe-2	2013	China	Kylin Linux
Titan	2012	America	Linux
Sequoia	2011	America	Linux
K-computer	2011	Japan	Linux
Mira	2010	America	Linux
Piz Diant	2009	Switzerland	Linux
Stampede	2008	America	Linux
JU Queen	2007	Germany	Linux
VULCAN	2005	America	Linux
SuperMUC	2002	Germany	Linux



GENERAL KNOWLEDGE

First in World (Male)

First Secretary General of United Nations	Trygve Lie (Norway)
First President of United States of America	George Washington
First President of the Republic of China	Dr Sun Yat-Sen (1912)
First Prime Minister of Great Britain	Robert Walpole (1715)
Pakistan's first Governor-General	Mohammed Ali Jinnah
First Ethnic-Indian Prime Minister of Fiji	Mahendra Choudhary
First American President to visit India	Dwight David Eisenhower
First Russian (Soviet) Prime Minister to visit India	Nikolai Bulganin
First European invader of Indian soil	Alexander, The Great
First man to go into space	Major Yuri Gagarin (USSR)
First man to walk in space	Alexai Leonov (Russia)
First space tourist	Dennis Tito (USA)
First man to set foot on the Moon	Neil Armstrong (USA)
First man to fly an aeroplane	Wright Brothers
First man to reach North Pole	Robert Peary
First man to reach South Pole	Roald Amundsen
First man to climb Mount Everest	Sherpa Tenzing Norgay and Sir Edmund Hillary (29th May, 1953)
First man to climb Mount Everest twice	Nawang Gombu (1965)
First blind man to scale Mount Everest	Erik Weihenmayer (25th May, 2001)
First person to sail around the world	Ferdinand Magellan
First deaf and dumb to cross the Strait of Gibralter	Taranath Shenoy (India)
First European to visit China	Marco Polo
First man to draw the map of Earth	Anaximander
First man to compile Encyclopaedia	Aspheosis (Athens)
First man to win Nobel Prize for Literature	Sully Prudhomme (France)
First man to win Nobel Prize for Peace	Henry Dunant (Switzerland) and Frederic Passy (France)
First man to win Nobel Prize for Physics	WK Roentgen (Germany)
First man to win Nobel Prize for Chemistry	Jacobus H Van't Hoff (Holland)
First man to win Nobel Prize for Medicine	AE Von Behring (Germany)
First man to win Nobel Prize for Economics	Ragnar Frisch (Norway) and Jan Tinbergen (Holland)
First and only black man ever to win singles Wimbledon Trophy	Arthur Ashe (USA)
First Asian to head the International Cricket Council	Jagmohan Dalmiya
First man to hit double century in One Day International match	Sachin Tendulkar (India)

First in World (Female)

First woman President of UN General Assembly	Vijaya Lakshmi Pandit (1953)
First woman President of a Country	Maria Estela Peron (*Argentina*)
First woman Prime Minister of a country	S Bhandarnayake (*Sri Lanka*)
First woman Prime Minister of England	Margaret Thatcher
First woman Prime Minister of any Muslim country	Benazir Bhutto (*Pakistan*)
First woman cosmonaut in space	Valentina Tereshkova (USSR)
First woman space tourist	Mrs Anousheh Ansari (*Irani American*)
First woman to reach the North Pole	Mrs Fran Phipps
First woman to set foot on North Pole	Ann Bancroft, USA
First woman to reach Antarctica	Caroline Mikkelsen
First woman to climb Mount Everest	Junko Tabei (*Japan*)
First woman in the world to cross the Strait of Gibraltar	Sophie Psilolignou (*Greek*)
First woman to chair US Central Bank 'Federal Reserve'	Janet Yellen
First UN Deputy Secretary-General	Louise Frechette (Canada)
First female Amputee to climb Mount Everest	Arunima Sinha
First woman CFO and MD of World Bank	Anshula Kant
First woman Vice President of USA	Kamla Harris
First woman chief economist for IMF	Gita Gopinath
First female ICC match raferee	GS Lakshmi
First astronauts to complete historic all female spacewalk	Christina Koch and Jessica Meir

First in World (Miscellaneous)

First country to make a written Constitution	United States of America
First country to appoint lokpal	Sweden
First country to ban capital punishment	Venezuela
First country to give voting right to woman	New Zealand
First country to impose carbon tax	New Zealand
First country to start VAT (Value Added Tax)	Brazil, Germany & Denmark
First country to issue plastic currency	Australia
First country to issue paper currency	China
First country to give constitutional status to animal rights	Switzerland
First country to implement family planning	India
First country to start Civil Services Competition	China
First country to ban Sunscreen (to protect Coral reef)	Palau
First country to print books	China
First country to sign nuclear agreement with India	France
First country to send human to Moon	United States of America
First space ship landed on Mars	Viking-I (August 20, 1975)
First space shuttle launched	Columbia
First country to launch satellite into space	Russia (former USSR)
First Radio Telescope Satellite launched into space	Japan
First city to be attacked with Atom Bomb	Hiroshima (Japan)
First country to win the Football World Cup	Uruguay (1930)
First country to host the Modern Olympic Games	Greece
First lamb created using DNA from an Adult Sheep	Dolly
First heart bypass operation by a Robot was carried out in	Germany
World's first cloned human baby	Eve
First country to declare a climate emergency	United Kingdom
First university of the world	Taxila University

Superlatives of the World
(The Largest, Biggest, Smallest, Longest, Highest)

Continent	Smallest—Australia, Largest—Asia
Country	Largest (in area)—Russia Largest (in population)—China
Mountain Range	Longest—Andes, South America Highest—Himalayas
Mountain Peak	Highest—Mount Everest (Nepal) (8848 m)
Desert	Largest hot desert—Sahara, Africa Largest cold desert (outside the poles)—Gobi Desert (Mongolia and China)
River Longest	Nile (6690 km)
Busiest River	Rhine (Germany)
Largest Basin	Amazon Basin (South America)
Gorge	Largest—Grand Canyon, on the Colorado river, USA
Waterfall	Highest—Salto Angel Falls (*Venezuela*)
Delta	Largest—Sundarbans, India
Gulf	Longest—Gulf of Mexico
Island	Largest—Greenland (*renamed Kalaallit Nunaat*)
Bay	Largest—Hudson Bay, Canada
Peninsula	Largest—Arabia
Volcano	Largest—Mauna Loa (Hawaii Islands); Highest-ojos del Salado, Andes, Argentina-Chile (6885m)
Sea (Inland)	Largest—Philippines sea
Ocean	Deepest and Biggest—The Pacific
Lake	Deepest—Baikal (Siberia) Highest—Titicaca (Bolivia) Largest (Fresh Water)—Lake Superior, USA Largest (Artificial)—Lake Kariba (between Zambia and Zimbabwe)
World's Rainiest Spot	Mawsynram—(Meghalaya)
Sea Port	Largest—Shanghai (China)
Airport	Largest—Daxing International Airport, Beijing, (China) Highest—Daocheng Yading Airport, Garzi, Tibet (*China*)
Airliner	Largest — Airbus A380
Dam	Tallest—Jinping-I Dam (China) Longest—Hirakud Dam (*Odisha*), India Largest (Concrete)—Three Gorges Dam (China)
Coral formation	Largest—The Great Barrier Reef (*Australia*)
Canal	Busiest—Baltic White Sea Canal Longest—Beijing-Hangzhou Grand Canal
Animal	Most Intelligent—Chimpanzee
Archipelago	Largest (area)—Malay archipelago
	Largest (number)—Norway archipelago
Asteroid impact zone	Largest—Australia
Dome	Longest—World Peace Monument Dome (Pune)
Country with the longest length of Coastline	Canada

Plateau	Largest—Pamir Plateau (Asia)
Motorable Road	Highest—Umling La Road, Ladakh (19300 ft)
Country sharing border with others	Maximum—China (14) and Russia (14)
Country with population density	Highest—Macau
Bird	Wingless—Kiwi. Largest—Ostrich Largest (Sea)—Albatross Fastest—Spine-tailed swift Smallest—Humming bird
Mammal	Largest—Blue Whale Smallest—Bumblebee Bat
Park	Largest—National Park, Greenland
Railway	Longest—Trans-Siberian Railway
Railway Platform	Longest—Gorakhpur, India
	Largest—Grand Central Terminal, New York (USA)
Tunnel	Longest and Largest (Canal)—Le Rove Tunnel (*South of France*) Longest (Railway)—Gotthard Base Tunnel (Switzerland)
Bridge	Longest—Danyang-Kunshan Grand Bridge (*Beijing-Shanghai High-speed Railway*)
Building	Tallest—Burj Khalifa, Dubai (828 m)
Minar (Free standing)	Tallest—Great Hassan II Mosque, Casablanca, Morocco
Statue	Tallest—Statue of Unity, Gujarat, India
Tower	Tallest—Tokyo Skytree, Tokyo (*Japan*)
Mosque	Largest—Masjid at Haram (*Mecca*)
Temple	Largest—Angkor Vat (*Combodia*)
Church	Longest—Basilica of St Peter, Vatican City, Rome (*Italy*)
Museum	Biggest—Smithsonian Institution, Washington DC
Place	Coldest—Verkhoyansk (Siberia) Temperature 85°C. Hottest—Al-Aziziyah (Libya, Africa) 136°F Driest—Atacama Desert Chile (*South America*)
Stadium	Largest—Strahov Stadium in Prague (*the Czech Republic*)
Wall	Longest—Great Wall of China
Capital City	Highest—La Paz (*Bolivia*)
City	Highest—Wen Chuan (*Tibet, China*) Largest (in population)—Tokyo Biggest (in area)—Hulunbuir, China
Day	Longest—21st June (*in Northern Hemisphere*) Shortest—22nd December (*in Northern Hemisphere*)
Substance	Hardest—Wurtzite boron nitride
Metal	Lightest—Lithium
	Heaviest—Osmium
	Costliest—Californium 252 ($27 million per gram)
Diamond	Largest—The Cullinan (over 1.5 lb) Largest (Mine)—Kimberley (South Africa)
Melting Point	Highest—Tungsten, 34100°C
Gas	Lightest—Hydrogen

Countries with Capitals and Currencies

Country	*Capital*	*Currency*
Afghanistan	Kabul	Afghani
Albania	Tirana	Lek
Algeria	Algiers	Algerian Dinar
Angola	Luanda	New Kwanza
Argentina	Buenos Aires	Peso
Armenia	Yerevan	Dram
Australia	Canberra	Australian Dollar
Austria	Vienna	Euro
Azerbaijan	Baku	Manat
Bangladesh	Dhaka	Taka
Belarus	Minsk	Ruble
Belgium	Brussels	Euro
Bhutan	Thimphu	Ngultrum
Brazil	Brasilia	Real
Cambodia	Phnom-Penh	Riel
Canada	Ottawa	Canadian Dollar
Chile	Santiago	Peso
China	Beijing	Yuan/Renminbi
Colombia	Santa Fede Bogota	Peso
Denmark	Copenhagen	Krone
East Timor	Dili	US Dollar
Egypt	Cairo	Egyptian Pound
Ethiopia	Adis Ababa	Birr
Finland	Helsinki	Euro (Formerly Morka)
France	Paris	Euro
Georgia	Tbilisi	Lari
Germany	Berlin	Euro
Greece	Athens	Euro
Hungary	Budapest	Florint
Iceland	Reykjavik	Krona
India	New Delhi	Rupee
Indonesia	Jakarta	Rupiah
Iran	Tehran	Toman
Iraq	Baghdad	Dinar
Ireland	Dublin	Euro
Israel	Jerusalem	Shekel
Italy	Rome	Euro
Japan	Tokyo	Yen
Kazakhstan	Nur-Sultan	Tenge
Kenya	Nairobi	Shilling
Kuwait	Kuwait City	Kuwait Dinar
Kyrgystan	Bishkek	Som
Laos	Vientiane	Kip
Libya	Tripoli	Libyan Dinar
Latvia	Riga	Euro
Lebanon	Beirut	Pound
Malaysia	Kuala Lumpur	Ringg it
Maldives	Male	Rufiyaa
Mauritius	Port Louis	Rupee
Mexico	Mexico City	Mexican Peso
Mongolia	Ulaanbaatar	Tugrik
Morocco	Rabat	Dirham
Mozambique	Maputo	Metical
Myanmar	Naypyidaw	Kyat
Namibia	Windhoek	Namibian Dollar
Nepal	Kathmandu	Rupee
Netherlands	Amsterdam	Euro
New Zealand	Wellington	New Zealand Dollar
Nigeria	Abuja	Naira
North Korea	Pyongyang	Won (WPW)
Norway	Oslo	Krone
Pakistan	Islamabad	Rupee
Phillippines	Manila	Peso
Poland	Warsaw	Zloty
Portugal	Lisbon	Euro
Qatar	Doha	Riyal
Russia	Moscow	Ruble
Saudi Arabia	Riyadh	Riyal
Singapore	Singapore	Dollar
Somalia	Mogadishu	Somali Shilling
South Africa	Pretoria, Cape Town, Bloemfortein	Rand
South Korea	Seoul	Won (KRW)
South Sudan	Juba	Sudanese Pound
Spain	Madrid	Euro
Sri Lanka	Colombo, Sri Jayawardene-pura Kotte	Rupee
Sudan	Khartoum	Pound
Sweden	Stockholm	Krona
Switzerland	Bern	Swiss Franc
Taiwan	Taipei	New Taiwan Dollar
Tajikistan	Dushanbe	Tajikistani Somani
Thailand	Bangkok	Baht
Turkey	Ankara	Lira
UK	London	Pound Sterling
Ukraine	Kiev	Hryvnia
USA	Washington DC	Dollar
Venezuela	Caracas	Bolivar
Zambia	Lusaka	Kwacha
Zimbabwe	Harare	US Dollar

TOP 5

Largest and Smallest Countries

Largest Country (Area wise)	*Largest Country* (Population wise)	*Smallest Country* (Area wise)	*Smallest Country* (Population wise)
Russia	China	Vatican City	Vatican city
Canada	India	Monaco	Tuvalu
United States	USA	Nauru	Nauru
China	Indonesia	Tuvalu	Palau
Brazil	Brazil	San Marino	San Marino

Languages and Religions

Languages Spoken

Language	*Speaker*
Mandarin Chinese	918 million
Spanish	480 million
English	312-380 million
Hindi	341 million
Bengali	228 million

Religions of the World

Religion	*Member*	*Percentage*
Christianity	2.4 billion	31.5%
Islam	1.9 billion	22.32%
Hinduism	1.1 billion	14%
Buddhism	0.52 billion	6%
Sikhism	0.30 billion	0.36%

Geographical Epithets

Geographical Epithet	*Location*
Blue Mountains	Nilgiri Hills
City Beautiful	Chandigarh
City of Golden Gate	San Francisco
City of Magnificent Buildings	Washington
City of Smoke	Chicago
City of Seven Hills	Rome
City of Skyscrapper	New York
Cockpit of Europe	Belgium
Dark Continent	Africa
Uranium City	Saskatchewan (Canada)
Forbidden City	Lhasa (Tibet)
Windy City	Chicago
Land of Golden Pagoda	Myanmar
Garden City	Chicago
Gift of the Nile	Egypt
Granite City	Aberdeen
Hermit Kingdom	North Korea
Holy Land	Palestine
Island Continent	Australia
Island of Cloves	Zanzibar

Geographical Epithet	*Location*
Island of Pearls	Bahrain
Key to the Mediterranean	Gibraltar
Continent of Birds	South America
Land of Golden Fleece	Australia
Land of Maple	Canada
Land of Morning Calm	South Korea
Land of the Midnight Sun	Norway
Land of the Rising Sun	Japan
Land of the Thunderbolt	Bhutan
Land of Thousand Lakes	Finland
Land of White Elephant	Thailand
Coffee Port of the world	Santos
Pearl of the Pacific	Guayaquil Port of Ecuador
Roof of the World	The Pamirs, Central Asia
Spice Garden of India	Kerala
Sugar Bowl of the World	Cuba
Whiteman's Grave	Guinea Coast of Africa
Playground of Europe	Switzerland
Land of Kangaroo	Australia

Geographical Discoveries

Discovery	*Discoverer*
America	Christopher Columbus
China	Marco Polo
Australia	James Cook
New Foundland	Cabot Sebastian
Hudson Bay	Henry Hudson
Brazil	Pedros Alvares Cabral
Hawaian Island	James Cook
Cape of the Good Hope	Bartolomeu Dias

Discovery	*Discoverer*
North Pole	Robert Peary
South Pole	Roald Amundsen
Mount Everest	Edmund Hillary
Sailed Around the World	Magellan
Sea route to India via Cape of Good Hope	Vasco-da-Gama
Planets	Kepler
Solar System	Copernicus

Important Monuments of Major Countries

Monument	*Country*
Statue of Liberty (*New York*)	*USA*
Kremlin (*Moscow*)	*Russia*
Great Wall of China	*China*
Emperial Palace (*Tokyo*)	*Japan*
Eiffel Tower (*Paris*)	*France*
Leaning Tower of Pisa	*Italy*
Opera House (*Sydney*)	*Australia*
Kinder Disk	*Denmark*

Monument	*Country*
Christ the Redeemer	*Brazil*
Machu Picchu	*Peru*
Taj Mahal (*Agra*)	*India*
Tugu Negara (*Kuala Lumpar*)	*Malaysia*
The Great Sphinx (*Giza*)	*Egypt*
Pyramid (*Giza*)	*Egypt*

Major News Agencies of the World

Agency	*Country*
Associated Press (AP)	*USA*
Novosti	*Russia*
Interfax	*Russia*
China News Service, Xin Hua	*China*
Allgemeiner Deutschar Nachrichtendienst	*Germany*
Deutsche Presse Agentur	*Germany*
Agence France Presse (AFP)	*France*
Agence Parisienne de Presse	*France*
Kyodon Tsushin	*Japan*
Reuters	*UK*
Exchange and Telegraph Company	*UK*
Australian Associated Press	*Australia*
Australian United Press	*Australia*
Agenzia Nazionale Stampa Associate (ANSA)	*Italy*
Europa Press	*Spain*

Agency	*Country*
Algemeen Nederlands Persbureau	*Netherlands*
Associated Israel Press (AIP)	*Israel*
Press Trust of India (PTI)	*India*
United News of India (UNI)	*India*
Islamic Republic News Agency	*Iran*
Petra	*Jordan*
WAFA	*Palastine*
Middle East News Agency	*Egypt*
Anadol Ajansi	*Turkey*
Antara	*Indonesia*
Malaysian National News Agency	*Malaysia*
Associated Press of Pakistan	*Pakistan*
Bangladesh Sangbad Sangstha	*Bangladesh*
Indonesian National News Agency	*Indonesia*
Kenya News Agency	*Kenya*

National Emblem of Major Countries

Country	*Emblem*
Australia	Kangaroo
Bangladesh	Water Lily
Canada	White Lily, Maple Leaf
Chile	Candor and Huemul
Denmark	Beach
France	Lily
Germany	Corn Flower
India	Lion Capital of Ashoka
Iran	Perso Arabic Script of Arabic word 'Allah'
Italy	White Lily
Japan	Chrysanthemum
Netherlands	Lion
New Zealand	Southern Cross, Kiwi, Fern
Norway	Lion
Pakistan	Crescent and Star
Spain	Eagle
UK	Rose
USA	Great Seal of United States

Legislatures of the World

Country Name	*Legislature Name*
Afghanistan	Shora
Albania	People's Assembly
Algeria	National People's Assembly
Australia	Parliament
Austria	National Assembly
Bangladesh	Jatia Sansad
Bhutan	Tasongadu
Botswana	National Assembly
Brazil	National Congress
Britain	Parliament (*House of Commons and House of Lords*)
Canada	Parliament
China, Mainland	National People Congress
Colombia	Congress
Denmark	Folketing
Egypt	People's Assembly
France	Parliament
Germany	Bundestag
India	Sansad (Parliament)
Indonesia	People's Consultative Assembly
Iran	Majilis
Iraq	National Assembly
Ireland	Orieachtas
Israel	Knesset
Japan	Diet
Kenya	National Assembly
Korea (North)	Supreme People's Assembly
Korea (South)	National Assembly
Libya	General People's Congress
Malaysia	Parliament (*Dewan Rakyat, Dewan Negara*)
Mongolia	Great People's Khural
Nepal	Rashtriya Panchayat
Netherlands	Staten Generaal
New Zealand	Parliament (*House of Representatives*)
Norway	Storting
Pakistan	National Assembly
Romania	Grand National Assembly
Russia	Duma
South Africa	Parliament
Spain	Cortes, Gernerales
Switzerland	Federal Assembly
Syria	People's Council
Turkey	Grand National Assembly
USA	Congress
Zambia	National Assembly

Major Newspapers of the World

Newspaper	*Country*	*Newspaper*	*Country*
The Sydney Morning Herald	*Australia*	People's Daily	*China*
The Age	*Australia*	Yomiuri Shimbun	*Japan*
Globe and Mail	*Canada*	The New Zealand Herald	*New Zealand*
The Gazette	*Canada*	The Press	*New Zealand*
Le Monde Dawn	*Paris (France)*	The Hindu	*India*
Dawn	*Pakistan*	The Scotsman	*UK*
Die Welt	*Germany*	The Guardian	*UK*
The Times	*Britain*	The Herald	*UK*
The Sun	*Britain*	The Courier	*UK*
New York Times	*USA*	Merdeka	*Indonesia*
Washington Post	*USA*	Pravada	*Russia*

Official Books of Major Countries

Book	*Description*
Blue Book	An official report of the British Government.
Green Book	An official publications of Italy and Iran.
Grey Book	An official reports of the Governments of Japan and Belgium.
Orange Book	An official publications of the Government of Netherlands.
White Book	An official publications of China, Germany and Portugal.
Yellow Book	French official book.
White Paper	An official paper of the Governments of Britain and India on a particular issue.
Joint Paper	The point report of two or more than two governments.

Some Important Symbols or Signs

Thing	*Symbol*
Pen	Symbol of culture and civilisation
Lotus	Culture and civilization
Red Cross	Medical aid and Hospital
Red Flag	Revolution; also sign of danger
Black Flag	Symbol of protest
Yellow Flag	Flown on ships or vehicles carrying patients suffering from infectious diseases
White Flag	Sign of truce, Symbol of peace
Flag flown upside down	Symbol of distress
Flag flown at half mast	Symbol of national mourning
Pigeon or Dove	Symbol of peace
A blindfolded woman holding a balanced scale	Symbol of justice
Black strip on face arm	Sign of mourning or protest
One skull on two bones crossing each other diagonally	Sign of danger
Wheel (Chakra)	Symbol of progress
Olive Branch	Symbol of peace

Fathers of Various Fields

Field	*Father*
Atom Bomb	Dr Robert Oppenheimer
Aviation	Alberto Santos Dumont
Railways	George Stephenson
Chemistry	Robert Boyle
Modern Chemistry	Antoine Lavoisier
Economics	Adam Smith
Geography	Eratosthenes
Sociology	Auguste Comte
Political Science	Aristotle
Modern Philosophy	Rene Descartes
Psychology	Wilhelm Maximilian Wundt
English Poetry	Geoffrey Chaucer
Greek Tragedy	Aeschylus
Immunology	Edward Jenner
Genetics	Gregor Johann Mendel
Biology	Aristotle
Microbiology	Antonie Van Leeuwenhock

Field	*Father*
Anatomy	Andreas Vesatius
Nuclear Physics	Ernest Rutherford
Modern Physics	Galileo Galilei
Sanskrit Grammar	Panini
Geometry	Euclid
Mathematics	Archimedes
Internet	Vint Cerfond Bob Kahn
Robotics	Al-Jazari
Computers	Charles Babbage
Artificial Intelligence	John Mc Carthy
Modern Observational Astronomy	Galileo Galilei
Science	Galileo Galilei
Modern Science	Galileo Galilei
Indian Nuclear Science	Homi Jahangir Bhabha
Nanotechnology	Richard Errett Smalley

Intelligence/Detective Agencies of the World

Detective Agency	*Country*	*Detective Agency*	*Country*
Central Intelligence Agency (CIA),Federal Bureau of Investigation (FBI)	*USA*	Research and Analysis Wing (RAW), Intelligence Bureau (IB)	*India*
Komitet Gosudarst Vennoy Bezopashosti (KGB)/Main Intelligence Agency (GRO)	*Russia*	National Intelligence Agency	*South Africa*
Central External Liaison Department	*China*	Australian Security and Intelligence Organisation	*Australia*
Naikaku Joho Chosashitu (NAICHO)	*Japan*	SAVAK (Sazamane Etelaat va Amniyate Kechvar)	*Iran*
DGSE (Director General de la Securite Exterieure)	*France*	General Security Directorate	*Iraq*
MI (Military Intelligence)-5 and 6, Special Branch, Joint Intelligence Organisation	*UK*	Al-Mukhabarat Al-Ammah	*Egypt*
MOSSAD	*Israel*	Inter Services Intelligence (ISI)	*Pakistan*

The Seven Wonders of the World

Ancient World	*Medieval World*	*Modern World*
Colossus of Rhodes	Colossus of Rome	Great Pyramid of Giza
Lighthouse of Alexandria	Catacombs of Alexandria	Hagia Sophia (*Istanbul*)
Hanging Gardens of Babylon	The Great Wall of China	Leaning Tower of Pisa (*Italy*)
Temple of Artemis at Ephesus	The Pagoda of Nanking	Taj Mahal (*Agra, India*)
Great Pyramid of Giza	Leaning Tower of Pisa	Washington Monument
Mausoleum at Halicarnassus	Sancta Sophia of Constantinople	Eiffel Tower (*Paris, France*)
Statue of Zeus at Olympia	Stonehenge	Empire State Building

The 'New' Wonders of Modern World

Chichen Itza, Mexico	Petra, Jordan
Christ Redeemer, Brazil	The Roman Colosseum, Italy
The Great Wall of China	The Taj Mahal, India
Machu Picchu, Peru	Great Pyramid of Giza, Egypt (Honorary Status)

First in India (Male)

First Governor-General of India	William Bentinck (1833-35)
Governor-General of Independent India	Lord Lewis Mountbatten
First and Last Indian Governor-General of Free India	C. Rajagopalachari (1948-1956)
The first Muslim President of India	Dr Zakir Hussain
The first Prime Minister of India who resigned before the full term	Morarji Desai
First person to stay in Rashtrapati Bhawan	Lord Irwin
First President of National Congress	Womesh Chandra Bonnerjee
The first Home Minister of India	Sardar Vallabhbhai Patel
The first Indian Judge of International Court of Justice	Dr Nagendra Singh
Indian Managing Director of World Bank	Gautam Kazi
First Ex-CJI appointed as governor	P Sathasivam, as Governor of Kerala
Governor of Reserve Bank of India	Sir Osborne Smith
The first Indian to join the ICS	Satyendra Nath Tagore
The first Field Marshal of India	SHFJ Manekshaw
The first Indian Commander-in-Chief of India	General KM Karippa
The first Indian Naval Chief	Vice Admiral RD Katari
The first Indian to win the Noble Prize	Rabindranath Tagore (1913)
The first Indian to get Noble Prize in Physics	CV Raman
The first Indian to receive Noble Prize in Economics	Amartya Sen
The first person of Indian origin to get Noble Prize in Medicine	Hargobind Khorana
The first Indian to receive Bharat Ratna Award	Dr Radhakrishnan
First Sportsperson to receive Bharat Ratna	Sachin Tendulkar
The first person to receive Magsaysay Award	Acharya Vinoba Bhave
First cricketer to get Padma Bhushan	C.K. Nayudu
First Indian to get the Grammy Award	Pandit Ravishankar
The first person to receive Stalin Peace Prize	Saifuddin Kitchlew
The first person to receive Jnanpith Award	Sri Shankar Kurup
The first Indian Pilot	JRD Tata (1929)
First Indian to swin across the English Channel	Mihir Sen
First Indian to score triple century in test cricket	Virender Sehwag
First viceroy of India	Lord Canning (1858-62)
First minister of education in India	Dr. Abul Kalam Azad
First speaker of Lok Sabha	G.V. Mavalankar (1952-57)
First Indian member of British House of Commons	Dadabhai Naoroji
FIrst Indian cricketer to score a test century	Lala Amarnath (1933)
First chief election commissioner of India	Sukumar Sen
First chief of Defence Staff India	General Bipin Rawat
First Lokpal of India	Pinaki Chandra Ghose

First in India (Female)

The first woman Prime Minister	Indira Gandhi
The first woman Minister in Government	Rajkumari Amrit Kaur
The first woman Judge in Supreme Court	Fathima Beevi
The first woman Chief Justice of High Court	Leela Seth
The first woman President of United Nations General Assembly	Vijaya Laxmi Pandit
The first woman Chief Minister of an Indian State	Sucheta Kripalani
The first woman Governor of a State in free India	Sarojini Naidu
The first woman Chairman of Union Public Service Commission	Rose Millian Bethew
The first woman Director General of Police	Kanchan C Bhattacharya
The first woman IPS Officer	Kiran Bedi
The first woman President of Indian National Congress	Mrs Annie Besant
The first woman Judge	Anna Chandy
The first woman Barrister	Cornelia Sorabjee
The first woman Honours Graduate	Kamini Roy
The first lady to become Miss World	Reita Faria
The first woman Chairperson of Indian Airlines	Sushma Chawla
The first woman to receive Noble Prize	Mother Teresa
The first woman Airline Pilot	Durga Banerjee
The first woman to receive Bharat Ratna	Mrs Indira Gandhi
The first woman to receive Jnanpith Award	Ashapurna Devi
The first woman to receive Ashoka Chakra	Nirja Bhanot
The first woman Asian Games Gold Medal Winner	Kamaljit Sandhu
The first woman Olympic Medal Winner	Karnam Malleswari
The first woman to climb Mount Everest	Bachhendri Pal
The first woman to climb Mount Everest twice	Santosh Yadav
The first woman to cross English Channel	Aarti Saha
First woman doctor	Kadambini Ganguly
First test tube baby	Durga (Kanupriya Agarwal 1978)
First woman to reach Seven Summits of the World	Premlata Agrawal
First woman chief election commissioner of India	VS Ramadevi (1990)
First woman speaker of Lok Sabha	– Meira Kumar (2009)
First woman full time Defence Minister	– Nirmala Sitharaman
First woman to go into space	Kalpana Chawla
First to win Silver in Olympics (Badminton)	P V Sindhu
First woman President	Pratibha Patil
First Indian woman fighter pilot to fly a fighter jet	– Avani Chaturvedi
First Indian naval woman pilot	Shubangi Swaroop
First Indian to win gold medal on International Track	Hima Das

First in India (Miscellaneous)

The first Bank	Bank of Hindustan (1770)
First general Post Office of India	Madras (GPO 1786)
First Railway and Transportation University	Vadodara, Gujarat
First telephone line introduced in India	1851
First talkie film of India	Alam Ara (1931)
First battle tank of India	Arjun
First satellite of India	Aryabhatta
The first Indian State to implement the Panchayati Raj System	Rajasthan
District to become India's first totally electrified district	Palakkad (Kerala)
City to have an e-court	Ahmedabad
First Indian to win individual olympic gold	Abhinav Bindra
First Lunar Probe	Chandrayaan–1 (22nd October, 2008)
First Dedicated Military Satellite	Rukmini (GSAT-7)
First Mars Orbiter Mission	Mangalyaan
First Central Agricultural University	Imphal (Manipur)
First District with complete rural Broadband coverage	Idukki (Kerala)
First state to attain 100%. Primary literacy level	Kerala
First Open Defecation Free (ODF) State	Sikkim

Superlatives India
(Biggest, Highest, Largest, Longest, Smallest etc)

The highest peak	Godwin Austin, K-2 (8611 m)
The longest river	The Ganga (2525 km)
The longest river of Southern India	Godavari
The longest canal	Indira Gandhi Canal or Rajasthan Canal (*Rajasthan*)
The longest dam	Hirakud Dam (*Odisha*)
The highest dam	Tehri Dam (*Uttarakhand*)
The longest sea beach	Marina Beach (*Chennai*)
The highest lake	Devtal Lake (*Uttarakhand*)
The largest lake	Wular Lake (*Jammu and Kashmir*)
The largest saline water lake	Chilka Lake (*Odisha*)
The largest fresh water lake	Wular Lake (*J & K*)
The largest artificial lake	Govind Sagar (*Rihand Dam*)
The biggest river islands	Majuli, Brahmaputra river (*Assam*)
The highest waterfall	Kunchikal falls, Shimoga (*Karnataka*)
The deepest river valley	Bhagirathi and Alaknanda
The longest river bridge	Dhola-Sadiya Bridge, Lohit River (9.15 km)
The longest sea bridge	Bandra-Worli Sea Link
The largest cantilever bridge	Rabindra Setu or Howrah Bridge (*Kolkata*)
The state with longest coastline	Gujarat
The longest river without delta	Narmada
The largest delta	Sunderbans (West Bengal)
The longest railway platform	Gorakhpur
The longest tunnel	Dr. Syama Prasad Mukherjee Tunnel (*Chenani-Nashiri*), Jammu and Kashmir

The highest road	Road at Umling La (*Ladakh*)
The longest corridor	Corridor of Ramnathswami Temple at Rameshwaram (*Tamil Nadu*)
The highest airport	Leh Airport (*Ladakh*)
The largest desert	Thar (*Rajasthan*)
The largest zoo	Zoological Garden (*Kolkata*)
The biggest stadium	Yuva Bharti (Salt Lake) Stadium Kolkata
The largest auditorium	Sri Shanmukhananda Hall (*Mumbai*)
The tallest TV tower	Rameswaram TV Tower (1060 ft) (*Tamil Nadu*)
The largest gurudwara	Golden Temple, Amritsar
The largest cave temple	Kailash Temple (*Ellora, Maharashtra*)
The largest mosque	Jama Masjid (*Delhi*)
The longest train	Vivek Express (Dibrugarh to Kanyakumari)
The largest animal fair	Sonepur (*Bihar*)
The largest cave	Amarnath (J & K)
The highest gateway	Buland Darwaza, Fatehpur Sikri (*UP*)
The tallest statue	Statue of Unity, Gujarat, India
The most populous city	Mumbai (*Maharashtra*)
The biggest church	Saint Cathedral at Old Goa (*Goa*)
The highest battle field	Siachen Glacier
The longest national highway	NH-44 (*Sri Nagar to Kanyakumari*)

Famous Nicknames of Important Persons

Person	*Nickname*
Abdul Gaffar Khan	Badshah Khan; Frontier Gandhi
Bal Gangadhar Tilak	Lokmanya
CF Andrews	Deenabandhu
CN Annadurai	Anna
CR Das	Deshabandhu
C Rajagopalachari	Rajaji
Dadabhai Naoroji	Grand Old Man of India
Jawaharlal Nehru	Chacha, Panditji
Jayaprakash Narayan	Loknayak
Lal Bahadur Shastri	Man of Peace
Lala Lajpat Rai	Punjab Kesari; Lion of Punjab (Sher-i-Punjab)

Person	*Nickname*
MK Gandhi	Bapu; Mahatma, Father of the Nation
MS Golwalkar	Shri Guruji
Madan Mohan Malaviya	Mahamana
Rabindranath Tagore	Gurudev
Rajinder Singh	Sparrow
Sarojini Naidu	Nightingale of India
Sheikh Mohammad Abdullah	Lion of Kashmir (Sher-i-Kashmir)
Sheikh Mujibur Rahman	Bangabandhu
Subhash Chandra Bose	Netaji
T Prakasam	Andhra Kesari
Vallabhbhai Patel	Iron Man of India, Bismarck of India

Crematorium of Famous Person (India)

Crematorium	*Persons*
Samata Sthal	Jagjeevan Ram
Karma Bhumi	Dr Shankar Dayal Sharma
Mahaprayan Ghat	Dr Rajendra Prasad
Raj Ghat	Mahatma Gandhi
Vijay Ghat	Lal Bahadur Shastri
Kisan Ghat	Chaudhary Charan Singh
Smriti Sthal	IK Gujral
Sadaiv Atal	Atal Bihari Vajpayee

Crematorium	*Persons*
Veer Bhumi	Rajiv Gandhi
Ekta Sthal	Giani Zail Singh, Chandra Shekhar, P.V. Narasimha
Uday Bhoomi	KR Narayanan
Shanti Van	Jawaharlal Nehru
Shakti Sthal	Indira Gandhi
Abhay Ghat	Morarji Desai
Chaitya Bhoomi	Dr BR Ambedkar

Some Great Works Associated with Famous Persons (India)

Father of Sanskrit Grammar	Panini
Founder of Anand Van	Baba Amte
Founder of 'Auroville Ashram' (Puducherry)	Aurobindo Ghosh
Founder of Shantiniketan	Rabindranath Tagore
Founder of Vishwabharati	Rabindranath Tagore
Founder of Paunar Ashram	Vinoba Bhave
Founder of Bhudan Movement	Vinoba Bhave
Founder of Golden Temple	Guru Arjan Dev
Founder of Khalsa Panth	Guru Gobind Singh

Major Newspapers in India

Newspaper	*Published From*	*Language*
Economic Times	Mumbai, Delhi	English
Hindustan Times	Delhi, Patna	English
Deccan	Bangalore	English
Mid Day	Mumbai	English
National Herald	Lucknow, Delhi, Kolkata	English
Pioneer	Lucknow, Delhi, Kanpur	English
Search Light	Patna	English
The Hindu	Bangaluru, Chennai, Coimbatore, Delhi	English
The Indian Express	Delhi, Mumbai, Chennai, Lucknow, Madurai, Ahmedabad	English
The Statesman	Kolkata, New Delhi	English
The Times of India (*Largest English circulating newspaper in the world*)	New Delhi/Mumbai/ Ahmedabad	English
The Tribune	Ambala, Chandigarh	English
Aaj	Kanpur, Varanasi	Hindi
Amar Ujala	Allahabad/Agra/Jhansi/ Meerut	Hindi
Hindustan	Delhi/Patna	Hindi
Nav Bharat Times	New Delhi/Mumbai	Hindi
Amrit Bazar Patrika	Kolkata	Bengali
Anand Bazar Patrika	Kolkata	Bengali
Yugantar	Kolkata	Bengali
Matrabhoomi	Cuttack	Odiya
Akali Patrika	Jallandhar	Punjabi
Tej	Delhi	Urdu
Dainik Bhaskar	Bhopal	Hindi

PRINCIPAL LANGUAGES OF INDIA

India has 22 languages which have been given the offficial language status.

Assamese It is an Indo-Aryan language and is the official language of Assam.

Bengali It is one of the leading Indo-Aryan language and is the official language of West Bengal.

Gujarati It is an Indo-Aryan language and is the official language of Gujarat.

Hindi The largest spoken Indo-Aryan language.
It is the official language of the Government of India.
Various dialects of Hindi are Khariboli, Brajbhasha, Bundeli, Awadhi, Marwari, Maithili and Bhojpuri.
In 6 states and UTs, Hindi is the official language.

Kannada It belongs to the Dravidian family and is the official langauge of Karnataka.

Kashmiri It is an Indo-Aryan language.
It is the official language of Jammu and Kashmir.

Konkani It is the official language of Goa and is spoken by thousands of Konkanis in Maharashtra, Karnataka and Kerala. It was added in 1992 by 71st Amendment.

Malayalam Belong to the Dravidian family and is the official language of Kerala.

Manipuri It is the official language of Manipur. It was added in 1992 by 71st Amendment.

Marathi It is an Indo-Aryan language and is the official language of Maharashtra.

Nepali It is spoken in parts of Uttar Predesh, Bihar, West Bengal, Assam etc. It was added in 1992 by 71st Amendment.

Oriya It is an Indo-Aryan language and is the official language of Odisha.

Punjabi It is an Indo-Aryan language and is the official language of Punjab.

Sanskrit It is one of the earliest languages of the world. Early Sanksrit is known as Vedic Sanskrit and covers the period between 2000 and 500 BC.

Sindhi It is an Indo-Aryan language. It was added in 1967 by 21st Amendment.

Tamil It is the oldest of the Dravidian languages and is the official language of Tamil Nadu.

Telugu It is numerically the biggest of the Dravidian languages and is the official language of Andhra Pradesh.

Urdu It is the official langauge of Jammu and Kashmir.
Modern Urdu developed due to the efforts of Sir Sayyed Ahmed Khan (1817-1898).

Dogri It is generally spoken in Himachal Pradesh and Jammu. It is a combination of ancient Sanskrit and Pahari Dogri languages.
It has been added by the 92nd Constitutional Amendment Act, 2003.

Maithili It is chiefly spoken in the Maithilianchal region of Bihar. It is the second state language of Bihar.
It has been added by the 92nd Constitutional Amendment Act, 2003.

Santhali It is chiefly spoken in the area of Chhotanagpur Plateau in Jharkhand and Bihar. It has been added by the 92nd Constitutional Amendment Act, 2003.

Bodo It is chiefly spoken in Assam and its adjoining North-East States.
It has been added by the 92nd Constitutional Amendment Act, 2003.

NATIONAL INSIGNIA OF INDIA

Emblem The National Emblem of India is an adaptation of the Buddhist Lion Capital of Ashoka at Sarnath, near Varanasi in Uttar Pradesh.

Animal The Tiger is the National Animal of India. The magnificent tiger, *Panthera tigris*, is a striped animal.

Bird The Peacock *(Pavo cristatus)*, is the National Bird of India. It is symbol of qualities like beauty and grace.

Flag The **National Flag** is a horizontal tricolour of deep saffron (kesari) at the top, white in the middle, dark green at the bottom and a blue wheel (chakra) with 24 sticks at the centre.

Fruit The Mango (*Mangiferra indica*) is the National Fruit. It has been cultivated in India since time immemorial.

Tree The National Tree of India is The Banyan *(Ficus bengalensis)* Tree. This huge tree towers over its neighbours and has the widest trunk.

Sport Field Hockey, in which India has an impressive record with eight Olympic medals, is considered as the National Sport. However, Home Ministry has said that officially, no sport has been accorded, the status of National Sport.

Anthem Jana-Gana-Mana.... The song was composed originally in Bengali by Rabindranath Tagore, adopted in its Hindi version is our National Anthen.

Song The song **Vande Mataram**, composed in Sanskrit by Bankimchandra Chatterji, is our National Song.

Calendar The National Calendar based on the Saka Era with Chaitra as its first month and a normal year of 365 days was adopted from 22nd March, 1957.

Flower Lotus scientifically known as *Nelumbo nucifera* is the National Flower of India

Other National Symbols

National Motto	Satyamev Jayate
Father of the Nation	Mahatma Gandhi
National Foreign Policy	Non-Alignment
National Information Letter	White Letter
National Currency	Rupee (₹)
National Festivals	– Republic Day (26th January) – Independence Day (15th August) – Gandhi Jayanti (2nd October)
National River	Ganga
National Aquatic Animal	Ganga Dolphin (*Platanista gangetica*)
National Heritage Animal	Elephant

INDIAN DEFENCE

Indian Army Commands

Command	*HQ Location*
Central Command	Lucknow
Eastern Command	Kolkata
Northern Command	Udhampur
Southern Command	Pune
S W Command	Jaipur
Western Command	Chandigarh
Training Command (ARTRAC)	Shimla

Indian Air Force Commands

Command (Operational)	*HQ Location*
Central Air Command	Allahabad
Eastern Air Command	Shillong
Southern Air Command	Thiruvananthapuram
South Western Air Command	Gandhi Nagar
Western Air Command	New Delhi

Command (Functional)	*HQ Location*
Training Command	Bengaluru
Maintenance Command	Nagpur

Indian Navy Commands

Command	*HQ Location*
Western Naval Command	Mumbai
Eastern Naval Command	Vishakhapatnam
Southern Naval Command	Kochi
Andaman Nicobar Command	Port Blair

Ranks of Commissioned Officers

Army	*Air Force*	*Navy*
General	Air Chief Marshal	Admiral
Lt General	Air Marshal	Vice Admiral
Major General	Air Vice Marshal	Rear Admiral
Brigadier	Air Commodore	Commodore
Colonel	Group Captain	Captain
Lt Colonel	Wing Commander	Commander
Major	Squadron Leader	Lt Commander
Captain	Flight Lieutenant	Lieutenant
Lieutenant	Flying Officer	Sub-Lieutenant

Note *In December, 2019 Cabinet Committee on security has established the post of Chief of Defence Staff, to bring coordination between the three services i.e. Army, Navy and Air force. General Bipin Rawat is the first Chief of Defence Staff.*

Indian Defence Training Institutions

Training Institution	*Place*	*Estd in*
Rashtriya Indian Military College (RIMC)	Dehradun	1922
Indian Military Academy (IMA)	Dehradun	1932
High Altitude Warfare School (HAWS)	Gulmarg	1948
National Defence Academy (NDA)	Khadakwasla	1949
National Defence College (NDC)	New Delhi	1960
Officers Training Academy (OTA)	Chennai	1963

Training Institution	*Place*	*Estd in*
Counter Insurgency and Jungle Warfare School	Veirangte (Mizoram)	1970
College of Defence Management	Secunderabad	1970
College of Combat/Army War College	Mhow (MP)	1971
Army Cadet College (ACC)	Dehradun	1977
Army Air Defence College (AADC)	Gopalpur (Odisha)	1989
Army School of Physical Training (ASPT)	Pune	

Internal Security of India

Organisation	*Established*
Assam Rifles	▪ It was established in 1835 and is the oldest paramilitary force in the country. ▪ Its main objective is to keep vigilance of international borders in North-East and countering insurgency operations in Arunachal Pradesh, Manipur, Mizoram and Nagaland.
Intelligence Bureau (IB)	▪ It was set-up in 1920. ▪ Its objective is to collect secret information relating to country's security. ▪ It was originally set-up as Central Special Branch (CSB) in 1887 and renamed IB in 1920.
Central Reserve Police Force (CRPF)	▪ It was set-up in 1939. ▪ Its main objective is to assist the State / Union Territory Police in maintenance of law and order.
National Cadet Corps (NCC)	▪ It was established in 1948. ▪ Its main objective is to stimulate interest among the youth in the defence of the country in order to build up a reserve manpower to expand armed forces.
Sashastra Seema Bal (SSB)	▪ It was set up in 1963. ▪ Its objective are to prevent cross border crime and smuggling along the Indian borders with Nepal and Bhutan.
Indo-Tibetan Border Police (ITBP)	▪ It was established in 1962, after the Chinese attack. ▪ It is basically employed in the Northern borders for monitoring the borders and also to stop smuggling and illegal immigration.
Home Guard	▪ It was established in 1962, to assist the police in maintaining security, to help defence forces and to help local authorities in case of any eventuality.
Central Bureau of Investigation (CBI)	▪ It was established in 1941 and renamed CBI on 1963. ▪ Its objective is to investigate cases of misconduct by public servants, cases of cheating, embezzlement and fraud. CBI is also entrusted with the investigation of international crime cases in collaboration with Interpol.
Border Security Force (BSF)	▪ It was established in 1965. ▪ It keeps a vigil over the international borders with Pakistan and Bangladesh against the intrusion in the country.
Central Industrial Security Force (CISF)	▪ It was set-up in 1969 after the recommendations of Justice B Mukherji. ▪ Its objective is to monitor the industrial complexes of Central Government.
Indian Coast Guard	▪ It was set-up in 1978. ▪ Its objective is to protect the maritime and other national interests in the maritime zones of India.

Organisation	*Established*
National Security Guards (NSG)	▪ It was established in 1984 to counter the surge of militancy in the country. ▪ It is a highly trained force which deals with the militants effectively.
National Investigation Agency (NIA)	▪ It was formed in 2009 after Mumbai terror attack. ▪ It is the professional investigation agency mandated with counter terrorism operation.

DEFENCE RESEARCH IN INDIA

DRDO

Defence Research and Development Organisation (DRDO) is responsible for development of technology for use by the military and is headquartered in New Delhi, India.

It was formed in 1958, by the merger of Technical Development Establishment and the Directorate of Technical Development and Production with the Defence Science Organisation.

IGMDP

The Integrated Guided Missile Development Programme (IGMDP) was started in 1983, for the development of a comprehensive range of missiles including the intermediate range, Agni Missile (surface to surface) and short range missiles Prithvi Ballistic Missile (surface to surface), Akash Missile (surface to air), Trishul Missile (surface to air) and Nag Missile (anti tank).

Indian Missiles

Name	*Range*
Agni-I	It is a medium-range ballistic missile with a range of 700-800 km.
Agni-II	It is an intermediate-range ballistic missile with a range of 2000-3000 km.
Agni-III	It is an intermediate-range ballistic missile with a range of 3000-5000 km.
Agni-IV	It is an intermediate-range ballistic missile with a range of at a distance of 4000 km.
Agni-V	It is an intercontinental ballistic missile with a range of 5000-8000 km.
Agni-VI	It is an intercontinental ballistic missile with a range of 8000-12000 km.
Prithvi	Prithvi is a tactical surface to surface Short-Range Ballistic Missile (SRBM). It has three versions.
Prithvi-I (SS-150)	Army Version (150 km range with a payload of 1000 kg).
Prithvi-II (SS-250)	Air Force Version (250 km range with a payload of 500 kg).
Prithvi-III (SS-350)	Naval Version (350 km range with a payload of 1000 kg).
Dhanush (SS-350)	Dhanush is reportedly a naval version of Prithvi-III, which can be launched from ships. It can carry 500 kg of conventional or nuclear warhead, to a distance of 600 km.
Astra	Astra is Beyond Visual Range (BVR) air-to-air missile. Astra is designed to be capable of engaging targets at varying range and altitudes allowing for both short-range targets (upto 20 km) and long range targets (upto 80 km) using alternative propulsion modes.
Akash	▪ Akash is a medium range surface-to-air missile defense system developed by DRDO and BEL as part of the IGMDP. The missile can target aircraft up to 30 km away, at altitudes of 18000 m. Akash can be fired from both tracked and wheeled platforms. ▪ Akash is said to be capable of both conventional and nuclear warheads, with a reported payload of 60 kg.
Trishul	Trishul is a short range surface-to-air missile. The range of the missile is 12 km and is fitted with a 15 kg warhead. The weight of the missile is 130 kg.

Name	*Range*
Nag	Nag is India's third generation "Fire-and-Forget" anti-tank missile. It is an all weather, top attack missile with a range of 3 to 7 km.
Sagarika	K-15 Sagarika is a nuclear-capable submarine launched ballistic missile with a range of 750 km. It belongs to the K Missile family. The latest test of the K-15 Missile was done on 11th March, 2012.
Brahmos	▪ Brahmos is a stealth supersonic cruise missile that can be launched from submarines, ships, aircraft or land. It is a joint venture between Republic of India's DRDO and Russian Federation's NPO Mashinostroeyenia, who have together formed Brahmos Aerospace Private Limited. ▪ It is the world's fastest cruise missile in operation. The missile travels at a speed of mach 2.8 to 3.0. It has a range of 290 km. It can carry 300 kg of conventional or nuclear warhead. ▪ It is a hypersonic cruise missile with a range of 290 km. It is under development.
Shaurya	▪ The Shaurya Missile is a short-range surface-to-surface ballistic missile developed by DRDO for use by the Indian Army capable of hypersonic speeds, it has a range of 600 km and is capable of carrying a payload of one-tonne conventional or nuclear warhead. ▪ Shaurya Missile is a land version of the under-water launched K-15 Missile, Sagarika.
Nirbhay	Nirbhay is a long range, subsonic cruise missile being developed in India. The missile has a range of 1000 km. The missile will have a speed of 0.8 mach. The Nirbhay could launched from multiple platforms on land, sea and air.
Rudram	It is India's first indigenous anti-radiation missile. This air to surface missile are design to detect, track and neutralise the adversary radar and communication assets.

Indian Military Aircrafts and Helicopters

Aircraft	*Origin*	*Type*	*Versions*
Il-76 Phalcon	Russia	Airborne Early Warning	Il-76
Sukhoi Su-30 MKI	Russia, India	Multirole Air Superiority Fighter	Su-30 MKI
Antonov An-32 Cline	Soviet Union	Cargo Aircraft	An-32
HAL Tejas	India	Fighter	Mark I
SEPECAT Jaguar	France, United Kingdom	Ground-Attack	SI
Mikoyan-Gurevich Mig-27	Soviet Union	Ground-Attack	MIG-27 ML
Mil Mi-35 Hind-E	Soviet Union	Attack Helicopter	Mi-35
Mi-17	Soviet Union	Transport Helicopter	Mi-17
HAL Dhruv	India	Utility Helicopter	Dhruv
Nishant	India	Unmanned Aerial Vehicles	UAV
HTT40	Indian (HAL)	Replacement of HDT-32	–
Rafale	France (Dassault Aviation)	Fighter Plane	–
Chinook	USA (Boeing)	Multi-Mission Helicopter	–

Submarines of the Indian Navy

Submarines	*Description*
INS Chakra	▪ The INS Chakra II (K-152 Nerpa) is a 8140 tonne project 518 (NATO Akula II) type nuclear-powered attack submarine. ▪ Constituted in 1993, but suspended due to lack of funding. ▪ K-152 Nerpa was launched in October 2008 and entered service with the Russian Navy in late 2009. ▪ The submarine was leased to the Indian Navy in 2011 and was formally commissioned into service as the INS Chakra II at a ceremony in Vishakhapatnam on 4th April, 2012.

INS Kalvari	▪ It is the first of the Indian Navy's Six Kalvari-class submarine being built in India. ▪ It is a diesel-electric attack submarine which is designed by French naval defence and energy company DCNS and being manufactured at Mazagon Dock Limited in Mumbai.
INS Arihant	▪ It is a class of nuclear-powered balistic missile submarines built for the Indian Navy. ▪ It was first launched in 2009 and began sea trials in December 2014, and has been commissioned in August 2016.
INS Arighat	▪ It is the second Arihant-class submarine and the second nuclear-powered ballistic missile submarine being built by India. It was launched on 19th November, 2017.
INS Khanderi	▪ It is the second of the Indian Navy's six Kalvari-class submarines being built in India. It is a diesel-electric attack submarine. It was launched on 12th January, 2017.
INS Karanj	▪ It is a diesel-ectric attack submarine that was launched on 31 January, 2018.

Warships/ Base of the Indian Navy

Warships	*Description*
INS Vikrant	▪ The first Aircraft carrier of Indian Navy was INS Vikrant. ▪ India purchased the INS Vikrant from the United Kingdom in 1957. (Decommissioned)
INS Viraat	▪ INS Viraat is a centaur class aircraft carrier currently in service with the Indian Navy. ▪ The Viraat was completed and commissioned in 1959, as the Royal Navy's HMS Hermes and transferred to India in 1987. (Decommissioned)
INS Prahar	▪ It is world's fastest missile ship commissioned in 1997. (Decommissioned)
INS Delhi (Destroyer)	▪ It is India's largest and most sophisticated indigenously built guided missile destroyer warship. ▪ It was launched in 1991 at Mazgaon Docks and Commissioned in 1997.
INS Mysore (Destroyer)	▪ It is a Delhi class guided missile destroyer currently in active service with the Indian Navy. INS Mysore was built at Mazgaon Dock Limited in Mumbai and it was commissioned in 1999.
INS Prabal (Corvette)	▪ It is a veer class corvette, currently in active service with the Indian Navy. ▪ It was built at Mazgaon Dock Limited and Commissioned in 2002.
INS Talwar (Frigate)	▪ It is the lead ship of the Talwar class frigates of the Indian Navy. ▪ Its name means ''Sword'' in Hindi. ▪ It was built in Russia and commissioned into the Indian Navy in 2003.
INS Beas (Frigate)	▪ It is a Brahmaputra class frigate of the Indian Navy. ▪ She was built at the Garden Reach Shipbuilders and Engineers (GRSE) Kolkata, and it was commissioned in 2005.
INS Kadamba (Naval Base)	▪ It is India's largest naval base located near Karwar in Karnataka. ▪ It was commissioned in 2005, under the project Seabird.
INS Vikramaditya (Surface Fleet)	▪ It is a kiev class aircraft carrier ▪ India Purchased the INS Vikramaditya from Russia. ▪ On 14th June, 2014 inducted into Indian Navy.
INS Vibhuti (Corvette)	▪ It was India's first indigenously built missile boat launched at Mazgaon Docks in Mumbai. ▪ It is a veer class corvette.
INS Savitri (Offshore Petrol Vessel)	▪ It was India's first warship fabricated at Hindustan shipyard Limited in 1990. ▪ It is a Sukanya class patrol vessel of the Indian Navy.
INS Shakti (Auxillary Fleet)	▪ It was India's first indigenously built submarine. ▪ The ship is one of the largest in the Indian Navy as it is 175 m in length and 32 m in width.

INS Kolkata (Destroyer)	▪ It is the lead ship of the Kolkata-class guided missile destroyers of Indian Navy. ▪ It was handed over to the Navy on 10th July, 2014.
INS Visakhapatnam (Destroyer)	▪ It is the lead ship of the Visakhapatnam-class stealth guided-missile destroyers of the Indian Navy. ▪ It is being constructed at Mazagon Dock Limited (MDL) and has been launched on 20 April, 2015. The ship is expected to get commissioned by 2018.
INS Kavaratti (Anti-submarine Warfare Corvette)	▪ It is an anti-submarine warfare corvette of the Indian Navy. ▪ It is the last of four Kamorta-class corvettes under various stages of induction with the Indian Navy. ▪ The ship was built by Garden Reach Shipbuilders and Engineers, Kolkata and launched on 19 May, 2015.
INS Kochi (Destroyer)	▪ It is the second ship of the Kolkata-class destroyers built for the Indian Navy. ▪ It is built at Mazagon Dock Limited, Mumbai. It was commissioned to Indian Navy on 30 September, 2015. ▪ It has been built under the code name of Project 15 Alpha.
INS Astradharani (Torpedo Recovery Vessel)	▪ It is first totally indigenously-designed and built torpedo launch and recovery vessel, commissioned on 6 october, 2015. ▪ The unique design of the ship was a collaborative effort of Naval Science and Technological Laboratory, IIT Kharagpur and shoft shipyard. ▪ It is the advanced replacement of the INS Astravahini which was decommissioned on 17 July, 2015.

INDIA'S ATOMIC RESEARCH

India's atomic energy research started with the establishment of the Atomic Energy Commission on 10th August, 1948. Department of Atomic Energy (DAE) was established in 1954, for implementation of atomic energy programmes.

- **Bhabha Atomic Research Centre** (BARC) It was set-up in 1954, as the Atomic Energy establishment at Trombay, near Mumbai and in 1967, was renamed as BARC in the memory of its founder Dr Homi Bhabha.
- **Centre for Advance Technology** (CAT) It was established in 1984, at Indore to coordinate research in high technology fields like lasers, fusion and accelerators.
- **Indira Gandhi Centre for Atomic Research** (IGCAR) It was set-up in 1971, at Kalpakkam in Madras for research and development of fast breeder technology.
- **Atomic Mineral Directorate** (AMD) It is located in Hyderabad and carries out surveys, exploration and evaluation of the resources required for the atomic energy programmes of the country.
- **Variable Energy Cyclotron Centre** (VECC) It was set-up in 1977, at Kolkata as a national centre for advanced work in nuclear chemistry, nuclear physics, production of isotopes for various applications and radiation damage studies on reactor materials.

Nuclear Power Stations

Location	*State*	*Number of Reactors*
Tarapur	Maharashtra	4
Kaiga	Karnataka	4
Kalpakkam	Tamil Nadu	2
Kakrapar	Gujarat	3
Rawatbhata	Rajasthan	5
Narora	Uttar Pradesh	2
Kudankulan	Tamil Nadu	2

There are now 22 operating nuclear power reactors (two boiling water reactors and twenty PHWRs (Pressurised Heavy Water Reactors) with a total installed capacity of 6780 MW.

SPACE PROGRAMME OF INDIA

- The Indian space programme was launched in 1962, when the Indian National Committee for Space Research was formed. To this were added the Indian Space Research Organisation in 1969 and the space commission and department of space in 1972.
- The Indian Space Research Organisation (ISRO) is responsible for the planning, execution and management of space research activities and space application programmes.
- The ISRO has headquarters at Bengaluru.
- India's first satellite communication Earth station was set-up at Arvi near Pune. The first Indian Satellite Aryabhatta was launched on 19th April, 1975 from Baikonur (erstwhile USSR).
- The first Indian Communication Satellite, APPLE was launched on 19th June, 1981 from Kourou in French Guyana (South America). It was the first Indian satellite that was placed in geostationary orbit.
- India is developing its second Space port (after Sriharikota) in Thoothukudi district of Tamil Nadu. It will be utilised for launching small satellites.

Indian Space Programme *(At a Glance)*

Satellite	*Date*	*Launch Vehicle*	*Place*	*Type*
Aryabhatta	19th April, 1975	Cosmos	Baikonur	Experimental
Bhaskara I	7th June, 1979	Cosmos	Baikonur	Earth Observation
Rohini	10th August, 1979	SLV-3	Sriharikota	Experimental
Rohini RS-1	18th July, 1980	SLV-3	Sriharikota	Experimental
Rohini D1	31st May, 1981	SLV-3	Sriharikota	Earth Observation
Apple	19th June, 1981	Ariane	Kourou	Communication
Bhaskara II	20th November, 1981	Cosmos	Baikonur	Earth Observation
INSAT-IA	10th April, 1982	Delta	America	Communication
SROSS I	24th March, 1987	ASLV D-1	Sriharikota	Science & Exploration
RISAT-1	26th April, 2012	PSLV-C19	Sriharikota	Earth Observation Satellite
GSAT-10	29th September, 2012	Ariane-5 VA-209	Kourou	Communication
SARAL	25the February, 2013	PSLV-C20	Sriharikota	Earth Observation Satellite
IRNSS-1A	1st July, 2013	PSLV-C22	Sriharikota	Navigation
GSAT-7	30th August, 2013	Ariane-5 VA-215	Kourou	Communication
MOM	5th November, 2013	PSLV-C25	Sriharikota	Scientific & Exploration
GSAT-14	5th November, 2014	GSLV-D5	Sriharikota	Communication
IRNSS-1B	4th April, 2014	PSLV-C24	Sriharikota	Navigation
GSAT-15	11th November, 2015	Ariane-5 VA-227	Kourou	Communication
GSAT-18	5th October, 2016	Ariane-5	Kourou	Communication
Cartosat-2	15th February, 2017	PSLV-C37	Sriharikota	Remote Sensing Satellite
GSAT9 (also called South Asia Satellite)	5th May, 2017	GSLV Mark-2	Sriharikota	Communication Satellite
Cartosat 2	23rd June, 2017	PSLV-C-38	Sriharikota	Remote Sensing Satellite
GSAT-17	29th June, 2017	Ariane 5	Kourou	Communication Satellite

Satellite	*Date*	*Launch Vehicle*	*Place*	*Type*
Cartosat 2	12th January, 2018	PSLV-C40	Sriharikota	Remote Sensing Satellite
GSAT-7A	19th December, 2018	GSLV-F11	Sriharikota	Communication Satellite
Kalamsat V2	25th January, 2019	PSLV-C44	Sriharikota	Communication Satellite (Lightest and smallest satellite to be ever launched)
Chandrayaan-2	22nd July, 2019	GSLV MK-III MI rocket	Sriharikota	Lunar Mission
GSAT-30	17th January, 2020	Ariane-5VA-251	Kourou	Communication Satellite
EOS/RISAT -2BR2	7th November, 2020	PSLV-DL-C49	Sriharikota	Earth Observation
GSAT-12R	17th December, 2020	PSLV-XL-C50	Sriharikota	Communication Satellite

Gaganyaan Mission

It is the first indigenous mission designed by ISRO, to be launched in 2022.

Nuclear and Space Research Centres in India

Research Centre	*Place*
Indian Rare Earths Limited	Alwaye (Kerala)
Uranium Corporation of India Limited	Jadugoda (Jharkhand)
Atomic Energy Commission (AEC)	Mumbai
Electronics Corporation of India Limited	Hyderabad
Bhabha Atomic Research Centre (BARC)	Trombay (Mumbai)
Radio Astronomy Centre	Udhagamandalam (Tamil Nadu)
Tata Institute of Fundamental Research	Mumbai
Saha Institute of Nuclear Physics	Kolkata
Centre of Earth Sciences Studies	Thiruvananthapuram (Kerala)
Physical Research Laboratory	Ahmedabad
Space Commission	Bengaluru
Vikram Sarabhai Space Centre	Thiruvananthapuram
Indian Space Research Organisation (ISRO)	Bengaluru
Space Application Centre	Ahmedabad
Thumba Equatorial Rocket Launching Station	Thumba (Kerala)
College of Satellite Communication Technology	Ahmedabad
Nuclear Fuel Complex	Hyderabad
Indira Gandhi Centre For Atomic Research	Tamil Nadu
National Chemical Laboratory (NCL)	Pune (Maharashtra)

AWARDS AND HONOURS

INTERNATIONAL AWARDS

Nobel Prize

- The Nobel Prizes are given under the will of **Alfred Bernhard Nobel** who died in 1896. He was a noted Swedish chemist and engineer who discovered Nitroglycerine and its use in the manufacture of dynamite.
- Nobel Prizes are given each year in the six fields. The Nobel Prizes for peace, physics, chemistry, medicine and literature were started in 1901. The Nobel Prize for economics was started in 1968.
- *The prize awarding bodies are*
  - The Swedish Academy of Literature awards the prize in **Literature**.
  - The Royal Swedish Academy of Sciences awards the prize in **Physics** and **Chemistry**.
  - The Bank of Sweden awards the prize in **Economics**.
  - The Nobel Assembly of Karolinska Chirugical (Swedan) awards the prize in **Medicine** (Physiology).
  - The Committee of the Norwegian Parliament awards the prize for **Peace**.
- Each recipient or laureate, receives a gold medal, a diploma and a sum of money which depends on the Nobel Foundations; income that year.
- The awards are presented in stockholm in an annual ceremony on 10th December, the anniversary of Nobel's death.
- The awards can be given to maximum **three persons** in the same field at the same time.
- The Nobel prize was not awarded between 1940 and 1942 due to the outbreak of World War II.

Indian Nobel Laureates

Name	*Field*	*Year*
Rabindranath Tagore	Literature (Gitanjali)	1913
CV Raman	Physics (Raman Effect)	1930
Har Gobind Khorana (of Indian Descent)	Medicine (Genetic Code)	1968
Mother Teresa	Peace	1979
Subramaniam Chandrasekhar *(of Indian Descent)*	Physics *(Chadrasekhar's Limit)*	1983
Amartya Sen	Economics *(Welfare Economics)*	1998
Sir VS Naipaul *(of Indian Descent)*	Literature	2001
Venkatraman Ramakrishna *(of Indian Descent)*	Chemistry *(Ribosomes)*	2009
Kailash Satyarthi	Peace *(Struggle against Child Labour)*	2014
Abhijeet Banerjee *(of Indian Descent)*	Economics *(Welfare Economics)*	2019

The Right Livelihood Award

Awarded for	"Practical and exemplary solutions to the most urgent challenges facing the world today"
Presented by	Right Livelihood Award Foundation
Country	Sweden
First awarded	1980

- The Right Livelihood Award is also referred as the "Alternative Nobel Prize".
- Colin Gonsalves was an Indian laureate won the award in 2017.

Ramon Magsaysay Award

Awarded for	Outstanding contributions in six categories (Government Service, Public Service and other Fields)
Presented by	Ramon Magsaysay Award Foundation
Presented to	Asian Individuals
Country	Philippines
First awarded	1958

Prominent Indian Magsaysay Award Winners

- Vinoba Bhave (1958)
- CD Deshmukh (1959)
- Verghese Kurien (1963)
- Jayaprakash Narayan (1965)
- Satyajit Ray (1967)
- Mother Teresa (1962)
- MS Swaminathan (1971)
- Ela Bhatt (1977)
- Pramod Karan Sethi (1981)
- Kiran Bedi (1994)
- TN Seshan (1996)
- RK Laxman (1984)
- Ravi Shankar (1992)
- Arvind Kejriwal (2006)
- P Sainath (2007)
- Prakash and Mandakini Amte (2008)
- Neelima Mishra (2011)
- Kulandei Francis (2012)
- TM Krishna, Bezwada Wilson (2016)
- Bharat Vatwani (2018)
- Sonam Wangchuk (2018)
- Ravish Kumar (2019)

Oscar Awards or Academy Awards

Awarded for	Excellence in cinematic achievements
Presented by	Academy of Motion Picture Arts and Sciences
Country	United States
First awarded	16th May, 1929

- The Oscar statuette is officially named "The Academy Award of Merit".
- The prominent Indian film *nominated for Oscar are*
  - Mother India (1957)
  - Salam Bombay (1988)
  - Lagaan (2001)
- The Oscar award is given every year in the month of February at Hollywood Kodek Theatre.
- **Bhanu Athaiya** is the first Indian to win an Oscar award in 1982 for costume design in film Gandhi.
- A.R. Rahman and Gulzar also won Oscar award in 2009.

Grammy Awards

Awarded for	Outstanding achievements in the music industry
Presented by	National Academy of Recording Arts and Sciences
Country	United States
First awarded	1959

- *Indian Winning Grammy Awards*

Pandit Ravi Shankar 3 times	1967, 1972, 2001
Zakir Hussain	1992 and 2009
Vikku Vinayak	1991
Vishwa Mohan Bhatt	1993
AR Rahman	2009
Ricky Kej	2015
Sandip Das	2017

Golden Globe Awards

Awarded for	Excellence in film and television
Presented by	Hollywood Foreign Press Association
Country	United States
First awarded	1944

- AR Rahman is the first Indian to win Golden Globe Award.
- 77rd Golden Globe Best Film (2019) is Awarded to '1917' (Drama).

Pulitzer Prize

Awarded for	Excellence in newspaper journalism, literary achieve ments and musical composition
Presented by	Columbia University
Country	United States
First awarded	1917

Indian Winning Pulitzer Prize

- Gobind Behari Lal (for Reporting) 1937
- Jhumpa Lahiri 2000
- Geeta Anand (for Journalism) 2003
- Dr Siddhartha Mukherjee (for his book on cancer "The Emperor of All Maladies A Biography of Cancer") 2011
- Vijay Seshadri Poet Category 2014
- Yasin Dar, Mukhtar Khan and Channi Anand (feature photography) 2020

Man Booker Prize

Awarded for	Best full-length English novel
Presented by	Man Group
Country	Commonwealth of Nations, Ireland and Zimbabwe
First awarded	1969

Indian winning Man Booker prize

- Arundhati Roy (Novel-The God of Small Things) 1997
- Kiran Desai (Novel- The Inheritance of Loss) 2006
- Aravind Adiga (Novel-the White Tiger) 2008

Man Booker International

Awarded for	Best English (or available for translation into English) fiction
Presented by	Man Group
Country	United Kingdom
First awarded	2005

- The award is worth US £ 50000 and an author can win only once. It is awarded for best English (or available translation into English) fiction.
- Olga Tokarczuk has won man booker prize for his work 'FLIGHT' in 2018.
- In 2019, Jokha al-Harthi of Oman won this prize for her book 'celestial bodies'.
- The 2020 Man Booker International was awarded to the Dutch writer Marieke Lucas Rijneveld for the Discomfort of Evening.

Beauty Contest

Miss Universe

- It is an annual international beauty contest that is run by the Miss Universe Organisation. The contest was founded in 1952, by California Clothing Company Pacific Mills. Its headquarters is at New York City (US).
- **Sushmita Sen** was the first Indian woman to win the Miss Universe contest in 1994.
- Catriona Gray of Philippines has been crowned as Miss Universe 2018.
- Zozibani Tunzi of South Africa has won Miss Universe, 2019 title.

Miss World

- The Miss World pageant is the oldest surviving international beauty pageant. It was created in the United Kingdom by Eric Morley in 1951. Since his death in 2000, Morley's wife, Julia Morley, co-chairs the pageant.
- Its headquarter is at London (UK).
- **Reita Faria Powell** (born Bombay) became the first Indian to win the Miss World title in 1966.
- India's Manushi Chillar won the Miss World 2017 contest. She crowned Vanessa Ponce of Mexico as Miss World 2018.
- Toni-Ann Singh from Jamaica won the Miss World title, 2019.

Miss India

- Miss India or Femina Miss India is a national beauty pageant in India. It is organised by Femina, a women's magazine published by Bennett, Coleman and Co Ltd.
- Its headquarters is at Mumbai and it was formed in 1963.
- Miss India beauty contest started on 1947.
- Pramila was the first woman to win the Miss India contest in 1947.
- Suman Rao won the title of Femina Miss India 2019.
- The 2020 Miss India event was postponed due to COVID-19 Pandemic.

India's International Awards

Mahatma Gandhi Peace Prize

- It was instituted in 1995 and awarded by Government of India to encourage and promote Gandhian values world wide.
- The award carries ₹ crore in cash, convertible in any currency in the world, a plaque and citation.
- First recipient (1995) Julius Nyerere
- Last recipient (2018) Yohei Sasakawa

Indira Gandhi Prize

- It was instituted in 1986 and awarded by Indira Gandhi Memorial Trust.
- It is awarded for peace, disarmament and development.
- The prize carries a cash award of 25 lakh Indian rupees and a citation.

- First recipient (1986) *Parliamentarians for Global Action* (2017)
- Last recipient David Attenborough (2019)

Jawaharlal Nehru Award

- It was insituted in 1965 and awarded by Government of India for international understanding, Goodwill and Friendship.
- The money constituent of this award is 25 Lakh Indian rupees.

First recipient (1965) *U Thant (3rd UN Secretary-General)*

Last recipient (2015) Former President of Egypt- Hosni Mubarak

NATIONAL AWARDS

Bharat Ratna

- Bharat Ratna is India's highest Civilian Award. It was first awarded in 1954.
- The actual award is designed in the shape of a Peepal leaf with *Bharat Ratna* inscribed in Devanagri script in the Sun Figure.
- The reverse side of the decoration *Satyamev Jayate* has been written in Hindi with an inscription of state emblem.
- The emblem, the Sun and the rim are of platinum. The inscriptions are in burnished bronze.
- Indira Gandhi was the youngest person alive at the time of receiving the award and Rajiv Gandhi is the Youngest overall to receive the Bharat Ratna
- DK Karve was the eldest person alive at the time of receiving the award and Vallabhbhai Patel is the eldest overall.
- First Naturalised Indian Citizen to be awarded with Bharat Ratna is Mother Teresa (1980).
- Abdul Ghaffar Khan was the first foreigner to be honoured with this award in 1987 and Later Nelson Mandela (1990).

Padma Awards

There are three padma awards given on republic day (26th January).

Padma Vibhushan

- This is the second highest National Award. Instituted on 2nd January, 1954 by the Government of India.
- It is awarded to recognise exceptional and distinguished service to the nation in any field.

Padma Bhushan

- This is the third highest National Award. Instituted on 2nd January, 1954 by the Government of India.

Winners of Bharat Ratna

Year	*Winners*
2019	Pranab Mukherjee
2019	Bhupen Hazarika
2019	Nanaji Deshmukh
2015	Madan Mohan Malaviya
2015	Atal Bihari Bajpayee
2014	CNR Rao
2014	Sachin Tendulkar
2008	Pandit Bhimsen Joshi
2001	Ustad Bismillah Khan
2001	Lata Dinanath Mangeshkar
1999	Lokpriya Gopinath Bordoloi
1999	Professor Amartya Sen
1999	Loknayak Jayprakash Narayan
1999	Pandit Ravi Shankar
1998	Smt Madurai Shanmukhavadivu Subbulakshmi
1998	Shri Chidambaram Subramaniam
1997	Shri (Dr) Avul Pakir Jainulabdeen Abdul Kalam
1997	Smt Aruna Asaf Ali
1997	Shri Gulzari Lal Nanda
1992	Shri Maulana Abul Kalam Azad
1992	Shri Satyajit Ray
1992	Shri Jehangir Ratanji Dadabhai Tata
1991	Shri Rajiv Gandhi
1991	Sardar Vallabhbhai Patel

Year	*Winners*
1991	Shri Morarji Ranchhodji Desai
1990	Dr Nelson Rolihlahla Mandela
1990	Dr Bhimrao Ramji Ambedkar
1988	Shri Marudur Gopalan Ramachandran
1987	Khan Abdul Ghaffar Khan
1983	Shri Acharya Vinoba Bhave
1980	Mother Teresa (Agnes Gonxha Bojaxhiu)
1976	Shri Kumaraswamy Kamraj
1975	Shri Varahagiri Venkata Giri
1971	Smt Indira Gandhi
1966	Shri Lal Bahadur Shastri
1963	Dr Zakir Hussain
1963	Dr Pandurang Vaman Kane
1962	Dr Rajendra Prasad
1961	Dr Bidhan Chandra Roy
1961	Shri Purushottam Das Tandon
1958	Dr Dhonde Keshav Karve
1957	Pt Govind Ballabh Pant
1955	Pandit Jawaharlal Nehru
1955	Dr Mokshagundam Vivesvaraya
1955	Dr Bhagwan Das
1954	Dr Chandrasekhara Venkata Raman
1954	Dr Sarvapalli Radhakrishnan
1954	Shri Chakravarti Rajagopalachari

- It is awarded to recognise distinguished service of a high order to the nation, in any field.

Padma Shri

- This is the fourth highest National Award instituted on 2nd January, 1954 by Government of India.
- In 1960 Dr MG Ramachandran refused to accept the award as the wordings of the award is in Hindi.
- It recognises the contribution of Indian citizens (generally) in various spheres of activity including Arts, Education, Industry, Literature, Science, Sports, Social Service and Public life.

MILITARY AWARDS

Wartime Valour Awards (awarded to officers or enlisted personnel from all branches of the Indian Military).

Award	*Year of Institution*	*Awarded by*	*Awarded for*
Paramvir Chakra	established on 26th January, 1950 with effect from 15th August, 1947	Government of India	Most conspicuous bravery or some during or pre-eminent valour or self-sacrifice.
Mahavir Chakra	established on 26th January, 1950 with effect from 15th August 1947	Government of India	National bravery in the presence of enemy
Vir Chakra	established on 26th January, 1950 with effect from 15th August, 1947	Government of India	Acts of bravery in the battle field

Note *All the three awards are also posthumously.*

Subedar Major Bana Singh of the 8th Jammu and Kashmir light infantry was the only serving personnel of the Indian defence establishment with a Param Vir Chakra.

Peacetime Gallantry Awards

Awarded Either to Military or Civilian Personnel

Award	*Year of Institution*	*Awarded by*	*Awarded for*
Ashoka Chakra	4th January, 1952 with effect from 15th August, 1947	Government of India	Most conspicuous bravery or some act of daring or pre-eminent act of valour or self-sacrifice otherwise than in the face of the enemy.
Kirti Chakra	4th January, 1952 with effect from 15th August, 1947	Government of India	Conspicuous gallantry otherwise than in the face of the enemy.
Shaurya Chakra	4th January, 1952 with effect from 15th August, 1947	Government of India	National bravery generally for counter insurgence operations and action against the enemy during peace time

Note *All the three awards are also given posthumously.*

Other Military Awards

Category	*Award*
Wartime/ Peacetime Service and Gallantry	Sena Medal (Army), Nao Sena Medal (Navy), Vayusena Medal (Air Force)
Wartime Distinguished Service	Sarvottam Yudh Seva Medal, Uttam Yudh Seva Medal, Yudha Seva Medal
Peacetime Distinguished Service	Param Vishisht Seva Medal, Ati Vishisht Seva Medal, Vishisht Seva Medal

SPORTS AWARDS

Rajiv Gandhi Khel Ratna Award

- The Rajiv Gandhi Khel Ratna Award (RGKR) is India's highest honour given for achievement in sports. The words 'Khel Ratna' literally means 'Sports Gem'. The award is named after the late Rajiv Gandhi, former Prime Minister of India.
- The award was instituted in the year 1991-1992 and was awarded by the Ministry of Youth Affairs and Sports.
- The prize money in this award has been increased from ₹ 7.5 lakh to ₹ 25 lakh. First Awardee - Viswanathan Anand, Chess (1991).
- In 2020, Rohit Sharma (Cricket), Mariyappan Thangavelu (Paralympic-high Jumper), Manika Batra (Table Tennis) Vinesh Phogat (Wrestling) and Rani Rampal (Hockey) were awarded Rajiv Gandhi Khel Ratna Award.

Arjuna Award

- The Arjuna Awards were instituted in 1961 by the Government of India to recognise outstanding achievement in National Sports. The award carries a cash prize of ₹ 15 lakh, a bronze statuette of Arjuna and a scoll.
- From the year 2001, the award is given only in disciplines *falling under the following categories*.
  - Olympic Games
  - Asian Games
  - Commonwealth Games
  - World Cup
  - World Championship Disciplines and Cricket.
  - Indigenous Games
  - Sports for the physically challenged.

Dronacharya Award

Dronacharya Award is an award presented by the Government of India for excellence in sports coaching. The award comprises a bronze statuette of Dronacharya a scroll of honour and a cash prize of ₹ 10 lakh. The award was instituted in 1985. In 2020, Dharmendra Tiwari (Archery), Purushottam Rai (Athletics), Shiv Singh (Boxing), Romesh Pathania (Hockey), Krishan Kumar Hooda (Kabaddi), Vijay Bhalchandra Munishwar (Power lifting), Naresh Kumar (Tennis) and Prakash Dahiya (Wrestling) were awarded.

Dhyanchand Award

Dhyanachand Award is India's highest award for lifetime achievement in sports and games, given by the Government of India. The award is named after the legendary Indian hockey player Dhyanchand. The award was initiated in 2002. The award carries a cash prize of ₹ 10 lakh. In 2020, it was awarded to 15 people for their exemplary performance in different sports. Prominent among them were kuldip Singh Bhullar (Athletics), Jincy Philips (Athletics), Trupti Murgunde (Badminton), Ajit Pal Singh (Hockey), Sachin Nag (Swimming), etc.

FILM AWARDS

National Film Awards

The National Film Awards are the most prominent film award ceremony in India, established in 1954 and it is administered, along with the international film festival of India and the Indian Panorama, by the Indian government's Directorate of Film Festivals since 1973. Due to the national character of the National Film Awards, it is considered to be the equivalent of the American Academy Awards.

Dada Saheb Phalke Award

The Dada Saheb Phalke Award is India's highest award in cinema given anually by the Government of India for lifetime contri- bution to Indian Cinema. It was instituted in 1969, the birth Centenary year of Dada Saheb Phalke, considered as the father of Indian Cinema. Phalke award carries a 'Swarna Kamal, a shawl and a cash prize of ₹ 1000000. Vinod Khanna won the 2017 Dada Saheb Phalke award, while Amitabh Bachhan was awarded 2018, Dada Saheb Phalke Award.

Filmfare Awards

- The filmfare awards are presented annually by the times group to honour both artistic and technical excellence of professionals in the Hindi language film industry of India. The awards were first introduced in 1954, the same year as the National Film Awards.
- They were initially referred to as the Clare Awards after the editor of the Times of India, Clare Mendonca.

LITERARY AND CULTURAL AWARDS

Jnanpith Award

- The Jnanpith Award is a literary award in India. The award was instituted in 1961. It is presented by the Bharatiya Jnanpith, a trust founded by the Sahu Jain family, the publishers of the Times of India newspaper.
- It carries a check of ₹ 7 lakh, a citation plaque and a bronze replica of Saraswati, the Indian Goddess of knowledge, music and arts. First awarded in 1965. Amitav Ghosh won the 2018 Jnanpith Award. (First English writer to be honoured with this Award.)
- Akkitham Achuthan Namboothiri (Malyalam) was awarded Jnanpith Award, 2019.

Sahitya Akademi Award

It is a literary honour in India instituted in 1954 which Sahitya Akademi, India's National Academy of Letters, annually confers on writers of outstanding works in twenty-four major Indian languages.

The award consists of a casket containing an engraved copper plague, a shawl and a cheque of 1 lakh rupees.

Bhasha Samman

Sahitya Akademi gives these special awards to writers for significant contribution to Indian languages other than the above 24 major ones and also for contribution to classical and medieval literature.

Sangeet Natak Akademi Puraskar (Akademi Award)

- Awarded by the Sangeet Natak Akademi India's National Academy of Music, Dance and Drama.
- It is the highest Indian recognition given to practicing artists in the categories of music, dance, theatre, other traditional/ folk/ tribal/ dance/ music/theatre and Puppetry and contribution / scholarship in performing arts.

Lalit Kala Akademi Ratna

- Instituted in 1955 by the Government of India is an honour for the fine arts given to eminent artists for their lifetime achievements in the field of visual arts.
- It is awarded by the Lalit Kala Akademi, India's National Academy of Art, it is the highest honour in the fine arts conferred by the Government of India.

Saraswati Samman

The Saraswati Samman is an annual award for outstanding prose or poetry literary works in any Indian language listed in Schedule VIII of the Constitution of India. The Saraswati Samman was instituted in 1991 by the KK Birla foundation. The award consist of ₹ 10 lakh, a citation and a plaque.

- First Awarded – Ismat Chughtai (Urdu-1990)
- Last Awarded– Vasdev Mohi (Sindhi) 2019

Vyas Samman

The Vyas Samman is a literary award in India, first awarded in 1991. It is awarded annually by the KK Birla Foundation and includes a cash amount of ₹ 2.5 lakh.

Hindi scholar and writer Nasira Sharma has been conferred with this award for 2019 for her novel 'Kagaz ki Nav'.

Some Famous Awards by State Governments

Award	*Awarded/Instituted by*	*Field of Honour*
Mahatama Gandhi Award	Madhya Pradesh Government	To an institution working according to Gandhi philosophy and ideology
Tansen Award	Madhya Pradesh Government	Indian classical music
Kalidas Award	Madhya Pradesh Government	Classical dance and music, theatre, painting and sculpture
Tulsi Award	Madhya Pradesh Government	Folk and traditional tribal art (only for male artist)

Award	*Awarded/Instituted by*	*Field of Honour*
Lata Mangeshkar Award	Madhya Pradesh Government	For music direction and playback singing in field of light music
Santhala Dance Award	Karnataka Government	Santhal dance (a tribal dance form)
Konark Samman	Odisha state Council of Culture	Literature, art, sculpture, music, dance and socio - cultural work
Moortidevi Award	Indian Jnanpith Trust	Literature
Yash Bharti Samman	Uttar Pradesh State Culture Department	Literature, fine arts, folk music, classical music and sports

SCIENCE AWARDS

Shanti Swarup Bhatnagar Awards

- This prize for science and technology is awarded annually by the Council of Scientific and Industrial Research (CSIR) for notable and outstanding research.
- It is the highest award for science in India. It was first awarded in 1958.
- The prize comprises a citation, a plaque and a cash award of ₹ 5 lakh.

Borlaug Award

- Instituted in 1971 and given for excellent work in Agricultural field.
- This, award comprises a certificate, Gold Medal and ₹ 1 lakh.

Dhanvantari Award

Instituted in 1971 and given for "Excellence in medical services''. This, award is given by *Dhanvantari Foundation*.

Dr BC Roy Award

- Medical Council of India instituted Dr BC Roy Award in 1976, in memory of Bharat Ratna Dr BC Roy. The award is given annually in each of the following categories: Statesmanship of the highest order in India, Medical man-cum-statesman, Eminent medical person, Eminent person in philosophy and arts.
- It is presented by the President of India in New Delhi on 1st July, the National Doctor's Day.
- This award comprises silver medal, certificate and ₹1 lakh.

Jamnalal Bajaj Award

It is a prestigious Indian award, for promoting Gandhian values, social work and social development. It is established in 1978 by the Jamnalal Bajaj Foundation of Bajaj Group. This award contains ₹5 lakh, certificate and a trophy.

GD Birla Award

GD Birla award for scientific research is conferred by KK Birla Foundation, was established in 1991, by Krishna Kumar Birla. This award comprises ₹ 1.45 lakh and a certificate.

This award is given by Birla foundation. This award comprises medals, certificate and ₹ 50000.

Homi Bhabha Award

Instituted in 1990, and given for excellence in the field of Atomic energy. This award comprises ₹ 50000 and certificate.

Vikram Sarabhai Award

Instituted in 1990 and given for excellence in field of Space Research.

Important Dates and Days of the Year

January

1 Global Family Day
4 Louis Braille Day
9 NRI Day (Pravasi Bhartiya Diwas)
12 National Youth Day (Birthday of Swami Vivekanand)
15 Army Day
25 National Tourism Day, Voter's Day
26 Indian Republic Day, International Customs Day
28 Data Protection Day
30 Martyr's Day (Mahatma Gandhi's Martyrdom), World Leprosy Eradication Day

February

4 World Cancer Day
20 World Day of Social Justice
24 Central Excise Day
28 National Science Day

March

8 International Women's Day
15 World Consumer Rights Day, World Disabled Day
21 World Forestry Day, International Day for the Elimination of Racial Discrimination
22 World Water Day
23 World Meteorological Day
24 World TB Day

April

5 National Maritime Day, International Day for Mine Awareness
7 World Health Day
10 World Homeopathy Day
17 World Haemophilia Day
18 World Heritage Day
21 Civil Services Day
22 World Earth Day

May

1 International Labour Day (May Day)
3 World Press Freedom Day
8 World Red Cross Day
12 International Nurses Day
17 World Telecommunications Day
21 Anti-Terrorism Day

June

5 World Environment Day
7 World Food Safety Day
12 World Day Against Labour
20 World Refugee Day
21 International Yoga Day
23 International Olympic Day

July

4 American Independence Day
7 International Day of Cooperatives
11 World Population Day

August

6 Hiroshima Day
8 World Senior Citizen's Day
9 August Kranti Divas, Nagasaki Day
12 International Youth Day
15 India's Independence Day
18 Day of the World's Indigenous Persons
29 National Sports Day (Dhyanchand's birthday)

September

5 Teacher's Day, Radhakrishnan's Birthday
8 International Literacy Day (UNESCO)
14 Hindi Day, World First Aid Day
15 Engineers Day, International Day of Democracy
16 World Ozone Day
18 International Day of Peace
21 Biosphere Day, World Alzheimer's Day
27 World Tourism Day

October

2 International Non-violence Day, Lal Bahadur Shastri and Mahatma Gandhi's Birthday
3 World Habitat Day
8 Indian Air Force Day
16 World Food Day
24 United Nations Day
31 National Integration Day

November

9 Legal Services Day
14 Children's Day, World Diabetes Day
17 National Epilepsy Day
20 Universal Children's Day (UN), Africa Industrialisation Day
26 Samvidhan Divas/Constitutional Day, World Environment Profection Day

December

1 World AIDS Day
4 Navy Day
7 Armed Forces Flag Day
10 Human Rights Day
11 UNICEF Day, International Mountain Day
14 National Energy Conservation Day
16 Vijay Divas
23 Kisan Divas (Farmer's Day)
25 National Good Governance Day

Important UNO Decades

2021-2030	International Decade of Ocean Science for Sustainable Development
2019-2028	United Nations Decade of Family Farming
2018-2028	International Decade for Action 'Water for Sustainable Development'
2016-2025	Third Industrial Development Decade for Africa
2016-2025	United Nations Decade of Action on Nutrition
2015-2024	International Decade for People of African Descent
2014-2024	United Nations Decade of Sustainable Energy for all
2013-2022	International Decade for the Reapproachment of Cultures (UNESCO)
2011-2020	Third International Decade for the Eradication of Colonialism, UN Decade on Biodiversity, Decade of Action for Road Safety
2010-2020	UN Decade for Deserts and the Fight against Desertification

UN International Year

2024	International Year of Camelids.
2022	International Year of Artisanal Fisheries and Aquaculture
2021	International Year of Peace and Trust, International Year of Creative Economy for Sustainable Development, International Year of Fruits and Vegetables.
2020	International Year of Plant Health, International Year of the Nurse and Midwife
2019	International Year of Indigenous Language, International Year of Moderation, International Year of the Periodic Table of Chemical Elements
2017	International Year of Sustainable Tourism for Development
2016	International Year of Pulses
2015	International Year of Light and Light-based Technologies, International Year of Soils

ABBREVIATIONS

A

ABM	Anti Ballistic Missiles
AD	Anno Domini (After the birth of Jesus)
ADB	Asian Development Bank
AFSPA	Armed Forces Special Power Act
AGOC	Asian Games Organising Committee
AHVS	Animal Husbandry and Veterinary Services
ALH	Advanced Light Helicopter
AIMPLB	All India Muslim Personal Law Board
AMRVT	Atal Mission for Rejuvenation and Urban Transformation
APPLE	Ariane Passenger Payload Experiment
ASLV	Augmented Satellite Launch Vehicle
ASI	Archaeological Survey of India
ATM	Automated Teller Machine
AWACS	Airborne Early Warning and Control System

B

BCSBI	Banking Codes and Standard Board of India
BARC	Bhabha Atomic Research Centre
BC	Before Christ (Before the birth of Jesus Christ)
BCTT	Bank Cash Transaction Tax
BDCA	Border Defence Cooperation Agreement
BIMSTEC	Bay of Bengal Initiative for Multisectoral Technical and Economic Cooperation
BIFR	Board of Industrial and Financial Reconstruction
BIS	Bureau of Indian Standards
BMDS	Ballistic Missile Defence System
BOSS	Bharat Operating System Solutions
BWA	Broadband Wireless Access

C

CAA	Civil Aviation Authority
CAD	Computer Aided Design
CAG	Comptroller and Auditor General
CAIR	Centre for Artificial Intelligence and Robotics
CAPES	Computer Aided Paperless Examination System
CAZRI	Central Arid Zone Research Institute
CBI	Central Bureau of Investigation
CECA	Comprehensive Economic Cooperation Agreement
CHOGM	Commonwealth Heads of Government Meeting
CID	Criminal Investigation Department
CISF	Central Industrial Security Force
CITES	Convention on International Trade in Endangered Species

CLASS Computer Literacy and Studies in School
CPCB Central Pollution Control Board
CPRI Central Power Research Institute
COIA Cellular Operators Association of India
COFEPOSA Conservation of Foreign Exchange and Prevention of Smuggling Activities Act
CPU Committee on Public undertakings
CSIR Council of Scientific and Industrial Research
CVC Central Vigilance Commission

D

DAVP Directorate of Advertising and Visual Publicity
DDT Dichlorodiphenyl trichloroethane
DFDR Digital Flight Data Recorder (BlackBox)
DLF Delhi Land and Finance
DPSA Deep Penetration Strike Aircraft
DRDO Defence Research and Development Organisation

E

EAS Employment Asssurance Scheme
ECT Electro-Convulsive Therapy
EEG Electro-Encephalography
EET Exempt Exempt Taxation
ESCAP Economic and Social Commission for Asia and the Pacific
EXIM Bank Export-Import Bank of India

F

FAO Food and Agriculture Organisation
FBI Federal Bureau of Investor
FBTR Fast Breeder Test Reactor
FEMA Foreign Exchange Management Act
FERA Foreign Exchange Regulation Act
FII Foreign Institutional Investor
FICCI Federation of India Chamber of Commerce and Industry
FLAG Fibre Optic Link Around the Globe
FRIBA Fellow of the Royal Institute of British Architects
FTII Film and Television Institute of India

G

GAAR General Anti Avoidance Rules
GAIN Global Alliance for Improved Nutrition
GANDHI Green Action for National Dandi Heritage Initiative
GATS General Agreement on Trade in Services
GATT General Agreement on Tariffs and Trade
GCHQ Government Communications Headquarters
GNSS Global Navigation Satellite System
GPS Global Positioning System
GSLV Geosynchronous Satellite Launch Vehicle
GST Goods and Service Tax

H

HAC Hindustan Aluminium Corporation
HAL Hindustan Aeronautics Limited
HESCO Himalayan Environmental Studies and Conservation Organisation
HRIDAY National Heritage City Development and Augmentation Yojana
HYVS High Yielding Variety Seeds

I

IAEA International Atomic Energy Agency
IBRD International Bank for Reconstruction and Development
ICAO International Civil Aviation Organisation
ICAR Indian Council of Agricultural Research
ICMR Indian Council of Medical Research
ICRC International Committee of the Red Cross
IGNOAPS Indira Gandhi National Old Age Pension Scheme
IMO International Maritime Organisation
INMAS Institute of Nuclear Medicines and Allied Sciences
INSAS Indian Small Arms System
IOC International Olympic Committee /Indian Oil Corporation
IPC Indian Penal Code
IPCC Intergovernmental Panel on Climate Change
IRBM Intermediate Range Ballistic Missile
IRS Indian Remate Sensing Satellite
IRNSS Indian Regional Nevigation Setellite System
ISTRAC ISRO Telemetry, Tracking and Command Network
IRDP Integrated Rural Development Programme

J, L, K

JAM Jandhan Yojana, Aadhar, Mobile numbers
JNNURM Jawaharlal Nehru National Urban Renewal Mission
LCA Light Combat Aircraft
LOAC Line of Actual Control
LTA Light Transport Aircraft

M

MAT Minimum Alternative Tax
METSAT Meteorological Satellite
MGC Mekong-Ganga Coorporation
MMS Multimedia Messaging Service
MISA Maintenance of Internal Security
MRTS Mass Rapid Transit System
MTCR Missile Technology Control Regime

N

NACO	National AIDS Control Organisation
NADA	National Anti-Doping Agency
NAFTA	North American Free Trade Agreement
NAFED	National Agricultural Cooperative Marketing Federation of India
NASA	National Aeronautics and Space Administration
NATO	North Atlantic Treaty Organisation
NCERT	National Council of Education Research and Training
NeGP	National e-Governance Plan
NEPA	National Environment Protection Authority
NHDP	National Highways Development Project
NSA	National Security Agency
NTPC	National Thermal Power Corporation
NALSA	National Legal Service Authority
NASSCOM	National Association of Software and Services Companies
NSAP	National Special Assistance Programme
NWDPRA	National Watershed Development Project for Rainfed Areas

O

OANA	The Organisation of ASIA-Pacific News Agencies
OCTOPUS	Organisation for Counter Terrorist Operations
OCI	Overseas Citizenship of India
OAS	Organisation of American States
OAU	Organisation of African Unity
ODS	Ozone Depleting Substances
OIC	Organisation of Islamic Countries
OSCE	Organisation for Security and Cooperation in Europe
OSCAR	Orbiting Satellite Carrying Amateur Radio

P

PAN	Permanent Account Number
PETA	People For Ethical Treatment of Animals
PURA	Providing Urban Amenities in Rural Areas
PN	Participatory Note
POTA	Prevention of Terrorism Act
PPE	Personal Protection Equipment
PRAGATI	Pro-Active Governance And Timely Implementation
PSLV	Polar Satellite Launch Vehicle
PWD	Public Works Department

Q, R

QIB	Qualified Institutional Buyer
QIP	Qualified Institutional Placement
RAF	Rapid Action Force
RBI	Reserve Bank of India
RCC	Reinforced Concrete Cement
RDSS	Radio Determination Satellite Service
RLV	Reusable Launch Vehicle
RNTCP	Revised National Tuberculosis Control Programme
RTGS	Real Time Gross Settlement System

S

SACHI	State Agency for comprehensive Health Insurance
SAFTA	South Asian Free Trade Area
SALT	Strategic Arms Limitation Talks
SAPTA	SAARC Prefencial Trading Agreement
SAVE	SAARC Audio Visual Exchange
SCOPE	Standing Committee of Public Enterprises
SIDBI	Small Industries Development Bank of India
SMART	Simple, Moral, Accountable Responsive and Transparent
START	Strategic Arms Reduction Treaty
SWIFT	Society for Worldwide Financial Telecommunication

T

TAPS	Tarapur Atomic Power Station
TIN	Tax Identification Number
TRAI	Telecom Regulatory Authority of India
TRIPS	Trade Related Intellectual Property Rights
TRYSEM	Training Rural Youth for Self Employment
TVoA	Tourist Visa on Arrival

U

UAV	Unmanned Aerial Vehicle
UIDAI	Unique Identification Authority of India
UNCTAD	United Nations Conference on Trade and Development
UNFPA	United Nations Fund for Population Activities
UNCTAD	United Nations conference on Trade and Development

V

VAT	Value Added Tax
VOIP	Voice Over Internet Protocol
VSAT	Very Small Aperture Terminals

W

WADA	World Anti-Doping Agency
WAVE	Wireless Access for Virtual Enterprise
WLL	Wireless in Local Loop
WWF	World Wide Fund for Nature
WWW	World Wide Web

Y

YMCA	Young Men's Christian Association
YWCA	Young Women's Christian Association

Z

ZSI	Zoological Survey of India
ZUPO	Zimbabwe United Peoples Organisation

BOOKS AND AUTHORS

Famous Books of Foreign Writers

Author	*Book*
Abdul Salam Zaeef	▪ My Life With the Taliban
Adam Smith	▪ Wealth of Nations
AG Noorani	▪ India— China Boundary Problems, 1846 to 1947
Arthur Conan Doyle	▪ Adventures of Sherlock Holmes
Ayad Akhtar	▪ Disgraced
Andy Marino	▪ Narendra Modi A Political Biography
Arthur I Miller	▪ Empire of the Stars
Arthur Stanley Eddington	▪ Expanding Universe
Alexander Bard	▪ Syntheism : Creating God in the Internet Age
Al Gore (former US Vice-President)	▪ An Inconvenient Truth
Adam Gilchrist	▪ True Colour (Autobiography)
Barack Obama	▪ Dreams from My Father : A Story of Race and Inheritance ▪ The Audacity of Hope: Thoughts on Reclaiming the American Dream
Benazir Bhutto	▪ Reconciliation : Islam Democracy and the West
Charles Dickens	▪ A Tale of Two Cities ▪ Oliver Twist ▪ Origin of Species ▪ David Copperfield ▪ Great Expectations
Catherin O' Flynn	▪ The News Where you are
Charles Darwin	▪ Descent of Man
Chester Bowles	▪ A View from Delhi
Cherie Blair	▪ Speaking for Myself
Dan Brown	▪ The Lost Symbol
David Loshak	▪ Pakistan Crisis
Desmond Tutu	▪ No Future without Forgiveness

Author	*Book*
DH Lawrence	▪ Sons and Lovers
Dominque Lapierre	▪ "A Rainbow in the Night-Nelson Mandela and the Tumultuous Birth of South Africa''
DP Lawrence	▪ Lady Chatterley's Lover
EM Forster	▪ A Passage to India
Edward Gibbon	▪ Decline and Fall of the Roman Empires
Eric Segal	▪ Love Story
Ernest Hemingway	▪ The Old Man and the Sea
Fyodor Dostoevsky	▪ The Idiot
George Bernard Shaw	▪ Back to Methuselah ▪ Man of Destiny ▪ Arms and the Man ▪ Man and Superman
George Eliot	▪ Mill on the Floss ▪ Middle March
GB Shaw	▪ Apple Cart
George Orwell	▪ Ninteen Eighty Four
George W Bush	▪ Decision Point (Autobiography)
HG Wells	▪ The War of the Worlds ▪ Time Machine ▪ Invisible Man
HB Stowe	▪ Uncle Tom's Cabin
Herta Muller	▪ The Appointment
Homer	▪ Odyssey
Herschell Gibbs	▪ To the Point (Autobiography)
Isaac Newton	▪ Principia Mathematica
John Milton	▪ Paradise Regained ▪ Paradise Lost
Jane Austen	▪ Pride and Prejudice ▪ Sense and Sensibility
Johann Goethe	▪ Faust
Jasper F Forde	▪ Shades of Grey

Author	*Book*
Jean Paul Sartre	▪ Iron in the Soul
Jeffrey Archer	▪ First Among Equals
Javier Moro	▪ The Red Sari
JK Rowling	▪ The Tales of Beedle the Bard
John Masefield	▪ Nine Days Wonder
John Ruskin	▪ Unto this Last
Jonathan Swift	▪ Gulliver's Travels
Joy Adamson	▪ Born Free
Jules Verne	▪ Around the World in Eighty days
Karl Marx	▪ Das Kapital
Katherine Mayo	▪ Mother India
Leo Tolstoy	▪ Resurrection ▪ War and Peace ▪ Anna Karenina
Lewis Carroll	▪ Through the Looking Glass ▪ Alice in Wonderland
L Fischer	▪ A Week with Gandhi
Lapierre and Collins	▪ Freedom at Midnight
Larry Collins and Dominique Lapierre	▪ Mountbatten and Independent India
Lord Byron	▪ Don Juan
Lord Curzon	▪ Problems of The East
Michele Kelley, Deepika D´Souza	▪ "The World Bank in India—Undermining Sovereignty, Distorting Development''
Machiavelli	▪ The Prince
Margaret Mitchell	▪ Gone with the Wind
Mark Twain	▪ Adventures of Tom Sawyer
Marlon James	▪ A Brief History of Seven Killings
Martin Amis	▪ The Pregnant Widow
Mathew Arnold	▪ Sohrab and Rustum
Maxim Gorky	▪ Mother
Michael Jackson	▪ Moon Walk (Autobiography)

Author	*Book*
Michael Phelps	▪ No Limits : The Will to Succeed
Nelson Mandela	▪ The Conversations with Myself
Oliver Goldsmith	▪ She Stoops to Conquer
Oscar Wilde	▪ Importance of Being Earnest
Pallava Bagla	▪ ''Destination Moon—India's quest for the Moon, Mars and Beyond''
Pearl S Buck	▪ The Rainbow
Philip Pullmen	▪ The Good Man Jesus and The Scoundrel Christ
Plato	▪ The Republic
Parvez Musharraf	▪ In the Line of Fire
Rober Louis Stevenson	▪ Kidnapped ▪ Treasure Island
Rudyard Kipling	▪ Jungle Book
Ricky Ponting	▪ At the Close of Play
Sydney Sheldon	▪ The Naked Face
Shoaib Akhtar	▪ Controversially Yours (Autobiography)
Salman Khurshid	▪ The other side of the Mountain
TS Eliot	▪ Murder in the Cathedral ▪ The Wasteland
TC Boyle	▪ Wild Child
Thomas Moor	▪ Utopia
Toni Morrison	▪ What Moves at the Margin : Selected Non Fiction
Thomos Weber	▪ Gandhi at First Sight
Tony Blair	▪ A Journey
U Thant	▪ Viem from the UN
Walter Scott	▪ Ivanhoe
Winston Churchill	▪ Gathering Storm
William Shakespeare	▪ All's Well That Ends Well ▪ Twelfth Night ▪ Comedy of Errors ▪ Romeo and Juliet ▪ The Tempest ▪ Macbeth , Julius Caesar
ZA Bhutto	▪ If I am Assassinated
Mala Yousafzai	▪ We Are Displaced

Famous Books of Indian Writers

Author	*Book*
Dr APJ Abdul Kalam	▪ Ignited Minds : Unleashing the Power within India ▪ You are Born to Blossom ▪ My Journey ▪ India 2020-A vision for the New Millennium ▪ Wings of Fire
Amrita Pritam	▪ Kora Kagaz ▪ Death of a City ▪ Kagaz Te Kanwas
Amartya Sen	▪ Development of Freedom ▪ The Idea of Justice
Abul Kalam Azad	▪ India Wins Freedom
Amish Tripathi	▪ The Immortals of Meluha ▪ The Secret of the Nagas ▪ The oath of the Vayuqutras
Anuradha Roy	▪ The Folded Earth
Aravind Adiga	▪ The White Tiger
Asha Purna Devi	▪ Subarnalata
Aurobindo Ghosh	▪ Life Divine
Bankim Chandra Chattopadhyay	▪ Devi Chaudharani ▪ Anand Math
Balwant Gargi	▪ Naked Triangle
BG Tilak	▪ Gita Rahasya
BR Ambedkar	▪ What Congress and Gandhi have done to Untouchables
Brig JP Dalvi	▪ Himalayan Blunder
Chetan Bhagat	▪ One Night @ the Call Centre ▪ Making India Awesome ▪ Revolution 2020, Half Girlfriend ▪ One Indian Girl
Dada Bhai Naoroji	▪ Poverty and Un-British Rule in India
Daisy Mason	▪ The To-Let House
Din Bandhu Mitra	▪ Neel Darpan
Dr Radha Krishnan	▪ Hindu View of Life
Dr Rajendra Prasad	▪ India Divided
Pranab Mukherjee	▪ The Congress and The Making of Indian Nation ▪ The Dramatic Decade : The Indira Gandhi Year's ▪ Selected Speech (Vol. I, II and III)
Gen Ayub Khan	▪ Friends not Master
Gopal Krishna Gandhi	▪ Of a Certain Age : Twenty Life Sketches

Author	*Book*
Gopinath Mohanty	▪ Moti Mahal
Govind Mishra	▪ Dhool Paudho Par
Hamid Ansari	▪ Travelling Through Conflict
IK Gujral	▪ Matters of Discretion (Autobiography)
Indira Gandhi	▪ My Truth
Imran Hashmi	▪ The Kiss of Life
Janardan Thakur	▪ All the Prime Minister's Men
Jaswant Singh	▪ Jinnah—India, Partition, Independence
Jawaharlal Nehru	▪ Bunch of Old Letters ▪ Glimpses of World History ▪ Letters from a Father to his Daughter ▪ The Discovery of India
JP Narayan	▪ Prison Diary ▪ To all Fighters of Freedom, Why Socialism?
Jyotiba Phule	▪ Ghulam Giri and other Stories
Khushwant Singh	▪ The Sunset Club ▪ Sahibs Who Loved India ▪ Why I Supported the Emergency ▪ Truth, Love and A Little Malice ▪ We Indians ▪ A Bride for the Sahib
K Natwar Singh	▪ My China Diary
KP Mathur	▪ The Unseen Indira Gandhi
Kailash Satyarthi	▪ Azad Bachpan Ki Aor
Kapil Dev	▪ Straight from the Heart
Kapil Sibal	▪ I Witness—Partial Observation
Kuldip Nayar	▪ The Judgement
Kingshuk Nag	▪ The Namo Story : A Political Life
Khan Abdul Ghaffar Khan	▪ Pakhtoon
KM Munshi	▪ I Follow the Mahatama
Lala Lajpat Rai	▪ Unhappy India
Lal Krishna Advani	▪ My Country My Life
Mahadevi Verma	▪ Yama
Mahatma Gandhi	▪ My Experiments with Truth ▪ Hind Swaraj
M Veerappa Moily	▪ Shree Ramayana Mahanveshanam
Maj HPS Ahluwalia	▪ Face of Everest
Manohar Malgaonkar	▪ A Bend in the Ganges
Maithili Sharan Gupt	▪ Saket ▪ Yashodhara

Author	*Book*
Meghnad Desai	▪ The Rediscovery of India
MJ Akbar	▪ Nehru : The Making of India
Morarji Desai	▪ A Minister and his Responsibilities
Mulk Raj Anand	▪ The Village ▪ Seven Summers ▪ Two Leaves and a Bud ▪ Coolie
Munshi Prem Chand	▪ Rang Bhoomi ▪ Godan
Nandan Nilekani	▪ Imagining India : Ideas for a New Century
Narayan Lakshman	▪ Patrons of the Poor : Caste Politics and Policy Making in India
Narayan Shehgal	▪ A Voice of Freedom
Narendra Modi	▪ "Convenient Action : Gujarat's Response to Climate Change''
Nilanjan Mukho Padhaya	▪ Narendra Modi : The Man, the Times
Govind Mishra	▪ Dhool Pandho Par
Nirad C Choudhary	▪ An Unknown Indian
NR Narayan Murthy	▪ A Better India A Better World
PM Nayar	▪ The Kalam Effect: My Years with the President
PVR Rao	▪ Defence without Drift
Rabindra Nath Tagore	▪ Chitra ▪ The Post Office ▪ Sadhana: The Realisation of Life
Ramachandra Guha	▪ "Makers of Modern India'' ▪ "India after Gandhi : The History of the World's Largest Democracy'' ▪ Patriots and Partisans ▪ Gandhi before India
RK Narayan	▪ Guide ▪ Dark Room ▪ The Vendor of Sweets
Raghunath Mashelkar	▪ Timeless Inspirator—Reliving Gandhi
Rajmohan Gandhi	▪ The Good Boat Man : A Portrait of Gandhi
Ramdhari Singh Dinkar	▪ Rashmirathi
Ramesh Chandra Shah	▪ Vinayak
Rashika Chaube and Chhaya Mahajan	▪ An Inspirational Journey: Pratibha Devi Singh Patil; The First Women President of India
Ramesh Chandra Dutta	▪ Economic History of India

Author	*Book*
Ravi Shankar	▪ My Music, My Life
Ronald Segal	▪ Crisis of India
Ram Mohan Roy	▪ Percepts of Jesus ▪ A Gift of Monotheists
Ruskin Bond	▪ Notes from a Small Room ▪ A Little Book of Happiness
Russ M Lala	▪ For the Love of India : The Life and Times of Jamsetji Tata
S Nihal Singh	▪ My India
Sachin Tendulkar	▪ Playing it my way
Shashi Tharoor	▪ The Elephant, the Tiger and the Cell Phone : Reflections on India in the 21st century ▪ Pax Indica, Paradoxical Prime Minister
Shashi Tharoor and Shaharyar Khan	▪ Shadows Across the Playing Field : 60 years of India-Pak Cricket
Shobha De	▪ Superstar India : From Incredible to Unstoppable
Shyam Bhatia	▪ Good Bye Shahzadi (A Biography of Benazir Bhutto)
SK Nandi	▪ Ramacharit
Subramanian Swamy	▪ "Electronic Voting Machines—Unconstituti onal and Tamperable''
Sunil Gavaskar	▪ Sunny Days
Surjit Singh Barnala	▪ My Other Two Daughters
Sharat Chandra Chatterjee	▪ Devdas ▪ Parineeta
Shatrughan Singha	▪ Anything But Khamosh
Sania Mirza	▪ Ace Against Odds
Sarojini Naidu	▪ The Song of India ▪ Broken Wing
Syyed Amir Ali	▪ The Spirit of Islam
SC Bose	▪ The Indian Struggle
Sir Syyed Ahmed Khan	▪ Causes of the Indian Mutiny
Sri Aurobindo Ghosh	▪ Savitri
Swami Dayanand	▪ Satyarth Prakash
TS Krishnanmurthy	▪ The Miracle of Democracy : India's Amazing Journey
Upinder Singh	▪ A History of Ancient and Early Medieval India from Stone Age to the 12th century
VD Savarkar	▪ Indian War of Independence

Author	*Book*
V V Giri	▪ Voice of Conscience
Vinita Kamte	▪ To the Last Bullet
Vinod Mehta	▪ Lucknow Boy (Autobiography)
Vijay Lokpally	▪ Driven
Yuvraj Singh	▪ Test of My Life Autobiography
YV Reddy	▪ Global Crisis Recession and Uneven Recovery
Navin Chawla	▪ Every Vote Counts

Author	*Book*
Jairam Ramesh	▪ Indira Gandhi : A Life in Nature, A Chequered Brilliance : The Many Lives of VK Krishna Menon
D.G. Tendulkar	▪ Gandhi in Champaran
Shri Narendra Modi	▪ Exam Warriors
Ramchandra Guha	▪ Gandhi the year that changed the World
VVS Laxman	▪ 281 and Beyond
Raghuram Rajan	▪ I do What, I do The Third Pillar
Abhijeet Banerjee & Estner Duflo	▪ Good Economics for Hard Times, Poor Economics

Award Winning Authors of Indian Origin and their Writings

Author	*Writing*	*Award*
Salman Rushdie	▪ Midnight Children ▪ Shame ▪ The Moor's Last Sigh ▪ Fury ▪ The Satanic Verses	Booker Prize 1981 (Midnight Children)
Vikram Seth	▪ The Golden Gate ▪ A Suitable Boy ▪ An Equal Music ▪ A Suitable Girl (2013) ▪ Summer Requiem	Padma Shri in Literature and Education 2007
Arundhati Roy	▪ The God of Small things ▪ The Algebra at Infinite Justice	Booker Prize 1997 (The God of Small Things)
Rohinton Mistry	▪ Such a Long Journey ▪ Family Matters ▪ A Fine Balance	Booker Prize 1991 (Such a Long Journey)
VS Naipaul	▪ A House for Mr Biswas ▪ India A Wounded Civilization ▪ An Area of Darkness ▪ India : a Million Mutinies now ▪ The Masque of Africa ▪ A Bend in The River	Nobel Prize in Literature 2001 (for having united perceptive narrative and incorruptible scrutiny in works that compel)
Amitav Ghosh	▪ The Circle of Reason ▪ River of Smoke ▪ The Glass Palace ▪ Shadow Line ▪ The Calcutta Chromosome ▪ The Hungry Tide ▪ Sea of Poppies ▪ In an Antique Land ▪ Flood of Fire ▪ The Great Derangement	Padma Shri by Government of India 2007 (for his best work in English language)
Jhumpa Lahiri	▪ The Namesake ▪ The Interpreter of Maladies ▪ The Unaccustomed Earth	Pulitzer Prize 2000 (The Namesake)
Shashi Tharoor	▪ The Great Indian Novel ▪ Show Business ▪ Indira Shastra: Reflections on the Nation in Our Time ▪ India : From Midnight to Millenium ▪ An Era of Darkness : British Empire in India ▪ Why I am Hindu	Commonwealth Writer's Prize (The Great Indian Novel). Sahitya Academy Award 2019 (for An Era of Darkness : British Empire in India)
Upamanyu Chatterjee	▪ The Mammaries of the Welfare State ▪ English, August : Indian Story ▪ Way to Go ▪ Weight Loss ▪ The Last Burden	Sahitya Akademi Award 2004 (The Mammaries of the Welfare State)
Arvind Adiga	▪ The White Tiger ▪ Last Man in Tower ▪ Between the Assassinations	Man Booker Prize 2008
Kiran Desai	▪ The Inheritance of Loss ▪ Hullabaloo in the Guava Island	Man Booker Prize 2006

UNITED NATIONS (UN)

- The United Nations (UN) is a world level organisation formed in 1945. It came into existence after World War II, when the leaders of the world, including American President Roosevelt and British Prime Minister Churchill, decided to create a world level organisation that would help to ensure peace.
- The original membership of 51 nations has grown to 193 members. The 193rd member being the newly created South Sudan. The United **Nation's** Headquarters is in New York City. The UN also has offices in Geneva, (Switzerland), and Vienna, (Austria).
- Six official languages are spoken and used in documents at the United Nations: Arabic, Chinese, English, French, Russian and Spanish.
- **The General Assembly** The General Assembly is the main place for discussions and policy making in the United Nations. All the members of the UN are the members of General Assembly.
- **The Security Council** The Security Council has primary responsibility for the maintenance of international peace and security. The Security Council is made up of 15 members.
- This comprise of 5 permanent and 10 non permanent member. India is a non-permanent members. There are five permanent members of the Security Council-China, France, Russia, United Kingkdom and USA.
- The International Court of Jusitce (ICJ), located in The Hague, Netherlands, is the primary judicial organ of the United Nations, established in 1945 by the United Nations Charter, the Court began work in 1946, as the successor to the Permanent Court of International Justice.

United Nations Secretary Generals

Name	*Tenure*	*Country*	*Resigned/Retired*
Trygve Lie	1946-53	Norway	Resigned
Dag Hammarsk jold	1953-61	Sweden	Died in plane crash in Northern Rhodesia (now Zambia)
U Thant	1961-71	Myanmar	Declined to consider a third term
Kurt Waldheim	1972-81	Austria	China vetoed his third term
Javier Perez	1982-91	Peru	Refused to be considered for a third term
Boutros Ghali	1992-96	Egypt	The United States vetoed his second term
Kofi Annan	1997-06	Ghana	Retired after two full terms
Ban ki- Moon	2007- 2016	South Korea	Retired after two full terms
Antonio Gueterres	2017-Present	Portugal	

Important United Nation Agencies

Name of Agency	*Estd in*	*Headquarter*	*Objective*
International Labour Organisation (ILO)	1919	Geneva	To improve conditions and living standard of workers.
International Monetary Fund (IMF)	1945	Washington DC	Promotes international monetary cooperation.
Food and Agricultural Organisation (FAO)	1945	Rome	To improve living conditions of rural population.
International Bank for Reconstruction and Development (IBRD)	1944	Washington DC	To provide funds from different sources.
United Nations International Children's Emergency Fund (UNICEF)	1946	New York	To promote children's welfare all over the world.

Name of Agency	*Estd in*	*Headquarter*	*Objective*
United Nations Educational, Scientific and Cultural Organisation (UNESCO)	1945	Paris	To promote collaboration among nations through education, science and culture.
International Telecommunication Union (ITU)	1865	Geneva	Sets international regulations for radio telegraph, telephone and space radio communications.
International Civil Aviation Organisation (ICAO)	1947	Montreal, Canada	It codifies the principles and techniques of international air navigation and fosters the planning and deve- lopment of international air transport to ensure safe and orderly growth.
World Health Organisation (WHO)	1948	Geneva	Attainment of highest possible level of health by all people.
International Atomic Energy Agency (IAEA)	1957	Vienna	To promote peaceful uses of atomic energy.
International Development Association (IDA)	1960	Washington DC	An affiliate of the World Bank, aims to help under-developed countries raise living standards.
United Nations Development Programme (UNDP)	1965	New York	Helps developing countries increase the wealth producing capabilities of their natural and human resources.
United Nations Environmental Programme (UNEP)	1972	Nairobi	Promotes international cooperation in human environment.
World Trade Organisation (WTO)	1995	Geneva	Setting rules for world trade to reduce tariffs.
UN Women	2010	New York City, USA	It works for the empowerment of women and girls.

International Organisations and Groups

Organisation and HQ	*Member*	*Objective*
Red Cross Estd in 1863 (Geneva, Switzerland)	192	International Humanitarian Movement for relief of suffering in time of war/disaster.
International Olympic Committee (IOC) Estd in 1894 (Switzerland)	95	To promote the Olympic ideals and administer olympic games.
International Criminal Police Organisation (INTERPOL) Estd in 1923 (Lyon, France)	194	To promote international cooperation among criminal police authorities.
The Commonwealth (London) Estd. in 1926	54	It was originally known as 'The British Commonwealth of Nations'. It is an association of sovereign and independent states which formally made up the British empire.
Arab League (AL) Estd in 1945 [Cairo (Egypt)]	22	To promote economic, social, political and military cooperation.
International Organisation for Standardisation (ISO) Estd in 1947 (Switzerland)	163	To promote the development of international standards.
North Atlantic Treaty Organisation (NATO) Estd in 1949 (Brussels)	29	Mutual defence and cooperation
Colombo Plan Estd in 1950 (Sri Lanka)	25	To promote economic development in South and South-East Asia.
South-East Asia Treaty Organisation (SEATO) Estd in 1954 (Bangkok)	8	To provide for collective and economic cooperation in South-East Asia.
Organisation of Petroleum Exporting Countries (OPEC) Estd in 1959 [Vienna (Austria)]	13	Attempts to set world prices by controlling oil production and also pursues member interest in trade and development.

Organisation and HQ	*Member*	*Objective*
World Wildlife Fund For Nature (WWF) (NGO) Estd in 1961 (Switzerland)	All countries	To save the wildlife from extinction.
Amnesty International (NGO) Estd in 1961 (London)	—	To keep a watch over human rights violation worldwide. Got Nobel Prize in 1977 for Peace.
Non-Aligned Movement (NAM) Estd. in 1961	120	Political cooperation and establishment of separate identity from both USA and USSR (in the Cold-War era).
Group of Seventy Seven (G-77) Estd in 1964	134	To promote economic cooperation among developing nations.
Asian Development Bank (ADB) Estd in 1966 (Manila)	68	To promote socioeconomic development in Asia
Association of South-East Asian Nations (ASEAN) Estd in 1967 (Jakarta)	10	Regional economic, social and cultural cooperation among the non-communist countries of South-East Asia.
Organisation of Arab Petroleum Exporting Countries (OAPEC) Estd in 1968 [Safat (Kuwait)]	11	To build an integrated petroleum industry as a cornerstone for future economic integration amongst Arab countries.
Organisation of Islamic Cooperation (OIC) Estd in 1969 (Saudi Arabia)	57	To promote Islamic solidarity among member states and to consolidate cooperation among members.
Gulf Cooperation Council (GCC)Estd in 1981	6	It is a political and economic union of the Arab states.
Nordic Council Estd in 1952	8	Geo-political, inter-parliamentary forum for cooperation among Nordic countries.
South Asian Association for Regional Cooperation (SAARC) Estd in 1985 (Kathmandu)	8	To promote economic, social and cultural cooperation in South Asia
Group of 15 (G-15) Estd in 1989	17	To promote economic cooperation among developing nations.
Asia Pacific Economic Cooperation (APEC) Estd in 1989 (Singapore)	21	To promote trade and investment in the Pacific basin.
Commonwealth of Independent States (CIS) Estd in 1991 [Kirava (Belarus)]	9	To coordinate inter-common wealth relations and to provide a mechanism for the orderly dissolution of the USSR.
European Union Estd. in 1993 [Brussels (Belgium)]	28	To create a United Europe in which member countries would have such strong economic and political bonds that war would cease to be a recurring fact.
Sanghai Cooperation Organisation (SCO) Estd in 1996 (Beijing, China)	8	To develop mutual cooperation.
Group of Twenty (G-20) Estd in 1999	20	For cooperation and consultation or matters pertaining to the International Financial System.
African Union Estd in 2002	55	To accelerate the political and socio-economic integration of the continent.
BRICS (*Brazil, Russia, India, China and South Africa*) Estd in 2006	5	To improve the economical condition of the country.
BASIC Estd in 2009	4	To coordinate the policies of developing countries regarding climate change.
Bay of Bengal Initiative for Multi-sectoral Technical and Economic Cooperation (BIMSTEC) (Estd. in 1997 at Bangkok)	7	Multi-sectoral Technical and Economic Cooperation among Members
International Solar Alliance (ISA) (Estd. in Paris (2015))	121	Promotion of Solar Energy among members countries.

SPORTS

OLYMPICS

- Olympics games were started in 776 BC on Mount Olympics in the honour of Greek God 'Zeus'.
- They were stopped by a royal order of the Roman Emperor Theodosius in AD 394.
- The modern Olympic Games were started in **Athens** the capital of Greece on **6th April, 1896** with great afforts made by French nobleman, **Baron Pierre de Coubertin.**
- Separate Winter Olympic Games began in 1924.
- Women have been participating in the Olympics since 1912.
- The Olympic Games are organised after every four years.
- The Olympic Flag is made up of White Silk and contains five interwined rings as the Olympic Emblem.
- The five interlaced rings are arranged in 3-2 pattern on a white background, with the blue ring to the extreme left, followed by yellow black, green and red. The blue represented Europe, Black for Africa red for Americas (North and South America both), yellow for Asia and green for Oceania (Australia and New Zealand).
- The official Olympic Motto is *Citius, Altius, Fortius*, a Latin phrase meaning swifter, higher and stronger.
- The Head Office of International Olympic Committee (IOC) is at Lausanne (Switzerland).
- 31st Olympic Games of 2016 was conducted at Rio-de-janeiro.
- 2018 winter olympics were held at Pyeonchang county, South Korea. Norway topped the medal tally. The 2022 winter olympic will be held in Beijing, China.
- Tokyo, will host 2020 summer olympic (postponed) to 2021 due to COVID-19 pandemic.

Indian Medal Winners at 2016 Rio Olympics

Silver	PV Sindhu	Badminton	Women's single
Bronze	Sakshi Malik	Wrestling	Women's free style 58 kg

Olympic Flame

- The Olympic flame symbolises the continuity between ancient and modern games.
- It was at the Amsterdam Games in 1928 that for the first time an Olympic flame was ceremonially lighted and burned in a giant torch at the entrance of the stadium. The modern version of the flame was adopted in 1936 at the Berlin Games.

Some Facts Related to Olympics

- ''Norman Pritchard'' was the first Indian player to participate in Olympic (2nd Olympic Games in 1900) and won two silver medals in athletics.
- Marrie Lila Rao is 1st Indian woman participant in the Olympics after independence.
- India officially participated in the Olympics for the first time in the Sixth Olympic Games (1920) at Antwerp (Belgium).
- The Indian Olympic Council was established in 1927.
- Sir Dorabji Tata was its first President.
- Sonia Denancord (Canada) was the 1st woman referee in football in Atlanta Olympics.
- Maximum number of medal winner sports women is Larissa Lavyanina (18 medals including 9 gold medals).
- Maximum gold medal winner sports woman is Christina Otty. She got 6 gold medals in swimming in Seoul Olympic of 1986.

Commonwealth Games

- The Commonwealth games are a festival of sports of the Commonwealth countries.
- The first Commonwealth Games were held in 1930 in Hamilton, Canada.
- Since 1930, the games have been conducted every four years except for 1942 and 1946.
- The Commonwealth Games Federation (CGF) is the organisation which is responsible for the direction and control of the Commonwealth Games.
- There are currently 53 members of the Commonwealth of Nations and 71 team participated in the games.
- The 2010 Commonwealth Games were held in New Delhi, India.
- 20th Commonwealth Games of 2014 were held in Glasgow (Scotland-UK).
- 21st Commonwealth Games of 2018 was held in Gold Coast Australia (where Australia topped the medal tally while India ranked 3rd with 26 Gold).
- 22nd Commonwealth Games 2022 will be held in Birmingham, England.

Asian Games

- The Asian Games, also called the Asiad, are a multi-sport event held every four years among athletes from all over Asia.
- The games are regulated by the Olympic Council of Asia (OCA), under the supervision of the International Olympic Committee (IOC).
- The first Asian Games were held in 1951 in New Delhi (India). The 2010 Asian Games were held in Guangzhou, China.
- 2014 Asian Games were held in Incheon, South Korea.
- 2018 Asian Games were held in Jakarta (Indonesia). India finished at 8th position with 15 golds.
- The 2022 Asian Games is scheduled to be held in Hangzhou, China.

South Asian Federation (SAF) Games

- Like the Commonwealth Games, the South Asian Federation (SAF) Games is a sports festival of South Asian countries.
- The South Asian Sports Federation was formed in New Delhi on 26th November, 1982.
- The first South Asian Federation Games were held at Kathmandu (Nepal) in 1984.
- No SAF Game were staged in 1986 as it was the year of commonwealth and Asian Games.
- The SAF Games have been rechristened South Asian Games on 2nd April, 2004. Afghanistan joined the Games in 2006.
- The 2016, SAF Games were held in, India.
- SAF game 2019 was held in Nepal (Winner-India) while SAF games, 2022 is scheduled to be held in Lahore, Pakistan.

Cricket World Cup

- The first Cricket World Cup was organised in England in 1975. A separate women's Cricket World Cup has been held every four years since 1973.
- The Cricket World Cup Tournament is organised by the International Cricket Council (ICC).
- The ICC was founded in 1909 and its headquarter is located in Dubai, United Arab Emirates.
- In 1877, the first cricket test match was played in Melbourne between England and Australia.
- The first one day International cricket match was played in the year 1971 between England and Australia in Melbourne.
- The Board of Control for Cricket in India (BCCI) was formed in 1927.

List of Cricket World Cups

Year	Final Match	Result
1975	Lord's, England	West Indies beat Australia
1979	Lord's, England	West Indies beat England
1983	Lord's, England	India beat West Indies
1987	Kolkata, India	Australia beat England
1992	Melbourne, (Australia)	Pakistan beat England
1996	Lahore, Pakistan	Sri Lanka beat Australia
1999	Lord's, England	Australia beat Pakistan
2003	Johannesburg, SA	Australia beat India
2007	Bridgetown, WI	Australia beat Sri Lanka
2011	Mumbai, India	India beat Sri Lanka
2015	Australia/New Zealand	Australia beat New Zealand
2019	England	England beat New Zealand

T20 World Cup

It is organised by the International Cricket Council (ICC). It is held every two years.

ICC Twenty-20 Cricket World Cup

Year	Host Nation	Final Venue	Winner	Runner-up
2007	South Africa	Johannesburg	India	Pakistan
2009	England	Lord's, London	Pakistan	Sri Lanka
2010	West Indies	Barbados	England	Australia
2012	Sri Lanka	R Premdasa Stadium, Colombo	West Indies	Sri Lanka
2014	Bangladesh	Sher-e-Bangla Cricket Stadium, Dhaka	Sri Lanka	India
2016	India	Kolkata, Eden Garden	West Indies	England
2021	India*	—	—	—

- The 2020 ICC Twenty-20 Cricket World Cup (scheduled to held in Australia) was postponed to 2021 and now India will host the tournament in October-November 2021.

IPL (Indian Premier League)

It was initiated by Board of Control for Cricket in India (BCCI). It is Headquartered in Mumbai. The first IPL tournament was started in 2008. Seventh IPL Tournament held in Bangalore, Kolkata Knight Riders defeated. King XI Punjab.

Football World Cup

- The Football World Cup is organised by FIFA (Federation of International Football Association).
- The headquarters of FIFA is located in Zurich, Switzerland.
- The World Cup is called 'Jules Rimet Cup' named after the name of FIFA President Jules Rimet.
- The first Football World Cup was organised in Uruguay in 1930.
- In 1942 and 1946, the Football World Cup was not played because of the II World War.
- Brazil is the only nation to have parti- cipated in every World Cup so far.
- The 20th Fifa World Cup held in Brazil. Germany defeated Argentina.
- The 2018 Football World Cup was hosted by Russia in 2018 (winner France) and Qatar will be the host city in 2022.

Hockey World Cup

- The Hockey World Cup is organised by the International Hockey Federation (FIH) once in four years. The headquarters of FIH is located in Lausanne, Switzerland.
- The first Hockey World Cup was organised in Barcelona (Spain) wa 1971 and winner was Pakistan.
- Women's Hockey World Cup is being held since 1974.
- The 13th Women's Hockey World Cup in 2014, was held in The Hague, Netherlands. Netherlands won the tournament for a record seventh time after defeating Australia. The 14th Women's Hockey World Cup of 2018, was held in London, England.
- The 14th Men's Hockey World Cup 2018 was held at Kalinga stadium in Bhubaneswar, India. Belgium became champion defeating Netherland.
- The 15th Men's Hockey World Cup 2023, will be held at the Kalinga Stadium in Bhubaneswar and in Rourkela, Odisha.

Nickname of Players

Player	*Nickname*
Major Dhyanchand	Hockey ka Jadugar
Milkha Singh	Flying Sikh
PT Usha	Payyoli Express
PT Usha	Golden Girl
Shoaib Aktar	Rawalpindi Express
Rahul Dravid	The Wall
Harbhajan singh	Bhajji, The Turbanator
Javagal Srinath	Mysore Express
Paes and Bhupati	Indian Express

National Sports of Some Countries

Country	*National Sport*
Australia	Cricket
Canada	Lacrose (Ice Hockey)
China	Table Tennis
England	Cricket
India	Hockey
Japan	Sumo
Malaysia	Badminton
Scotland	Rugby/Football
Spain	Bull Fighting
USA	Baseball

Important Sport Terms

Sport	*Terms*
Basketball	Dunk, Front court, Lay up, Held ball, Pivot, Rebound, Steal
Cricket	Bye, Draw, Googly, Topspin, Over throw, Duck, Hit wicket, Cut, Pull, Full Toss
Football	Bend dribble, Dissent, Dummy, Feint, Free kick, Header, Red card, Throwins
Hockey	Bully, Striking, Circle, Post back
Chess	Castle, Diagonaes, Files, Pawns, Peices, Promote, Gambit, Pawn
Boxing	Jab, laying on Knock, Second out habbit punch, Upper cut
Badminton	Loab, Let, Drive, Drop, Love
Polo	Chuker, Bunker
Baseball	Diamond, Home run, Put out, Strike, Ant-rubber.
Rifle Shooting	Target, Muzzle fulb, Bulls eye
Wrestling	Hold Sager, Rebuts, Hal, Nelson, Free Style
Golf	Fore some, Stymie, T, put hole, Caddy, Nib lick, Iron, The green, Bunkeer
Billiards	Jigger, Pot, Break pot, In off, Cans, Bolting, Long, Hazard, Cue
Swimming	Breast Stroke, Twist, Butterfly, Crawl, Spring Board
Volley Ball	Antennae, Attack hit, Libero, Service, Set up, Blocking, Dribbling
Lawn Tennis	Advantage, Ace, Dence, Volley, Foot Foult, Smash, Grand-Slam, Slice, Love
Table Tennis	End line, Flat hit, Foil, Service, Phnholder grip, Reverse, Top-spin, Couter-hitting, Let

Important Cups and Trophies

Sport	*Cup and Trophy*
Cricket	Irani Trophy, Duleep Trophy, Ranji Trophy, Vijay Hazare Trophy, Asia Cup, Deodhar Trophy, CK Naidu Trophy, Cooch-Behar Trophy, Gandhi-Mandela Series, The Ashes Series etc.
Football	Durand Cup, Nizam Gold Cup, Rovers Cup, Sanjay Gold Cup, Santosh Trophy, Subroto Mukherjee Cup, Vittal Trophy, Nehru Gold Cup
Hockey	Agha Khan Cup, Azlan Shah Cup, Nehru Trophy, Dhyanchand Trophy, Beighton Cup, Scindia Gold Cup, Modi Gold Cup, Indira Gandhi Gold Cup, Rangaswami Cup, Khan Abdul Gaffar Cup.
Golf	Canada Cup, Muthian Gold Cup, Ryder Cup, Walker Cup
Table Tennis	Corbillion Cup (women), Jayalaxmi Cup (women), Swaythling Cup (men)
Lawn Tennis	Davis Cup, Hamlet Cup, Australian Open, French Open, Wimbledon, US open
Badminton	Thomas Cup (men), Uber Cup (women), Narang Cup, Malaysian open
Boxing	Aspy Adjania Trophy
Rowing	Wellington Trophy
Bridge	Ruia Trophy
Polo	Ezra Cup, Winchestor Cup, Radha Mohan Cup, West Chester Cup

Sports Organisations

Organisation		*Game*	*Headquarter*
FIDE	Federation International des Echecs (*World Chess Federation*)	Chess	Athens (*Greece*)
FIFA	Federation International de Football Association	Football	Zurich (*Switzerland*)
IHF	Indian Hockey Federation	Hockey	Lausanne (*Switzerland*)
ICC	International Cricket Council	Cricket	Dubai (*UAE*)
ITTF	International Table Tennis Federation	Table Tennis	Lausanne (*Switzerland*)